

MICROCOMPUTER MN101E

MN101EF56K/57G/76K
LSI User's Manual Vol.1

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About This Manual

■Objective

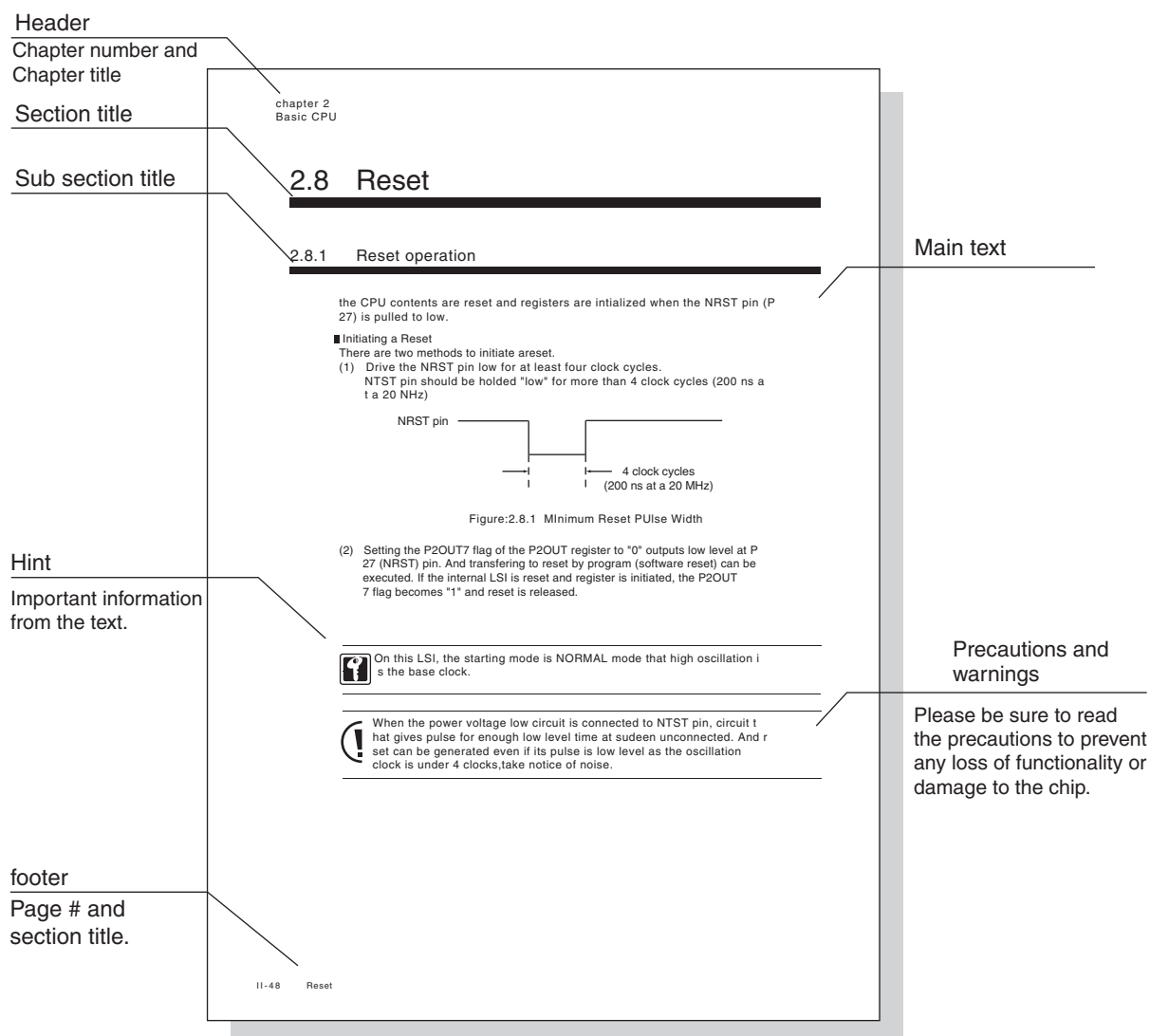
The primary objective of this LSI manual is to describe the features of this product including an overview, CPU basic functions, interrupt, port, timer, serial interface, and other peripheral hardware functions.

Each section consists of brief functional information, block diagrams and the details of control registers including operation methods and setting examples.

■Structure of This Manual

Each section of this manual consists of a title, summary, main text, hint, precautions and warnings, and references.

The layout and definition of each section are shown below.



This page serves as an example to the explanations above. It may be different on an actual page.

■Finding Desired Information

This manual provides three methods for finding the desired information quickly and easily.

- 1.Refer to the index at the front of the manual to locate the beginning of each section.
- 2.Refer to the table of contents at the front of the manual to locate the desired titles.
- 3.The chapter number and chapter title are located at the top corner of each page, and the section titles are located at the bottom corner of each page.

■Related Manuals

Note that the following documents related to MN101E series are available.

- "MN101E Series Instruction Manual"
<Describes the instruction set.>
- "MN101C/MN101E Series C Compiler User's Manual Usage Guide"
<Describes the installation, commands and options of the C Compiler.>
- "MN101C/MN101E Series C Compiler User's Manual Language Description"
<Describes the syntax of the C Compiler.>
- "MN101C/MN101E Series C Compiler User's Manual Library Reference"
<Describes the standard library of the C Compiler.>
- "MN101C/MN101E Series Cross-assembler User's Manual"
<Describes the assembler syntax and notation.>
- "PanaX EX Installation Manual"
<Describes the steps to install the Integrated Development Environment (DebugFactory Builder), C compiler and the real-time OS. It also describes the procedure to setup the on-board environment.>
- "In-Circuit Emulator Installation Manual"
<Describes the steps to install the Integrated Development Environment (DebugFactory Builder), C compiler and the real-time OS. It also describes the procedure to setup the in-circuit emulator.>
- "PanaX EX Commander User's Manual"
< Describes the usage and notes on designing target board.>

■Caution

The related documents listed above are subject to change without notice.
Use the latest version of each document for designing.

■Contact Information

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Chapter 1 Overview

1.1 Overview

1.1.1 Overview

The MN101E series of 8-bit single-chip microcomputers (the memory expansion version of MN101C series) incorporate multiple types of peripheral functions. This chip series is well suited for camera, VCR, MD, TV, CD, LD, printer, telephone, home automation, pager, air conditioner, PPC, fax machine, music instrument and other applications.

This LSI brings to embedded microcomputer applications flexible, optimized hardware configurations and a simple efficient instruction set. MN101EF57 series has an internal 128 KB of ROM and 6 KB of RAM. Peripheral functions include 5 external interrupts, 29 internal interrupts including NMI, 12 timer counters, 4 types of serial interfaces, A/D converter, D/A converter, LCD driver, 2 types of watchdog timer, data automatic function and buzzer output. The system configuration is suitable for in camera, timer selector for VCR, CD player, or mini-component.

With 5 oscillation systems (high-speed (internal frequency: 20 MHz), high-speed (crystal/ceramic frequency: max. 10 MHz) / low-speed (internal frequency: 30 kHz), low-speed (crystal/ceramic frequency: 32.768 kHz) and PLL: frequency multiplier of high frequency) contained on the chip, the system clock can be switched to high-speed frequency input (NORMAL mode), PLL input (PLL mode), or to low-speed frequency input (SLOW mode). The system clock is generated by dividing the oscillation clock or PLL clock. The best operation clock for the system can be selected by switching its frequency ratio by programming. High speed mode has the normal mode which is based on the clock dividing f_{pll} , (f_{pll} is generated by original oscillation and PLL), by 2 ($f_{pll}/2$), and the double speed mode which is based on the clock not dividing f_{pll} .

A machine cycle (minimum instruction execution time) in the normal mode is 200 ns when the original oscillation f_{osc} is 10 MHz (PLL is not used). A machine cycle in the double speed mode, in which the CPU operates on the same clock as the external clock, is 100 ns when f_{osc} is 10 MHz. A machine cycle in the PLL mode is 50 ns (maximum).

1.1.2 Product Summary

This manual describes the following model.

Table:1.1.1 Product Summary

Model	ROM Size	RAM Size	Classification
MN101EF76K	256 KB	10 KB	Flash EEPROM version
MN101EF57G	128 KB	6 KB	Flash EEPROM version
MN101EF56K	256 KB	10 KB	Flash EEPROM version



DMOD internal pull-up resistor is in only Flash EEPROM version.
When using In-circuit Emulator, it is necessary to connect the pull-up resistor on the circuit board.

1.2 Hardware Functions

■ Feature

- ROM / RAM capacity:
 - MN101EF76K: ROM 256 KB / RAM 10 KB
 - MN101EF57G: ROM 128 KB / RAM 6 KB
 - MN101EF56K: ROM 256 KB / RAM 10 KB
- Package:
 - MN101EF76K: LQFP128-P-1818C (18 mm × 18 mm / 0.5 mm pitch)
 - MN101EF57G: LQFP080-P-1414A (14 mm × 14 mm / 0.65 mm pitch)
 - TQFP080-P-1212F (12 mm × 12 mm / 0.5 mm pitch)
 - MN101EF56K: QFP100-P-1818B (18 mm × 18 mm / 0.65 mm pitch)
- Machine Cycle:
 - High-speed mode
 - 0.05 μs/20 MHz (2.7 V to 5.5 V)
 - 0.125 μs/8 MHz (1.8 V to 5.5 V)
 - Low-speed mode
 - 62.5 μs/ 32 kHz (1.8 V to 5.5 V)
- Clock Gear Circuit:
 - Internal system clock speed is changeable by selecting division ratio of oscillation clock.
(Divided by 1, 2, 4, 16, 32, 64, 128)
- Oscillation Circuit: 4 types
 - High-speed (Internal oscillation: frc), High-speed (crystal/ceramic: fosc),
 - Low-speed (Internal oscillation: frcs), Low-speed (crystal/ceramic: fx)
 - High-speed internal oscillation 20 MHz / 16 MHz (selectable)
 - Low-speed internal oscillation 30 kHz
- Clock Multiplication Circuit:
 - PLL circuit output clock (fppll) fosc multiplied by 2, 3, 4, 5, 6, 8, 10,
1/2 × frc multiplied by 4, 5 enabled
 - * When clock multiplication circuit is not used, fppll = fosc or fppll = frc
 - * Selectable from high-speed clock for peripheral functions (fppll-div) fppll, fppll divided by 2, 4, 8, 16
- Memory bank
 - Data memory space is expanded by the bank system.
 - Bank for the source address / Bank for the destination address.
- Operation Mode
 - NORMAL mode (high-speed mode)
 - PLL mode
 - SLOW mode (low-speed mode)
 - HALT mode
 - STOP mode
 - and operation clock switching
- Operating Voltage: 1.8 V to 5.5 V

- Operation ambient temperature: -40 °C to +85 °C

- Interrupts:

Interrupts	MN101EF76K 36 sets	MN101EF57G 34 sets	MN101EF56K 36 sets
<Overrun interrupt>			
Non-maskable interrupt (NMI)	√	√	√
<Timer interrupt>			
Timer 0 interrupt	√	√	√
Timer 1 interrupt	√	√	√
Timer 2 interrupt	√	√	√
Timer 3 interrupt	√	√	√
Timer 4 interrupt	√	√	√
Timer 6 interrupt	√	√	√
Timer 7 interrupt	√	√	√
Time-base interrupt	√	√	√
Timer 7 compare register 2 match interrupt	√	√	√
Timer 8 interrupt	√	√	√
Timer 8 compare register 2 match interrupt	√	√	√
PWM overflow interrupt	√	√	√
PWM under flow interrupt	√	√	√
Timer 9 compare register 2 match interrupt	√	√	√
24H timer interrupt	√	√	√
Alarm match interrupt	√	√	√
<Serial interrupt>			
LIN interrupt	√	√	√
Serial 0 interrupt	√	√	√
Serial 0 UART reception interrupt	√	√	√
Serial 1 interrupt	√	√	√
Serial 1 UART reception interrupt	√	√	√
Serial 2 interrupt	√	√	√
Serial 2 UART reception interrupt	√	√	√
Serial 3 interrupt	√	-	√
Serial 3 UART reception interrupt	√	-	√
Serial 4 interrupt	√	√	√
Serial 4 stop condition interrupt	√	√	√
<A/D interrupt>			
A/D conversion interrupt	√	√	√
<Data automatic transfer interrupt>			
ATC1 interrupt	√	√	√
<Low voltage detection interrupt>			
Low voltage detection interrupt	√	√	√
<External interrupt>			
IRQ0 (Edge selection, noise filter connectable)	√	√	√
IRQ1 (Edge selection, noise filter connectable)	√	√	√
IRQ2 (Edge selection, both edge interrupt, noise filter connectable)	√	√	√
IRQ3 (Edge selection, both edge interrupt, noise filter connectable)	√	√	√
IRQ4 (Edge selection, both edge interrupt, noise filter connectable, KEY scan interrupt)	√	√	√

- Timer Counter: 12 sets

- General-purpose 8-bit timer × 5 sets
- General-purpose 16-bit timer × 2 sets
- General-purpose 16-bit timer × 2 sets
- Motor control 16-bit timer × 1 set
- 8-bit free-run timer × 1 set

- Time-base timer × 1 set
- Baud rate timer × 1 set
- 24H timer × 1 set

Timer 0 (General-purpose 8-bit timer)

- Square wave output (Timer pulse output), added pulse (2 bits) type PWM output can be output to large current pin TM0IOB, event count, simple pulse width measurement
- Double-buffered compare register (× 1) * Function in MN101EF76K and MN101EF56K
- Clock source
fppll-div, fppll-div/4, fppll-div/16, fppll-div/32, fppll-div/64, fppll-div/128,
fs/2, fs/4, fs/8, fslow, external clock, timer A output
- Real-time control
Timer (PWM) output is controlled among the three values: "Fixed to High", "Fixed to Low", or "Hi-Z" at falling edge of external interrupt 0 (IRQ0)

Timer 1 (General-purpose 8-bit timer)

- Square wave output (Timer pulse output), event count
16-bit cascade connection (connected with timer 0)
- Double-buffered compare register (× 1) * Function in MN101EF76K and MN101EF56K
- Clock source
fppll-div, fppll-div/4, fppll-div/16, fppll-div/32, fppll-div/64, fppll-div/128,
fs/2, fs/4, fs/8, fslow, external clock, timer A output

Timer 2 (General-purpose 8-bit timer)

- Square wave output (Timer pulse output), added pulse (2 bits) type PWM output can be output to large current pin TM2IOB, event count, simple pulse width measurement,
24-bit cascade connection (connected with timer 0, 1), timer synchronous output
- Double-buffered compare register (× 1)
- Clock source
fppll-div, fppll-div/4, fppll-div/16, fppll-div/32, fppll-div/64, fppll-div/128,
fs/2, fs/4, fs/8, fslow, external clock, timer A output
- Real-time control
Timer (PWM) output is controlled among the three values: "Fixed to High", "Fixed to Low", or "Hi-Z" at falling edge of external interrupt 0 (IRQ0)

Timer 3 (General-purpose 8-bit timer)

- Square wave output (Timer pulse output), event count
16-bit cascade connection (connected with timer 2),
32-bit cascade connection (connected with timer 0, 1, 2)
- Double-buffered compare register (× 1)
- Clock source
fppll-div, fppll-div/4, fppll-div/16, fppll-div/32, fppll-div/64, fppll-div/128,
fs/2, fs/4, fs/8, fslow, external clock, timer A output

Timer 4 (General-purpose 8-bit timer)

- Square wave output (Timer pulse output), added pulse (2bit) type PWM output,
event count, simple pulse width measurement
- Clock source
fppll-div, fppll-div/4, fppll-div/16, fppll-div/32, fppll-div/64, fppll-div/128,
fs/2, fs/4, fs/8, fslow, external clock, timer A output

Timer 6 (8-bit free-run timer, time-base timer)

8-bit free-run timer

- Clock source
 fppll-div , $\text{fppll-div}/2^2$, $\text{fppll-div}/2^3$, $\text{fppll-div}/2^{12}$, $\text{fppll-div}/2^{13}$, fs , fslow ,
 $\text{fslow}/2^2$, $\text{fslow}/2^3$, $\text{fslow}/2^{12}$, $\text{fslow}/2^{13}$

Time-base timer

- Interrupt generation cycle
 $\text{fppll-div}/2^7$, $\text{fppll-div}/2^8$, $\text{fppll-div}/2^9$, $\text{fppll-div}/2^{10}$, $\text{fppll-div}/2^{13}$,
 $\text{fppll-div}/2^{15}$, $\text{fslow}/2^7$, $\text{fslow}/2^8$, $\text{fslow}/2^9$, $\text{fslow}/2^{10}$, $\text{fslow}/2^{13}$, $\text{fslow}/2^{15}$

Timer 7 (General-purpose 16-bit timer)

- Clock source
 fppll-div , fs , external clock, timer A output, serial 0 transfer clock output,
timer 6 compare match cycle divided by 1, 2, 4, 16
- Hardware configuration
Double-buffered compare register ($\times 2$)
Double-buffered input capture register ($\times 2$)
Timer interrupt ($\times 2$ vector)
- Timer function
Square wave output (Timer pulse output), high-precision PWM output (cycle/duty continuous changeable) can be output to large current pin TM7IOB, timer synchronous output, event count, input capture function (both edges operable)
- Real-time control
Timer (PWM) output is controlled among the three values: "Fixed to High", "Fixed to Low", or "Hi-Z" at falling edge of external interrupt 0 (IRQ0)

Timer 8 (General-purpose 16-bit timer)

- Clock source
 fppll-div , fs , external clock, timer A output, timer 6 compare match cycle divided by 1, 2, 4, 16
- Hardware configuration
Double-buffered compare register ($\times 2$)
Double-buffered input capture register ($\times 1$)
Timer interrupt ($\times 2$ vector)
- Timer function
Square wave output (Timer pulse output), high-precision PWM output (cycle/duty continuous changeable) can be output to large current pin TM8IOB, event count, pulse width measurement, input capture function (both edges operable)
32-bit cascade connection (connected with timer 7), 32-bit PWM output, input capture is available in 32-bit cascade

Timer 9 (Motor control 16-bit timer)

- Clock source
 fppll-div , fs , external clock, Timer A output divided by 1, 2, 4, 16
- Hardware configuration
Double-buffered compare register ($\times 2$)
Timer interrupt ($\times 3$ vector)
- Timer function

Square wave output (Timer pulse output) can be changed to large current output, complementary 3-phase PWM output, triangle wave and saw tooth wave are supported, dead time insertion available, event count

- Pin output control

PWM output control is possible by external interrupt 0 to 4 (IRQ 0 to 4) ("Hi-z", output data fixed)

Timer A (baud rate timer)

- Clock output for peripheral functions

- Clock source

fppll-div divided by 1/1, 2, 4, 8, 16, 32, and fs divided by 2, 4

24H timer

- Clock source (Usable frequency)

fppll (4 MHz, 4.19 MHz, 5 MHz, 8 MHz, 8.38 MHz, 10 MHz, 16 MHz, 16.77 MHz, 20 MHz),
fx (32.768 kHz), frc (20 MHz, 16 MHz), frcs (30 kHz)

- Hardware configuration

0.5 seconds counter, minute counter, hour counter

Alarm compare register (in 0.5 seconds, in minutes, in hours) ($\times 1$)

Timer interrupt ($\times 2$ vector)

- Timer function

Interval function (interrupts every 0.5 seconds, 1 second, 1 minute, 1 hour, 24 hours)

Alarm function

- Watchdog timer

Overrun detection cycle is selectable from $fs/2^{16}$, $fs/2^{18}$, $fs/2^{20}$

Forced to reset inside LSI by hardware when a software processing error is detected twice

- Watchdog timer2

Overrun detection cycle is selectable from $frcs/2^4$, $frcs/2^5$, $frcs/2^6$, $frcs/2^7$, $frcs/2^8$, $frcs/2^9$, $frcs/2^{10}$, $frcs/2^{11}$, $frcs/2^{12}$, $frcs/2^{13}$, $frcs/2^{14}$, $frcs/2^{15}$

Forced to reset inside LSI by hardware when a software processing error is detected twice

- Synchronous output function (Timer synchronous output, interrupt synchronous output)

Latch data is output from port 8 at the event timing of synchronous output signal of timer 1, timer 2, timer 7, or external interrupt2 (IRQ2)

- Buzzer Output

Output frequency can be selected from $fp\text{ll-div}/2^9$, $fp\text{ll-div}/2^{10}$, $fp\text{ll-div}/2^{11}$, $fp\text{ll-div}/2^{12}$, $fp\text{ll-div}/2^{13}$, $fp\text{ll-div}/2^{14}$, $fs\text{low}/2^3$, $fs\text{low}/2^4$

- A/D converter

MN101EF76K: 10-bit $\times$ 24 channels

MN101EF57G: 10-bit $\times$ 12 channels

MN101EF56K: 10-bit $\times$ 24 channels

- D/A converter

MN101EF76K: 8-bit $\times$ 4 channels

MN101EF57G: 8-bit $\times$ 2 channels

MN101EF56K: 8-bit $\times$ 4 channels

- Data automatic transfer: 1 system

Data is automatically transferred in all memory space

- External interrupt activation/internal event activation/software activation
- Max. 255 byte continuous transfer
- Serial continuous transmission and reception is supported
- Burst transfer function (Including interrupt emergency stop)

- Serial interface

MN101EF76K: 5 systems

MN101EF57G: 4 systems

MN101EF56K: 5 systems

Serial interface 0 (Hardware LIN / Full duplex UART / Synchronous serial interface)

Synchronous serial interface

- Transfer clock source
 $fp\text{ll-div}/2$, $fp\text{ll-div}/4$, $fp\text{ll-div}/16$, $fp\text{ll-div}/64$, $fs/2$, $fs/4$,
Timer 0 to 4, Timer A output divided by 1, 2, 4, 8, 16, External clock
- MSB/LSB first selectable, 1 to 8 bits of arbitrary transfer
- Continuous transmission, continuous reception, continuous transmission and reception are available.

Full duplex UART (Baud rate timer: selected from timer 0 to 4, or timer A)

- Parity check, overrun error/framing error are detected
- Transfer bits 7 to 8 are selectable
- Hardware LIN
Synch Break generation, Wake-up detection, Synch Break detection, Synch Field measurement are available

Serial interface 1 (Full duplex UART / Synchronous serial interface)

Synchronous serial interface

- Transfer clock source
fppll-div/2, fppll-div/4, fppll-div/16, fppll-div/64, fs/2, fs/4,
Timer 0 to 4, Timer A output divided by 1, 2, 4, 8, 16, External clock
- MSB/LSB first selectable, 1 to 8 bits of arbitrary transfer
- Continuous transmission, continuous reception, continuous transmission and reception are available.

Full duplex UART (Baud rate timer: selected from timer 0 to 4, or timer A)

- Parity check, overrun error/framing error are detected
- Transfer bits 7 to 8 are selectable

Serial interface 2 (Full duplex UART / Synchronous serial interface)

Synchronous serial interface

- Transfer clock source
fppll-div/2, fppll-div/4, fppll-div/16, fppll-div/64, fs/2, fs/4,
Timer 0 to 4, Timer A output divided by 1, 2, 4, 8, 16, External clock
- MSB/LSB first selectable, 1 to 8 bits of arbitrary transfer
- Continuous transmission, continuous reception, continuous transmission and reception are available.

Full duplex UART (Baud rate timer: selected from timer 0 to 4, or timer A)

- Parity check, overrun error/framing error are detected
- Transfer bits 7 to 8 are selectable

Serial interface 3 (Full duplex UART / Synchronous serial interface)

* Function in MN101EF76K and MN101EF56K

Synchronous serial interface

- Transfer clock source
fppll-div/2, fppll-div/4, fppll-div/16, fppll-div/64, fs/2, fs/4,
Timer 0 to 4, Timer A output divided by 1, 2, 4, 8, 16, External clock
- MSB/LSB first selectable, 1 to 8 bits of arbitrary transfer
- Continuous transmission, continuous reception, continuous transmission and reception are available.

Full duplex UART (Baud rate timer: selected from timer 0 to 4, or timer A)

- Parity check, overrun error/framing error are detected
- Transfer bits 7 to 8 are selectable

Serial interface 4 (Multi master IIC / Synchronous serial interface)

Synchronous serial interface

- Transfer clock source
fppll-div/2, fppll-div/4, fppll-div/8, fppll-div/32, fs/2, fs/4,
Timer 0 to 4, Timer A output divided by 1, 2, 4, 8, 16, External clock
- MSB/LSB first selectable, 1 to 8 bits of arbitrary transfer
- Continuous transmission, continuous reception, continuous transmission and reception are available.

Multi master IIC

- 7, 10-bit slave address is settable
 - General call communication mode is supported
-
- Auto reset circuit
 - Low voltage detection circuit

- Clock Monitoring Function

- LED driver: 8 sets

- LCD driver

Segment output

MN101EF76K: Max. 55 pins (SEG0 to SEG54)

MN101EF57G: Max. 41 pins (SEG0 to SEG40)

MN101EF56K: Max. 55 pins (SEG0 to SEG54)

Segment output pins can be switched to I/O ports in 1 bit.

* At reset, Segment outputs are input ports.

Common output: 4 pins

COM0 to 3 can be switched to I/O ports in 1 bit.

Display mode selection

Static

1/2 duty, 1/2 bias

1/3 duty, 1/3 bias

1/4 duty, 1/3 bias

LCD driver clock

When the source clock is the main clock (fp11)

$1/2^{18}, 1/2^{17}, 1/2^{16}, 1/2^{15}, 1/2^{14}, 1/2^{13}, 1/2^{12}, 1/2^{11}$

When the source clock is the sub clock (fslow)

$1/2^9, 1/2^8, 1/2^7, 1/2^6$

Timer 0 to 4, Timer A output

LCD power supply

LCD power supply is separated from V_{DD5} . (can be used when $V_{LC1} \leq V_{DD5}$)

External power supply voltage can be selectable.

(Supply voltage is supplied from V_{LC1} , V_{LC2} , and V_{LC3})

Internal dividing resistors

(External power supply voltage is divided the voltage input to VLC1 by internal resistors.)

- Ports

Ports	MN101EF76K (pins)	MN101EF57G (pins)	MN101EF56K (pins)
<I/O ports>	104	70	90
LCD segment	55	41	55
LCD common	4	4	4
Serial interface communication	30	21	30
Timer I/O	34	21	28
Buzzer output	4	2	4
A/D input	24	16	24
External interrupt	10	5	5
LCD power supply	3	3	3
LED driver (high-current)	8	8	8
High-speed oscillation	2	2	2
Low-speed oscillation	2	2	2
D/A output	4	2	4
<Special function pins>	10	10	10
Operation mode input	3	3	3
Reset input	1	1	1
Analog reference voltage input	1	1	1
Power supply	4	4	4

1.3 Pin Description

1.3.1 Pin configuration

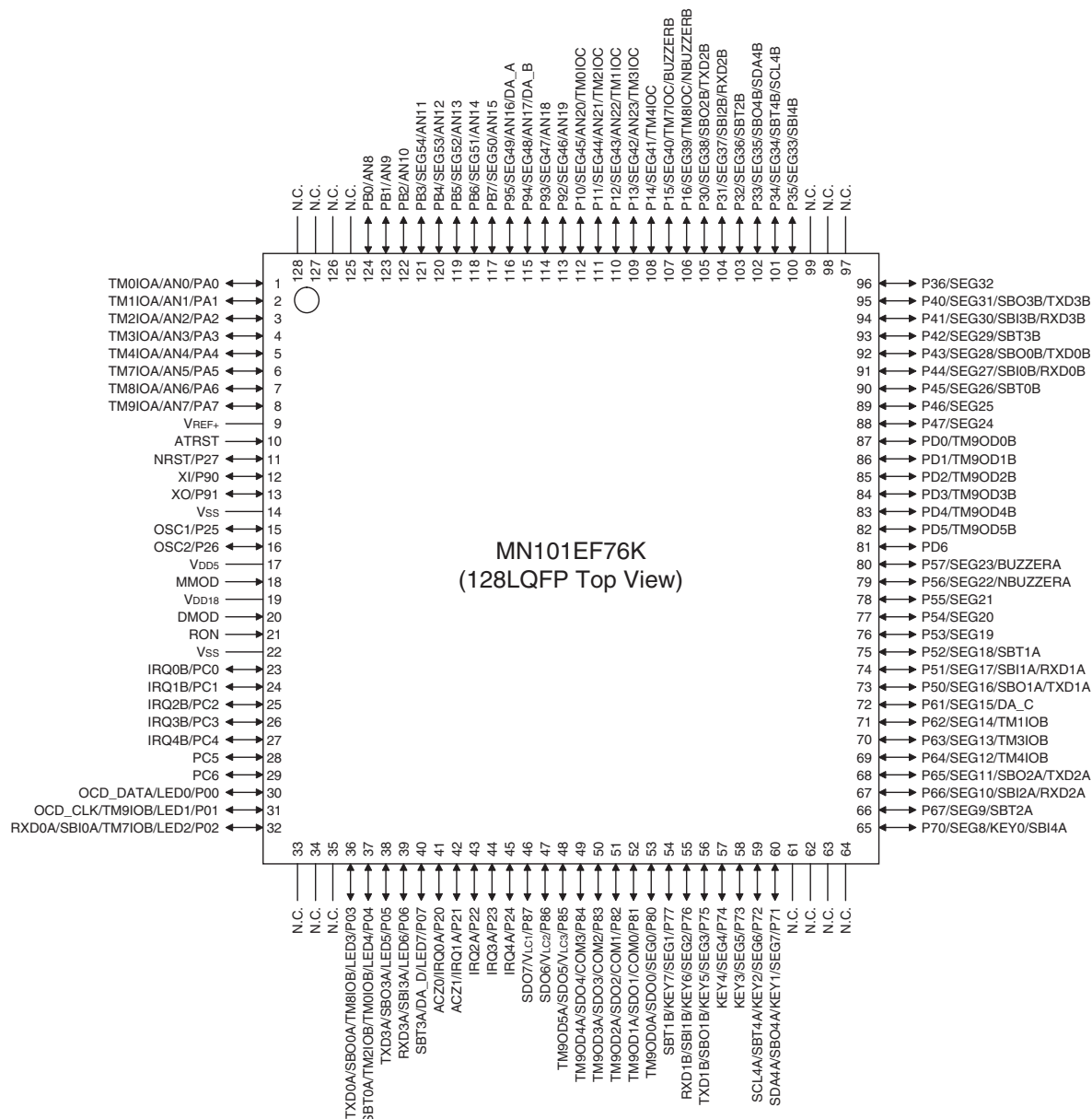


Figure:1.3.1 Pin Configuration (MN101EF76K)

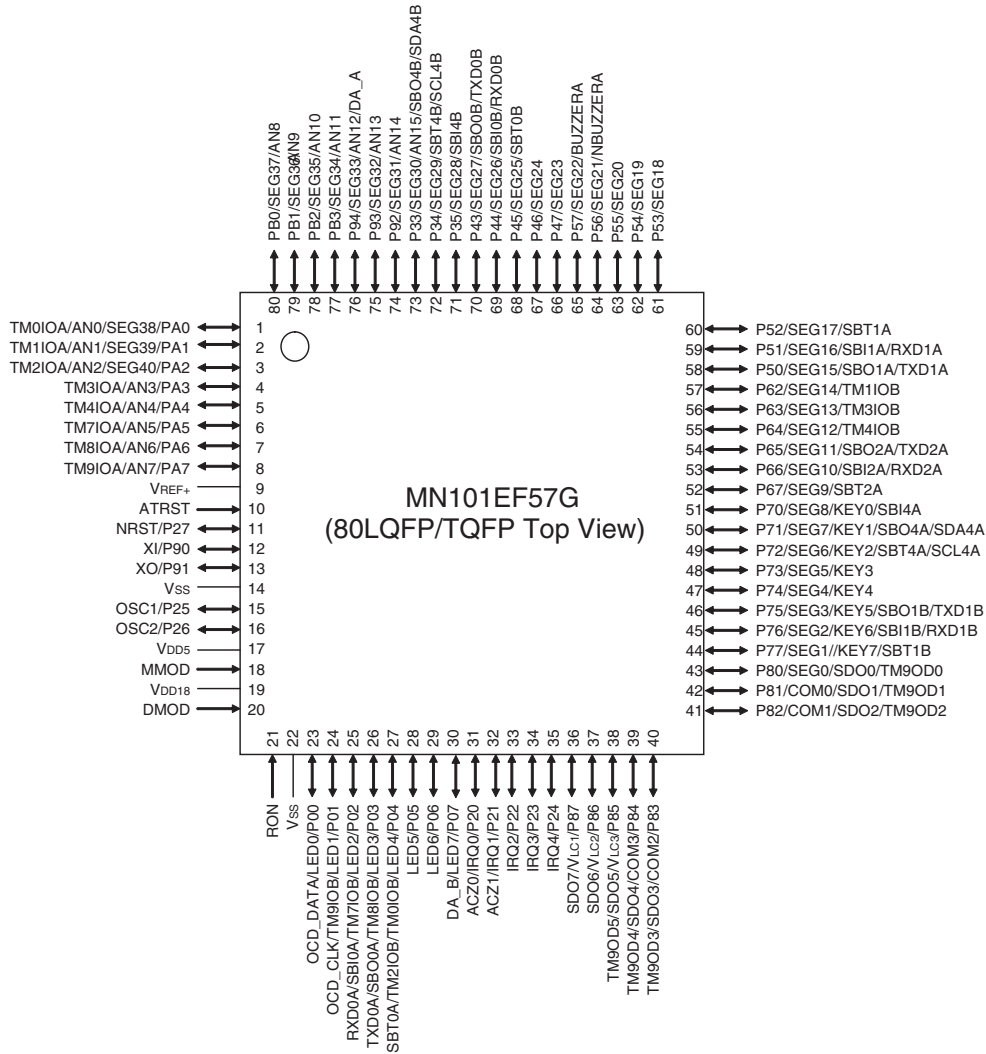


Figure:1.3.2 Pin Configuration (MN101EF57G)

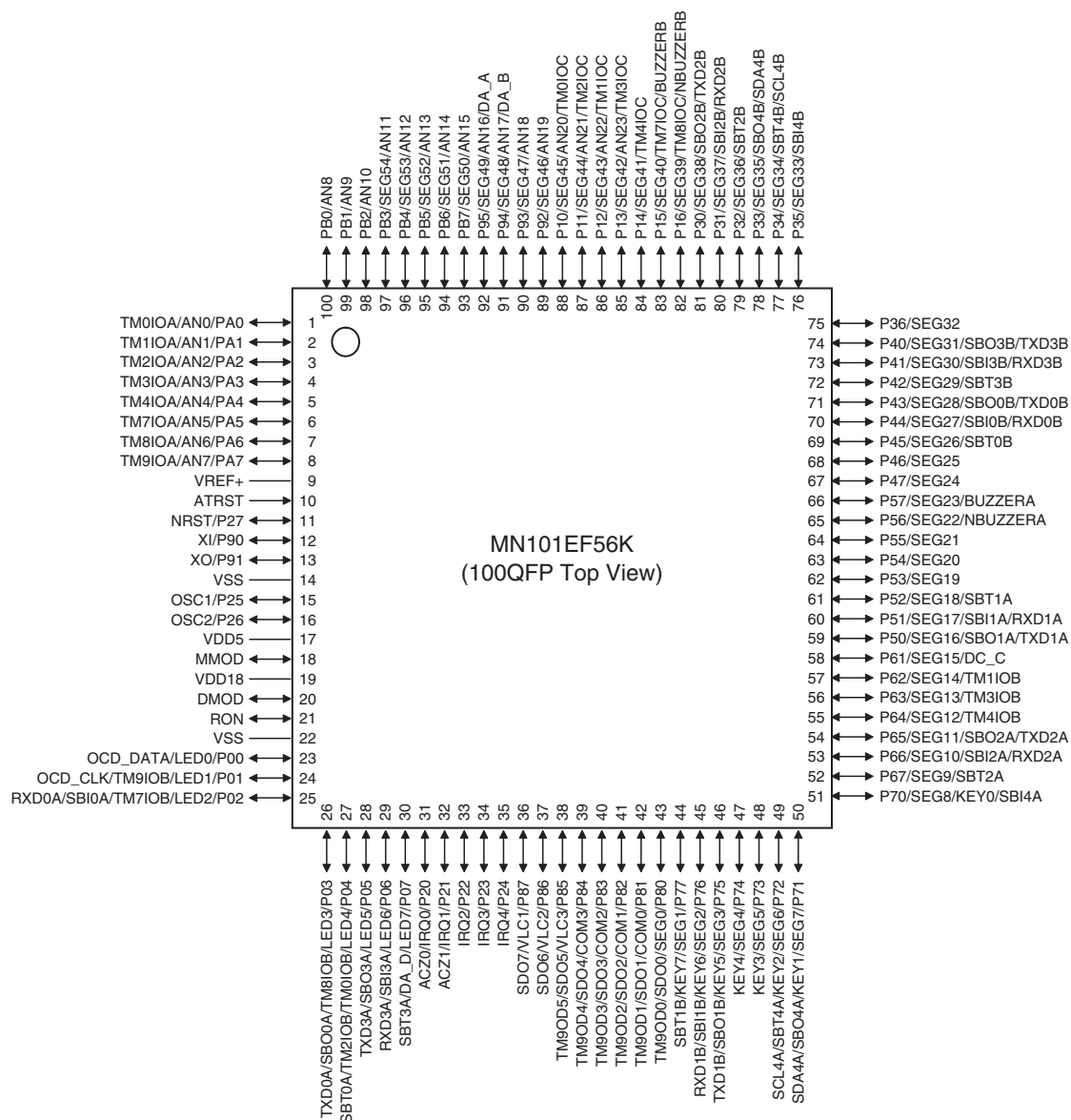


Figure:1.3.3 Pin Configuration (MN101EF56K)

1.3.2 Pin Specification

Table:1.3.1 Pin Specification (MN101EF76K)

MN101EF76K		I/O	Direction control	Pin control	Function Description
Pins	Special functions				
P00	LED0 OCD_DATA	in/out	P0DIR0	P0PLUD0	LED0: LED driver pin 0 OCD_DATA: On-board programmer Data pin
P01	LED1 TM9IOB	in/out	P0DIR1	P0PLUD1	LED1: LED driver pin 1 TM9IOB: Timer 9 I/O
	OCD_CLK				OCD_CLK: On-board programmer Clock supply pin
P02	LED2 TM7IOB	in/out	P0DIR2	P0PLUD2	LED2: LED driver pin 2 TM7IOB: Timer7 I/O
	SBI0A RXD0A				SBI0A: Serial 0 data input RXD0A: UART0 data input
P03	LED3 TM8IOB	in/out	P0DIR3	P0PLUD3	LED3: LED driver pin 3 TM8IOB: Timer 8 I/O
	SBO0A TXD0A				SBO0A: Serial 0 data output TXD0A: UART0 data output
P04	LED4 TM0IOB	in/out	P0DIR4	P0PLUD4	LED4: LED driver pin 4 TM0IOB: Timer0 I/O
	TM2IOB SBT0A				TM2IOB: Timer 2 I/O SBT0A: Serial 0 clock I/O
P05	LED5	in/out	P0DIR5	P0PLUD5	LED5: LED driver pin 5
P06	LED6	in/out	P0DIR6	P0PLUD6	LED6: LED driver pin 6
P07	LED7 DA_B	in/out	P0DIR7	P0PLUD7	LED7: LED driver pin 7 DA_B: Analog B output
P10	AN20 TM0IOC	in/out	P1DIR0	P1PLUD0	AN20: Analog 20 input TM0IOC: Timer 0 I/O
	SEG45				SEG45: Segment 45 output
P11	AN21 TM2IOC	in/out	P1DIR1	P1PLUD1	AN21: Analog 21 input TM2IOC: Timer 2 I/O
	SEG44				SEG44: Segment 44 output
P12	AN22 TM1IOC	in/out	P1DIR2	P1PLUD2	AN22: Analog 22 input TM1IOC: Timer 1 I/O
	SEG43				SEG43: Segment 43 output
P13	AN23 TM3IOC	in/out	P1DIR3	P1PLUD3	AN23: Analog 23 input TM3IOC: Timer 3 I/O
	SEG42				SEG42: Segment 42 output
P14	TM4IOC SEG41	in/out	P1DIR4	P1PLUD4	TM4IOC: Timer 4 I/O SEG41: Segment 41 output
P15	TM7IOC BUZZERB	in/out	P1DIR5	P1PLUD5	TM7IOC: Timer 7 I/O BUZZERB: Buzzer output
	SEG40				SEG40: Segment 40 output
P16	TM8IOC NBUZZERB	in/out	P1DIR6	P1PLUD6	TM8IOC: Timer 8 I/O NBUZZERB: Buzzer reverse output
	SEG39				SEG39: Segment 39 output
P20	IRQ0A ACZ0	in/out	P2DIR0	P2PLU0	IRQ0A: External interrupt 0 ACZ0: Zero cross 0 input
P21	IRQ1A ACZ1	in/out	P2DIR1	P2PLU1	IRQ1A: External interrupt 1 ACZ1: Zero cross 1 input
P22	IRQ2A	in/out	P2DIR2	P2PLU2	IRQ2A: External interrupt 2
P23	IRQ3A	in/out	P2DIR3	P2PLU3	IRQ3A: External interrupt 3
P24	IRQ4A	in/out	P2DIR4	P2PLU4	IRQ4A: External interrupt 4
P25	OSC1	in/out	P2DIR5	P2PLU5	OSC1: Ceramic / crystal high-speed clock input
P26	OSC2	in/out	P2DIR6	P2PLU6	OSC2: Ceramic / crystal high-speed clock output
P27	NRST	in/out	P2DIR7	P2PLU7	NRST: Reset
P30	SBO2B TXD2B	in/out	P3DIR0	P3PLU0	SBO2B: Serial 2 transmission data output TXD2B: UART 2 transmission data output
	SEG38				SEG38: Segment 38 output
P31	SBI2B RXD2B	in/out	P3DIR1	P3PLU1	SBI2B: Serial 2 reception data input RXD2B: UART 2 reception data input
	SEG37				SEG37: Segment 37 output
P32	SBT2B SEG36	in/out	P3DIR2	P3PLU2	SBT2B: Serial 2 clock I/O SEG36: Segment 36 output
P33	AN15 SBO4B	in/out	P3DIR3	P3PLU3	AN15: Analog 15 input SBO4B: Serial 4 transmission data output
	SDA4B SEG35				SDA4B: IIC 4 data I/O SEG35: Segment 35 output
P34	SBT4B SCL4B	in/out	P3DIR4	P3PLU4	SBT4B: Serial 4 clock I/O SCL4B: IIC 4 clock I/O
	SEG34				SEG34: Segment 34 output
P35	SBI4B SEG33	in/out	P3DIR5	P3PLU5	SBI4B: Serial 4 reception data input SEG33: Segment 33 output
P36	SEG32	in/out	P3DIR6	P3PLU6	SEG32: Segment 32 output

MN101EF76K			I/O	Direction control	Pin control	Function Description	
Pins	Special functions						
P40	SBO3B	TXD3B	in/out	P4DIR0	P4PLUD0	SBO3B: Serial 3 transmission data output	TXD3B: UART 3 transmission data output
	SEG31					SEG31: Segment 31 output	
P41	SBI3B	RXD3B	in/out	P4DIR1	P4PLUD1	SBI3B: Serial 3 reception data input	RXD3B: UART 3 reception data input
	SEG30					SEG30: Segment 30 output	
P42	SBT3B	SEG29	in/out	P4DIR2	P4PLUD2	SBT3B: Serial 3 clock I/O	SEG29: Segment 29 output
P43	SBO0B	TXD0B	in/out	P4DIR3	P4PLUD3	SBO0B: Serial 0 transmission data output	TXD0B: UART 0 transmission data output
	SEG28					SEG28: Segment 28 output	
P44	SBI0B	RXD0B	in/out	P4DIR4	P4PLUD4	SBI0B: Serial 0 reception data input	RXD0B: UART 0 reception data input
	SEG27					SEG27: Segment 27 output	
P45	STB0B	SEG26	in/out	P4DIR5	P4PULD5	STB0B: Serial 0 clock I/O	SEG26: Segment 26 output
P46	SEG25		in/out	P4DIR6	P4PULD6	SEG25: Segment 25 output	
P47	SEG24		in/out	P4DIR7	P4PULD7	SEG24: Segment 24 output	
P50	SBO1A	TXD1A	in/out	P5DIR0	P5PLUD0	SBO1A: Serial 1 transmission data output	TXD1: UART1 transmission data output
	SEG16					SEG16: Segment 16 output	
P51	SBI1A	RXD1A	in/out	P5DIR1	P5PLUD1	SBI1A: Serial 1 reception data input	RXD1A: UART1 reception data input
	SEG17					SEG17: Segment 17 output	
P52	SBT1A	SEG18	in/out	P5DIR2	P5PLUD2	SBT1A: Serial 1 clock I/O	SEG18: Segment 18 output
P53	SEG19		in/out	P5DIR3	P5PLUD3	SEG19: Segment 19 output	
P54	SEG20		in/out	P5DIR4	P5PLUD4	SEG20: Segment 20 output	
P55	SEG21		in/out	P5DIR5	P5PLUD5	SEG21: Segment 21 output	
P56	SEG22	NBUZZERA	in/out	P5DIR6	P5PLUD6	SEG22: Segment 22 output	NBUZZERA: Buzzer reverse output
P57	SEG23	BUZZERA	in/out	P5DIR7	P5PLUD7	SEG23: Segment 23 output	BUZZERA: Buzzer output
P61	DA_C	SEG15	in/out	P6DIR1	P6PLU1	DA_C: Analog C output	SEG15: Segment 15 output
P62	TM1IOB	SEG14	in/out	P6DIR2	P6PLU2	TM1IOB: Timer 1 I/O	SEG14: Segment 14 output
P63	TM3IOB	SEG13	in/out	P6DIR3	P6PLU3	TM3IOB: Timer 3 I/O	SEG13: Segment 13 output
P64	TM4IOB	SEG12	in/out	P6DIR4	P6PLU4	TM4IOB: Timer 4 I/O	SEG12: Segment 12 output
P65	SBO2A	TXD2A	in/out	P6DIR5	P6PLU5	SBO2A: Serial 2 transmission data output	TXD2A: UART2 transmission data output
	SEG11					SEG11: Segment 11 output	
P66	SBI2A	RXD2A	in/out	P6DIP6	P6PLU6	SBI2A: Serial 2 reception data input	RXD2A: UART2 reception data input
	SEG10					SEG10: Segment 10 output	
P67	SBT2A	SEG9	in/out	P6DIR7	P6PLU7	SBT2A: Serial 2 clock I/O	SEG9: Segment 9 output
P70	KEY0	SBI4A	in/out	P7DIR0	P7PLUD0	KEY0: KEY interrupt input 0	SBI4A: Serial 4 reception data input
	SEG8					SEG8: Segment 8 output	
P71	KEY1	SBO4A	in/out	P7DIR1	P7PLUD1	KEY1: KEY interrupt input 1	SBO4A: Serial 4 transmission data output
	SDA4A	SEG7				SDA4A: IIC 4 data I/O	SEG7: Segment 7 output
P72	KEY2	SBT4A	in/out	P7DIR2	P7PLUD2	KEY2: KEY interrupt input 2	SBT4A: Serial 4clock I/O
	SCL4A	SEG6				SCL4A: IIC 4 clock I/O	SEG6: Segment 6 output
P73	KEY3	SEG5	in/out	P7DIR3	P7PLUD3	KEY3: KEY interrupt input 3	SEG5: Segment 5 output
P74	KEY4	SEG4	in/out	P7DIR4	P7PLUD4	KEY4: KEY interrupt input 4	SEG4: Segment 4 output
P75	KEY5	SBO1B	in/out	P7DIR5	P7PLUD5	KEY5: KEY interrupt input 5	SBO1B: Serial 1data output
	TXD1B	SEG3				TXD1B: UART 1 transmission data output	SEG3: Segment 3 output
P76	KEY6	SBI1B	in/out	P7DIR6	P7PLUD6	KEY6: KEY interrupt input 6	SBI1B: Serial 1 reception data input
	RXD1B	SEG2				RXD1B: UART 1 reception data input	SEG2: Segment 2 output
P77	KEY7	SBT1B	in/out	P7DIR7	P7PLUD7	KEY7: KEY interrupt input7	SBT1B: Serial 1clock I/O
	SEG1					SEG1: Segment 1 output	

MN101EF76K			I/O	Direction control	Pin control	Function Description	
Pins	Special functions						
P80	SDO0	TM9OD0A	in/out	P8DIR0	P8PLU0	SDO0: Timer synchronous output 0	TM9OD0A: Timer 9 output
P81	SEG0					SEG0: Segment 0 output	
	SDO1	TM9OD1A	in/out	P8DIR1	P8PLU1	SDO1: Timer synchronous output 1	TM9OD1A: Timer 9 output
P82	COM0					COM0: LCD common output	
	SDO2	TM9OD2A	in/out	P8DIR2	P8PLU2	SDO2: Timer synchronous output 2	TM9OD2A: Timer 9 output
P83	COM1					COM1: LCD common output	
	SDO3	TM9OD3A	in/out	P8DIR3	P8PLU3	SDO3: Timer synchronous output 3	TM9OD3A: Timer 9 output
P84	COM2					COM2: LCD common output	
	SDO4	TM9OD4A	in/out	P8DIR4	P8PLU4	SDO4: Timer synchronous output 4	TM9OD4A: Timer 9 output
P85	COM3					COM3: LCD common output	
	SDO5	TM9OD5A	in/out	P8DIR5	P8PLU5	SDO5: Timer synchronous output 5	TM9OD5A: Timer 9 output
P86	V _{LC3}					V _{LC3} : LCD power supply	
	SDO6		in/out	P8DIR6	P8PLU6	SDO6: Timer synchronous output 6	
P87	V _{LC2}					V _{LC2} : LCD power supply	
	SDO7		in/out	P8DIR7	P8PLU7	SDO7: Timer synchronous output 7	
V _{LC1}						V _{LC1} : LCD power supply	
P90	XI		in/out	P9DIR0	P9PLU0	XI: Ceramic / crystal low-speed clock input	
P91	XO		in/out	P9DIR1	P9PLU1	XOI: Ceramic / crystal low-speed clock output	
P92	AN19	SEG46	in/out	P9DIR2	P9PLU2	AN19: Analog 14 input	SEG46: Segment 46 output
P93	AN18	SEG47	in/out	P9DIR3	P9PLU3	AN18: Analog 13 input	SEG47: Segment 47 output
P94	AN17	SEG48	in/out	P9DIR4	P9PLU4	AN17: Analog 12 input	SEG48: Segment 48 output
	DA_B					DA_B: Analog B output	
P95	AN16	SEG49	in/out	P9DIR4	P9PLU4	AN16: Analog 16 input	SEG49: Segment 49 output
	DA_A					DA_A: Analog A output	
PA0	AN0	TM0IOA	in/out	PADIR0	PAPLU0	AN0: Analog 0 input	TM0IOA: Timer 0 I/O
PA1	AN1	TM1IOA	in/out	PADIR1	PAPLU1	AN1: Analog 1 input	TM1IOA: Timer 1 I/O
PA2	AN2	TM2IOA	in/out	PADIR2	PAPLU2	AN2: Analog 2 input	TM2IOA: Timer 2 I/O
PA3	AN3	TM3IOA	in/out	PADIR3	PAPLU3	AN3: Analog 3 input	TM3IOA: Timer 3 I/O
PA4	AN4	TM4IOA	in/out	PADIR4	PAPLU4	AN4: Analog 4 input	TM4IOA: Timer 4 I/O
PA5	AN5	TM7IOA	in/out	PADIR5	PAPLU5	AN5: Analog 5 input	TM7IOA: Timer 7 I/O
PA6	AN6	TM8IOA	in/out	PADIR6	PAPLU6	AN6: Analog 6 input	TM8IOA: Timer 8 I/O
PA7	AN7	TM9IOA	in/out	PADIR7	PAPLU7	AN7: Analog 7 input	TM9IOA: Timer 9 I/O
PB0	AN8		in/out	PBDIR0	PBPLU0	AN8: Analog 8 input	
PB1	AN9		in/out	PBDIR1	PBPLU1	AN9: Analog 9 input	
PB2	AN10		in/out	PBDIR2	PBPLU2	AN10: Analog 10 input	
PB3	AN11	SEG54	in/out	PBDIR3	PBPLU3	AN11: Analog 11 input	SEG54: Segment 54 output
PB4	AN12	SEG53	in/out	PBDIR4	PBPLU4	AN12: Analog 12 input	SEG53: Segment 53 output
PB5	AN13	SEG52	in/out	PBDIR5	PBPLU5	AN13: Analog 13 input	SEG52: Segment 52 output
PB6	AN14	SEG51	in/out	PBDIR6	PBPLU6	AN14: Analog 14 input	SEG51: Segment 51 output
PB7	AN15	SEG50	in/out	PBDIR7	PBPLU7	AN15: Analog 15 input	SEG50: Segment 50 output
PC0	IRQ0B		in/out	PCDIR0	PCPLU0	IRQ0B: External interrupt 0	
PC1	IRQ1B		in/out	PCDIR1	PCPLU1	IRQ1B: External interrupt 1	
PC2	IRQ2B		in/out	PCDIR2	PCPLU2	IRQ2B: External interrupt 2	
PC3	IRQ3B		in/out	PCDIR3	PCPLU3	IRQ3B: External interrupt 3	
PC4	IRQ4B		in/out	PCDIR4	PCPLU4	IRQ4B: External interrupt 4	
PC5			in/out	PCDIR5	PCPLU5		
PC6			in/out	PCDIR6	PCPLU6		
PD0	TM9OD0B		in/out	PDDIR0	PDPLU0	TM9OD0B: Timer 9 output	
PD1	TM9OD1B		in/out	PDDIR1	PDPLU1	TM9OD1B: Timer 9 output	
PD2	TM9OD2B		in/out	PDDIR2	PDPLU2	TM9OD2B: Timer 9 output	
PD3	TM9OD3B		in/out	PDDIR3	PDPLU3	TM9OD3B: Timer 9 output	
PD4	TM9OD4B		in/out	PDDIR4	PDPLU4	TM9OD4B: Timer 9 output	
PD5	TM9OD5B		in/out	PDDIR5	PDPLU5	TM9OD5B: Timer 9 output	
PD6			in/out	PDDIR6	PDPLU6		

Table:1.3.2 Pin Specification (MN101EF57G)

MN101EF57G			I/O	Direction control	Pin control	Function Description	
Pins	Special functions						
P00	LED0	OCD_DATA	in/out	P0DIR0	P0PLUD0	LED0: LED driver pin 0	OCD_DATA: On-board programmer Data pin
P01	LED1	TM9IOB	in/out	P0DIR1	P0PLUD1	LED1: LED driver pin 1	TM9IOB: Timer 9 I/O
P02	OCD_CLK		in/out	P0DIR2	P0PLUD2	OCD_CLK: On-board programmer Clock supply pin	
	LED2	TM7IOB				LED2: LED driver pin 2	TM7IOB: Timer7 I/O
P03	SBI0A	RXD0A	in/out	P0DIR3	P0PLUD3	SBI0A: Serial 0 data input	RXD0A: UART0 data input
	LED3	TM8IOB				LED3: LED driver pin 3	TM8IOB: Timer 8 I/O
P04	SBO0A	TXD0A	in/out	P0DIR4	P0PLUD4	SBO0A: Serial 0 data output	TXD0A: UART0 data output
	LED4	TM0IOB				LED4: LED driver pin 4	TM0IOB: Timer0 I/O
P05	TM2IOB		in/out	P0DIR5	P0PLUD5	TM2IOB: Timer 2 I/O	
	LED5	SBT0A				LED5: LED driver pin 5	SBT0A: Serial 0 clock I/O
P06	LED6		in/out	P0DIR6	P0PLUD6	LED6: LED driver pin 6	
P07	LED7	DA_B	in/out	P0DIR7	P0PLUD7	LED7: LED driver pin 7	DA_B: Analog B output
P20	IRQ0	ACZ0	in/out	P2DIR0	P2PLU0	IRQ0: External interrupt 0	ACZ0: Zero cross 0 input
P21	IRQ1	ACZ1	in/out	P2DIR1	P2PLU1	IRQ1: External interrupt 1	ACZ1: Zero cross 1 input
P22	IRQ2		in/out	P2DIR2	P2PLU2	IRQ2: External interrupt 2	
P23	IRQ3		in/out	P2DIR3	P2PLU3	IRQ3: External interrupt 3	
P24	IRQ4		in/out	P2DIR4	P2PLU4	IRQ4: External interrupt 4	
P25	OSC1		in/out	P2DIR5	P2PLU5	OSC1: Ceramic / crystal high-speed clock input	
P26	OSC2		in/out	P2DIR6	P2PLU6	OSC2: Ceramic / crystal high-speed clock output	
P27	NRST		in/out	P2DIR7	P2PLU7	NRST: Reset	
P33	AN15	SBO4B	in/out	P3DIR3	P3PLU3	AN15: Analog 15 input	SBO4B: Serial 4 transmission data output
P34	SDA4B	SEG30	in/out	P3DIR4	P3PLU4	SDA4B: IIC4 data I/O	SEG30: Segment 30 output
	SBT4B	SCL4B				SBT4B: Serial 4 clock I/O	SCL4B: IIC4 clock I/O
P35	SEG29		in/out	P3DIR5	P3PLU5	SEG29: Segment 29 output	
P43	SBI4B	SEG28	in/out	P4DIR3	P4PLUD3	SBI4B: Serial 4 reception data input	SEG28: Segment 28 output
	SBO0B	TXD0B				SBO0B: Serial 0 transmission data output	TXD0B: UART0 transmission data output
P44	SEG27		in/out	P4DIR4	P4PLUD4	SEG27: Segment 27 output	
	SBI0B	RXD0B				SBI0B: Serial 0 reception data input	RXD0B: UART0 reception data input
P45	SEG26		in/out	P4DIR5	P4PULD5	SEG26: Segment 26 output	
	STB0B	SEG25				STB0B: Serial 0 clock I/O	SEG25: Segment 25 output
P46	SEG24		in/out	P4DIR6	P4PULD6	SEG24: Segment 24 output	
P47	SEG23		in/out	P4DIR7	P4PULD7	SEG23: Segment 23 output	
P50	SBO1A	TXD1A	in/out	P5DIR0	P5PLUD0	SBO1A: Serial 1 transmission data output	TXD1: UART1 transmission data output
P51	SEG15		in/out	P5DIR1	P5PLUD1	SEG15: Segment 15 output	
	SBI1A	RXD1A				SBI1A: Serial 1 reception data input	RXD1A: UART1 reception data input
P52	SEG16		in/out	P5DIR2	P5PLUD2	SEG16: Segment 16 output	
	SBT1A	SEG17				SBT1A: Serial 1 clock I/O	SEG17: Segment 17 output
P53	SEG18		in/out	P5DIR3	P5PLUD3	SEG18: Segment 18 output	
P54	SEG19		in/out	P5DIR4	P5PLUD4	SEG19: Segment 19 output	
P55	SEG20		in/out	P5DIR5	P5PLUD5	SEG20: Segment 20 output	
P56	SEG21	NBUZZERA	in/out	P5DIR6	P5PLUD6	SEG21: Segment 21 output	NBUZZERA: Buzzer reverse output
P57	SEG22	BUZZERA	in/out	P5DIR7	P5PLUD7	SEG22: Segment 22 output	BUZZERA: Buzzer output
P62	TM1IOB	SEG14	in/out	P6DIR2	P6PLU2	TM1IOB: Timer 1 I/O	SEG14: Segment 14 output
P63	TM3IOB	SEG13	in/out	P6DIR3	P6PLU3	TM3IOB: Timer 3 I/O	SEG13: Segment 13 output
P64	TM4IOB	SEG12	in/out	P6DIR4	P6PLU4	TM4IOB: Timer 4 I/O	SEG12: Segment 12 output
P65	SBO2A	TXD2A	in/out	P6DIR5	P6PLU5	SBO2A: Serial 2 transmission data output	TXD2A: UART2 transmission data output
P66	SEG11		in/out	P6DIP6	P6PLU6	SEG11: Segment 11 output	
	SBI2A	RXD2A				SBI2A: Serial 2 reception data input	RXD2A: UART2 reception data input
P67	SEG10		in/out	P6DIR7	P6PLU7	SEG10: Segment 10 output	
	SBT2A	SEG9				SBT2A: Serial 2 clock I/O	SEG9: Segment 9 output

MN101EF57G			I/O	Direction control	Pin control	Function Description	
Pins	Special functions						
P70	KEY0	SBI4A	in/out	P7DIR0	P7PLUD0	KEY0: KEY interrupt input 0	SBI4A: Serial 4 reception data input
	SEG8					SEG8: Segment 8 output	
P71	KEY1	SBO4A	in/out	P7DIR1	P7PLUD1	KEY1: KEY interrupt input 1	SBO4A: Serial 4 transmission data output
	SDA4A	SEG7				SDA4A: IIC4 data I/O	SEG7: Segment 7 output
P72	KEY2	SBT4A	in/out	P7DIR2	P7PLUD2	KEY2: KEY interrupt input 2	SBT4A: Serial 4clock I/O
	SCL4A	SEG6				SCL4A: IIC4 clock I/O	SEG6: Segment 6 output
P73	KEY3	SEG5	in/out	P7DIR3	P7PLUD3	KEY3: KEY interrupt input 3	SEG5: Segment 5 output
P74	KEY4	SEG4	in/out	P7DIR4	P7PLUD4	KEY4: KEY interrupt input 4	SEG4: Segment 4 output
P75	KEY5	SBO1B	in/out	P7DIR5	P7PLUD5	KEY5: KEY interrupt input 5	SBO1B: Serial 1data output
	TXD1B	SEG3				TXD1B: UART1 transmission data output	SEG3: Segment 3 output
P76	KEY6	SBI1B	in/out	P7DIR6	P7PLUD6	KEY6: KEY interrupt input 6	SBI1B: Serial 1 reception data input
	RXD1B	SEG2				RXD1B: UART1 reception data input	SEG2: Segment 2 output
P77	KEY7	SBT1B	in/out	P7DIR7	P7PLUD7	KEY7: KEY interrupt input7	SBT1B: Serial 1clock I/O
	SEG1					SEG1: Segment 1 output	
P80	SDO0	TM9OD0	in/out	P8DIR0	P8PLU0	SDO0: Timer synchronous output 0	TM9OD0: Timer 9 output
	SEG0					SEG0: Segment output0	
P81	SDO1	TM9OD1	in/out	P8DIR1	P8PLU1	SDO1: Timer synchronous output 1	TM9OD1: Timer 9 output
	COM0					COM0: LCD common output	
P82	SDO2	TM9OD2	in/out	P8DIR2	P8PLU2	SDO2: Timer synchronous output 2	TM9OD2: Timer 9 output
	COM1					COM1: LCD common output	
P83	SDO3	TM9OD3	in/out	P8DIR3	P8PLU3	SDO3: Timer synchronous output 3	TM9OD3: Timer 9 output
	COM2					COM2: LCD common output	
P84	SDO4	TM9OD4	in/out	P8DIR4	P8PLU4	SDO4: Timer synchronous output 4	TM9OD4: Timer 9 output
	COM3					COM3: LCD common output	
P85	SDO5	TM9OD5	in/out	P8DIR5	P8PLU5	SDO5: Timer synchronous output 5	TM9OD5: Timer 9 output
	VLC3					VLC3: LCD power	
P86	SDO6		in/out	P8DIR6	P8PLU6	SDO6: Timer synchronous output 6	
	VLC2					VLC2: LCD power	
P87	SDO7		in/out	P8DIR7	P8PLU7	SDO7: Timer synchronous output 7	
	VLC1					VLC1: LCD power	
P90	XI		in/out	P9DIR0	P9PLU0	XI: Ceramic / crystal low-speed clock input	
P91	XO		in/out	P9DIR1	P9PLU1	XOI: Ceramic / crystal low-speed clock output	
P92	AN14	SEG31	in/out	P9DIR2	P9PLU2	AN14: Analog 14 input	SEG31: Segment 31 output
P93	AN13	SEG32	in/out	P9DIR3	P9PLU3	AN13: Analog 13 input	SEG32: Segment 32 output
P94	AN12	SEG33	in/out	P9DIR4	P9PLU4	AN12: Analog 12 input	SEG33: Segment 33 output
	DA_A					DA_A: Analog A output	
PA0	AN0	TM0IOA	in/out	PADIR0	PAPLU0	AN0: Analog 0 input	TM0IOA: Timer 0 I/O
	SEG38					SEG38: Segment 38 output	
PA1	AN1	TM1IOA	in/out	PADIR1	PAPLU1	AN1: Analog 1 input	TM1IOA: Timer 1 I/O
	SEG39					SEG39: Segment 39 output	
PA2	AN2	TM2IOA	in/out	PADIR2	PAPLU2	AN2: Analog 2 input	TM2IOA: Timer 2 I/O
	SEG40					SEG40: Segment 40 output	
PA3	AN3	TM3IOA	in/out	PADIR3	PAPLU3	AN3: Analog 3 input	TM3IOA: Timer 3 I/O
PA4	AN4	TM4IOA	in/out	PADIR4	PAPLU4	AN4: Analog 4 input	TM4IOA: Timer 4 I/O
PA5	AN5	TM7IOA	in/out	PADIR5	PAPLU5	AN5: Analog 5 input	TM7IOA: Timer 7 I/O
PA6	AN6	TM8IOA	in/out	PADIR6	PAPLU6	AN6: Analog 6 input	TM8IOA: Timer 8 I/O
PA7	AN7	TM9IOA	in/out	PADIR7	PAPLU7	AN7: Analog 7 input	TM9IOA: Timer 9 I/O
PB0	AN8	SEG37	in/out	PBDIR0	PBPLU0	AN8: Analog 8 input	SEG37: Segment 37 output
PB1	AN9	SEG36	in/out	PBDIR1	PBPLU1	AN9: Analog 9 input	SEG36: Segment 36 output
PB2	AN10	SEG35	in/out	PBDIR2	PBPLU2	AN10: Analog 10 input	SEG35: Segment 35 output
PB3	AN11	SEG34	in/out	PBDIR3	PBPLU3	AN11: Analog 11 input	SEG34: Segment 34 output

Table:1.3.3 Pin Specification (MN101EF56K)

MN101EF56K			I/O	Direction control	Pin control	Function Description	
Pins	Special functions						
P00	LED0	OCD_DATA	in/out	P0DIR0	P0PLUD0	LED0: LED driver pin 0	OCD_DATA: On-board programmer Data pin
P01	LED1	TM9IOB	in/out	P0DIR1	P0PLUD1	LED1: LED driver pin 1	TM9IOB: Timer 9 I/O
	OCD_CLK					OCD_CLK: On-board programmer Clock supply pin	
P02	LED2	TM7IOB	in/out	P0DIR2	P0PLUD2	LED2: LED driver pin 2	TM7IOB: Timer7 I/O
	SBI0A	RXD0A				SBI0A: Serial 0 data input	RXD0A: UART0 data input
P03	LED3	TM8IOB	in/out	P0DIR3	P0PLUD3	LED3: LED driver pin 3	TM8IOB: Timer 8 I/O
	SBO0A	TXD0A				SBO0A: Serial 0 data output	TXD0A: UART0 data output
P04	LED4	TM0IOB	in/out	P0DIR4	P0PLUD4	LED4: LED driver pin 4	TM0IOB: Timer0 I/O
	TM2IOB	SBT0A				TM2IOB: Timer 2 I/O	SBT0A: Serial 0 clock I/O
P05	LED5		in/out	P0DIR5	P0PLUD5	LED5: LED driver pin 5	
P06	LED6		in/out	P0DIR6	P0PLUD6	LED6: LED driver pin 6	
P07	LED7	DA_B	in/out	P0DIR7	P0PLUD7	LED7: LED driver pin 7	DA_B: Analog B output
P10	AN20	TM0IOC	in/out	P1DIR0	P1PLUD0	AN20: Analog 20 input	TM0IOC: Timer 0 I/O
	SEG45					SEG45: Segment 45 output	
P11	AN21	TM2IOC	in/out	P1DIR1	P1PLUD1	AN21: Analog 21 input	TM2IOC: Timer 2 I/O
	SEG44					SEG44: Segment 44 output	
P12	AN22	TM1IOC	in/out	P1DIR2	P1PLUD2	AN22: Analog 22 input	TM1IOC: Timer 1 I/O
	SEG43					SEG43: Segment 43 output	
P13	AN23	TM3IOC	in/out	P1DIR3	P1PLUD3	AN23: Analog 23 input	TM3IOC: Timer 3 I/O
	SEG42					SEG42: Segment 42 output	
P14	TM4IOC	SEG41	in/out	P1DIR4	P1PLUD4	TM4IOC: Timer 4 I/O	SEG41: Segment 41 output
P15	TM7IOC	BUZZERB	in/out	P1DIR5	P1PLUD5	TM7IOC: Timer 7 I/O	BUZZERB: Buzzer output
	SEG40					SEG40: Segment 40 output	
P16	TM8IOC	NBUZZERB	in/out	P1DIR6	P1PLUD6	TM8IOC: Timer 8 I/O	NBUZZERB: Buzzer reverse output
	SEG39					SEG39: Segment 39 output	
P20	IRQ0	ACZ0	in/out	P2DIR0	P2PLU0	IRQ0: External interrupt 0	ACZ0: Zero cross 0 input
P21	IRQ1	ACZ1	in/out	P2DIR1	P2PLU1	IRQ1: External interrupt 1	ACZ1: Zero cross 1 input
P22	IRQ2		in/out	P2DIR2	P2PLU2	IRQ2: External interrupt 2	
P23	IRQ3		in/out	P2DIR3	P2PLU3	IRQ3: External interrupt 3	
P24	IRQ4		in/out	P2DIR4	P2PLU4	IRQ4: External interrupt 4	
P25	OSC1		in/out	P2DIR5	P2PLU5	OSC1: Ceramic / crystal high-speed clock input	
P26	OSC2		in/out	P2DIR6	P2PLU6	OSC2: Ceramic / crystal high-speed clock output	
P27	NRST		in/out	P2DIR7	P2PLU7	NRST: Reset	
P30	SBO2B	TXD2B	in/out	P3DIR0	P3PLU0	SBO2B: Serial 2 transmission data output	TXD2B: UART 2 transmission data output
	SEG38					SEG38: Segment 38 output	
P31	SBI2B	RXD2B	in/out	P3DIR1	P3PLU1	SBI2B: Serial 2 reception data input	RXD2B: UART 2 reception data input
	SEG37					SEG37: Segment 37 output	
P32	SBT2B	SEG36	in/out	P3DIR2	P3PLU2	SBT2B: Serial 2 clock I/O	SEG36: Segment 36 output
P33	AN15	SBO4B	in/out	P3DIR3	P3PLU3	AN15: Analog 15 input	SBO4B: Serial 4 transmission data output
	SDA4B	SEG35				SDA4B: IIC 4 data I/O	SEG35: Segment 35 output
P34	SBT4B	SCL4B	in/out	P3DIR4	P3PLU4	SBT4B: Serial 4 clock I/O	SCL4B: IIC 4 clock I/O
	SEG34					SEG34: Segment 34 output	
P35	SBI4B	SEG33	in/out	P3DIR5	P3PLU5	SBI4B: Serial 4 reception data input	SEG33: Segment 33 output
P36	SEG32		in/out	P3DIR6	P3PLU6	SEG32: Segment 32 output	
P40	SBO3B	TXD3B	in/out	P4DIR0	P4PLUD0	SBO3B: Serial 3 transmission data output	TXD3B: UART 3 transmission data output
	SEG31					SEG31: Segment 31 output	
P41	SBI3B	RXD3B	in/out	P4DIR1	P4PLUD1	SBI3B: Serial 3 reception data input	RXD3B: UART 3 reception data input
	SEG30					SEG30: Segment 30 output	
P42	SBT3B	SEG29	in/out	P4DIR2	P4PLUD2	SBT3B: Serial 3 clock I/O	SEG29: Segment 29 output
P43	SBO0B	TXD0B	in/out	P4DIR3	P4PLUD3	SBO0B: Serial 0 transmission data output	TXD0B: UART 0 transmission data output
	SEG28					SEG28: Segment 28 output	
P44	SBI0B	RXD0B	in/out	P4DIR4	P4PLUD4	SBI0B: Serial 0 reception data input	RXD0B: UART 0 reception data input
	SEG27					SEG27: Segment 27 output	
P45	STB0B	SEG26	in/out	P4DIR5	P4PULD5	STB0B: Serial 0 clock I/O	SEG26: Segment 26 output
P46	SEG25		in/out	P4DIR6	P4PULD6	SEG25: Segment 25 output	
P47	SEG24		in/out	P4DIR7	P4PULD7	SEG24: Segment 24 output	

MN101EF56K			I/O	Direction control	Pin control	Function Description	
Pins	Special functions						
P50	SBO1A	TXD1A	in/out	P5DIR0	P5PLUD0	SBO1A: Serial 1 transmission data output	TXD1: UART1 transmission data output
	SEG16					SEG16: Segment 16 output	
P51	SBI1A	RXD1A	in/out	P5DIR1	P5PLUD1	SBI1A: Serial 1 reception data input	RXD1A: UART1 reception data input
	SEG17					SEG17: Segment 17 output	
P52	SBT1A	SEG18	in/out	P5DIR2	P5PLUD2	SBT1A: Serial 1 clock I/O	SEG18: Segment 18 output
P53	SEG19		in/out	P5DIR3	P5PLUD3	SEG19: Segment 19 output	
P54	SEG20		in/out	P5DIR4	P5PLUD4	SEG20: Segment 20 output	
P55	SEG21		in/out	P5DIR5	P5PLUD5	SEG21: Segment 21 output	
P56	SEG22	NBUZZERA	in/out	P5DIR6	P5PLUD6	SEG22: Segment 22 output	NBUZZERA: Buzzer reverse output
P57	SEG23	BUZZERA	in/out	P5DIR7	P5PLUD7	SEG23: Segment 23 output	BUZZERA: Buzzer output
P61	DA_C	SEG15	in/out	P6DIR1	P6PLU1	DA_C: Analog C output	SEG15: Segment 15 output
P62	TM1IOB	SEG14	in/out	P6DIR2	P6PLU2	TM1IOB: Timer 1 I/O	SEG14: Segment 14 output
P63	TM3IOB	SEG13	in/out	P6DIR3	P6PLU3	TM3IOB: Timer 3 I/O	SEG13: Segment 13 output
P64	TM4IOB	SEG12	in/out	P6DIR4	P6PLU4	TM4IOB: Timer 4 I/O	SEG12: Segment 12 output
P65	SBO2A	TXD2A	in/out	P6DIR5	P6PLU5	SBO2A: Serial 2 transmission data output	TXD2A: UART2 transmission data output
	SEG11					SEG11: Segment 11 output	
P66	SBI2A	RXD2A	in/out	P6DIP6	P6PLU6	SBI2A: Serial 2 reception data input	RXD2A: UART2 reception data input
	SEG10					SEG10: Segment 10 output	
P67	SBT2A	SEG9	in/out	P6DIR7	P6PLU7	SBT2A: Serial 2 clock I/O	SEG9: Segment 9 output
P70	KEY0	SBI4A	in/out	P7DIR0	P7PLUD0	KEY0: KEY interrupt input 0	SBI4A: Serial 4 reception data input
	SEG8					SEG8: Segment 8 output	
P71	KEY1	SBO4A	in/out	P7DIR1	P7PLUD1	KEY1: KEY interrupt input 1	SBO4A: Serial 4 transmission data output
	SDA4A	SEG7				SDA4A: IIC 4 data I/O	SEG7: Segment 7 output
P72	KEY2	SBT4A	in/out	P7DIR2	P7PLUD2	KEY2: KEY interrupt input 2	SBT4A: Serial 4clock I/O
	SCL4A	SEG6				SCL4A: IIC 4 clock I/O	SEG6: Segment 6 output
P73	KEY3	SEG5	in/out	P7DIR3	P7PLUD3	KEY3: KEY interrupt input 3	SEG5: Segment 5 output
P74	KEY4	SEG4	in/out	P7DIR4	P7PLUD4	KEY4: KEY interrupt input 4	SEG4: Segment 4 output
P75	KEY5	SBO1B	in/out	P7DIR5	P7PLUD5	KEY5: KEY interrupt input 5	SBO1B: Serial 1data output
	TXD1B	SEG3				TXD1B: UART 1 transmission data output	SEG3: Segment 3 output
P76	KEY6	SBI1B	in/out	P7DIR6	P7PLUD6	KEY6: KEY interrupt input 6	SBI1B: Serial 1 reception data input
	RXD1B	SEG2				RXD1B: UART 1 reception data input	SEG2: Segment 2 output
P77	KEY7	SBT1B	in/out	P7DIR7	P7PLUD7	KEY7: KEY interrupt input7	SBT1B: Serial 1clock I/O
	SEG1					SEG1: Segment 1 output	
P80	SDO0	TM9OD0	in/out	P8DIR0	P8PLU0	SDO0: Timer synchronous output 0	TM9OD0: Timer 9 output
	SEG0					SEG0: Segment 0 output	
P81	SDO1	TM9OD1	in/out	P8DIR1	P8PLU1	SDO1: Timer synchronous output 1	TM9OD1: Timer 9 output
	COM0					COM0: LCD common output	
P82	SDO2	TM9OD2	in/out	P8DIR2	P8PLU2	SDO2: Timer synchronous output 2	TM9OD2: Timer 9 output
	COM1					COM1: LCD common output	
P83	SDO3	TM9OD3	in/out	P8DIR3	P8PLU3	SDO3: Timer synchronous output 3	TM9OD3: Timer 9 output
	COM2					COM2: LCD common output	
P84	SDO4	TM9OD4	in/out	P8DIR4	P8PLU4	SDO4: Timer synchronous output 4	TM9OD4: Timer 9 output
	COM3					COM3: LCD common output	
P85	SDO5	TM9OD5	in/out	P8DIR5	P8PLU5	SDO5: Timer synchronous output 5	TM9OD5: Timer 9 output
	VLC3					VLC3: LCD power supply	
P86	SDO6		in/out	P8DIR6	P8PLU6	SDO6: Timer synchronous output 6	
	VLC2					VLC2: LCD power supply	
P87	SDO7		in/out	P8DIR7	P8PLU7	SDO7: Timer synchronous output 7	
	VLC1					VLC1: LCD power supply	
P90	XI		in/out	P9DIR0	P9PLU0	XI: Ceramic / crystal low-speed clock input	
P91	XO		in/out	P9DIR1	P9PLU1	XOI: Ceramic / crystal low-speed clock output	
P92	AN19	SEG46	in/out	P9DIR2	P9PLU2	AN19: Analog 14 input	SEG46: Segment 46 output
P93	AN18	SEG47	in/out	P9DIR3	P9PLU3	AN18: Analog 13 input	SEG47: Segment 47 output
P94	AN17	SEG48	in/out	P9DIR4	P9PLU4	AN17: Analog 12 input	SEG48: Segment 48 output
	DA_B					DA_B: Analog B output	
P95	AN16	SEG49	in/out	P9DIR4	P9PLU4	AN16: Analog 16 input	SEG49: Segment 49 output
	DA_A					DA_A: Analog A output	

MN101EF56K			I/O	Direction control	Pin control	Function Description	
Pins	Special functions						
PA0	AN0	TM0IOA	in/out	PADIR0	PAPLU0	AN0: Analog 0 input	TM0IOA: Timer 0 I/O
PA1	AN1	TM1IOA	in/out	PADIR1	PAPLU1	AN1: Analog 1 input	TM1IOA: Timer 1 I/O
PA2	AN2	TM2IOA	in/out	PADIR2	PAPLU2	AN2: Analog 2 input	TM2IOA: Timer 2 I/O
PA3	AN3	TM3IOA	in/out	PADIR3	PAPLU3	AN3: Analog 3 input	TM3IOA: Timer 3 I/O
PA4	AN4	TM4IOA	in/out	PADIR4	PAPLU4	AN4: Analog 4 input	TM4IOA: Timer 4 I/O
PA5	AN5	TM7IOA	in/out	PADIR5	PAPLU5	AN5: Analog 5 input	TM7IOA: Timer 7 I/O
PA6	AN6	TM8IOA	in/out	PADIR6	PAPLU6	AN6: Analog 6 input	TM8IOA: Timer 8 I/O
PA7	AN7	TM9IOA	in/out	PADIR7	PAPLU7	AN7: Analog 7 input	TM9IOA: Timer 9 I/O
PB0	AN8		in/out	PBDIR0	PBPLU0	AN8: Analog 8 input	
PB1	AN9		in/out	PBDIR1	PBPLU1	AN9: Analog 9 input	
PB2	AN10		in/out	PBDIR2	PBPLU2	AN10: Analog 10 input	
PB3	AN11	SEG54	in/out	PBDIR3	PBPLU3	AN11: Analog 11 input	SEG54: Segment 54 output
PB4	AN12	SEG53	in/out	PBDIR4	PBPLU4	AN12: Analog 12 input	SEG53: Segment 53 output
PB5	AN13	SEG52	in/out	PBDIR5	PBPLU5	AN13: Analog 13 input	SEG52: Segment 52 output
PB6	AN14	SEG51	in/out	PBDIR6	PBPLU6	AN14: Analog 14 input	SEG51: Segment 51 output
PB7	AN15	SEG50	in/out	PBDIR7	PBPLU7	AN15: Analog 15 input	SEG50: Segment 50 output

1.3.3 Pin Functions

Table:1.3.4 Pin Functions (MN101EF76K)

MN101EF76K		I/O	Other Functions	Functions	Descriptions
Pins	Pin No.				
V _{SS}	14, 22	-		Power supply pins	Supply 1.8 V to 5.5 V to V _{DD5} , and 0 V to V _{SS} . Connect 0.1 μF and more than 1 μF of bypass capacitor for internal power stabilization
V _{DD5}	17				
V _{DD18}	19	-		Internal power output pin	Outputs internal power voltage 1.8 V. Connect 0.1 μF and more than 1 μF of bypass capacitor between V _{DD18} and V _{SS} pins for internal power stabilization.
OSC1	15	Input	P25	High-speed operation clock input pin	Connect these oscillation pins to ceramic oscillator or crystal oscillator for high-speed operation clock.
OSC2	16	Output	P26	High-speed operation clock output pin	For external clock input, input to OSC1 and open OSC2. The chip will not operate with an external clock when using either STOP mode or SLOW mode
XI	12	Input	P90	Low-speed operation clock input pin	Connect these oscillation pins to ceramic oscillator or crystal oscillator for low-speed operation clock.
XO	13	Output	P91	Low-speed operation clock output pin	For external clock input, input to XI and open XO. The chip will not operate with an external clock when using STOP mode.
NRST	11	Input	P27	Reset pin [Active low]	This pin resets the chip at power on, is allocated as P27 and contains an internal pull-up resistor (Typ. 50 kΩ). Setting this pin low initializes the internal state of the LSI. Thereafter, setting the input to high releases the reset. The hardware waits for the system clock to stabilize, then processes the reset interrupt. Also, if "0" is written to P27 and the reset is initiated by software, a low level will be output. The output has an Nch open-drain configuration. If a capacitor is to be inserted between NRST and V _{SS} , it is recommended that a discharge diode be placed between NRST and V _{DD5} .
ATRST	10	Input		Auto reset setting pin	Input "High" to enable auto reset function and "Low" to disable this function.
P00	30	I/O	LED0	OCD_DATA	8-bit CMOS tri-state I/O port. Each bit can be set individually as either an input or output by the P0DIR register. A pull-up /pull-down resistor for each bit can be selected individually by the P0PLUD register. A pull-up/down resistor connection for each port can be selected individually by the SELUD register. (However, pull-up and pull-down resistors cannot be mixed.) At reset, the input mode is selected and pull-up resistors are disabled (high impedance).
P01	31		LED1	TM9IOB	
			OCD_CLK		
P02	32		LED2	TM7IOB	
			SBI0A	RXD0A	
P03	36		LED3	TM8IOB	
			SBO0A	TXD0A	
P04	37		LED4	TM0IOB	
		I/O	TM2IOB	SBT0A	7-bit CMOS tri-state I/O port. Each bit can be set individually as either an input or output by the P1DIR register. A pull-up /pull-down resistor for each bit can be selected individually by the P1PLUD register. A pull-up/down resistor connection for each port can be selected individually by the SELUD register. (However, pull-up and pull-down resistors cannot be mixed.) At reset, the input mode is selected and pull-up resistors are disabled (high impedance).
P05	38		LED5	SBO3A	
			TXD3A		
P06	39		LED6	SBI3A	
			RXD3A		
P07	40		LED7	DA_D	
			SBT3A		
P10	112		TM0IOC	AN20	
P11	111	I/O	TM2IOC	AN21	7-bit CMOS tri-state I/O port. Each bit can be set individually as either an input or output by the P2DIR register. A pull-up resistor for each bit can be selected individually by the P2PLU register. At reset, the input mode is selected and pull-up resistors are disabled (high impedance).
P12	110		TM1IOC	AN22	
P13	109		TM3IOC	AN23	
P14	108		TM4IOC		
P15	107		TM7IOC	BUZZERB	
P16	106		TM8IOC	NBUZZERB	
P20	41		IRQ0A	ACZ0	
P21	42		IRQ1A	ACZ1	
P22	43	I/O	IRQ2A		Input port 2
P23	44		IRQ3A		
P24	45		IRQ4A		
P25	15		OSC1		
P26	16	Input	OSC2		P27 is an Nch open-drain port. When "0" is written and the reset is initiated by software, a low level will be output.
P27	11		NRST		

MN101EF76K		I/O	Other Functions		Functions	Descriptions
Pins	Pin No.					
P30	105	I/O	SBO2B	TXD2B	I/O port 3	7-bit CMOS tri-state I/O port. Each bit can be set individually as either an input or output by the P3DIR register. A pull-up resistor for each bit can be selected individually by the P3PLU register. At reset, the input mode is selected and pull-up resistors are disabled (high impedance).
P31	104		SEG38			
			SBI2B	RXD2B		
P32	103		SEG37			
P33	102		SBT2B	SEG36		
P34	101		SBO4B	SDA4B		
			SEG35			
P35	100	I/O	SBT4B	SCL4B	I/O port 4	8-bit CMOS tri-state I/O port. Each bit can be set individually as either an input or output by the P4DIR register. A pull-up /pull-down resistor for each bit can be selected individually by the P4PLUD register. A pull-up/down resistor connection for each port can be selected individually by the SELUD register. (However, pull-up and pull-down resistors cannot be mixed.) At reset, the input mode is selected and pull-up resistors are disabled (high impedance).
P36	96		SEG34			
			SBI4B	SEG33		
			SEG32			
P40	95		SBO3B	TXD3B		
P41	94		SEG31			
			SBI3B	RXD3B		
P42	93	I/O	SEG30		I/O port 5	8-bit CMOS tri-state I/O port. Each bit can be set individually as either an input or output by the P5DIR register. A pull-up /pull-down resistor for each bit can be selected individually by the P5PLUD register. A pull-up/down resistor connection for each port can be selected individually by the SELUD register. (A pull-up/pull down can not be mixed.) At reset, the input mode is selected and pull-up resistors are disabled (high impedance).
P43	92		SBT3B	SEG29		
			SBO0B	TXD0B		
P44	91		SEG28			
			SBI0B	RXD0B		
P45	90		SEG27			
P46	89		SBT0B	SEG25		
P47	88	I/O	SEG25		I/O port 6	7-bit CMOS tri-state I/O port. Each bit can be set individually as either an input or output by the P6DIR register. A pull-up resistor for each bit can be selected individually by the P6PLU register. At reset, the input mode is selected and pull-up resistors are disabled (high impedance).
			SEG24			
P50	73		SBO1A	TXD1A		
P51	74		SEG16			
			SBI1A	RXD1A		
P52	75		SEG17			
P53	76		SBT1A	SEG18		
P54	77	I/O	SEG19		I/O port 7	8-bit CMOS tri-state I/O port. Each bit can be set individually as either an input or output by the P7DIR register. A pull-up /pull-down resistor for each bit can be selected individually by the P7PLUD register. A pull-up/down resistor connection for each port can be selected individually by the SELUD register. (A pull-up/pull down can not be mixed.) At reset, the input mode is selected and pull-up resistors are disabled (high impedance).
P55	78		SEG20			
P56	79		SEG21			
P57	80		SEG22	NBUZZERA		
			BUZZERA			
P61	72		DA_C	SEG15		
P62	71		TM1IOB	SEG14		
P63	70	I/O	TM3IOB	SEG13	I/O port 7	8-bit CMOS tri-state I/O port. Each bit can be set individually as either an input or output by the P7DIR register. A pull-up /pull-down resistor for each bit can be selected individually by the P7PLUD register. A pull-up/down resistor connection for each port can be selected individually by the SELUD register. (A pull-up/pull down can not be mixed.) At reset, the input mode is selected and pull-up resistors are disabled (high impedance).
P64	69		TM4IOB	SEG12		
P65	68		SBO2A	TXD2A		
			SEG11			
P66	67		SBI2A	RXD2A		
			SEG10			
P67	66		SBT2A	SEG9		
P70	65	I/O	KEY0	SBI4A	I/O port 7	8-bit CMOS tri-state I/O port. Each bit can be set individually as either an input or output by the P7DIR register. A pull-up /pull-down resistor for each bit can be selected individually by the P7PLUD register. A pull-up/down resistor connection for each port can be selected individually by the SELUD register. (A pull-up/pull down can not be mixed.) At reset, the input mode is selected and pull-up resistors are disabled (high impedance).
			SEG8			
P71	60		KEY1	SBO4A		
			SDA4A	SEG7		
P72	59		KEY2	SBT4A		
			SCL4A	SEG6		
P73	58		KEY3	SEG5		
P74	57	I/O	KEY4	SEG4	I/O port 7	8-bit CMOS tri-state I/O port. Each bit can be set individually as either an input or output by the P7DIR register. A pull-up /pull-down resistor for each bit can be selected individually by the P7PLUD register. A pull-up/down resistor connection for each port can be selected individually by the SELUD register. (A pull-up/pull down can not be mixed.) At reset, the input mode is selected and pull-up resistors are disabled (high impedance).
P75	56		KEY5	SBO1B		
			TXD1B	SEG3		
P76	55		KEY6	SBI1B		
			RXD1B	SEG2		
P77	54		KEY7	SBT1B		
			SEG1			

MN101EF76K		I/O	Other Functions	Functions	Descriptions
Pins	Pin No.				
P80	53	I/O	SDO0 TM9OD0	I/O port 8	8-bit CMOS tri-state I/O port. Each bit can be set individually as either an input or output by the P8DIR register. A pull-up resistor for each bit can be selected individually by the P8PLU register. At reset, the input mode is selected and pull-up resistors are disabled (high impedance).
P81	52		SEG0 TM9OD1		
P82	51		SDO1 COM0		
P83	50		SDO2 TM9OD2		
P84	49		COM1 TM9OD3		
P85	48		SDO3 COM2		
P86	47		SDO4 TM9OD4		
P87	46		COM3 TM9OD5		
			V _{LC3}		
			SDO6		
			V _{LC2}		
			SDO7		
			V _{LC1}		
P90	12	I/O	XI	I/O port 9	6-bit CMOS tri-state I/O port. Each bit can be set individually as either an input or output by the P9DIR register. A pull-up resistor for each bit can be selected individually by the P9PLU register. At reset, the input mode is selected and pull-up resistors are disabled (high impedance).
P91	13		XO		
P92	113		AN19 SEG46		
P93	114		AN18 SEG47		
P94	115		AN17 SEG48		
P95	116		DA_B		
			AN16 SEG49		
			DA_A		
PA0	1	I/O	AN0 TM0IOA	I/O port A	8-bit CMOS tri-state I/O port. Each bit can be set individually as either an input or output by the PADIR register. A pull-up resistor for each bit can be selected individually by the PAPLU register. At reset, the input mode is selected and pull-up resistors are disabled (high impedance).
PA1	2		AN1 TM1IOA		
PA2	3		AN2 TM2IOA		
PA3	4		AN3 TM3IOA		
PA4	5		AN4 TM4IOA		
PA5	6		AN5 TM7IOA		
PA6	7		AN6 TM8IOA		
PA7	8		AN7 TM9IOA		
PB0	124	I/O	AN8	I/O port B	8-bit CMOS tri-state I/O port. Each bit can be set individually as either an input or output by the PBDIR register. A pull-up /pull-down resistor for each bit can be selected individually by the PBPLUD register. At reset, the input mode is selected and pull-up resistors are disabled (high impedance).
PB1	123		AN9		
PB2	122		AN10		
PB3	121		AN11 SEG54		
PB4	120		AN12 SEG53		
PB5	119		AN13 SEG52		
PB6	118		AN14 SEG51		
PB7	117		AN15 SEG50		
PC0	23	I/O	IRQ0B	I/O port C	7-bit CMOS tri-state I/O port. Each bit can be set individually as either an input or output by the PCDIR register. A pull-up resistor for each bit can be selected individually by the PCPLU register. At reset, the input mode is selected and pull-up resistors are disabled (high impedance).
PC1	24		IRQ1B		
PC2	25		IRQ2B		
PC3	26		IRQ3B		
PC4	27		IRQ4B		
PC5	28				
PC6	29				
PD0	87	I/O	TM9OD0B	I/O port D	7-bit CMOS tri-state I/O port. Each bit can be set individually as either an input or output by the PDDIR register. A pull-up resistor for each bit can be selected individually by the PDPLU register. At reset, the input mode is selected and pull-up resistors are disabled (high impedance).
PD1	86		TM9OD1B		
PD2	85		TM9OD2B		
PD3	84		TM9OD3B		
PD4	83		TM9OD4B		
PD5	82		TM9OD5B		
PD6	81				

MN101EF76K		I/O	Other Functions		Functions	Descriptions
Pins	Pin No.					
SBO0A	36	I/O	P03	TXD0A	Serial interface transmission data output pins	Transmission data output pins for serial interface 0, 1, 2, 3, and 4. The output configuration, either COMS push-pull or Nch open-drain can be selected by the P0ODC, P3ODC, P4ODC, P5ODC, P6ODC, and P7ODC registers. Pull-up resistor can be selected by the P0PLUD, P3PLU, P4PLUD, P5PLUD, P6PLU and P7PLUD registers. Select the output mode by the P0DIR, P3DIR, P4DIR, P5DIR, P6DIR and P7DIR registers to select serial data output by the serial mode register 1 (SC0MD1, SC1MD1, SC2MD1, SC3MD1, and SC4MD1). These can be used as normal I/O pins when the serial interface is not used.
SBO0B	92		TM8IOB	LED3		
			P43	SEG8		
SBO1A	73		TXD0B			
			P50	SEG16		
SBO1B	56		TXD1A			
			P75	SEG3		
			KEY5			
			TXD1B			
SBO2A	68		P65	SEG11		
			TXD2A			
SBO2B	105		P30	SEG38		
			TXD2B			
SBO3A	38	I/O	P05	LED5	Serial interface reception data input pins	Reception data input pins for serial interface 0, 1, 2, 3, and 4. Pull-up resistor can be selected by the P0PLUD, P3PLU, P4PLUD, P5PLUD, P6PLU and P7PLUD registers. Select input mode by the P0DIR, P3DIR, P4DIR, P5DIR, P6DIR and P7DIR registers to select serial data input by the serial mode register 1 (SC0MD1, SC1MD1, SC2MD1, SC3MD1, and SC4MD1). These can be used as normal I/O pins when the serial interface is not used.
			TXD3A			
SBO3B	95		P40	SEG31		
			TXD3B			
SBO4A	60		P71	SEG7		
			KEY1	SDA4A		
SBO4B	102		P33	SEG35		
			AN15	SDA4B		
SBI0A	32		P02	RXD0A		
			TM7IOB	LED2		
SBI0B	91		P44	SEG27		
			RXD0B			
SBI1A	74	I/O	P51	SEG17	Serial interface clock I/O pins	Clock I/O pins for serial interface 0, 1, 2, 3, and 4. The output configuration, either CMOS push-pull or Nch open-drain can be selected with the P0ODC, P3ODC, P4ODC, P5ODC, P6ODC, and P7ODC registers. Pull-up resistor can be selected by the P0PLUD, P3PLU, P4PLUD, P5PLUD, P6PLU and P7PLUD registers. Select input mode or output mode by the P0DIR, P3DIR, P4DIR, P5DIR, P6DIR and P7DIR registers and serial data output mode by serial mode register 1 (SC0MD1, SC1MD1, SC2MD1, SC3MD1, and SC4MD1) according to the communication mode. These can be used as normal I/O pins when the serial interface is not used.
			RXD1A			
SBI1B	55		P76	SEG2		
			KEY6	RXD1B		
SBI2A	67		P66	SEG10		
			RXD2A			
SBI2B	104		P31	SEG37		
			RXD2B			
SBI3A	39		P06	LED6		
			RXD3A			
SBI3B	94		P41	SEG30		
			RXD3B			
SBI4A	65	I/O	P70	SEG8	Serial interface clock I/O pins	Clock I/O pins for serial interface 0, 1, 2, 3, and 4. The output configuration, either CMOS push-pull or Nch open-drain can be selected with the P0ODC, P3ODC, P4ODC, P5ODC, P6ODC, and P7ODC registers. Pull-up resistor can be selected by the P0PLUD, P3PLU, P4PLUD, P5PLUD, P6PLU and P7PLUD registers. Select input mode or output mode by the P0DIR, P3DIR, P4DIR, P5DIR, P6DIR and P7DIR registers and serial data output mode by serial mode register 1 (SC0MD1, SC1MD1, SC2MD1, SC3MD1, and SC4MD1) according to the communication mode. These can be used as normal I/O pins when the serial interface is not used.
			KEY0			
SBI4B	100		P35	SEG33		
SBT0A	37		P04	TM2IOB		
			TM0IOB	LED4		
SBT0B	90		P45	SEG26		
SBT1A	75		P52	SEG18		
SBT1B	54		P77	SEG1		
			KEY7			
SBT2A	66		P67	SEG9		
SBT2B	103		P32	SEG36		
SBT3A	40		P07	LED7		
			DA_D			
SBT3B	93	I/O	P42	SEG29	Serial interface clock I/O pins	Clock I/O pins for serial interface 0, 1, 2, 3, and 4. The output configuration, either CMOS push-pull or Nch open-drain can be selected with the P0ODC, P3ODC, P4ODC, P5ODC, P6ODC, and P7ODC registers. Pull-up resistor can be selected by the P0PLUD, P3PLU, P4PLUD, P5PLUD, P6PLU and P7PLUD registers. Select input mode or output mode by the P0DIR, P3DIR, P4DIR, P5DIR, P6DIR and P7DIR registers and serial data output mode by serial mode register 1 (SC0MD1, SC1MD1, SC2MD1, SC3MD1, and SC4MD1) according to the communication mode. These can be used as normal I/O pins when the serial interface is not used.
SBT4A	59		P72	SEG6		
			KEY2	SCL4A		
SBT4B	101		P34	SEG34		
			SCL4B			

MN101EF76K		I/O	Other Functions		Functions	Descriptions
Pins	Pin No.					
TXD0A	36	Output	P03	SBO0A	UART transmission data output pins	In the serial interface 0, 1, 2, and 3 in UART mode, these pins are configured as the transmission data output pin. The output configuration, either CMOS push-pull or Nch open-drain can be selected with the P0ODC, P3ODC, P4ODC, P5ODC, P6ODC, and P7ODC registers. Pull-up resistor can be selected by the P0PLUD, P3PLU, P4PLUD, P5PLUD, P6PLU, and P7PLUD registers. Select output mode by the P0DIR, P3DIR, P4DIR, P5DIR, P6DIR and P7DIR registers to select serial data output by the serial mode register 1 (SC0MD1, SC1MD1, SC2MD1, and SC3MD1). These can be used as normal I/O pins when the serial interface is not used.
TXD0B	92		TM8IOB	LED3		
TXD1A	73		P43	SEG28		
TXD1B	56		SBO0B			
TXD2A	68		P50	SEG16		
TXD2B	105		SBO1A			
TXD3A	38		P75	SEG3		
TXD3B	95		KEY5	SBO1B		
RXD0A	32	Input	P65	SEG11	UART reception data input pins	In the serial interface 0, 1, 2, and 3 in UART mode, these pins are configured as the reception data input pin. Pull-up resistor can be selected by the P0PLUD, P3PLU, P4PLUD, P5PLUD, P6PLU, and P7PLUD registers. Select input mode by the P0DIR, P3DIR, P4DIR, P5DIR, P6DIR and P7DIR registers to select serial data input by the serial mode register 1 (SC0MD1, SC1MD1, SC2MD1, and SC3MD1). These can be used as normal I/O pins when the serial interface is not used.
RXD0B	91		SBO2A	TXD2A		
RXD1A	74		P30	SEG38		
RXD1B	55		SBO2B			
RXD2A	67		P05	LED5		
RXD2B	104		SBO3A			
RXD3A	39		P40	SEG31		
RXD3B	94		SBO3B			
SDA4A	60	I/O	P02	SBI0A	IIC data I/O pins	In the serial interface 4 in IIC mode, these pins are configured as the data input / output pin. For the output configuration, select Nch open-drain by the P3ODC and P7ODC register to select pull-up resistor by the P3PLU and P7PLUD register. Select output mode by the P3DIR and P7DIR register to select serial data input / output by the serial mode register 1 (SC4MD1). These can be used as normal I/O pins when the serial interface is not used.
SDA4B	102		TM7IOB	LED2		
SCL4A	59	I/O	P44	SEG27	IIC clock I/O pins	In the serial interface 4 in IIC mode, these pins are configured as the clock input / output pin. For the output configuration, select Nch open-drain by the P3ODC and P7ODC register to select pull-up resistor by the P3PLU and P7PLUD register. Select output mode by the P3DIR and P7DIR register to select clock data input / output by the serial mode register 1 (SC4MD1). These can be used as normal I/O pins when the serial interface is not used.
SCL4B	101		SBI0B	RXD0B		
OCD_DATA	30	I/O	P51	SEG17	On-board programmer I/O pins	Data I/O pin and clock input pin for the on-board programmer. Refer to [Chapter 24 24.3 Serial Programmer Rewriting]. These can be used as normal I/O pins when the on-board programmer is not used.
OCD_CLK	31		SBI1A			
			P76	SEG2		
			SBI1B	KEY6		
			P66	SEG10		
			SBI2A			
			P31	SEG37		
			SBI2B			
			P06	LED6		
			SBI3A			
			P41	SEG30		
			SBI3B			

MN101EF76K		I/O	Other Functions		Functions	Descriptions			
Pins	Pin No.								
TM0IOA	1	I/O	PA0	AN0	Timer I/O pins	Event count clock input, timer output, and PWM signal output pins for 8-bit timer 0 to 4. To use these pins for event clock input, input mode can be selected by the P0DIR, P1DIR, P6DIR, PADIR, TMCKSEL1, TMINSEL1, and TMINSEL2 registers. In input mode, pull-up resistors can be selected by the P0PLUD, P1PLUD, P6PLU, and PAPLU register. To use these pins for timer output or PWM signal output, select special function pins by the port 0 output mode register, port 1 output mode register, port 6 output mode register, and port A output mode register (P0OMD, P1OMD, P6OMD, and PAOMD) to select output mode by the P0DIR, P1DIR, P6DIR, and PADIR registers. These can be used as normal I/O pins when not used as timer I/O pins.			
TM0IOB	37		P04	SBT0A					
TM0IOC	112		TM2IOB	LED4					
			P10	AN20					
TM1IOA	2		SEG45						
			PA1	AN1					
TM1IOB	71		P62	SEG14					
TM1IOC	110		P12	AN22					
TM2IOA	3		SEG43						
			PA2	AN2					
TM2IOB	37		P04	SBT0A					
TM2IOC	111		TM0IOB	LED4					
			P11	AN21					
TM3IOA	4		SEG44						
			PA3	AN3					
TM3IOB	70		P63	SEG13					
TM3IOC	109	TM3IOB							
		P13	AN23						
TM4IOA	5	SEG42							
TM4IOB	69	PA4	AN4						
TM4IOC	108	P64	SEG12						
		P14	SEG41						
BUZZERA	80	Output	P57	SEG23	Buzzer output pins	Piezoelectric buzzer driving pin. Buzzer output is available to port 1, and port 5. The driving frequency can be set in the DLYCTR register. In order to select buzzer output to port 1 and port 5, select the special function pin in the output mode registers (P1OMD1, P1OMD2, P5OMD), and set the direction control registers (P1DIR, and P5DIR) to the output mode. At the same time, select buzzer output in the oscillation stabilization wait control register (DLYCTR). These can be used as normal I/O pins when the buzzer output function is not used.			
NBUZZERA	79		P56	SEG22					
BUZZERB	107		P15	SEG40					
NBUZZERB	106		TM7IOC						
		P16	SEG39						
TM7IOA	6	I/O	PA5	AN5	Timer I/O pins	Event count clock input, timer output, and PWM signal output pins for 16-bit timer 7, 8, and 9. To use these pins for event clock input, input mode can be selected by the P0DIR, P1DIR, and PADIR registers. In input mode, pull-up resistors can be selected by the P0PLUD, P1PLUD, and PAPLU register. To use these pins for timer output or PWM signal output, select special function pins by the output mode registers (P0OMD, P1OMD1, and PAOMD) to select output mode by the direction control registers (P0DIR, P1DIR, and PADIR). These can be used as normal I/O pins when not used as timer I/O pins.			
			TM7IOB	32			P02	RXD0A	
TM7IOC	107		SBI0A	LED2					
			P15	BUZZERB					
TM8IOA	7		SEG40						
			PA6	AN6					
TM8IOB	36		P03	TXD0A					
TM8IOC	106		SBO0A	LED3					
			P16	NBUZZERB					
TM9IOA	8		SEG39						
			PA7	AN7					
TM9IOB	31		P01	OCD_CLK					
TM9OD0A	53		Output	LED1				Timer output pins	Timer output and PWM signal output pins for 16-bit timer. To use these pins for timer output or PWM signal output, select special function pins by the output mode registers (P8OMD, and PDOMD) to select output mode by the direction control registers (P8DIR, and PDDIR). These can be used as normal I/O pins when not used as timer output pins.
				P80			SEG0		
				SD00					
				TM9OD1A			52		
		TM9OD2A		51	SD01				
					P82	COM1			
		TM9OD3A		50	SD02				
					P83	COM2			
		TM9OD4A		49	SDO3				
					P84	COM3			
		TM9OD5A		48	SDO4				
					P85	V _{LC3}			
		TM9OD0B		87	SDO5				
					PD0				
		TM9OD1B		86	PD1				
		TM9OD2B		85	PD2				
TM9OD3B	84	PD3							
TM9OD4B	83	PD4							
TM9OD5B	82	PD5							

MN101EF76K		I/O	Other Functions		Functions	Descriptions
Pins	Pin No.					
SDO0	53	Output	P80	SEG0	Synchronous output pins	8-bit synchronous output pins. Synchronous output for each bit can be selected individually by the port 8 synchronous output control register (P8SYO). To use these pins for synchronous output, set output mode by the P8DIR register. These pins can be used as normal I/O pins when not used as synchronous output pins.
SDO1	52		TM9OD0A	COM0		
SDO2	51		P81	COM0		
SDO3	50		TM9OD1A	COM1		
SDO4	49		P82	COM1		
SDO5	48		TM9OD2A	COM2		
SDO6	47		P83	COM2		
SDO7	46		TM9OD3A	COM3		
			P84	COM3		
			TM9OD4A			
			P85	V _{LC3}		
			TM9OD5A			
			P86	V _{LC2}		
			P87	V _{LC1}		
V _{REF+}	9	-			Reference power supply pin (+) for A/D converter	Reference power supply pin for the A/D converter. This pin is generally used as V _{REF+} = V _{DD5} .
AN0	1	Input	PA0	TM0IOA	Analog input pins	Analog input pins for 24-channel, 10-bit A/D converter. These pins can be used as normal input pins when not used as analog input pins.
AN1	2		PA1	TM1IOA		
AN2	3		PA2	TM2IOA		
AN3	4		PA3	TM3IOA		
AN4	5		PA4	TM4IOA		
AN5	6		PA5	TM7IOA		
AN6	7		PA6	TM8IOA		
AN7	8		PA7	TM9IOA		
AN8	124		PB0			
AN9	123		PB1			
AN10	122		PB2			
AN11	121		PB3	SEG54		
AN12	120		PB4	SEG53		
AN13	119		PB5	SEG52		
AN14	118		PB6	SEG51		
AN15	117		PB7	SEG50		
AN16	116		P95	SEG49		
			DA_A			
AN17	115		P94	SEG48		
			DA_B			
AN18	114		P93	SEG47		
AN19	113		P92	SEG46		
AN20	112		P10	SEG45		
			TM0IOC			
AN21	111		P11	SEG44		
			TM2IOC			
AN22	110		P12	SEG43		
			TM1IOC			
AN23	109		P13	SEG42		
			TM3IOC			
DA_A	116	Output	P95	SEG49	Analog output pins	Analog input pins for 4-channel, 8-bit D/A converter. These pins can be used as normal input pins when not used as analog input pins.
DA_B	115		AN16			
DA_C	72		P94	SEG48		
DA_D	40		AN17			
			P61	SEG15		
			P07	LED7		
			SBT3A			
IRQ0A	41	Input	P20	ACZ0	External interrupt input pins	External interrupt input pins. The valid edge for IRQ0A to IRQ4A and IRQ0B to IRQ4B can be selected by the IRQnICR register. IRQ0A and IRQ1A are able to determine AC zero cross. Both edge and pin voltage level for IRQ2A, IRQ3A, IRQ4A, IRQ2B, IRQ3B, and IRQ4B are valid for interrupt. These pins can be used as normal input pins when not used as external interrupt pins.
IRQ1A	42		P21	ACZ1		
IRQ2A	43		P22			
IRQ3A	44		P23			
IRQ4A	45		P24			
IRQ0B	23		PC0			
IRQ1B	24		PC1			
IRQ2B	25		PC2			
IRQ3B	26		PC3			
IRQ4B	27		PC4			

MN101EF76K		I/O	Other Functions	Functions	Descriptions
Pins	Pin No.				
ACZ0 ACZ1	41 42	Input	P20 IRQ0A P21 IRQ1A	AC zero-cross detection input pins	Input pins for an AC zero-cross detection circuit. The AC zero-cross detection circuit outputs "High-level" when the input is at an intermediate level. It outputs "Low-level" at all other times. ACZ input signal is connected to the P20 input circuit and the IRQ0A interrupt circuit or the P21 input circuit and the IRQ1A interrupt circuit. When the AC zero-cross detection circuit is not used, this pin can be used as normal input port.
KEY0 KEY1 KEY2 KEY3 KEY4 KEY5 KEY6 KEY7	65 60 59 58 57 56 55 54	Input	P70 SEG8 SBI4A P71 SEG7 SBO4A SDA4A P72 SEG6 SBT4A SCL4A P73 SEG5 P74 SEG4 P75 SEG3 SBI01B TXD1B P76 SEG2 SBI1B RXD1B P77 SEG1 SBT1B	Key interrupt input pins	Input pins for interrupt based on ORed result of pin inputs. These can be set as key input pins in increments of one bit by the key interrupt control registers (KEYT3_1IMD, KEYT3_2IMD). These pins can be used as normal I/O pins when not used as the key input pins.
LED0 LED1 LED2 LED3 LED4 LED5 LED6 LED7	30 31 32 36 37 38 39 40	Output	P00 OCD_DATA P01 TM9IOB OCD_CLK P02 TM7IOB RXD0A SBI0A P03 TM8IOB TXD0A SBO0A P04 TM0IOB SBT0A TM2IOB P05 SBO3A TXD3A P06 SBI3A RXD3A P07 SBT3A DA_D	LED driver pins	Large current output pins. These pins can be used as normal I/O pins when not used as the LED driver pins.
COM0 COM1 COM2 COM3	52 51 50 49	Output	P81 SDO1 TM9OD1A P82 SDO2 TM9OD2A P83 SDO3 TM9OD3A P84 SDO4 TM9OD4A	LCD common output pin	These pins output common signal of required timing for LCD display. Connect to the common pins of LCD display panel. When the LCD functions are not used, these pins can be used as normal ports by the setting of the LCD output control register (LCCTR0).
V _{LC1} V _{LC2} V _{LC3}	46 47 48	-	P87 SDO7 P86 SDO6 P85 SDO5 TM9OD5A	LCD power supply pins	Apply voltage of $5.5\text{ V} \geq V_{LC1} \geq V_{LC2} \geq V_{LC3} \geq 0\text{ V}$. When LCD is not used, V _{LC1} to V _{LC3} can be used as normal ports by the setting of the LCD output control register 0 (LCCTR0).
SEG0 SEG1 SEG2 SEG3 SEG4 SEG5 SEG6 SEG7 SEG8 SEG9	53 54 55 56 57 58 59 60 65 66	Output	P80 SDO0 TM9OD0A P77 KEY7 SBT1B P76 KEY6 SBI1B RXD1B P75 KEY5 SBO1B TXD1B P74 KEY4 P73 KEY3 P72 KEY2 SBT4A SCL4A P71 KEY1 SBO4A SDA4A P70 KEY0 SBI4A P67 SBT2A	LCD segment output pins	These pins output segment signal of required timing for LCD display. Connect to the segment pins of the LCD panel. When LCD display is turned off, V _{SS} -level is output. These pins can be used as normal ports by the setting of the LCD output control registers (LCCTR1 to LCCTR7). SEG for each bit can be individually set as a segment pin or a normal port.

MN101EF76K		I/O	Other Functions	Functions	Descriptions
Pins	Pin No.				
SEG10	67		P66	LCD segment output pins	These pins output segment signal of required timing for LCD display. Connect to the segment pins of the LCD panel. When LCD display is turned off, V_{SS}-level is output. These pins can be used as normal ports by the setting of the LCD output control registers (LCCTR1 to LCCTR7). SEG for each bit can be individually set as a segment pin or a normal port.
			SBI2A RXD2A		
SEG11	68		P65		
			SBO2A TXD2A		
SEG12	69		P64		
SEG13	70		P63		
SEG14	71		P62		
SEG15	72		P61		
SEG16	73		P50		
			SBO1A		
			TXD1A		
SEG17	74		P51		
			SBI1A		
			RXD1A		
SEG18	75		P52		
SEG19	76		P53		
SEG20	77		P54		
SEG21	78		P55		
SEG22	79		P56		
SEG23	80		P57		
			NBUZZERA		
SEG24	88		P47		
SEG25	89		P46		
SEG26	90		P45		
SEG27	91		P44		
			SBI0B		
			RXD0B		
SEG28	92		P43		
			SBO0B		
			TXD0B		
SEG29	93		P42		
SEG30	94		P41		
			SBI3B		
			RXD3B		
SEG31	95		P40		
			SBO3B		
			TXD3B		
SEG32	96		P36		
SEG33	100		P35		
SEG34	101		P34		
			SBI4B		
			SBT4B		
			SCL4B		
SEG35	102		P33		
			SBO4B		
			SDA4B		
SEG36	103		P32		
SEG37	104		P31		
			SBI2B		
			RXD2B		
SEG38	105		P30		
			SBO2B		
			TXD2B		
SEG39	106		P16		
			TM8IOC		
			NBUZZERB		
SEG40	107		P15		
			TM7IOC		
			BUZZERB		
SEG41	108		P14		
SEG42	109		P13		
			AN23		
			TM3IOC		
SEG43	110		P12		
			AN22		
			TM1IOC		
SEG44	111		P11		
			AN21		
			TM2IOC		
SEG45	112		P10		
			AN20		
			TM0IOC		
SEG46	113		P92		
SEG47	114		P93		
SEG48	115		P94		
			AN17		
			DA_B		
SEG49	116		P95		
			AN16		
			DA_A		

MN101EF76K		I/O	Other Functions	Functions	Descriptions
Pins	Pin No.				
SEG50	117		PB7 AN15	LCD segment output pins	These pins output segment signal of required timing for LCD display. Connect to the segment pins of the LCD panel. When LCD display is turned off, V_{SS} -level is output. These pins can be used as normal ports by the setting of the LCD output control registers (LCCTR1 to LCCTR7). SEG for each bit can be individually set as a segment pin or a normal port.
SEG51	118		PB6 AN14		
SEG52	119		PB5 AN13		
SEG53	120		PB4 AN12		
SEG54	121		PB3 AN11		
MMOD	18	Input		Memory mode switch input pins	Set to V_{DD5} -level or V_{SS} -level.
DMOD	20	Input		Mode switch input pins	Set always to V_{DD5} -level. Only flash EEPROM version, DMOD contains an internal pull-up resistor.
RON	21	Input		Regulator control pin	When connecting the pull-up resistor with this pin, make it to 100 Ω or less. Set always to V_{DD5} -level.

Table:1.3.5 Pin Functions (MN101EF57G)

MN101EF57G		I/O	Other Functions	Functions	Descriptions
Pins	Pin No.				
V _{SS}	14, 22	-		Power supply pins	Supply 1.8 V to 5.5 V to V _{DD5} , and 0 V to V _{SS} . Connect 0.1 μF and more than 1 μF of bypass capacitor for internal power stabilization.
V _{DD18}	17	-		Internal power output pin	Outputs internal power voltage 1.8 V. Connect 0.1 μF and more than 1 μF of bypass capacitor between V _{DD18} and V _{SS} pins for internal power stabilization.
OSC1	15	Input	P25	High-speed operation clock input pin	Connect these oscillation pins to ceramic oscillator or crystal oscillator for high-speed operation clock.
OSC2	16	Output	P26	High-speed operation clock output pin	For external clock input, input to OSC1 and open OSC2. The chip will not operate with an external clock when using either STOP mode or SLOW mode.
XI	12	Input	P90	Low-speed operation clock input pin	Connect these oscillation pins to ceramic oscillator or crystal oscillator for low-speed operation clock.
XO	13	Output	P91	Low-speed operation clock output pin	For external clock input, input to XI and open XO. The chip will not operate with an external clock when using STOP mode.
NRST	11	Input	P27	Reset pin [Active low]	This pin resets the chip at power on, is allocated as P27 and contains an internal pull-up resistor (Typ. 50 kΩ). Setting this pin low initializes the internal state of the LSI. Thereafter, setting the input to high releases the reset. The hardware waits for the system clock to stabilize, then processes the reset interrupt. Also, if "0" is written to P27 and the reset is initiated by software, a low level will be output. The output has an Nch open-drain configuration. If a capacitor is to be inserted between NRST and V _{SS} , it is recommended that a discharge diode be placed between NRST and V _{DD5} .
ATRST	10	Input		Auto reset setting pin	Input "High" to enable auto reset function and "Low" to disable this function.
P00 P01 P02 P03 P04 P05 P06 P07	23 24 25 26 27 28 29 30	I/O	LED0 OCD_DATA LED1 TM9IOB OCD_CLK LED2 TM7IOB SBI0A RXD0A LED3 TM8IOB SBO0A TXD0A LED4 TM0IOB TM2IOB SBT0A LED5 LED6 LED7 DA_B	I/O port 0	8-bit CMOS tri-state I/O port. Each bit can be set individually as either an input or output by the P0DIR register. A pull-up /pull-down resistor for each bit can be selected individually by the P0PLUD register. A pull-up/down resistor connection for each port can be selected individually by the SELUD register. (However, pull-up and pull-down resistors cannot be mixed.) At reset, the input mode is selected and pull-up resistors are disabled (high impedance).
P20 P21 P22 P23 P24 P25 P26	31 32 33 34 35 15 16	I/O	IRQ0 ACZ0 IRQ1 ACZ1 IRQ2 IRQ3 IRQ4 OSC1 OSC2	I/O port 2	7-bit CMOS tri-state I/O port. Each bit can be set individually as either an input or output by the P2DIR register. A pull-up resistor for each bit can be selected individually by the P2PLU register. At reset, the input mode is selected and pull-up resistors are disabled (high impedance).
P27	11	Input	NRST	Input port 2	P27 is an Nch open-drain port. When "0" is written and the reset is initiated by software, a low level will be output.
P33 P34 P35	73 72 71	I/O	AN15 SBO4B SDA4B SEG30 SBT4B SCL4B SEG29 SBI4B SEG28	I/O port 3	3-bit CMOS tri-state I/O port. Each bit can be set individually as either an input or output by the P3DIR register. A pull-up resistor for each bit can be selected individually by the P3PLU register. At reset, the input mode is selected and pull-up resistors are disabled (high impedance).
P43 P44 P45 P46 P47	70 69 68 67 66	I/O	SBO0B TXD0B SEG27 SBI0B RXD0B SEG26 SBT0B SEG25 SEG24 SEG23	I/O port 4	5-bit CMOS tri-state I/O port. Each bit can be set individually as either an input or output by the P4DIR register. A pull-up /pull-down resistor for each bit can be selected individually by the P4PLUD register. A pull-up/down resistor connection for each port can be selected individually by the SELUD register. (However, pull-up and pull-down resistors cannot be mixed.) At reset, the input mode is selected and pull-up resistors are disabled (high impedance).

MN101EF57G		I/O	Other Functions		Functions	Descriptions
Pins	Pin No.					
P50	58	I/O	SBO1A	TXD1A	I/O port 5	8-bit CMOS tri-state I/O port. Each bit can be set individually as either an input or output by the P5DIR register. A pull-up /pull-down resistor for each bit can be selected individually by the P5PLUD register. A pull-up/down resistor connection for each port can be selected individually by the SELUD register. (A pull-up/pull down can not be mixed.) At reset, the input mode is selected and pull-up resistors are disabled (high impedance).
P51	59		SEG15	RXD1A		
P52	60		SBI1A	SEG17		
P53	61		SEG18			
P54	62		SEG19			
P55	63		SEG20			
P56	64		SEG21	NBUZZERA		
P57	65		SEG22	BUZZERA		
P62	57	I/O	TM1IOB	SEG14	I/O port 6	6-bit CMOS tri-state I/O port. Each bit can be set individually as either an input or output by the P6DIR register. A pull-up resistor for each bit can be selected individually by the P6PLU register. At reset, the input mode is selected and pull-up resistors are disabled (high impedance).
P63	56		TM3IOB	SEG13		
P64	55		TM4IOB	SEG12		
P65	54		SBO2A	TXD2A		
P66	53		SEG11	RXD2A		
P67	52		SBI2A	SEG10		
P70	51	I/O	KEY0	SBI4A	I/O port 7	8-bit CMOS tri-state I/O port. Each bit can be set individually as either an input or output by the P7DIR register. A pull-up /pull-down resistor for each bit can be selected individually by the P7PLUD register. A pull-up/down resistor connection for each port can be selected individually by the SELUD register. (A pull-up/pull down can not be mixed.) At reset, the input mode is selected and pull-up resistors are disabled (high impedance).
P71	50		SEG8	SBO4A		
P72	49		KEY1	SEG7		
P73	48		KEY2	SEG6		
P74	47		KEY3	SEG5		
P75	46		KEY4	SEG4		
P76	45		KEY5	SBO1B		
P77	44		TXD1B	SEG3		
			KEY6	SBI1B		
			RXD1B	SEG2		
P80	43	I/O	KEY7	SBT1B	I/O port 8	8-bit CMOS tri-state I/O port. Each bit can be set individually as either an input or output by the P8DIR register. A pull-up resistor for each bit can be selected individually by the P8PLU register. At reset, the input mode is selected and pull-up resistors are disabled (high impedance).
P81	42		SDO0	TM9OD0		
P82	41		SEG0	TM9OD1		
P83	40		COM0	TM9OD2		
P84	39		SDO1	TM9OD3		
P85	38		COM1	TM9OD4		
P86	37		SDO2	TM9OD5		
P87	36		COM2			
			SDO3			
P90	12	I/O	SDO4		I/O port 9	5-bit CMOS tri-state I/O port. Each bit can be set individually as either an input or output by the P9DIR register. A pull-up resistor for each bit can be selected individually by the P9PLU register. At reset, the input mode is selected and pull-up resistors are disabled (high impedance).
P91	13		XI			
P92	74		AN14	SEG31		
P93	75		AN13	SEG32		
P94	76		AN12	SEG33		
PA0	1	I/O	DA_A		I/O port A	8-bit CMOS tri-state I/O port. Each bit can be set individually as either an input or output by the PADIR register. A pull-up resistor for each bit can be selected individually by the PAPLU register. At reset, the input mode is selected and pull-up resistors are disabled (high impedance).
PA1	2		AN0	TM0IOA		
PA2	3		SEG38	TM1IOA		
PA3	4		AN1	TM2IOA		
PA4	5		SEG39	TM3IOA		
PA5	6		AN2	TM4IOA		
PA6	7		SEG40	TM7IOA		
PA7	8		AN3	TM8IOA		
			AN4	TM9IOA		

MN101EF57G		I/O	Other Functions		Functions	Descriptions
Pins	Pin No.					
PB0	80	I/O	AN8	SEG37	I/O port B	4-bit CMOS tri-state I/O port. Each bit can be set individually as either an input or output by the PBDIR register. A pull-up /pull-down resistor for each bit can be selected individually by the PBPLUD register. At reset, the input mode is selected and pull-up resistors are disabled (high impedance).
PB1	79		AN9	SEG36		
PB2	78		AN10	SEG35		
PB3	77		AN11	SEG34		
SBO0A	26	I/O	P03	TXD0A	Serial interface transmission data output pins	Transmission data output pins for serial interface 0, 1, 2, and 4. The output configuration, either COMS push-pull or Nch open-drain can be selected by the P0ODC, P3ODC, P4ODC, P5ODC, P6ODC, and P7ODC registers. Pull-up resistor can be selected by the P0PLUD, P3PLU, P4PLUD, P5PLUD, P6PLU and P7PLUD registers. Select the output mode by the P0DIR, P3DIR, P4DIR, P5DIR, P6DIR and P7DIR registers to select serial data output by the serial mode register 1 (SC0MD1, SC1MD1, SC2MD1 to SC4MD1). These can be used as normal I/O pins when the serial interface is not used.
SBO0B	70		TM8IOB	LED3		
SBO1A	58		P43	SEG27		
SBO1B	46		TXD0B			
SBO2A	54		P50	SEG15		
SBO4A	50		TXD1A			
SBO4B	73		P75	SEG3		
			KEY5			
		I/O	TXD1B		Serial interface reception data input pins	Reception data input pins for serial interface 0, 1, 2, and 4. Pull-up resistor can be selected by the P0PLUD, P3PLU, P4PLUD, P5PLUD, P6PLU and P7PLUD registers. Select input mode by the P0DIR, P3DIR, P4DIR, P5DIR, P6DIR and P7DIR registers to select serial data input by the serial mode register 1 (SC0MD1, SC1MD1, SC2MD1 to SC4MD1). These can be used as normal I/O pins when the serial interface is not used.
			P65	SEG11		
			TXD2A			
			P71	SEG7		
			KEY1	SDA4A		
			P33	SEG30		
			AN15	SDA4B		
SBI0A	25	I/O	P02	RXD0A	Serial interface clock I/O pins	Clock I/O pins for serial interface 0, 1, 2, and 4. The output configuration, either CMOS push-pull or Nch open-drain can be selected with the P0ODC, P3ODC, P4ODC, P5ODC, P6ODC, and P7ODC registers. Pull-up resistor can be selected by the P0PLUD, P3PLU, P4PLUD, P5PLUD, P6PLU and P7PLUD registers. Select input mode or output mode by the P0DIR, P3DIR, P4DIR, P5DIR, P6DIR and P7DIR registers and serial data output mode by serial mode register 1 (SC0MD1, SC1MD1, SC2MD1 to SC4MD1) according to the communication mode. These can be used as normal I/O pins when the serial interface is not used.
SBI0B	69		TM7IOB	LED2		
SBI1A	59		P44	SEG26		
SBI1B	45		RXD0B			
SBI2A	53		P51	SEG16		
SBI4A	51		RXD1A			
SBI4B	71		P76	SEG2		
			KEY6	RXD1B		
SBT0A	27	I/O	P66	SEG10	UART transmission data output pins	In the serial interface 0, 1, and 2 in UART mode, these pins are configured as the transmission data output pin. The output configuration, either CMOS push-pull or Nch open-drain can be selected with the P0ODC, P4ODC, P5ODC, P6ODC, and P7ODC registers. Pull-up resistor can be selected by the P0PLUD, P4PLUD, P5PLUD, P6PLU, and P7PLUD registers. Select output mode by the P0DIR, P4DIR, P5DIR, P6DIR and P7DIR registers to select serial data output by the serial mode register 1 (SC0MD1 to SC2MD1). These can be used as normal I/O pins when the serial interface is not used.
SBT0B	68		RXD2A			
SBT1A	60		P70	SEG8		
SBT1B	44		KEY0			
SBT2A	52		P35	SEG28		
SBT4A	49					
SBT4B	72					
TXD0A	26	Output	P04	TM2IOB	UART reception data input pins	In the serial interface 0, 1, and 2 in UART mode, these pins are configured as the reception data input pin. Pull-up resistor can be selected by the P0PLUD, P4PLUD, P5PLUD, P6PLU, and P7PLUD registers. Select input mode by the P0DIR, P4DIR, P5DIR, P6DIR and P7DIR registers to select serial data input by the serial mode register 1 (SC0MD1 to SC2MD1). These can be used as normal I/O pins when the serial interface is not used.
TXD0B	70		TM0IOB	LED4		
TXD1A	58		P45	SEG25		
TXD1B	46		P52	SEG17		
TXD2A	54		P77	SEG1		
			KEY7			
		Input	P67	SEG9	UART reception data input pins	In the serial interface 0, 1, and 2 in UART mode, these pins are configured as the reception data input pin. Pull-up resistor can be selected by the P0PLUD, P4PLUD, P5PLUD, P6PLU, and P7PLUD registers. Select input mode by the P0DIR, P4DIR, P5DIR, P6DIR and P7DIR registers to select serial data input by the serial mode register 1 (SC0MD1 to SC2MD1). These can be used as normal I/O pins when the serial interface is not used.
			P72	SEG6		
			KEY2	SCL4A		
			P34	SEG29		
			SCL4B			
		Input	P02	SBI0A	UART reception data input pins	In the serial interface 0, 1, and 2 in UART mode, these pins are configured as the reception data input pin. Pull-up resistor can be selected by the P0PLUD, P4PLUD, P5PLUD, P6PLU, and P7PLUD registers. Select input mode by the P0DIR, P4DIR, P5DIR, P6DIR and P7DIR registers to select serial data input by the serial mode register 1 (SC0MD1 to SC2MD1). These can be used as normal I/O pins when the serial interface is not used.
			TM7IOB	LED2		
			P44	SEG26		
			SBI0B	RXD0B		
			P51	SEG16		
			SBI1A			
		Input	P76	SEG2	UART reception data input pins	In the serial interface 0, 1, and 2 in UART mode, these pins are configured as the reception data input pin. Pull-up resistor can be selected by the P0PLUD, P4PLUD, P5PLUD, P6PLU, and P7PLUD registers. Select input mode by the P0DIR, P4DIR, P5DIR, P6DIR and P7DIR registers to select serial data input by the serial mode register 1 (SC0MD1 to SC2MD1). These can be used as normal I/O pins when the serial interface is not used.
			SBI1B	KEY6		
			P66	SEG10		
			SBI2A			

MN101EF57G		I/O	Other Functions	Functions	Descriptions
Pins	Pin No.				
SDA4A	50	I/O	P71 SEG7	IIC data I/O pins	In the serial interface 4 in IIC mode, these pins are configured as the data input / output pin. For the output configuration, select Nch open-drain by the P3ODC and P7ODC register to select pull-up resistor by the P3PLU and P7PLUD register. Select output mode by the P3DIR and P7DIR register to select serial data input / output by the serial mode register 1 (SC4MD1). These can be used as normal I/O pins when the serial interface is not used.
SDA4B	73		SBO4A KEY1 P33 SEG30 AN15 SBO4B		
SCL4A	49	I/O	P72 SEG6	IIC clock I/O pins	In the serial interface 4 in IIC mode, these pins are configured as the clock input / output pin. For the output configuration, select Nch open-drain by the P3ODC and P7ODC register to select pull-up resistor by the P3PLU and P7PLUD register. Select output mode by the P3DIR and P7DIR register to select clock data input / output by the serial mode register 1 (SC4MD1). These can be used as normal I/O pins when the serial interface is not used.
SCL4B	72		SBT4A KEY2 P34 SEG29 SBT4B		
OCD_DATA	23	I/O	P00 LED0	On-board programmer I/O pins	Data I/O pin and clock input pin for the on-board programmer. Refer to [Chapter 24 24.3 Serial Programmer Rewriting]. These can be used as normal I/O pins when the on-board programmer is not used.
OCD_CLK	24		P01 LED1 TM9IOB		
TM0IOA	1	I/O	PA0 AN0	Timer I/O pins	Event count clock input, timer output, and PWM signal output pins for 8-bit timer 0 to 4. To use these pins for event clock input, input mode can be selected by the P0DIR, P6DIR, PADIR, TMCKSEL1, TMINSEL1, and TMINSEL2 registers. In input mode, pull-up resistors can be selected by the P0PLUD, P6PLU, and PAPLU register. To use these pins for timer output or PWM signal output, select special function pins by the port 0 output mode register, port 6 output mode register, and port A output mode register (P0OMD, P6OMD, and PAOMD) to select output mode by the P0DIR, P6DIR, and PADIR registers. These can be used as normal I/O pins when not used as timer I/O pins.
TM0IOB	27		SEG38 P04 SBT0A		
TM1IOA	2		TM2IOB LED4 PA1 AN1		
TM1IOB	57		SEG39		
TM2IOA	3		P62 SEG14 PA2 SEG40		
TM2IOB	27		AN2 P04 SBT0A		
TM3IOA	4		TM0IOB LED4 PA3 AN3		
TM3IOB	56		P63 SEG13		
TM4IOA	5		TM3IOB		
TM4IOB	55		PA4 AN4 P64 SEG12		
BUZZERA	65	Output	P57 SEG22	Buzzer output pins	Piezoelectric buzzer driving pin. Buzzer output is available to port 5. The driving frequency can be set in the DLYCTR register. In order to select buzzer output to port 5, select the special function pin in the port 5 output mode register (P5OMD), and set the P5DIR register to the output mode. At the same time, select buzzer output in the oscillation stabilization wait control register (DLYCTR). These can be used as normal I/O pins when the buzzer output function is not used.
NBUZZERA	64		P56 SEG21		
TM7IOA	6	I/O	PA5 AN5	Timer I/O pins	Event count clock input, timer output, and PWM signal output pins for 16-bit timer 7, 8, and 9. To use these pins for event clock input, input mode can be selected by the P0DIR and PADIR registers. In input mode, pull-up resistors can be selected by the P0PLUD and PAPLU register. To use these pins for timer output or PWM signal output, select special function pins by the port A output mode register to select output mode by the P0DIR and PADIR registers. These can be used as normal I/O pins when not used as timer I/O pins.
TM7IOB	25		P02 RXD0A		
TM8IOA	7		SBIOA LED2		
TM8IOB	26		PA6 AN6		
TM9IOA	8		P03 TXD0A		
TM9IOB	24		SBO0A LED3 PA7 AN7 P01 OCD_CLK LED1		
TM9OD0	43	Output	P80 SEG0	Timer output pins	Timer output and PWM signal output pins for 16-bit timer. To use these pins for timer output or PWM signal output, select special function pins by the P8OMD register to select output mode by the P8DIR register. These can be used as normal I/O pins when not used as timer output pins.
TM9OD1	42		SD00 P81 COM0		
TM9OD2	41		SD01 P82 COM1		
TM9OD3	40		SD02 P83 COM2		
TM9OD4	39		SD03 P84 COM3		
TM9OD5	38		SD04 P85 V _{LC3} SD05		

MN101EF57G		I/O	Other Functions	Functions	Descriptions
Pins	Pin No.				
SDO0	43	Output	P80 SEG0	Synchronous output pins	8-bit synchronous output pins. Synchronous output for each bit can be selected individually by the port 8 synchronous output control register (P8SYO). To use these pins for synchronous output, set output mode by the P8DIR register. These pins can be used as normal I/O pins when not used as synchronous output pins.
SDO1	42		TM9OD0 P81 COM0		
SDO2	41		TM9OD1 P82 COM1		
SDO3	40		TM9OD2 P83 COM2		
SDO4	39		TM9OD3 P84 COM3		
SDO5	38		TM9OD4 P85 V _{LC3}		
SDO6	37		TM9OD5 P86 V _{LC2}		
SDO7	36		P87 V _{LC1}		
V _{REF+}	9	-		Reference power supply pin (+) for A/D converter	Reference power supply pin for the A/D converter. This pin is generally used as V _{REF+} = V _{DD5} .
AN0	1	Input	PA0 TM0IOA	Analog input pins	Analog input pins for 16-channel, 10-bit A/D converter. These pins can be used as normal input pins when not used as analog input pins.
AN1	2		SEG38 PA1 TM1IOA		
AN2	3		SEG39 PA2 TM2IOA		
AN3	4		SEG40 PA3 TM3IOA		
AN4	5		PA4 TM4IOA		
AN5	6		PA5 TM7IOA		
AN6	7		PA6 TM8IOA		
AN7	8		PA7 TM9IOA		
AN8	80		PB0 SEG37		
AN9	79		PB1 SEG36		
AN10	78		PB2 SEG35		
AN11	77		PB3 SEG34		
AN12	76		P94 SEG33		
AN13	75		DA_A P93 SEG32		
AN14	74		P92 SEG31		
AN15	73		P33 SEG30		
DA_A	76	Output	SBO4B SDA4B	Analog output pins	Analog input pins for 2-channel, 8-bit D/A converter. These pins can be used as normal input pins when not used as analog input pins.
DA_B	30		P94 SEG33 AN12 P07 LED7		
IRQ0	31	Input	P20 ACZ0	External interrupt input pins	External interrupt input pins. The valid edge for IRQ0 to 4 can be selected by the IRQnICR register. IRQ0, 1 are able to determine AC zero cross. Both edge and pin voltage level for IRQ2, 3, 4 are valid for interrupt. These pins can be used as normal input pins when not used as external interrupt pins.
IRQ1	32		P21 ACZ1		
IRQ2	33		P22		
IRQ3	34		P23		
IRQ4	35		P24		
ACZ0	31	Input	P20 IRQ0	AC zero-cross detection input pins	Input pins for an AC zero-cross detection circuit. The AC zero-cross detection circuit outputs "High-level" when the input is at an intermediate level. It outputs "Low-level" at all other times. ACZ input signal is connected to the P20 input circuit and the IRQ0 interrupt circuit or the P21 input circuit and the IRQ1 interrupt circuit. When the AC zero-cross detection circuit is not used, this pin can be used as normal input port.
ACZ1	32		P21 IRQ1		
KEY0	51	Input	P70 SEG8	Key interrupt input pins	Input pins for interrupt based on ORed result of pin inputs. These can be set as key input pins in increments of one bit by the key interrupt control registers (KEYT3_1IMD, KEYT3_2IMD). These pins can be used as normal I/O pins when not used as the key input pins.
KEY1	50		SBI4A P71 SEG7		
KEY2	49		SBO4A SDA4A		
KEY3	48		P72 SEG6		
KEY4	47		SBT4A SCL4A		
KEY5	46		P73 SEG5		
KEY6	45		P74 SEG4		
KEY7	44		P75 SEG3		
			SBI01B TXD1B		
			P76 SEG2		
			SBI1B RXD1B		
			P77 SEG1		
			SBT1B		

MN101EF57G		I/O	Other Functions	Functions	Descriptions
Pins	Pin No.				
LED0	23	Output	P00	LED driver pins	Large current output pins. These pins can be used as normal I/O pins when not used as the LED driver pins.
LED1	24		P01		
			OCD_CLK		
LED2	25		P02		
			RXD0A		
LED3	26		P03		
			TXD0A		
LED4	27		P04		
			SBT0A		
LED5	28		P05		
LED6	29		P06		
LED7	30		P07		
			DA_B		
COM0	42	Output	P81	LCD common output pin	These pins output common signal of required timing for LCD display. Connect to the common pins of LCD display panel. When the LCD functions are not used, these pins can be used as normal ports by the setting of the LCD output control register (LCCTR0).
			TM9OD1		
COM1	41		P82		
			TM9OD2		
COM2	40		P83		
			TM9OD3		
COM3	39		P84		
			TM9OD4		
V _{LC1}	36	-	P87	LCD power supply pins	Apply voltage of 5.5 V $\geq V_{LC1} \geq V_{LC2} \geq V_{LC3} \geq 0$ V. When LCD is not used, V _{LC1} to V _{LC3} can be used as normal ports by the setting of the LCD output control register 0 (LCCTR0).
V _{LC2}	37		P86		
V _{LC3}	38		P85		
			TM9OD5		
SEG0	43	Output	P80	LCD segment output pins	These pins output segment signal of required timing for LCD display. Connect to the segment pins of the LCD panel. When LCD display is turned off, V _{SS} -level is output. These pins can be used as normal ports by the setting of the LCD output control registers (LCCTR1 to LCCTR7). SEG for each bit can be individually set as a segment pin or a normal port.
			SDO0		
SEG1	44		P77		
			KEY7		
SEG2	45		P76		
			KEY6		
SEG3	46		P75		
			KEY5		
SEG4	47		P74		
			KEY4		
SEG5	48		P73		
			KEY3		
SEG6	49		P72		
			KEY2		
			SBT4A		
SEG7	50		P71		
			KEY1		
			SBO4A		
SEG8	51		P70		
			KEY0		
			SBI4A		
SEG9	52		P67		
			SBT2A		
SEG10	53		P66		
			SBI2A		
			RXD2A		
SEG11	54		P65		
			SBO2A		
			TXD2A		
SEG12	55		P64		
			TM4IOB		
SEG13	56		P63		
			TM3IOB		
SEG14	57		P62		
			TM1IOB		
SEG15	58		P50		
			SBO1A		
			TXD1A		
SEG16	59		P51		
			SBI1A		
			RXD1A		
SEG17	60		P52		
			SBT1A		
SEG18	61		P53		
SEG19	62		P54		
SEG20	63		P55		
SEG21	64		P56		
			NBUZZERA		
SEG22	65		P57		
			BUZZERA		
SEG23	66		P47		
SEG24	67		P46		
SEG25	68		P45		
			SBT0B		
SEG26	69		P44		
			SBI0B		
			RAD0B		
SEG27	70		P43		
			SBO0B		
			TXD0B		
SEG28	71		P35		
			SBI4B		

MN101EF57G		I/O	Other Functions	Functions	Descriptions
Pins	Pin No.				
SEG29	72		P34 SBT4B	LCD segment output pins	These pins output segment signal of required timing for LCD display. Connect to the segment pins of the LCD panel. When LCD display is turned off, V_{SS} -level is output. These pins can be used as normal ports by the setting of the LCD output control registers (LCCTR1 to LCCTR7). SEG for each bit can be individually set as a segment pin or a normal port.
SEG30	73		P33 AN15		
SEG31	74		SBO4B SDA4B		
SEG32	75		P92 AN14		
SEG33	76		P93 AN13		
			P94 AN12		
			DA_A		
SEG34	77		PB3 AN11		
SEG35	78		PB2 AN10		
SEG36	79		PB1 AN9		
SEG37	80		PB0 AN8		
SEG38	1		PA0 AN0		
			TM0IOA		
SEG39	2		PA1 AN1		
			TM1IOA		
SEG40	3		PA2 AN2		
			TM2IOA		
MMOD	18	Input		Memory mode switch input pins	Set to V_{DD5} -level or V_{SS} -level.
DMOD	20	Input		Mode switch input pins	Set always to V_{DD5} -level. Only flash EEPROM version, DMOD contains an internal pull-up resistor.
RON	21	Input		Regulator control pin	When connecting the pull-up resistor with this pin, make it to 100 Ω or less. Set always to V_{DD5} -level.



DMOD internal pull-up resistor is in only Flash EEPROM version.
When using In-circuit Emulator, it is necessary to connect the pull-up resistor on the circuit board.

Table:1.3.6 Pin Functions (MN101EF56K)

MN101EF56K		I/O	Other Functions	Functions	Descriptions
Pins	Pin No.				
V _{SS} V _{DD5}	14, 22 17	-		Power supply pins	Supply 1.8 V to 5.5 V to V _{DD5} , and 0 V to V _{SS} . Connect 0.1 μF and more than 1 μF of bypass capacitor for internal power stabilization.
V _{DD18}	19	-		Internal power output pin	Outputs internal power voltage 1.8 V. Connect 0.1 μF and more than 1 μF of bypass capacitor between V _{DD18} and V _{SS} pins for internal power stabilization.
OSC1	15	Input	P25	High-speed operation clock input pin	Connect these oscillation pins to ceramic oscillator or crystal oscillator for high-speed operation clock.
OSC2	16	Output	P26	High-speed operation clock output pin	For external clock input, input to OSC1 and open OSC2. The chip will not operate with an external clock when using either STOP mode or SLOW mode.
XI	12	Input	P90	Low-speed operation clock input pin	Connect these oscillation pins to ceramic oscillator or crystal oscillator for low-speed operation clock.
XO	13	Output	P91	Low-speed operation clock output pin	For external clock input, input to XI and open XO. The chip will not operate with an external clock when using STOP mode.
NRST	11	Input	P27	Reset pin [Active low]	This pin resets the chip at power on, is allocated as P27 and contains an internal pull-up resistor (Typ. 50 kΩ). Setting this pin low initializes the internal state of the LSI. Thereafter, setting the input to high releases the reset. The hardware waits for the system clock to stabilize, then processes the reset interrupt. Also, if "0" is written to P27 and the reset is initiated by software, a low level will be output. The output has an Nch open-drain configuration. If a capacitor is to be inserted between NRST and V _{SS} , it is recommended that a discharge diode be placed between NRST and V _{DD5} .
ATRST	10	Input		Auto reset setting pin	Input "High" to enable auto reset function and "Low" to disable this function.
P00 P01 P02 P03 P04 P05 P06 P07	23 24 25 26 27 28 29 30	I/O	LED0 OCD_DATA LED1 TM9IOB OCD_CLK LED2 TM7IOB SBI0A RXD0A LED3 TM8IOB SBO0A TXD0A LED4 TM0IOB TM2IOB SBT0A LED5 SBO3A TXD3A LED6 SBI3A RXD3A LED7 DA_D SBT3A	I/O port 0	8-bit CMOS tri-state I/O port. Each bit can be set individually as either an input or output by the P0DIR register. A pull-up /pull-down resistor for each bit can be selected individually by the P0PLUD register. A pull-up/down resistor connection for each port can be selected individually by the SELUD register. (However, pull-up and pull-down resistors cannot be mixed.) At reset, the input mode is selected and pull-up resistors are disabled (high impedance).
P10 P11 P12 P13 P14 P15 P16	88 87 86 85 84 83 82	I/O	TM0IOC AN20 TM2IOC AN21 TM1IOC AN22 TM3IOC AN23 TM4IOC TM7IOC BUZZERB TM8IOC NBUZZERB	I/O port 1	7-bit CMOS tri-state I/O port. Each bit can be set individually as either an input or output by the P1DIR register. A pull-up /pull-down resistor for each bit can be selected individually by the P1PLUD register. A pull-up/down resistor connection for each port can be selected individually by the SELUD register. (However, pull-up and pull-down resistors cannot be mixed.) At reset, the input mode is selected and pull-up resistors are disabled (high impedance).
P20 P21 P22 P23 P24 P25 P26	31 32 33 34 35 15 16	I/O	IRQ0 ACZ0 IRQ1 ACZ1 IRQ2 IRQ3 IRQ4 OSC1 OSC2	I/O port 2	7-bit CMOS tri-state I/O port. Each bit can be set individually as either an input or output by the P2DIR register. A pull-up resistor for each bit can be selected individually by the P2PLU register. At reset, the input mode is selected and pull-up resistors are disabled (high impedance).
P27	11	Input	NRST	Input port 2	P27 is an Nch open-drain port. When "0" is written and the reset is initiated by software, a low level will be output.

MN101EF56K		I/O	Other Functions		Functions	Descriptions
Pins	Pin No.					
P30	81	I/O	SBO2B	TXD2B	I/O port 3	7-bit CMOS tri-state I/O port. Each bit can be set individually as either an input or output by the P3DIR register. A pull-up resistor for each bit can be selected individually by the P3PLU register. At reset, the input mode is selected and pull-up resistors are disabled (high impedance).
P31	80		SEG38			
			SBI2B	RXD2B		
P32	79		SEG37			
P33	78		SBT2B	SEG36		
			SBO4B	SDA4B		
P34	77		SEG35			
			SBT4B	SCL4B		
P35	76	I/O	SEG34		I/O port 4	8-bit CMOS tri-state I/O port. Each bit can be set individually as either an input or output by the P4DIR register. A pull-up /pull-down resistor for each bit can be selected individually by the P4PLUD register. A pull-up/down resistor connection for each port can be selected individually by the SELUD register. (However, pull-up and pull-down resistors cannot be mixed.) At reset, the input mode is selected and pull-up resistors are disabled (high impedance).
P36	75		SBI4B	SEG33		
			SEG32			
P40	74		SBO3B	TXD3B		
			SEG31			
P41	73		SBI3B	RXD3B		
			SEG30			
P42	72		SBT3B	SEG29		
P43	71	I/O	SBO0B	TXD0B	I/O port 5	8-bit CMOS tri-state I/O port. Each bit can be set individually as either an input or output by the P5DIR register. A pull-up /pull-down resistor for each bit can be selected individually by the P5PLUD register. A pull-up/down resistor connection for each port can be selected individually by the SELUD register. (A pull-up/pull down can not be mixed.) At reset, the input mode is selected and pull-up resistors are disabled (high impedance).
			SEG28			
P44	70		SBI0B	RXD0B		
			SEG27			
P45	69		SBT0B	SEG25		
P46	68		SEG25			
P47	67		SEG24			
P50	59	I/O	SBO1A	TXD1A	I/O port 6	7-bit CMOS tri-state I/O port. Each bit can be set individually as either an input or output by the P6DIR register. A pull-up resistor for each bit can be selected individually by the P6PLU register. At reset, the input mode is selected and pull-up resistors are disabled (high impedance).
			SEG16			
P51	60		SBI1A	RXD1A		
			SEG17			
P52	61		SBT1A	SEG18		
P53	62		SEG19			
P54	63		SEG20			
P55	64		SEG21			
P56	65	I/O	SEG22	NBUZZERA	I/O port 7	8-bit CMOS tri-state I/O port. Each bit can be set individually as either an input or output by the P7DIR register. A pull-up /pull-down resistor for each bit can be selected individually by the P7PLUD register. A pull-up/down resistor connection for each port can be selected individually by the SELUD register. (A pull-up/pull down can not be mixed.) At reset, the input mode is selected and pull-up resistors are disabled (high impedance).
P57	66		SEG23	BUZZERA		
P61	58		DA_C	SEG15		
P62	57		TM1IOB	SEG14		
P63	56		TM3IOB	SEG13		
P64	55		TM4IOB	SEG12		
P65	54		SBO2A	TXD2A		
			SEG11			
P66	53	I/O	SBI2A	RXD2A	I/O port 7	8-bit CMOS tri-state I/O port. Each bit can be set individually as either an input or output by the P7DIR register. A pull-up /pull-down resistor for each bit can be selected individually by the P7PLUD register. A pull-up/down resistor connection for each port can be selected individually by the SELUD register. (A pull-up/pull down can not be mixed.) At reset, the input mode is selected and pull-up resistors are disabled (high impedance).
			SEG10			
P67	52		SBT2A	SEG9		
P70	51		KEY0	SBI4A		
			SEG8			
P71	50		KEY1	SBO4A		
			SDA4A	SEG7		
P72	49		KEY2	SBT4A		
			SCL4A	SEG6		
P73	48	I/O	KEY3	SEG5	I/O port 7	8-bit CMOS tri-state I/O port. Each bit can be set individually as either an input or output by the P7DIR register. A pull-up /pull-down resistor for each bit can be selected individually by the P7PLUD register. A pull-up/down resistor connection for each port can be selected individually by the SELUD register. (A pull-up/pull down can not be mixed.) At reset, the input mode is selected and pull-up resistors are disabled (high impedance).
P74	47		KEY4	SEG4		
P75	46		KEY5	SBO1B		
			TXD1B	SEG3		
P76	45		KEY6	SBI1B		
			RXD1B	SEG2		
P77	44		KEY7	SBT1B		
			SEG1			

MN101EF56K		I/O	Other Functions		Functions	Descriptions
Pins	Pin No.					
P80	43	I/O	SDO0	TM9OD0	I/O port 8	8-bit CMOS tri-state I/O port. Each bit can be set individually as either an input or output by the P8DIR register. A pull-up resistor for each bit can be selected individually by the P8PLU register. At reset, the input mode is selected and pull-up resistors are disabled (high impedance).
P81	42		SDO1	TM9OD1		
P82	41		COM0	TM9OD2		
P83	40		SDO2	TM9OD3		
P84	39		COM1	TM9OD4		
P85	38		SDO3	TM9OD5		
P86	37		COM2			
P87	36		SDO4			
P90	12	I/O	SDO5		I/O port 9	6-bit CMOS tri-state I/O port. Each bit can be set individually as either an input or output by the P9DIR register. A pull-up resistor for each bit can be selected individually by the P9PLU register. At reset, the input mode is selected and pull-up resistors are disabled (high impedance).
P91	13		XL3			
P92	89		SDO6			
P93	90		XL2			
P94	91		SDO7			
P95	92		XL1			
PA0	1	I/O	AN0	TM0IOA	I/O port A	8-bit CMOS tri-state I/O port. Each bit can be set individually as either an input or output by the PADIR register. A pull-up resistor for each bit can be selected individually by the PAPLU register. At reset, the input mode is selected and pull-up resistors are disabled (high impedance).
PA1	2		AN1	TM1IOA		
PA2	3		AN2	TM2IOA		
PA3	4		AN3	TM3IOA		
PA4	5		AN4	TM4IOA		
PA5	6		AN5	TM7IOA		
PA6	7		AN6	TM8IOA		
PA7	8		AN7	TM9IOA		
PB0	100	I/O	AN8		I/O port B	8-bit CMOS tri-state I/O port. Each bit can be set individually as either an input or output by the PBDIR register. A pull-up /pull-down resistor for each bit can be selected individually by the PBPLUD register. At reset, the input mode is selected and pull-up resistors are disabled (high impedance).
PB1	99		AN9			
PB2	98		AN10			
PB3	97		AN11	SEG54		
PB4	96		AN12	SEG53		
PB5	95		AN13	SEG52		
PB6	94		AN14	SEG51		
PB7	93		AN15	SEG50		
SBO0A	26	I/O	P03	TXD0A	Serial interface transmission data output pins	Transmission data output pins for serial interface 0, 1, 2, 3, and 4. The output configuration, either COMS push-pull or Nch open-drain can be selected by the P0ODC, P3ODC, P4ODC, P5ODC, P6ODC, and P7ODC registers. Pull-up resistor can be selected by the P0PLUD, P3PLU, P4PLUD, P5PLUD, P6PLU and P7PLUD registers. Select the output mode by the P0DIR, P3DIR, P4DIR, P5DIR, P6DIR and P7DIR registers to select serial data output by the serial mode register 1 (SC0MD1, SC1MD1, SC2MD1, SC3MD1, and SC4MD1). These can be used as normal I/O pins when the serial interface is not used.
SBO0B	71		TM8IOB	LED3		
SBO1A	59		P43	SEG8		
SBO1B	46		TXD0B			
SBO2A	54		P50	SEG16		
SBO2B	81		TXD1A			
SBO3A	28		P75	SEG3		
SBO3B	74		KEY5			
SBO4A	50		TXD1B			
SBO4B	78		P65	SEG11		
			TXD2A			
			P30	SEG38		
			TXD2B			
			P05	LED5		
			TXD3A			
			P40	SEG31		
			TXD3B			
			P71	SEG7		
			KEY1	SDA4A		
			P33	SEG35		
			AN15	SDA4B		

MN101EF56K		I/O	Other Functions		Functions	Descriptions
Pins	Pin No.					
SBI0A	25	I/O	P02	RXD0A	Serial interface reception data input pins	Reception data input pins for serial interface 0, 1, 2, 3, and 4. Pull-up resistor can be selected by the P0PLUD, P3PLU, P4PLUD, P5PLUD, P6PLU and P7PLUD registers. Select input mode by the P0DIR, P3DIR, P4DIR, P5DIR, P6DIR and P7DIR registers to select serial data input by the serial mode register 1 (SC0MD1, SC1MD1, SC2MD1, SC3MD1, and SC4MD1). These can be used as normal I/O pins when the serial interface is not used.
SBI0B	70		TM7IOB	LED2		
			P44	SEG27		
SBI1A	60		RXD0B			
			P51	SEG17		
SBI1B	45		RXD1A			
			P76	SEG2		
SBI2A	53		KEY6	RXD1B		
			P66	SEG10		
SBI2B	80		RXD2A			
SBI3A	29	I/O	P31	SEG37	Serial interface clock I/O pins	Clock I/O pins for serial interface 0, 1, 2, 3, and 4. The output configuration, either CMOS push-pull or Nch open-drain can be selected with the P0ODC, P3ODC, P4ODC, P5ODC, P6ODC, and P7ODC registers. Pull-up resistor can be selected by the P0PLUD, P3PLU, P4PLUD, P5PLUD, P6PLU and P7PLUD registers. Select input mode or output mode by the P0DIR, P3DIR, P4DIR, P5DIR, P6DIR and P7DIR registers and serial data output mode by serial mode register 1 (SC0MD1, SC1MD1, SC2MD1, SC3MD1, and SC4MD1) according to the communication mode. These can be used as normal I/O pins when the serial interface is not used.
SBI3B	73		RXD2B			
			P06	LED6		
SBI4A	51		RXD3A			
			P41	SEG30		
SBI4B	76		RXD3B			
SBT0A	27	I/O	P70	SEG8	Serial interface clock I/O pins	Clock I/O pins for serial interface 0, 1, 2, 3, and 4. The output configuration, either CMOS push-pull or Nch open-drain can be selected with the P0ODC, P3ODC, P4ODC, P5ODC, P6ODC, and P7ODC registers. Pull-up resistor can be selected by the P0PLUD, P3PLU, P4PLUD, P5PLUD, P6PLU and P7PLUD registers. Select input mode or output mode by the P0DIR, P3DIR, P4DIR, P5DIR, P6DIR and P7DIR registers and serial data output mode by serial mode register 1 (SC0MD1, SC1MD1, SC2MD1, SC3MD1, and SC4MD1) according to the communication mode. These can be used as normal I/O pins when the serial interface is not used.
SBT0B	69		KEY0			
SBT1A	61		P35	SEG33		
SBT1B	44		P04	TM2IOB		
SBT2A	52		TM0IOB	LED4		
SBT2B	79		P45	SEG26		
SBT3A	30		P52	SEG18		
			P77	SEG1		
SBT3B	72		KEY7			
SBT4A	49		P67	SEG9		
SBT4B	77	Output	P32	SEG36	UART transmission data output pins	In the serial interface 0, 1, 2, and 3 in UART mode, these pins are configured as the transmission data output pin. The output configuration, either CMOS push-pull or Nch open-drain can be selected with the P0ODC, P3ODC, P4ODC, P5ODC, P6ODC, and P7ODC registers. Pull-up resistor can be selected by the P0PLUD, P3PLU, P4PLUD, P5PLUD, P6PLU, and P7PLUD registers. Select output mode by the P0DIR, P3DIR, P4DIR, P5DIR, P6DIR and P7DIR registers to select serial data output by the serial mode register 1 (SC0MD1, SC1MD1, SC2MD1, and SC3MD1). These can be used as normal I/O pins when the serial interface is not used.
			P07	LED7		
			DA_D			
			P42	SEG29		
			P72	SEG6		
			KEY2	SCL4A		
			P34	SEG34		
			SCL4B			
TXD0A	26		P03	SBO0A		
TXD0B	71		TM8IOB	LED3		
TXD1A	59	Input	P43	SEG28	UART reception data input pins	In the serial interface 0, 1, 2, and 3 in UART mode, these pins are configured as the reception data input pin. Pull-up resistor can be selected by the P0PLUD, P3PLU, P4PLUD, P5PLUD, P6PLU, and P7PLUD registers. Select input mode by the P0DIR, P3DIR, P4DIR, P5DIR, P6DIR and P7DIR registers to select serial data input by the serial mode register 1 (SC0MD1, SC1MD1, SC2MD1, and SC3MD1). These can be used as normal I/O pins when the serial interface is not used.
TXD1B	46		SBO0B			
TXD2A	54		P50	SEG16		
TXD2B	81		SBO1A			
TXD3A	28		P75	SEG3		
TXD3B	74		KEY5	SBO1B		
			P65	SEG11		
			SBO2A	TXD2A		
RXD0A	25	Input	P30	SEG38	UART reception data input pins	In the serial interface 0, 1, 2, and 3 in UART mode, these pins are configured as the reception data input pin. Pull-up resistor can be selected by the P0PLUD, P3PLU, P4PLUD, P5PLUD, P6PLU, and P7PLUD registers. Select input mode by the P0DIR, P3DIR, P4DIR, P5DIR, P6DIR and P7DIR registers to select serial data input by the serial mode register 1 (SC0MD1, SC1MD1, SC2MD1, and SC3MD1). These can be used as normal I/O pins when the serial interface is not used.
RXD0B	70		SBO2B			
RXD1A	60		P05	LED5		
RXD1B	45		SBO3A			
RXD2A	53		P40	SEG31		
RXD2B	80		SBO3B			
RXD3A	29		P02	SBI0A		
RXD3B	73		TM7IOB	LED2		
			P44	SEG27		
			SBI0B	RXD0B		

MN101EF56K		I/O	Other Functions	Functions	Descriptions
Pins	Pin No.				
SDA4A	50	I/O	P71 SEG7	IIC data I/O pins	In the serial interface 4 in IIC mode, these pins are configured as the data input / output pin. For the output configuration, select Nch open-drain by the P3ODC and P7ODC register to select pull-up resistor by the P3PLU and P7PLUD register. Select output mode by the P3DIR and P7DIR register to select serial data input / output by the serial mode register 1 (SC4MD1). These can be used as normal I/O pins when the serial interface is not used.
SDA4B	78		SBO4A KEY1 P33 SEG35 SBO4B		
SCL4A	49		P72 SEG6		
SCL4B	77		SBT4A KEY2 P34 SEG34 SBT4B		
OCD_DATA	23	I/O	P00 LED0	On-board programmer I/O pins	Data I/O pin and clock input pin for the on-board programmer. Refer to [Chapter 24 24.3 Serial Programmer Rewriting]. These can be used as normal I/O pins when the on-board programmer is not used.
OCD_CLK	24		P01 LED1 TM9IOB		
TM0IOA	1	I/O	PA0 AN0	Timer I/O pins	Event count clock input, timer output, and PWM signal output pins for 8-bit timer 0 to 4. To use these pins for event clock input, input mode can be selected by the P0DIR, P1DIR, P6DIR, PADIR, TMCKSEL1, TMINSEL1, and TMINSEL2 registers. In input mode, pull-up resistors can be selected by the P0PLUD, P1PLUD, P6PLU, and PAPLU register. To use these pins for timer output or PWM signal output, select special function pins by the port 0 output mode register, port 1 output mode register, port 6 output mode register, and port A output mode register (P0OMD, P1OMD, P6OMD, and PAOMD) to select output mode by the P0DIR, P1DIR, P6DIR, and PADIR registers. These can be used as normal I/O pins when not used as timer I/O pins.
TM0IOB	27		P04 SBT0A TM2IOB LED4		
TM0IOC	88		P10 AN20 SEG45		
TM1IOA	2		PA1 AN1		
TM1IOB	57		P62 SEG14		
TM1IOC	86		P12 AN22 SEG43		
TM2IOA	3		PA2 AN2		
TM2IOB	27		P04 SBT0A TM0IOB LED4		
TM2IOC	87		P11 AN21 SEG44		
TM3IOA	4		PA3 AN3		
TM3IOB	56		P63 SEG13 TM3IOB		
TM3IOC	85		P13 AN23 SEG42		
TM4IOA	5		PA4 AN4		
TM4IOB	55		P64 SEG12		
TM4IOC	84		P14 SEG41		
BUZZERA	66	Output	P57 SEG23	Buzzer output pins	Piezoelectric buzzer driving pin. Buzzer output is available to port 1, and port 5. The driving frequency can be set in the DLYCTR register. In order to select buzzer output to port 1 and port 5, select the special function pin in the output mode registers (P1OMD1, P1OMD2, P5OMD), and set the direction control registers (P1DIR, and P5DIR) to the output mode. At the same time, select buzzer output in the oscillation stabilization wait control register (DLYCTR). These can be used as normal I/O pins when the buzzer output function is not used.
NBUZZERA	65		P56 SEG22		
BUZZERB	83		P15 SEG40 TM7IOC		
NBUZZERB	82		P16 SEG39 TM8IOC		
TM7IOA	6	I/O	PA5 AN5	Timer I/O pins	Event count clock input, timer output, and PWM signal output pins for 16-bit timer 7, 8, and 9. To use these pins for event clock input, input mode can be selected by the P0DIR, P1DIR, and PADIR registers. In input mode, pull-up resistors can be selected by the P0PLUD, P1PLUD, and PAPLU register. To use these pins for timer output or PWM signal output, select special function pins by the output mode registers (P0OMD, P1OMD1, and PAOMD) to select output mode by the direction control registers (P0DIR, P1DIR, and PADIR). These can be used as normal I/O pins when not used as timer I/O pins.
TM7IOB	25		P02 RXD0A SBI0A LED2		
TM7IOC	83		P15 BUZZERB SEG40		
TM8IOA	7		PA6 AN6		
TM8IOB	26		P03 TXD0A SBO0A LED3		
TM8IOC	82		P16 NBUZZERB SEG39		
TM9IOA	8		PA7 AN7		
TM9IOB	24		P01 OCD_CLK LED1		

MN101EF56K		I/O	Other Functions		Functions	Descriptions
Pins	Pin No.					
TM9OD0	43	Output	P80	SEG0	Timer output pins	Timer output and PWM signal output pins for 16-bit timer. To use these pins for timer output or PWM signal output, select special function pins by the output mode registers (P8OMD, and PDOMD) to select output mode by the direction control registers (P8DIR, and PDDIR). These can be used as normal I/O pins when not used as timer output pins.
TM9OD1	42		SD00			
			P81	COM0		
TM9OD2	41		SD01			
			P82	COM1		
TM9OD3	40		SD02			
		Output	P83	COM2	Synchronous output pins	8-bit synchronous output pins. Synchronous output for each bit can be selected individually by the port 8 synchronous output control register (P8SYO). To use these pins for synchronous output, set output mode by the P8DIR register. These pins can be used as normal I/O pins when not used as synchronous output pins.
TM9OD4	39		SDO3			
			P84	COM3		
TM9OD5	38		SDO4			
			P85	V _{LC3}		
			SDO5			
SDO0	43	Output	P80	SEG0	Synchronous output pins	8-bit synchronous output pins. Synchronous output for each bit can be selected individually by the port 8 synchronous output control register (P8SYO). To use these pins for synchronous output, set output mode by the P8DIR register. These pins can be used as normal I/O pins when not used as synchronous output pins.
SDO1	42		TM9OD0			
			P81	COM0		
SDO2	41		TM9OD1			
			P82	COM1		
SDO3	40		TM9OD2			
			P83	COM2		
SDO4	39		TM9OD3			
		Output	P84	COM3	Reference power supply pin (+) for A/D converter	Reference power supply pin for the A/D converter. This pin is generally used as V _{REF+} = V _{DD5} .
SDO5	38		TM9OD4			
			P85	V _{LC3}		
SDO6	37		TM9OD5			
			P86	V _{LC2}		
SDO7	36		P87	V _{LC1}		
V _{REF+}	9	-				
AN0	1	Input	PA0	TM0IOA	Analog input pins	Analog input pins for 24-channel, 10-bit A/D converter. These pins can be used as normal input pins when not used as analog input pins.
AN1	2		PA1	TM1IOA		
AN2	3		PA2	TM2IOA		
AN3	4		PA3	TM3IOA		
AN4	5		PA4	TM4IOA		
AN5	6		PA5	TM7IOA		
AN6	7		PA6	TM8IOA		
AN7	8		PA7	TM9IOA		
AN8	100		PB0			
AN9	99		PB1			
AN10	98		PB2			
AN11	97		PB3	SEG54		
AN12	96		PB4	SEG53		
AN13	95		PB5	SEG52		
AN14	94		PB6	SEG51		
AN15	93		PB7	SEG50		
AN16	92		P95	SEG49		
			DA_A			
AN17	91		P94	SEG48		
			DA_B			
AN18	90		P93	SEG47		
AN19	89		P92	SEG46		
AN20	88		P10	SEG45		
		Output	TM0IOC		Analog output pins	Analog input pins for 4-channel, 8-bit D/A converter. These pins can be used as normal input pins when not used as analog input pins.
AN21	87		P11	SEG44		
			TM2IOC			
AN22	86		P12	SEG43		
			TM1IOC			
AN23	85		P13	SEG42		
			TM3IOC			
DA_A	92	Output	P95	SEG49	Analog output pins	Analog input pins for 4-channel, 8-bit D/A converter. These pins can be used as normal input pins when not used as analog input pins.
DA_B	91		AN16			
			P94	SEG48		
DA_C	58		AN17			
DA_D	30		P61	SEG15		
			P07	LED7		
			SBT3A			

MN101EF56K		I/O	Other Functions	Functions	Descriptions
Pins	Pin No.				
IRQ0 IRQ1 IRQ2 IRQ3 IRQ4	31 32 33 34 35	Input	P20 ACZ0 P21 ACZ1 P22 P23 P24	External interrupt input pins	External interrupt input pins. The valid edge for IRQ0 to IRQ4 can be selected by the IRQn/CR register. IRQ0 and IRQ1 are able to determine AC zero cross. Both edge and pin voltage level for IRQ2, IRQ3 and IRQ4 are valid for interrupt. These pins can be used as normal input pins when not used as external interrupt pins.
ACZ0 ACZ1	31 32	Input	P20 IRQ0 P21 IRQ1	AC zero-cross detection input pins	Input pins for an AC zero-cross detection circuit. The AC zero-cross detection circuit outputs "High-level" when the input is at an intermediate level. It outputs "Low-level" at all other times. ACZ input signal is connected to the P20 input circuit and the IRQ0 interrupt circuit or the P21 input circuit and the IRQ1 interrupt circuit. When the AC zero-cross detection circuit is not used, this pin can be used as normal input port.
KEY0 KEY1 KEY2 KEY3 KEY4 KEY5 KEY6 KEY7	51 50 49 48 47 46 45 44	Input	P70 SEG8 SBI4A P71 SEG7 SBO4A SDA4A P72 SEG6 SBT4A SCL4A P73 SEG5 P74 SEG4 P75 SEG3 SBI01B TXD1B P76 SEG2 SBI1B RXD1B P77 SEG1 SBT1B	Key interrupt input pins	Input pins for interrupt based on ORed result of pin inputs. These can be set as key input pins in increments of one bit by the key interrupt control registers (KEYT3_1IMD, KEYT3_2IMD). These pins can be used as normal I/O pins when not used as the key input pins.
LED0 LED1 LED2 LED3 LED4 LED5 LED6 LED7	23 24 25 26 27 28 29 30	Output	P00 OCD_DATA P01 TM9IOB OCD_CLK P02 TM7IOB RXD0A SBI0A P03 TM8IOB TXD0A SB00A P04 TM0IOB SBT0A TM2IOB P05 SBO3A TXD3A P06 SBI3A RXD3A P07 SBT3A DA_D	LED driver pins	Large current output pins. These pins can be used as normal I/O pins when not used as the LED driver pins.
COM0 COM1 COM2 COM3	42 41 40 39	Output	P81 SDO1 TM9OD1 P82 SDO2 TM9OD2 P83 SDO3 TM9OD3 P84 SDO4 TM9OD4	LCD common output pin	These pins output common signal of required timing for LCD display. Connect to the common pins of LCD display panel. When the LCD functions are not used, these pins can be used as normal ports by the setting of the LCD output control register (LCCTR0).
V _{LC1} V _{LC2} V _{LC3}	36 37 38	-	P87 SDO7 P86 SDO6 P85 SDO5 TM9OD5	LCD power supply pins	Apply voltage of 5.5 V $\geq$ V _{LC1} $\geq$ V _{LC2} $\geq$ V _{LC3} $\geq$ 0 V. When LCD is not used, V _{LC1} to V _{LC3} can be used as normal ports by the setting of the LCD output control register 0 (LCCTR0).

MN101EF56K		I/O	Other Functions	Functions	Descriptions
Pins	Pin No.				
SEG0	43	Output	P80	LCD segment output pins	These pins output segment signal of required timing for LCD display. Connect to the segment pins of the LCD panel. When LCD display is turned off, V_{SS} -level is output. These pins can be used as normal ports by the setting of the LCD output control registers (LCCTR1 to LCCTR7). SEG for each bit can be individually set as a segment pin or a normal port.
			SDO0 TM9OD0		
SEG1	44		P77 KEY7		
			SBT1B		
SEG2	45		P76 KEY6		
			SBI1B RXD1B		
SEG3	46		P75 KEY5		
			SBO1B TXD1B		
SEG4	47		P74 KEY4		
SEG5	48		P73 KYE3		
SEG6	49		P72 KEY2		
			SBT4A SCL4A		
SEG7	50		P71 KEY1		
			SBO4A SDA4A		
SEG8	51		P70 KEY0		
			SBI4A		
SEG9	52		P67 SBT2A		
SEG10	53		P66		
			SBI2A RXD2A		
SEG11	54		P65		
			SBO2A TXD2A		
SEG12	55		P64 TM4IOB		
SEG13	56		P63 TM3IOB		
SEG14	57		P62 TM1IOB		
SEG15	58		P61 DA_C		
SEG16	59		P50 SBO1A		
			TXD1A		
SEG17	60		P51 SBI1A		
			RXD1A		
SEG18	61		P52 SBT1A		
SEG19	62		P53		
SEG20	63		P54		
SEG21	64		P55		
SEG22	65		P56 NBUZZERA		
SEG23	66		P57 BUZZERA		
SEG24	67		P47		
SEG25	68		P46		
SEG26	69		P45 SBT0B		
SEG27	70		P44 SBI0B		
			RXD0B		
SEG28	71		P43 SBO0B		
			TXD0B		
SEG29	72		P42 SBT3B		
SEG30	73		P41 SBI3B		
			RXD3B		
SEG31	74		P40 SBO3B		
			TXD3B		
SEG32	75		P36		
SEG33	76		P35 SBI4B		
SEG34	77		P34 SBT4B		
			SCL4B		
SEG35	78		P33 SBO4B		
			SDA4B		
SEG36	79		P32 SBT2B		
SEG37	80		P31 SBI2B		
			RXD2B		
SEG38	81		P30 SBO2B		
			TXD2B		
SEG39	82		P16 TM8IOC		
			NBUZZERB		

MN101EF56K		I/O	Other Functions	Functions	Descriptions
Pins	Pin No.				
SEG40	83	Output	P15 TM7IOC BUZZERB	LCD segment output pins	These pins output segment signal of required timing for LCD display. Connect to the segment pins of the LCD panel. When LCD display is turned off, V _{SS} -level is output. These pins can be used as normal ports by the setting of the LCD output control registers (LCCTR1 to LCCTR7). SEG for each bit can be individually set as a segment pin or a normal port.
SEG41	84		P14 TM4IOC		
SEG42	85		P13 AN23 TM3IOC		
SEG43	86		P12 AN22 TM1IOC		
SEG44	87		P11 AN21 TM2IOC		
SEG45	88		P10 AN20 TM0IOC		
SEG46	89		P92 AN19		
SEG47	90		P93 AN18		
SEG48	91		P94 AN17 DA_B		
SEG49	92		P95 AN16 DA_A		
SEG50	93		PB7 AN15		
SEG51	94		PB6 AN14		
SEG52	95		PB5 AN13		
SEG53	96		PB4 AN12		
SEG54	97		PB3 AN11		
MMOD	18	Input		Memory mode switch input pins	Set to V _{DD5} -level or V _{SS} -level.
DMOD	20	Input		Mode switch input pins	Set always to V _{DD5} -level. Only flash EEPROM version, DMOD contains an internal pull-up resistor.
RON	21	Input		Regulator control pin	When connecting the pull-up resistor with this pin, make it to 100 Ω or less. Set always to V _{DD5} -level.

1.4 Block Diagram

1.4.1 Block Diagram

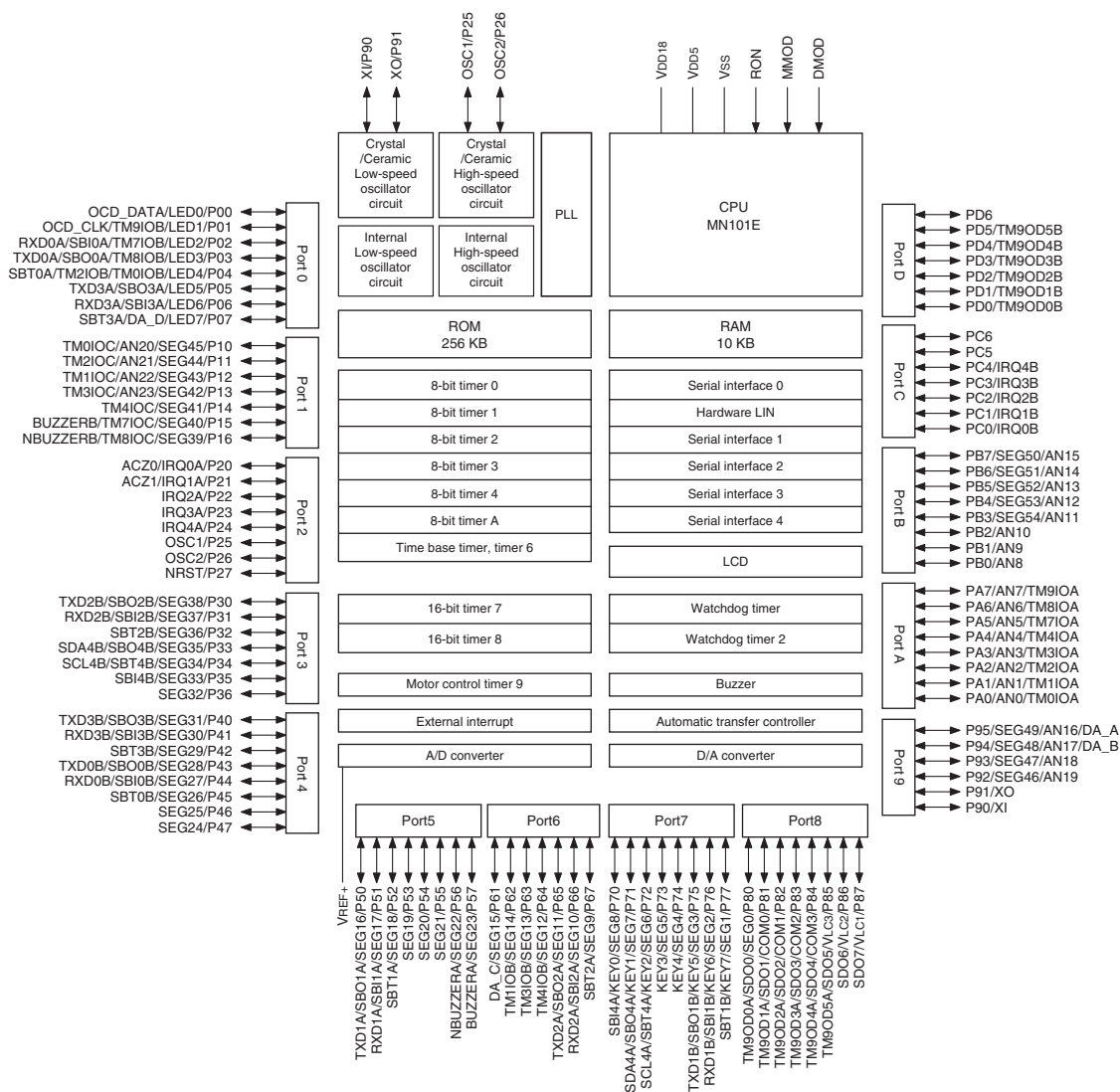


Figure:1.4.1 Block Diagram (MN101EF76K)

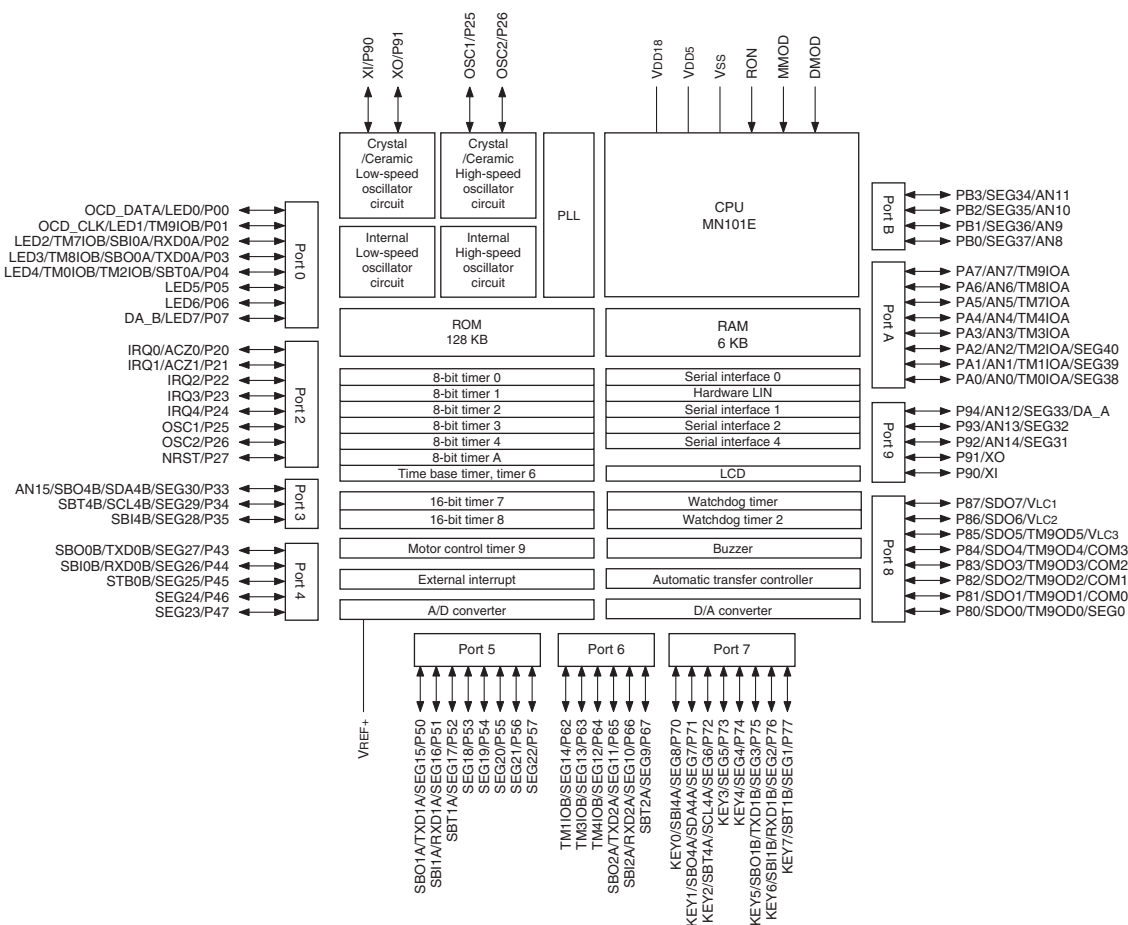


Figure:1.4.2 Block Diagram (MN101EF57G)

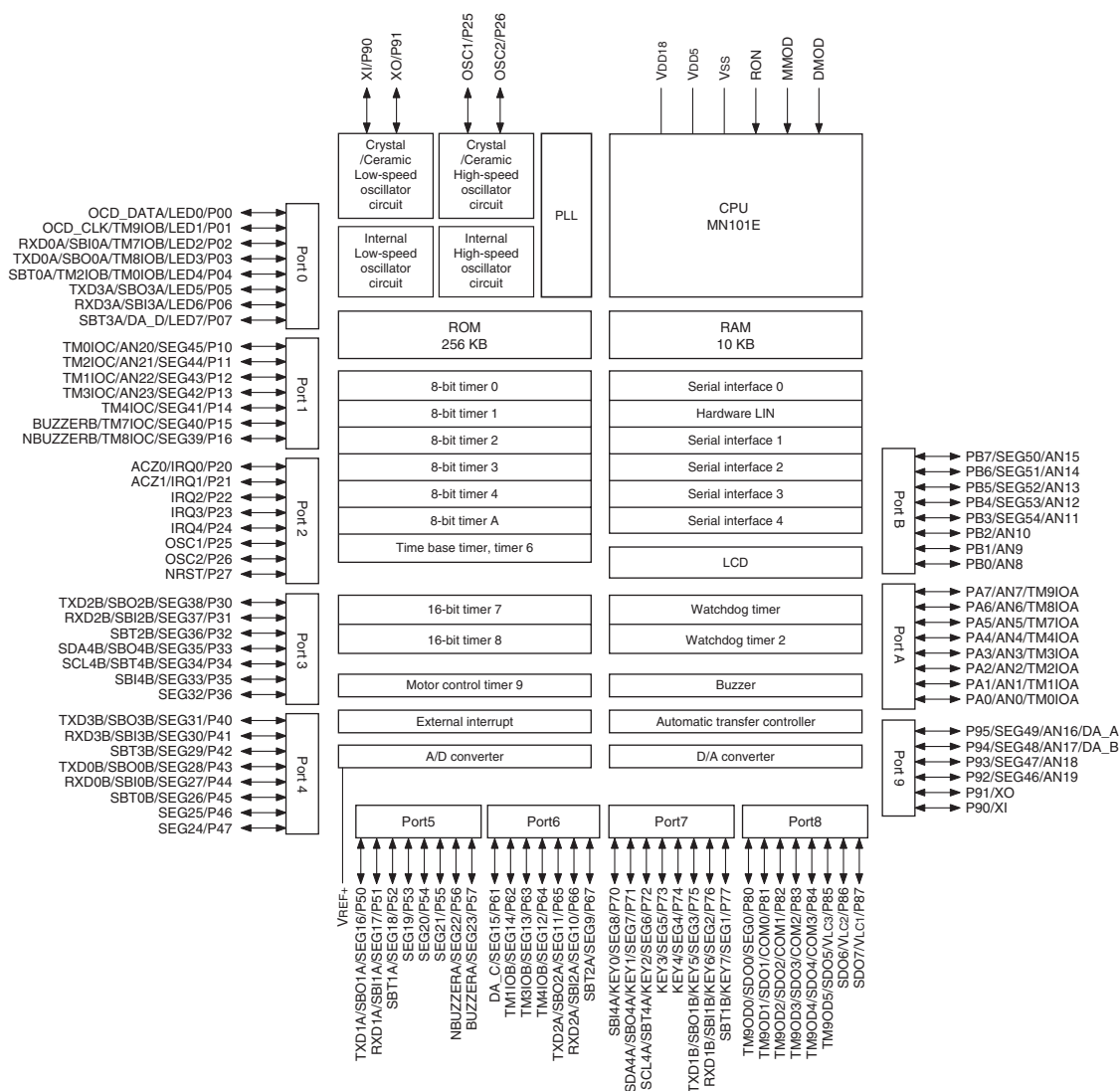


Figure:1.4.3 Block Diagram (MN101EF56K)

1.5 Electrical Characteristics

This LSI manual describes standard specifications.

When using this LSI, consult our sales offices for the product specifications.

Structure	CMOS integrated circuit
Application	General-purpose
Function	CMOS 8-bit single chip microcomputer

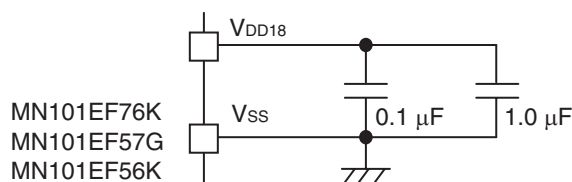
1.5.1 Absolute Maximum Ratings

A. D Absolute Maximum Ratings *2 *3 *4

 $V_{SS} = 0\text{ V}$

Parameter			Symbol	Rating	Unit
A1	Power supply voltage		V _{DD5}	-0.3 to +7.0	V
A2	Power supply voltage		V _{DD18}	-0.3 to +2.5	
A3	Input clamp current (ACZ)		I _C	-500 to +500	μA
A4	Input pin voltage		V _I	-0.3 to V _{DD5} +0.3 (upper limit 7.0 V)	V
A5	Output pin voltage		V _O	-0.3 to V _{DD5} +0.3 (upper limit 7.0 V)	
A6	I/O pin voltage		V _{IO1}	-0.3 to V _{DD5} +0.3 (upper limit 7.0 V)	
A7	Peak output current	LED output	I _{OL1} (peak)	30	mA
A8		Other than LED output	I _{OL2} (peak)	20	
A9		All pins	I _{OH} (peak)	-10	
A10	Average output current *1	LED output	I _{OL1} (avg)	20	
A11		Other than LED output	I _{OL2} (avg)	15	
A12		All pins	I _{OH} (avg)	-5	
A13	Power dissipation		P _T	400	mW
A14	Operating ambient temperature		T _{opr}	-40 to +85	°C
A15	Storage temperature		T _{STG}	-55 to +125	

*1 Applied to any 100 ms period.

*2 Connect at least one bypass capacitor of 1.0 μF or larger between V_{DD5} pin and GND for the internal power voltage stabilization.*3 Connect appropriate 0.1 μF or 1.0 μF capacitor between V_{DD18} pin and V_{SS} pin, near the microcontroller according to the figure shown below for the internal power supply stabilization.Figure:1.5.1 Capacitor Connection between V_{DD18} and V_{SS} Pins

*4 The absolute maximum ratings are the limit values beyond which the LSI may be damaged.

1.5.2 Operating Conditions

B. Operating Conditions

$V_{SS} = 0\text{ V}$
 $T_a = -40\text{ }^{\circ}\text{C to } +85\text{ }^{\circ}\text{C}$

Parameter	Symbol	Conditions	Rating			Unit
			MIN	TYP	MAX	

Power supply voltage *5

B1	Power supply voltage	V_{DD1}	$f_s \leq 20\text{ MHz}$ *7	2.7		5.5	V
B2		V_{DD2}	$f_s \leq 10\text{ MHz}$ *8	2.7		5.5	
B3		V_{DD3}	$f_s \leq 8\text{ MHz}$ *7	1.8		5.5	
B4		V_{DD4}	$f_s \leq 8\text{ MHz}$ *7, *9	2.0		5.5	
B5		V_{DD5}	$f_s \leq 4\text{ MHz}$ *8	1.8		5.5	
B6		V_{DD6}	$f_s \leq 4\text{ MHz}$ *8, *10	2.0		5.5	
B7		V_{DD7}	$f_s = 16.384\text{ kHz}$	1.8		5.5	
B8	RAM retention power supply voltage	V_{DD8}	During STOP mode	1.8		5.5	

Operating speed *6

B9	Instruction execution time f_s	t_{c1}	$V_{DD5} = 2.7\text{ V to } 5.5\text{ V}$ *7	0.05			μs
B10		t_{c2}	$V_{DD5} = 2.7\text{ V to } 5.5\text{ V}$ *8	0.10			
B11		t_{c3}	$V_{DD5} = 1.8\text{ V to } 5.5\text{ V}$ *7	0.125			
B12		t_{c4}	$V_{DD5} = 2.0\text{ V to } 5.5\text{ V}$ *7, *9	0.125			
B13		t_{c5}	$V_{DD5} = 1.8\text{ V to } 5.5\text{ V}$ *8	0.25			
B14		t_{c6}	$V_{DD5} = 2.0\text{ V to } 5.5\text{ V}$ *8, *10	0.25			
B15		t_{c7}	$V_{DD5} = 1.8\text{ V to } 5.5\text{ V}$	61			

*5 f_s : Machine clock frequency

*6 t_{c1} to 2: when the machine clock is selected from external high-speed oscillation, internal high-speed oscillation, or both the oscillations multiplied by PLL.
 t_{c3} : when the machine clock is selected from external high-speed oscillation or internal high-speed oscillation.

*7 When bp2 of the HANDSHAKE register (0x03F06) is set to "1'b1"

*8 When bp2 of the HANDSHAKE register (0x03F06) is set to "1'b0"

*9 When setting $f_{rc}=16\text{ MHz}$, $f_s=f_{rc}/2$

*10 When setting $f_{rc}=16\text{ MHz}$, $f_s=f_{rc}/4$

$V_{SS} = 0 \text{ V}$
 $T_a = -40 \text{ }^{\circ}\text{C}$ to $+85 \text{ }^{\circ}\text{C}$

Parameter	Symbol	Conditions	Rating			Unit
			MIN	TYP	MAX	

External Oscillator 1 Figure:1.5.2

B16	Frequency	f_{hosc1}	V_{DD5} is within the specified operating power supply voltage range. (Refer to the ratings B1 to B6 for the specified operating power supply voltage range)			2.0	10	MHz
B17	Internal feedback resistor	R_{f10}	$V_{DD5} = 5.0 \text{ V}$				980	$\text{k}\Omega$

External Oscillator 2 Figure:1.5.3

B18	Frequency	f_{sosc1}	$V_{DD5} = 1.8 \text{ V}$ to 5.5 V				32.768	kHz
B19	Internal feedback resistor	R_{f20}	$V_{DD5} = 5.0 \text{ V}$				6.2	$\text{M}\Omega$

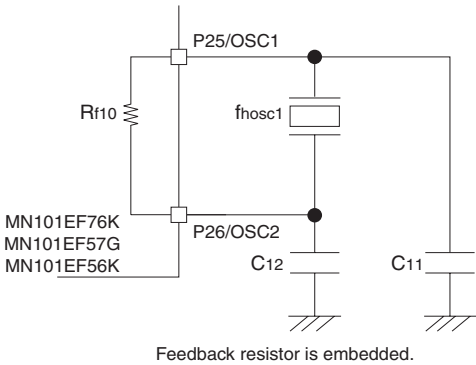


Figure:1.5.2 External Oscillator 1

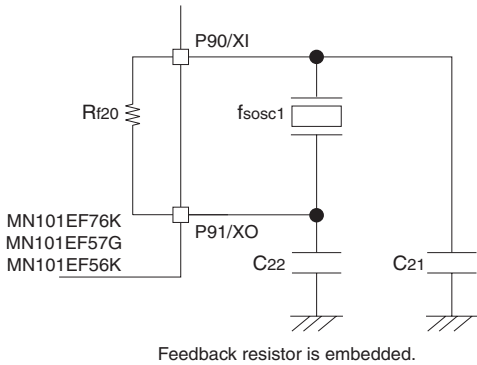


Figure:1.5.3 External Oscillator 2



Connect external capacitors suited for the used oscillator.
The reference value denotes external capacity value based on our matching result.
When crystal oscillator or ceramic oscillator is used, the oscillation frequency is changed depending on the value of capacitor. For external capacity value, please consult the oscillator manufacturer and perform matching tests enough for determining appropriate values.

$$V_{DD5} = 1.8 \text{ V to } 5.5 \text{ V}$$

$$V_{SS} = 0 \text{ V}$$

$$T_a = -40 \text{ }^{\circ}\text{C to } +85 \text{ }^{\circ}\text{C}$$

Parameter	Symbol	Conditions	Rating			Unit
			MIN	TYP	MAX	

External clock input 1 OSC1 (OSC2 is unconnected)

B20	Clock frequency	f_{hosc2}		2		10.0	MHz
B21	High-level pulse width *11	t_{wh1}	Figure:1.5.4	45			ns
B22	Low-level pulse width *11	t_{wl1}		45			
B23	Rising time *12	t_{wr1}	Figure:1.5.4	0		5.0	
B24	Falling time *12	t_{wf1}		0		5.0	

External clock input 2 XI (XO is unconnected)

B25	Clock frequency	f_{sosc2}			32.768		kHz
B26	High-level pulse width *11	t_{wh2}	Figure:1.5.5		4.5		μs
B27	Low-level pulse width *11	t_{wl2}			4.5		
B28	Rising time *12	t_{wr2}	Figure:1.5.5	0		20	ns
B29	Falling time *12	t_{wf2}		0		20	

*11 The clock duty ratio should be 45% to 55%

*12 Rising time and falling time differ depending on the oscillation frequency.
The max value is not a specified value but a rough value.
Please consult the oscillator manufacturer and perform matching tests enough for determining appropriate values.

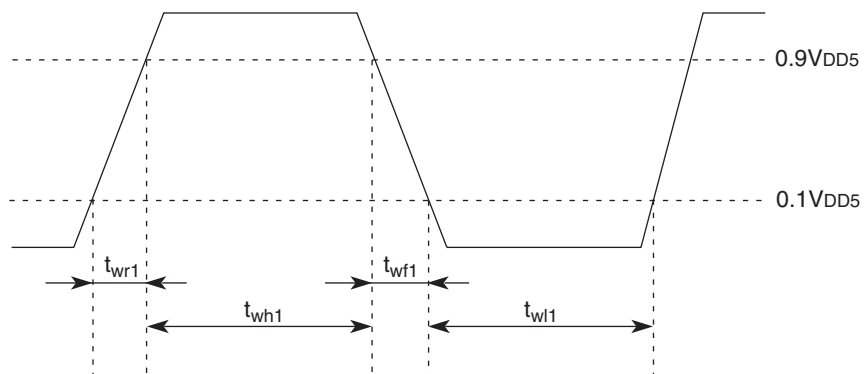


Figure:1.5.4 OSC1 Timing Chart

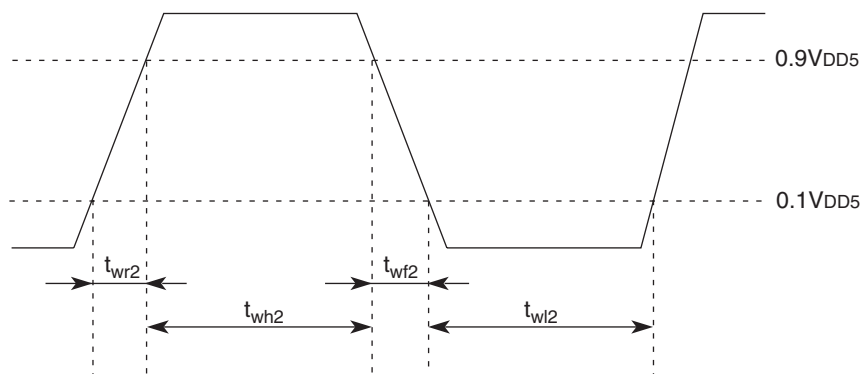


Figure:1.5.5 XI Timing Chart

1.5.3 DC Characteristics

C. DC Characteristics

$V_{SS} = 0\text{ V}$
 $T_a = -40\text{ }^{\circ}\text{C to } +85\text{ }^{\circ}\text{C}$

Parameter	Symbol	Conditions	MN101EF76K /MN101EF56K			MN101EF57G			Unit		
			Rating			Rating					
			MIN	TYP	MAX	MIN	TYP	MAX			
Power supply current *13											
C1	Power supply current during operation	I _{DD1}	fosc=10 MHz [Double-speed mode:fs=fosc] V _{DD5} =5 V (PLL is not used) *14			5.2	14.4		5	14	mA
C2		I _{DD2}	fosc=10 MHz [Double-speed mode:fs=fosc] V _{DD5} =3 V (PLL is not used) *14			5.7			5.5		
C3		I _{DD3}	fosc=8 MHz [Double-speed mode:fs=fosc] V _{DD5} =5 V (PLL is not used) *14			4.7	13.4		4.5	13	
C4		I _{DD4}	fosc=8 MHz [Double-speed mode:fs=fosc] V _{DD5} =3 V (PLL is not used) *14			4.7			4.5		
C5		I _{DD5}	fosc=4 MHz [Double-speed mode:fs=fosc] V _{DD5} =5 V (PLL is not used) *15			3.7	11.4		3.5	11	
C6		I _{DD6}	fosc=4 MHz [Double-speed mode:fs=fosc] V _{DD5} =3 V (PLL is not used) *15			3.7			3.5		
C7		I _{DD7}	fosc=4 MHz [Multiplied by 10:fs=20 MHz] V _{DD5} =5 V (PLL is used) *14			8.2	18.4		8	18	
C8		I _{DD8}	fosc=4 MHz [Multiplied by 10:fs=20 MHz] V _{DD5} =3 V (PLL is used) *14			9.2			9		
C9		I _{DD9}	frc=20 MHz [Double-speed mode:fs=frc] V _{DD5} =5 V (PLL is not used) *14			7.7	16.4		7.5	16	
C10		I _{DD10}	frc=20 MHz [Double-speed mode:fs=frc] V _{DD5} =3 V (PLL is not used) *14			8.7			8.5		
C11		I _{DD11}	frcs=30 kHz [fs=frcs/2] V _{DD5} =3 V Ta=25 °C ROM is executed.			75	90		50	65	μA
C12			frcs=30 kHz [fs=frcs/2] V _{DD5} =3 V Ta=85 °C ROM is executed.				250			150	
C13		I _{DD12}	frcs=30 kHz [fs=frcs/2] V _{DD5} =3 V Ta=25 °C RAM is executed. *16			10	25		10	25	
C14			frcs=30 kHz [fs=frcs/2] V _{DD5} =3 V Ta=85 °C RAM is executed. *16				65			65	
C15	Power supply current during HALT1	I _{DD13}	frcs=30 kHz, V _{DD5} =3 V Ta=25 °C			6	15		6	15	
C16			frcs=30 kHz, V _{DD5} =3 V Ta=85 °C				60			55	
C17	Power supply current during STOP	I _{DD14}	V _{DD5} =5 V, Ta=25 °C			1	5		1	5	
C18			V _{DD5} =5 V, Ta=85 °C				45			40	

*13 Measured without loading (pull-up and pull-down resistors are not connected.)

To measure the power supply current in operation I_{DD1} to I_{DD10} :

1. Set the all I/O pins to input mode.
2. Set the CPU mode to <NORMAL>.
3. Fix the MMOD pin at V_{SS} -level and input pin at V_{DD5} -level.
4. And input the square wave of 10 MHz (8 MHz, 4 MHz), which has amplitude of V_{DD5} and V_{SS} potential, from the OSC1 pin.

To measure the power supply current in operation I_{DD11} , and I_{DD12} :

1. Set the all I/O pins to input mode.
 2. Set the CPU mode to <SLOW>.
 3. Fix the MMOD pin at V_{SS} -level and input pin at V_{DD5} -level.
- Clock is supplied from the internal low-speed oscillation circuit.

To measure the power supply current in HALT1 mode I_{DD13} :

1. Set the all I/O pins to input mode.
2. Set the CPU mode to <HALT1>.
3. And fix the MMOD pin at V_{SS} -level and input pin at V_{DD5} -level.

To measure the power supply current in STOP mode I_{DD14} :

1. Set the CPU mode to <STOP>.
2. Fix the MMOD pin at V_{SS} -level and input pin at V_{DD5} -level.
3. And open the OSC1 pin.

*14 When bp2 of the HANDSHAKE register (0x03F06) is set to "1'b1"

*15 When bp2 of the HANDSHAKE register (0x03F06) is set to "1'b0"

*16 When bp3 of the FEWSPD register (0x03FBF) to "1'b1"

$V_{DD5} = 1.8 \text{ V to } 5.5 \text{ V}$ $V_{SS} = 0 \text{ V}$
 $T_a = -40 \text{ }^{\circ}\text{C to } +85 \text{ }^{\circ}\text{C}$

Parameter	Symbol	Conditions	Rating			Unit
			MIN	TYP	MAX	

Input pin 1 RON

C19	Input high voltage	V_{IH1}	$0.8V_{DD5}$		V_{DD5}	V
C20	Input low voltage	V_{IL1}	0		$0.2V_{DD5}$	

Input pin 2 ATRST, MMOD

C21	Input high voltage	V_{IH2}	$0.8V_{DD5}$		V_{DD5}	V
C22	Input low voltage	V_{IL2}	0		$0.2V_{DD5}$	
C23	Input leakage current	I_{LK1}	$V_{IN} = 0 \text{ V to } V_{DD5}$		± 2	μA

Input pin 3 DMOD *17

C24	Input high voltage	V_{IH3}	$0.8V_{DD5}$		V_{DD5}	V
C25	Input low voltage	V_{IL3}	0		$0.2V_{DD5}$	
C26	Pull-up resistor	R_{RH1}	$V_{DD5} = 5 \text{ V}$, $V_{IN} = V_{SS}$ Pull-up resistor ON		10 50 100	$k\Omega$

Input pin 4 P27/NRST

C27	Input high voltage	V_{IH4}	$0.8V_{DD5}$		V_{DD5}	V
C28	Input low voltage	V_{IL4}	0		$0.15V_{DD5}$	
C29	Pull-up resistor	R_{RH2}	$V_{DD5} = 5 \text{ V}$, $V_{IN} = V_{SS}$ Pull-up resistor ON		10 50 100	$k\Omega$

I/O pin 1 P00 to P07

C30	Input high voltage	V_{IH5}	$0.8V_{DD5}$		V_{DD5}	V
C31	Input low voltage	V_{IL5}	Flash option = select port input voltage level A		0 $0.2V_{DD5}$	
C32	Input low voltage	V_{IL6}	Flash option = select port input voltage level B $V_{DD5} = 4.5 \text{ V to } 5.5 \text{ V}$		0 $0.45V_{DD5}$	
C33	Input leakage current	I_{LK2}	$V_{IN} = 0 \text{ V to } V_{DD5}$		± 2	μA
C34	Pull-up resistor	R_{RH3}	$V_{DD5} = 5 \text{ V}$, $V_{IN} = V_{SS}$ Pull-up resistor ON		10 50 100	$k\Omega$
C35	Pull-down resistor	R_{RL1}	$V_{DD5} = 5 \text{ V}$, $V_{IN} = V_{DD5}$ Pull-down resistor ON		10 50 100	
C36	Output high voltage	V_{OH1}	$V_{DD5} = 5.0 \text{ V}$, $I_{OH} = -0.5 \text{ mA}$		4.5	V
C37	Output low voltage 1	V_{OL1}	$V_{DD5} = 5.0 \text{ V}$, $I_{OL} = 1.0 \text{ mA}$ LED output OFF		0.5	
C38	Output low voltage 2	V_{OL2}	$V_{DD5} = 5.0 \text{ V}$, $I_{OL} = 15.0 \text{ mA}$ LED output ON		1.0	

*17 DMOD internal pull-up resistor is in only Flash EEPROM version.

When using In-circuit Emulator, it is necessary to connect the pull-up resistor on the circuit board.

$$V_{DD5} = 1.8 \text{ V to } 5.5 \text{ V} \quad V_{SS} = 0 \text{ V}$$

$$T_a = -40 \text{ }^{\circ}\text{C to } +85 \text{ }^{\circ}\text{C}$$

Parameter	Symbol	Conditions	Rating			Unit
			MIN	TYP	MAX	

I/O pin 2

MN101EF76K: P20 to P26, P33 to P36, P61 to P67, P80 to P87, P90 to P95, PB0 to PB7

MN101EF57G: P20 to P26, P33 to P35, P62 to P67, P80 to P87, P90 to P94, PB0 to PB3

MN101EF56K: P20 to P26, P33 to P36, P61 to P67, P80 to P87, P90 to P95, PB0 to PB7

C39	Input high voltage	V_{IH6}		$0.8V_{DD5}$		V_{DD5}	V
C40	Input low voltage	V_{IL7}	Flash option = select port input voltage level A	0		$0.2V_{DD5}$	
C41	Input low voltage	V_{IL8}	Flash option = select port input voltage level B $V_{DD5} = 4.5 \text{ V to } 5.5 \text{ V}$	0		$0.45V_{DD5}$	
C42	Input leakage current	I_{LK3}	$V_{IN} = 0 \text{ V to } V_{DD5}$			± 2	μA
C43	Pull-up resistor	R_{RH4}	$V_{DD5} = 5 \text{ V}$, $V_{IN} = V_{SS}$ Pull-up resistor ON	10	50	100	$\text{k}\Omega$
C44	Output high voltage	V_{OH2}	$V_{DD5} = 5.0 \text{ V}$, $I_{OH} = -0.5 \text{ mA}$	4.5			V
C45	Output low voltage 1	V_{OL3}	$V_{DD5} = 5.0 \text{ V}$, $I_{OL} = 1.0 \text{ mA}$			0.5	

I/O pin 3

MN101EF76K: P10 to P16, P40 to P47, P50 to P57, P70 to P77

MN101EF57G: P43 to P47, P50 to P57, P70 to P77

MN101EF56K: P10 to P16, P40 to P47, P50 to P57, P70 to P77

C46	Input high voltage	V_{IH7}		$0.8V_{DD5}$		V_{DD5}	V
C47	Input low voltage	V_{IL9}	Flash option = select port input voltage level A	0		$0.2V_{DD5}$	
C48	Input low voltage	V_{IL10}	Flash option = select port input voltage level B $V_{DD5} = 4.5 \text{ V to } 5.5 \text{ V}$	0		$0.45V_{DD5}$	
C49	Input leakage current	I_{LK4}	$V_{IN} = 0 \text{ V to } V_{DD5}$			± 2	μA
C50	Pull-up resistor	R_{RH5}	$V_{DD5} = 5 \text{ V}$, $V_{IN} = V_{SS}$ Pull-up resistor ON	10	50	100	$\text{k}\Omega$
C51	Pull-down resistor	R_{RL2}	$V_{DD5} = 5 \text{ V}$, $V_{IN} = V_{DD5}$ Pull-down resistor ON	10	50	100	
C52	Output high voltage	V_{OH3}	$V_{DD5} = 5.0 \text{ V}$, $I_{OH} = -0.5 \text{ mA}$	4.5			V
C53	Output low voltage 1	V_{OL4}	$V_{DD5} = 5.0 \text{ V}$, $I_{OL} = 1.0 \text{ mA}$			0.5	

$V_{DD5} = 1.8 \text{ V to } 5.5 \text{ V}$ $V_{SS} = 0 \text{ V}$
 $T_a = -40 \text{ }^{\circ}\text{C to } +85 \text{ }^{\circ}\text{C}$

Parameter	Symbol	Conditions	Rating			Unit
			MIN	TYP	MAX	

I/O pin 4 PA0 to PA7

C54	Input high voltage	V_{IH8}	*18	$0.8V_{DD5}$		V_{DD5}	V
C55	Input high voltage	V_{IH9}	*19	$0.54V_{DD5}$		V_{DD5}	
C56	Input low voltage	V_{IL11}	Flash option = select port input voltage level A	0		$0.2V_{DD5}$	
C57	Input low voltage	V_{IL12}	Flash option = select port input voltage level B $V_{DD5} = 4.5 \text{ V to } 5.5 \text{ V}$	0		$0.45V_{DD5}$	
C58	Input leakage current	I_{LK5}	$V_{IN} = 0 \text{ V to } V_{DD5}$			± 2	μA
C59	Pull-up resistor	R_{RH6}	$V_{DD5} = 5 \text{ V}$, $V_{IN} = V_{SS}$ Pull-up resistor ON	10	50	100	$\text{k}\Omega$
C60	Output high voltage	V_{OH4}	$V_{DD5} = 5.0 \text{ V}$, $I_{OH} = -0.5 \text{ mA}$	4.5			V
C61	Output low voltage	V_{OL5}	$V_{DD5} = 5.0 \text{ V}$, $I_{OL} = 1.0 \text{ mA}$			0.5	

I/O pin 5

MN101EF76K: PC0 to PC6, PD0 to PD6

C62	Input high voltage	V_{IH10}		$0.8V_{DD5}$		V_{DD5}	V
C63	Input low voltage	V_{IL13}		0		$0.2V_{DD5}$	
C64	Input leakage current	I_{LK6}	$V_{IN} = 0 \text{ V to } V_{DD5}$			± 2	μA
C65	Pull-up resistor	R_{RH7}	$V_{DD5} = 5 \text{ V}$, $V_{IN} = V_{SS}$ Pull-up resistor ON	10	50	100	$\text{k}\Omega$
C66	Output high voltage	V_{OH5}	$V_{DD5} = 5.0 \text{ V}$, $I_{OH} = -0.5 \text{ mA}$	4.5			V
C67	Output low voltage	V_{OL6}	$V_{DD5} = 5.0 \text{ V}$, $I_{OL} = 1.0 \text{ mA}$			0.5	

*18 When bp2 of the SWCNT register (0x03E8F) is set to "1'b0".

*19 When bp2 of the SWCNT register (0x03E8F) is set to "1'b1".

$$V_{DD5} = 1.8 \text{ V to } 5.5 \text{ V } V_{SS} = 0 \text{ V}$$

$$T_a = -40 \text{ }^{\circ}\text{C to } +85 \text{ }^{\circ}\text{C}$$

Parameter	Symbol	Conditions	Rating			Unit
			MIN	TYP	MAX	

Input pin 5 P20, P21 (during used as ACZ)

C68	Input high voltage 1	V_{DHH}	Figure:1.5.6	4.5			V
C69	Input high voltage 2	V_{DHL}		1.5			
C70	Input low voltage 1	V_{DLH}				3.5	
C71	Input low voltage 2	V_{DLL}				0.5	
C72	Input clamp current	I_{C1}	$V_{IN} > V_{DD5}, V_{IN} < 0 \text{ V}$			± 500	μA

Display output pin 1 COM0 to COM3 (At V_{LC1} , V_{SS} voltage output) *20

C73	Output impedance	Z_{OCOM1}	$V_{DD5} = V_{LC1} = 5.0 \text{ V } I_{com} = 10 \mu\text{A}$			0.6	V
-----	------------------	-------------	---	--	--	-----	---

Display output pin 2

MN101EF76K: SEG0 to SEG54 (At V_{LC1} , V_{SS} voltage output) *21

MN101EF57G: SEG0 to SEG40 (At V_{LC1} , V_{SS} voltage output) *21

MN101EF56K: SEG0 to SEG54 (At V_{LC1} , V_{SS} voltage output) *21

C74	Output impedance	Z_{OSEG1}	$V_{DD5} = V_{LC1} = 5.0 \text{ V } I_{seg} = 2 \mu\text{A}$			0.6	V
-----	------------------	-------------	--	--	--	-----	---

Display power pin 1 V_{LC1} , V_{LC2} , V_{LC3}

C75	Internal dividing resistor	R_{VL1}	$T_a = +25 \text{ }^{\circ}\text{C}$ (Impedance between V_{LC1} and V_{SS}) *22	15	30	60	$k\Omega$
C76		R_{VL2}		30	60	120	
C77		R_{VL3}		145	300	570	
C78		R_{VL4}		320	660	1260	

*20 COM0 to COM3 are also used as P81 to P84.

*21 MN101EF76K: SEG0 to SEG54 are also used as P10 to P16, P30 to P36, P40 to P47, P50 to P57, P61 to P67, P70 to P77, P80, P92 to P95 and PB3 to PB7.

MN101EF57G: SEG0 to SEG40 are also used as P33 to P35, P43 to P47, P50 to P57, P62 to P67, P70 to P77, P80, P92 to P94, PA0 to PA2 and PB0 to PB3.

MN101EF56K: SEG0 to SEG54 are also used as P10 to P16, P30 to P36, P40 to P47, P50 to P57, P61 to P67, P70 to P77, P80, P92 to P95 and PB3 to PB7.

*22 Total of 3 resistor values between V_{LC1} and V_{LC2} , V_{LC2} and V_{LC3} , V_{LC3} and V_{SS} , respectively.

1.5.4 AC Characteristics

D. AC Characteristics

$V_{DD5} = 5.0\text{ V}$ $V_{SS} = 0\text{ V}$
 $T_a = -40\text{ }^{\circ}\text{C}$ to $+85\text{ }^{\circ}\text{C}$

Parameter	Symbol	Conditions	Rating			Unit
			MIN	TYP	MAX	

D. ACZ pin

D1	Rising time	t_{rs}	Figure:1.5.6	30			μs
D2	Falling time	t_{fs}		30			

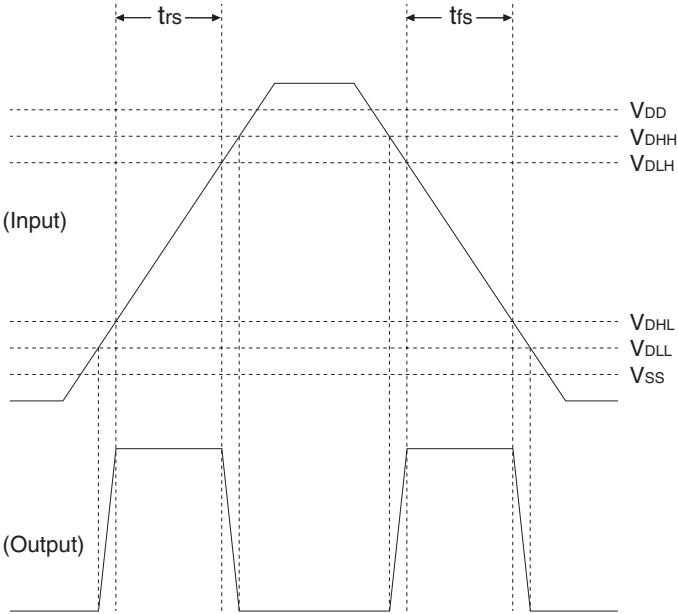


Figure:1.5.6 AC zero-volt detection circuit operation

1.5.5 A/D Converter Characteristics

E. A/D Converter Characteristics *23

 $V_{DD5} = 5.0\text{ V}$ $V_{SS} = 0\text{ V}$ $T_a = -40\text{ }^{\circ}\text{C}$ to $+85\text{ }^{\circ}\text{C}$

Parameter		Symbol	Conditions	Rating			Unit
				MIN	TYP	MAX	
E1	Resolution					10	Bits
E2	Non-linearity error 1		$V_{DD5} = 5.0\text{ V}$, $V_{SS} = 0\text{ V}$			± 3	LSB
E3	Differential linearity error 1		$V_{REF+} = 5.0\text{ V}$ $T_{AD} = 800\text{ ns}$			± 3	
E4	Zero transition voltage		$V_{DD5} = 5.0\text{ V}$, $V_{SS} = 0\text{ V}$		10	30	mV
E5	Full-scale transition voltage		$V_{REF+} = 5.0\text{ V}$ $T_{AD} = 800\text{ ns}$	4970	4990		
E6	A/D conversion time		$T_{AD} = 800\text{ ns}$	12.93			μs
E7			$f_x = 32.768\text{ kHz}$, $T_{AD} = 15.26\text{ }\mu\text{s}$	427.25			
E8	Sampling time		$T_{AD} = 800\text{ ns}$	1.6			
E9			$f_x = 32.768\text{ kHz}$, $T_{AD} = 15.26\text{ }\mu\text{s}$	30.52			
E10	Reference voltage	V_{REF+}		1.8		V_{DD5}	V
E11	Analog input voltage			V_{SS}		V_{REF+}	
E12	Analog input leakage current		Channel OFF $V_{ADIN} = V_{SS}$ to V_{DD5}			± 2	μA
E13	Reference voltage pin input leakage current		Ladder resistance OFF $V_{SS} \leq V_{REF+} \leq V_{DD5}$			± 5	
E14	Ladder resistance	R_{LADD}	$V_{DD5} = 5.0\text{ V}$	15	40	80	$\text{k}\Omega$

*23 T_{AD} is A/D conversion clock cycle.The values of E2 to E5 are guaranteed on the condition of $V_{DD5} = V_{REF+} = 5\text{ V}$, $V_{SS} = 0\text{ V}$.

1.5.6 D/A Converter Characteristics

F. D/A Converter Characteristics

$V_{DD5} = 5.0 \text{ V}$ $V_{SS} = 0 \text{ V}$

$T_a = 25 \text{ }^{\circ}\text{C}$

Parameter	Symbol	Conditions	Rating			Unit
			MIN	TYP	MAX	
F1 Resolution					8	Bits
F2 Reference voltage low level	D_{AVSS}		V_{SS}			V
F3 Reference voltage high level	D_{AVDD}				V_{DD5}	
F4 Zero scale output voltage	V_{ZS}	$D_{AVSS} = 0 \text{ V}$, $D_{AVDD} = 5.0 \text{ V}$ D7 to D0 = ALL "Low"		0	0.05	
F5 Full scale output voltage	V_{FS}	$D_{AVSS} = 0 \text{ V}$, $D_{AVDD} = 5.0 \text{ V}$ D7 to D0 = ALL "High"	4.93	4.98		
F6 Analog output resistance (Minimum reference resistance)	R_{OAT}		5	10	15	k Ω
F7 Non-linearity error	N_{LE}	$D_{AVSS} = 0 \text{ V}$, $D_{AVDD} = 5.0 \text{ V}$	-	± 2.0	± 3.0	LSB
F8 Differential non-linearity error	D_{NLE}	$D_{AVSS} = 0 \text{ V}$, $D_{AVDD} = 5.0 \text{ V}$	-	± 2.0	± 3.0	
F9 Setting time	T_{SET}	External capacitor $C_L = 15 \text{ pF}$ All bits are set to ON or OFF	-	1.5	3.0	μs

Ratings of F7 and F8 are guaranteed at $V_{DD5} = D_{AVDD} = 5.0 \text{ V}$, $V_{SS} = D_{AVSS} = 0 \text{ V}$

1.5.7 Auto Reset Characteristics

G. Auto Reset Characteristics

$V_{DD5} = V_{RST}$ to 5.5 V $V_{SS} = 0$ V
 $T_a = -40$ °C to +85 °C

Parameter	Symbol	Conditions	Rating			Unit
			MIN	TYP	MAX	

Power supply voltage

G1	Operating supply voltage	V_{DD9}	Auto reset is used	V_{RST}		5.5	V
----	--------------------------	-----------	--------------------	-----------	--	-----	---

Power supply voltage

G2	Power detection level	V_{RST1}	At rising	1.90	2.20	2.45	V
G3	Power detection level	V_{RST2}	At falling	1.80	1.90	2.00	V
G4	Supply voltage change rate	$\Delta t/\Delta V$		2			ms/V

Consumption current

G5	Auto reset power consumption	I_{DD15}	$V_{DD5} = 5$ V		1.5	3	μA
----	------------------------------	------------	-----------------	--	-----	---	----

1.5.8 Power Supply Voltage Detection Circuit

$V_{DD5} = 1.8 \text{ V to } 5.5 \text{ V}$ $V_{SS} = 0 \text{ V}$

$T_a = -40 \text{ }^{\circ}\text{C to } +85 \text{ }^{\circ}\text{C}$

Parameter		Symbol	Conditions	Rating			Unit
				MIN	TYP	MAX	
Power supply voltage							
H1	Power supply voltage detection level 1-1	V _{LVI11}	At rising	3.8	4.0	4.2	V
H2	Power supply voltage detection level 1-2	V _{LVI12}	At falling	3.7	3.9	4.1	
H3	Power supply voltage detection level 2-1	V _{LVI21}	At rising	3.6	3.8	4.0	
H4	Power supply voltage detection level 2-2	V _{LVI22}	At falling	3.5	3.7	3.9	
H5	Power supply voltage detection level 3-1	V _{LVI31}	At rising	3.4	3.6	3.8	
H6	Power supply voltage detection level 3-2	V _{LVI32}	At falling	3.3	3.5	3.7	
H7	Power supply voltage detection level 4-1	V _{LVI41}	At rising	3.2	3.4	3.6	
H8	Power supply voltage detection level 4-2	V _{LVI42}	At falling	3.1	3.3	3.5	
H9	Power supply voltage detection level 5-1	V _{LVI51}	At rising	3.0	3.2	3.4	
H10	Power supply voltage detection level 5-2	V _{LVI52}	At falling	2.9	3.1	3.3	
H11	Power supply voltage detection level 6-1	V _{LVI61}	At rising	2.8	3.0	3.2	
H12	Power supply voltage detection level 6-2	V _{LVI62}	At falling	2.7	2.9	3.1	
H13	Power supply voltage detection level 7-1	V _{LVI71}	At rising	2.7	2.8	2.9	
H14	Power supply voltage detection level 7-2	V _{LVI72}	At falling	2.6	2.7	2.8	
H15	Power supply voltage detection level 8-1	V _{LVI81}	At rising	2.5	2.6	2.7	
H16	Power supply voltage detection level 8-2	V _{LVI82}	At falling	2.4	2.5	2.6	
H17	Power supply voltage detection level 9-1	V _{LVI91}	At rising	2.3	2.4	2.5	
H18	Power supply voltage detection level 9-2	V _{LVI92}	At falling	2.2	2.3	2.4	
H19	Power supply voltage detection level 10-1	V _{LVI101}	At rising	2.1	2.2	2.3	
H20	Power supply voltage detection level 10-2	V _{LVI102}	At falling	2.0	2.1	2.2	
H21	Minimum pulse width	T _W		20	60		μs
H22	Supply voltage change rate	Δt/ΔV		2			ms/V
Consumption current							
H23	Consumption current in power supply voltage detection circuit	I _{DD16}	V _{DD5} = 5.0 V		2	4	μA

1.5.9 Internal Oscillation Circuit

I. Internal High-speed Oscillation Circuit

$$V_{DD5} = 2.0 \text{ V to } 5.5 \text{ V } V_{SS} = 0 \text{ V}$$

Parameter		Symbol	Conditions	MN101EF76K			MN101EF57G			MN101EF56K			Unit
				Rating			Rating			Rating			
				MIN	TYP	MAX	MIN	TYP	MAX	MIN	TYP	MAX	
I1	Internal high-speed oscillation circuit frequency	f _{rc20}	Ta = -40 °C to +85 °C		20			20			20		MHz
I2		f _{rc16}			16			16			16		
I3	Temperature dependence of oscillation frequency *24	f _{rc1}	Ta = 25 °C	T.B.D.		T.B.D.	-1.0		+1.0	-1.0		+1.0	%
I4		f _{rc2}	Ta = -40 °C to +85 °C	T.B.D.		T.B.D.	-1.6		+1.6	-1.7		+1.7	

*24 The specification values described in I3 and I4 are for standard application.
For special application (such as for automotive equipment) has different value.
Oscillation characteristic improvement product ($\pm 1 \%$ $T_a = -40 \text{ }^{\circ}\text{C to } +85 \text{ }^{\circ}\text{C}$) is under development.
When using this LSI, consult our sales offices for the product specifications.

J. Internal Low-speed Oscillation Circuit

$$V_{DD5} = 1.8 \text{ V to } 5.5 \text{ V } V_{SS} = 0 \text{ V}$$

$$T_a = -40 \text{ }^{\circ}\text{C to } +85 \text{ }^{\circ}\text{C}$$

Parameter		Symbol	Conditions	Rating			Unit
				MIN	TYP	MAX	
J1	Internal low-speed oscillation circuit frequency	f_{rcs}		27	30	33	kHz

1.5.10 Flash EEPROM Program Conditions

K. Flash EEPROM Program Conditions

$V_{DD5} = 2.7 \text{ V to } 5.5 \text{ V}$ $V_{SS} = 0 \text{ V}$
 $T_a = -40 \text{ }^{\circ}\text{C to } +85 \text{ }^{\circ}\text{C}$

Parameter		Symbol	Conditions	Rating			Unit
				MIN	TYP	MAX	
K1	Data retention period *25		Guaranteed programming times 1000 times	10			Year

*25 The specification value described in K1 is for standard application.
 For special application (such as for automotive equipment) has different value.
 When using this LSI, consult our sales offices for the product specifications.

1.6 Package Dimension

■ Package code: LQFP128-P-1818C (MN101EF76G)

Unit: mm

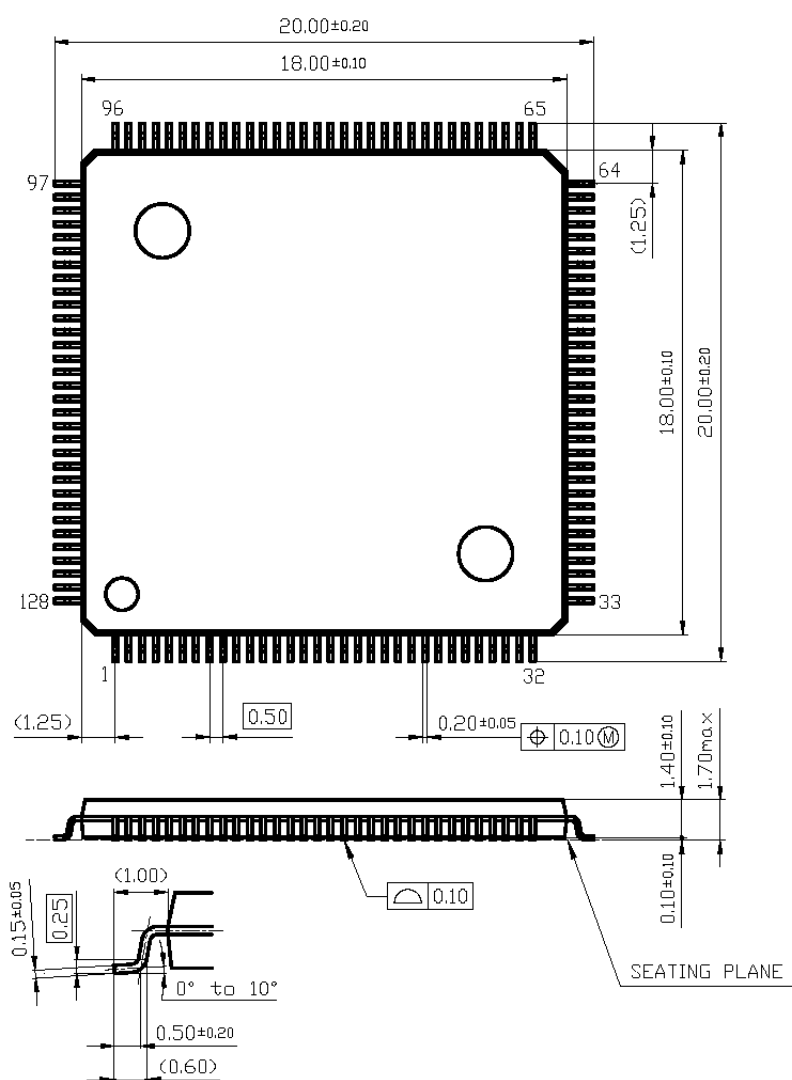


Figure:1.6.1 128-pin LQFP Package Dimension



This package dimension is subject to change. Before using this product, please obtain product specifications from our sales offices.

■ Package code: LQFP080-P-1414A (MN101EF57G)

Unit: mm

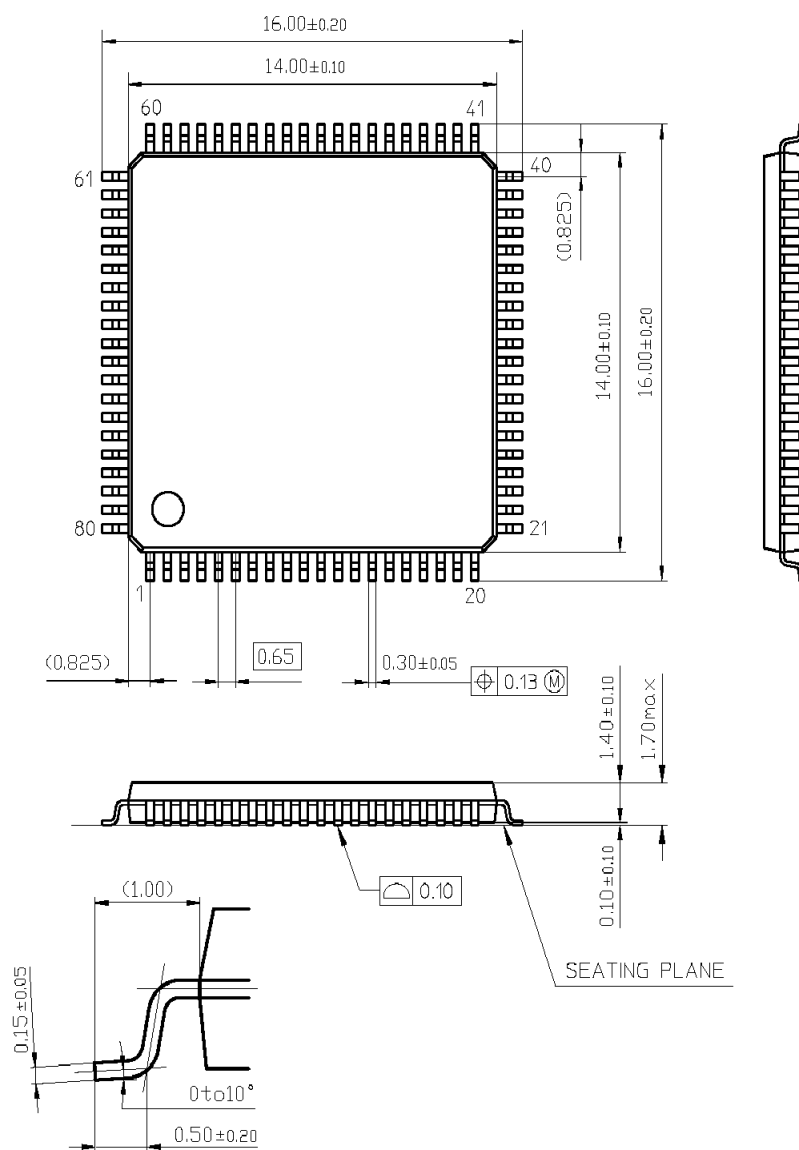


Figure:1.6.2 80-pin LQFP Package Dimension



This package dimension is subject to change. Before using this product, please obtain product specifications from our sales offices.

■ Package code: TQFP080-P-1212F (MN101EF57G)

Unit: mm

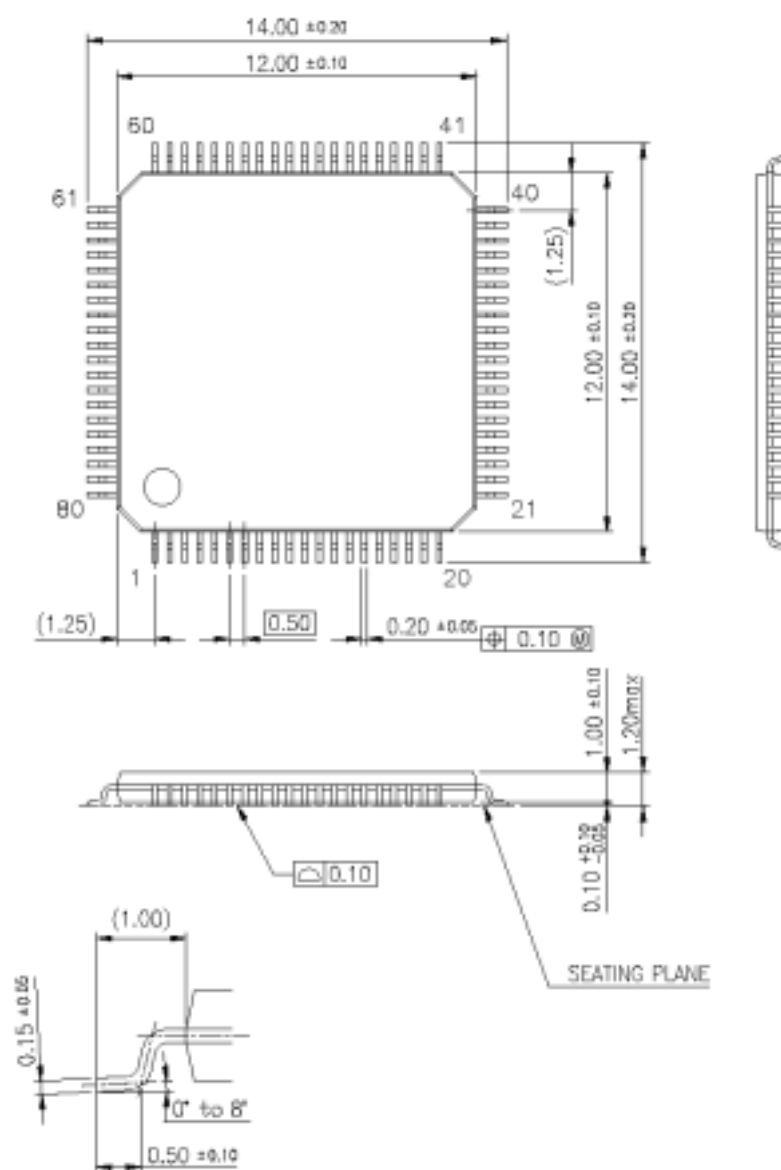


Figure:1.6.3 80-pin TQFP Package Dimension



This package dimension is subject to change. Before using this product, please obtain product specifications from our sales offices.

- Package code: QFP100-P-1818B (MN101EF56K) Unit: mm

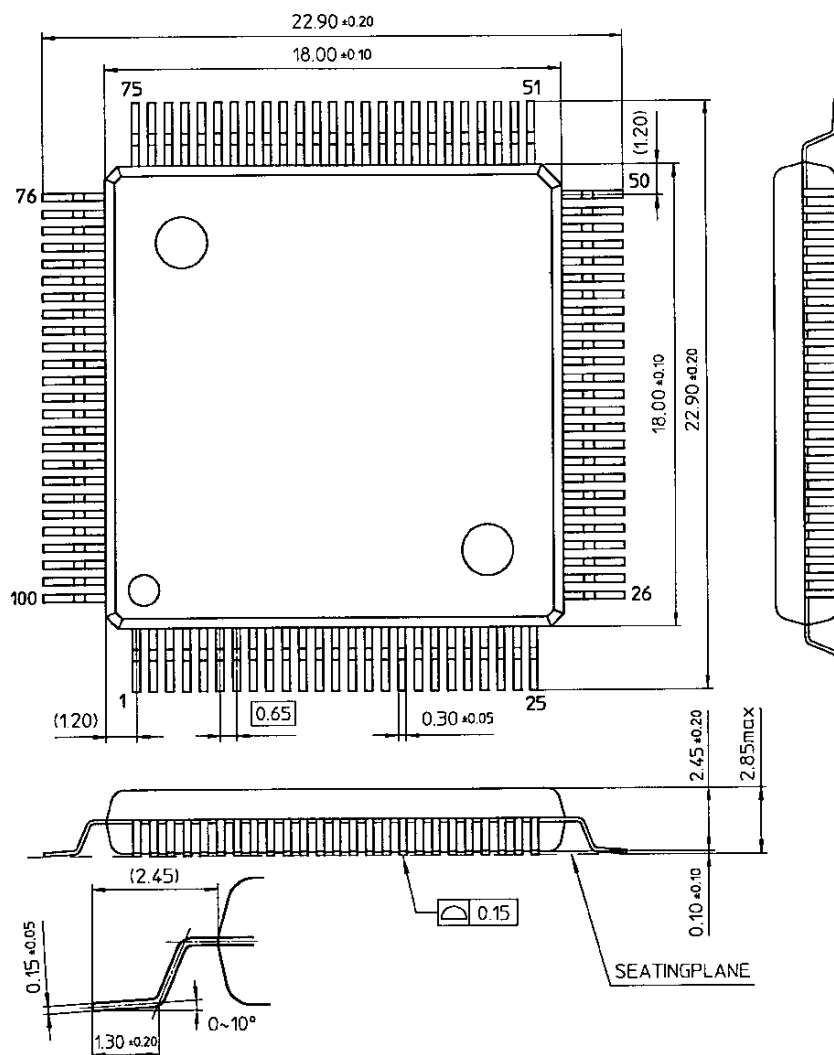


Figure:1.6.4 100-pin QFP Package Dimension



This package dimension is subject to change. Before using this product, please obtain product specifications from our sales offices.

1.7 Cautions for Circuit Setup

1.7.1 General Usage

■ Connection of V_{DD5} pin and V_{SS} pin

All of the V_{DD5} and V_{SS} pins should be connected directly to the power source and ground in the external. Put them on printed circuit board after the location of LSI (package) pin is confirmed. Connection error may lead a fusion and breakdown of a micro controller.

■ V_{REF+} pin Connection

When using V_{REF+} pin and V_{DD5} pin in the same potential, separate V_{REF+} pin at the root of the power supply.

■ Cautions for Operation

1. If you install the product close to high-field emissions (under the cathode ray tube, etc.), shield the package surface to ensure normal performance.
2. Operation temperature should be well considered. Each product has different condition. For example, if the operation temperature is over the condition, improper operation could be occurred.
3. Operation voltage should be also well considered. Each product has different operating range.
 - If the operation voltage is over the operating range, duration of the product could be shortened.
 - If the operation voltage is below the operating range, improper operation could be occurred.

1.7.2 Oscillator

This LSI's oscillation clock can be used with a ceramic and crystal oscillator.

■ Recommended oscillators

Figure:1.7.1 show basic configuration connected with a ceramic oscillator, and Table:1.7.1 shows recommended oscillators and the circuit constants.

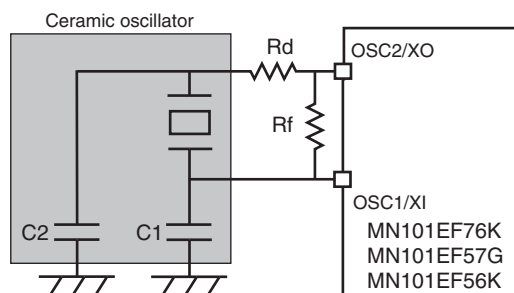


Figure:1.7.1 Basic configuration of oscillator connection

Table:1.7.1 Recommended Ceramic Oscillators and Circuit Constants for MN101EF57 Series

Product name	Manufacturer	Frequency [Hz]	Type	Oscillator Product Number	Recommended circuit constant				Recommended operation condition		
					C1 [pF]	C2 [pF]	Dumping resistor Rd [Ω]	Feedback resistor Rf [Ω]	Power supply voltage [V]		Temperature range [°C]
									Oscillation Power "Low"	Oscillation Power "High"	
MN101EF57G	Murata Manufacturing Company, Ltd.	2.000M	SMD	CSTCC2M00G56-R0	(47)	(47)	Open	2.2k	2.2 to 5.5	1.8 to 3.3	-40 to 85
		4.000M	SMD	CSTCR4M00G55-R0	(39)	(39)	Open	1.5k			
			Lead	CSTLS4M00G56-B0	(47)	(47)	Open	1k			
		8.000M	SMD	CSTCE8M00G52-R0	(10)	(10)	Open	470			
			Lead	CSTLS8M00G53-B0	(15)	(15)	Open	680			
		10.000M	SMD	CSTCE10M0G52-R0	(10)	(10)	Open	330			
			Lead	CSTLS10M0G53-B0	(15)	(15)	Open	330			

Note): () denotes internal capacity.
The end of product name, after "-", denotes package specification.
-SMD type [-R0: plastic taping (φ =180mm), -B0: Individual]
-Read type [-A0: Flat pack (Ho=18mm), -B0: Individual]
Oscillation power "High" or "Low" is controlled by the oscillation switching register (CHDRVSEL).

The above recommended value is the result of oscillator evaluation only on this LSI. After an evaluation on a set board, insert dumping resistor if needed.

Crystal oscillator is not evaluated. So consult the oscillator manufacturer for the appropriate circuit constants.



Circuit constant of each ceramic or crystal oscillator, which is connected to OSC1/OSC2 or XI/XO, differs depending on stray capacitance of the oscillator or on the mounting circuit. So consult the oscillator manufacturer for the appropriate circuit constant.



When switching the product, matching evaluation with each product and oscillator is necessary.

1.7.3 Unused pins

■ Unused Pins (only for output)

Unconnect the unused output pins.

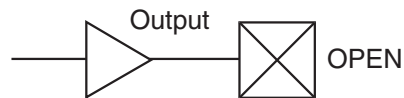


Figure:1.7.2 Unused Pins (only for output)

■ Unused Pins (only for input)

Insert some 10 k Ω resistor to unused pins (only for input) for pull-up or pull-down.

If the input is unstable, Pch transistor and Nch transistor of input inverter are on, and through current goes to the input circuit. That increases current consumption and becomes noise sources to power supply.

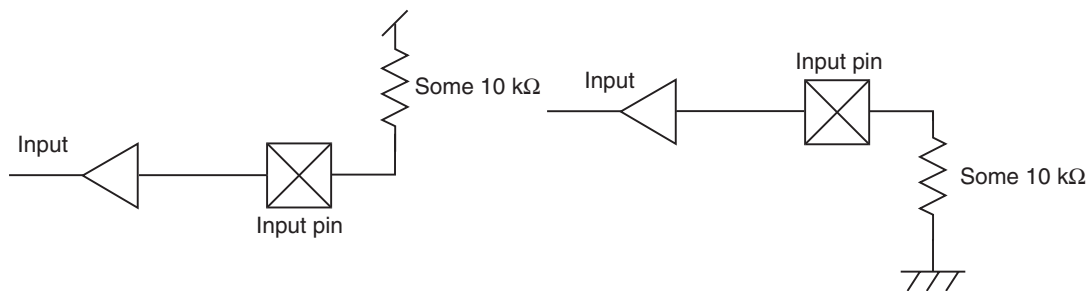


Figure:1.7.3 Unused Pins (only for input)

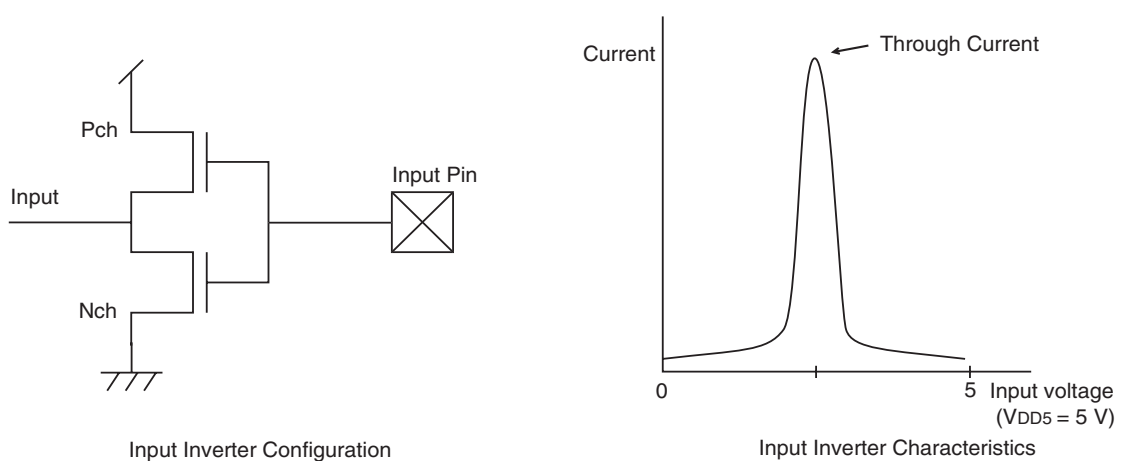


Figure:1.7.4 Input Inverter Organization and Characteristics

■ Unused Pins (for I/O)

Unused I/O pins should be set according to port direction control. In the input mode, set some 10 kΩ resistor to be pull-up in V_{DD5} or pull-down in V_{SS} . In the output mode, set output pins to unconnect.

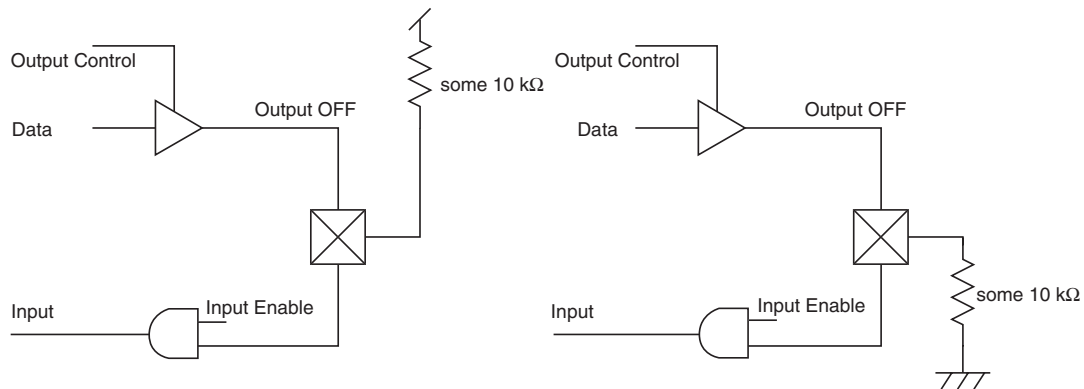


Figure:1.7.5 Unused I/O Pins

■ Recommended condition of each pins

Table:1.7.2 Recommended condition of unused pins *4

Pin Name		Input/Output	Recommended condition of unused pins	
MN101EF76K/ MN101EF56K	MN101EF57G			
P00 to P07 P20 to P26 P30 to P36 P40 to P47 P50 to P57 P61 to P67 P70 to P77 P80 to P87 P90 to P95 PA0 to PA7 PB0 to PB7	P00 to P07 P20 to P26 P33 to P35 P43 to P47 P50 to P57 P62 to P67 P70 to P77 P80 to P87 P90 to P94 PA0 to PA7 PB0 to PB3	Input/Output	Input	Set some 10 kΩ resistor for pull-up in V_{DD5} or pull-down in V_{SS} . *1, *2
			Output	Set open.
P27		Reset		If a capacitor is to be inserted between P27 and V_{SS} , it is recommended that a discharge diode be placed between P27 and V_{DD5} .
V_{REF+}		Reference power supply		Set open or connect directly V_{DD5} .
ATRST		Input		Set some 10 kΩ resistor for pull-up in V_{DD5} (using auto reset function) or pull-down in V_{SS} (not using auto reset function).
MMOD		Input		Set some 100 Ω resistor for pull-up in V_{DD5} (using BOOT mode) or pull-down in V_{SS} (not using BOOT mode).
DMOD		Input		Set some 10 kΩ resistor for pull-up in V_{DD5} .
RON		Input		Set some 100 Ω resistor for pull-up in V_{DD5} .

*1 When unused pins are not connected, the microcomputer does not have the problem. However, it is easily influenced by the surge or the noise.

Evaluate enough for determining the appropriate configuration.

*2 When pins are unused, set them to the normal port function.

*3 The condition of unused pins should be determined by evaluating enough in consideration of the exogenous noise.

*4 Evaluate enough in consideration of the influence of the exogenous noise etc. and decide the method of processing each pin.

1.7.4 Power Supply

■ The Relation between Power Supply and Input Pin Voltage

Input pin voltage should be supplied only after power supply is on. If this order is reversed the destruction of microcontroller by a large current flow could be occurred.

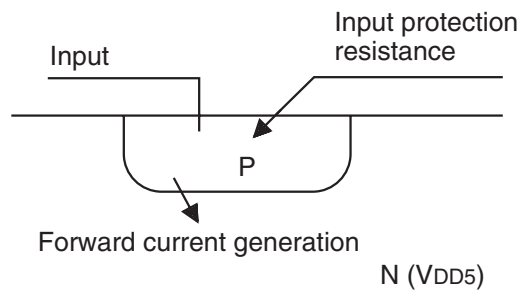


Figure:1.7.6 V_{DD5} and Input Pin Voltage

■ The Relation between V_{DD} and Reset Input Voltage (Don't use Auto Reset function)

This LSI operates in the internal power supply (V_{DD18}) generated with internal regulator.

After internal power supply is over 1.65V, the relation between V_{DD} and Reset input voltage may observe the following conditions and design the substrate.

- [The time of V_{DD18} is over 1.65 V (tV_{DD18})] $\times 2 \leq$ [The time of Reset pin voltage should be low ($tNRSTL$)]

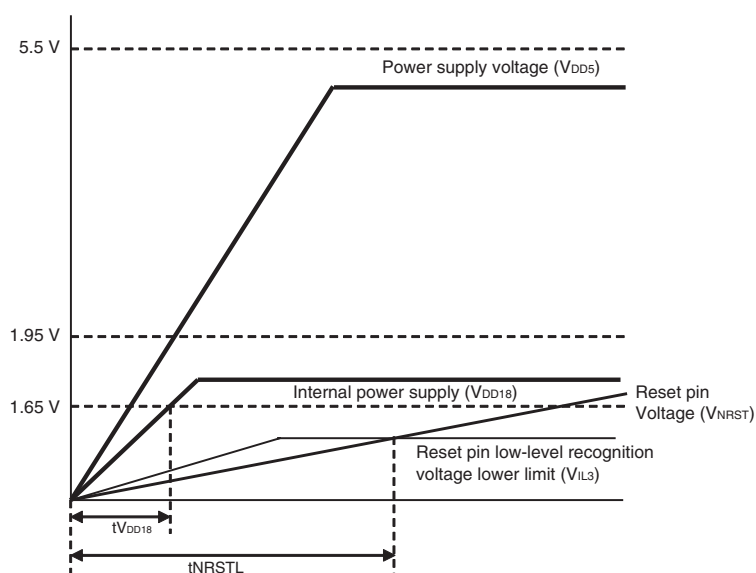
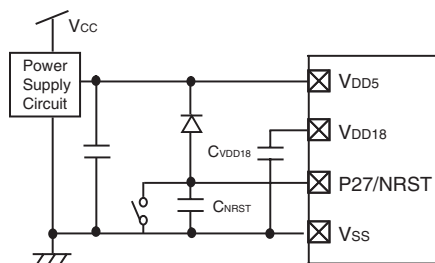


Figure:1.7.7 Power Supply and Reset Input Voltage



When,
 Power supply (V_{DD5}) = 5.0 V
 Ability to supply power = 50 mA
 $C_{VDD18} = 1 \mu F + 0.1 \mu F$
 Input current of reset pin = $5.0 V \div 50 k\Omega = 100 \mu A$
 C_{NRST} connects more than the following values.

$$tV_{DD18} \approx (1.65 V \times 1.1 \mu F) \div 50 mA = 36.5 \mu s$$

$$C_{NRST} \approx (36.5 \mu s \times 2 \times 100 \mu A) \div 5.0 V \times 0.2 = 7.3 nF$$

Figure:1.7.8 Power Circuit Example



The circuit connected with the reset pin should be evaluated enough noting the reset generation caused by the exogenous noise for determining the appropriate configuration.

■ The Relation between V_{DD} and Reset Input Voltage (Use Auto Reset function, reset pin is open)

This LSI operates in the internal power supply (V_{DD18}) generated with internal regulator.

After internal power supply is over 1.65V, the relation between V_{DD} and power detection level of Auto Reset lower limit (V_{RST1}) may observe the following conditions and design the substrate.

- Supply voltage change rate $\geq 2 \text{ ms/V}$
- [The time of V_{DD18} is over 1.65 V (tV_{DD18})] < [The time of power detection level of Auto Reset lower limit ($tATRSTH$)]
- Low level pulse width of reset pin ($tNRSTW$) > 1 μs

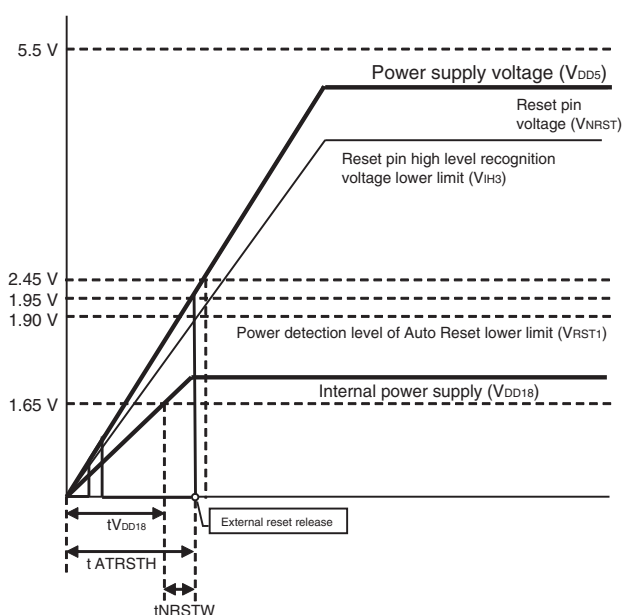


Figure:1.7.9 Power Supply and Reset Input Voltage

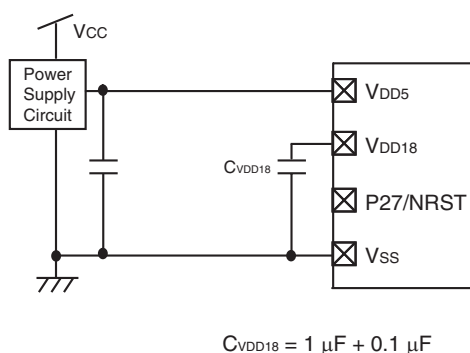


Figure:1.7.10 Power Circuit Example



The circuit connected with the reset pin should be evaluated enough noting the reset generation caused by the exogenous noise for determining the appropriate configuration.

■ The Relation between V_{DD} and Reset Input Voltage
(Use Auto Reset function, reset pin is connect Condenser)

This LSI operates in the internal power supply (V_{DD18}) generated with internal regulator.
After internal power supply is over 1.65V, the relation between V_{DD} and power detection level of Auto Reset lower limit (V_{RST1}) may observe the following conditions and design the substrate.

- [The time of V_{DD18} is over 1.65 V (t_{VDD18})] $\times 2 <$ [The time of power detection level of Auto Reset lower limit (t_{ATRSTH})]
- Low level pulse width of reset pin (t_{NRSTW}) $> 1 \mu s$

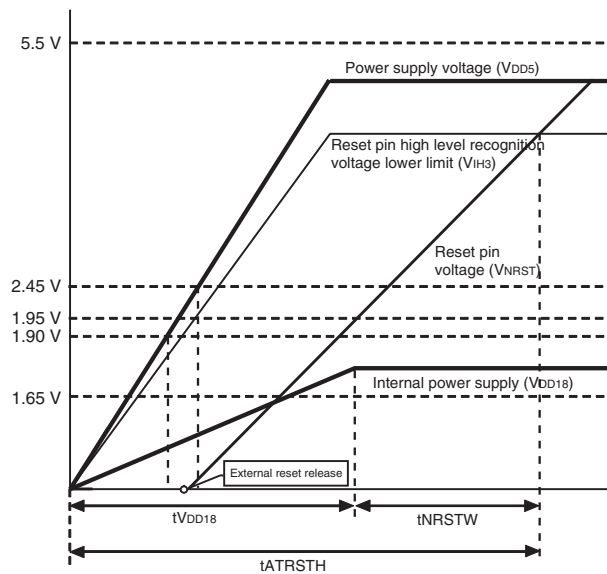


Figure:1.7.11 Power Supply and Reset Input Voltage

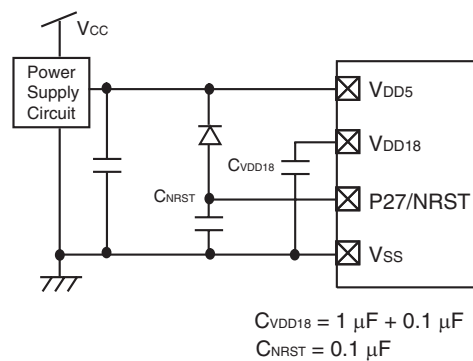


Figure:1.7.12 Power Circuit Example



The circuit connected with the reset pin should be evaluated enough noting the reset generation caused by the exogenous noise for determining the appropriate configuration.

1.7.5 Power Supply Circuit

■ Cautions for Power Circuit Design

The MOS logic such a microcomputer is high speed and high density. So, the power circuit should be designed, taking into consideration of AC line noise, ripple caused by LED driver.

Figure:1.7.13 shows an example for a circuit with V_{DD} (Emitter follower type).

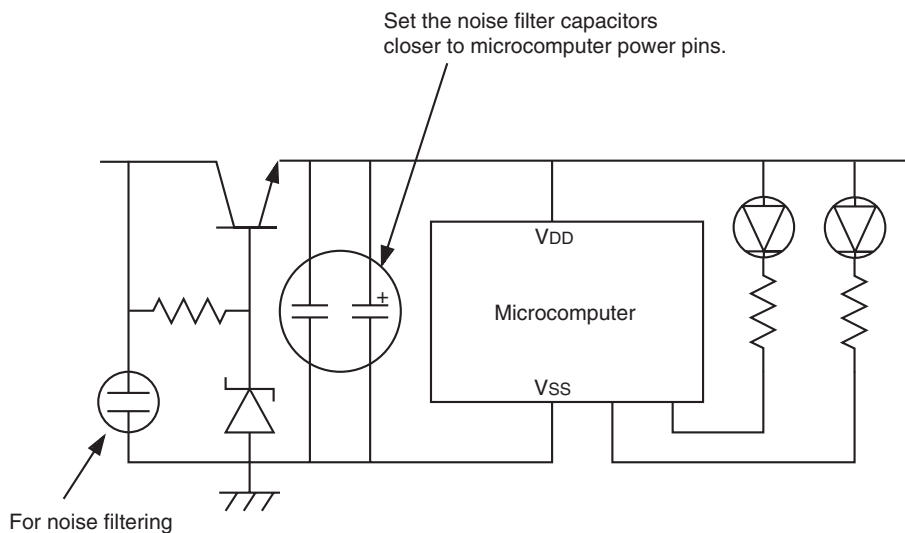


Figure:1.7.13 Power Circuit Example (Emitter follower type)

Chapter 2 CPU Basics

2

2.1 Overview

The MN101E has a flexible and optimized hardware configuration. It is CPU which realizes coexistence of economical efficiency and high-speed operation with a simple and efficient instruction set. Specific features are as follows:

1. Minimized code sizes with instruction lengths based on 4-bit increments:

The series keeps code sizes down by adopting a basic instruction length of one byte and variable instruction lengths based on 4-bit increments.

2. Minimum execution instruction time is one system clock cycle. (50 ns)

3. Minimized register set that simplifies the architecture and supports C language:

The instruction set has been determined, depending on the size and capacity of hardware, after on analysis of embedded application programming code and creation code by C language compiler. Therefore, the set is simple instruction using the minimal register set required for C language compiler.

Table:2.1.1 Basic Specifications

Structure	Structure	
	Six registers	Data : 8-bit × 4 Address : 16-bit × 2
	Others	PC:21-bit PSW:8-bit SP:16-bit
Instructions	Number of instructions	39
	Addressing modes	9
	Instruction length	Basic portion : 1 byte (min.) Extended portion : 0.5-byte × n (0≤n≤9)
Basic performance	Internal operating frequency (max)	20 MHz
	Instruction execution	Min. 1 cycle
	Inter-register operation	Min. 2 cycle
	Load / store	Min. 2 cycle
	Conditional branch	2 to 3 cycles
Pipeline	3-stage (instruction fetch, decode, execution)	
Address space	1 MB (Data area: 64 KB (MAX))	
	Instruction/data space	
External bus	Address	20-bit (Max)
	Data	8-bit
	Minimum bus cycle	1 system clock cycle
Interrupt	Vector interrupt	3 interrupt levels
Low-power consumption mode	STOP mode	
	HALT mode	

2.1.1 Block Diagram

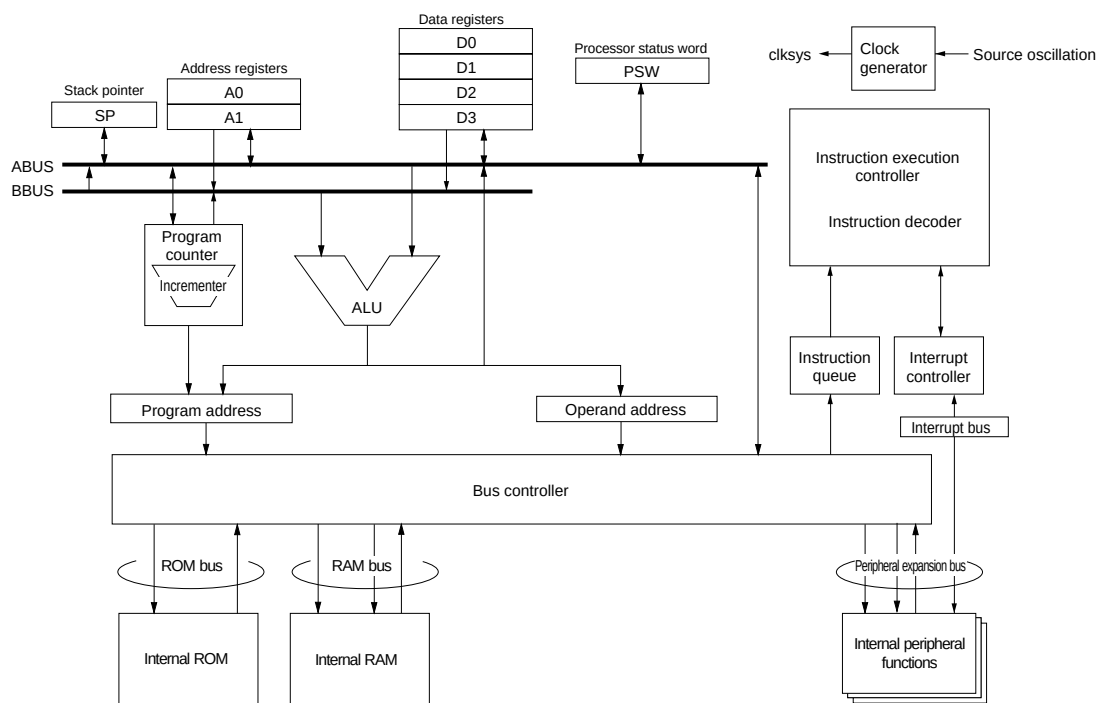


Figure:2.1.1 CPU Block Diagram

Table:2.1.2 Block Diagram and Function

Clock generator	Uses a clock oscillator circuit driven by an external crystal or ceramic resonator to supply clock signals to CPU blocks.
Program counter	Generates addresses for the instructions to be inserted into the instruction queue. Normally incremented by sequencer indication, but may be set to branch destination address or ALU operation result when branch instructions or interrupts occur.
Instruction queue	Stores up to 2 bytes of pre-fetched instructions.
Instruction decoder	Decodes the instruction queue, sequentially generates the control signals needed for instruction execution, and executes the instruction by controlling the blocks within the chip.
Instruction execution controller	Controls CPU block operations in response to the result decoded by the instruction decoder and interrupt requests.
ALU	Executes arithmetic operations, logic operations, shift operations, and calculates operand addresses for register relative indirect addressing mode.
Internal ROM, RAM	Assigned to the execution program, data and stack region.
Address register	Stores the addresses specifying memory for data transfer. Stores the base address for register relative indirect addressing mode.
Data register	Holds data for operations. Two 8-bit registers can be connected to form a 16-bit register.
Interrupt controller	Detects interrupt requests from peripheral functions and requests CPU shift to interrupt processing.
Bus controller	Controls connection of CPU internal bus and CPU external bus. Includes bus usage arbitration function.
Internal peripheral functions	Includes peripheral functions (timer, serial interface, A/D converter, D/A converter, etc.). Peripheral functions vary depending on the model.

2.1.2 CPU Control Registers

This LSI locates the peripheral circuit registers in memory space (0x03DF0 to 0x03FFF) with memory mapped I/O. CPU control registers are also located in this memory space.

Table:2.1.3 CPU Control Registers

Registers	Address	R/W	Function	Pages
CPUM	0x03F00	R/W *1	CPU mode control register	III-22
MEMCTR	0x03F01	R/W	Memory control register	II-38
Reserved	0x03F04	-	(For test)	-
Reserved	0x03F0F	-	(for test)	-
Reserved	0x03FE0	-	(For debugger)	-
NMICR	0x03FE1	R/W	Non - maskable interrupt control register	VII-23
xxxICR	0x03FE2 to 0x03FFD	R/W	Maskable interrupt control register	VII-24 VII-36
Reserved	0x03FFF	-	(For reading interrupt vector data on interrupt process)	-

*1 a part of bit is for read only



Please do not access the reserved address (read / write).

2.1.3 Instruction Execution Controller

The instruction execution controller consists of four blocks: memory, instruction queue, instruction registers, and instruction decoder.

Instructions are fetched in 1-byte units, and temporarily stored in the 2-byte instruction queue. Transfer is made in 1-byte or half-byte units from the instruction queue to the instruction register to be decoded by the instruction decoder.

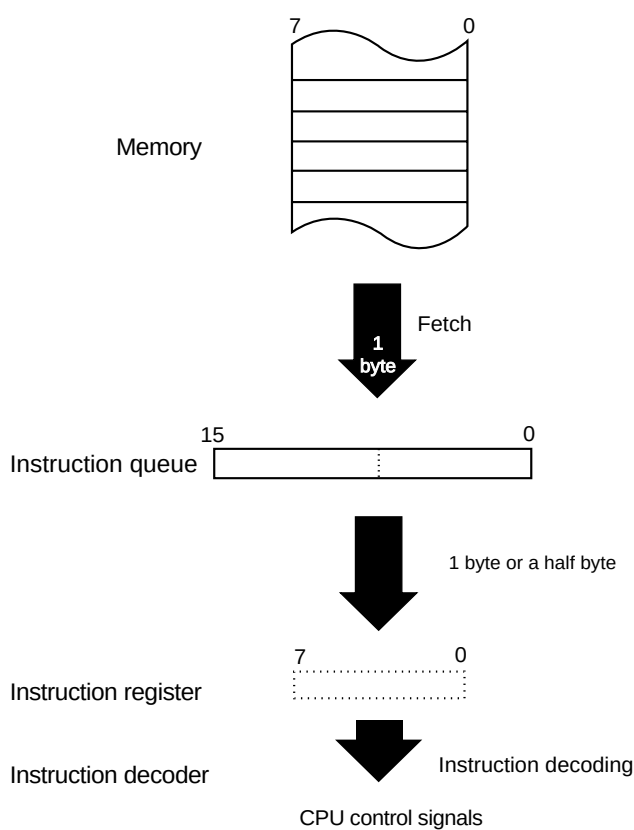


Figure:2.1.2 Instruction Execution Controller Configuration

2.1.4 Pipeline Process

Pipeline process means that reading and decoding are executed at the same time on different instructions, then instructions are executed without stopping. Pipeline process enables instruction execution continuously and faster. This process is executed with instruction queue and instruction decoder.

Instruction queue is buffer that fetches the second instruction in advance. That is controlled to fetch the next instruction when instruction queue is empty at each cycle on execution. At the last cycle of instruction execution, the first word (operation code) of executed next instruction is stored to instruction register. At that time, the next operand or operation code is fetched to instruction queue, so that the next instruction can be executed immediately, even if register direct (da) or immediate data (imm) is needed at the first cycle of the next instruction execution. But on some other instruction such as branch instruction, instruction queue becomes empty on the time that the next operation code to be executed is stored to instruction register at the last cycle. Therefore, only when instruction queue is empty, and direct address (da) or immediate data (imm) are needed, instruction queue keeps waiting for a cycle.

Instruction queue is controlled automatically by hardware so that there is no need to be controlled by software. But when instruction execution time is estimated, operation of instruction queue should be into consideration. Instruction decoder generates control signal at each cycle of instruction execution by micro program control. Instruction decoder uses pipeline process to decode instruction queue at one cycle before control signal is needed.

2.1.5 Registers for Address

Registers for address include program counter (PC), address registers (A0, A1), and stack pointer (SP)

■ Program Counter (PC)

This register gives the address of the currently executing instruction. It is 21 MB bits wide to provide access to a 21 address space in half byte(4-bit increments). The LSB of the program counter is used to indicate half byte instruction. The program counter after reset is stored from the value of vector table at the address of 0x04000.

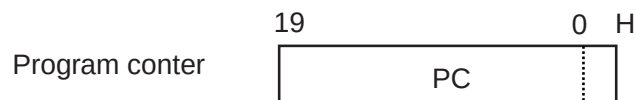


Figure:2.1.3 Program Counter

■ Address Registers (A0, A1)

These registers are used as address pointers specifying data locations in memory. They support the operations involved in address calculations (i.e. addition, subtraction and comparison). Those pointers are 2 bytes data. Transfers between these registers and memory are always in 16-bit units. Either odd or even address can be transferred. At reset, the value of address register is undefined.

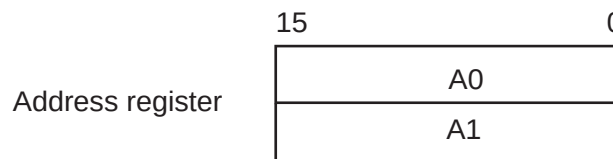


Figure:2.1.4 Address Registers

■ Stack Pointer (SP)

This register gives the address of the byte at the top of the stack. It is decremented during push operations and incremented during pop operations. At reset, the value of SP is undefined.

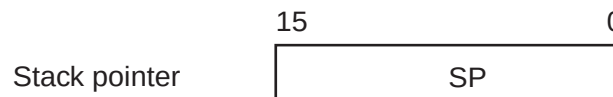


Figure:2.1.5 Stack Pointer

2.1.6 Registers for Data

Registers for data include four data registers (D0, D1, D2, D3).

■ Data Registers (D0, D1, D2, D3)

Data registers D0 to D3 are 8-bit general-purpose registers that support all arithmetic, logical and shift operations. All registers can be used for data transfers with memory. The four data registers may be paired to form the 16-bit data registers DW0 (D0+D1) and DW1 (D2+D3). At reset, the value of Dn is undefined.

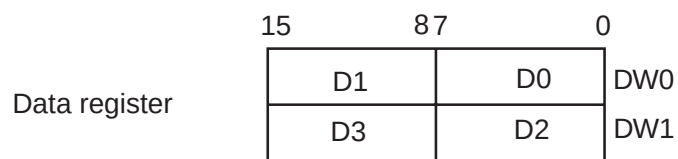


Figure:2.1.6 Data Registers

2.1.7 Processor Status Word

Processor status word (PSW) is an 8-bit register that stores flags for operation results, interrupt mask level, and maskable interrupt enable flag. PSW is automatically pushed onto the stack when an interrupt occurs and is automatically popped when return from the interrupt service routine.

■ Processor Status Word (PSW)

bp	7	6	5	4	3	2	1	0
Flag	BKD	MIE	IM1	IM0	VF	NF	CF	ZF
At reset	0	0	0	0	0	0	0	0

bp	Flag	Description
7	BKD	Bank disable flag 0: Bank addressing is enabled. 1: Bank addressing is disabled.
6	MIE	Maskable interrupt enable 0: All maskable interrupts are disabled. 1: (xxxLVn,xxxIE) for each interrupt is enabled.
5-4	IM1 IM0	Interrupt mask level Controls maskable interrupt acceptance.
3	VF	Overflow flag 0: No overflow occurred. 1: Overflow occurred.
2	NF	Negative flag 0: MSB of operation results is "0". 1: MSB of operation results is "1".
1	CF	Carry flag 0: No carry or borrow from MSB occurred. 1: A carry or a borrow from MSB occurs.
0	ZF	Zero flag 0: Operation result is not "0". 1: Operation result is "0".

■ Zero Flag (ZF)

Zero flag (ZF) is set to "1", when all bits are '0' in the operation result. Otherwise, zero flag is cleared to "0".

■ Carry Flag (CF)

Carry flag (CF) is set to "1", when a carry from or a borrow to the MSB occurs. Carry flag is cleared to "0", when no carry or borrow occurs.

■ Negative Flag (NF)

Negative flag (NF) is set to "1" when MSB is '1' and reset to "0" when MSB is '0'. Negative flag is used to handle a signed value.

■ Overflow Flag (VF)

Overflow flag (VF) is set to "1", when the operation results overflow as a signed value. Otherwise, overflow flag is cleared to "0". Overflow flag is used to handle a signed value.

■ Interrupt Mask Level (IM1 and IM0)

Interrupt mask level (IM1 and IM0) controls the maskable interrupt acceptance in accordance with the interrupt factor interrupt priority for the interrupt control circuit in the CPU. The two-bit control flag defines levels '0' to '3'. Level 0 is the highest mask level. The interrupt request will be accepted only when the level set in the interrupt level flag (xxxLV1 and xxxLV0) of the interrupt control register (xxxICR) is higher than the interrupt mask level. When the interrupt is accepted, the value of accepted interrupt level flag is set to IM1-IM0, and interrupts whose mask levels are the same or lower are rejected during the accepted interrupt processing.

Table:2.1.4 Interrupt Mask Level and Interrupt Acceptance

	Interrupt mask level		Priority	Acceptable interrupt level
	IM1	IM0		
Mask level 0	0	0	High	Non-maskable interrupt (NMI) only
Mask level 1	0	1	.	NMI, level 0
Mask level 2	1	0	.	NMI, level 0 to 1
Mask level 3	1	1	Low	NMI, level 0 to 2

■ Maskable Interrupt Enable (MIE)

Maskable interrupt enable flag (MIE) enables/disables acceptance of maskable interrupts by the CPU's internal interrupt acceptance circuit. A '1' enables maskable interrupts; a '0' disables all maskable interrupts regardless of the interrupt mask level (IM1-IM0) setting in PSW. This flag is not changed by interrupts.

■ Bank disable flag (BKD)

Bank disable flag (BKD) enables/disables bank addressing of 64 KB unit. When this flag is set to "0", bank addressing is enabled and you can access to total 16 banks by setting the bank register value. When this flag is set to "1", bank addressing is disabled and the only area you can access is the first 64 KB. On an interrupt generation, BKD flag is automatically set to "1" and bank addressing is disabled. At returning from interrupt service routine, the value of BKD flag is returned to the previous one. (before the interrupt generation)



To enable bank addressing in an interrupt service routine, reset the BKD flag to "0" before accessing to data.



Write to the interrupt control register (xxxICR) after setting the maskable interrupt enable flag (MIE) of processor status word (PSW) to disable all maskable interrupts. (MIE flag is set to "0".) There's no guarantee of proper operation when writing is executed to the interrupt control register with setting MIE flag to "1".

2.1.8 Address Space

The address space of this LSI is 1 MB. (max.) The instruction and data areas are in the same area.

The instruction area can be used as linear address space. The data area needs bank specification in every 64 KB. (The initial value is first 64 KB space). The data described in this section includes RAM data and ROM table data.

The data area consists of an area of 256 bytes that supports efficient accesses with RAM short addressing and an area of 256 bytes that supports efficient accesses with I/O short addressing.

The memory control register controls the operation of the memory to be expanded.

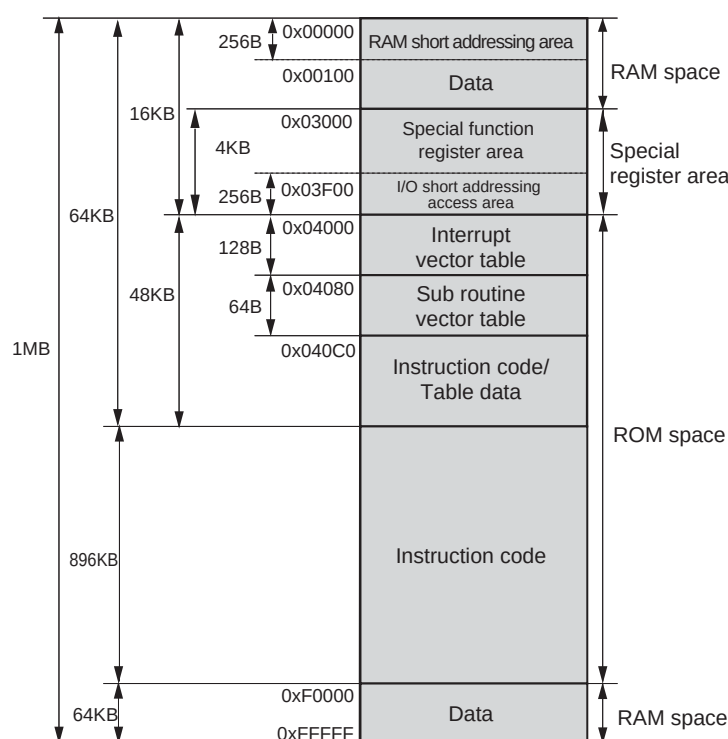


Figure:2.1.7 Address Space

2.1.9 Addressing Modes

This LSI supports the nine addressing modes. Each instruction uses a combination of the following addressing

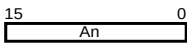
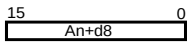
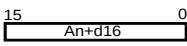
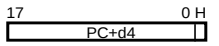
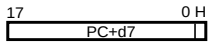
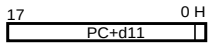
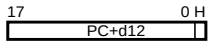
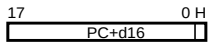
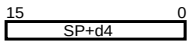
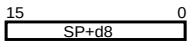
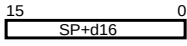
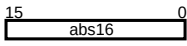
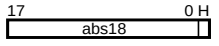
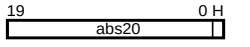
1. Register direct
2. Immediate
3. Register indirect
4. Register relative indirect
5. Stack relative indirect
6. Absolute
7. RAM short
8. I/O short
9. Handy

These addressing modes are well-suited for C language compilers. All of the addressing modes can be used for data transfer instructions. In modes that allow half-byte addressing, the relative value can be specified in half-byte (4-bit) increments, so that instruction length can be shorter. Handy addressing reuses the last memory address accessed. There are three instructions that can use this mode: MOV Dn, (HA), MOVW DWn, (HA), MOVW An, (HA). Combining handy addressing with absolute addressing reduces code size. For transfer data between memory, 8 addressing modes; register indirect, register relative indirect, stack relative indirect, absolute, RAM short, I/O short, handy can be used. For operation instruction, register direct and immediate can be used. Refer to instruction's manual for the MN101E series.



This LSI is designed for 8-bit data access. When 16-bit data access is carried out, 8-bit data access is performed twice from the lower address.
It is possible to transfer data in 16-bit increments with odd or all even addresses.

Table:2.1.5 Address Space

Addressing mode		Effective address	Explanation
Register direct	Dn/DWn An/SP PSW	-	Directly specifies the register. Only internal registers can be specified.
Immediate	imm4/imm8 imm16	-	Directly specifies the operand or mask value appended to the instruction code.
Register indirect	(An)		Specifies the address using an address register.
Register relative indirect	(d8,An)		Specifies the address using an address register with 8-bit displacement.
	(d16,An)		Specifies the address using an address register with 16-bit displacement.
	(d4,PC) (branch instructions only)		*1 Specifies the address using the program counter with 4-bit displacement and H bit.
	(d7,PC) (branch instructions only)		*1 Specifies the address using the program counter with 7-bit displacement and H bit.
	(d11,PC) (branch instructions only)		*1 Specifies the address using the program counter with 11-bit displacement and H bit.
	(d12,PC) (branch instructions only)		*1 Specifies the address using the program counter with 12-bit displacement and H bit.
	(d16,PC) (branch instructions only)		*1 Specifies the address using the program counter with 16-bit displacement and H bit.
Stack relative indirect	(d4,SP)		Specifies the address using the stack pointer with 4-bit displacement.
	(d8,SP)		Specifies the address using the stack pointer with 8-bit displacement.
	(d16,SP)		Specifies the address using the stack pointer with 16-bit displacement.
Absolute	(abs8)		Specifies the address using the operand value appended to the instruction code. Optimum operand length can be used to specify the address.
	(abs12)		
	(abs16)		
	(abs18) (branch instructions only)		
	(abs20) (branch instructions only)		
RAM short	(abs8)		Specifies an 8-bit offset from the address x'00000'.
I/O short	(io8)		Specifies an 8-bit offset from the top address (x'03F00') of the special function register area.
Handy	(HA)	-	Reuses the last memory address accessed and is only available with the MOV and MOVW instructions. Combined use with absolute addressing reduces code size.

*1 H: half-byte bit

2.1.10 Machine Clock

Machine clock is generated based on the system clock (fs) dividing the source oscillation frequency. The machine clock is the base timing for control of CPU.

■ Internal Memory Access

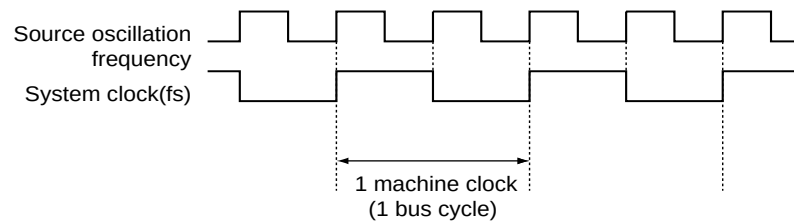


Figure:2.1.8 Machine Clock of Internal Memory Access



Division ratio of system clock (fs) differs depending on the CPUM register settings.
Refer to [Chapter 3 Clock Control].

2.2 Memory Space

2.2.1 Memory Mode

ROM is the read only area and RAM is the memory area which is readable/writable of data. In addition to these, peripheral resources such as memory-mapped special registers are allocated. The MN101E series supports two memory modes (single chip mode, memory expansion mode) in its memory model.

This LSI supports only single chip mode. In single chip mode, the system consists of only internal memory.



Always set bp4 of the MEMCTR register to "0". When setting "1", there's no guarantee of proper operation.

Table:2.2.1 Memory Mode Setup

Memory mode	MMOD
Single chip mode	Low



Do not change the settings of MMOD pin even after reset is released.



For MMOD setup to rewrite the flash memory, refer to [Chapter 24 24.5 User Mode Microcontroller Rewriting], [Chapter 24 24.6 BOOT Mode Microcontroller Rewriting], [Chapter 24 24.7 Appendix]

2.2.2 RAM Space

■ RAM Space

MN101E series has maximum 64 KB of RAM space.

RAM space is divided to be allocated to the address space. Mirror RAM space is provided for effective utilization of the divided RAM spaces.

RAM space: 0x00000 to 0x02FFF (12 KB) + 0xF3000 to 0xFFFFF (52 KB) (maximum 64KB)

Mirror RAM space: 0xF0000 to 0xF3DFF = 0x00000 to 0x03DFF (Mapped to same RAM space)

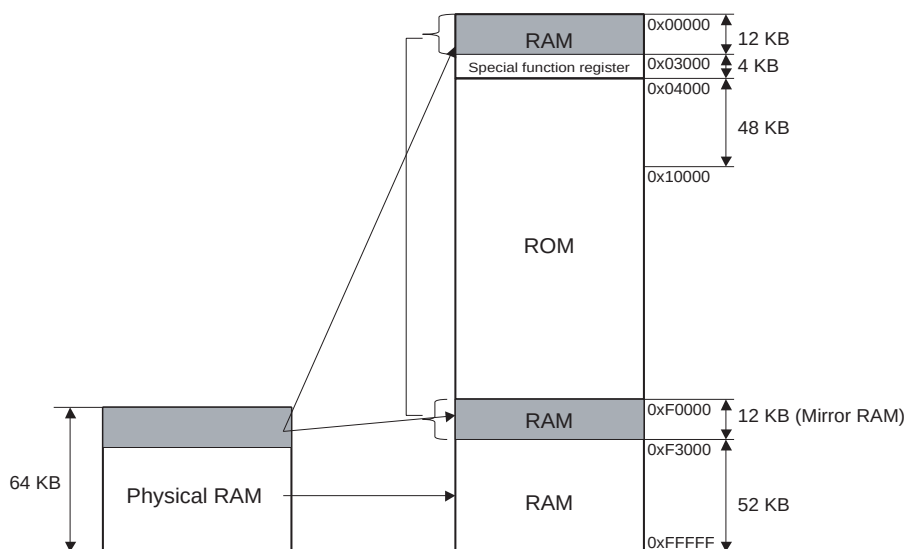


Figure:2.2.1 RAM Space

■ How to use mirror RAM Space

Sub routine A

Address bank 15

mov x'0F', (SBNKR): Source side

mov x'0F', (DBNKR): Destination side

Transfer data 15 between memories (1)

mov (x'XYZZ'), dn

mov dn, (x'ABCD'): x'XYZZ' → x'ABCD'

(x'ABCD', x'XYZZ' are address of abs16.)

Sub routine B (Address bank 0)

mov (x'ABCD'), d1: Use mirror function (2)

Execute the same access ignoring the upper 4 bits.

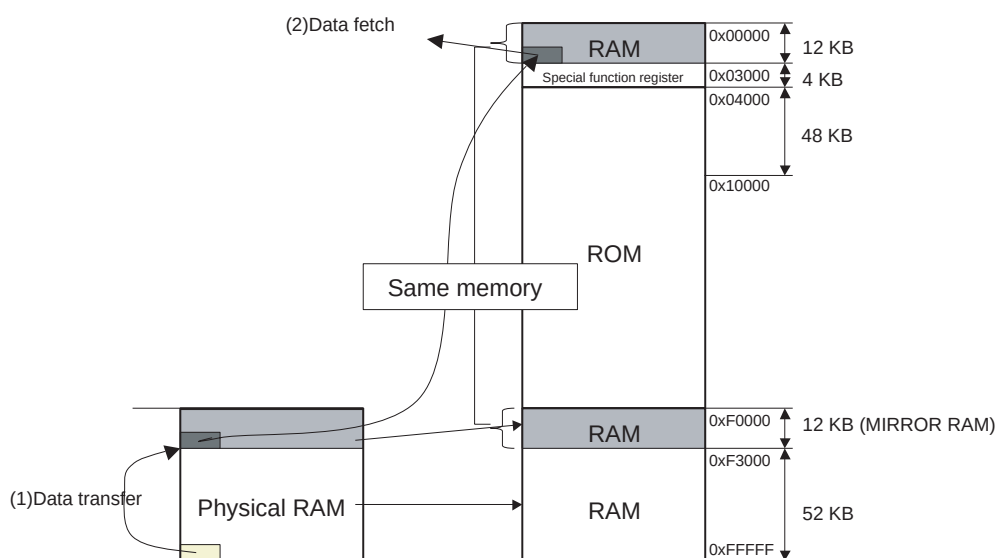


Figure:2.2.2 How to use mirror RAM Space (Example)

2.2.3 Single-chip Mode

In single-chip mode, the system consists of only internal memory. This is the optimized memory mode and allows construction of systems with the highest performance. The single-chip mode uses only internal ROM and internal RAM. The MN101E series devices offer up to 64 KB of RAM and up to 944 KB of ROM. MN101EF76K and MN101EF56K offer 10 KB of RAM and 256 KB of ROM. MN101EF57G offers 6 KB of RAM and 128 KB of ROM.

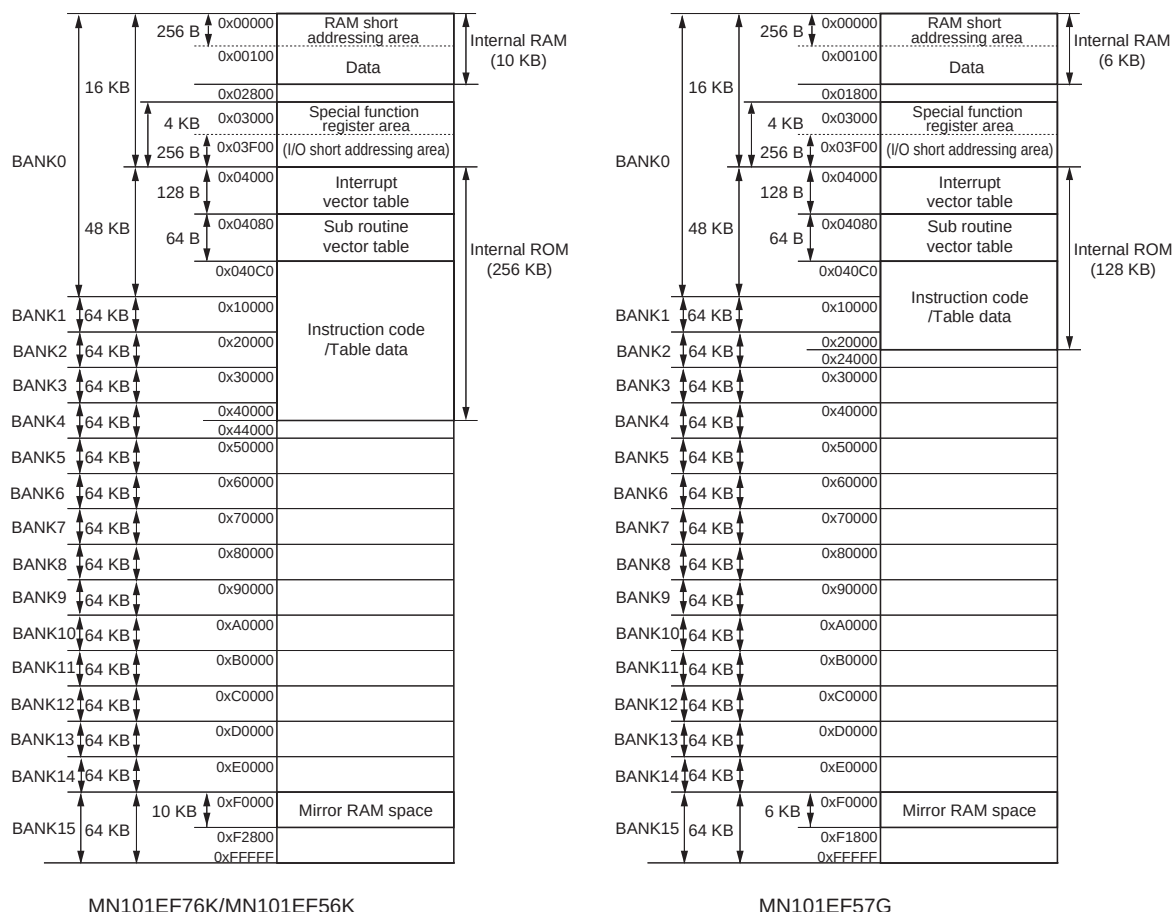


Figure:2.2.3 Single-chip Mode



The value of internal RAM is uncertain at power-on. It needs to be initialized before using.



There's no guarantee of proper operation when an access is executed to the non-implemented space where a memory (ROM/RAM), a special function register, or others are not arranged.

2.2.4 Bank Function

CPU of MN101E series has basically 64 KB data area. On this LSI, data area can be expanded up to 16 banks (1 MB) based on units of 64KB with bank function.

Bank function can be used by setting the proper bank area to the bank register for source address (SBNKR) or the bank register for destination address (DBNKR). At reset, both of the SBNKR register and the DBNKR register indicate bank 0. Bank function is valid after setting bank disable flag (BKD), which is on the seventh bit of the processor status word (PSW), to 0.

Table:2.2.2 Address Range

SBA3 (DBA3)	SBA2 (DBA2)	SBA1 (DBA1)	SBA0 (DBA0)	Bank area	Address Range
0	0	0	0	bank 0	0x00000 to 0x0FFFF
0	0	0	1	bank 1	0x10000 to 0x1FFFF
0	0	1	0	bank 2	0x20000 to 0x2FFFF
0	0	1	1	bank 3	0x30000 to 0x3FFFF
0	1	0	0	bank 4	0x40000 to 0x4FFFF
0	1	0	1	bank 5	0x50000 to 0x5FFFF
0	1	1	0	bank 6	0x60000 to 0x6FFFF
0	1	1	1	bank 7	0x70000 to 0x7FFFF
1	0	0	0	bank 8	0x80000 to 0x8FFFF
1	0	0	1	bank 9	0x90000 to 0x9FFFF
1	0	1	0	bank 10	0xA0000 to 0xAFFFF
1	0	1	1	bank 11	0xB0000 to 0xBFFFF
1	1	0	0	bank 12	0xC0000 to 0xCFFFF
1	1	0	1	bank 13	0xD0000 to 0xDFFFF
1	1	1	0	bank 14	0xE0000 to 0xEFFFF
1	1	1	1	bank 15	0xF0000 to 0xFFFFF



When bank area is changed at interrupt processing, pushing onto the stack or popping must be done by program, if needed.



While bank function is valid, I/O short instruction should be used for access to 0x03F00 to 0x03FFF in the special function register area (0x03000 to 0x03FFF). For access to the memory space 0x13F00 to 0x13FFF, 0x23F00 to 0x23FFF, 0x33F00 to 0x33FFF, 0x43F00 to 0x43FFF, 0x53F00 to 0x53FFF, 0x63F00 to 0x63FFF, 0x73F00 to 0x73FFF, 0x83F00 to 0x83FFF, 0x93F00 to 0x93FFF, 0xA3F00 to 0xA3FFF, 0xB3F00 to 0xB3FFF, 0xC3F00 to 0xC3FFF, 0xD3F00 to 0xD3FFF, 0xE3F00 to 0xE3FFF, 0xF3F00 to 0xF3FFF, both instructions of register indirect and register relative indirect should be used.
[Chapter 2 2.1.9 Addressing Modes]



Set the stack area to bank 0. The provided C-compiler for this series does not support the bank function.



Our linker supports the function that prevents data from straddling over bank boundaries. See "MN101E Series Cross-assembler User's Manual" for details.

■ Bank register for source address (SBNKR: 0x03F0A)

The SBNKR register is used to specify bank area for loading instruction from memory to register. Once this register is specified, bank control is valid for all addressing modes except I/O short instruction and stack relative indirect instruction.

[Chapter 2 2.1.9 Addressing Modes]

bp	7	6	5	4	3	2	1	0
Flag	-	-	-	-	SBA3	SBA2	SBA1	SBA0
At reset	-	-	-	-	0	0	0	0
Access	-	-	-	-	R/W	R/W	R/W	R/W

bp	Flag	Description
7 to 4	-	-
3 to 0	SBA3 SBA2 SBA1 SBA0	Bank for source address selection 0000: bank 0 0001: bank 1 0010: bank 2 0011: bank 3 0100: bank 4 0101: bank 5 0110: bank 6 0111: bank 7 1000: bank 8 1001: bank 9 1010: bank 10 1011: bank 11 1100: bank 12 1101: bank 13 1110: bank 14 1111: bank 15

■ Bank Register for Destination Address (DBNKR: 0x03F0B)

The DBNKR register is used to specify bank area for storing instruction from register to memory. Once this register is specified, bank control is valid for all addressing modes except I/O short instruction, stack relative indirect instruction and bit manipulation instruction.

[Chapter 2 2.1.9 Addressing Modes]

bp	7	6	5	4	3	2	1	0
Flag	-	-	-	-	DBA3	DBA2	DBA1	DBA0
At reset	-	-	-	-	0	0	0	0
Access	-	-	-	-	R/W	R/W	R/W	R/W

bp	Flag	Description
7 to 4	-	-
3 to 0	DBA3 DBA2 DBA1 DBA0	Bank selection for destination address 0000: bank 0 0001: bank 1 0010: bank 2 0011: bank 3 0100: bank 4 0101: bank 5 0110: bank 6 0111: bank 7 1000: bank 8 1001: bank 9 1010: bank A 1011: bank B 1100: bank C 1101: bank D 1110: bank E 1111: bank F



Read, modify, write instruction such as bit manipulation (BSET, BCLR, BTST) depends on the value of the SBNKR register, both of for reading and writing.

2.2.5 Special Function Registers

The MN101E series locates the special function registers (I/O spaces) at the addresses 0x03DF0 to 0x03FFF in memory space. The special function registers of this LSI are located as shown below.
Address 0x03C00 to 0x03DEF are reserved.

[illegible]

Figure:2.2.4 Register Map (MN101EF76K)

	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
03DFX	CHDRVSEL	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved
03EDX	PWMMDL	PWMMDH	PWMSEL	PWMSEL	PWMSEL	PWMSEL	PWMSEL	TCMPAH	TCMPBL	TCMPBH	TCMPL	TCMPCH	OUTMD	Reserved	DTMSET1	Reserved
03E0X	PWMSEL	PWMSEL	PWMSEL	PWMSEL	PWMSEL	PWMSEL	PWMSEL	RELCTR	RELCTR	RELCTR	RELCTR	PWMCMF1	PWMCMF2	Reserved	Reserved	Reserved
03E1X	PWMSEL	PWMSEL	PWMSEL	PWMSEL	PWMSEL	PWMSEL	PWMSEL	RELCTR	RELCTR	RELCTR	RELCTR	PWMCMF1	PWMCMF2	Reserved	Reserved	Reserved
03E2X	LC00BUF	LC01BUF	LC02BUF	LC03BUF	LC04BUF	LC05BUF	LC06BUF	LC07BUF	LC08BUF	LC09BUF	LC10BUF	LC11BUF	LC12BUF	LC13BUF	LC14BUF	LC15BUF
03E3X	LC10BUF	LC11BUF	LC12BUF	LC13BUF	LC14BUF	LC15BUF	LC16BUF	LC17BUF	LC18BUF	LC19BUF	LC20BUF	LC21BUF	LC22BUF	LC23BUF	LC24BUF	LC25BUF
03E4X	LC10MD1	LC11MD1	LC12MD1	LC13MD1	LC14MD1	LC15MD1	LC16MD1	LC17MD1	LC18MD1	LC19MD1	LC20MD1	LC21MD1	LC22MD1	LC23MD1	LC24MD1	LC25MD1
03E5X	PRKEY	CKTRMD	CKMONEN	CKMONSET	CKMONSET	CKMONSET	CKMONSET	CKMONSET	CKMONSET	CKMONSET	CKMONSET	CKMONSET	CKMONSET	CKMONSET	CKMONSET	CKMONSET
03E6X	ATCNT0	ATCNT1	ATCNT2	ATCNT3	ATCNT4	ATCNT5	ATCNT6	ATCNT7	ATCNT8	ATCNT9	ATCNT10	ATCNT11	ATCNT12	ATCNT13	ATCNT14	ATCNT15
03E7X	P0OUT	Reserved	P2OUT	P3OUT	P4OUT	P5OUT	P6OUT	P7OUT	P8OUT	P9OUT	PAOUT	PBOUT	Reserved	Reserved	Reserved	Reserved
03E8X	P0IN	Reserved	P2IN	P3IN	P4IN	P5IN	P6IN	P7IN	P8IN	P9IN	PAIN	PBIN	Reserved	Reserved	Reserved	Reserved
03E9X	P0DIR	Reserved	P2DIR	P3DIR	P4DIR	P5DIR	P6DIR	P7DIR	P8DIR	P9DIR	PADIR	PBDIR	Reserved	Reserved	Reserved	Reserved
03EAX	P0PLUD	Reserved	P2PLU	P3PLU	P4PLUD	P5PLUD	P6PLU	P7PLUD	P8PLU	P9PLU	PAPLU	PBPLU	Reserved	Reserved	Reserved	Reserved
03EBX	P0MD1	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved
03ECX	P0MD2	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved
03EDX	P0LED	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved
03EEF	P0DC	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved
03EFX	CPUM	MEMCTR	WDCTR	DLYCTR	P4ODC	P5ODC	P6ODC	P7ODC	P8ODC	P9ODC	PADODC	PBDODC	Reserved	Reserved	Reserved	Reserved
03F1X	SC0SEL	SC0MD0	SC0MD1	SC0MD2	SC0MD3	SC0MD4	SC0MD5	SC0MD6	SC0MD7	SC0MD8	SC0MD9	SC0MD10	SC0MD11	SC0MD12	SC0MD13	SC0MD14
03F2X	SC1SEL	SC1MD0	SC1MD1	SC1MD2	SC1MD3	SC1MD4	SC1MD5	SC1MD6	SC1MD7	SC1MD8	SC1MD9	SC1MD10	SC1MD11	SC1MD12	SC1MD13	SC1MD14
03F3X	SC2SEL	SC2MD0	SC2MD1	SC2MD2	SC2MD3	SC2MD4	SC2MD5	SC2MD6	SC2MD7	SC2MD8	SC2MD9	SC2MD10	SC2MD11	SC2MD12	SC2MD13	SC2MD14
03F4X	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved
03F5X	SC4MD0	SC4MD1	SC4MD2	SC4MD3	SC4MD4	SC4MD5	SC4MD6	SC4MD7	SC4MD8	SC4MD9	SC4MD10	SC4MD11	SC4MD12	SC4MD13	SC4MD14	SC4MD15
03F6X	TM0BC	TM1BC	TM2BC	TM3BC	TM4BC	TM5BC	TM6BC	TM7BC	TM8BC	TM9BC	TM10BC	TM11BC	TM12BC	TM13BC	TM14BC	TM15BC
03F7X	TM4BC	TM5BC	TM6BC	TM7BC	TM8BC	TM9BC	TM10BC	TM11BC	TM12BC	TM13BC	TM14BC	TM15BC	TM16BC	TM17BC	TM18BC	TM19BC
03F8X	TM7BCL	TM7BCH	TM7BCL	TM7BCH	TM7BCL	TM7BCH	TM7BCL	TM7BCH	TM7BCL	TM7BCH	TM7BCL	TM7BCH	TM7BCL	TM7BCH	TM7BCL	TM7BCH
03F9X	TM8BCL	TM8BCH	TM8BCL	TM8BCH	TM8BCL	TM8BCH	TM8BCL	TM8BCL	TM8BCH	TM8BCL	TM8BCL	TM8BCH	TM8BCL	TM8BCH	TM8BCL	TM8BCH
03FAX	TM7ICL1	TM7ICL1	TM7ICL1	TM7ICL1	TM7ICL1	TM7ICL1	TM7ICL1	TM7ICL1	TM7ICL1	TM7ICL1	TM7ICL1	TM7ICL1	TM7ICL1	TM7ICL1	TM7ICL1	TM7ICL1
03FBX	TMCKSEL1	TMCKSEL2	TMCKSEL3	TMCKSEL4	TMCKSEL5	TMCKSEL6	TMCKSEL7	TMCKSEL8	TMCKSEL9	TMCKSEL10	TMCKSEL11	TMCKSEL12	TMCKSEL13	TMCKSEL14	TMCKSEL15	TMCKSEL16
03FCX	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved
03FDX	IRQCNT	NF0CTR	NF1CTR	NF2CTR	NF3CTR	NF4CTR	NF5CTR	NF6CTR	NF7CTR	NF8CTR	NF9CTR	NF10CTR	NF11CTR	NF12CTR	NF13CTR	NF14CTR
03FEX	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved
03FFX	TBICR	TM7ICR	TM8ICR	TM9ICR	TM10ICR	TM11ICR	TM12ICR	TM13ICR	TM14ICR	TM15ICR	TM16ICR	TM17ICR	TM18ICR	TM19ICR	TM20ICR	TM21ICR

Figure:2.2.5 Register Map (MN101EF57G)

	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
03DFX	CHDRVSEL	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved
03EDX	PWMMDL	PWMMDH	PWMSEL	PWMSEL	PWMSEL	PWMSEL	PWMSEL	TCMPAH	TCMPBL	TCMPBH	TCMPCB	TCMPCB	OUTMD	DTMSET	DTMSET1	
03E1X	PWMSEL	PWMSEL	PWMSEL	PWMSEL	PWMSEL	PWMSEL	PWMSEL	RELCTR	RELCTR	RELCTR	PWMCMF1	PWMCMF2	LODBUF	LODBUF	LODBUF	
03E2X	LC00BUF	LC01BUF	LC02BUF	LC03BUF	LC04BUF	LC05BUF	LC06BUF	LC07BUF	LC08BUF	LC09BUF	LC10BUF	LC11BUF	LC12BUF	LC13BUF	LC14BUF	
03E3X	LC10BUF	LC11BUF	LC12BUF	LC13BUF	LC14BUF	LC15BUF	LC16BUF	LC17BUF	LC18BUF	LC19BUF	LC20BUF	LC21BUF	LC22BUF	LC23BUF	LC24BUF	
03E4X	LC0MD1	LC0MD2	LC0MD3	LC0MD4	LC0MD5	LC0MD6	LC0MD7	LC0MD8	LC0MD9	LC0MD10	LC0MD11	LC0MD12	LC0MD13	LC0MD14	LC0MD15	
03E5X	PRKEY	CKTRMD	CKMONEN	CKMONSET	CKMONSET1	CKMONSET2	CKMONSET3	CKMONSET4	CKMONSET5	CKMONSET6	CKMONSET7	CKMONSET8	CKMONSET9	CKMONSET10	CKMONSET11	
03E6X	ATLCNT0	ATLCNT1	ATLCNT2	ATLCNT3	ATLCNT4	ATLCNT5	ATLCNT6	ATLCNT7	ATLCNT8	ATLCNT9	ATLCNT10	ATLCNT11	ATLCNT12	ATLCNT13	ATLCNT14	
03E7X	P0OUT	P1OUT	P2OUT	P3OUT	P4OUT	P5OUT	P6OUT	P7OUT	P8OUT	P9OUT	PAOUT	PBOUT	Reserved	Reserved	Reserved	
03E8X	P0IN	P1IN	P2IN	P3IN	P4IN	P5IN	P6IN	P7IN	P8IN	P9IN	PAIN	PBIN	Reserved	Reserved	Reserved	
03E9X	P0DIR	P1DIR	P2DIR	P3DIR	P4DIR	P5DIR	P6DIR	P7DIR	P8DIR	P9DIR	PADIR	PBDIR	Reserved	Reserved	Reserved	
03EAX	P0PLUD	P1PLUD	P2PLUD	P3PLUD	P4PLUD	P5PLUD	P6PLUD	P7PLUD	P8PLUD	P9PLUD	PAPLUD	PBPLUD	Reserved	Reserved	Reserved	
03EAX	P0MD1	P1MD1	P2MD1	P3MD1	P4MD1	P5MD1	P6MD1	P7MD1	P8MD1	P9MD1	PA0MD1	PB0MD1	Reserved	Reserved	Reserved	
03ECX	P0MD2	P1MD2	P2MD2	P3MD2	P4MD2	P5MD2	P6MD2	P7MD2	P8MD2	P9MD2	PA0MD2	PB0MD2	Reserved	Reserved	Reserved	
03EDX	P0MD	P1MD	P2MD	P3MD	P4MD	P5MD	P6MD	P7MD	P8MD	P9MD	PA0MD	PB0MD	Reserved	Reserved	Reserved	
03EEF	P0ED	P1ED	P2ED	P3ED	P4ED	P5ED	P6ED	P7ED	P8ED	P9ED	PA0ED	PB0ED	Reserved	Reserved	Reserved	
03EFX	P0DC	P1DC	P2DC	P3DC	P4DC	P5DC	P6DC	P7DC	P8DC	P9DC	PA0DC	PB0DC	Reserved	Reserved	Reserved	
03F0X	CPUM	MEMCTR	WDCTR	DLYCTR	WD2CTR	HANDSHAKE	WD2CTR	AUCTR	WD2CTR	WD2CTR	WD2CTR	WD2CTR	WD2CTR	WD2CTR	WD2CTR	
03F1X	SC0SEL	SC0MD0	SC0MD1	SC0MD2	SC0MD3	SC0MD4	SC0MD5	SC0MD6	SC0MD7	SC0MD8	SC0MD9	SC0MD10	SC0MD11	SC0MD12	SC0MD13	
03F2X	SC1SEL	SC1MD0	SC1MD1	SC1MD2	SC1MD3	SC1MD4	SC1MD5	SC1MD6	SC1MD7	SC1MD8	SC1MD9	SC1MD10	SC1MD11	SC1MD12	SC1MD13	
03F3X	SC2SEL	SC2MD0	SC2MD1	SC2MD2	SC2MD3	SC2MD4	SC2MD5	SC2MD6	SC2MD7	SC2MD8	SC2MD9	SC2MD10	SC2MD11	SC2MD12	SC2MD13	
03F4X	SC3SEL	SC3MD0	SC3MD1	SC3MD2	SC3MD3	SC3MD4	SC3MD5	SC3MD6	SC3MD7	SC3MD8	SC3MD9	SC3MD10	SC3MD11	SC3MD12	SC3MD13	
03F5X	SC4MD0	SC4MD1	SC4MD2	SC4MD3	SC4MD4	SC4MD5	SC4MD6	SC4MD7	SC4MD8	SC4MD9	SC4MD10	SC4MD11	SC4MD12	SC4MD13	SC4MD14	
03F6X	TM0BC	TM1BC	TM2BC	TM3BC	TM4BC	TM5BC	TM6BC	TM7BC	TM8BC	TM9BC	TM10BC	TM11BC	TM12BC	TM13BC	TM14BC	
03F7X	TM4BC	TM5BC	TM6BC	TM7BC	TM8BC	TM9BC	TM10BC	TM11BC	TM12BC	TM13BC	TM14BC	TM15BC	TM16BC	TM17BC	TM18BC	
03F8X	TM7BCL	TM7BCH	TM7CCL	TM7CHH	TM7CCL	TM7CHH	TM7CCL	TM7CHH	TM7CCL	TM7CHH	TM7CCL	TM7CHH	TM7CCL	TM7CHH	TM7CCL	
03F9X	TM8BCL	TM8BCH	TM8CCL	TM8CHH	TM8CCL	TM8CHH	TM8CCL	TM8CHH	TM8CCL	TM8CHH	TM8CCL	TM8CHH	TM8CCL	TM8CHH	TM8CCL	
03FAX	TM7ICL1	TM7ICH1	TM7ICL2	TM7ICH2	TM7ICL3	TM7ICH3	TM7ICL4	TM7ICH4	TM7ICL5	TM7ICH5	TM7ICL6	TM7ICH6	TM7ICL7	TM7ICH7	TM7ICL8	
03FBX	TMCKSEL1	TMCKSEL2	TMCKSEL3	TMCKSEL4	TMCKSEL5	TMCKSEL6	TMCKSEL7	TMCKSEL8	TMCKSEL9	TMCKSEL10	TMCKSEL11	TMCKSEL12	TMCKSEL13	TMCKSEL14	TMCKSEL15	
03FCX	RCAPL	RCAPM	RCAPN	RCAPH	RCAPH	RCAPH	RCAPH	RCAPH	RCAPH	RCAPH	RCAPH	RCAPH	RCAPH	RCAPH	RCAPH	
03FDX	IRQONT	NF1CTR	NF2CTR	NF3CTR	NF4CTR	NF5CTR	NF6CTR	NF7CTR	NF8CTR	NF9CTR	NF10CTR	NF11CTR	NF12CTR	NF13CTR	NF14CTR	
03FEX	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	
03FFX	TBICR	TM7ICR	TM8ICR	TM9ICR	TM10ICR	TM11ICR	TM12ICR	TM13ICR	TM14ICR	TM15ICR	TM16ICR	TM17ICR	TM18ICR	TM19ICR	TM20ICR	

Figure:2.2.6 Register Map (MN101EF56K)



Do not access (read/write) to the "Reserved" address. In that case, proper operation is not guaranteed.



Each register has a different setting according to bp. Refer to the detailed description for its usage.

2.2.6 Flash Option

This LSI allocates 0x040C0 and 0x040C1 of memory area to an flash option area.

When internal reset release is completed after oscillation stabilization wait time, the flash option area is referred and the setting value of this area is automatically read out by hardware to set functions.

When using this LSI, please always configure the flash option settings.

Flash option in this LSI is allocated as below.

■ Flash Option 0 (FLOP0: 0x040C0)

bp	7	6	5	4	3	2	1	0
Flag	P4_VIL	P3_VIL	P2_VIL	P1_VIL	P0_VIL	WD2OP2	WD2OP1	WD2OP0

bp	Flag	Description
7	P4_VIL	Select Port 4 input voltage level 1: Input voltage level A 0: Input voltage level B
6	P3_VIL	Select Port 3 input voltage level 1: Input voltage level A 0: Input voltage level B
5	P2_VIL	Select Port 2 input voltage level 1: Input voltage level A 0: Input voltage level B
4	P1_VIL	Select Port 1 input voltage level 1: Input voltage level A 0: Input voltage level B
3	P0_VIL	Select Port 0 input voltage level 1: Input voltage level A 0: Input voltage level B
2 to 0	WD2OP2 WD2OP1 WD2OP0	Setting control of WDT2 operation mode setting register (WD2MDSEL) 101 : Setting enabled for WD2MDSEL register Other than 101 : Setting disabled for other than WD2MDSEL register (WDT2 is always operating.)



For details of the P4_VIL to P0_VIL flags, refer to [Chapter 8 8.19 Input Voltage Level Switching Function].



For details of the WD2OP2 to WD2OP0 flags, refer to [Chapter 16 16.2.2 Watchdog Timer 2 Control Register].



The flash option function is determined before internal reset release. Therefore, the settings will not be reflected if rewritten during operation.

■ Flash Option 1 (FLOP1: 0x040C1)

bp	7	6	5	4	3	2	1	0
Flag	-	PB_VIL	PA_VIL	P9_VIL	P8_VIL	P7_VIL	P6_VIL	P5_VIL

bp	Flag	Description
7	-	-
6	PB_VIL	Select Port B input voltage level 1: Input voltage level A 0: Input voltage level B
5	PA_VIL	Select Port A input voltage level 1: Input voltage level A 0: Input voltage level B
4	P9_VIL	Select Port 9 input voltage level 1: Input voltage level A 0: Input voltage level B
3	P8_VIL	Select Port 8 input voltage level 1: Input voltage level A 0: Input voltage level B
2	P7_VIL	Select Port 7 input voltage level 1: Input voltage level A 0: Input voltage level B
1	P6_VIL	Select Port 6 input voltage level 1: Input voltage level A 0: Input voltage level B
0	P5_VIL	Select Port 5 input voltage level 1: Input voltage level A 0: Input voltage level B



For details of the PB_VIL to P5_VIL flags, refer to [Chapter 8 8.19 Input Voltage Level Switching Function].

2.3 ROM Correction (MN101EF76K/ MN101EF56K)

2.3.1 Overview

MN101EF76K and MN101EF56K can correct and change max 7 parts in a program on ROM both internal and external with ROM correction function. The correct program is read from the external to the internal RAM space by using the external EEPROM or by using the serial transmission. This function is valid to the system with the external EEPROM.

2.3.2 Correction Sequence

Program is corrected as following steps.

1. The instruction execution address is compared to the correction address.
2. Program counter is branched indirectly to the internal RAM address (the head address of the correct program) stored to the RC vector table (RCnV(L), RCnV(H)), after matching the above address. It is necessary to set the head address of the correct program as the address by a byte. This process needs 6 cycle.
3. The corrected program at the RAM area is executed.
4. Program counter is branched back to the program at ROM area.

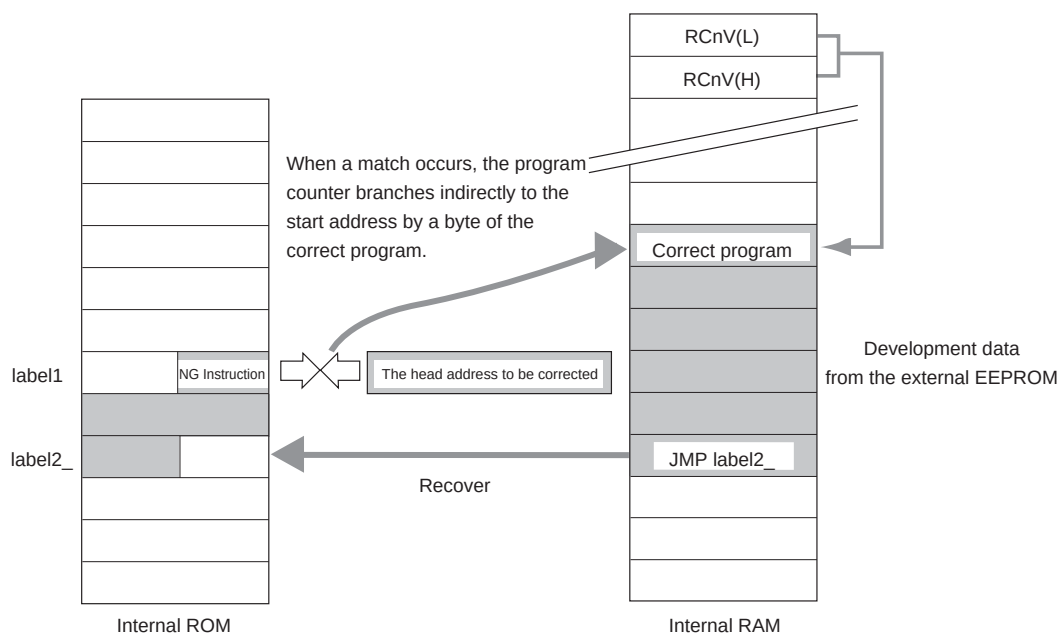


Figure:2.3.1 ROM Correction

The ROM correction setup procedure is as follows.

- Specify the ROM correction pointer specification register (RCPSR).
- Set the head address of the program to be corrected to the ROM correction address setting register (RCAP(H/M/L)).
- Set the correct program at RAM area.
- Set the head address of the correct program to RC vector table (RCnV(L), RCnV(H)).
- Set the RCnEN flag of ROM correction control register (RCCTR0) to enable the ROM correction.



When the head address of the program to be corrected is the half-byte instruction, the ROM correction checks the execution instruction of the half-byte. Therefore, set the address by a byte to the ROM correction address setting register.



When the next instruction of the program to be corrected program last address is the instruction placed in half-byte address, the recover address should be set by half byte.

2.3.3 ROM Correction Control Register

ROM correction is controlled by ROM correction pointer specification register (RCPSR), ROM correction control register (RCCTR0) and ROM correction address setting register (RCnAPL, RCnAPM, RCnAPH).

ROM correction pointer specification register (RCPSR) is the register which specifies the pointer of the ROM correction address register (RCAP (L/M/H)). ROM correction address setting register becomes effective by setting the value of the pointer n (RCnAP(L/M/H)) which is specified by RCPSR.

When setting a ROM correction address setting register, set up the ROM correction pointer specification register (RCPSR) first.

■ ROM Correction Pointer Specification Register (RCPSR: 0x03FC3)

bp	7	6	5	4	3	2	1	0
Flag	-	-	-	-	Reserved	RCPSR2	RCPSR1	RCPSR0
At reset	-	-	-	-	0	0	0	0
Access	-	-	-	-	R/W	R/W	R/W	R/W

bp	Flag	Description
7 to 4	-	-
3	Reserved	Always set to "0".
2 to 0	RCPSR2 RCPSR1 RCPSR0	Specification of ROM correction pointer 000: Address pointer 0 (RC0AP(L/M/H)) 001: Address pointer 1 (RC1AP(L/M/H)) 010: Address pointer 2 (RC2AP(L/M/H)) 011: Address pointer 3 (RC3AP(L/M/H)) 100: Address pointer 4 (RC4AP(L/M/H)) 101: Address pointer 5 (RC5AP(L/M/H)) 110: Address pointer 6 (RC6AP(L/M/H)) 111: Setting prohibited

ROM correction control register (RCCTR0) enables/disables the ROM correction function to max 7 parts of the program to be corrected. When the RCnEN flag is set, the ROM correction is activated. And when the ROM address (the instruction execution address) reaches the set address to the ROM correction address setting register, it branches indirectly to the RAM address set on the RC vector table (RCnV(L), RCnV(H)). Set the RCnEN flag after setting the ROM correction address setting register.

■ ROM Correction Control Register 0 (RCCTR0: 0x03FC4)

bp	7	6	5	4	3	2	1	0
Flag	Reserved	RC6EN	RC5EN	RC4EN	RC3EN	RC2EN	RC1EN	RC0EN
At reset	0	0	0	0	0	0	0	0
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description
7	Reserved	Always set to "0".
6	RC6EN	Control of ROM correction of 6th address (RC6AP(L/M/H)) 0: Disable 1: Enable
5	RC5EN	Control of ROM correction of 5th address (RC5AP(L/M/H)) 0: Disable 1: Enable
4	RC4EN	Control of ROM correction of 4th address (RC4AP(L/M/H)) 0: Disable 1: Enable
3	RC3EN	Control of ROM correction of 3rd address (RC3AP(L/M/H)) 0: Disable 1: Enable
2	RC2EN	Control of ROM correction of 2nd address (RC2AP(L/M/H)) 0: Disable 1: Enable
1	RC1EN	Control of ROM correction of 1st address (RC1AP(L/M/H)) 0: Disable 1: Enable
0	RC0EN	Control of ROM correction of 0th address (RC0AP(L/M/H)) 0: Disable 1: Enable



Set the ROM correction address to the 4th or later instructions from the instruction which enables the ROM correction control.

■ ROM correction address setting register (RCAPL, RCAPM, RCAPH)

The address in which the command to be corrected is stored is set up. The value to set up is written in the register specified by the ROM correction pointer specification register (RCPSR). Register of the pointer specified by RCPSR if the execution address of the set-up value and a command is in agreement, indirect branch will be carried out at the address set as RC vector table (RCnV (L), RCnV (H)). In enabling a ROM correction function, please set the RCnEN flag of RCCTR0 register as this register after setting up a desired address.

■ ROM Correction Address Setting Register for Lower 8bits (RCAPL: 0x03FC0)

bp	7	6	5	4	3	2	1	0
Flag	RCAPL7	RCAPL6	RCAPL5	RCAPL4	RCAPL3	RCAPL2	RCAPL1	RCAPL0
At reset	0	0	0	0	0	0	0	0
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

■ ROM Correction Address Setting Register for Middle 8 bits (RCAPM: 0x03FC1)

bp	7	6	5	4	3	2	1	0
Flag	RCAPM7	RCAPM6	RCAPM5	RCAPM4	RCAPM3	RCAPM2	RCAPM1	RCAPM0
At reset	0	0	0	0	0	0	0	0
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

■ ROM Correction Address Setting Register for Upper 4 bits (RCAPH: 0x03FC2)

bp	7	6	5	4	3	2	1	0
Flag	-	-	-	-	RCAPH3	RCAPH2	RCAPH1	RCAPH0
At reset	-	-	-	-	0	0	0	0
Access	-	-	-	-	R/W	R/W	R/W	R/W



Please do not perform the same address setup to two or more RCnAP (H/M/L) registers. When the same address is set up, the priority shown below is followed.
RC0AP>RC1AP>RC2AP>.....>RC6AP.



Set the ROM correction address to the 4th or later instructions from the instruction which enables the ROM correction control.

Here is the correspondence of ROM correction pointer specification register, the ROM correction address setting register, the ROM correction control register of ROM correction control flag and RC-vector table.

	ROM correction pointer specification register	ROM correction address			ROM correction control flag	RC-vector table	
	RCPSR	RCAPL	RCAPM	RCAPH	RCCTR0	RCnV(L)	RCnV(H)
0th correction address	0000	RC0APL	RC0APM	RC0APH	RC0EN	0x0010	0x0011
1st correction address	0001	RC1APL	RC1APM	RC1APH	RC1EN	0x0012	0x0013
2nd correction address	0010	RC2APL	RC2APM	RC2APH	RC2EN	0x0014	0x0015
3rd correction address	0011	RC3APL	RC3APM	RC3APH	RC3EN	0x0016	0x0017
4th correction address	0100	RC4APL	RC4APM	RC4APH	RC4EN	0x0018	0x0019
5th correction address	0101	RC5APL	RC5APM	RC5APH	RC5EN	0x001A	0x001B
6th correction address	0110	RC6APL	RC6APM	RC6APH	RC6EN	0x001C	0x001D

2.3.4 ROM Correction Setup Example

■ Initial Routine with ROM Correction

The following routine should be set to correct the program. Also store the ROM correction setup and correct program to the external EEPROM, in advance.

Here is the steps for ROM correction execution.

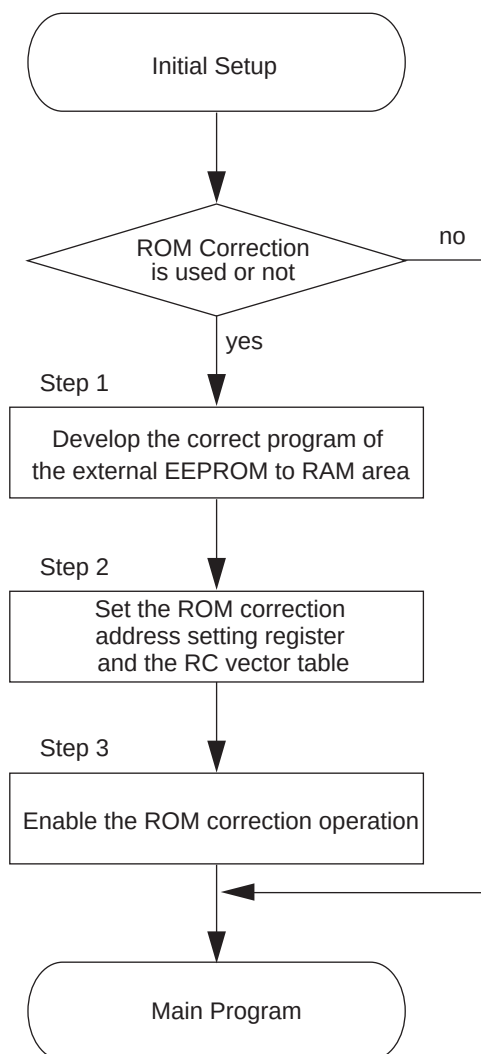


Figure:2.3.2 Initial Routine for ROM Correction

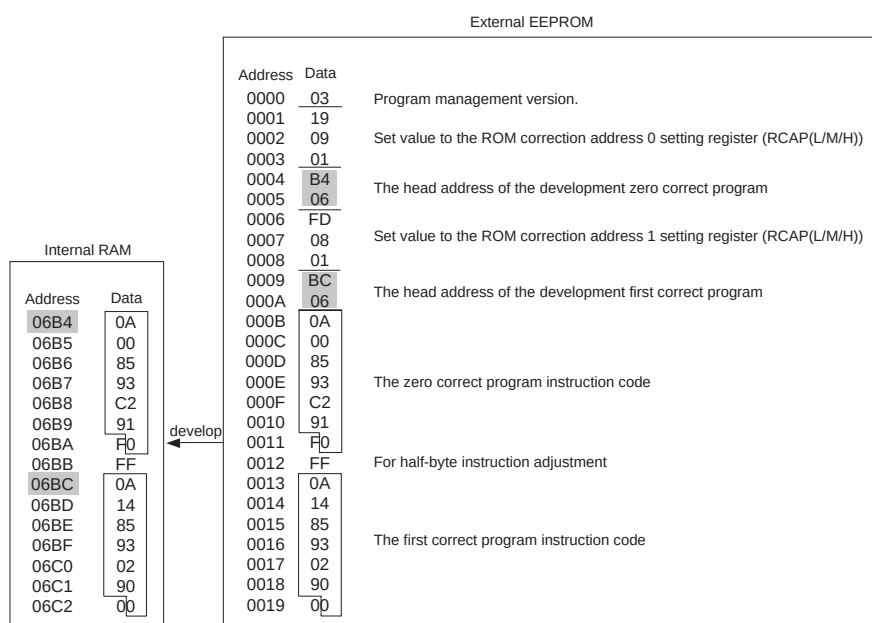
■ ROM Correction Setup Example

The setup procedure with ROM correction to correct two parts of program is shown below. For the step to execute the ROM correction, refer to Figure:2.3.2 Initial Routine for ROM correction on the previous page.

(STEP 1)

Develop the correct program of the external EEPROM to internal RAM area.

It is necessary to set the head address of the correct program as the address by a byte.



(STEP 2)

Set the ROM correction address setting register and the RC-vector table.

[Setup for the 0th correction]

Specify the ROM correction pointer specification register (RCPSR).

RCPSR = 0x00

Set the head address of the program to be corrected to the ROM correction address setting register (RCAP)

RCAPL = 0x19

RCAPM = 0x09

RCAPH = 0x01

(When the address is RCPSR=0x00, the value is set as the ROM correction address pointer 0 (RC0AP (L/M/H)).

Set the internal RAM address 0x06B4 that stored zeroth correct program to the RC-vector table address (RC0V(L), RC0V(H)).

RC0V(L) = 0xB4

RC0V(H) = 0x06

The zero program to be corrected (internal ROM)

The head address of the correction (the set value of RC0AP)	
Address	Data
10916	D900A0 cbne 0, d1, 1091E
10919	A005 mov 50, d0
1091B	58 mov d0, (a0)
1091C	8940 bra 10920
1091E	B4 sub d0, d0

The address for recover

The zero correct program (internal RAM)

The head address of the correction program (the set value of RC0V)	
Address	Data
006B4	A000 mov 0, d0
006B6	58 mov d0, (a0)
006B7	392C190 bra 1091C

The address for recover

[Setup for the 1st correction]

Specify the ROM correction pointer specification register (RCPSR).

RCPSR = 0x01

Set the head address of the program to be corrected to the ROM correction address setting register (RCAP)

RC1APL = 0xFD

RC1APM = 0x08

RC1APH = 0x01

Set the internal RAM address 0x06BC that stored first correct program to the RC-vector table address (RC1V(L), RC1V(H)).

RC1V(L) = 0xBC

RC1V(H) = 0x06

The first program to be corrected (internal ROM)

Address	Data		
108FC	85	sub	d1, d1
108FD	A011	mov	11, d0
108FF	58	mov	d0, (a0)
10900	EC1	addw	1, a0
10901	A081	mov	_Msyscom_edge, 0

The head address of the correction (the set value of RC1AP)

The address for recover

The first correct program (internal RAM)

Address	Data		
006BC	A041	mov	14, d0
006BE	58	mov	d0, (a0)
006BF	3920090	jmp	10900

The head address of the correction program (the set value of RC1V)

The address for recover

(STEP 3)

Please set bit 0 (RC0EN), bit 1 (RC1EN) of ROM correction control register (RCCTR0) to "1."

For the 4th or later instructions from the instruction which enables the ROM correction control, the instruction fetched address and the set address to ROM correction address setting register (RCnAP) are always compared, then once they are matched program counter indirectly branches to the address in RAM area, that are stored to the RC vector table (RCnV).

Execute the correction program in RAM area.

Program counter recovers to the program in ROM area.

2.4 Bus Interface

2.4.1 Bus Controller

The MN101E series provides separate buses to the internal memory and internal peripheral circuits to reduce bus line loads and thus realize faster operation.

There are three such buses: ROM bus, RAM bus, and peripheral expansion bus. They connect to the internal ROM, internal RAM, and internal peripheral circuits respectively. The bus control block controls the parallel operation of instruction read and data access. A functional block diagram of the bus controller is given below.

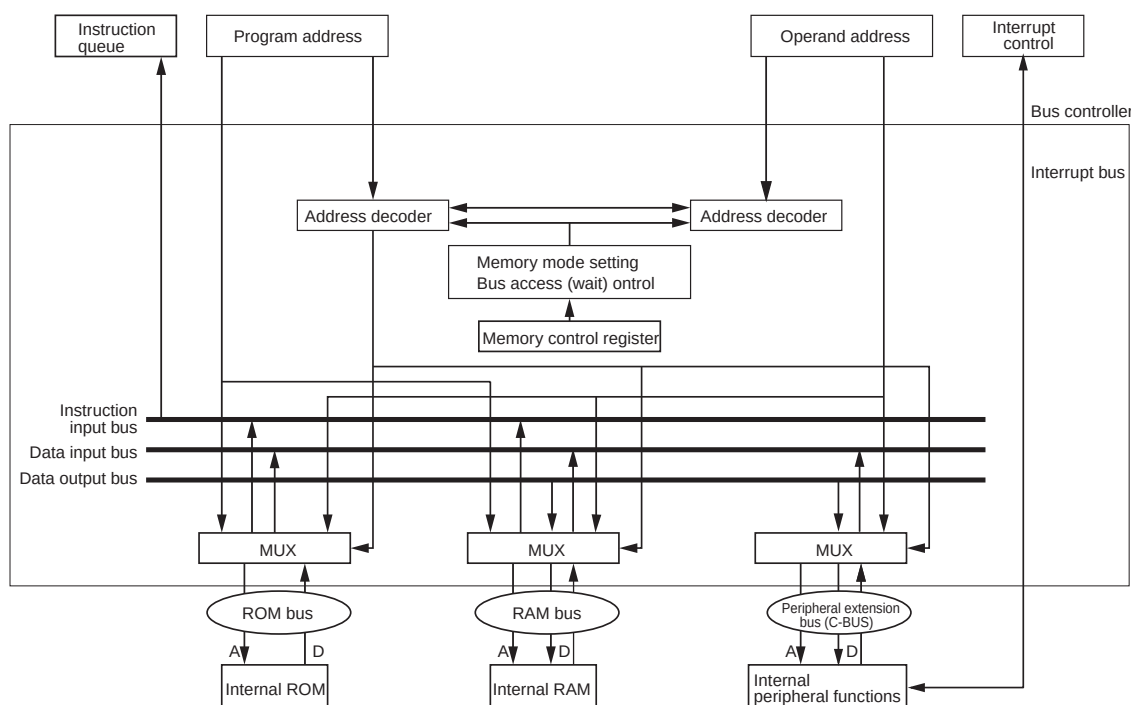


Figure:2.4.1 Functional Block Diagram of the Bus Controller

Memory control register (MEMCTR) can be used to set wait cycle to peripheral expansion bus (C-BUS) connected to internal peripheral circuits.

2.4.2 Control Registers

Bus interface is controlled by the memory control register (MEMCTR).

■ Memory Control Register (MEMCTR: 0x03F01)

bp	7	6	5	4	3	2	1	0
Flag	IOW1	IOW0	IVBM	Reserved	Reserved	IRWE	Reserved	Reserved
At reset	1	1	0	0	1	0	1	1
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description
7 to 6	IOW1 IOW0	Wait cycles when accessing special register area 00: No wait cycle 01: 1 wait cycle 10: 2 wait cycles 11: 3 wait cycles
5	IVBM	Base address specification for interrupt vector table 0: Interrupt vector base = 0x04000 1: Interrupt vector base = 0x00100
4	Reserved	Always fix to "0"
3	Reserved	Always fix to "1"
2	IRWE	Software write setting for interrupt request flag 0: Even if data is written to each interrupt control register (xxxICR), the state of the interrupt request flag (xxxIR) will not change. 1: Software write enable
1 to 0	Reserved	Always fix to "1"



bp0, 1, 3 and 4 of the MEMCTR register are reserved. If accessing these flags, make sure to write the specified value.



The IOW1 to IOW0 wait settings affect accesses to the special registers located at the addresses 0x03000 to 0x03FFF. After reset, MEMCTR specifies the fixed wait cycle mode with three wait cycles. Wait setting of IOW is a function, which CPU supports for special use, for example, when special function register or I/O is expanded to external. For this LSI, wait cycle setting is not always necessary. Select "no-wait cycle" in this LSI.



Always set IRWE flag of memory control register (MEMCTR) to “0” except in writing IR with software.

If operating interrupt control register xxxICR with software in setting IRWE flag to “1”, the interrupt request flag which is set to “1” by interrupt source may be cleared to “0”.

For example, if giving the bit operation order to interrupt control register xxxICR, it executes bit operation to read-out 1 byte and writes it back.

When the interrupt source occurs between reading out and writing back, IR flag may be cleared to “0” by mistake and then the interrupt source is missing. If IRWE flag is set to “0”, the interrupt source will not be missing.

2.5 Extended Calculation Instruction

2.5.1 Overview

This LSI contains the functions of 16-bit x 16-bit and 32-bit/16-bit calculation in addition to the existing calculation functions which can be executed by MN101C series. The executable calculation and execution cycles are as follows:

Table:2.5.1 List of Extended Calculation Functions

Calculation	Instruction	Operation	Execution cycle	PSW Flag variation			
				VF	NF	CF	ZF
16-bit x 16-bit multiplication (unsigned)	MOV 1,(0x3F07) Extended calculation macro instruction MULWU	DW0 * DW1→{DW1, DW0}	13 cycles	0	√	0	√
16-bit x 16-bit multiplication (signed)	MOV 2,(0x3F07) Extended calculation macro instruction MULW	DW0 * DW1→{DW1, DW0}	13 cycles	0	√	0	√
32-bit/16-bit division (unsigned)	MOV 4,(0x3F07) Extended calculation macro instruction DIVWU	{DW1, DW0}/A0→DW0...DW1	14 cycles	√	√	0	√

√: flag varies.

VF: The value is 0 when the division result can be expressed with the unsigned 16-bit, and otherwise is 1. When zero divide is executed, the value is 1. When the value is 1, the calculation ends in 6 executing cycles.

The value is always 0 in case of multiplication.

NF: The value is 1 when the MSB of multiplication/division result is 1. It is 0 when the MSB of the result is 0.

However, when VF is 1, the value is undefined.

CF: Always 0.

ZF: The value is 1 when the multiplication/division result is 0, and otherwise is 0.

However, when VF is 1, the value is undefined.

For details, refer to [Chapter 2 2.6 Extended Calculation Macro Instruction].

2.5.2 Extended Calculation Control Register

Extended calculation can be executed by setting the extended calculation control flag after setting the multiplier to DW0, the multiplicand to DW1, the divisor to A0, and the dividend to DW1 and DW0 respectively.

■ Extended Calculation Control Register (AUCTR: 0x03F07)

bp	7	6	5	4	3	2	1	0
Flag	-	-	-	-	-	AUDIVU	AUMUL	AUMULU
At reset	-	-	-	-	-	0	0	0
Access	-	-	-	-	-	W	W	W

bp	Flag	Description
7 to 3	-	-
2	AUDIVU	Unsigned division execution 0: Disabled 1: Enabled
1	AUMUL	Signed multiplication execution 0: Disabled 1: Enabled
0	AUMULU	Unsigned multiplication execution 0: Disabled 1: Enabled

When calculation is finished, each flag is cleared to "0" by hardware.



Do not set a number of bits at the same time.

2.5.3 Execution of Extended Calculation

■ Execution of 16-bit x 16-bit multiplication (unsigned)

1. Store the multiplier to DW0 register and the multiplicand to DW1 register.
2. Execute MOV 1, (0x03F07) (Extended calculation macro instruction MULWU).
3. The value of the unsigned 16-bit of DW0 register is multiplied by the unsigned 16-bit of DW1 register. Then the upper 16-bit of the result (32-bit) is stored in DW1 register and the lower 16-bit is stored in DW0 register.

■ Execution of 16-bit x 16-bit multiplication (signed)

1. Store the multiplier to DW0 register and the multiplicand to DW1 register.
2. Execute MOV 2, (0x03F07) (Extended calculation macro instruction MULW).
3. The value of the signed 16-bit of DW0 register is multiplied by the signed 16-bit of DW1 register. Then the upper 16-bit of the results (32-bit) is stored in DW1 register and the lower 16-bit register is stored in DW0 register.

■ Execution of 32-bit / 16-bit division (unsigned)

1. Store the upper 16-bit of the dividend to DW1 register, the lower 16-bit of the dividend to DW0 register, and the divisor to A0 register.
2. Execute MOV 4, (0x03F07) (Extended calculation macro instruction DIVWU).
3. The value of the unsigned 32-bit which is stored in the DW1 register (upper 16-bit) and DW0 register (lower 16-bit) is divided by the value of the unsigned 16-bit of A0 register. Then the quotient 16-bit of the result is stored in DW0 register and the residue 16-bit of the result is stored in DW1 register.



VF is 1 when the division result cannot be expressed with the unsigned 16-bit or zero divide is executed.



When VF is 1, the calculation ends in 6 execution cycles. In this case, the result is undefined.

2.6 Extended Calculation Macro Instruction

Extended calculation macro instruction can be generated by specifying machine dependence option -mmuldivw by compiler of MN101C/MN101E series.

2.6.1 About Extended Calculation Macro Instruction

■ About this Table

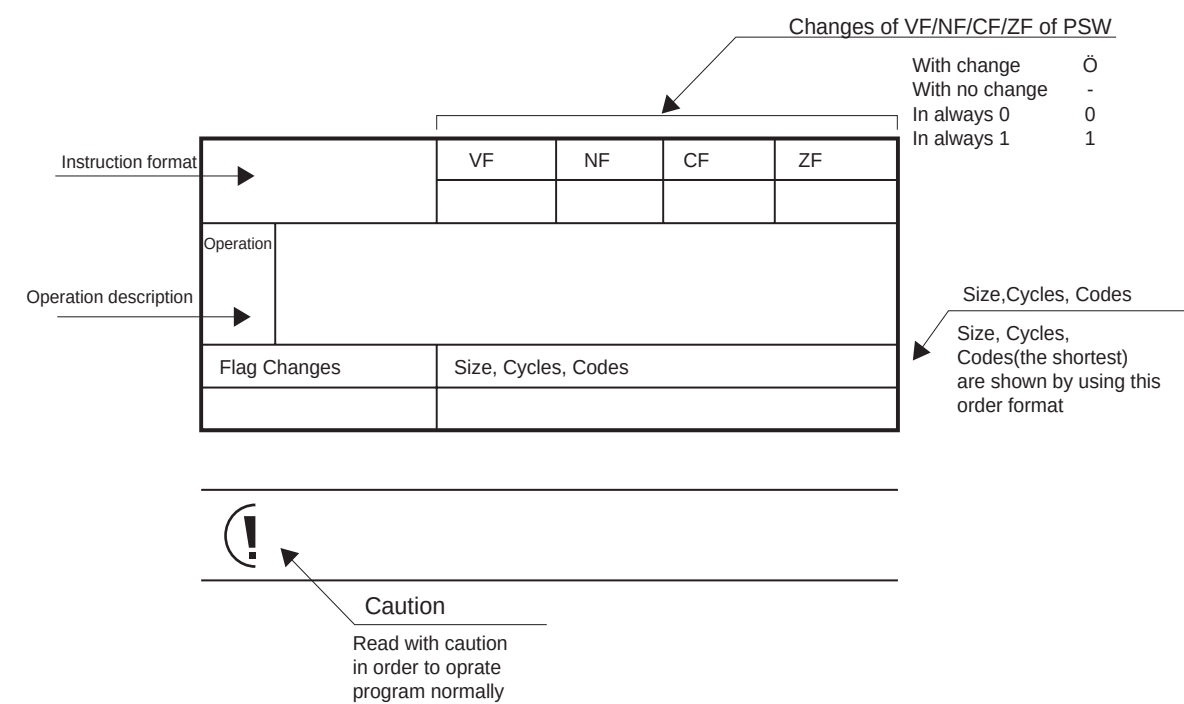


Figure:2.6.1 About this Table

■ Sign

- * Multiplication
- / Division
- Substitution
- ... Residue
- {DW1,DW0} 32 bit data (high 16 bits in DW1 resister and low 16 bits in DW0 resister are stored)

2.6.2 MULWU

MULWU		VF	NF	CF	ZF
		0	√	0	√
Operation	DW0 *DW1 → {DW1, DW0} Multiplies the unsigned 16-bit value of DW0 register by the unsigned 16-bit value or DW1 register, and store the upper 16-bit of the result (32-bit) in the DW1 register and the lower 16-bit of the result in the DW0 register.				
Flag Changes		Size, Cycles, Codes			
VF: 0 NF: Set if the MSB of the result is 1, reset otherwise. CF: 0 ZF: Set if the result is 0, reset otherwise.		6 nibbles 13 cycles 0000 0010 0111 0000 0001 0000			



Store the multiplier to DW0 register and the multiplicand to DW1 register before executing this instruction.



This instruction is a macro instruction. The following operation is actually executed:

MOV 1, (0x03F07); Set 1 to AUCTR register (address 0x03F07)

With this instruction, the address used in the handy addressing mode is updated to 0x03F07.

The updating example of handy addresses is as follows:

```

_CODE section code, pulic, 0

    mov    (0x100), d0
    mulwu
    mov    d0, (HA)      ; HA value is updated and
                        ; 0x03F07 address is accessed

    mov    (0x100), d0
    add    1, d0
    mov    d0, (HA)      ; access to HA value 0x100 address
  
```

2.6.3 MULW

MULW		VF	NF	CF	ZF
		0	√	0	√
Operation	DW0 *DW1 → {DW1, DW0} Multiplies the unsigned 16-bit value of DW0 register by the unsigned 16-bit value or DW1 register, and store the upper 16-bit of the result (32-bit) in the DW1 register and the lower 16-bit of the result in the DW0 register.				
Flag Changes		Size, Cycles, Codes			
VF: 0 NF: Set if the MSB of the result is 1, reset otherwise. CF: 0 ZF: Set if the result is 0, reset otherwise.		6 nibbles 13 cycles 0000 0010 0111 0000 0010 0000			



Store the multiplier to DW0 register and the multiplicand to DW1 register before executing this instruction.



This instruction is a macro instruction. The following operation is actually executed:

MOV 2, (0x03F07); set 2 to AUCTR register (address 0x03F07)

With this instruction, the address used in the handy addressing mode is updated to 0x03F07.

The updating example of handy addresses is as follows:

```

_CODE section code, pulic, 0

    mov    (0x100), d0
    mulwu
    mov    d0, (HA)      ; HA value is updated and
                        ; 0x03F07 address is accessed

    mov    (0x100), d0
    add    1, d0
    mov    d0, (HA)      ; access to HA value 0x100 address
  
```

2.6.4 DIVWU

DIVWU		VF	NF	CF	ZF
		√	√	0	√
Operation	{DW1, DW0}/A0 → DW0...DW1 Divides the unsigned 32-bit value which is stored in the DW1 register (upper 16-bit) and DW0 register (lower 16-bit) by the unsigned 16-bit value of A0 register, and stores the quotient 16-bit of the result in DW0 register and the residue 16-bit of the result in DW1 register.				
Flag Changes		Size, Cycles, Codes			
If VF is 0 VF: 0 (if the quotient is an unsigned 16-bit value) NF: Set if the MSB of the quotient is 1, reset otherwise. CF: 0 ZF: Set if the MSB of the quotient is 0, reset otherwise.		If VF is 1 VF: 1 (if the quotient is not an unsigned 16-bit value) NF: Undefined CF: 0 ZF: Undefined		6 nibbles 14 cycles 0000 0010 0111 0000 0100 0000	



Set the upper 16-bit of the dividend to DW1 register, the lower 16-bit of the dividend to DW0 register, and the divisor to A0 register.



This instruction is a macro instruction. The following operation is actually executed:

MOV 4, (0x03F07); Set 4 to AUCTR register (address 0x03F07)

With this instruction, the address used in the handy addressing mode is updated to 0x03F07. The updating example of handy addresses is as follows:

```

_CODE section code, pulic, 0

    mov    (0x100), d0
    mulwu
    mov    d0, (HA)      ; HA value is updated and
                        ; 0x03F07 address is accessed

    mov    (0x100), d0
    add    1, d0
    mov    d0, (HA)      ; access to HA value 0x100 address
  
```



VF is 1 when the division result cannot be expressed with the unsigned 16-bit or zero divide is executed.



When VF is 1, the calculation ends in 6 execution cycles. In this case, the result is undefined.

2.7 Reset

2.7.1 Reset operation

The CPU contents are reset and registers are initialized when NRST pin (P27) is pulled to low.

■ Initiating a Reset

There are two methods to initiate a reset.

1. Drive NRST pin low.

NRST pin should be held "low" for more than 1 μ s.

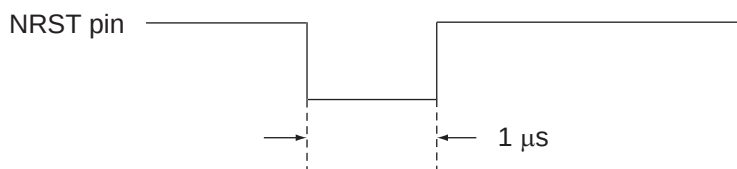


Figure:2.7.1 Minimum Reset Pulse Width

2. Setting the P2OUT7 flag of P2OUT register to "0" outputs low level at P27 (NRST) pin. And transferring to reset by program (software reset) can be executed.

If the internal LSI is reset and register is initiated, the P2OUT7 flag becomes "1" and reset is released.



This LSI is activated in NORMAL mode (RC mode) in which the base clock is internal high frequency output.



In this LSI, the internal oscillation (high-speed oscillation, low-speed oscillation) is operated during reset.



Refer to [Chapter 1 1.7.4 Power Supply] for reset control at power supply on.

■ Reset Sequence

1. When reset pin comes to high level from low level, the internal binary counter (also used as watchdog timer) starts counting the system clock (fs). The length of time required from the start of counting to overflow is called "oscillation stabilization wait".
2. Internal registers and special function registers are initiated during the reset period.

Register	Address	R/W	Description	Initial value
PSW	-		Processor status word	0x00
PC	-		Program counter	Address stored in 0x04000
SP	-		Stack Pointer	undefined
An	-		Address register	undefined
Dn	-		Data register	undefined
CPUM	0x03F00	R/W	CPU mode control register	0x40
MEMCTR	0x03F01	R/W	Memory control register	0xCB
xxxICR	0x03FE2 to 0x03FFE	R/W	Maskable interrupt control register	0x00

3. Internal resets are cleared when the oscillation stabilization wait ends and program execution begins from the address written in the vector table at address 0x4000.

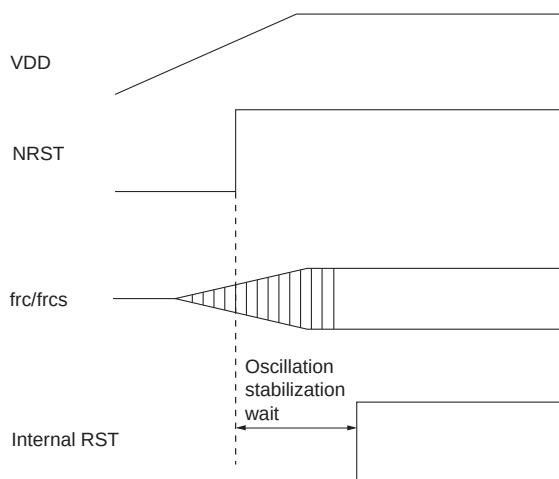


Figure:2.7.2 Reset Release Sequence



The value of internal RAM is uncertain at power-on because it is not initialized by reset. The internal RAM needs to be initialized before used.

2.7.2 Oscillation Stabilization Wait time

The oscillation stabilization wait time is the period required for a stopped oscillation circuit to reach stable oscillation. An oscillation stabilization wait time is automatically inserted when a reset is cleared or when recovering from STOP mode. When recovering from STOP mode, the oscillation stabilization wait time control register (DLYCTR) is set to select the oscillation stabilization wait time. When the reset is released, oscillation stabilization wait time is fixed.

The timer which counts oscillation stabilization wait time is also used as a watchdog timer except when a reset is cleared or returning from STOP mode.

The watchdog timer is initiated at reset or in STOP mode, and starts counting from the initial value (0x0000) when system clock (fs) is selected as a clock source. After oscillation stabilization wait time, the timer continues counting as a watchdog timer if the overrun detection is enabled.

■ Oscillation Stabilization Wait (watchdog timer) Functions Block Diagram

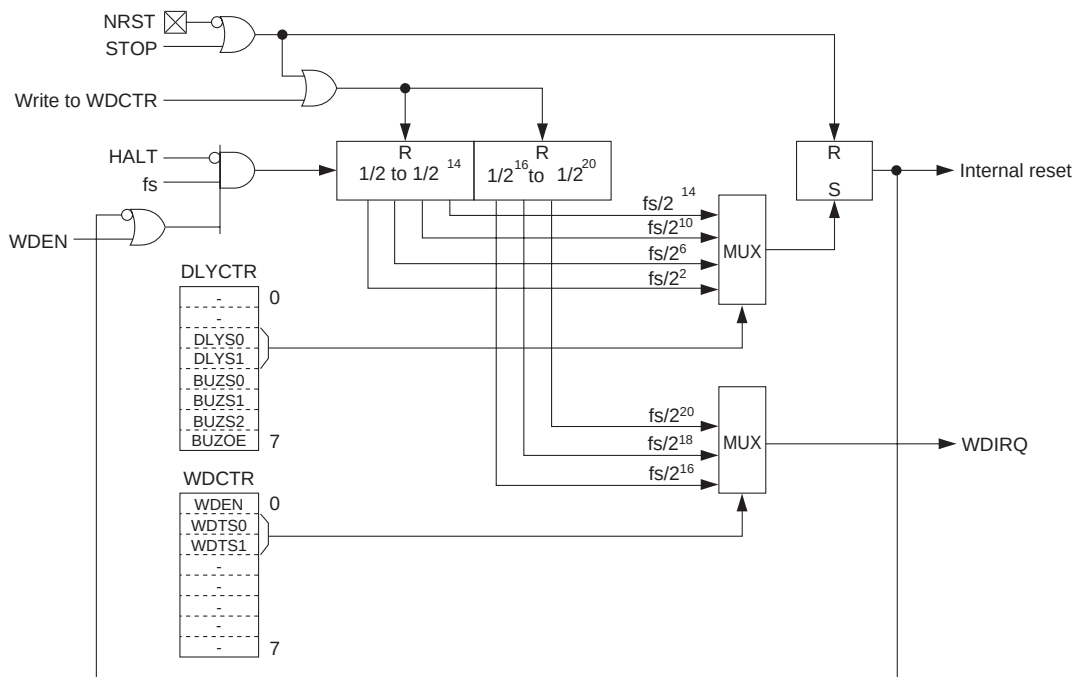


Figure:2.7.3 Block Diagram of Oscillation Stabilization Wait Time (Watchdog Timer)

■ Oscillation Stabilization Wait Control Register (DLYCTR: 0x03F03)

bp	7	6	5	4	3	2	1	0
Flag	BUZOE	BUZS2	BUZS1	BUZS0	DLYS1	DLYS0	-	-
At reset	0	0	0	0	0	1	-	-
Access	R/W	R/W	R/W	R/W	R/W	R/W	-	-

bp	Flag	Description
7	BUZOE	Buzzer output selection 0: Buzzer output disable 1: Buzzer output enable
6 to 4	BUZS2 BUZS1 BUZS0	Buzzer output frequency selection 000: $f_{pll} \div 2^{14}$ 001: $f_{pll} \div 2^{13}$ 010: $f_{pll} \div 2^{12}$ 011: $f_{pll} \div 2^{11}$ 100: $f_{pll} \div 2^{10}$ 101: $f_{pll} \div 2^9$ 110: $f_{slow} / 2^4$ 111: $f_{slow} / 2^3$
3 to 2	DLYS1 DLYS0	Oscillation stabilization wait period selection 00: $f_s / 2^{14}$ 01: $f_s / 2^{10}$ 10: $f_s / 2^6$ 11: $f_s / 2^2$
1 to 0	-	-



For the oscillation stabilization wait cycle required for high-speed/low-speed frequency, which is set by DLYS1 and DLYS0 flags, it is recommended to consult the oscillator manufacturer for determining appropriate values.



When recovering from STOP mode, more than 100 μ s of oscillation stabilization wait cycle must be set for internal regulator output stabilization wait.



See [Chapter 17 Buzzer] for setup bp7 to bp4 flags of the oscillation stabilization wait time control register (DLYCTR)

■ Oscillation Stabilization Wait Time Control

When recovering from STOP mode, the oscillation stabilization wait time can be selected from 2^{14} , 2^{10} , 2^6 , $2^2 \times$ system clock by setting the bit 3 and bit 2 (DLYS1, DLYS0) of the oscillation stabilization wait control register (DLYCTR).

At reset release, the oscillation stabilization wait time is fixed to " $2^{10} \times$ system clock". System clock (fs) is determined by the CPU mode control register (CPUM).

Table:2.7.1 Oscillation Stabilization Wait Time

DLYS1	DLYS0	Oscillation stabilization wait time
0	0	$2^{14} \times$ System clock
0	1	$2^{10} \times$ System clock
1	0	$2^6 \times$ System clock
1	1	$2^2 \times$ System clock

For oscillation stabilization wait cycle, set the enough value to stabilize the oscillation circuit.

2.8 Auto Reset Functions

2.8.1 Overview

This LSI has includes one type of Auto Reset circuit to detects power supply voltage decrease.

- When using Auto reset --- ATRST pin (10 pin): fixed at V_{DD5} -level
- When not using auto reset --- ATRST pin (10 pin): fixed at V_{SS} -level

■ When power supply is failed to 0V.

When power supply voltage is failed into the power supply change rate is upper 2 ms/V, Auto reset circuit outputs reset signal, if the power supply voltage becomes power detection level (V_{RST2}) or less.

After the internal reset reaction time (max: 400 μ s) passes, LSI becomes a reset state.

■ When power supply rises after detecting the power detection level (V_{RST2}) or less.

LSI can't become a reset state from detect power detection level (V_{RST2}) or less to the internal reset reaction time.

The power supply is risen after the internal reset reaction time passes, or the power supply change rate is lower 20 ms/V, because this state of LSI can't be guaranteed.

And, when the power supply (V_{DD5}) rises 1.8V or less, observe [Chapter 1 1.7.4 Power Supply].

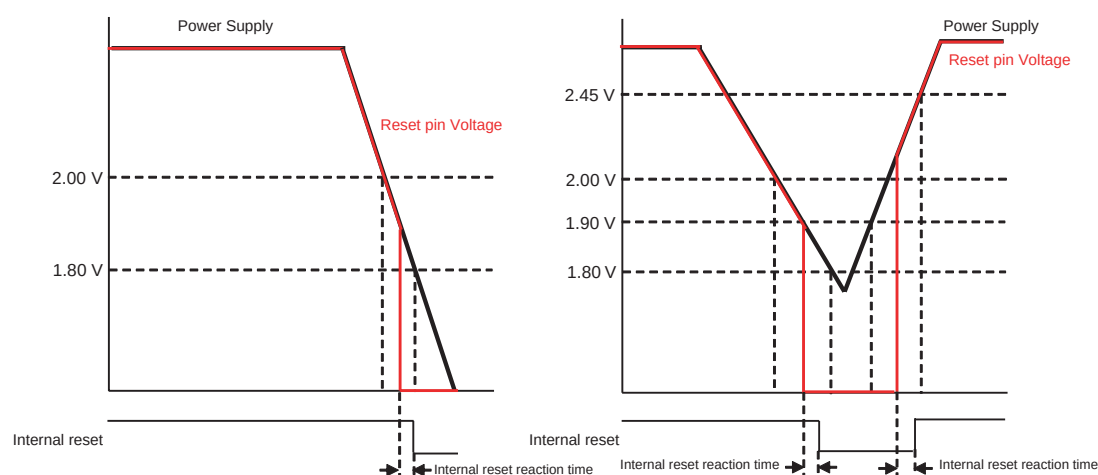


Figure:2.8.1 Power Supply and Reset Input Voltage



Refer to [Chapter 1 1.5.7 Auto Reset Characteristics] for $V_{RST1,2}$.

2.9 Reset Factor Determination

2.9.1 Overview

This LSI can detect reset factors (Hardware reset or Software reset).

- Hardware reset --- External reset (P27), Auto reset
- Software reset --- Watch dog reset, Watch dog 2 reset, Reset with P2OUT7 register programming, Reset with clock monitoring function

The reset factors at the last reset can be determined by reading registers after releasing reset.

2.9.2 Registers

- Reset Factor Determination register (RSTFACT: 0x03F19)

bp	7	6	5	4	3	2	1	0
Flag	-	-	-	-	-	-	-	RSTMON
At reset	-	-	-	-	-	-	-	0
Access	-	-	-	-	-	-	-	R

bp	Flag	Description
7-1	-	-
0	RSTMON	Reset Determination factor 0: Hardware reset occurs 1: Software reset occurs



If both hardware reset and software reset are detected at the same time, the RSTMON flag will be "0" (hardware reset).

Chapter 3 Clock Control

3.1 Overview

3.1.1 Overview

This LSI has following five types of clock generation circuit which supplies to CPU and peripheral circuits.

- 1) Internal high-speed oscillation circuit
Generates 20 MHz / 16 MHz (Typ.) of clock inside a chip.
At reset releasing, oscillation circuit is operated at 20 MHz.
- 2) Internal low-speed oscillation circuit
Generates 30 kHz (Typ.) of clock inside a chip.
At reset releasing, oscillation circuit is operated.
- 3) External high-speed oscillation circuit
Generates clocks by connecting crystal/ceramic oscillators to P25/OSC1 pin and P26/OSC2 pin.
And, the oscillation performance can be switched by CHDRVSEL register.
At reset releasing, oscillation circuit is stopped.
- 4) External low-speed oscillation circuit
Generates clocks by connecting crystal/ceramic oscillators to P90/XI pin and P91/XO pin.
And, the oscillation performance can be switched by CHDRVSEL register.
At reset releasing, oscillation circuit is stopped.
- 5) PLL circuit
Generates clocks by multiplying clocks which generated in the internal high-speed oscillation circuit or the external high-speed oscillation circuit.
At reset releasing, PLL circuit is stopped.

When using this clock generation circuit as a system clock, the division ratio can be switched by the program.

At reset release, this product starts in RC mode (internal high-speed oscillation operation) $f_s = f_{rcs}/8$.
And the internal low-speed oscillator starts in operating condition.

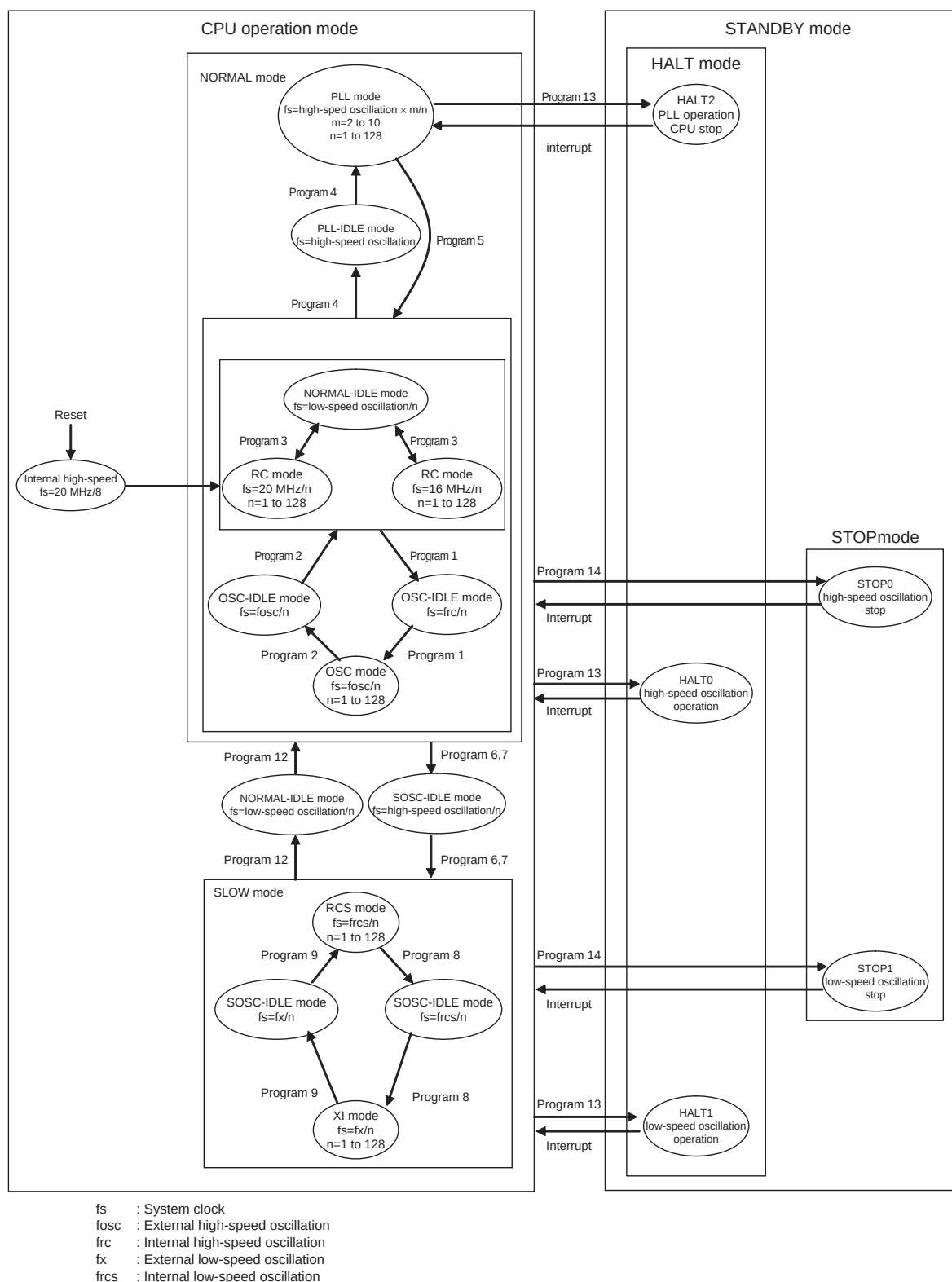


Figure:3.1.1 Transition Between Operation Modes

This LSI has two modes for CPU operation mode (NORMAL and SLOW), and two modes for standby mode (HALT and STOP).

■ NORMAL modes

There are three modes in NORMAL mode.

- RC mode
In this mode, a clock which generated in the internal high-speed oscillation circuit is used as a system clock. 20 MHz and 16 MHz can be switched by controlling RCCNT register.
At reset release, LSI starts with this mode (20 MHz, divided by 8).
- OSC mode
In this mode, a clock which generated in the external high-speed oscillation circuit is used as a system clock. CPU enters this mode by controlling the OSCCNT register.
- PLL mode
In this mode, a clock which generated in the PLL circuit is used as a system clock.
CPU enters this mode by controlling the PLLCNT register.
Which the internal high-speed oscillation and the external high-speed oscillation can be selected for an input clock to the PLL circuit. And the selection clock before transiting to PLL mode is used for an input clock.

■ SLOW modes

This mode executes the program using the low-speed operation clock. Low-power consumption while executing the program is possible because the high-speed oscillation circuit is stopped.

There are two modes in SLOW mode.

- RCS mode
In this mode, a clock which generated in the internal low-speed oscillation circuit is used as a system clock. CPU enters this mode by SLOW mode setting by the CPUM register after the SOSCCNT register is controlled.
At reset release, this mode is selected by the SOSCCNT register.
- XI mode
In this mode, a clock which generated in the external low-speed oscillation circuit is used as a system clock. CPU enters this mode by SLOW mode setting by the CPUM register after the SOSCCNT register is controlled.

■ HALT modes

In this mode, CPU stops but the oscillation circuit and PLL circuit are operating. An interrupt immediately returns CPU to operating mode. There are three modes in HALT mode.

- HALT0 mode
Both the high-speed oscillation and the low-speed oscillation circuits are operating. When an interrupt occurs, CPU enters RC/OSC mode.
- HALT1 mode
Only the low-speed oscillation circuit is operating. When an interrupt occurs, CPU enters SLOW mode.
- HALT2 mode
The high-speed oscillation, the low-speed oscillation and PLL circuit are operating. When an interrupt occurs, CPU enters PLL mode.

■ STOP Modes

Both CPU and the oscillation circuits stops operating. The transition to the operation modes is performed after the operation of the oscillation circuit is started by an interrupt and wait the oscillation becomes stable. There are two modes in STOP mode.

- STOP0 mode
When an interrupt occurs, CPU enters NORMAL mode.
- STOP1 mode
When an interrupt occurs, CPU enters SLOW mode.

■ IDLE Modes

There are four modes in IDLE mode.

- NORMAL - IDLE mode
This mode waits the program to stabilize the high-frequency oscillation clock when CPU enters from SLOW to NORMAL mode, and changing the frequency of the internal high-speed oscillation circuit. CPU enters this mode by controlling the CPUM register.
- OSC - IDLE mode
This mode waits the program to stabilize the high-frequency oscillation clock when CPU enters from RC to OSC mode, and from OSC to RC mode. In this mode, the transition by register setting is not needed. The stabilization waiting is executed by the program in the status of operation mode before transition.
- PLL - IDLE mode
This mode waits the program to stabilize the PLL clock when CPU enters from RC to OSC mode. In this mode, the transition by register setting is not needed. The stabilization waiting is executed by the program in the status of operation mode before transition.
- SOSC - IDLE mode
This mode waits the program to stabilize the low-frequency oscillation clock when CPU enters from NORMAL to XI or RCS mode, from RCS to XI mode, and from XI to RCS mode. In this mode, the transition by register setting is not needed. The stabilization waiting is executed by the program in the status of operation mode before transition.

To reduce power consumption in STOP and HALT modes, it is necessary to check the stability of both the output current from pins and port level of input pins. For output pins, the output level should match the external level or direction control should be changed to input mode. For input pins, the external level should be fixed.

Transition between NORMAL and SLOW modes or to standby mode is controlled by the CPU mode control register (CPUM). Reset and interrupts are the return factors from standby mode. A wait period is inserted for oscillation stabilization at reset and when returning from STOP mode, but not when returning from HALT mode. High/low-speed oscillation mode is automatically returned to the same state as existed before entering standby mode.



To stabilize the synchronization at the moment of switching clock frequency between high-speed oscillation (fpll) and low-speed oscillation (fx), fpll should be set to 2.5 times higher than fx.



In IDLE mode, the clock for the high-speed operation (fpll) is oscillated, however, do not operate the peripheral functions with fpll. Enable fpll operation of the peripheral functions after the high-speed operation state is changed to NORMAL mode.



To realize low power consumption, set A/D ladder resistance, D/A ladder resistance to OFF before transition to STOP/HALT mode.

3.1.2 Clock Control Function Block Diagram

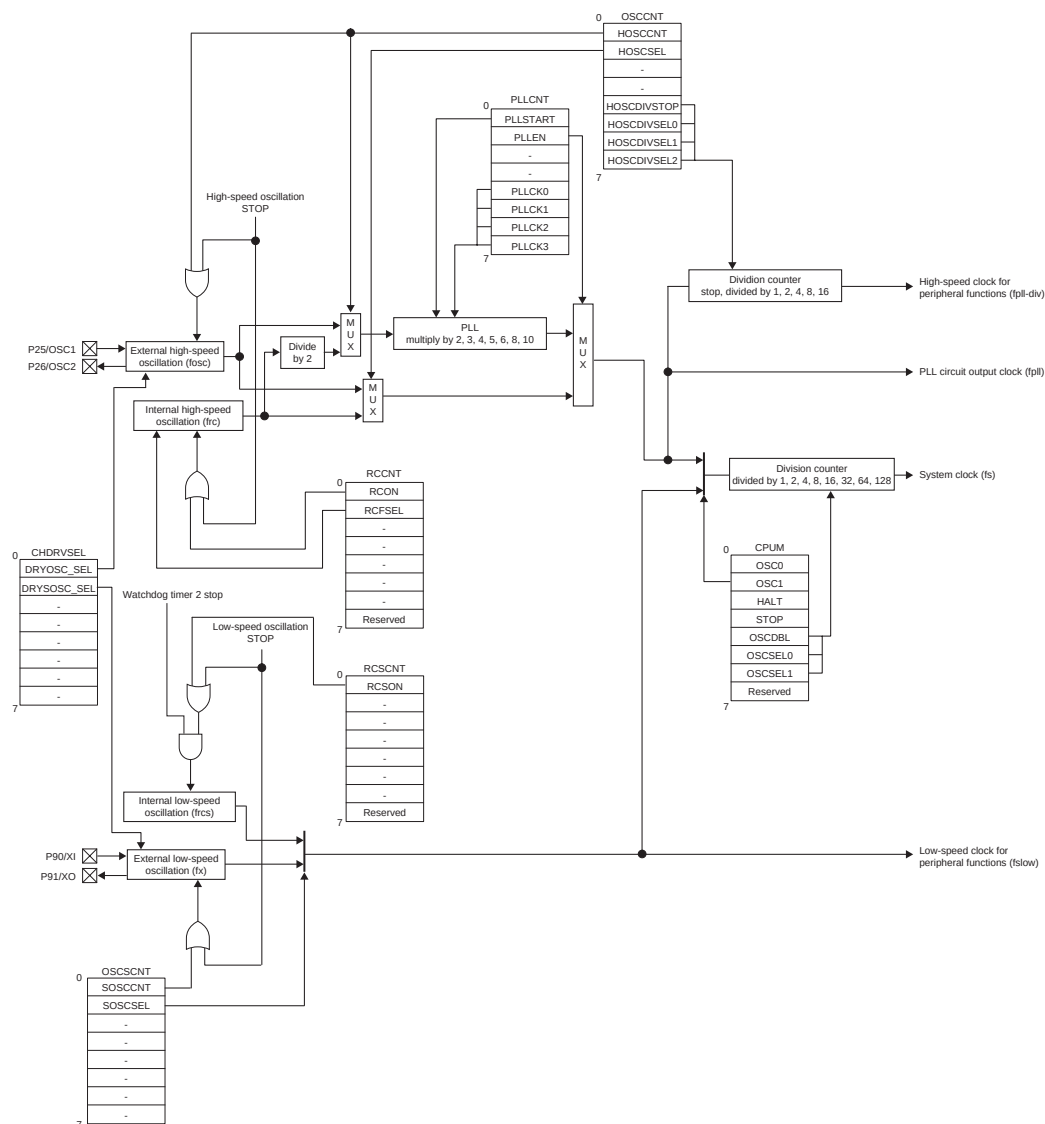


Figure 3.1.2 Clock Control Function Block Diagram



The clock which supplies to PLL circuit is selected by P25/P26 function selection flag HOSC-CNT of the external high-speed oscillation control register (OSCCNT). When the HOSC-CNT is set to "1", this clock is used to supply to PLL circuit by the external high-speed oscillation. And when the HOSC-CNT is set to "0", this clock is used to supply to PLL circuit by the internal high-speed oscillation. This clock is different from the clock selected by the operation clock selection flag (HOSC-SEL) of the external high-speed oscillation control register (OSCCNT).

Table:3.1.1 Blocks which the high-speed clock is supplied for peripheral functions

Peripheral functions	PLL circuit output clock (fpll)	High-speed clock for peripheral functions (fpll-div)
8-bit timer	-	√
Simple 8-bit timer	-	√
16-bit timer	-	√
16-bit timer for motor control	-	√
24 hours timer	√	-
Timer base timer	-	√
Free-running timer	-	√
Buzzer	-	√
Serial interface	-	√
LCD	√	-



To set system clock (f_s) > 10 MHz, set ROMHND flag of the HANDSHAKE register in advance.



When the reset is released, LSI starts with the following conditions:
Internal high-speed oscillation, $f_s = f_{pll}/8$, Internal ROM access method = HANDSHAKE

3.2 Control Registers

3.2.1 Registers List

Table shows the registers to control the clock generation function.

Table:3.2.1 Clock Generation Function Control Registers List

Register	Address	R/W	Function	Page
CHDRVSEL	0x03DF0	R/W	External oscillation control register	III-18
OSCLOCK	0x03E56	R/W	Oscillation control register protect register	III-19
CPUM	0x03F00	R/W	CPU mode control register	III-22
HANDSHAKE	0x03F06	R/W	Internal ROM access method control register	III-24
OSCCNT	0x03F1A	R/W	External high-speed oscillation control register	III-10
RCCNT	0x03F1B	R/W	Internal high-speed oscillation control register	III-12
OSCSCNT	0x03F1C	R/W	External low-speed oscillation control register	III-14
RCSCNT	0x03F1D	R/W	Internal low-speed oscillation control register	III-16
PLLCNT	0x03F1F	R/W	Clock multiplication circuit control register	III-20
FEWSPD	0x03FBF	R/W	Internal flash control register	III-26

3.2.2 Oscillation Control Registers

■ External High-speed Oscillation Control Register (OSCCNT: 0x03F1A)

This register is used to control the external high-speed oscillation, select between the external high-speed oscillation (fosc) and the internal high-speed oscillation (frc), control and divide the high-speed oscillation for peripheral functions.

bp	7	6	5	4	3	2	1	0
Flag	HOSCDIVSEL2	HOSCDIVSEL1	HOSCDIVSEL0	HOSCDIVSTOP	-	-	HOSCSEL	HOSCCNT
At reset	0	0	0	0	-	-	0	0
Access	R/W	R/W	R/W	R/W	-	-	R/W	R/W

bp	Flag	Description
7 to 5	HOSCDIVSEL2 HOSCDIVSEL1 HOSCDIVSEL0	High-speed oscillation dividing selection for peripheral functions 000: Not divide 001: Divide by 2 010: Divide by 4 011: Divide by 8 1XX: Divide by 16
4	HOSCDIVSTOP	High-speed oscillation operating selection for peripheral functions 0: Active 1: Stop
3 to 2	-	-
1	HOSCSEL	Operating clock selection 0: Internal high-speed oscillation 1: External high-speed oscillation
0	HOSCCNT *1	P25/P26 function selection 0: General port 1: High-speed oscillation pin

*1 Rewriting is enabled when HOSCSEL flag is "0".



HOSCSEL flag and HOSCCNT flag must be written in NORMAL mode.
The operation switching in SLOW/IDLE/PLL mode is not guaranteed.



When changing the operation clock from the internal high-speed oscillation to the external high-speed oscillation, wait for the oscillation to stable after setting HOSCCNT flag before switching the operating clock by setting HOSCSEL flag.



When changing the operation clock from the external high-speed oscillation to the internal high-speed oscillation, wait for 30 μ s or longer after setting the RCON flag of RCCNT register to "0". Then set the HOSCSEL flag of OSCCNT register to "0".



The oscillation stabilization wait time of the external high-speed oscillation should be set after operates the oscillation matching on the board and consults the manufacturers.



The clock which supplies to PLL circuit is selected by P25/P26 function selection flag HOSCCNT of the external high-speed oscillation control register (OSCCNT).
When the HOSCCNT is set to "1", this clock is used to supply to PLL circuit by the external high-speed oscillation.
And when the HOSCCNT is set to "0", this clock is used to supply to PLL circuit by the internal high-speed oscillation.
This clock is different from the clock selected by the operation clock selection flag (HOSCSEL) of the external high-speed oscillation control register (OSCCNT).



When the HOSCCNT flag is cleared to set P25/P26 to a general port after the operating clock is changed from the external high-speed oscillation to the internal high-speed oscillation, insert two NOP instructions and set the HOSCCNT flag.



Peripheral function operations must be stopped before changing the division setting and the operation setting of the high-speed oscillation for peripheral functions



Do not set the HOSCCNT flag to "0" when the external high-speed oscillation is selected as fosc.



In OSCCNT register, the writing processing can be masked by setting of the OSCLOCK flag of the OSCLOCK register.
By this operation, unexpected register writing can be restricted.

■ Internal High-speed Oscillation Control Register (RCCNT: 0x03F1B)

This register is used to control the operation of the internal high-speed oscillation.

bp	7	6	5	4	3	2	1	0
Flag	Reserved	-	-	-	-	-	RCFSEL	RCON
At reset	1	-	-	-	-	-	0	0
Access	R/W	-	-	-	-	-	R/W	R/W

bp	Flag	Description
7	Reserved	Always set to "1".
6 to 2	-	-
1	RCFSEL	Internal high-speed oscillation frequency selection 0: 20 MHz 1: 16 MHz
0	RCON *1	Internal high-speed oscillation control 0: Active 1: Stop

*1 Rewriting is enabled when HOSCSEL flag is "0".



RCON flag must be written in NORMAL mode.
The operation switching in SLOW/IDLE/PLL mode is not guaranteed.



When the RCON flag is set to stop the internal high-speed oscillation after the operation clock is changed to the external high-speed oscillation, set it after inserting two NOP instructions.



Do not set the RCON flag to "1" when the HOSCSEL flag of OSCCNT register is set to "0", and the operating clock is selected as the internal high-speed oscillation.



When changing the operating clock from the external high-speed oscillation to the internal high-speed oscillation, wait for 30 μ s or longer after setting the RCON flag of RCCNT register to "0". Then set the HOSCSEL flag of OSCCNT register to "0".



When changing the RCFSEL flag, always set OSC0 flag of CPUM register and switch to NORMAL-IDLE mode before setting this flag.
And wait more than 30 μ s after setting of the RCFSEL flag. Then switch to NORMAL mode.



When setting the HOSCSEL flag of OSCCNT register to change the operation clock from the internal high-speed oscillation to the external high-speed oscillation, the internal high-speed oscillation becomes STOP by setting the RCON flag. And the operating current can be reduced.



In RCCNT register, the writing processing can be masked by setting of the OSCLOCK flag of OSCLOCK register.
By this operation, unexpected register writing can be restricted.

■ External Low-speed Oscillation Control Register (OSCSCNT: 0x03F1C)

This register is used to control the external low-speed oscillation and select the external high-speed oscillation (fx), the internal low-speed oscillation (frcs).

bp	7	6	5	4	3	2	1	0
Flag	-	-	-	-	-	-	SOSCSEL	SOSCCNT
At reset	-	-	-	-	-	-	0	0
Access	-	-	-	-	-	-	R/W	R/W

bp	Flag	Description
7 to 2	-	-
1	SOSCSEL	Operation clock selection 0: Internal low-speed oscillation 1: External low-speed oscillation
0	SOSCCNT *1	P90/P91 function selection 0: General port 1: Low-speed oscillation pin

*1 Rewriting is enabled when SOSCSEL flag is "0".



SOSCSEL flag and SOSCCNT flag must be written in NORMAL mode.
The operation switching in SLOW/IDLE/PLL mode is not guaranteed.



When changing the operation clock from the internal low-speed oscillation to the external low-speed oscillation, wait the oscillation stabilization wait time enough after the setting SOSCCNT flag. Then set the SOSCSEL flag and switch the operation clock.



The oscillation stabilization wait time of the external low-speed oscillation should be set after operates the oscillation matching on the board and consults the manufacturers.



When changing the operation clock from the external low-speed oscillation to the internal low-speed oscillation, wait more than 200 μ s after setting the RCON flag of RCSCNT register to "0". Then set the SOSCSEL flag of OSCSCNT register to "0".



When the SOSCCNT flag is cleared to set P90/P91 to a general port after the operation clock is changed from the external low-speed oscillation to the internal low-speed oscillation, set the SOSCCNT flag after waiting two external low-speed oscillations.



Do not set the SOSCCNT flag to "0" when the external low-speed oscillation is selected as fx.



In OSCSCNT register, the writing processing can be masked by setting of the OSCLOCK flag of OSCLOCK register.
By this operation, unexpected register writing can be restricted.

■ Internal Low-speed Oscillation Control Register (RCSCNT: 0x03F1D)

This register is used to control operations of the internal low-speed oscillation.

bp	7	6	5	4	3	2	1	0
Flag	-	-	-	-	-	-	-	RCSON
At reset	-	-	-	-	-	-	-	0
Access	-	-	-	-	-	-	-	R/W

bp	Flag	Description
7 to 1	-	-
0	RCSON *1	Internal low-speed oscillation control 0: Active 1: Stop

*1 Rewriting is enabled when WD2MD7 to WD2MD0 flags of WD2MDSEL register are "0x00".



RCSON flag must be written in NORMAL mode.
The operation switching in SLOW/IDLE/PLL mode is not guaranteed.



To stop the internal low-speed oscillation circuit, the watchdog timer 2 must be stopped.
Refer to [Chapter 16 Watchdog Timer 2] for details.



When the operation clock is changed to the external low-speed oscillation to stop the internal low-speed oscillation after the RCSON flag is set, set it after waiting two external low-speed oscillations.



Do not set the RCSON flag to "1" when the SOSSEL flag of OSCSCNT register is set to "0", and the operation clock is selected as the internal low-speed oscillation.



When changing the operation clock from the external low-speed oscillation to the internal low-speed oscillation, wait more than 200 μ s after setting the RCSON flag of RCSCNT register to "0". Then set the SOSSEL flag of OSCSCNT register to "0".



When setting the SOSSEL flag of the OSCSCNT register to change the operation clock from the internal high-speed oscillation to the external high-speed oscillation, the internal high-speed oscillation becomes STOP by setting the RCON flag. And the operating current can be reduced.



In RCSCNT register, the writing processing can be masked by setting of the OSCLOCK flag of the OSCLOCK register.
By this operation, unexpected register writing can be restricted.

■ External Oscillation Control Register (CHDRVSEL: 0x03DF0)

This register is used to control the oscillation performance of the external high-speed oscillation and the external low-speed oscillation.

bp	7	6	5	4	3	2	1	0
Flag	-	-	-	-	-	-	DRVS OSC_SEL	DRV OSC_SEL
At reset	-	-	-	-	-	-	0	0
Access	-	-	-	-	-	-	R/W	R/W

bp	Flag	Description
7 to 2	-	-
1	DRVS OSC_SEL	External low-speed oscillation performance selection 0: Normal performance 1: High performance
0	DRV OSC_SEL	External high-speed oscillation performance selection 0: Normal performance 1: High performance



Change the oscillation performance before selecting the external oscillation as the operating clock. The operation switching the oscillation performance while setting the external oscillation as the operating clock can not be guaranteed.

■ Oscillation Control Register Protect Register (OSCLOCK: 0x03E56)

This register is used to control the writing to OSCCNT register, RCCNT register and OSCSCNT register.

bp	7	6	5	4	3	2	1	0
Flag	-	-	-	-	-	-	-	LOCKEN
At reset	-	-	-	-	-	-	-	0
Access	-	-	-	-	-	-	-	R/W

bp	Flag	Description
7 to 1	-	-
0	LOCKEN	Oscillation control register writing control 0: Writing enable 1: Writing disable



When the LOCKEN flag is set to "1", writing to OSCCNT, RCCNT, OSCSCNT and RCSCNT registers is masked. But when reading, the setting values are read-out.

■ Clock Multiplication Circuit Control Register (PLLCNT: 0x03F1F)

This register is used to control the clock generation of 2 to 10 multiplications by using the oscillation selected by OSCCNT register.

bp	7	6	5	4	3	2	1	0
Flag	PLLCK3	PLLCK2	PLLCK1	PLLCK0	-	-	PLLEN	PLL START
At reset	0	0	0	0	-	-	0	0
Access	R/W	R/W	R/W	R/W	-	-	R/W	R/W

bp	Flag	Description
7 to 4	PLLCK3 PLLCK2 PLLCK1 PLLCK0	Multiplication setting 0000: Multiply by 2 (input frequency 4 to 7.5 MHz) 0001: Multiply by 2 (input frequency 7.5 to 10 MHz) 0010: Multiply by 3 (input frequency 4 to 5 MHz) 0011: Multiply by 4 (input frequency 4 to 7.5 MHz) 0100: Multiply by 4 (input frequency 7.5 to 10 MHz) 0101: Multiply by 5 (input frequency 4 to 6 MHz) 0110: Multiply by 5 (input frequency 6 to 8 MHz) 0111: Multiply by 6 (input frequency 4 to 5 MHz) 1000: Multiply by 8 (input frequency 4 to 5 MHz) 1001: Multiply by 10 (input frequency 4 MHz) 1010: Setting prohibited 1011: Setting prohibited 1100: Setting prohibited 1101: Setting prohibited 1110: Setting prohibited 1111: Setting prohibited
3 to 2	-	-
1	PLLEN	PLL clock enable 0: High-speed oscillation clock (fosc) operation 1: PLL clock operation
0	PLLSTART	PLL operation control 0: PLL stop 1: PLL operation



Switch the operation clock to PLL output after setting the PLLSTART flag and wait for 100 μ s or longer. Then set the PLEN flag and switch the operating clock.



Do not change the multiplication setting by changing the PLLCK3 to PLLCK0 flags on PLL operation. Change the multiplication setting when the PLLSTART flag is "0".



Do not switch the PLL input clock by changing the HOSCSEL flag of OSCCNT register. Make sure to change the PLL input clock when the PLLSTART flag is "0".



When stopping the PLL circuit by clearing the PLLSTART flag after the transition from PLL mode to RC mode or OSC mode, make sure to insert more than two NOP instructions.



The clock which supplies to PLL circuit is selected by P25/P26 function selection flag HOSC-CNT of external high-speed oscillation control register (OSCCNT).
When the HOSCCNT is set to "1", this clock is used to supply to PLL circuit by the external high-speed oscillation.
And when the HOSCCNT is set to "0", this clock is used to supply to PLL circuit by the internal high-speed oscillation.
This clock is different from the clock selected by the operation clock selection flag (HOSC-SEL) of external high-speed oscillation control register (OSCCNT).



Directly transition between PLL mode and SLOW mode can not be operated.
Make sure to transit through RC mode or OSC mode.



The operation is not guaranteed when the system clock exceeds 20 MHz.
The division setting must be operated by CPU mode control register before transition to PLL mode. And transit to PLL mode before the system clock is set to 20 MHz or less.



When the internal high-speed oscillation is set as PLL input clock, 1/2 of frequency set by RCCNT register (20 MHz/16 MHz) is used to an input clock.

■ CPU Mode Control Register (CPUM: 0x03F00)

This register is used to control the operating mode and division ratio of system clock.

bp	7	6	5	4	3	2	1	0
Flag	Reserved	OSCSEL1	OSCSEL0	OSCDBL	STOP	HALT	OSC1	OSC0
At reset	0	0	1	0	0	0	0	0
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description
7	Reserved	Always set to "0".
6 to 5	OSCSEL1 OSCSEL0	Division ratio 00: 1 01: 4 10: 16 11: 64
4	OSCDBL	Internal system clock (fs) 0: Normal (fpll/2) 1: Double-speed (fpll)
3 to 0	STOP HALT OSC1 OSC0	CPU mode selection 0000: NORMAL 0001: NORMAL-IDLE 0011: SLOW 0100: HALT0/HALT2 0111: HALT1 1000: STOP0 1011: STOP1 Others: Setting prohibited



Do not change both settings of the clock switching function (OSCDBL, OSCSEL1 to 0 flags) and the STANDBY function (STOP, HALT, OSC1 and OSC0 flags) at the same time.

Table:3.2.2 Division Ratio Setting by Combination of OSCSEL and OSCDBL

OSCSEL1	OSCSEL0	OSCDBL	Division ratio
0	0	0	2
0	0	1	1
0	1	0	8
0	1	1	4
1	0	0	32
1	0	1	16
1	1	0	128
1	1	1	64



Set the division ratio of the oscillation clock and the operating mode transition within the range satisfying LSI operating condition and operating speed.
Refer to [Chapter 1 1.5 Electrical Characteristics]



OSCDBL, OSCSEL1 and OSCSEL0 flags can be changed simultaneously.

Table:3.2.3 Operating Mode Control and Clock Oscillation/Halt

Operating mode					Status			
	STOP	HALT	OSC1	OSC0	OSC/RC	XI/RCS	System clock	CPU
NORMAL	0	0	0	0	Oscillate	Oscillate	OSC/RC/PLL	Operate
NORMAL-IDLE	0	0	0	1	Oscillate	Oscillate	XI/RCS	Operate
SLOW	0	0	1	1	Halt	Oscillate	XI/RCS	Operate
HALT0/HALT2	0	1	0	0	Oscillate	Oscillate	OSC/RC/PLL	Halt
HALT1	0	1	1	1	Halt	Oscillate	XI/RCS	Halt
STOP0	1	0	0	0	Halt	Halt	Halt	Halt
STOP1	1	0	1	1	Halt	Halt	Halt	Halt

The procedure for transition from NORMAL to HALT or STOP mode is shown below.

1. Clear the interrupt request flag (xxxIR) in the maskable interrupt control register (xxxICR) and set the interrupt enable flag (xxxIE) for the return factor.
2. To return using an maskable interrupt, set the MIE flag of PSW to "1" and set the interrupt mask (IM) to a level permitting the interrupt to be accept.
3. Set CPUM to HALT or STOP mode.



Set the IRWE flag of the memory control register (MEMCTR) to clear the interrupt request flag by software. Be sure to clear the IRWE flag after the interrupt request flag is cleared.

■ Internal ROM Access Method Control Register (HANDSHAKE: 0x03F06)

This register is used to control the access method for the internal ROM area. In this LSI, the access method for the internal ROM area should be set by relation between power supply voltage (V_{DD5}) and system clock frequency (fs).

Table:3.2.4 ROMHND Flag Setting List

Power supply voltage (V_{DD5})	System clock (fs)	ROMHND flag
$1.8\text{ V} \leq V_{DD5} < 2.7\text{ V}$	$fs \leq 4\text{ MHz}$	0
	$4\text{ MHz} < fs \leq 8\text{ MHz}$	1
$2.7\text{ V} \leq V_{DD5} \leq 5.5\text{ V}$	$fs \leq 10\text{ MHz}$	0
	$10\text{ MHz} < fs \leq 20\text{ MHz}$	1

When HANDSHAKE is set to the internal ROM access method, access cycles increases at the conditions of below table.

Table:3.2.5 Extension Conditions of Access Cycle at HANDSHAKE access

Previous conditions	Conditions of access extension	Cycles
Reading instructions whose branch destination is located in odd address	Reading branch instructions located in consecutive even address	2
Reading branch instructions located in consecutive odd address	Reading instructions whose branch destination is located in odd address	3
Reading branch instructions located in consecutive even address	Reading instructions whose branch destination is located in odd address	2
Reading branch instructions located in consecutive odd address	Reading instructions whose branch destination is located in even address	3
Reading branch instructions located in consecutive even address	Reading instructions whose branch destination is located in even address	2
Reading instructions located in odd address	Data access to odd address	3
Reading instructions located in even address	Data access to odd address	2
Reading instructions located in odd address	Data access to even address	3
Reading instructions located in even address	Data access to even address	2
-	DMA transfer (one-word transfer)	2
-	DMA transfer (burst mode, even address reading)	2

bp	7	6	5	4	3	2	1	0
Flag	-	-	-	-	-	ROMHND	-	-
At reset	-	-	-	-	-	1	-	-
Access	-	-	-	-	-	R/W	-	-

bp	Flag	Description
7 to 3	-	-
2	ROMHND	Internal ROM access method selection 0: Normal access 1: HANDSHAKE access
1 to 0	-	-



Set the ROMHND flag in advance to select HANDSHAKE access when transition to the power supply voltage and the system clock frequency of the ROMHND flag settings described in Table:3.2.4.



Clear the ROMHND flag and select normal access when operating with the power supply voltage and the system clock frequency of the ROMHND flag clearing described in Table:3.2.4. The access timing to the internal ROM area improves.



When the reset is released, LSI starts with the internal ROM access method = HANDSHAKE.



On the condition of $f_s \leq 4 \text{ MHz}$ ($1.8 \text{ V} \leq V_{DD5} < 2.7 \text{ V}$), $f_s \leq 10 \text{ MHz}$ ($2.7 \text{ V} \leq V_{DD5} \leq 5.5 \text{ V}$), CPU processing performance in HANDSHAKE access (ROMHND = 1) is inferior to that in NORMAL access (ROMHND = 0).

■ Internal Flash Control Register (FEWSPD: 0x03FBBF)

This register is used to control operation status of the internal low-speed oscillation in SLOW mode.
When the program in SLOW mode can execute on the RAM, the power consumption can be reduced by setting the internal flash to STOP using this function.

bp	7	6	5	4	3	2	1	0
Flag	Reserved	-	-	-	NSTOP	-	-	Reserved
At reset	0	-	-	-	0	-	-	1
Access	R/W	-	-	-	R/W	-	-	R

bp	Flag	Description
7	Reserved	Always set to "0".
6 to 4	-	-
3	NSTOP	Internal flash control 0: Operating 1: STOP
2 to 1	-	-
0	Reserved	-



Settings of the NSTOP flag is valid only in SLOW mode.



Make sure to set the NSTOP flag while the program is executed on the RAM after transition to SLOW mode.

When the NSTOP flag becomes "1" in SLOW mode and executing ROM, error occurs in CPU because an access to the internal flash can not be operated.



After the NSTOP flag is cleared, switch to the program on the internal flash after waiting for 20 μ s or longer.



BSET/BCLR instruction must be used for setting of NSTOP flag.

3.3 Mode Transition

The sample programs for transition to each mode is shown below.

3.3.1 NORMAL Mode

■ Transition from RC mode to OSC mode

Transition from RC mode to OSC mode requires passing through IDLE mode. In OSC-IDLE mode, CPU operates with the internal high-speed oscillation clock before transition to OSC mode.



The frequency used as the system clock varies depending on the power supply voltage. Therefore, confirm the power supply voltage by using the power supply voltage detection function before setting the system clock.

Refer to [Chapter 6 Power Supply Voltage Detection] for details of the power supply voltage detection function.

Confirm “ $V_{DD5} \geq 2.7 \text{ V}$ ” by using the power supply voltage detection function.

MOV x'10', (CPUM); system clock is set double speed ($f_s = f_{rc}$).

Program1

	BSET (OSCCNT)0	Set P25/P26 to the high-speed oscillation pin
	MOV X'FF', D0	
LOOP	ADD -1, D0	Loop to wait 76 μs at 20 MHz operation
	BNE LOOP	
	BSET (OSCCNT)1	Set OSC mode
	NOP	
	NOP	
	BSET (RCCNT)0	Stop the internal high-speed oscillation circuit



The oscillation stabilization wait time of the external low-speed oscillation should be set after operates the oscillation matching on the board and consults other manufacturers.



When the internal high-speed oscillation is stopped to set the RCON flag after the operating clock is changed to the external high-speed oscillation, insert two NOP instruction and set the RCON flag.



When setting the HOSCSEL flag of the OSCCNT register to change the operation clock from the internal high-speed oscillation to the external high-speed oscillation, the internal high-speed oscillation is disabled by setting the RCON flag. As a result, the operating current can be reduced.

■ Transition from OSC mode to RC mode

Transition from OSC mode to RC mode requires passing through IDLE mode. In OSC-IDLE mode, CPU operates with the internal high-speed oscillation clock before transition to the modes.



The frequency used as a system clock varies depending on the power supply voltage. Therefore, confirm the power supply voltage by using the power supply voltage detection function before setting a system clock.

Refer to [Chapter 6 Power Supply Voltage Detection] for details of the power supply voltage detection function.

Confirm “ $V_{DD5} \geq 2.7 \text{ V}$ ” by using the power supply voltage detection function.

MOV x'10', (CPUM); system clock is set normal speed ($f_s = f_{rc}/2$).

Program2

	BCLR (RCCNT)0	Operate the internal high-speed oscillation circuit
	MOV X'46', D0	
LOOP	ADD -1, D0	Loop to wait 42 μs at 10 MHz operation
	BNE LOOP	
	BSET (HANDSHAKE)2	Set the internal ROM access method to HANDSHAKE
	BCLR (OSCCNT)1	Set RC mode
	NOP	
	NOP	
	BCLR (OSCCNT)0	Set P25/P26 pin to general pin



When changing the operation clock from the external high-speed oscillation to the internal high-speed oscillation, wait more than 30 μs after setting the RCON flag of RCCNT register to “0”. Then set the HOSCSEL flag of OSCCNT register to “0”.



When the HOSCCNT flag is cleared to set P25/P26 to a general port after the operation clock is changed from the external high-speed oscillation to the internal high-speed oscillation, insert two NOP instructions and set the HOSCCNT flag.



Set the ROMHND flag in advance to select HANDSHAKE access when transition to the power supply voltage and the system clock frequency of the ROMHND flag settings described in Table:3.2.4.

■ Switching from 20 MHz output of the internal high-speed oscillation to 16 MHz output in RC mode

Switching the internal high-speed oscillation output in RC mode requires passing through NORMAL-IDLE mode. In NORMAL-IDLE mode, CPU operates with the internal low-speed oscillation clock.



The frequency used as a system clock varies depending on the power supply voltage. Therefore, confirm the power supply voltage by using the power supply voltage detection function before setting a system clock.

Refer to [Chapter 6 Power Supply Voltage Detection] for details of the power supply voltage detection function.

Confirm “ $V_{DD5} \geq 2.7 \text{ V}$ ” by using the power supply voltage detection function.

MOV x'10', (CPUM); system clock is set double speed (fs = frc).

Program3

BSET (CPUM)0	Set NORMAL-IDLE mode
BSET (RCCNT)1	Set the internal high-speed oscillation output to 16 MHz
NOP	
BCLR (CPUM)0	Set NORMAL mode



When changing the RCFSEL flag, always set OSC0 flag of CPUM register and switch to IDLE mode before setting this flag.
And wait 30 μs and over after setting of the RCFSEL flag. Then switch to NORMAL mode.



Operate the low-speed oscillation before transition to NORMAL-IDLE mode.

■ Transition from RC mode or OSC mode to PLL mode

Transition to PLL mode requires passing through PLL-IDLE mode. In PLL-IDLE mode, CPU operates with the high-speed oscillation clock (RC, OSC).



The frequency used as the system clock varies depending on the power supply voltage. Therefore, confirm the power supply voltage by using the power supply voltage detection function before setting the system clock.

Refer to [Chapter 6 Power Supply Voltage Detection] for details of the power supply voltage detection function.

Confirm " $V_{DD5} \geq 2.7 \text{ V}$ " by using the power supply voltage detection function.

MOV x'10', (CPUM); system clock is set normal speed ($f_s = f_{rc}/2$).

Program4

	MOV x'40', (PLLCNT)	Set to 4 multiplication
	BSET (PLLCNT)0	Operate PLL circuit
	MOV X'A6', D0	
LOOP	ADD -1, D0	Loop to wait 100 μs at 10 MHz
	BNE LOOP	
	BSET (HANDSHAKE)2	Set the internal ROM access method to HANDSHAKE
	BSET (PLLCNT)1	Set PLL mode



When changing the operating clock to PLL output, set the PLLSTART flag and wait 100 μs and over. Then set the PLEN flag.



Set the ROMHND flag in advance to select HANDSHAKE access when transition to the power supply voltage and the system clock frequency of the ROMHND flag settings described in Table:3.2.4.



The clock which supplies to PLL circuit is selected by P25/P26 function selection flag HOSCCNT of external high-speed oscillation control register (OSCCNT). When the HOSCCNT is set to "1", this clock is used to supply to PLL circuit by the external high-speed oscillation. And when the HOSCCNT is set to "0", this clock is used to supply to PLL circuit by the internal high-speed oscillation. This clock is different from the clock selected by the operation clock selection flag (HOSCSEL) of external high-speed oscillation control register (OSCCNT).

■ Transition from PLL mode to RC mode or OSC mode

The transition from PLL mode to RC mode or OSC mode can be operated by writing to PLLCNT register. In this case, transition through IDLE state is not needed.

Program5

BCLR (PLLCNT)1	Set RC mode or OSC mode
NOP	
NOP	
BCLR (PLLCNT)0	Stop PLL circuit



When stopping the PLL circuit by clearing the PLLSTART flag after the transition from PLL mode to RC mode or OSC mode, make sure to insert more than two NOP instructions.

3.3.2 SLOW Mode

Transition from NORMAL to SLOW mode, when the low-speed clock (RCS, XI) has fully stabilized, can be done by writing to the CPU mode control register.

In this case, transition through IDLE state is not needed.

■ Transition from NORMAL mode to RCS mode

Program6		
	BCLR (RCSCNT)0	Operate the internal low-speed oscillation
	MOV X'FF', D0	
	MOVW X'0F42', A0	
LOOP2		
LOOP1	ADD -1, D0	Loop to wait 300 ms at 20 MHz operation
	BNE LOOP1	
	ADDW -1, A0	
	BNE LOOP2	
	MOV X'03', D0	Set SLOW mode
	MOV D0, (CPUM)	



After setting the RCSON flag of RCSCNT register to “0” to set the internal low-speed oscillation operation, transition to SLOW mode should be executed after waiting more than 200 μ s.



After the reset is released, wait for 30 ms or longer to change the operation mode to SLOW, HALT1 or STOP.

■ Transition from NORMAL mode to XI mode

Program7		
	BSET (OSCSCNT)0	Set P90/P91 to the low-speed oscillation pin
	MOV X'FF', D0	
	MOVW X'0F42', A0	
LOOP2		
LOOP1	ADD -1, D0	Loop to wait 300 ms at 20 MHz
	BNE LOOP1	
	ADDW -1, A0	
	BNE LOOP2	
	BSET (OSCSCNT)1	Set the low-speed clock to the external low-speed oscillation
	MOV X'03', D0	Set SLOW mode
	MOV D0, (CPUM)	



The oscillation stabilization wait time of the external low-speed oscillation should be set after operates the oscillation matching on the board and consults the manufacturer of your oscillator.



After the reset is released, wait for 30 ms or longer to change the operation mode to SLOW, HALT1 or STOP.

■ Transition from RCS mode to XI mode

Transition from RCS mode to XI mode requires passing through SOSC-IDLE mode. In SOSC-IDLE mode, CPU operates with the internal low-speed oscillation clock before transition to the modes.

Program8		
	BSET (OSCSCNT)0	Set P90/P91 to the low-speed oscillation pin
	MOV X'06', D0	
LOOP	ADD -1, D0	Loop to wait 1 ms at 30 kHz
	BNE LOOP	
	BSET (OSCSCNT)1	Set the low-speed clock to the external low-speed oscillation
	MOV X'88', (PRTKEY)	Release the protection of WD2MDSEL register
	MOV X'00', (WD2MDSEL)	Stop the watchdog timer 2
	BSET (RCSCNT)0	Stop the internal low-speed oscillation



To stop the internal low-speed oscillation circuit, the watchdog timer 2 must be stopped. Refer to [Chapter 16 Watchdog Timer 2] for details.



The oscillation stabilization wait time of the external low-speed oscillation should be set after operating the oscillation matching on the board and consulting with other manufacturers.

■ Transition from XI mode to RCS mode

Transition from XI mode to RCS mode requires passing through SOSC-IDLE mode. In SOSC-IDLE mode, CPU operates with the external low-speed oscillation clock before transition to the modes.

Program9		
	BCLR (RCSCNT)0	Operate the internal low-speed oscillation
	MOV X'01', D0	
LOOP	ADD -1, D0	Loop to wait 200 μ s at 32 kHz
	BNE LOOP	
	BCLR (OSCSCNT)1	Set RCS mode
	NOP	
	NOP	
	BCLR (OSCCNT)0	Set P90/P91 to general port



When changing the operation clock from the external low-speed oscillation to the internal low-speed oscillation, wait more than 200 μ s after setting the RCSON flag of RCSCNT register to "0". Then set the SOSCSEL flag of OSCSCNT register to "0".



When the SOSCCNT flag is cleared to set P25/P26 to a general port after the operation clock is changed from the external low-speed oscillation to the internal low-speed oscillation, set the SOSCCNT flag after waiting two external low-speed oscillations.

■ Disable the Internal Flash in SLOW mode

In SLOW mode, the power consumption can be reduced by executing the program on the RAM to disable the internal flash.

Program10

Transfer	
MOV X'30', D0	Set MIE is 0 and disable all maskable interrupts
MOV D0, PSW	
BSET (MEMCTR)5	Change the interrupt vector table base address on the RAM
Branch to the internal ROM area	
BSET (FEWSPD)3	Disable the internal flash
MOV X'70', D0	Set MIE is 1 and enable all maskable interrupts
MOV D0, PSW	

■ Enable the Internal Flash in SLOW mode

Program11

MOV X'30',D0	Set MIE is 0 and disable all maskable interrupts
MOV D0, PSW	
BCLR (FEWSPD)3	Enable the internal flash
Branch to the internal ROM area	
MOV X'70', D0	Set MIE is 1 and enable all maskable interrupts
MOV D0, PSW	



Settings of the NSTOP flag is valid only in SLOW mode.



Make sure to set the NSTOP flag while the program is executed on the RAM after transition to SLOW mode.

When the NSTOP flag becomes "1" in SLOW mode and executing ROM, error occurs in CPU because an access to the internal flash can not be operated.



After the NSTOP flag is cleared, switch to the program on the internal flash after waiting for 20 μ s or longer.



BSET/BCLR instruction must be used for setting of NSTOP flag.

■ Transition from SLOW mode to NORMAL mode

For transition from SLOW to NORMAL mode, the program must be hold at the IDLE state until the high-speed clock oscillation is fully stable. In IDLE mode, CPU operates with the low-speed clock.

Program12		
	MOV X'01', (CPUM)	Set NORMAL-IDLE mode
	MOV X'01', D0	
LOOP	ADD -1, D0	Loop to wait 200 μ s at 32 kHz
	BNE LOOP	
	MOV X'00', (CPUM)	Set NORMAL mode



When changing from SLOW mode to RC mode, make sure to wait more than 30 μ s at NORMAL-IDLE mode before switch to NORMAL mode.



When changing from SLOW to OSC mode, make sure to wait until the high-speed clock oscillation is fully stabilize at NORMAL-IDLE mode before switch to NORMAL mode. The oscillation stabilization wait time of the external low-speed oscillation should be set after operates the oscillation matching on the board and consults the manufacturers.

3.3.3 STANDBY Mode

The program operates the transition from CPU OPERATION mode to STANDBY mode. Interrupts operate the return to CPU OPERATION mode.

Before transition to STANDBY mode, the following settings are required.

1. Set the maskable interrupt enable flag (xxxIE) and the maskable interrupt control register (xxxICR) of the processor status word (PSW) to "0" to disable all maskable interrupts temporarily.
2. Specify an interrupt factor which enables returning from STANDBY mode to CPU OPERATION mode, and set the interrupt enable flag (xxxIE) of the maskable interrupt control register (xxxICR) only. Besides set MIE of PSW.

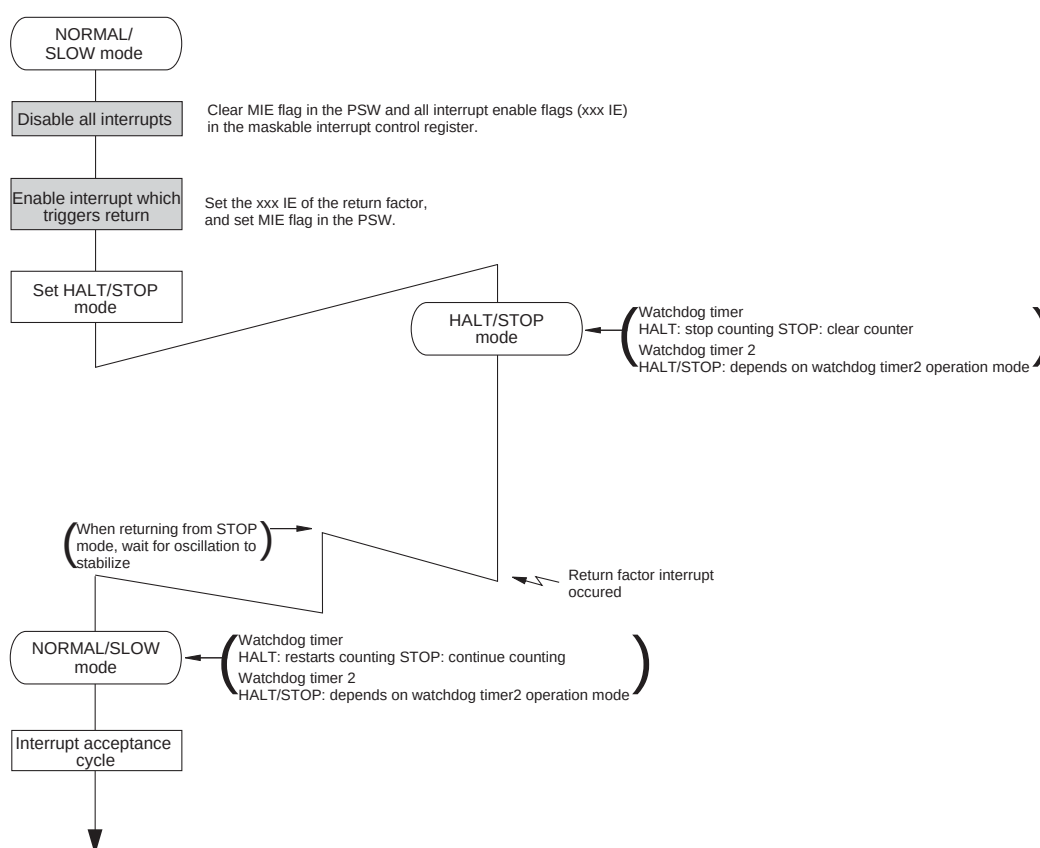


Figure:3.3.1 Transition to/from STANDBY mode



If an interrupt is enabled but interrupt priority level of the interrupt to be used is not equal to or higher than the mask level in PSW before transition to HALT or STOP mode, it is impossible to return to CPU operation mode by maskable interrupt.



After the reset is released, wait for 30 ms or longer to change the operation mode to SLOW, HALT1 or STOP.

■ Transition to HALT Modes

The system transfers from RC mode or OSC mode to HALT0 mode, and from SLOW mode to HALT1 mode, and from PLL mode to HALT2 mode. CPU stops operating, but the oscillators remain operational. There are two ways to return from a HALT mode: a reset or an interrupt. A reset produces a normal reset; an interrupt, an immediate return to the CPU state prior to the transition to HALT mode. The watchdog timer, if enabled, resumes counting. Watchdog timer 2 depends on watchdog timer 2 operation mode setting.

Program13

MOV X'04', D0	Set HALT mode.
MOV D0, (CPUM)	
NOP	After written in CPUM, some NOP instructions (three or less) are
NOP	executed depending on pipeline's condition.
NOP	

■ Transition to STOP mode

The system transfers from RC mode or OSC mode to STOP0 mode, and from SLOW mode to STOP1 mode. The transition from PLL mode to STOP mode can not be executed. In both cases, oscillation and CPU are both halted. There are two ways to leave a STOP mode: a reset or interrupt.

On the transition to STOP mode, the counter of watchdog timer is cleared. During recovery, the counter is started and oscillation stabilization wait is generated. After recover to CPU operation, the counting is continued. The watchdog timer 2 depends on the watchdog timer 2 operation mode setting.

Program 14

MOV	x'8',D0	Set STOP mode.
MOV	D0,(CPUM)	
NOP		After written in CPUM, some NOP instructions (three or less) are
NOP		executed depending on pipeline's condition.
NOP		



To stop the internal low-speed oscillation circuit, the watchdog timer 2 must be stopped. Refer to [Chapter 16 Watchdog Timer 2] for details.



Insert three NOP instructions right after the instruction of the transition to HALT and STOP mode.



Transition from PLL mode to STOP mode requires passing through RC mode or OSC mode.



If the generation of the return interrupt factor is not guaranteed after the request for the transition to STANDBY mode (HALT/STOP) by setting of CPUM register, refer to [3.3.4 Note for Transition to STANDBY Modes].

3.3.4 Note for Transition to Standby Modes

When the return interrupt factor is not guaranteed to occur after the request for the transition to standby mode by the setting for the CPUM register, the following process should be performed for not transiting to standby mode after the occurrence of the return interrupt.

```
#Program for transition to standby mode
.
.
    mov  x'08', D1                ; Preparation for setting stop mode
    mov  psw, D0
    or   x'40', D0
    mov  D0, psw                  ; Setting of MIE flag
label_standby:
    mov  D1, (CPUM)              ; Setting of stop mode
.
.

#Program for transition to standby mode
    push D0                      ; Save of register
    push D1                      ; Save of register
    mov  (x'5', sp), D0          ;
    and  x'0F', D0               ; Reduction of half byte information
    cmp  ah(label_standby), D0   ; Comparison of upper address for saving
    bne  next_program
    movw(x'3', sp), DW0          ;
    cmp  al(label_standby), DW0  ; Comparison of lower address for saving
    bne  next_program
    addw x'02', DW0              ; Instruction length of mov dn, (CPUM)
    movw DW0, (x'3', sp)         ; Change of address for saving
next_program:
    pop  D1
    pop  D0
```


Chapter 4 Clock Monitoring Function

4.1 Overview

MN101EF76K/MN101EF57G can monitor the system clock stop or clock frequency error.
NMI interrupt or hardware reset can be selected as the operation performed when a frequency error is detected.

4.1.1 Functions

Table:4.1.1 shows functions for clock monitoring.

Table:4.1.1 Clock Monitoring Functions

Function summary	It detects a higher frequency than the upper limit of the system clock or a lower frequency than the lower limit (including "stop").
Monitoring target	System clock using external frequency or PLL as a clock source.
Reference clock	Internal high-speed frequency : NORMAL mode, HALT0 mode, HALT2 mode Internal low-speed frequency : SLOW mode, HALT1 mode
Monitoring target mode	NORMAL mode / SLOW mode / HALT mode *1
Operation when a frequency error is detected	Selectable from - Hardware reset - Oscillation error interrupt (NMI) generation - Switching system clock to internal oscillation
Error source retention	Retains the clock causing the following error frequencies by the register flags. - External high-speed oscillation - External low-speed oscillation - PLL use
Detection frequency selection	Selectable from the upper/lower limit of the frequency to be detected. *2 NORMAL mode : 30 MHz to 0 Hz SLOW mode : 32 kHz to 0 Hz



- *1
Clock monitoring cannot be performed in the following cases.
- In STOP0 mode or STOP1 mode
 - Internal oscillation is treated as a clock source directly.
 - Internal oscillation for reference clock is stopped.



- *2
Select more than 150 % of the system clock as the upper limit frequency.
Select less than 50 % of the system clock as the lower limit frequency.

4.1.2 Block Diagram

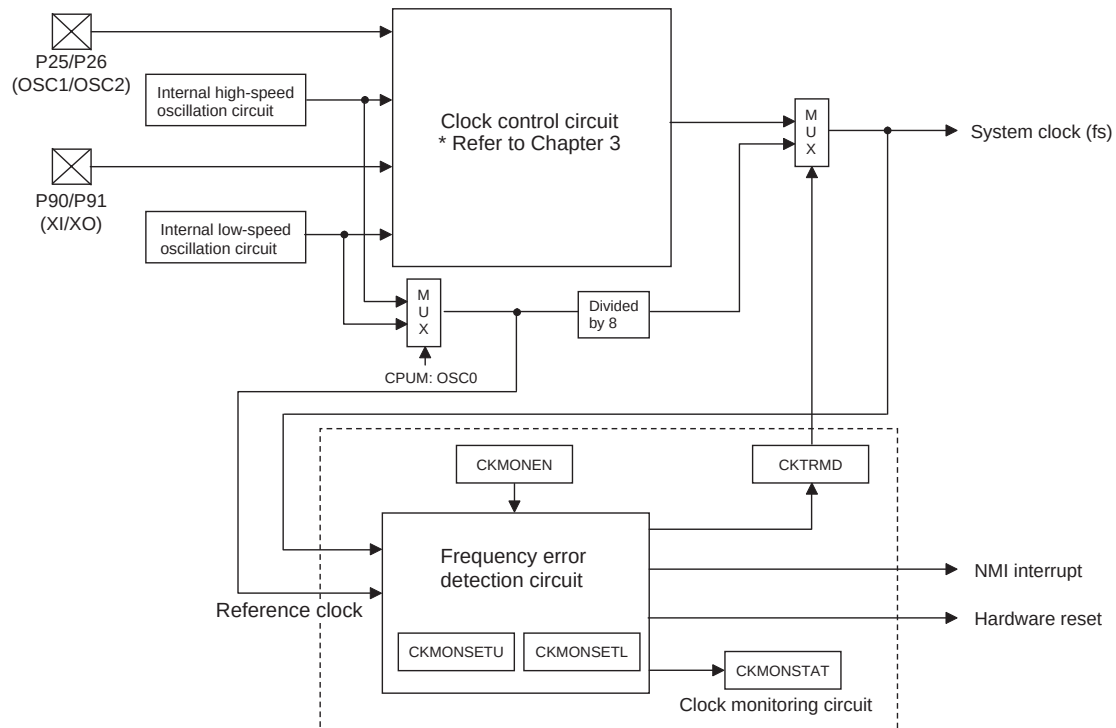


Figure:4.1.1 Clock Monitoring Circuit Block Diagram

4.2 Registers

4.2.1 Clock Monitoring Function Registers

Table:4.2.1 shows the clock monitoring function registers.

Table:4.2.1 Clock Monitoring Function Control Registers

Register	Address	R/W	Function	Page
CKMONEN	0x03E52	R/W	Clock monitoring enable register	IV-5
CKMONSETU	0x03E53	R/W	Clock monitoring frequency setting register (upper limit)	IV-6
CKMONSETL	0x03E54	R/W	Clock monitoring frequency setting register (lower limit)	IV-8
CKMONSTAT	0x03E55	R/W	Clock monitoring status register	IV-9
CKTRMD	0x03E51	R/W	Clock transition mode register	IV-10
PRTKEY	0x03E50	R/W	Register protect control register	V-4

R/W: Readable/Writable

4.2.2 Clock Monitoring Enable Register

Clock monitoring enable register is used to control start/stop of the clock monitoring function. This register is also used to select a behavior at frequency error detection.

■ Clock Monitoring Enable Register (CKMONEN: 0x03E52)

bp	7	6	5	4	3	2	1	0
Flag	-	-	-	-	-	MVCLK	DETIRQ	MONIEN
At reset	-	-	-	-	-	0	0	0
Access	-	-	-	-	-	R/W	R/W	R/W

bp	Flag	Description	Condition
7 to 3	-	-	-
2	MVCLK	Clock switching selection	1: CPU enters the clock transition mode when an error is detected. * Only when selecting a detection interrupt 0: No clock switching
1	DETIRQ	Detection interrupt selection	1: At error detection, an interrupt generates. 0: At error detection, reset generates.
0	MONIEN	Clock monitoring function enable	1: Enable 0: Disable



Before enabling clock monitoring, start internal high-speed oscillation in NORMAL/PLL mode or start internal low-speed oscillation in SLOW mode in advance.



When changing above registers, be sure to access the register protect control register (PRTKEY) to enable writing. Refer to [Chapter 5 Register Protect].



Even if clock monitoring is enabled, it does not work during clock transition mode.



The upper limit frequency in NORMAL mode is a detectable frequency. There is no guarantee of proper operation on the system clock of the microcontroller.

4.2.3 Clock Monitoring Frequency Setting Register

Clock monitoring frequency setting register is used to select upper/lower limit frequency which monitors system clock.

- Clock Monitoring Frequency Upper Limit Setting Register (CKMONSETU: 0x03E53)

bp	7	6	5	4	3	2	1	0
Flag	MNUSET7	MNUSET6	MNUSET5	MNUSET4	MNUSET3	MNUSET2	MNUSET1	MNUSET0
At reset	0	0	0	0	0	0	0	0
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description	Condition																																																																																																																					
7 to 0	MNUSET7 to MNUSET0	Clock monitoring frequency upper limit setting value	<table><tr><th>Setting value</th><th>NORMAL (MHz)</th><th>SLOW (kHz)</th></tr><tr><td>0x00</td><td>30</td><td>32</td></tr><tr><td>0x01</td><td>29</td><td>31</td></tr><tr><td>0x02</td><td>28</td><td>30</td></tr><tr><td>0x03</td><td>27</td><td>29</td></tr><tr><td>0x04</td><td>26</td><td>28</td></tr><tr><td>0x05</td><td>25</td><td>27</td></tr><tr><td>0x06</td><td>24</td><td>26</td></tr><tr><td>0x07</td><td>23</td><td>25</td></tr><tr><td>0x08</td><td>22</td><td>24</td></tr><tr><td>0x09</td><td>21</td><td>23</td></tr><tr><td>0x0a</td><td>20</td><td>22</td></tr><tr><td>0x0b</td><td>19</td><td>21</td></tr><tr><td>0x0c</td><td>18</td><td>20</td></tr><tr><td>0x0d</td><td>17</td><td>19</td></tr><tr><td>0x0e</td><td>16</td><td>18</td></tr><tr><td>0x0f</td><td>15</td><td>17</td></tr><tr><td>0x10</td><td>14</td><td>16</td></tr><tr><td>0x11</td><td>13</td><td>15</td></tr><tr><td>0x12</td><td>12</td><td>14</td></tr><tr><td>0x13</td><td>11</td><td>13</td></tr><tr><td>0x14</td><td>10</td><td>12</td></tr><tr><td>0x15</td><td>9</td><td>11</td></tr><tr><td>0x16</td><td>8</td><td>10</td></tr><tr><td>0x17</td><td>7</td><td>9</td></tr><tr><td>0x18</td><td>6</td><td>8</td></tr><tr><td>0x19</td><td>5</td><td>7</td></tr><tr><td>0x1a</td><td>4</td><td>6</td></tr><tr><td>0x1b</td><td>3</td><td>5</td></tr><tr><td>0x1c</td><td>2</td><td>4</td></tr><tr><td>0x1d</td><td>1</td><td>3</td></tr><tr><td>0x1e</td><td>0.9</td><td>2</td></tr><tr><td>0x1f</td><td>0.8</td><td>1</td></tr><tr><td>0x20</td><td>0.7</td><td rowspan="7">Setting prohibited</td></tr><tr><td>0x21</td><td>0.6</td></tr><tr><td>0x22</td><td>0.5</td></tr><tr><td>0x23</td><td>0.4</td></tr><tr><td>0x24</td><td>0.3</td></tr><tr><td>0x25</td><td>0.2</td></tr><tr><td>0x26</td><td>0.1</td></tr><tr><td>else</td><td>Setting prohibited</td><td></td></tr></table>	Setting value	NORMAL (MHz)	SLOW (kHz)	0x00	30	32	0x01	29	31	0x02	28	30	0x03	27	29	0x04	26	28	0x05	25	27	0x06	24	26	0x07	23	25	0x08	22	24	0x09	21	23	0x0a	20	22	0x0b	19	21	0x0c	18	20	0x0d	17	19	0x0e	16	18	0x0f	15	17	0x10	14	16	0x11	13	15	0x12	12	14	0x13	11	13	0x14	10	12	0x15	9	11	0x16	8	10	0x17	7	9	0x18	6	8	0x19	5	7	0x1a	4	6	0x1b	3	5	0x1c	2	4	0x1d	1	3	0x1e	0.9	2	0x1f	0.8	1	0x20	0.7	Setting prohibited	0x21	0.6	0x22	0.5	0x23	0.4	0x24	0.3	0x25	0.2	0x26	0.1	else	Setting prohibited	
			Setting value	NORMAL (MHz)	SLOW (kHz)																																																																																																																			
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A frequency which is faster than the upper limit frequency of the above register is considered as an error (having a margin of error).



When upper limit frequency = system clock, it is not considered as an error.



When changing above registers, be sure to access register protect control register (PRTKEY) to enable writing. Refer to [Chapter 5 Register Protect].



If changing upper limit frequency in other than clock transition mode, stop the clock monitoring function in advance.



The upper limit frequency in NORMAL mode is a detectable frequency. There is no guarantee of proper operation on the system clock of the microcontroller.

■ Clock Monitoring Frequency Lower Limit Setting Register (CKMONSETL: 0x03E54)

bp	7	6	5	4	3	2	1	0
Flag	MNLSET7	MNLSET6	MNLSET5	MNLSET4	MNLSET3	MNLSET2	MNLSET1	MNLSET0
At reset	0	0	0	0	0	0	0	0
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description	Condition																																																									
7 to 0	MNLSET7 to MNLSET0	Clock monitoring frequency lower limit setting value	<table><tr><th>Setting value</th><th>NORMAL (MHz)</th><th>SLOW (kHz)</th></tr><tr><td>0x00</td><td>0</td><td>0</td></tr><tr><td>0x01</td><td>0.1</td><td>1</td></tr><tr><td>0x02</td><td>0.2</td><td>2</td></tr><tr><td>0x03</td><td>0.3</td><td>3</td></tr><tr><td>0x04</td><td>0.4</td><td>4</td></tr><tr><td>0x05</td><td>0.5</td><td>5</td></tr><tr><td>0x06</td><td>0.6</td><td>6</td></tr><tr><td>0x07</td><td>0.7</td><td>7</td></tr><tr><td>0x08</td><td>0.8</td><td>8</td></tr><tr><td>0x09</td><td>0.9</td><td>9</td></tr><tr><td>0x0a</td><td>1</td><td rowspan="10">Setting prohibited</td></tr><tr><td>0x0b</td><td>2</td></tr><tr><td>0x0c</td><td>3</td></tr><tr><td>0x0d</td><td>4</td></tr><tr><td>0x0e</td><td>5</td></tr><tr><td>0x0f</td><td>6</td></tr><tr><td>0x10</td><td>7</td></tr><tr><td>0x11</td><td>8</td></tr><tr><td>0x12</td><td>9</td></tr><tr><td>0x13</td><td>10</td></tr><tr><td>else</td><td>Setting prohibited</td><td></td></tr></table>	Setting value	NORMAL (MHz)	SLOW (kHz)	0x00	0	0	0x01	0.1	1	0x02	0.2	2	0x03	0.3	3	0x04	0.4	4	0x05	0.5	5	0x06	0.6	6	0x07	0.7	7	0x08	0.8	8	0x09	0.9	9	0x0a	1	Setting prohibited	0x0b	2	0x0c	3	0x0d	4	0x0e	5	0x0f	6	0x10	7	0x11	8	0x12	9	0x13	10	else	Setting prohibited	
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A frequency which is slower than the lower limit frequency of the above register is considered as an error (having a margin of error).



When lower limit frequency = system clock, it is not considered as an error.



When changing above registers, be sure to access register protect control register (PRTKEY) to enable writing. Refer to [Chapter 5 Register Protect].



If changing lower limit frequency in other than clock transition mode, stop the clock monitoring function in advance.

4.2.4 Clock Monitoring Status Register

Clock monitoring status register is used to retain the system clock information selected at the error generation. By reading this register after generating NMI interrupt or hardware reset, it can be confirmed which oscillation had an error.

■ Clock Monitoring Status Register (CKMONSTAT: 0x03E55)

bp	7	6	5	4	3	2	1	0
Flag	-	-	-	-	-	PLLERR	XIERR	OSCERR
At pin reset (Initial state)	-	-	-	-	-	0	0	0
At software reset *	-	-	-	-	-	Retained	Retained	Retained
Access	-	-	-	-	-	R/(W)	R/(W)	R/(W)

bp	Flag	Description	Condition
7 to 3	-	-	-
2	PLLERR	Flag using PLL	1: Error is generated in PLL mode 0: Normal
1	XIERR	Flag using external low-speed	1: Error is generated in using external low-speed oscillation 0: Normal
0	OSCERR	Flag using external high-speed	1: Error is generated in using external high-speed oscillation 0: Normal



The system clock information selected when a frequency error is detected is set in the above flags. The selected information of the system clock cannot always be monitored.



Software reset is the reset by P2OUT7 register, the reset by overflow of Watchdog timer, or the reset at clock error detection.



The above register must be cleared by reset from the pin, or by writing an appropriate value. When changing above registers, be sure to access register protect control register (PRTKEY) to enable writing. Refer to [Chapter 5 Register Protect].



If an frequency error is detected in external high-speed oscillation use or PLL mode, both the PLLERR and OSCERR flags are set.

4.2.5 Clock Transition Mode Register

Clock transition mode register is used to set the clock transition mode, in which the system clock is forced to operate with internal oscillation. When the clock operation state changes to the clock transition mode, all of the clock settings of the clock control circuit and the clock monitoring settings cannot be reflected in the system clock.

■ Clock Transition Mode Register (CKTRMD: 0x03E51)

bp	7	6	5	4	3	2	1	0
Flag	-	-	-	-	-	-	-	TRMD
At reset	-	-	-	-	-	-	-	0
Access	-	-	-	-	-	-	-	R/W

bp	Flag	Description	Condition
7 to 1	-	-	-
0	TRMD	Clock transition mode	1: Clock transition mode 0: Normal mode



When changing above registers, be sure to access register protect control register (PRTKEY) to enable writing. Refer to [Chapter 5 Register Protect].



Unless the clock transition mode is released, the settings of the clock switching and clock monitoring are not valid.



In the clock transition mode, the system clock is operated with internal oscillation. When the internal oscillation is stopped, transition must not be performed.



If the clock operation state changes to the clock transition mode at error detection, the TRMD flag is set by hardware.

4.3 Operation

4.3.1 Operation in Each Mode

- NORMAL mode
 - Clock monitoring can be performed only when the external high-speed oscillation (fosc) is used as the system clock.
 - Clock monitoring settings should not be enabled when the internal high-speed oscillation (frc) is used.
 - Clock monitoring cannot be performed when the internal high-speed oscillation (frc) is stopped.
 - At the transition to SLOW or PLL mode, stop the clock monitoring function or transit via the clock transition mode.
- PLL mode
 - Clock monitoring can be performed even when either of the external high-speed oscillation (fosc) or internal high-speed oscillation (frc) is used as a clock source.
 - When the internal high-speed oscillation (frc) is used as a clock source, only the errors of the PLL circuit can be monitored. (Stop state or an error of the internal high-speed oscillation (frc) itself cannot be monitored.)
 - Clock monitoring cannot be performed when the internal high-speed oscillation (frc) is stopped.
 - At the transition to NORMAL mode, stop the clock monitoring function or transit via clock transition mode.
- SLOW mode
 - Clock monitoring can be performed only when the external low-speed oscillation (fx) is used as the system clock.
 - Clock monitoring cannot be performed when the internal low-speed oscillation (frcs) is stopped. Do not set the clock monitoring.
 - At the transition to NORMAL mode, stop the clock monitoring function or transit via clock transition mode.
- HALT mode
 - Clock monitoring can be performed in HALT0/ 1, 2 mode, except HALT0/1 (frc), when the internal high-speed oscillation (frc) is used as a clock source.
 - Clock monitoring does not need to be set anew in changing to HALT mode. (The clock monitoring set just before the transition to HALT mode is applied without change.)
 - When an clock error is detected in HALT mode and an error interrupt is selected, LSI is forced to return from the standby mode and performs interrupt processing.
- STOP mode
 - Clock monitoring cannot be performed.
- Clock transition mode
 - Clock monitoring can be performed with internal oscillation (NORMAL: internal high-speed oscillation (frc) divided by 8, SLOW: Internal low-speed oscillation (frcs) divided by 8).
 - Even if clock switching, clock dividing, and PLL use are set in the clock transition mode, the settings are not available.
 - When the mode transition between NORMAL and SLOW is performed, the operation clock is switched between internal high-speed oscillation (frc) and internal low-speed oscillation (frcs).



To activate the clock monitoring function, start internal oscillation (NORMAL: internal high-speed oscillation (frc), SLOW: internal low-speed oscillation (frcs) in advance. For the operating procedure of the internal high-speed oscillation and internal low-speed oscillation, refer to [Chapter 3 Clock Control].

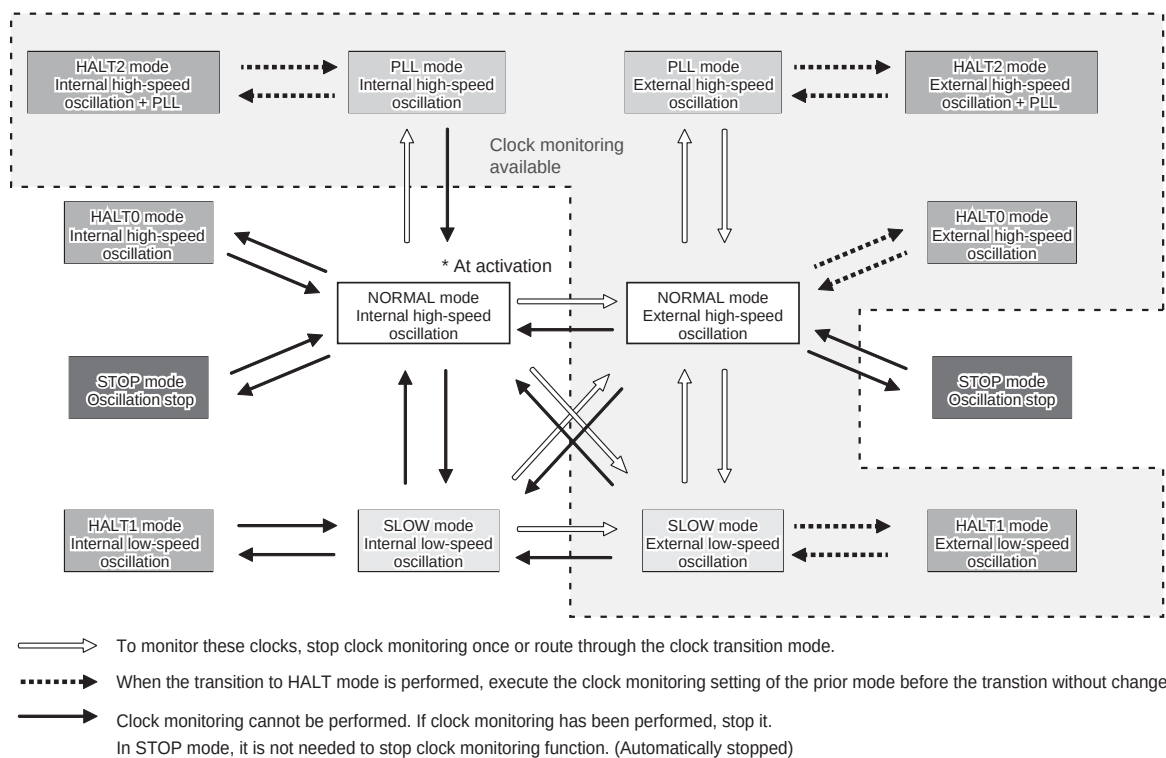


Figure:4.3.1 Mode Transition Diagram and Clock Monitoring Enable/Disable

4.3.2 Setup Example

■ Setup Example Between From Activation To Clock Setting: Clock Transition Mode is Not Used.

After activating this LSI, finish the clock setting and then perform clock monitoring. Check the flags of the clock monitoring status register at activation to confirm whether the LSI has restarted by a clock frequency error generation. The operating condition can be changed depending on the flags.

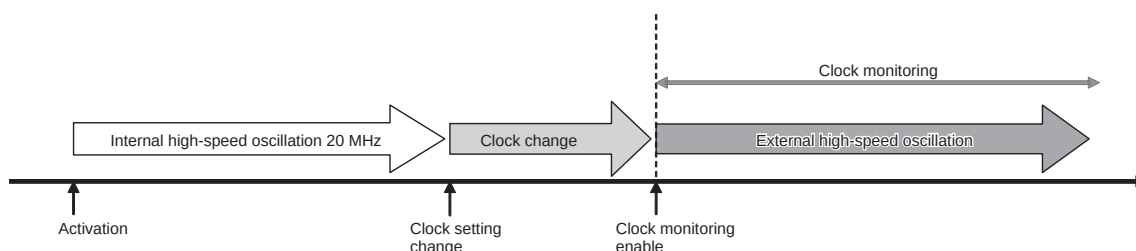


Figure:4.3.2 Clock Monitoring State Diagram (1)

The clock monitoring settings as shown in the setup example is follows.

Mode : NORMAL mode
 Upper limit frequency : 10 MHz
 Lower limit frequency : 1 MHz
 Operation on the error detection : Hardware reset

Setup Procedure	Description
(1) Disable interrupts * Recommended PSW bp6: MIE = 0 (2) Check the clock state CKMONSTAT (0x03E55) bp2: PLLERR bp1: XIERR bp0: OSCERR	(1) During operations of the clock settings and mode change, it is recommended to disable the other interrupts. If not, the operation speed inside the interrupt routine may be changed because the clock is being set. (2) Read out the clock monitoring status register. When the register is under the following condition, LSI has restarted after hardware reset caused by the error status of the corresponding oscillation output. It can be controlled to avoid selecting abnormal clocks by software. [PLLERR, XIERR, OSCERR] = [0,0,1] : Frequency error is generated at using external high-speed oscillation (PLL is not used) and the clock is restarted. [PLLERR, XIERR, OSCERR] = [1,0,1] : Frequency error is generated at using external high-speed oscillation (PLL is used) and the clock is restarted. [PLLERR, XIERR, OSCERR] = [1,0,0] : Frequency error is generated at using internal high-speed oscillation (PLL is not used) and the clock is restarted. [PLLERR, XIERR, OSCERR] = [0,1,0] : Frequency error is generated at using external low-speed oscillation and the clock is restarted. [PLLERR, XIERR, OSCERR] = [0,0,0] : Normal activation

Setup Procedure	Description
(3) Configure the clock settings OSCCNT RCCNT PLLCNT	(3) Change the clock settings. (Refer to [Chapter 3 Clock Control] for details.) - Select internal oscillation or external oscillation - Select PLL use or not
(4) Set the clock monitoring upper limit frequency PRTKEY (0x03E50) bp7 to 0: PRTK7 to 0 = 0x55 CKMONSETU (0x03E53) bp7 to 0: MNUSET7 to 0 = 0x14	(4) Set the upper limit of the clock frequency. Be sure to write "0x55" in PRTKEY to enable writing. Set "0x14" in CKMONSETU to set the upper limit frequency to 10 MHz.
(5) Set the clock monitoring lower limit frequency PRTKEY (0x03E50) bp7 to 0: PRTK[7:0] = 0x77 CKMONSETL (0x03E54) bp7 to 0: MNLSET7 to 0 = 0x0A	(5) Set the lower limit of the clock frequency. Be sure to write "0x77" in PRTKEY to enable writing. Set "0x0A" in CKMONSETL to set the lower limit frequency to 1 MHz.
(6) Clock monitoring function enable PRTKEY (0x03E50) bp7 to 0: PRTK7 to 0 = 0x33 CKMONEN (0x03E52) bp2: MVCLK = 0 bp1: DTIRQ = 0 bp0: MONIEN = 1	(6) Enable the clock monitoring function and select the action at clock error detection. Be sure to write "0x33" in PRTKEY to enable writing. Set "0x01" in CKMONEN to enable reset at error detection and clock monitoring function. Immediately, the clock monitoring function becomes valid.
(7) Set the register protect PRTKEY (0x03E50) bp7 to 0: PRTK7 to 0 = 0xFF	(7) Write any value (0xFF) in PRTKEY to set the register protect again.
(8) Enable interrupts PSW bp6: MIE = 1	(8) Enable interrupts.

■ Setup Example: Clock Transition Mode is Not Used.

When changing mode, stop the clock monitoring function once.

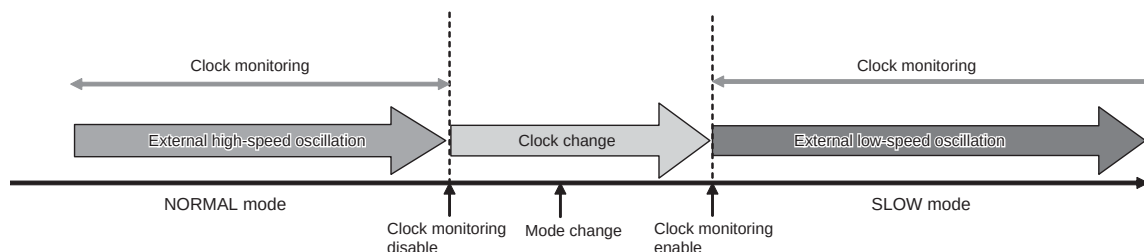


Figure:4.3.3 Clock Monitoring State Diagram (2)

The clock monitoring settings as shown in the setup example is follows.

Mode : NORMAL mode to SLOW mode
 Upper limit frequency : 27 kHz
 Lower limit frequency : 1 kHz
 Operation on the error detection : NMI interrupt generation

Setup Procedure	Description
(1) Disable interrupts * Recommended PSW bp6: MIE=0	(1) During operations of the clock settings and mode change, it is recommended to disable the other interrupts. If not, the operation speed inside the interrupt routine may be changed because the clock is being set.
(2) Stop the clock monitoring function PRTKEY (0x03E50) bp7 to 0: PRTK7 to 0 = 0x33 CKMONEN (0x03E52) bp0: MONIEN=0	(2) Stop the clock monitoring function if it is used. Be sure to write "0x33" in PRTKEY to enable writing.
(3) Execute clock setting SOSCCNT RCSCNT	(3) Change the clock setting. (Refer to [Chapter 3 Clock Control] for details.) - Select external oscillation.
(4) Execute mode setting CPUM	(4) Set the transition of the SLOW mode. (Refer to [Chapter 3 Clock Control] for details.)
(5) Set the clock monitoring upper limit frequency PRTKEY (0x03E50) bp7 to 0: PRTK7 to 0 = 0x55 CKMONSETU (0x03E53) bp7 to 0: MNUSET7 to 0 = 0x05	(5) Set the upper limit of the clock frequency. Be sure to write "0x55" in PRTKEY to enable writing. Set "0x05" in CKMONSETU to set the upper limit frequency to 27 kHz.
(6) Set the clock monitoring lower limit frequency PRTKEY (0x03E50) bp7 to 0: PRTK7 to 0 = 0x77 CKMONSETL(0x03E54) bp7 to 0: MNLSET7 to 0 = 0x01	(6) Set the lower limit of the clock frequency. Be sure to write "0x77" in PRTKEY to enable writing. Set "0x01" in CKMONSETL to set the lower limit frequency to 1 kHz.

Setup Procedure	Description
(7) Clock monitoring function enable PRTKEY (0x03E50) bp7 to 0: PRTK7 to 0 = 0x33 CKMONEN (0x03E52) bp2: MVCLK = 0 bp1: DTIRQ = 1 bp0: MONIEN = 1 (8) Set the register protect PRTKEY (0x03E50) bp7 to 0: PRTK7 to 0 = 0xFF (9) Enable interrupts PSW bp6: MIE = 1	(7) Enable the clock monitoring function and select the action at clock error detection. Be sure to write "0x33" in PRTKEY to enable writing. Set "0x03" in CKMONEN to select the error detection interrupt and enable the clock monitoring function. (8) Write any value (0xFF) in PRTKEY to set the register protect again. (9) Enable interrupts.

■ Setup Example Between From Activation To Clock Setting: Clock Transition Mode is Used.

After this LSI's activation, the CPU enters clock transition mode to perform clock setting. After completion of all the settings of clock change/monitoring, the clock transition mode is released. If clock settings are performed during clock transition mode, the period in which oscillation errors cannot be monitored is not generated because the LSI always operates with internal oscillation. Check the flags of the clock monitoring status register at activation to confirm whether the LSI has restarted by a clock frequency error generation. The operating condition can be changed depending on the flags.

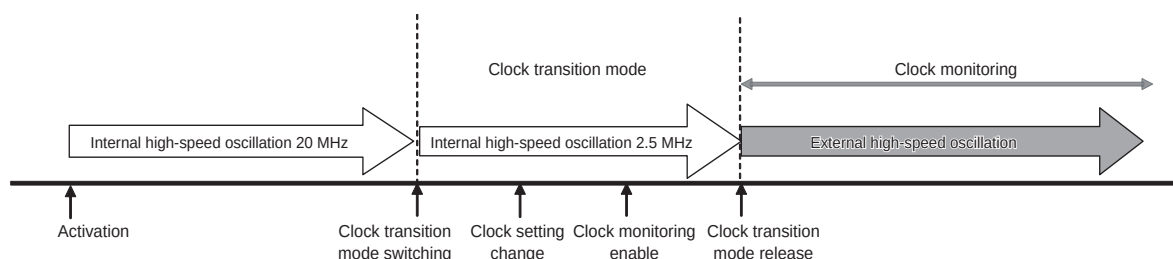


Figure:4.3.4 Clock Monitoring State Diagram (3)

The clock monitoring settings as shown in the setup example is follows.

Mode : NORMAL mode

Upper limit frequency : 10 MHz

Lower limit frequency : 1 MHz

Operation on the error detection : Hardware reset

Setup Procedure	Description
(1) Disable interrupts * Recommended PSW bp6: MIE = 0 (2) Set the clock transition mode PRTKEY (0x03E50) bp7 to 0: PRTK7 to 0 = 0x11 CKTRMD (0x03E51) bp0: TRMD = 1	(1) During operations of the clock settings and mode change, it is recommended to disable the other interrupts. If not, the operation speed inside the interrupt routine may be changed because the clock is being set. (2) Transit to the clock transition mode. Be sure to write "0x11" in PRTKEY to enable writing. The operation clock during clock transition mode is always internal oscillation, and the frequency is fixed. NORMAL model: Internal high-speed oscillation divided by 8 (20 MHz: 2.5 MHz, 16 MHz: 2 MHz) SLOW model: Internal low-speed oscillation divided by 8 (3.75 kHz)

Setup Procedure	Description
(3) Check the clock state CKMONSTAT (0x03E55) bp2: PLLERR bp1: XIERR bp0: OSCERR	(3) Read out the clock monitoring status register. When the register is under the following condition, LSI has restarted after hardware reset caused by the error status of the corresponding oscillation output. It can be controlled to avoid selecting abnormal clocks by software. [PLLERR, XIERR, OSCERR] = [0,0,1] : Frequency error is generated at using external high-speed oscillation (PLL is not used) and the clock is restarted. [PLLERR, XIERR, OSCERR] = [1,0,1] : Frequency error is generated at using external high-speed oscillation (PLL is used) and the clock is restarted. [PLLERR, XIERR, OSCERR] = [1,0,0] : Frequency error is generated at using internal high-speed oscillation (PLL is not used) and the clock is restarted. [PLLERR, XIERR, OSCERR] = [0,1,0] : Frequency error is generated at using external low-speed oscillation and the clock is restarted. [PLLERR, XIERR, OSCERR] = [0,0,0] : Normal activation
(4) Configure the clock settings OSCCNT RCCNT PLLCNT	(4) Change the clock settings (For details, refer to Chapter 3 clock control) - Select internal oscillation or external oscillation - Select PLL use or not Clock cannot be switched during clock transition mode even if selecting the above action.
(5) Set the clock monitoring upper limit frequency PRTKEY (0x03E50) bp7 to 0: PRTK7 to 0 = 0x55 CKMONSETU (0x03E53) bp7 to 0: MNUSET7 to 0 = 0x14	(5) Set the upper limit of the clock frequency. Be sure to write "0x55" in PRTKEY to enable writing. Set "0x14" in CKMONSETU to set the upper limit frequency to 10 MHz.
(6) Set the clock monitoring lower limit frequency PRTKEY (0x03E50) bp7 to 0: PRTK7 to 0 = 0x77 CKMONSETL (0x03E54) bp7 to 0: MNLSET7 to 0 = 0x0A	(6) Set the lower limit of the clock frequency. Be sure to write "0x77" in PRTKEY to enable writing. Set "0x0A" in CKMONSETL to set the lower limit frequency to 1 MHz.
(7) Clock monitoring function enable PRTKEY (0x03E50) bp7 to 0: PRTK7 to 0 = 0x33 CKMONEN (0x03E52) bp2: MVCLK = 0 bp1: DTIRQ = 0 bp0: MONIEN = 1	(7) Enable the clock monitoring function and select the action at clock error detection. Be sure to write "0x33" in PRTKEY to enable writing. Set "0x01" in CKMONEN to enable reset at error detection and clock monitoring function. Immediately, the clock monitoring function becomes valid.
(8) Release the clock transition mode PRTKEY (0x03E50) bp7 to 0: PRTK7 to 0 = 0x11 CKTRMD bp0: TRMD = 0	(8) Release the clock transition mode. Be sure to write "0x11" in PRTKEY to enable writing. All of the clock settings and clock monitoring function become valid.
(9) Set the register protect PRTKEY (0x03E50) bp7 to 0: PRTK7 to 0 = 0xFF	(9) Write any value (0xFF) in PRTKEY to set the register protect again.
(10) Enable interrupts PSW bp6: MIE = 1	(10) Enable interrupts.

■ Setup Example: Clock Transition Mode is Used.

When using clock transition mode, it is not necessary to stop the clock monitoring function. Monitoring oscillation errors can always be performed.

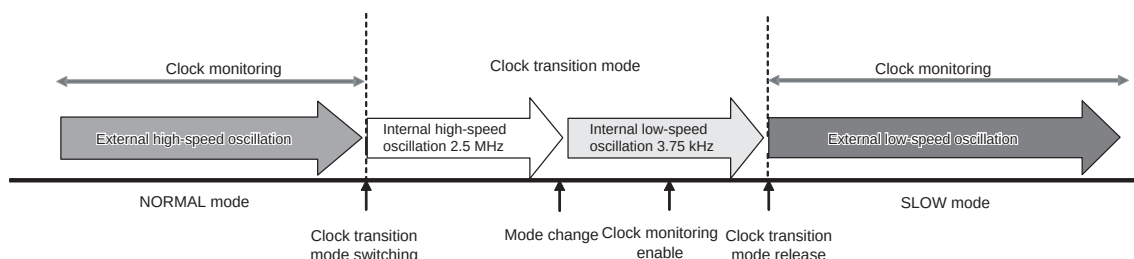


Figure:4.3.5 Clock Monitoring State Diagram (4)

The clock monitoring settings as shown in the setup example is follows.

Mode : NORMAL mode to SLOW mode
 Upper limit frequency : 27 kHz
 Lower limit frequency : 15 kHz
 Operation on the error detection : Enters clock transition mode and NMI interrupt generation

Setup Procedure	Description
(1) Disable interrupts * Recommended PSW bp6: MIE = 0 (2) Set the clock transition mode PRTKEY (0x03E50) bp7 to 0: PRTK7 to 0 = 0x11 CKTRMD (0x03E51) bp0: TRMD = 1 (3) Execute clock setting SOSCNT RCSCNT (4) Execute mode setting CPUM	(1) During operations of the clock settings and mode change, it is recommended to disable the other interrupts. If not, the operation speed inside the interrupt routine may be changed because the clock is being set. (2) Change to the clock transition mode. Be sure to write "0x11" in PRTKEY to enable writing. Operation clock during clock transition mode is always internal oscillation, and the frequency is fixed. NORMAL mode: Internal high-speed oscillation divided by 8 (20 MHz: 2.5 MHz, 16 MHz: 2 MHz) SLOW mode : Internal low-speed oscillation divided by 8 (3.75 kHz) (3) Change the clock setting. - Select external oscillation. Clock cannot be switched during clock transition mode even if selecting above action. (4) Set the transition of the SLOW mode. After setting, immediately, internal high-speed oscillation (20 MHz: 2.5 MHz, 16 MHz: 2 MHz) is switched to the internal low-speed oscillation (3.75 kHz).

Setup Procedure	Description
(5) Set the clock monitoring upper limit frequency PRTKEY (0x03E50) bp7 to 0: PRTK7 to 0 = 0x55 CKMONSETU (0x03E53) bp7 to 0: MNUSET7 to 0 = 0x05	(5) Set the upper limit of the clock frequency. Be sure to write "0x55" in PRTKEY to enable writing. Set "0x05" in CKMONSETU to set the upper limit frequency to 27 kHz.
(6) Set the clock monitoring lower limit frequency PRTKEY (0x03E50) bp7 to 0: PRTK7 to 0 = 0x77 CKMONSETL (0x03E54) bp7 to 0: MNLSET7 to 0 = 0x0F	(6) Set the lower limit of the clock frequency. Be sure to write "0x77" in PRTKEY to enable writing. Set "0x0F" in CKMONSETL to set the lower limit frequency to 15 kHz.
(7) Clock monitoring function enable PRTKEY (0x03E50) bp7 to 0: PRTK7 to 0 = 0x33 CKMONEN (0x03E52) bp2: MVCLK = 0 bp1: DTIRQ = 1 bp0: MONIEN = 1	(7) Enable the clock monitoring function and select the action at clock error detection. Be sure to write "0x33" in PRTKEY to enable writing. Set "0x03" in CKMONEN to select the error detection interrupt and enable the clock monitoring function. However, the clock monitoring function does not operate during clock transition mode.
(8) Release the clock transition mode PRTKEY (0x03E50) bp7 to 0: PRTK7 to 0 = 0x11 CKTRMD bp0: TRMD = 0	(8) Release the clock transition mode. Be sure to write "0x11" in PRTKEY to enable writing. All of the clock settings and clock monitoring function become valid.
(9) Set the register protect PRTKEY (0x03E50) bp7 to 0: PRTK7 to 0 = 0xFF	(9) Write any value (0xFF) in PRTKEY to set the register protect again.
(10) Enable interrupts PSW bp6: MIE=1	(10) Enable interrupts.

Chapter 5 Register Protect

5.1 Overview

This LSI has a register protect function to prevent writing errors to the control registers as fail-safe functions (watchdog timer, watchdog timer 2 and clock monitoring function).

If the registers to be protected are written without writing a specified value to the register protect control register, NMI interrupt is generated.

5.2 Control Registers

5.2.1 Registers List

The following table shows the list of registers which related the register protect functions.

Table:5.2.1 Register Protect Function Register

Register	Address	R/W	Function	Page
PRTKEY	0x03E50	R/W	Register protect control register	V-4
WDCTR	0x03F02	R/W	Watchdog timer control register	XV-5
WD2CTR	0x03F05	R/W	Watchdog timer 2 control register	XVI-5
WD2MSEL	0x03F4C	R/W	Watchdog timer 2 operation mode setup register	XVI-6
WD2CLR	0x03F4D	R/W	Watchdog timer 2 clear register	XVI-8
CKMONEN	0x03E52	R/W	Clock monitoring enable register	IV-5
CKMONSETU	0x03E53	R/W	Clock monitoring frequency setup register (upper limit)	IV-6
CKMONSETL	0x03E54	R/W	Clock monitoring frequency setup register (lower limit)	IV-8
CKMONSTAT	0x03E55	R/W	Clock monitoring status register	IV-9
CKTRMD	0x03E51	R/W	Clock transition mode register	IV-10

R/W: Readable/Writable

5.2.2 Register Protect Control Register

The register protect control register enables the writing to the registers to be protected (watchdog timer, watchdog timer 2 and the registers of clock monitoring function).

If the registers to be protected are accessed without writing a defined value to this register, it is judged an error writing and NMI interrupt is generated.

■ Register Protect Control Register (PRTKEY: 0x03E50)

bp	7	6	5	4	3	2	1	0
Flag	PRTK7	PRTK6	PRTK5	PRTK4	PRTK3	PRTK2	PRTK1	PRTK0
At reset	0	0	0	0	0	0	0	0
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description
7 to 0	PRTK7 to PRTK0	Register protect control 0x44: Enable writing to WDCTR register 0x66: Enable writing to WD2CTR register 0x88: Enable writing to WD2MSEL register 0xaa: Enable writing to WD2CLR register 0x11: Enable writing to CKTRMD register 0x33: Enable writing to CKMONEN register 0x55: Enable writing to CKMONSETU register 0x77: Enable writing to CKMONSETL register 0x99: Enable writing to CKMONSTAT register else: Disable writing to the registers to be protected



After the writing to the registers to be protected is enabled in PRTKEY register and that register is changed, it is recommended to write an arbitrary value to PRTKEY register to and to set the registers to be protected to “writing disable”.



We recommend that an interrupt is disabled between PRTKEY writing and appropriate register writing. If an interrupt is not be disable, save a value of PRTKEY at the front of the interrupt processing program, and recover it at the end of the program.
Refer to [Chapter 5 5.3.3 Points of an Interrupt Generation].



Refer to chapters of each function block for the specification of the registers to be protected ([Chapter 4 Clock Monitoring Function], [Chapter 15 Watchdog Timer] and [Chapter 16 Watchdog Timer 2]).

5.3 Operation

5.3.1 Operation

The register protect function enables the writing by writing a normal value, allocated to the registers to be protected, to the register protect control register.

If the registers to be protected are written without writing a normal value to the register protect control register, NMI interrupt is generated.

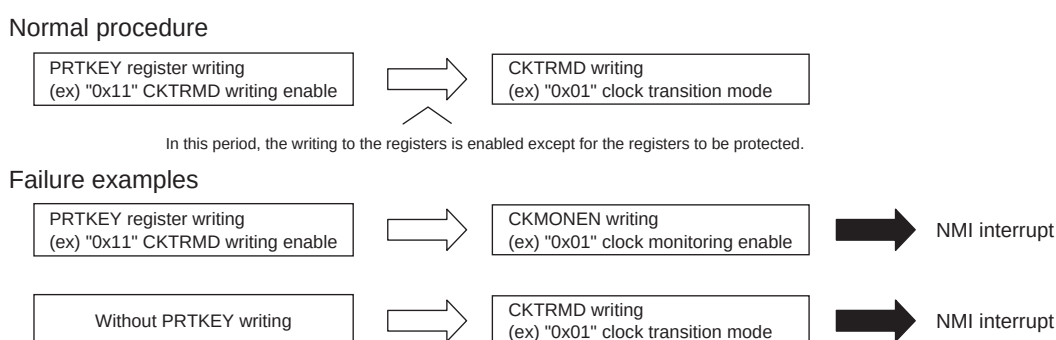


Figure:5.3.1 Register Protect Operation

5.3.2 Setup Example

A setup example of the registers to be protected (watchdog timer operation startup) is shown below.

Setup Procedure	Description
(1) Disable interrupt *recommended PSW bp6: MIE = 0	(1) An interrupt disable is recommended when changing the registers to be protected.
(2) Release the register protect PRTKEY (0x03E50) bp7 to 0: PRTK7 to 0 = 0x44	(2) Enable the writing the watchdog timer control register (WDCTR).
(3) Start the watchdog timer operation WDCTR (0x3F02) bp0: WDEN = 1	(3) Start the watchdog timer operation.
(4) Register protect setting PRTKEY (0x03E50) bp7 to 0: PRTK7 to 0 = 0xFF	(4) Write an arbitrary value (0xFF) to the register protect control register and set the protect setting again.
(5) Enable interrupt PSW bp6: MIE = 1	(5) Enable interrupt.

5.3.3 Points of an Interrupt Generation

If an interrupt is generated during writing to the register protect control register and the registers to be protected, NMI interrupt.

As shown Figure:5.3.2, save a value of the register protect control register at the front of the interrupt processing program and recover it at the end of the program.

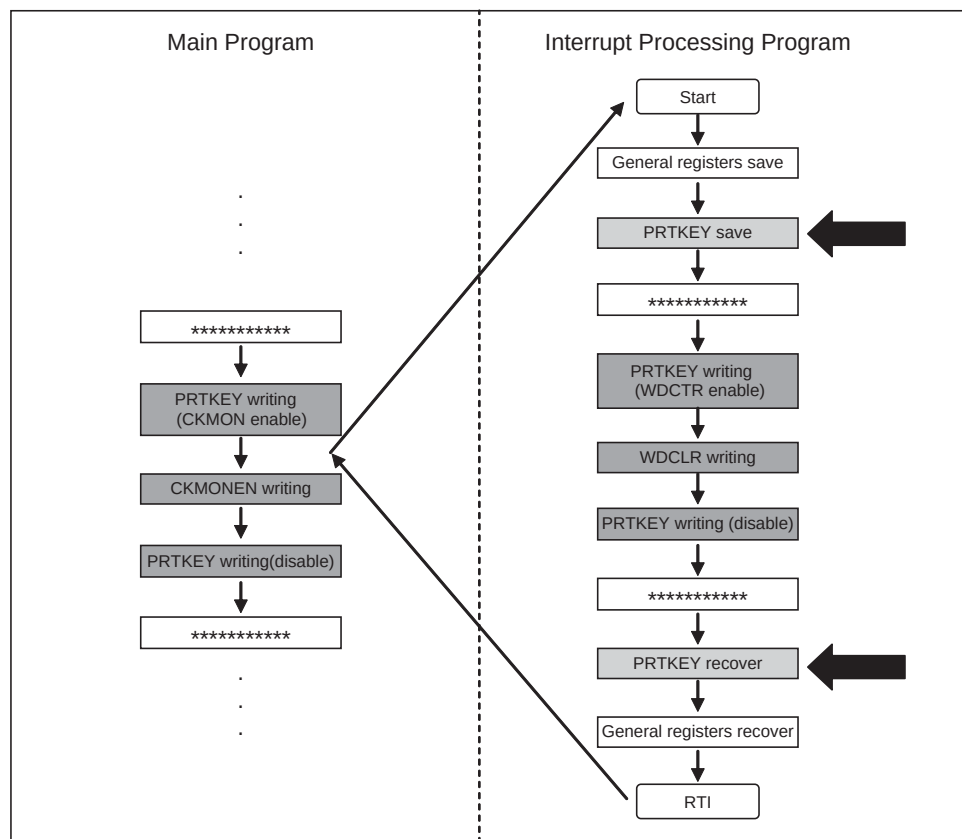


Figure:5.3.2 Example of Interrupt Processing Program

Chapter 6 Power Supply Voltage Detection

6

6.1 Overview

6.1.1 Power Supply Voltage Detection Overview

This LSI includes a set of power supply voltage detection circuit to detect power supply voltage and generate interrupts.

The power supply voltage detection function compares power supply (V_{DD5}) and detection voltage (V_{LVIH} , V_{LVIL}) to generate interrupts (LVIIRQ). The detection voltage (V_{LVIH} , V_{LVIL}) can be changed between from 2.2 V to 4.0 V (at rising) by the setting of the power supply detection control register (LVIMD). The value of falling detection voltage (V_{LVIL}) drops 0.1 V of that of the rising detection voltage (V_{LVIH}).

This function operates regardless of the value of the CPU control register.

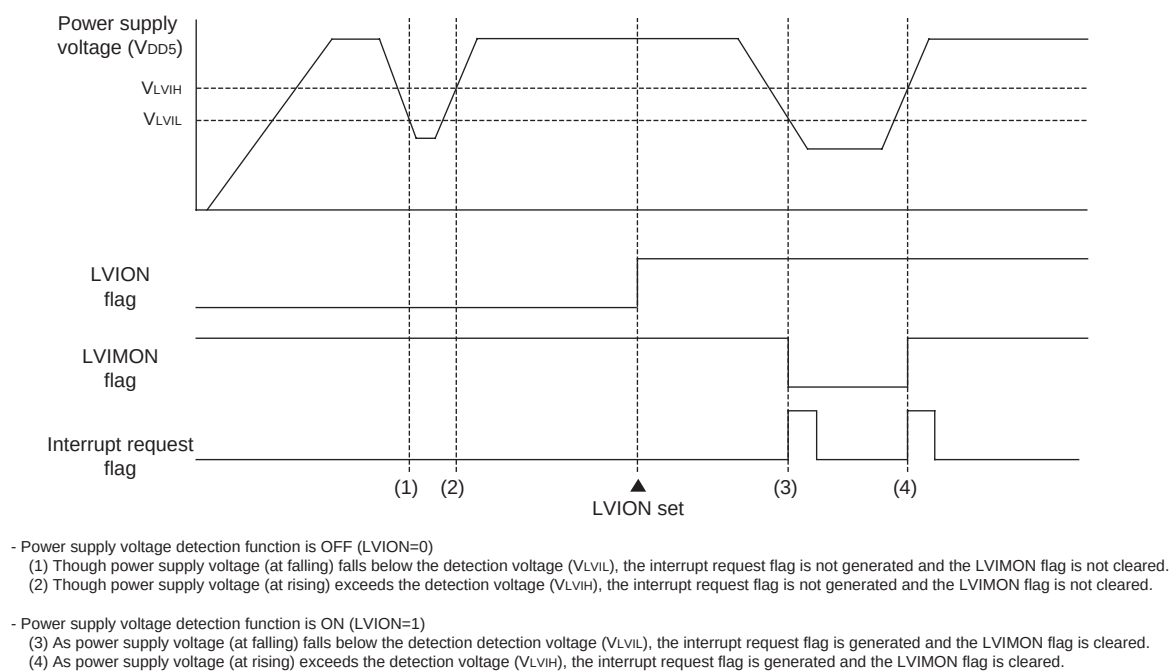


Figure:6.1.1 Power Supply Detection Waveform

6.2 Registers

6.2.1 Power Supply Detection Control Registers

■ Power Supply Detection Control Registers (LVIMD: 0x03F2F)

bp	7	6	5	4	3	2	1	0
Flag	LVIMOD	LVICTR2	LVICTR1	LVICTR0	-	-	LVIMON	LVION
At reset	0	0	0	0	-	-	1	0
Access	R/W	R/W	R/W	R/W	-	-	R	R/W

bp	Flag	Description
7 to 4	LVIMOD LVICTR2 LVICTR1 LVICTR0	Power supply voltage detection voltage setting Detection voltage at rising (standard) 0111 :4.0 V 0110 :4.0 V 0101 :4.0 V 0100 :3.8 V 0011 :3.6 V 0010 :3.4 V 0001 :3.2 V 0000 :3.0 V 1000 :3.0 V 1001 :2.8 V 1010 :2.6 V 1011 :2.4 V 1100 :2.2 V 1101 :2.2 V 1110 :2.2 V 1111 :2.2 V
3 to 2	-	-
1	LVIMON	Power supply voltage monitor flag 0: $V_{DD5} \leq$ detection voltage 1: $V_{DD5} >$ detection voltage
0	LVION	Power supply detection enable 0: Power supply detection function disabled 1: Power supply detection function enabled



The detection voltage which can be monitored by the LVION flag is the voltage detected by the power supply detection circuit when the register is read. It is not the voltage detected when the power supply detection interrupt is generated.



After the LVIMON flag is set, wait more than 2.0 ms to enable the power supply voltage detection interrupt (LVIICR register) or read the LVIMON flag.



The LVIMOD flag and the LVICTR0 to LVICTR2 flags must be set when the LVION flag is "0".

6.3 Setting Example

6.3.1 Power Supply Voltage Detection Setting Example

■ Mode Trans i on Operation with Power Supply Voltage Detection Setting Example

When the power supply voltage exceeds 3.8 V using an interrupt generated by the power supply voltage detection function, the operating mode transits from STOP mode to NORMAL mode.

An example setup procedure, with a description of each step is shown below.

Setup Procedure	Description
(1) Disable all maskable interrupts. PSW bp6: MIE=0	(1) Clear the MIE flag of PSW to disable all maskable interrupts. To change the interrupt control registers, this step must be performed.
(2) Set the detection voltage LVIMD (0x03F2F) bp7: LVIMOD = 0 bp6 to 4: LVICTR2 to LVICTR0 = 011	(2) Set the LVIMOD flag and the LVICTR2 to LVICTR0 flags of the power supply voltage control register to "0" and "011" to set the upper limit voltage of the power supply voltage detection level at rising to 3.8 V.
(3) Enable power supply detection function LVIMD (0x03F2F) bp0: LVION = 1	(3) Set the LVION flag of the power supply voltage control register to "1" to enable the power supply voltage detection function.
(4) Wait for power supply voltage detection function activation wait 2ms	(4) Wait for the power supply voltage function activation time (2 ms) or more.
(5) Check power supply voltage . . LVIMD (0x03F2F) bp1: LVIMON . .	(5) Monitor the LVIMON flag of the power supply voltage control register periodically.
(6) When LVIMON flag is "0", enable writing in the interrupt request flag MEMCTR (0x03F01) bp2: IRWE = 1	(6) Set the IRWE flag of the memory control register (MEMCTR) to enable writing in the interrupt request flag by software. This step is needed only when the interrupt request flag is changed by software.
(7) Clear the interrupt request flag LVIICR (0x03FF8) bp0: LVIIR = 0	(7) Rewrite the interrupt request flag (LVIIR) of the interrupt control register (LVIICR). (The interrupt request flag may have been already set. In such case, clear it in this way.)

Setup Procedure	Description
(8) Disable writing in the interrupt request flag. MEMCTR (0x03F01) bp2: IRWE = 0	(8) Clear the IRWE flag to disable writing in the interrupt request flag by software.
(9) Set the interrupt level LVIICR (0x03FF8) bp7 to 6: LVILV1 to LVILV0 PSW bp5 to 4: IM1 to IM0	(9) Set the interrupt level by the LVILV1 to LVILV0 flags of the interrupt control register (LVIICR). If the interrupt mask level of PSW needs to change, set the IM1 to IM0 flags of PSW.
(10) Enable interrupts LVIICR (0x03FF8) bp1: LVIIE = 1	(10) Set the LVIIE flag of the interrupt control register (LVIICR) to enable interrupts.
(11) Enable all maskable interrupts PSW bp6: MIE = 1	(11) Set the MIE flag of PSW to enable maskable interrupts.
(12) Check power supply voltage LVIMD (0x03F2F) bp1: LVIMON	(12) Monitor the LVIMON flag of the power supply voltage control register periodically.
(13) When LVIMON flag is "0", operating mode transits to STOP mode CPUM (0x03F00) bp3: STOP = 1	(13) In step (4) above, when the LVIMON flag is "0", the operating mode transits to STOP mode.
(14) Return from STOP mode by interrupt	(14) When the power supply voltage exceeds 3.8 V, an interrupt is generated to return from STOP mode.



An interrupt may be generated between the step (11) and step (12). In that case, after returning from the interrupt, transition to standby mode is performed with higher power supply voltage than detection voltage. It prevents to return from STOP mode at rising of the power supply voltage.

To avoid such operation, refer to the addresses saved to stack in interrupt processing. If it is considered that the operation as above is occurred, rewrite the value of stack address or saved data to prevent the standby transition program execution.

Chapter 7 Interrupts

7.1 Overview

This LSI speeds up interrupt response with circuitry that automatically loads the branch address to the corresponding interrupt service routine from an interrupt vector table: reset, non-maskable interrupts (NMI), 5 external interrupts and 22 internal interrupts (peripheral function interrupts).

For interrupts other than reset, the interrupts processing sequence consists of interrupt request, interrupt acceptance determination (only maskable interrupt), interrupt acceptance (hardware processing) and return (RTI instruction). After the interrupt is accepted (hardware processing), the program counter (PC) and processor status word (PSW) and handy addressing data (HA) are saved into the stack and program is branched to the address specified by the corresponding interrupt vector. Handy addressing data (HA) is saved into the stack so that it will not be affected by the interrupt.

An interrupts handler ends by restoring the contents of any registers used during processing by using the POP instruction and other means, then executing the return from interrupt (RTI) instruction to return to the point at which execution was interrupted. At least 12 machine cycles are needed before executing the interrupt request and 11 machine cycles are needed for the return from interrupt.

Each interrupt has a interrupt control register, which controls the interrupts. Interrupt control register consists of interrupt request flag (IR), interrupt enable flag (IE) and interrupt level specification flag (LV1 to 0).

Interrupt request flag (IR) is set to "1" when an event as an interrupt factor is occurred, and cleared to "0" by the interrupt acceptance. This flag is controlled by hardware, but can be rewritten by software.

Interrupt enable flag (IE) is the flag that enables specified interrupts. There is no interrupt enable flag in non-maskable interrupt (NMI). Once this interrupt request flag is set, it is accepted without any conditions. Interrupts enable flag is set in maskable interrupt. Interrupt enable flag of maskable interrupt is valid when the maskable interrupt enable flag (MIE flag) of PSW is "1".

Maskable interrupts have vector numbers by hardware, but their priority can be changed by user's program by setting interrupts level specification flags (LV1 to 0). There are three hierarchical interrupt levels. If multiple interrupts have the same priority, the one with the lowest vector number takes priority. Maskable interrupts are accepted when its level is higher than the interrupt mask level (IM1 to 0) of PSW. Non-maskable interrupts are always accepted, regardless of the interrupt mask level.

7.1.1 Functions

Table:7.1.1 Interrupt Functions

Interrupt type		Reset interrupt	Non-maskable interrupt	Maskable interrupt
Vector number		0	1	2 to 30
Table address	IVBM = 0	0x04000	0x04004	0x04008 to 0x04078
	IVBM = 1		0x00104	0x00108 to 0x00178
Starting address		Address specified by vector address		
Interrupt level		-	-	Can be set to level 0 to 2 (by software)
Interrupt factor		External Reset pin input	Errors detection, Program interrupt (PI)	External pin input internal peripheral function
Generated operation		Direct input to CPU	Input to CPU from non-maskable interrupt control register (NMICR)	Input interrupt request level set in interrupt level flag (xxxLVn) of maskable interrupt control register (xxxICR) to CPU.
Accept operation		Always accepts	Always accepts	Acceptance is determined by the interrupt control of the PSW interrupt enable flag (MIE), interrupt mask level (IM) and maskable interrupt control register (xxxICR).
Machine cycles until accepted		Least 6 Machine cycles + oscillation stabilization wait time	Least 12	Least 12
PSW status after acceptance		All flags are cleared to "0"	The interrupt mask level flag in PSW is cleared to "00"	Values of the interrupt level flag (xxxLVn) are set to the interrupt mask level (masking all interrupt requests with the same or the lower priority).

7.1.2 Block Diagram

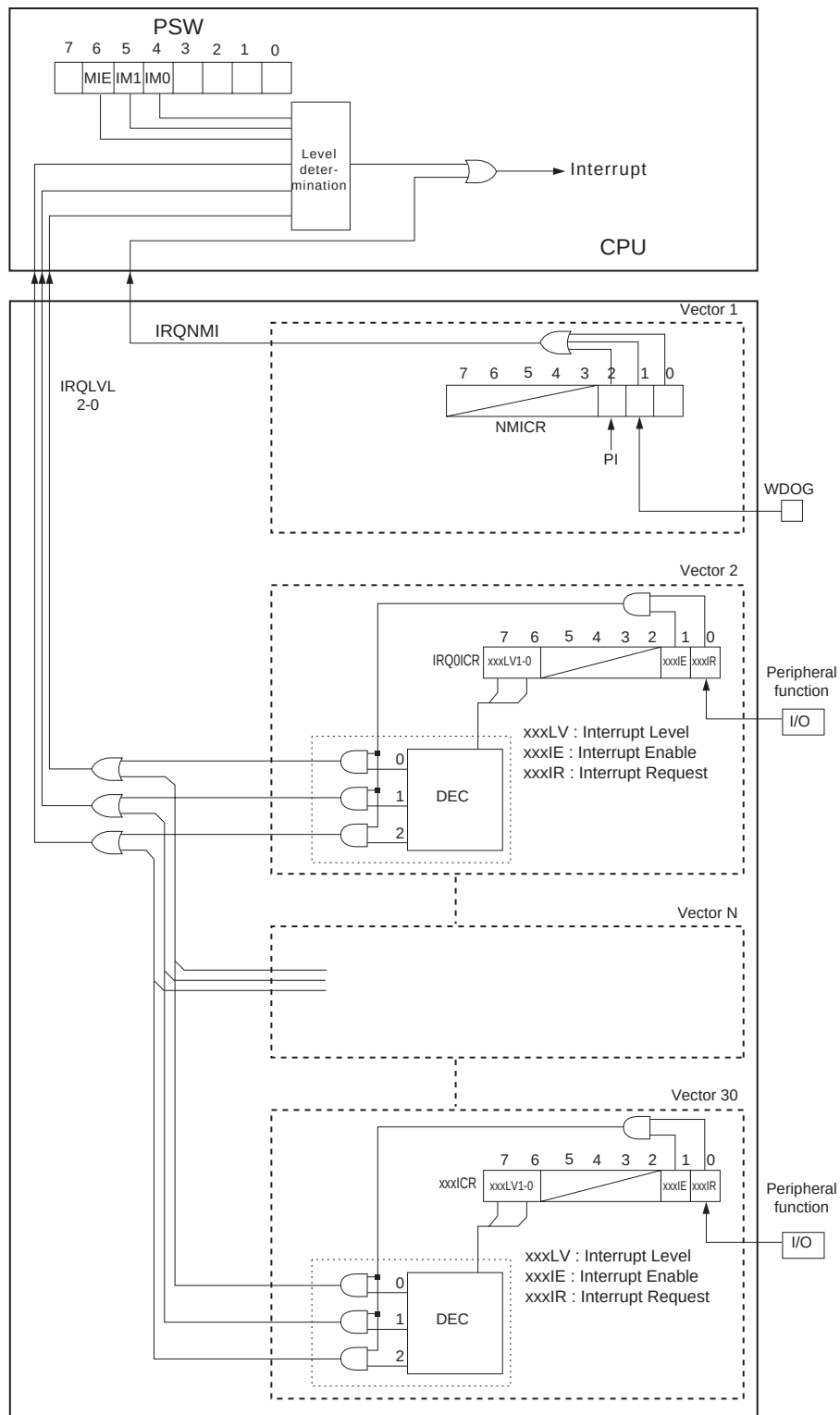


Figure:7.1.1 Interrupt Block Diagram

7.1.3 Operation

■ Interrupt Processing Sequence

For interrupts other than reset, the interrupt processing sequence consists of interrupt request, interrupt acceptance determination (only maskable interrupt), interrupt acceptance (hardware processing) and return (RTI instruction). The program counter (PC), processor status word (PSW) and hard addressing data (HA) are saved into the stack, and program is branched to the address specified by the corresponding interrupt vector. An interrupt handler ends by restoring the contents of the registers used during processing and then executing the return from interrupt (RTI) instruction to return to the point where the execution was interrupted.

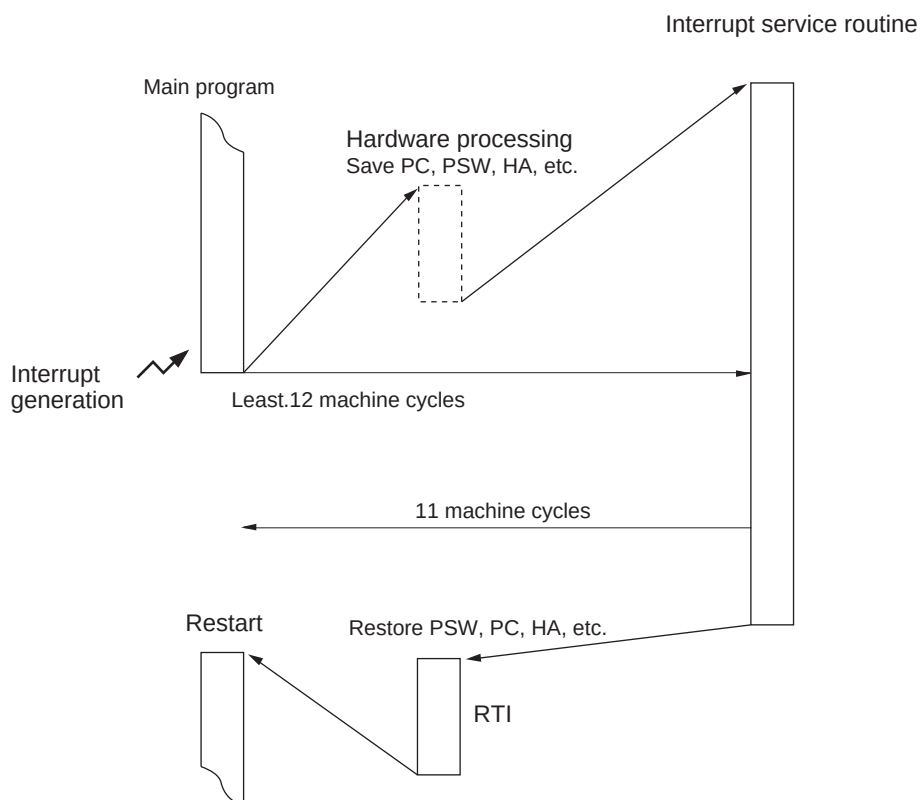


Figure:7.1.2 Interrupt Processing Sequence

The interrupt request flag (IR) of maskable interrupt control register (ICR) is cleared by hardware when an interrupt is accepted. The interrupt request flags of non-maskable interrupt control register (NMICR) is not cleared by hardware.

■ Interrupt Group and Vector Addresses

Table:7.1.2 shows the list of interrupt vector addresses and interrupt group.

Table:7.1.2 Interrupt Vector Addresses and Interrupt Group

Table remarks √: With function -: Without function

Vector number		Vector addresses		Interrupt group (interrupt factor)		Control register (address)		MN101 EF76K/ MN101 EF56K	MN101 EF57G
MN101 EF76K/ MN101 EF56K	MN101 EF57G	IVBM=0	IVBM=1						
0	0	0x04000		Reset	-	-	-	√	√
1	1	0x04004	0x00104	Non-maskable interrupt	NMI	NMICR	0x03FE1	√	√
2	2	0x04008	0x00108	External Interrupt 0	IRQ0	IRQ0ICR	0x03FE2	√	√
3	3	0x0400C	0x0010C	External Interrupt 1	IRQ1	IRQ1ICR	0x03FE3	√	√
4	4	0x04010	0x00110	External Interrupt 2	IRQ2	IRQ2ICR	0x03FE4	√	√
5	5	0x04014	0x00114	External Interrupt 3	IRQ3	IRQ3ICR	0x03FE5	√	√
6	6	0x04018	0x00118	External Interrupt 4	IRQ4	IRQ4ICR	0x03FE6	√	√
7	7	0x0401C	0x0011C	LIN interrupt	LINIRQ	LINICR	0x03FE7	√	√
8	8	0x04020	0x00120	Low power voltage detection interrupt	LV11IRQ	LV11ICR	0x03FE8	√	√
9	9	0x04024	0x00124	Timer 0 interrupt	TM0IRQ	TM0ICR	0x03FE9	√	√
10	10	0x04028	0x00128	Timer 1 interrupt	TM1IRQ	TM1ICR	0x03FEA	√	√
11	11	0x0402C	0x0012C	Timer 2 interrupt	TM2IRQ	TM2ICR	0x03FEB	√	√
12	12	0x04030	0x00130	Timer 3 interrupt	TM3IRQ	TM3ICR	0x03FEC	√	√
13	13	0x04034	0x00134	Timer 4 interrupt	TM4IRQ	TM4ICR	0x03FED	√	√
14	14	0x04038	0x00138	24H Timer interrupt	TMRIRQ	TMRICR	0x03FEE	√	√
15	15	0x0403C	0x0013C	Timer 6 interrupt	TM6IRQ	TM6ICR	0x03FEF	√	√
16	16	0x04040	0x00140	Time base interrupt	TBIRQ	TBICR	0x03FF0	√	√
17	17	0x04044	0x00144	Timer 7 interrupt	TM7IRQ	TM7ICR	0x03FF1	√	√
18	18	0x04048	0x00148	Timer 8 interrupt	TM8IRQ	TM8ICR	0x03FF2	√	√
19	19	0x0404C	0x0014C	Timer 9 compare register 2 match interrupt	TM9OC21IRQ	TM9OC2ICR	0x03FF3	√	√
20	20	0x04050	0x00150	Serial 0 interrupt	SC0TIRQ	SC0TICR	0x03FF4	√	√
21	21	0x04054	0x00154	Serial 0 UART reception interrupt	SC0RIRQ	SC0RICR	0x03FF5	√	√
22	22	0x04058	0x00158	Serial 1 interrupt	SC1TIRQ	SC1TICR	0x03FF6	√	√
23	23	0x0405C	0x0015C	Serial 1 UART reception interrupt	SC1RIRQ	SC1RICR	0x03FF7	√	√
24	24	0x04060	0x00160	Serial 2 interrupt	SC2TIRQ	SC2TICR	0x03FF8	√	√
25	25	0x04064	0x00164	Serial 2 UART reception interrupt	SC2RIRQ	SC2RICR	0x03FF9	√	√
26	-	0x04068	0x00168	Serial 3 interrupt	SC3TIRQ	SC3TICR	0x03FFA	√	-
27	-	0x0406C	0x0016C	Serial 3 UART reception interrupt	SC3RIRQ	SC3RICR	0x03FFB	√	-
28	26	0x04070	0x00170	Serial 4 interrupt	SC4IRQ	SC4ICR	0x03FFC	√	√
29	27	0x04074	0x00174	Serial 4 stop condition interrupt	SC4STPCIRQ	SC4STPCICR	0x03FFD	√	√
30	28	0x04078	0x00178	Peripheral function group interrupt	PERIIRQ	PERIILR	0x03FFE	√	√

■ Interrupt Level and Priority

In this LSI, vector numbers and interrupt control registers (except reset interrupt) are allocated to each interrupt. The interrupt level (except reset interrupt, non-maskable interrupt) can be set by software, for each interrupt group. There are three hierarchical interrupt levels. If multiple interrupts have the same priority, the one with the lowest vector number takes priority. For example, if a vector 3 set to level 1 and a vector 4 set to level 1 request interrupt simultaneously, vector 3 will be accepted.

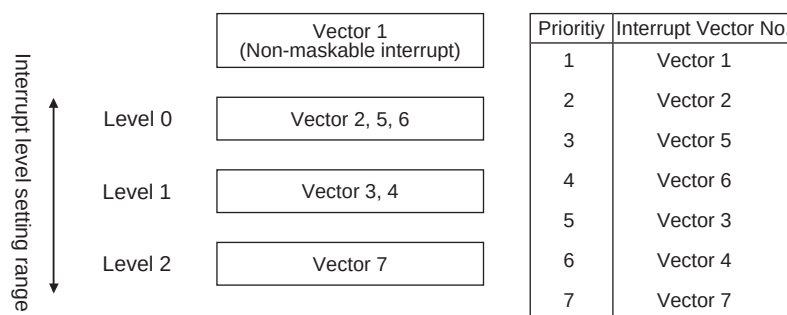


Figure:7.1.3 Interrupt Priority Outline

Table:7.1.3 Interrupt Mask Level and Interrupt Acceptance

	Interrupt mask level		Priority	Acceptable interrupt level
	IM1	IM0		
Mask level 0	0	0	Highest	Non-maskable interrupt (NMI) only
Mask level 1	0	1	:	NMI, level 0
Mask level 2	1	0	:	NMI, level 0 to 1
Mask level 3	1	1	Lowest	NMI, level 0 to 2

■ Determination of Maskable Interrupt Acceptance

The following is the procedure from interrupt request input to acceptance.

1. The interrupt request flag (xxxIR) in corresponding external interrupt control register (IRQnICR) and internal interrupt control register (xxxICR) are set to "1".
2. When the interrupt enable flag (xxxIE) corresponding to the internal request flag is "1", the information of the level set in the interrupt level flag (xxxLV1 to 0) is output to the CPU as an interrupt request signal.
3. The interrupt request is accepted if the output interrupt request signal has a higher priority than the level set in the internal mask level of the processor status word (PSW) and the interrupt enable flag of PSW (MIE) is "1" (enabled).
4. After acceptance of an interrupt, the interrupt request flag (xxxIR) is cleared by hardware. However, the interrupt enable flag (xxxIE) is not cleared.

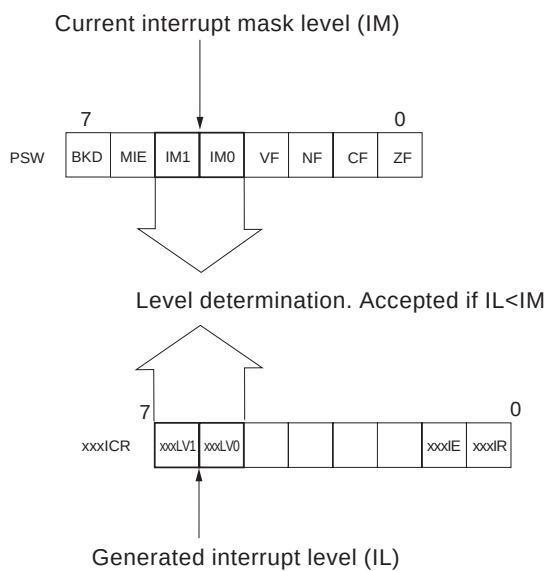


Figure:7.1.4 Determination of Interrupt Acceptance



After acceptance of an interrupt, the interrupt enable flag (xxxIE) is not cleared.



For the peripheral function group interrupt, the flag corresponding to each factor of peripheral function group interrupt input enable register (IRQEXPEN) is "1".



For the peripheral function group interrupt, when the flag of peripheral function group interrupt input enable register corresponding to the generated interrupt factor is "1", the information which has the level set in the interrupt level flag (ICR30LV1 to 0) is output to the CPU as an interrupt request signal.



After acceptance of an interrupt, the interrupt with the same factor will be ignored until the interrupt request flag (xxxIR) is cleared by hardware.



For the peripheral function group interrupt, after acceptance of an interrupt, peripheral function group interrupt factor retention register (IRQEXPDT) is not cleared by hardware. It must be cleared by software. Also, peripheral function group input enable register (IRQEXPEN) is not cleared.

■ Maskable Interrupt Enable (MIE) and Interrupt Mask level (IM1 to 0) in PSW

MIE = "0" and maskable interrupts are disabled when:

- MIE in the PSW is reset to "0" by a program
- BE instruction is executed. (BKD is reset and MIE is reset)
- Reset is detected.

MIE = "1" and maskable interrupts are enabled when:

- MIE in the PSW is reset to "1" by a program
- BE instruction is executed. (BKD is set and MIE is set)

The interrupt mask level (IM1 to 0) in the processor status word (PSW) changes when:

- The program alters it directly.
- Reset is detected. (IM1 to 0 = "00")
- Maskable interrupt is accepted (the interrupt level becomes the interrupt mask level. Execution of the RTI instruction at the end of an interrupt service routine restores the processor status word (PSW) and thus the previous interrupt mask level.).
- Non-Maskable interrupt is accepted (IM1 to 0 = (00b)).



When accepting interrupts, MIE of PSW is not cleared to "0".



When a non-maskable interrupt and a maskable interrupt are generated at the same time, the non-maskable interrupt has a priority to the maskable one.



As for the BE instruction and BD instruction, refer to the appendix [Chapter 25 25.2 Instruction Set] at the end of this manual.

■ Interrupt Acceptance Operation (Hardware Processing)

When accepting an interrupt, this LSI hardware saves the return address from the program counter and the processor status word (PSW) to the stack and branches program to the interrupt handler using the starting address in the vector table. The following is the hardware processing sequence invoked by interrupt acceptance.

1. the stack pointer (SP) is updated.
 $SP - 6 \rightarrow SP$
2. The contents of the program counter (PC)-i.e., the return address- are saved to the stack.
 $PC \text{ bit } 7 \text{ to } 0 \rightarrow \text{Address } (SP + 1)$
 $PC \text{ bits } 15 \text{ to } 8 \rightarrow \text{Address } (SP + 2)$
 $PC \text{ bits } 19 \text{ to } 16, \text{ and } H \rightarrow \text{Address } (SP + 3)$
3. The contents of the handy addressing data (HA) are saved to the stack.
 $\text{Lower half of HA} \rightarrow \text{Address } (SP + 4)$
 $\text{Upper half of HA} \rightarrow \text{Address } (SP + 5)$
4. The contents of the PSW are saved to the stack.
 $PSW \rightarrow \text{Address } (SP)$
5. The interrupt level (xxxLVn) for the interrupt is copied to the interrupt mask IMn in the PSW.
 $\text{Interrupt level (xxxLVn)} \rightarrow \text{IMn}$
6. BKD flag of the PSW is reset (When accepting interrupts, the bank register always address the first 64 KB.)
7. The hardware branches program to the address in the vector table.

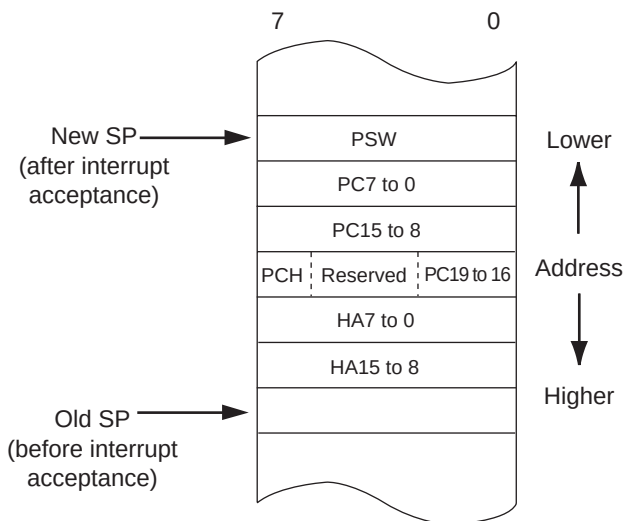


Figure:7.1.5 Stack Operation during Interrupt Acceptance

■ Interrupt Return Operation (RTI Instruction)

An interrupt handler ends by restoring the contents of the registers saved to the stack during processing by the POP instruction and other means, and the RTI instruction restores the program to the point where the execution was interrupted.

The following is the processing sequence invoked by the RTI instruction.

1. The contents of the PSW are restored from the stack. (SP)
2. The contents of the program counter (PC) -i.e., the return address- are restored from the stack. (SP + 1 to SP + 3)
3. The contents of the handy addressing data (HA) are restored from the stack. (SP + 4, SP + 5)
4. The stack pointer is updated. $SP + 6 \rightarrow SP$
5. Execution branches program to the address in the program counter.

The handy addressing data is an internal register used by the handy addressing function. The hardware saves its contents to the stack to prevent the interrupt from interfering with the operation of the function.



Registers including data registers and address registers are not saved. If needed, save these registers to stack using PUSH instructions.

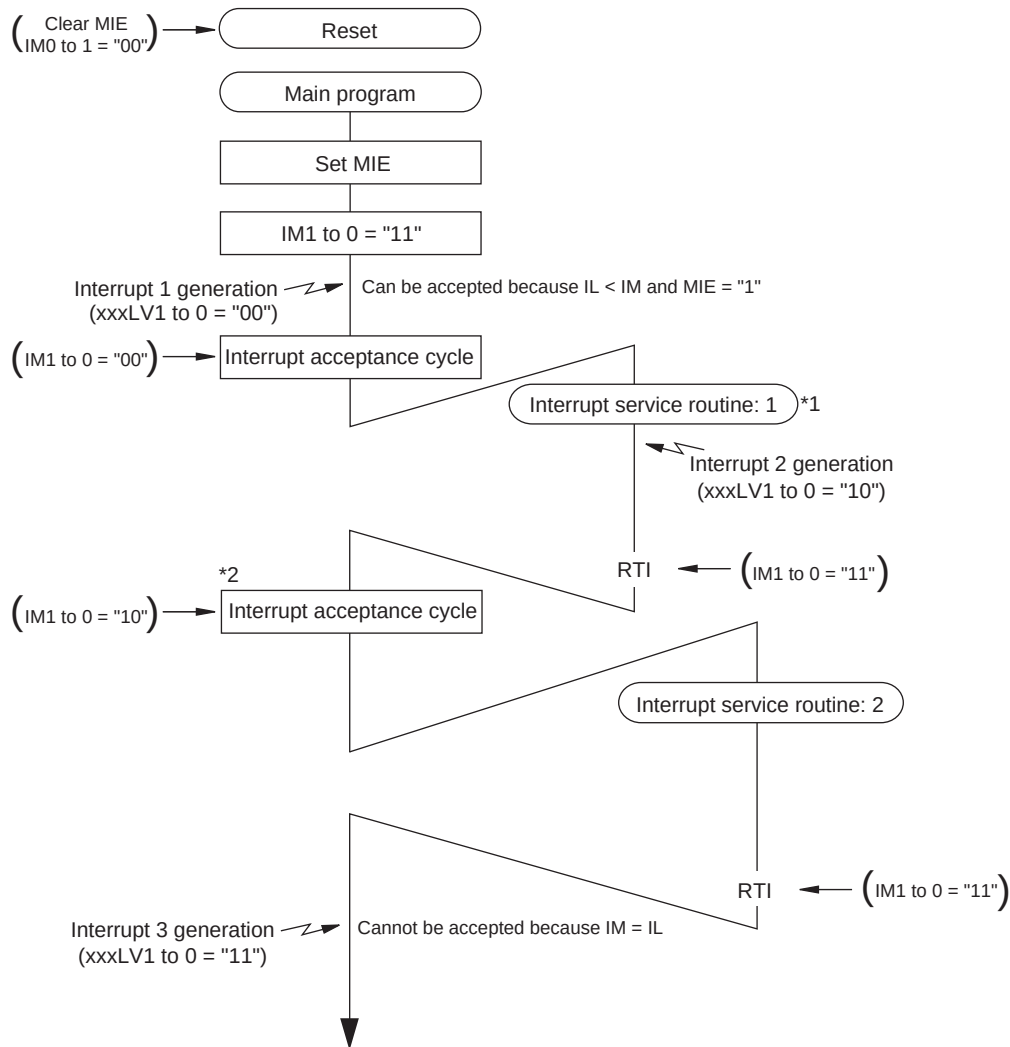


In Figure:7.1.5, bp6 to bp4 of the stack address where program counter (PC [bit19 to bit16, and bitH]) are saved, are reserved. Do not change it by program.

■ Maskable Interrupt

The following is the processing sequence when the lower priority level interrupt occurs while processing the higher priority level interrupt.

(Interrupt 1: xxxLV1 to 0 = "00", Interrupt 2: xxxLV1 to 0 = "10", Interrupt 3: xxxLV1 to 0 = "11")



Parentheses () indicates hardware processing.

*1: If $IL < IM$, an interrupt generated in the interrupt processing program 1 is accepted as multiple interrupt.
If $IL > IM$, however, the interrupt is not accepted.

*2: If the interrupt 2 generated in the interrupt processing program 1 is not accepted when $IL \geq IM$, the interrupt 2 will be accepted after completion of the interrupt processing program 1.

Figure:7.1.6 Processing Sequence for Maskable Interrupts

■ Multiple Maskable Interrupt

This LSI automatically disables acceptance of subsequent interrupts with the same or lower priority level when accepting an interrupt. The LSI copies the interrupt level (xxxLVn) for the interrupt to the interrupt mask (IM) in the PSW when the hardware accepts an interrupt. As a result, subsequent interrupts with the same or lower priority levels are automatically disabled. Only interrupts with higher priority levels are accepted as multiple interrupt. And also multiple interrupt can be controlled by the following procedure.

1. To disable interrupt nesting

- Reset the MIE bit in PSW to "0".
- Raise the priority level of the interrupt mask (IM) in PSW.

Execute either one of the operation above.

2. To enable interrupts with lower priority than the currently accepted interrupt

- Lower the priority level or the interrupt mask (IM) in PSW.



Multiple interrupt is enabled only for interrupts with levels higher than the PSW interrupt mask level (IM).



It is possible to forcibly rewrite IM to accept an interrupt with a priority lower than the interrupt being processed. In this case, be careful of stack overflow caused by multiple interrupts.



Do not operate maskable interrupt control register (xxxICR) when multiple interrupts are enabled. If the operation is necessary, clear the MIE flag of PSW once to disable the interrupts.

Figure:7.1.7 shows the processing sequence of multiple maskable interrupts.

(Multiple interrupt 1: xxxLV1 to 0 = "10", Multiple interrupt 2: xxxLV1 to 0 = "00")

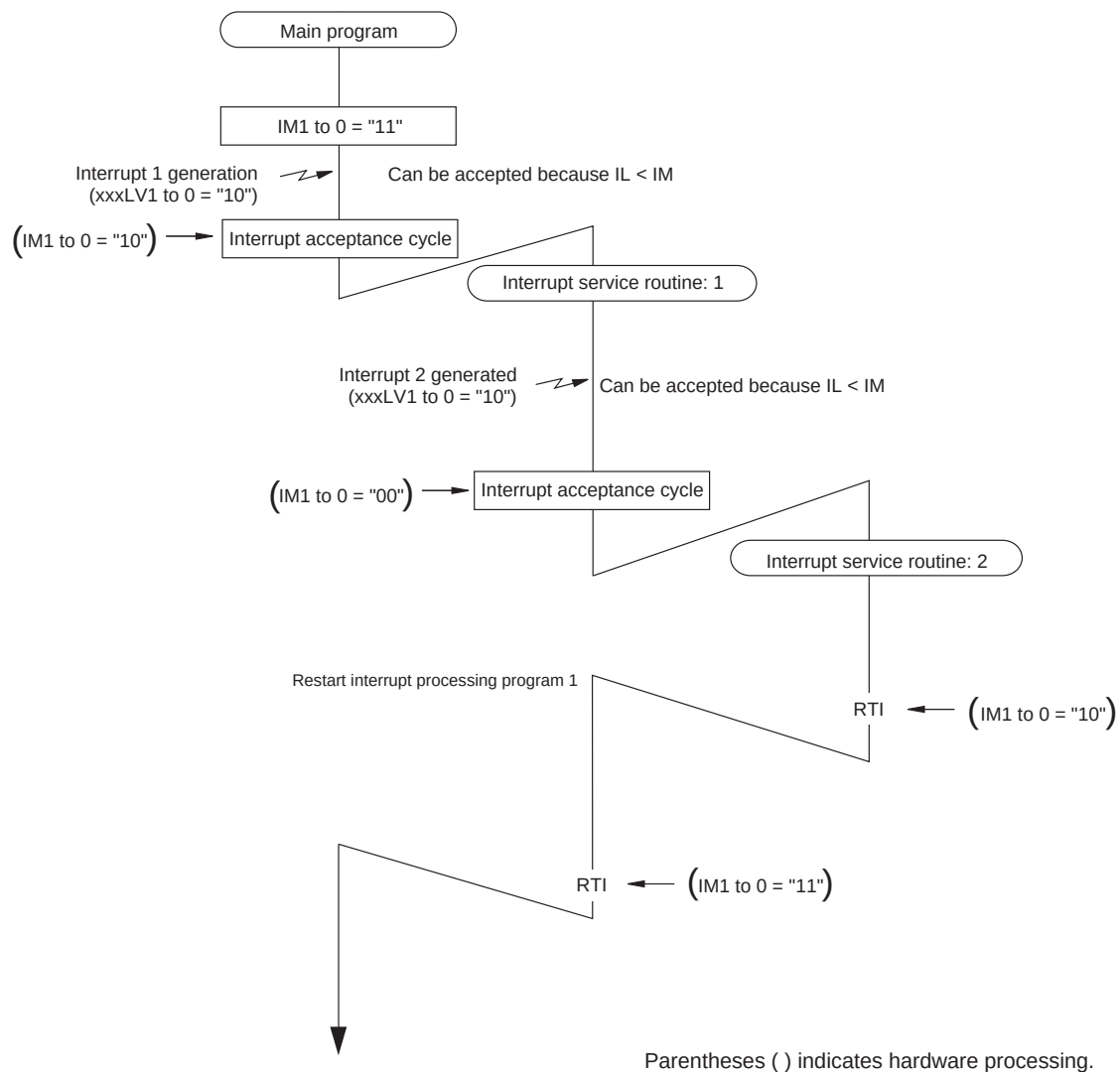


Figure:7.1.7 Processing Sequence for Multiple Maskable Interrupts

■ Multiple Non-maskable Interrupt (NMI)

On the acceptance of NMI, when other NMI factor is generated, this interrupt is processed immediately. Also, the same NMI factor is not accepted if it is generated before the NMI flag is cleared by the software. (Unless the NMI flag is cleared by software, otherwise, the next same NMI interrupt is not accepted and becomes invalid.)

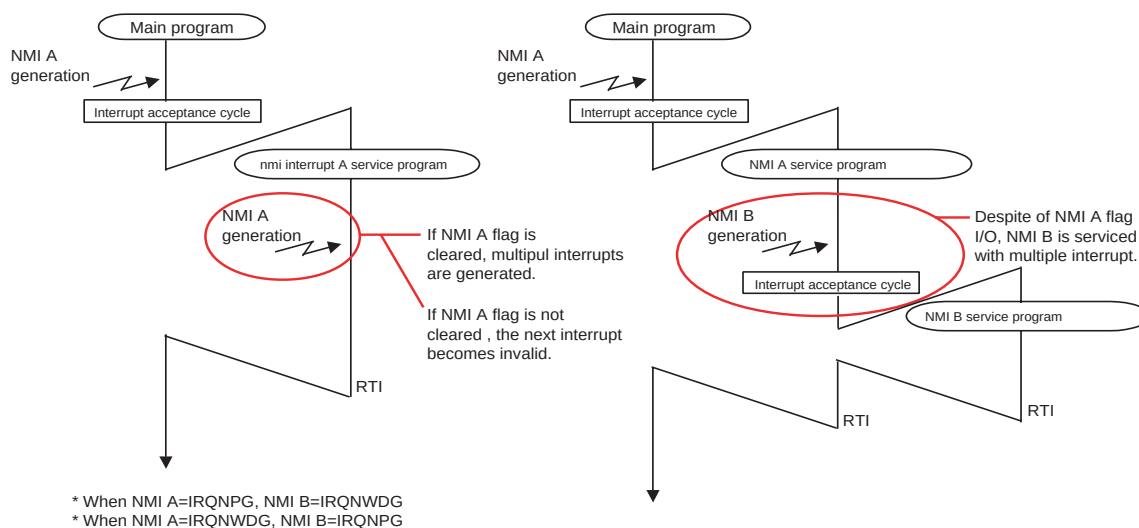


Figure:7.1.8 Processing Sequence of Multiple Interrupts for Non-maskable Interrupt

7.1.4 Maskable Interrupt Control Register Setup

■ Interrupt Request Flag (IR) Setup by Software

The interrupt request flag is operated by hardware. The flag is set to "1" when an interrupt factor is generated, and cleared to "0" when the interrupt is accepted. To rewrite the interrupt request flag by software, set the IRWE flag of MEMCTR register.

■ Maskable Interrupt Control Register Setup Procedure

A setup procedures of maskable interrupt control register including changing of the interrupt request flag by software are shown below.

Setup Procedure	Description
(1) Disable all maskable interrupts. PSW bp6: MIE = 0	(1) Clear the MIE flag of PSW to disable all maskable interrupts. This is necessary, especially when the interrupt control register is changed.
(2) Select the interrupt factor.	(2) Select the interrupt factor such as interrupt edge selection or timer interrupt cycle change.
(3) Enable writing in the interrupt request flag. MEMCTR (0x03F01) bp2: IRWE = 1	(3) Set the IRWE flag of memory control register (MEMCTR) to enable writing of the interrupt request flag. This is necessary only when the interrupt request flag is changed by the software.
(4) Rewrite the interrupt request flag. xxxICR bp0: xxxIR	(4) Rewrite the interrupt request flag (xxxIR) of interrupt control register (xxxICR). (The interrupt request flag may have been already set. In such case, clear it in this way.)
(5) Disable writing of the interrupt request flag. MEMCTR (0x03F01) bp2: IRWE = 0	(5) Clear the IRWE flag to disable writing of the interrupt request flag by the software.
(6) Set the interrupt level. xxxICR bp7 to 6: xxxLV1 to 0 PSW bp5 to 4: IM1 to 0	(6) Set the interrupt level by the xxxLV1 to 0 flags of interrupt control register (xxxICR). If it is necessary to change the interrupt mask level of PSW, set the IM1 to 0 flags of PSW.
(7) Enable the interrupt. xxxICR bp1: xxxIE = 1	(7) Set the xxxIE flag of interrupt control register (xxxICR) to enable the interrupt.
(8) Enable all maskable interrupts. PSW bp6: MIE = 1	(8) Enable all maskable interrupts.



The interrupt request flag of the interrupt control register is set by the interrupt generation, the edge switching and others, regardless of the xxxIE flag. Clear the flag following the setup procedures (3) to (5).



Be sure to set the IRWE flag of memory control register (MEMCTR) to "0" except in writing IR by software. If interrupt control register (xxxICR) is written by software while the IRWE flag is set to "1", the interrupt request flag which is set to "1" by interrupt factor may be cleared. For example, if xxxICR register is set using a bit operation instruction, the instruction executes "read-modify-write" for the value of 1-byte in the register. When an interrupt is generated between the reading and writing the register, the IR flag may be cleared to "0" accidentally and the interrupt processing is not executed. If the IRWE flag is set to "0", the IR flag is not modified when writing to xxxICR register.



Disable all maskable interrupt for the maskable interrupt enable flag (MIE) of processor status word (PSW) (Set the MIE flag to "0") before writing to interrupt control register (xxxICR). There is no guarantee of proper operation when writing to interrupt control register with the MIE flag set to "1".

7.1.5 Peripheral Function Group Interrupt Control Register Setup

■ Interrupt Request Flag (IRQEXPDT) Setup

The peripheral function group interrupt request flag is set by hardware/software, and cleared by hardware. When an interrupt factor is generated, the flag is set to "1" and the interrupt is accepted.

When an interrupt factor corresponding to the settings of peripheral function group interrupt input enable register (IRQEXPEN) is generated, the corresponding flag is set to "1" and it becomes the factor for maskable interrupt.

Interrupts can be generated by setting the above flag to "1" by software.

The flag differs from the xxxIR flag of interrupt control register (xxxICR), and cannot be cleared by hardware. Write "1" in the corresponding flag to clear it.

Following table shows the flag status, write data and flag status after written.

Flag status	Write data	Flag status after written	Flag change
0	0	0	Not changed
0	1	1	Set
1	0	1	Not changed
1	1	0	Cleared



To initialize (clear) all flags, read the data of the register once, then write the contents to the register.



The peripheral function group interrupt request flag cannot be cleared by hardware. To clear it, write "1" to the corresponding flag in interrupt processing program.



Do not change the value of PERILR when peripheral function group interrupt factor retention register (IRQEXPDT) corresponding to peripheral function group interrupt input enable register (IRQEXPEN) is set. If changed, an unintended interrupt may be processed.

■ Interrupt Flag Setup Procedure

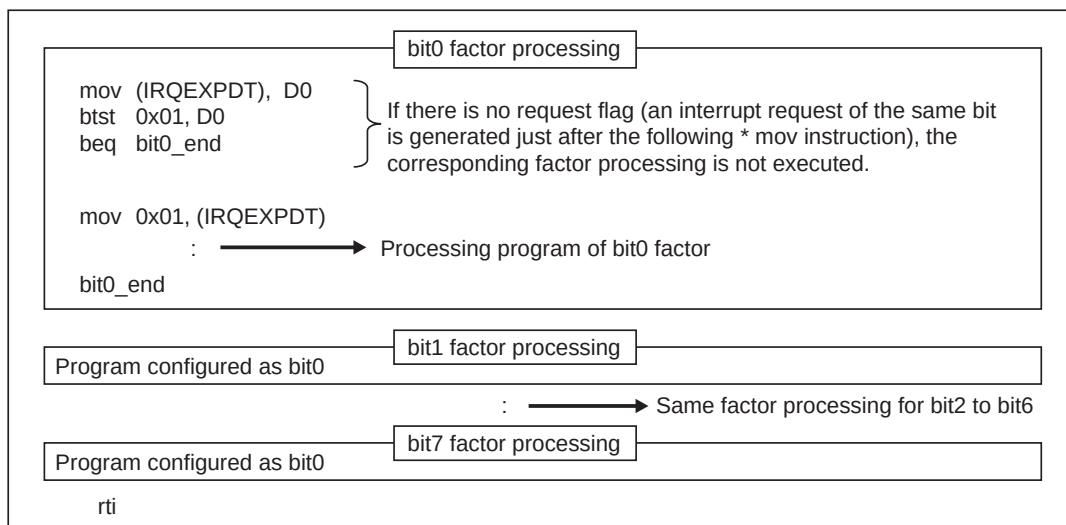
An example setup procedure, with a description of each step is shown below.

Setup Procedure	Description
(1) Disable all maskable interrupts. PSW bp6: MIE = 0	(1) Clear the MIE flag of PSW to disable all maskable interrupts. This is necessary, especially when the interrupt control register is changed.
(2) Clear the peripheral function group interrupt request flag IRQEXPDT (0x03F4F) bp6 to 0: IRQEXPDT6 to 0 = 0x00	(2) Read the IRQEXPDT6 to 0 flags of peripheral function group interrupt factor retention register (IRQEXPDT) and write the contents to clear the peripheral function group interrupt request flag. To execute interrupt processing before this setting, do not clear the corresponding flag of IRQEXPDT register.
(3) Set the peripheral function group interrupt input IRQEXPEN (0x03F4E) bp6 to 0: IRQEXPEN6 to 0 = 0xEF	(3) Set the IRQEXPEN6 to 0 flags of peripheral function group interrupt input enable register (IRQEXPEN) to enable the corresponding interrupt.
(4) Set the level of PERIILR.	(4) Set the level of PERIILR.
(5) Enable all maskable interrupts. PSW bp6: MIE = 1	(5) Enable all maskable interrupts.
(6) [Interrupt acceptance]	(6) Read the flag of peripheral function group interrupt factor retention register (IRQEXPDT) to determine the interrupt factors by software.
(7) Clear the peripheral function group interrupt request flag IRQEXPDT (0x03F4F) bp x: IRQEXPDTx = 1	(7) Write "1" to the corresponding flag of peripheral function group interrupt factor retention register (IRQEXPDT) to clear the request flag.



Peripheral function group interrupt request flag (IRQEXPDT) is set by the interrupt generation regardless of the settings of peripheral function group interrupt input enable register (IRQEXPEN). Clear the flag following the setup procedures (2) to (4).

■ Program Example for Peripheral Function Group Interrupt Process



7.2 Control Registers

7.2.1 Registers List

Table:7.2.1 Interrupt Control Registers

Table remarks √: With function -: Without function

Register	Address	R/W	Functions	Page	MN101 EF76K	MN101 EF57G	MN101 EF56K
NMICR	0x03FE1	R/W	Non-maskable interrupt control register	VII-23	√	√	√
IRQ0ICR	0x03FE2	R/W	External interrupt 0 control register	VII-24	√	√	√
IRQ1ICR	0x03FE3	R/W	External interrupt 1 control register	VII-24	√	√	√
IRQ2ICR	0x03FE4	R/W	External interrupt 2 control register	VII-24	√	√	√
IRQ3ICR	0x03FE5	R/W	External interrupt 3 control register	VII-24	√	√	√
IRQ4ICR	0x03FE6	R/W	External interrupt 4 control register	VII-24	√	√	√
LINICR	0x03FE7	R/W	LIN interrupt control register	VII-25	√	√	√
LVIICR	0x03FE8	R/W	Power supply detection interrupt control register	VII-26	√	√	√
TM0ICR	0x03FE9	R/W	Timer 0 interrupt control register	VII-27	√	√	√
TM1ICR	0x03FEA	R/W	Timer 1 interrupt control register	VII-27	√	√	√
TM2ICR	0x03FEB	R/W	Timer 2 interrupt control register	VII-27	√	√	√
TM3ICR	0x03FEC	R/W	Timer 3 interrupt control register	VII-27	√	√	√
TM4ICR	0x03FED	R/W	Timer 4 interrupt control register	VII-27	√	√	√
TMRICR	0x03FEE	R/W	24H Timer interrupt control register	VII-28	√	√	√
TM6ICR	0x03FEF	R/W	Timer 6 interrupt control register	VII-27	√	√	√
TBICR	0x03FF0	R/W	Time base interrupt control register	VII-29	√	√	√
TM7ICR	0x03FF1	R/W	Timer 7 interrupt control register	VII-30	√	√	√
TM8ICR	0x03FF2	R/W	Timer 8 interrupt control register	VII-31	√	√	√
TM9OC2ICR	0x03FF3	R/W	Timer 9 compare register 2 match interrupt control register	VII-32	√	√	√
SC0TICR	0x03FF4	R/W	Serial 0 interrupt control register	VII-33	√	√	√
SC0RICR	0x03FF5	R/W	Serial 0 UART reception interrupt control register	VII-34	√	√	√
SC1TICR	0x03FF6	R/W	Serial 1 interrupt control register	VII-33	√	√	√
SC1RICR	0x03FF7	R/W	Serial 1 UART reception interrupt control register	VII-34	√	√	√
SC2TICR	0x03FF8	R/W	Serial 2 interrupt control register	VII-33	√	√	√
SC2RICR	0x03FF9	R/W	Serial 2 UART reception interrupt control register	VII-34	√	√	√
SC3TICR	0x03FFA	R/W	Serial 3 interrupt control register	VII-33	√	-	√
SC3RICR	0x03FFB	R/W	Serial 3 UART reception interrupt control register	VII-34	√	-	√
SC4ICR	0x03FFC	R/W	Serial 4 interrupt control register	VII-35	√	√	√
SC4STPCICR	0x03FFD	R/W	Serial 4 stop condition interrupt control register	VII-36	√	√	√
PERILR	0x03FFE	R/W	Peripheral function group interrupt level setting register	VII-37	√	√	√
IRQEXPEN	0x03F4E	R/W	Peripheral function group interrupt input enable Register	VII-38	√	√	√
IRQEXPDT	0x03F4F	R/W	Peripheral function group interrupt factor retention register	VII-39	√	√	√



If the interrupt level flag (xxxLVn) is set to "level 3", its vector is disabled, regardless of the interrupt enable flag and the interrupt request flag.



Disable all maskable interrupt for the maskable interrupt enable flag (MIE) of processor status word (PSW) (Set the MIE flag to "0") before writing to interrupt control register (xxxICR). There is no guarantee of proper operation when writing to interrupt control register with the MIE flag set to "1".

7.2.2 Interrupt Control Registers

The interrupt control registers include non-maskable interrupt control register (NMICTR) and maskable interrupt control register. Maskable interrupt control register consists of external interrupt control register (IRQnICR) and internal interrupt control register (xxxICR).

■ Non-maskable Interrupt Control Register (NMICR: 0x03FE1)

The non-maskable interrupt request is stored in non-maskable interrupt control register (NMICTR). When the non-maskable interrupt request is generated, the interrupt is accepted regardless of the interrupt mask level (IMn) of PSW. The hardware then branches program to the address stored at location 0x04004 in the interrupt vector table. The watchdog timer overflow interrupt request flag (IRQNWGDG) is set to "1" when the watchdog timer overflows.

The program interrupt request flag (IRQNPG) is set to "1" when the undefined instruction is executed.

The peripheral function non maskable interrupt request flag IRQNPRI is set to be "1" when the watchdog timer overflows, an oscillation error interrupt is generated in the clock monitoring function, or a write error interrupt is generated in the register protect function.

bp	7	6	5	4	3	2	1	0
Flag	-	-	-	-	-	IRQNPG	IRQNWGDG	IRQNPRI
At reset	-	-	-	-	-	0	0	0
Access	-	-	-	-	-	R/W	R/W	R/W

bp	Flag	Description
7 to 3	-	-
2	IRQNPG	Program interrupt request flag 0: No interrupt request has been generated. 1: Interrupt request has been generated.
1	IRQNWGDG	Watchdog interrupt request flag 0: No interrupt request has been generated. 1: Interrupt request has been generated.
0	IRQNPRI	Peripheral function non-maskable interrupt request flag (Interrupt by watchdog timer 2, clock monitoring function or register protect function) 0: No interrupt request has been generated. 1: Interrupt request has been generated.



When the undefined instruction is going to be executed, this LSI generates the non-maskable interrupt at the same time as the program interrupt request flag (IRQNPG) is set. When the setting of the IRQNPG flag is confirmed by the non-maskable interrupt process program, the software reset is recommended by outputting "0" to the reset pin (P27).

- External Interrupt 0 to 4 Control Registers (IRQ0ICR: 0x03FE2, IRQ1ICR: 0x03FE3, IRQ2ICR: 0x03FE4, IRQ3ICR: 0x03FE5, IRQ4ICR: 0x03FE6)

External interrupt 0 to 4 control register (IRQ0ICR to IRQ4ICR) are used to control interrupt levels of external interrupts 0 to 4, active edge, interrupt enable and interrupt request. Interrupt control registers should be operated when the maskable interrupt enable flag (MIE) of PSW is "0". Refer to [Chapter 7 7.1.4 Maskable Interrupt Control Register Setup] for the setting procedure of the flags.

bp	7	6	5	4	3	2	1	0
Flag	IRQnLV1	IRQnLV0	REDGn	-	Reserved	-	IRQnIE	IRQnIR
At reset	0	0	0	-	0	-	0	0
Access	R/W	R/W	R/W	-	R/W	-	R/W	R/W

bp	Flag	Description
7 to 6	IRQnLV1 IRQnLV0	Interrupt level flag This flag sets the interrupt level 0 to 3 for interrupt requests.
5	REDGn	Interrupt valid edge flag (in STANDBY mode, level interrupt selection) 0: Falling edge (Low-level) 1: Rising edge (High-level)
4	-	-
3	Reserved	Always set to "0".
2	-	-
1	IRQnIE	Interrupt enable flag 0: Disabled 1: Enabled
0	IRQnIR	Interrupt request flag 0: No interrupt request 1: Generate interrupt request



For the interrupt generation factors of external interrupt 0 to 4/key interrupt in STANDBY mode, refer to [Chapter 7 7.3.9 External Interrupt Operation in STANDBY Mode]

■ LIN Interrupt Control Registers (LINICR: 0x03FE7)

LIN interrupt control register (LINICR) is used to control interrupt levels of LIN interrupt, interrupt enable and interrupt request. Interrupt control register should be operated when the maskable interrupt enable flag (MIE) of PSW is "0". Refer to [Chapter 7 7.1.4 Maskable Interrupt Control Register Setup] for the setting procedure of the flags.

bp	7	6	5	4	3	2	1	0
Flag	LINLV1	LINLV0	LINREDG	-	Reserved	-	LINIE	LINIR
At reset	0	0	0	-	0	-	0	0
Access	R/W	R/W	R/W	-	R/W	-	R/W	R/W

bp	Flag	Description
7 to 6	LINLV1 LINLV0	Interrupt level flag This flag sets the interrupt level 0 to 3 for interrupt requests.
5	LINREDG	Interrupt enable edge specification flag (In STANDBY mode) 0: Falling edge (Low-level) 1: Rising edge (High-level)
4	-	-
3	Reserved	Always set to "0".
2	-	-
1	LINIE	Interrupt enable flag 0: Disabled 1: Enabled
0	LINIR	Interrupt request flag 0: No interrupt request 1: Generate interrupt request



When using the hardware LIN function, it is necessary set the LINREDG flag of LIN interrupt control register (LINICR) to "1" to select the rising edge.

■ Power Supply Detection Interrupt Control Register (LVIICR: 0x03FE8)

Power supply detection interrupt control register (LVIICR) is used to control interrupt levels of power supply detection interrupts, interrupt enable and interrupt request. Interrupt control register should be operated when the maskable interrupt enable flag (MIE) of PSW is "0". Refer to [Chapter 7 7.1.4 Maskable Interrupt Control Register Setup] for the setting procedure of the flags.

bp	7	6	5	4	3	2	1	0
Flag	LVILV1	LVILV0	-	-	Reserved	-	LVIIIE	LVIIIR
At reset	0	0	-	-	0	-	0	0
Access	R/W	R/W	-	-	R/W	-	R/W	R/W

bp	Flag	Description
7 to 6	LVILV1 LVILV0	Interrupt level specified flag This flag sets the interrupt level 0 to 3 for interrupt requests.
5 to 4	-	-
3	Reserved	Always set to "0".
2	-	-
1	LVIIIE	Interrupt enable flag 0: Disabled 1: Enabled
0	LVIIIR	Interrupt request flag 0: No interrupt request 1: Generate interrupt request



When setting the LVIIIE flag, it is necessary to wait for 2.0 ms or longer after the LVION flag of LV IMD register is set.

- Timer 0 to 4, 6 Interrupt Control Registers (TM0ICR: 0x03FE9, TM1ICR: 0x03FEA, TM2ICR: 0x03FEB, TM3ICR: 0x03FEC, TM4ICR: 0x03FED, TM6ICR: 0x03FEF)

Timer 0 to 4, 6 interrupt control register (TM0ICR to TM4ICR, TM6ICR) are used to control interrupt levels of timer 0 to 4, 6 interrupts, interrupt enable and interrupt request. Interrupt control register should be operated when the maskable interrupt enable flag (MIE) of PSW is "0". Refer to [Chapter 7 7.1.4 Maskable Interrupt Control Register Setup] for the setting procedure of the flags.

bp	7	6	5	4	3	2	1	0
Flag	TMnLV1	TMnLV0	-	-	Reserved	-	TMnIE	TMnIR
At reset	0	0	-	-	0	-	0	0
Access	R/W	R/W	-	-	R/W	-	R/W	R/W

bp	Flag	Description
7 to 6	TMnLV1 TMnLV0	Interrupt level flag This flag sets the interrupt level 0 to 3 for interrupt requests.
5 to 4	-	-
3	Reserved	Always set to "0".
2	-	-
1	TMnIE	Interrupt enable flag 0: Disabled 1: Enabled
0	TMnIR	Interrupt request flag 0: No interrupt request 1: Generate interrupt request

■ 24H Timer Interrupt Control Register (TMRICR: 0x03FEE)

24H timer interrupt control register (TMRICR) is used to control interrupt levels of 24H timer interrupts, interrupt enable and interrupt request. Interrupt control register should be operated when the maskable interrupt enable flag (MIE) of PSW is "0". Refer to [Chapter 7 7.1.4 Maskable Interrupt Control Register Setup] for the setting procedure of the flags.

bp	7	6	5	4	3	2	1	0
Flag	TMRLV1	TMRLV0	-	-	Reserved	-	TMRIE	TMRIR
At reset	0	0	-	-	0	-	0	0
Access	R/W	R/W	-	-	R/W	-	R/W	R/W

bp	Flag	Description
7 to 6	TMRLV1 TMRLV0	Interrupt level flag This flag sets the interrupt level 0 to 3 for interrupt requests.
5 to 4	-	-
3	Reserved	Always set to "0".
2	-	-
1	TMRIE	Interrupt enable flag 0: Disabled 1: Enabled
0	TMRIR	Interrupt request flag 0: No interrupt request 1: Generate interrupt request

■ Time Base Interrupt Control Register (TBICR: 0x03FF0)

Time base interrupt control register (TBICR) is used to control interrupt level of time base interrupt, interrupt enable and interrupt request. Interrupt control register should be operated when the maskable interrupt enable flag (MIE) of PSW is "0". Refer to [Chapter 7 7.1.4 Maskable Interrupt Control Register Setup] for the setting procedure of the flags.

bp	7	6	5	4	3	2	1	0
Flag	TBLV1	TBLV0	-	-	Reserved	-	TBIE	TBIR
At reset	0	0	-	-	0	-	0	0
Access	R/W	R/W	-	-	R/W	-	R/W	R/W

bp	Flag	Description
7 to 6	TBLV1 TBLV0	Interrupt level flag This flag sets the interrupt level 0 to 3 for interrupt requests.
5 to 4	-	-
3	Reserved	Always set to "0".
2	-	-
1	TBIE	Interrupt enable flag 0: Disabled 1: Enabled
0	TBIR	Interrupt request flag 0: No interrupt request 1: Generate interrupt request

■ Timer 7 Interrupt Control Register (TM7ICR: 0x03FF1)

Timer 7 interrupt control register (TM7ICR) is used to control interrupt level of timer 7 interrupt, interrupt enable and interrupt request. Interrupt control register should be operated when the maskable interrupt enable flag (MIE) of PSW is "0". Refer to [Chapter 7 7.1.4 Maskable Interrupt Control Register Setup] for the setting procedure of the flags.

bp	7	6	5	4	3	2	1	0
Flag	TM7LV1	TM7LV0	-	-	Reserved	-	TM7IE	TM7IR
At reset	0	0	-	-	0	-	0	0
Access	R/W	R/W	-	-	R/W	-	R/W	R/W

bp	Flag	Description
7 to 6	TM7LV1 TM7LV0	Interrupt level flag This flag sets the interrupt level 0 to 3 for interrupt requests.
5 to 4	-	-
3	Reserved	Always set to "0".
2	-	-
1	TM7IE	Interrupt enable flag 0: Disabled 1: Enabled
0	TM7IR	Interrupt request flag 0: No interrupt request 1: Generate interrupt request

■ Timer 8 Interrupt Control Register (TM8ICR: 0x03FF2)

Timer 8 interrupt control register (TM8ICR) is used to control interrupt level of timer 8 interrupt, interrupt enable and interrupt request. Interrupt control register should be operated when the maskable interrupt enable flag (MIE) of PSW is "0". Refer to [Chapter 7 7.1.4 Maskable Interrupt Control Register Setup] for the setting procedure of the flags.

bp	7	6	5	4	3	2	1	0
Flag	TM8LV1	TM8LV0	-	-	Reserved	-	TM8IE	TM8IR
At reset	0	0	-	-	0	-	0	0
Access	R/W	R/W	-	-	R/W	-	R/W	R/W

bp	Flag	Description
7 to 6	TM8LV1 TM8LV0	Interrupt level flag This flag sets the interrupt level 0 to 3 for interrupt requests.
5 to 4	-	-
3	Reserved	Always set to "0".
2	-	-
1	TM8IE	Interrupt enable flag 0: Disabled 1: Enabled
0	TM8IR	Interrupt request flag 0: No interrupt request 1: Generate interrupt request

■ Timer 9 Compare Register 2 Match Interrupt Control Register (TM9OC2ICR: 0x03FF3)

Timer 9 compare register 2 match interrupt control register (TM9OC2ICR) is used to control interrupt level of timer 9 compare register 2 match interrupt, interrupt enable and interrupt request. Interrupt control register should be operated when the maskable interrupt enable flag (MIE) of PSW is "0". Refer to [Chapter 7 7.1.4 Maskable Interrupt Control Register Setup] for the setting procedure of the flags.

bp	7	6	5	4	3	2	1	0
Flag	TM9OC2LV1	TM9OC2LV0	-	-	Reserved	-	TM9OC2IE	TM9OC2IR
At reset	0	0	-	-	0	-	0	0
Access	R/W	R/W	-	-	R/W	-	R/W	R/W

bp	Flag	Description
7 to 6	TM9OC2LV1 TM9OC2LV0	Interrupt level flag This flag sets the interrupt level 0 to 3 for interrupt requests.
5 to 4	-	-
3	Reserved	Always set to "0".
2	-	-
1	TM9OC2IE	Interrupt enable flag 0: Disabled 1: Enabled
0	TM9OC2IR	Interrupt request flag 0: No interrupt request 1: Generate interrupt request

- Serial 0 to 3 Interrupt Control Registers (SC0TICR: 0x03FF4, SC1TICR: 0x03FF6, SC2TICR: 0x03FF8, SC3TICR: 0x03FFA)

Serial 0 to 3 interrupt control register (SC0TICR to SC3TICR) are used to control interrupt levels of Serial 0 to 3 interrupts, interrupt enable and interrupt request. Interrupt control registers should be operated when the maskable interrupt enable flag (MIE) of PSW is "0". Refer to [Chapter 7 7.1.4 Maskable Interrupt Control Register Setup] for the setting procedure of the flags.

bp	7	6	5	4	3	2	1	0
Flag	SCnTLV1	SCnTLV0	-	-	Reserved	-	SCnTIE	SCnTIR
At reset	0	0	-	-	0	-	0	0
Access	R/W	R/W	-	-	R/W	-	R/W	R/W

bp	Flag	Description
7 to 6	SCnTLV1 SCnTLV0	Interrupt level flag This flag sets the interrupt level 0 to 3 for interrupt requests.
5 to 4	-	-
3	Reserved	Always set to "0".
2	-	-
1	SCnTIE	Interrupt enable flag 0: Disabled 1: Enabled
0	SCnTIR	Interrupt request flag 0: No interrupt request 1: Generate interrupt request

- Serial 0 to 3 UART Reception Interrupt Control Register (SC0RICR: 0x03FF5, SC1RICR: 0x03FF7, SC2RICR: 0x03FF9, SC3RICR: 0x03FFB)

Serial 0 to 3 UART reception interrupt control registers (SC0RICR to SC3RICR) are used to control interrupt level of serial 0 to 3 UART reception interrupt, interrupt enable and interrupt request. Interrupt control register should be operated when the maskable interrupt enable flag (MIE) of PSW is "0". Refer to [Chapter 7 7.1.4 Maskable Interrupt Control Register Setup] for the setting procedure of the flags.

bp	7	6	5	4	3	2	1	0
Flag	SCnRLV1	SCnRLV0	-	-	Reserved	-	SCnRIE	SCnRIR
At reset	0	0	-	-	0	-	0	0
Access	R/W	R/W	-	-	R/W	-	R/W	R/W

bp	Flag	Description
7 to 6	SCnRLV1 SCnRLV0	Interrupt level flag This flag sets the interrupt level 0 to 3 for interrupt requests.
5 to 4	-	-
3	Reserved	Always set to "0".
2	-	-
1	SCnRIE	Interrupt enable flag 0: Disabled 1: Enabled
0	SCnRIR	Interrupt request flag 0: No interrupt request 1: Generate interrupt request

■ Serial 4 Interrupt Control Register (SC4ICR: 0x03FFC)

Serial 4 interrupt control register (SC4ICR) is used to control interrupt level of serial 4 interrupt, interrupt enable and interrupt request. Interrupt control register should be operated when the maskable interrupt enable flag (MIE) of PSW is "0". Refer to [Chapter 7 7.1.4 Maskable Interrupt Control Register Setup] for the setting procedure of the flags.

bp	7	6	5	4	3	2	1	0
Flag	SC4LV1	SC4LV0	-	-	Reserved	-	SC4IE	SC4IR
At reset	0	0	-	-	0	-	0	0
Access	R/W	R/W	-	-	R/W	-	R/W	R/W

bp	Flag	Description
7 to 6	SC4LV1 SC4LV0	Interrupt level flag This flag sets the interrupt level 0 to 3 for interrupt requests.
5 to 4	-	-
3	Reserved	Always set to "0".
2	-	-
1	SC4IE	Interrupt enable flag 0: Disabled 1: Enabled
0	SC4IR	Interrupt request flag 0: No interrupt request 1: Generate interrupt request

■ Serial 4 Stop Condition Interrupt Control Register (SC4STPCICR: 0x03FFD)

Serial 4 stop condition interrupt control register (SC4STPCICR) is used to control interrupt level of serial 4 stop condition interrupt, interrupt enable and interrupt request. Interrupt control register should be operated when the maskable interrupt enable flag (MIE) of PSW is "0". Refer to [Chapter 7 7.1.4 Maskable Interrupt Control Register Setup] for the setting procedure of the flags.

bp	7	6	5	4	3	2	1	0
Flag	SC4 STPCLV1	SC4 STPCLV0	-	-	Reserved	-	SC4 STPCIE	SC4 STPCIR
At reset	0	0	-	-	0	-	0	0
Access	R/W	R/W	-	-	R/W	-	R/W	R/W

bp	Flag	Description
7 to 6	SC4STPCLV1 SC4STPCLV0	Interrupt level flag This flag sets the interrupt level 0 to 3 for interrupt requests.
5 to 4	-	-
3	Reserved	Always set to "0".
2	-	-
1	SC4STPCIE	Interrupt enable flag 0: Disabled 1: Enabled
0	SC4STPCIR	Interrupt request flag 0: No interrupt request 1: Generate interrupt request

■ Peripheral Function Group Interrupt Level Setting Register (PERILR: 0x03FFE)

Peripheral function group interrupt level setting register (PERILR) is used to control interrupt level of peripheral function group interrupt. Interrupt control register should be operated when the maskable interrupt enable flag (MIE) of PSW is "0". Refer to [Chapter 7 7.1.5 Peripheral Function Group Interrupt Control Register Setup] for the setting procedure of the flags.

bp	7	6	5	4	3	2	1	0
Flag	PERILV1	PERILV0	-	-	Reserved	-	-	-
At reset	0	0	-	-	0	-	-	-
Access	R/W	R/W	-	-	R/W	-	-	-

bp	Flag	Description
7 to 6	PERILV1 PERILV0	Interrupt level flag This flag sets the interrupt level 0 to 3 for interrupt requests.
5 to 4	-	-
3	Reserved	Always set to "0".
2 to 0	-	-



Do not change the value of PERILR register when peripheral function group interrupt factor retention register (IRQEXPDT) corresponding to peripheral function group interrupt input enable register (IRQEXPEN) is set. If changed, an unintended interrupt may be processed.

■ Peripheral Function Group Interrupt Input Enable Register (IRQEXPEN: 0x03F4E)

Peripheral function group interrupt input enable register (IRQEXPEN) is used to control peripheral function group interrupt factors. Interrupt control register should be operated when the maskable interrupt enable flag (MIE) of PSW is "0". Refer to [Chapter 7 7.1.5 Peripheral Function Group Interrupt Control Register Setup] for the setting procedure of the flags.

bp	7	6	5	4	3	2	1	0
Flag	Reserved	IRQEXP EN6	IRQEXP EN5	IRQEXP EN4	IRQEXP EN3	IRQEXP EN2	IRQEXP EN1	IRQEXP EN0
At reset	0	0	0	0	0	0	0	0
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description
7	Reserved	Always set to "0".
6	IRQEXPEN6	ATC1 interrupt enable flag 0: Disabled 1: Enabled
5	IRQEXPEN5	A/D conversion interrupt enable flag 0: Disabled 1: Enabled
4	IRQEXPEN4	24H timer alarm interrupt enable flag 0: Disabled 1: Enabled
3	IRQEXPEN3	PWM underflow interrupt enable flag 0: Disabled 1: Enabled
2	IRQEXPEN2	PWM overflow interrupt enable flag 0: Disabled 1: Enabled
1	IRQEXPEN1	Timer 8 compare 2 match interrupt enable flag 0: Disabled 1: Enabled
0	IRQEXPEN0	Timer 7 compare 2 match interrupt enable flag 0: Disabled 1: Enabled

■ Peripheral Function Group Interrupt Factor Retention Register (IRQEXPDT: 0x03F4F)

Peripheral function group interrupt factor retention register (IRQEXPDT) is used to retain the peripheral function group interrupt factors. Interrupt control register should be operated when the maskable interrupt enable flag (MIE) of PSW is "0". Refer to [Chapter 7 7.1.5 Peripheral Function Group Interrupt Control Register Setup] for the setting procedure of the flags.

bp	7	6	5	4	3	2	1	0
Flag	Reserved	IRQEXP DT6	IRQEXP DT5	IRQEXP DT4	IRQEXP DT3	IRQEXP DT2	IRQEXP DT1	IRQEXP DT0
At reset	0	0	0	0	0	0	0	0
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description
7	Reserved	Always set to "0".
6	IRQEXPDT6	ATC1 interrupt request flag 0: No interrupt request 1: Generate interrupt request
5	IRQEXPDT5	A/D conversion interrupt request flag 0: No interrupt request 1: Generate interrupt request
4	IRQEXPDT4	24H timer alarm interrupt request flag 0: No interrupt request 1: Generate interrupt request
3	IRQEXPDT3	PWM underflow interrupt request flag 0: No interrupt request 1: Generate interrupt request
2	IRQEXPDT2	PWM overflow interrupt request flag 0: No interrupt request 1: Generate interrupt request
1	IRQEXPDT1	Timer 8 compare 2 match interrupt request flag 0: No interrupt request 1: Generate interrupt request
0	IRQEXPDT0	Timer 7 compare 2 match interrupt request flag 0: No interrupt request 1: Generate interrupt request

7.2.3 Block Diagram

■ Peripheral Function Group Interrupt Interface Block Diagram

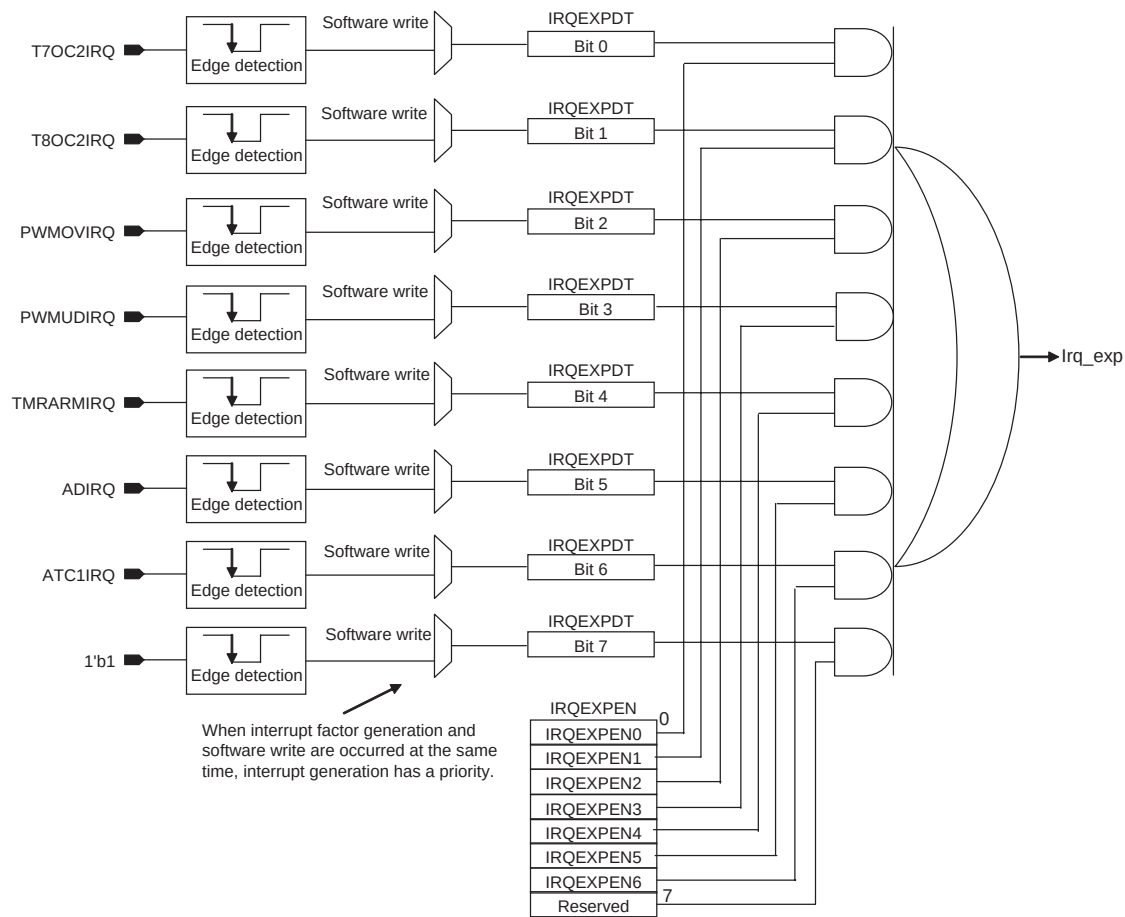
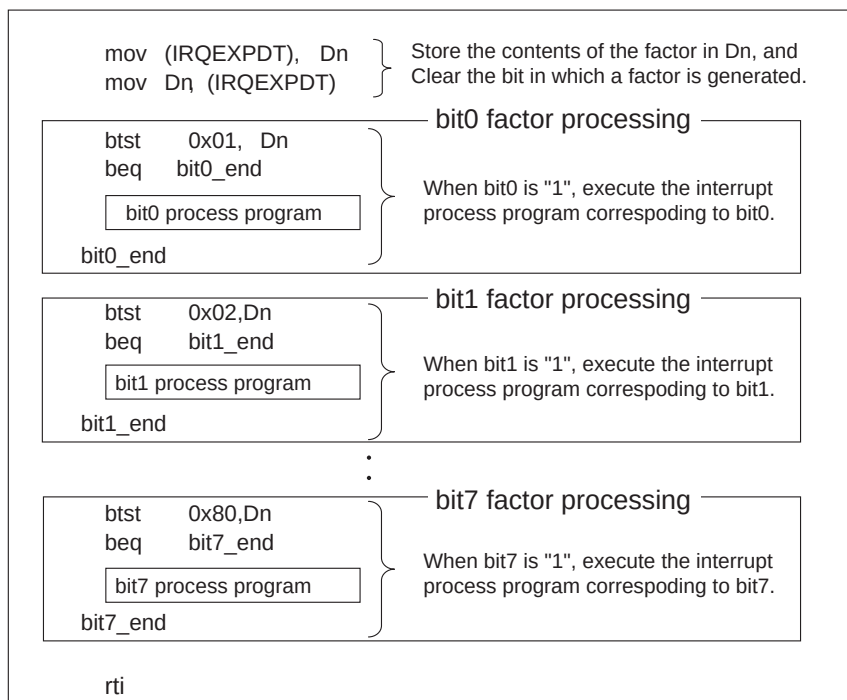


Figure:7.2.1 Peripheral Function Group Interrupt Interface Block Diagram

■ Program Example of Interrupt Process Routine



7.3 External Interrupts

This LSI has 5 external interrupts. The circuit (external interrupt interface), operates the external interrupt input signal, is built-in between the external interrupt input pin and the external interrupt block. This external interrupt interface can manage to do with any kind of external interrupts.

7.3.1 Overview

Table:7.3.1 shows the list of functions which external interrupts 0 to 4/key interrupt are used.

Table:7.3.1 External Interrupt Functions

Table remarks √: With function -: Without function

	External interrupt input pin	Programmable active edge interrupt	Both edges interrupt	Built-in Noise filter	Key input interrupt	Level interrupt	MN101 EF76K	MN101 EF57G	MN101 EF56K
External interrupt 0	P20	√	-	√	-	-	√	√	√
	PC0						√	-	-
External interrupt 1	P21	√	-	√	-	-	√	√	√
	PC1						√	-	-
External interrupt 2	P22	√	√	√	-	√	√	√	√
	PC2						√	-	-
External interrupt 3	P23	√	√	√	-	√	√	√	√
	PC3						√	-	-
External interrupt 4/ Key interrupt	P24	√	√	√	-	√	√	√	√
	PC4						√	-	-
	P70 to P77			-	√	-	√	√	√



The pulse which is shorter than the system clock (fs) cycle is neglected because the external interrupts from pins and the key input interrupt events are acknowledged at the rising of the system clock (fs).



Both external interrupt 4 and the key interrupt cannot be used at the same time because they share external interrupt 4 control register (IRQ4ICR). The KEYT3SEL flag of key interrupt control register 1 (KEYT3_1MD) is used to select either one to use.

7.3.2 Block Diagram

■ External Interrupt Interface 0 Block Diagram

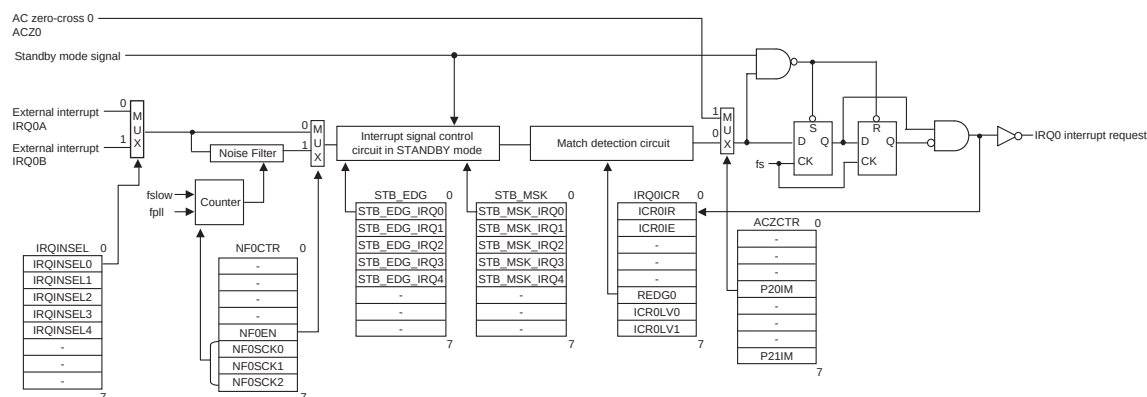


Figure:7.3.1 External Interrupt Interface 0 Block Diagram (MN101EF76K)

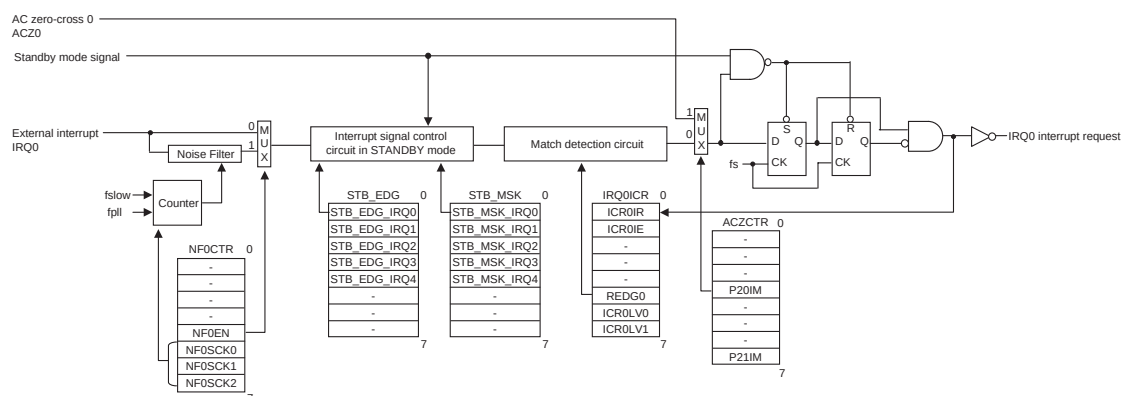


Figure:7.3.2 External Interrupt Interface 0 Block Diagram (MN101EF57G/MN101EF56K)

■ External Interrupt Interface 1 Block Diagram

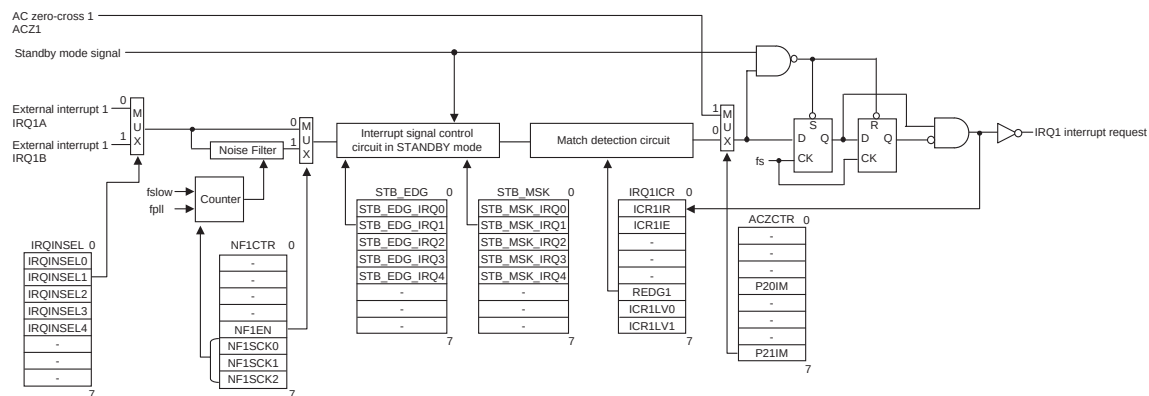


Figure:7.3.3 External Interrupt Interface 1 Block Diagram (MN101EF76K)

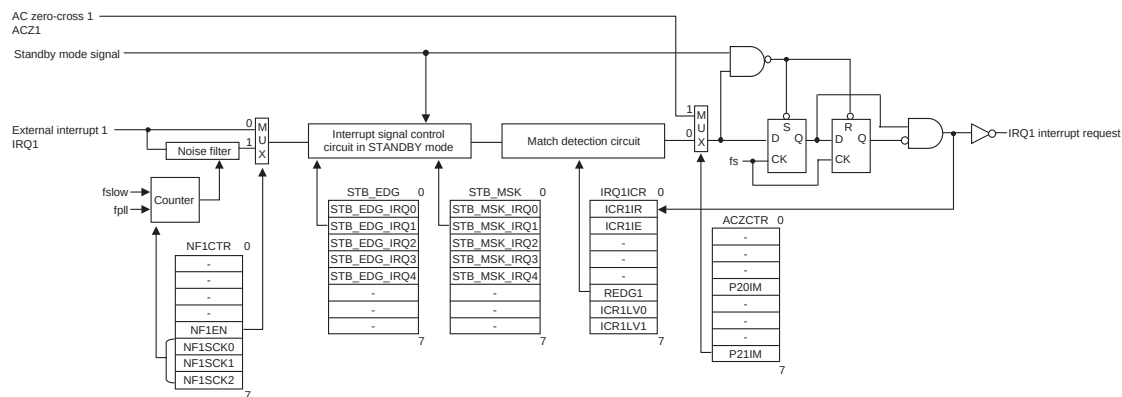


Figure:7.3.4 External Interrupt Interface 1 Block Diagram (MN101EF57G/MN101EF56K)

■ External Interrupt Interface 2 Block Diagram

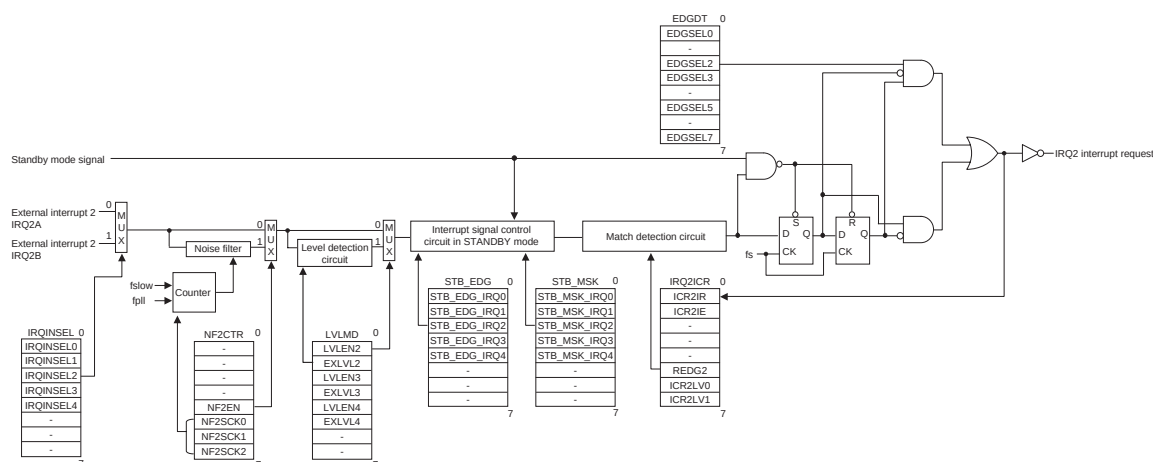


Figure:7.3.5 External Interrupt Interface 2 Block Diagram (MN101EF76K)

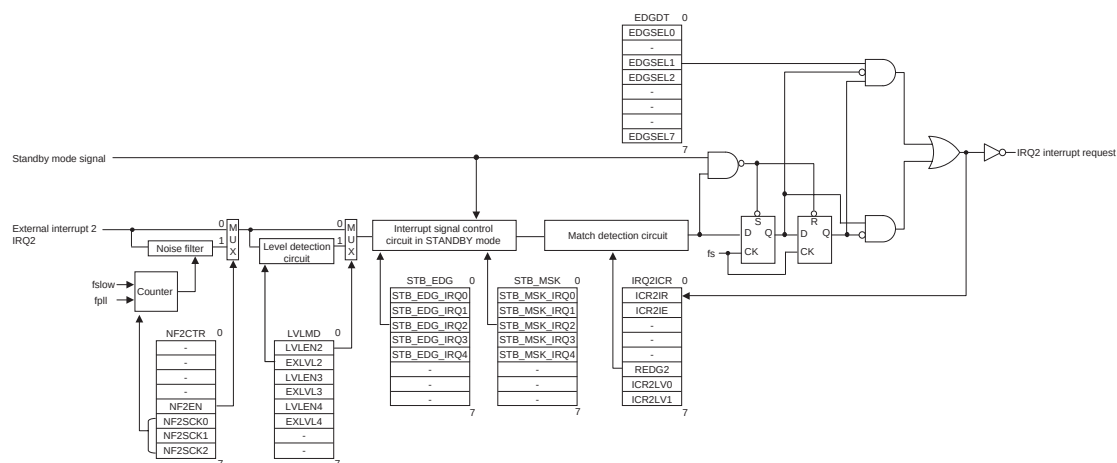


Figure:7.3.6 External Interrupt Interface 2 Block Diagram (MN101EF57G/MN101EF56K)

■ External Interrupt Interface 3 Block Diagram

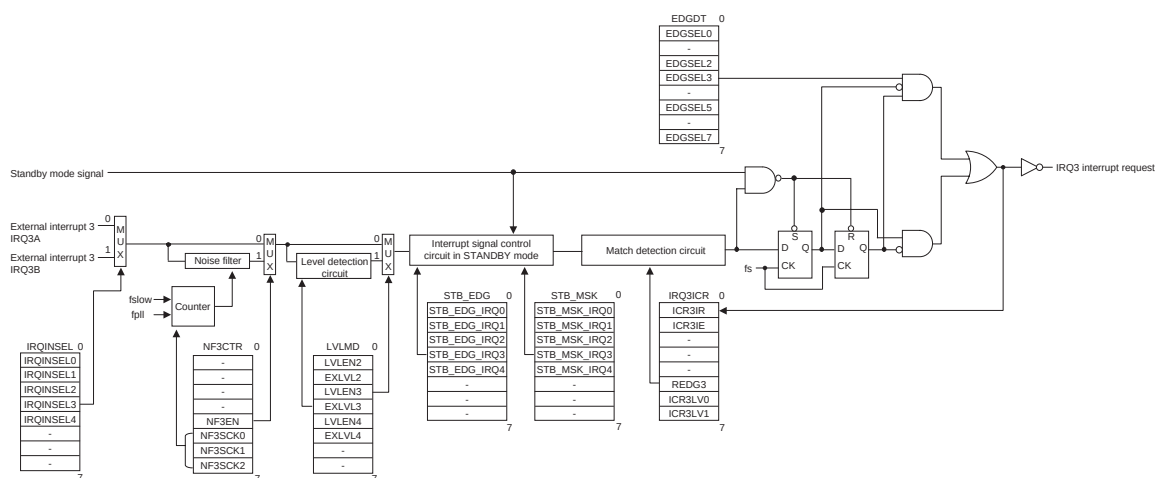


Figure:7.3.7 External Interrupt Interface 3 Block Diagram (MN101EF76K)

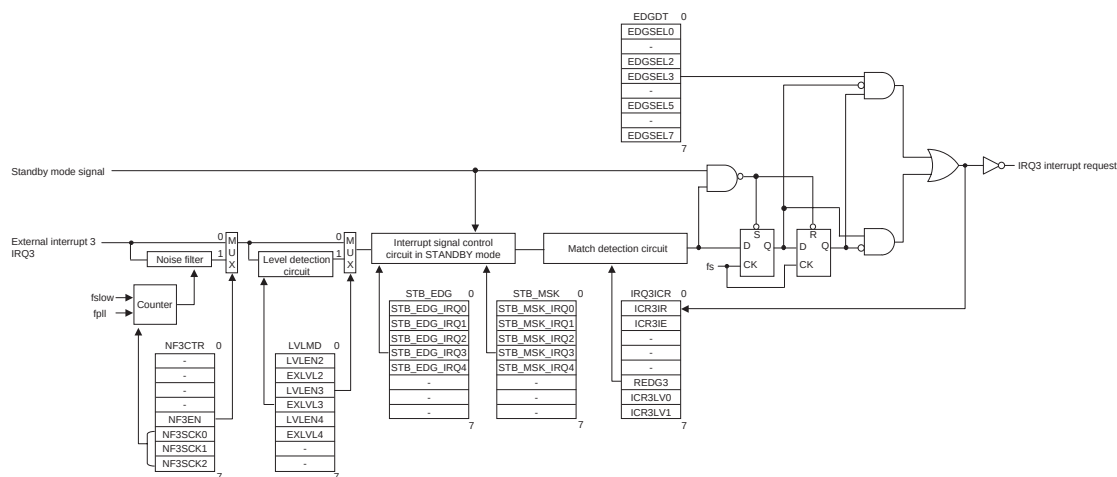


Figure:7.3.8 External Interrupt Interface 3 Block Diagram (MN101EF57G/MN101EF56K)

■ External Interrupt Interface 4/ Key Input Block Diagram

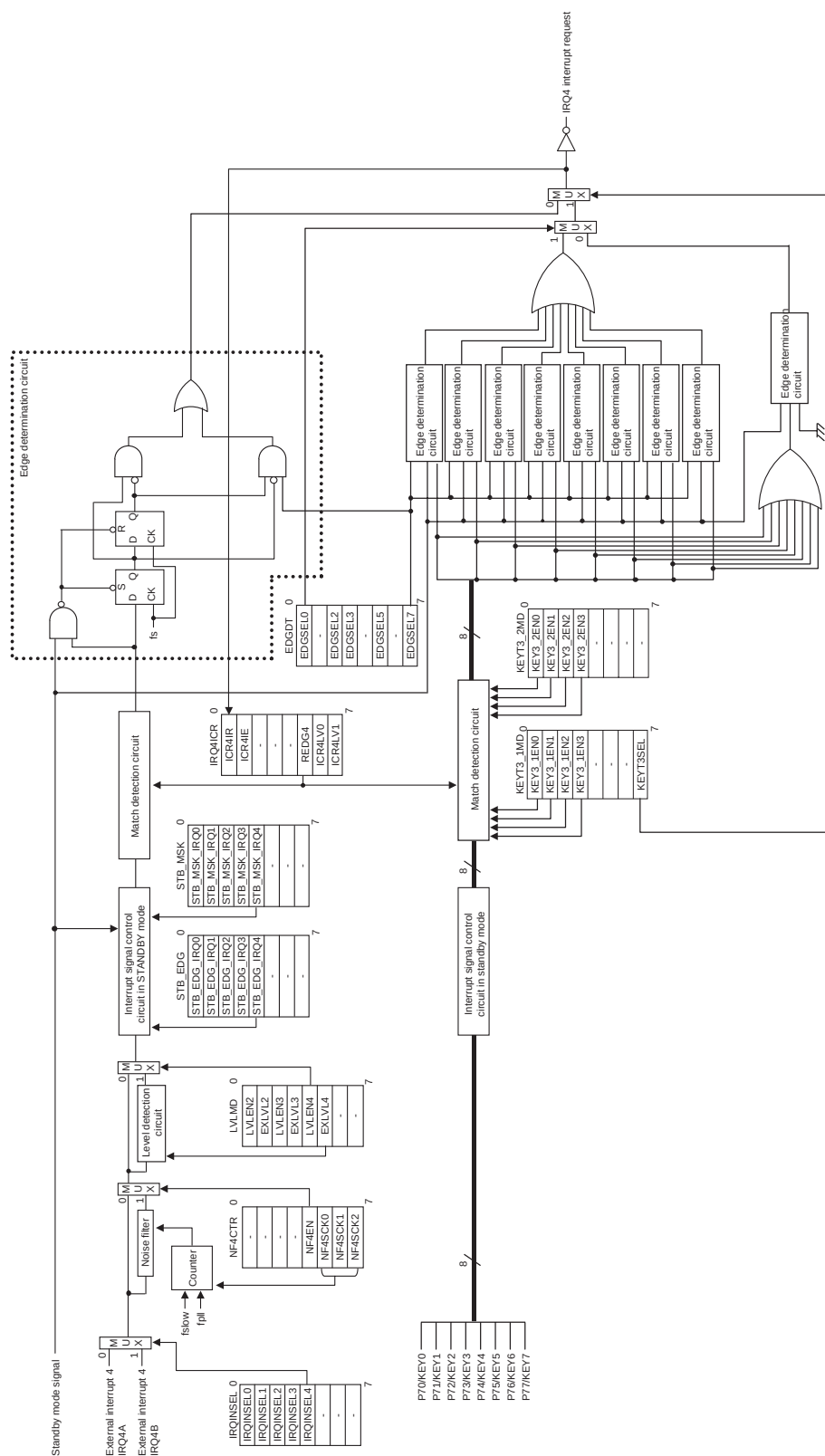


Figure:7.3.9 External Interrupt Interface 4 Block Diagram (MN101EF76K)

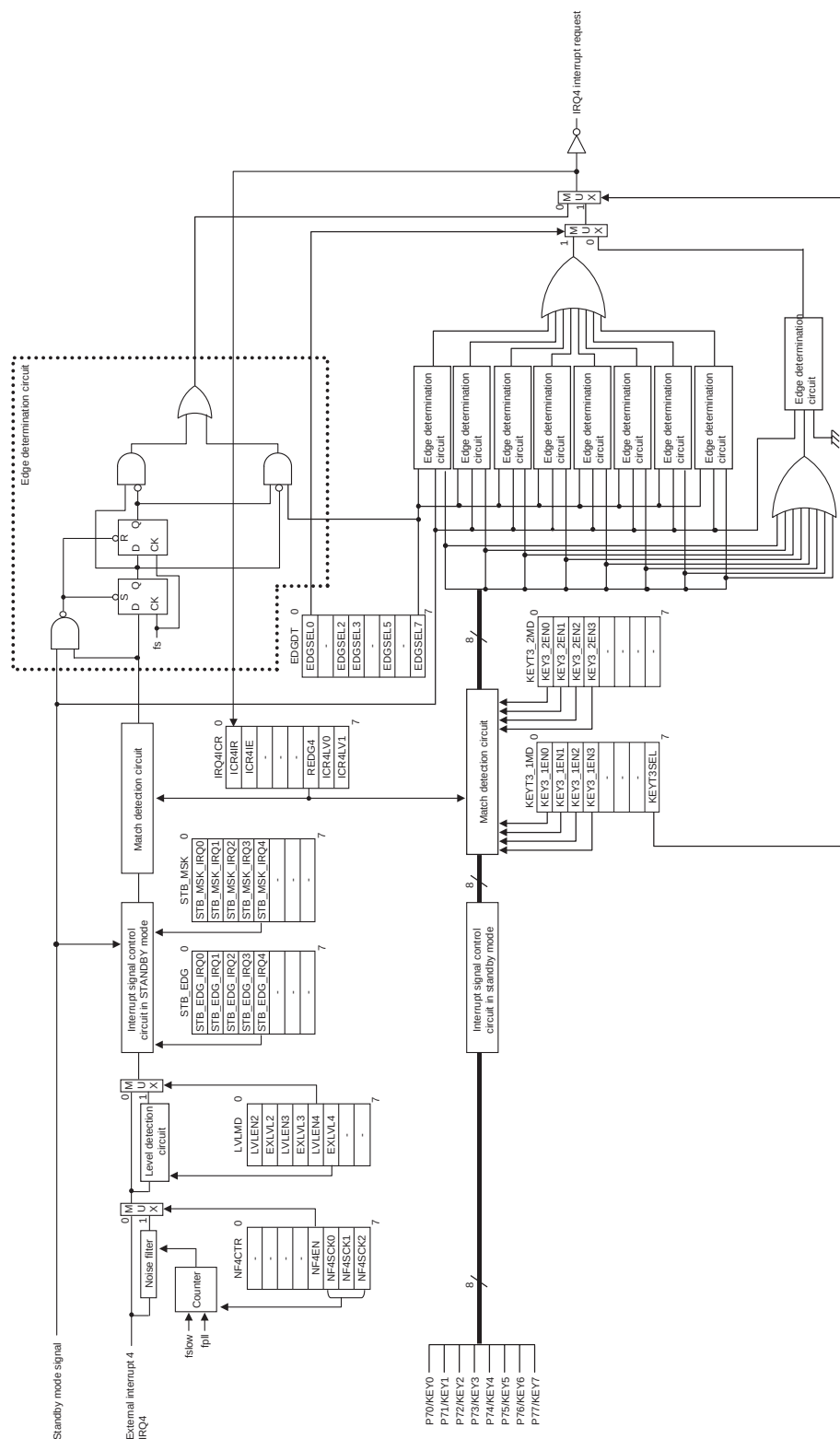


Figure:7.3.10 External Interrupt Interface 4 Block Diagram (MN101EF57G/MN101EF56K)

7.3.3 External Interrupt Control Registers

External interrupt input signals pass through each respective external interrupt interface to generate external interrupts 0 to 3 and external interrupt 4/key interrupt.

External interrupt interface 0 to 3 and the external interrupt interface 4/key interrupt are controlled by external interrupt control register (IRQnICR).

External interrupt interface 0 to 4 are controlled by noise filter function (NF0CTR to NF4CTR) and external interrupt pin setting register (IRQCNT).

External interrupt interface 2 to 3 are controlled by both edges interrupt control register (EDGDT), external interrupt enable input switch control register (LVLMD), noise filter function (NF2CTR to NF3CTR) and external interrupt pin setting register (IRQCNT).

External interrupt interface 4/key interrupt is controlled by both edges interrupt control register (EDGDT), external interrupt enable input switch control register (LVLMD), noise filter function (NF4CTR), external interrupt pin setting register (IRQCNT), key interrupt control register 1 (KEYT3_1IMD) and key interrupt control register 2 (KEYT3_2IMD).

In addition, external interrupt interface 0 to 4/key interrupt are used to control interrupt generation factors in CPU OPERATION mode/STANDBY mode by edge interrupt in STANDBY control register (STB_EDG) and interrupt in STANDBY control register (STB_MSK).

The following table shows registers which control external interrupt 0 to 3 and external interrupt 4/key interrupt.

Table:7.3.2 External Interrupt Control Register

External interrupt	Register	Address	R/W	Function	Page	MN101 EF76K	MN101 EF57G	MN101 EF56K
External interrupt 0	IRQ0ICR	0x03FE2	R/W	External interrupt 0 control register	VII-24	√	√	√
	IRQCNT	0x03FD0	R/W	External interrupt pin setting register	VII-51	√	√	√
	IRQINSEL	0x03F8D	R/W	External interrupt pin selection register	VII-53	√	-	-
	NF0CTR	0x03ED1	R/W	Noise filter 0 control register	VII-54	√	√	√
	ACZCTR	0x03F2E	R/W	AC zero-cross detection interrupt register	VII-55	√	√	√
	STB_EDG	0x03F5E	R/W	Edge interrupt in STANDBY control register	VII-56	√	√	√
	STB_MSK	0x03F5F	R/W	Interrupt in STANDBY control register	VII-61	√	√	√
External interrupt 1	IRQ1ICR	0x03FE3	R/W	External interrupt 1 control register	VII-24	√	√	√
	IRQCNT	0x03FD0	R/W	External interrupt pin setting register	VII-51	√	√	√
	IRQINSEL	0x03F8D	R/W	External interrupt pin selection register	VII-53	√	-	-
	NF1CTR	0x03FD2	R/W	Noise filter 1 control register	VII-54	√	√	√
	ACZCTR	0x03F2E	R/W	AC zero-cross detection interrupt register	VII-55	√	√	√
	STB_EDG	0x03F5E	R/W	Edge interrupt in STANDBY control register	VII-60	√	√	√
	STB_MSK	0x03F5F	R/W	Interrupt in STANDBY control register	VII-61	√	√	√
External interrupt 2	IRQ2ICR	0x03FE4	R/W	External interrupt 2 control register	VII-24	√	√	√
	IRQCNT	0x03FD0	R/W	External interrupt pin setting register	VII-51	√	√	√
	IRQINSEL	0x03F8D	R/W	External interrupt pin selection register	VII-53	√	-	-
	NF2CTR	0x03FD3	R/W	Noise filter 2 control register	VII-54	√	√	√
	EDGDT	0x03F1E	R/W	Both edges interrupt control register	VII-56	√	√	√
	LVLMD	0x03FD7	R/W	External interrupt valid input switching control register	VII-57	√	-	-
	STB_EDG	0x03F5E	R/W	Edge interrupt in STANDBY control register	VII-60	√	√	√
	STB_MSK	0x03F5F	R/W	Interrupt in STANDBY control register	VII-61	√	√	√

External interrupt	Register	Address	R/W	Function	Page	MN101 EF76K	MN101 EF57G	MN101 EF56K
External interrupt 3	IRQ3ICR	0x03FE5	R/W	External interrupt 3 control register	VII-24	√	√	√
	IRQCNT	0x03FD0	R/W	External interrupt pin setting register	VII-51	√	√	√
	IRQINSEL	0x03F8D	R/W	External interrupt pin selection register	VII-53	√	-	-
	NF3CTR	0x03FD4	R/W	Noise filter 3 control register	VII-54	√	√	√
	EDGDT	0x03F1E	R/W	Both edges interrupt control register	VII-56	√	√	√
	LVLMD	0x03FD7	R/W	External interrupt valid input switching control register	VII-57	√	√	√
	STB_EDG	0x03F5E	R/W	Edge interrupt in STANDBY control register	VII-60	√	√	√
	STB_MSK	0x03F5F	R/W	Interrupt in STANDBY control register	VII-61	√	√	√
External interrupt 4 /Key interrupt	IRQ4ICR	0x03FE6	R/W	External interrupt 4 control register	VII-24	√	√	√
	IRQCNT	0x03FD0	R/W	External interrupt pin setting register	VII-51	√	√	√
	IRQINSEL	0x03F8D	R/W	External interrupt pin selection register	VII-53	√	-	-
	NF4CTR	0x03FD5	R/W	Noise filter 4 control register	VII-54	√	√	√
	EDGDT	0x03F1E	R/W	Both edges interrupt control register	VII-56	√	√	√
	LVLMD	0x03FD7	R/W	External interrupt valid input switching control register	VII-57	√	√	√
	KEYT3_1IMD	0x03F3E	R/W	Key interrupt control register 1	VII-58	√	√	√
	KEYT3_2IMD	0x03F3F	R/W	Key interrupt control register 2	VII-59	√	√	√
	STB_EDG	0x03F5E	R/W	Edge interrupt in STANDBY control register	VII-60	√	√	√
	STB_MSK	0x03F5F	R/W	Interrupt in STANDBY control register	VII-61	√	√	√

R/W: Readable / Writable.

■ External Interrupt Pin Setting Register (IRQCNT: 0x03FD0)

External interrupt setting register (IRQCNT) is used to enable or disable External Interrupt 0 to 4. IRQCNT register should be operated when the maskable interrupt enable flag (MIE) of PSW is "0".

bp	7	6	5	4	3	2	1	0
Flag	-	-	-	P24EN	P23EN	P22EN	P21EN	P20EN
At reset	-	-	-	0	0	0	0	0
Access	-	-	-	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description
7 to 5	-	-
4	P24EN	P24 external interrupt 4 input 0: Disable 1: Enable
3	P23EN	P23 external interrupt 3 input 0: Disable 1: Enable
2	P22EN	P22 external interrupt 2 input 0: Disable 1: Enable
1	P21EN	P21 external interrupt 1 input 0: Disable 1: Enable
0	P20EN	P20 external interrupt 0 input 0: Disable 1: Enable

Change flags in accordance with (2) described in [Chapter 7 7.1.4 Maskable Interrupt Control Register Setup].



Set P2nEN flag to “0” to fix the input of external interrupts to “Low” except when P2IN register is being read.



To enable maskable interrupts after setting P2nEN flag to “1”, make sure to set IR flag of the corresponding interrupt to “0” beforehand.



When the external interrupt is not used, set IRQnIE flag to “0”.
An unintended interrupt may be generated when Port 2 input register (P2IN) is read while the IRQnIE flag is set to “1”, even if the P2nEN flag is set to “0”.



The interrupt request flag may have been set when the interrupt edge is switched. It is necessary to specify the interrupt valid edge before interrupts are enabled. Also, if the interrupt request flag has already been set, it must be cleared.

■ External Interrupt Pin Selection Register (IRQINSEL: 0x03FD8)

External interrupt selection register (IRQINSEL) is used to select type A or type B for the input pin of IRQ0 to IRQ4.

<MN101EF76K>

bp	7	6	5	4	3	2	1	0
Flag	-	-	-	IRQINSEL4	IRQINSEL3	IRQINSEL2	IRQINSEL1	IRQINSEL0
At reset	-	-	-	0	0	0	0	0
Access	-	-	-	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description
7 to 5	-	-
4	IRQINSEL4	IRQ4 pin selection 0: IRQ4A (P24) 1: IRQ4B (PC4)
3	IRQINSEL3	IRQ3 pin selection 0: IRQ3A (P23) 1: IRQ3B (PC3)
2	IRQINSEL2	IRQ2 pin selection 0: IRQ2A (P22) 1: IRQ2B (PC2)
1	IRQINSEL1	IRQ1 pin selection 0: IRQ1A (P21) 1: IRQ1B (PC1)
0	IRQINSEL0	IRQ0 pin selection 0: IRQ0A (P20) 1: IRQ0B (PC0)



To set IRQINSELn flags, make sure to set IRQCNT register to "00" beforehand.
Refer to [Chapter 7 7.1.4 Maskable Interrupt Control Register Setup] for changing flags.



PC0 (IRQ0B) and PC1 (IRQ1B) do not equip AC zero-cross function.
To enable AC zero-cross function by setting the P2nIM flag of ACZCTR register, make sure to set IRQINSELn flag to "0" beforehand.

- Noise Filter 0 to 4 Control Registers (NF0CTR: 0x03FD1, NF1CTR: 0x03FD2, NF2CTR: 0x03FD3, NF3CTR: 0x03FD4, NF4CTR: 0x03FD5)

Noise filter 0 to 4 control registers (NF0CTR to NF4CTR) are used to set the noise removal function on the external interrupt 0 to 4. They are also used to select the sampling cycle of noise removal function.

The NFnEN1 flags should be operated when the maskable interrupt enable flag (MIE) of PSW is "0".

bp	7	6	5	4	3	2	1	0
Flag	NFnSCK2	NFnSCK1	NFnSCK0	NFnEN1	-	-	-	-
At reset	0	0	0	0	-	-	-	-
Access	R/W	R/W	R/W	R/W	-	-	-	-

bp	Flag	Description
7 to 5	NFnSCK2 NFnSCK1 NFnSCK0	IRQn noise sampling frequency 000: fpll 001: fpll/2 ⁵ 010: fpll/2 ⁶ 011: fpll/2 ⁷ 100: fpll/2 ⁸ 101: fpll/2 ⁹ 110: fpll/2 ¹⁰ 111: fslow
4	NFnEN1	0: Noise filter OFF 1: Noise filter ON
3 to 0	-	-

Change flags in accordance with (2) described in [Chapter 7 7.1.4 Maskable Interrupt Control Register Setup].



To enable maskable interrupts after setting NFnEN1 flag to "1", make sure to set IR flag of the corresponding interrupt to "0" beforehand.

■ AC zero-cross Detection Interrupt Register (ACZCTR: 0x03F2E)

AC zero-cross detection interrupt register (ACZCTR) is used to set zero-cross detection function to IRQ0A and IRQ1A in MN101EF76K (to IRQ0 and IRQ1 in MN101EF57G and MN101EF56K).

ACZCTR register should be operated when the maskable interrupt enable flag (MIE) of PSW is "0".

bp	7	6	5	4	3	2	1	0
Flag	P21IM	-	-	-	P20IM	-	-	-
At reset	0	-	-	-	0	-	-	-
Access	R/W	-	-	-	R/W	-	-	-

bp	Flag	Description
7	P21IM	ACZ1 input enable flag 0: Disabled 1: Enabled
6 to 4	-	-
3	P20IM	ACZ0 input enable flag 0: Disabled 1: Enabled
2 to 0	-	-

Change flags in accordance with (2) described in [Chapter 7 7.1.4 Maskable Interrupt Control Register Setup].



To enable maskable interrupts after setting P2nIM flag to "1", make sure to set IR flag of the corresponding interrupt to "0" beforehand.

■ Both Edges Interrupt Control Register (EDGDT: 0x03F1E)

Both edges interrupt control register (EDGDT) is used to select interrupt edges of IRQ2 to 4 and LINIRQ by bp 2, 3, 5 and 7. With this register, the edge to generate interrupts is selected; both edges or the edge specified by external interrupt control register (IRQ2ICR, IRQ3ICR, IRQ4ICR) and LIN interrupt control register (LINICR). Both edges interrupt control register (EDGDT) is also used to select bit common or bit independent for the edge determination circuit in the key interrupt block.

bp	7	6	5	4	3	2	1	0
Flag	EDGSEL7	-	EDGSEL5	-	EDGSEL3	EDGSEL2	-	EDGSEL0
At reset	0	-	0	-	0	0	-	0
Access	R/W	-	R/W	-	R/W	R/W	-	R/W

bp	Flag	Description
7	EDGSEL7	IRQ4 both edges operation selection 0: Programmable active edge interrupt (Specified by IRQ4ICR bp:5 REDG4) 1: Both edges interrupt
6	-	-
5	EDGSEL5	LINIRQ both edges operation selection 0: Programmable active edge interrupt (Specified by LINICR bp:5 LINREDG) 1: Both edges interrupt
4	-	-
3	EDGSEL3	IRQ3 both edges operation selection 0: Programmable active edge interrupt (Specified by IRQ3ICR bp:5 REDG3) 1: Both edges interrupt
2	EDGSEL2	IRQ2 both edges operation selection 0: Programmable active edge interrupt (Specified by IRQ2ICR bp:5 REDG2) 1: Both edges interrupt
1	-	-
0	EDGSEL0	Key interrupt selection 0: Key input edge detection circuit bit common * 1: Key input edge detection circuit bit independent

Change flags in accordance with (2) described in [Chapter 7 7.1.4 Maskable Interrupt Control Register Setup].

* Refer to [Chapter 7 7.3.7 Key Input Interrupt] for operation details.



When “key input edge detection circuit bit common” is selected by the EDGSEL0 flag, an interrupt request signal is generated by inputting OR of the match detection circuit output for the valid key input. During the match detection by a single valid key input, no interrupt request signal is generated even if a match is detected by another valid key input.



When the key input edge determination circuit bit independent is selected by the EDGSEL flag, the edge determination circuit is used for each valid key input. Thus, even during match detection by a single valid key input, an interrupt request signal is generated by detecting a match by another valid key input.

■ External Interrupt Valid Input Switch Control Register (LVLMD: 0x03FD7)

bp	7	6	5	4	3	2	1	0
Flag	-	-	EXLVL4	LVLEN4	EXLVL3	LVLEN3	EXLVL2	LVLEN2
At reset	-	-	0	0	0	0	0	0
Access	-	-	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description
7 to 6	-	-
5	EXLVL4	External interrupt 4 valid Input Level setting (enabled at bp: 4 LVLEN4="1") 0: Low-level 1: High-level
4	LVLEN4	External interrupt 4 valid Input setting 0: Edge 1: Level
3	EXLVL3	External interrupt 3 valid Input Level setting (enabled at bp: 3 LVLEN3="1") 0: Low-level 1: High-level
2	LVLEN3	External interrupt 3 valid Input setting 0: Edge 1: Level
1	EXLVL2	External interrupt 2 valid Input Level setting (enabled at bp: 2 LVLEN2="1") 0: Low-level 1: High-level
0	LVLEN2	External interrupt 2 valid Input setting 0: Edge 1: Level

Change flags in accordance with (2) described in [Chapter 7 7.1.4 Maskable Interrupt Control Register Setup].

■ Key Interrupt Control Register 1 (KEYT3_1IMD: 0x03F3E)

Key interrupt control register (KEYT3_1IMD) is used to select whether the key interrupt or the external interrupt 4 is enabled. This register can also select which pin of Port 7 should be used to enable the key interrupt for each bit.

The KEYT3_1ENn flags should be operated when the maskable interrupt enable flag (MIE) of PSW is "0".

bp	7	6	5	4	3	2	1	0
Flag	KEYT3SEL	-	-	-	KEYT3_1EN3	KEYT3_1EN2	KEYT3_1EN1	KEYT3_1EN0
At reset	0	-	-	-	0	0	0	0
Access	R/W	-	-	-	R/W	R/W	R/W	R/W

bp	Flag	Description
7	KEYT3SEL	Interrupt source selection 0: External interrupt 4 1: Key interrupt
6 to 4	-	-
3	KEYT3_1EN3	KEY3 interrupt selection 0: Disabled 1: Enabled
2	KEYT3_1EN2	KEY2 interrupt selection 0: Disabled 1: Enabled
1	KEYT3_1EN1	KEY1 interrupt selection 0: Disabled 1: Enabled
0	KEYT3_1EN0	KEY0 interrupt selection 0: Disabled 1: Enabled

Change flags in accordance with (2) described in [Chapter 7 7.1.4 Maskable Interrupt Control Register Setup].



Both external interrupt 4 and the key interrupt cannot be used at the same time because they share external interrupt 4 control register (IRQ4ICR). The KEYT3SEL flag of key interrupt control register 1 (KEYT3_1IMD) is used to select which to use.



To enable maskable interrupts after setting KEYT3_1ENn flag to "1", make sure to set IR flag of the corresponding interrupt to "0" beforehand.

■ Key Interrupt Control Register 2 (KEYT3_2IMD: 0x03F3F)

Key interrupt control register 2 (KEYT3_2IMD) is used to select which pin of Port 7 should be used to enable the key interrupt for each bit.

KEYT3_2IMD register should be operated when the maskable interrupt enable flag (MIE) of PSW is "0".

bp	7	6	5	4	3	2	1	0
Flag	-	-	-	-	KEYT3_2EN3	KEYT3_2EN2	KEYT3_2EN1	KEYT3_2EN0
At reset	-	-	-	-	0	0	0	0
Access	-	-	-	-	R/W	R/W	R/W	R/W

bp	Flag	Description
7 to 4	-	-
3	KEYT3_2EN3	KEY7 interrupt selection 0: Disabled 1: Enabled
2	KEYT3_2EN2	KEY6 interrupt selection 0: Disabled 1: Enabled
1	KEYT3_2EN1	KEY5 interrupt selection 0: Disabled 1: Enabled
0	KEYT3_2EN0	KEY4 interrupt selection 0: Disabled 1: Enabled



To enable maskable interrupts after setting KEYT3_2ENn flag to "1", make sure to set IR flag of the corresponding interrupt to "0" beforehand.

■ Edge Interrupt in STANDBY Control Register (STB_EDG: 0x03F5E)

Edge interrupt in STANDBY control register (STB_EDG) is used to select the factors of the external interrupt 0 to 4/ key interrupt generation in STANDBY mode (HALT0/HALT1/HALT2/STOP0/STOP1) between edge detection and level detection of input signal.

bp	7	6	5	4	3	2	1	0
Flag	-	-	-	STB_EDG_IRQ4	STB_EDG_IRQ3	STB_EDG_IRQ2	STB_EDG_IRQ1	STB_EDG_IRQ0
At reset	-	-	-	0	0	0	0	0
Access	-	-	-	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description
7 to 5	-	-
4	STB_EDG_IRQ4	External interrupt 4/key interrupt Edge interrupt operation setting in STANDBY mode 0: Level interrupt 1: Edge interrupt
3	STB_EDG_IRQ3	External interrupt 3 Edge interrupt operation setting in STANDBY mode 0: Level interrupt 1: Edge interrupt
2	STB_EDG_IRQ2	External interrupt 2 Edge interrupt operation setting in STANDBY mode 0: Level interrupt 1: Edge interrupt
1	STB_EDG_IRQ1	External interrupt 1 Edge interrupt operation setting in STANDBY mode 0: Level interrupt 1: Edge interrupt
0	STB_EDG_IRQ0	External interrupt 0 Edge interrupt operation setting in STANDBY mode 0: Level interrupt 1: Edge interrupt



It is necessary to clear edge interrupt in STANDBY control register (STB_EDG) and interrupt in STANDBY control register (STB_MSK) to "00" before changing the value of external interrupt pin setting register (IRQCNT), noise filter n control register (NFnCTR) and external interrupt valid input control register (LVLMD).



"Edge interrupt selection" by the STB_EDG_IRQn flag of edge interrupt in STANDBY control register (STB_EDG) and "Level" settings by the EXLENn flag of external interrupt valid input control register (LVLMD) cannot be used at the same time. If using them together, an unintended interrupt may be generated.

■ Interrupt in STANDBY Control Register (STB_MSK: 0x03F5F)

Interrupt in STANDBY control register (STB_MSK) is used to select the execution timing of interrupt between before STANDBY mode transition and after STANDBY mode transition, when IRQ_n interrupt signal generates just before the transition to STANDBY mode.

bp	7	6	5	4	3	2	1	0
Flag	-	-	-	STB_MSK_IRQ4	STB_MSK_IRQ3	STB_MSK_IRQ2	STB_MSK_IRQ1	STB_MSK_IRQ0
At reset	-	-	-	0	0	0	0	0
Access	-	-	-	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description
7 to 5	-	-
4	STB_MSK_IRQ4	IRQ4 interrupt execution timing control until the transition to STANDBY mode 0: Before STANDBY mode transition (Normal operation) 1: After STANDBY mode transition (Masking operation)
3	STB_MSK_IRQ3	IRQ3 interrupt execution timing control until the transition to STANDBY mode 0: Before STANDBY mode transition (Normal operation) 1: After STANDBY mode transition (Masking operation)
2	STB_MSK_IRQ2	IRQ2 interrupt execution timing control until the transition to STANDBY mode 0: Before STANDBY mode transition (Normal operation) 1: After STANDBY mode transition (Masking operation)
1	STB_MSK_IRQ1	IRQ1 interrupt execution timing control until the transition to STANDBY mode 0: Before STANDBY mode transition (Normal operation) 1: After STANDBY mode transition (Masking operation)
0	STB_MSK_IRQ0	IRQ0 interrupt execution timing control until the transition to STANDBY mode 0: Before STANDBY mode transition (Normal operation) 1: After STANDBY mode transition (Masking operation)



It is necessary to clear edge interrupt in STANDBY control register (STB_EDG) and interrupt in STANDBY control register (STB_MSK) to "00" before changing the value of external interrupt pin setting register (IRQCNT), noise filter n control register (NFnCTR) and external interrupt valid input control register (LVLMD).



"Masking operation" by the STB_MSK_IRQ_n flag of interrupt in STANDBY control register (STB_MSK) and "Level" settings by the EXLEN_n flag of external interrupt valid input control register (LVLMD) cannot be used at the same time. If using them together, an unintended interrupt may be generated.



If the STB_MSK_IRQ_n flag of interrupt in STANDBY control register (STB_MSK) is set and LSI returns from STANDBY by external interrupt IRQ_n (including retention request) generated after the transition to STANDBY mode, the STB_MSK_IRQ_n flag is automatically turned to "0".

7.3.4 Programmable Active Edge Interrupt

■ Programmable Active Edge Interrupt (External Interrupt 0 to 4, Key interrupt)

Programmable active edge interrupt is the function which can select rising or falling edge for the signal input from the external interrupt input pin, then generate interrupts at the selected edge.



For external interrupt 0 to 4/key interrupt in STANDBY mode, the interrupt generation factor differs depending on the value of edge interrupt in STANDBY control register (STB_EDG) and interrupt in STANDBY control register (STB_MSK).
[Chapter 7 7.3.9 External Interrupt Operation in STANDBY Mode].

■ Programmable Active Edge Interrupt Setup Example (External Interrupt 0 to 4, Key Interrupt)

External interrupt 0 (IRQ0) is generated at rising edge of the input signal from P20 pin.
The table below shows a setup example.

Setup Procedure	Description
(1) Set the external interrupt IRQCNT (0x03FD0) bp0: P20EN = 1	(1) Set the P20EN flag of external interrupt pin set register (IRQCNT) to "1" to set P20 as an external interrupt.
(2) Specify the interrupt active edge IRQ0ICR (0x03FE2) bp5: REDG0 = 1	(2) Set the REDG0 flag of external interrupt 0 control register (IRQ0ICR) to "1" to specify rising edge as the active edge for interrupts.
(3) Set the interrupt level IRQ0ICR (0x03FE2) bp7 to 6: IRQ0LV1 to 0 = 10	(3) Set the interrupt priority level in the IRQ0LV1 to 0 flag of IRQ0ICR register. If the interrupt request flag (IRQ0IR) has already been set, make sure to clear it. [Chapter 7 7.1.4 Maskable Interrupt Control Register Setup]
(4) Enable interrupts IRQ0ICR (0x03FE2) bp1: IRQ0IE = 1	(4) Set the IRQ0IE flag of IRQ0ICR register to "1" to enable interrupts.



The interrupt request flag may have been set when the interrupt edge is switched. It is necessary to specify the interrupt valid edge before interrupts are enabled. Also, if the interrupt request flag (IRQ0IR) has already been set, it must be cleared.



To use the external interrupt pin as interrupt function, it needs to be pulled up in advance to prevent intermediate potential input.

7.3.5 Both Edges Interrupt

■ Both Edges Interrupt (External Interrupt 2 to 4/Key Interrupt)

Both edges interrupt is the function which can generate interrupt at both falling and rising edges for the signal input from the external interrupt input pin.



For the external interrupt 0 to 4/key interrupt in STANDBY mode, the interrupt generation factor differs depending on the value of edge interrupt in STANDBY control register (STB_EDG) and interrupt in STANDBY control register (STB_MSK).
[Chapter 7 7.3.9 External Interrupt Operation in STANDBY Mode].

■ Both Edges Interrupt Setup Example (External Interrupt 2 to 4/Key Interrupt)

External interrupt 2 (IRQ2) is generated at the both edges of the input signal from pin P22. The table below shows a setup example.

Setup Procedure	Description
(1) Set the external interrupt IRQCNT (0x03FD0) bp2: P22EN = 1	(1) Set the P22EN flag of external interrupt pin set register (IRQCNT) to "1" to set P22 to an external interrupt.
(2) Select the both edges interrupt EDGDT (0x03F1E) bp2: EDGSEL2 = 1	(2) Set the EDGSEL2 flag of both edges interrupt control register (EDGDT) to "1" to select the both edges interrupt.
(3) Set the interrupt level IRQ2ICR (0x03FE4) bp7 to 6: IRQ2LV1 to 0 = 10	(3) Set the interrupt level by the IRQ2LV1 to 0 flag of IRQ2ICR register. If the interrupt request flag (IRQ2IR) has already been set, make sure to clear it. [Chapter 7 7.1.4 Maskable Interrupt Control Register Setup]
(4) Enable the interrupt IRQ2ICR (0x03FE4) bp1: IRQ2IE = 1	(4) Set the IRQ2E flag of IRQ2ICR register to "1" to enable the interrupt.



When the both edges interrupt is selected, the interrupt request is generated at both edge, regardless of the REDGn flag of external interrupt control register (IRQnICR).



The interrupt request flag may have been set when the interrupt edge is switched. It is necessary to clear the interrupt request flag before interrupts are enabled. Also, the both edges interrupt must be selected before interrupts are enabled.



To use the external interrupt pin as interrupt function, it needs to be pulled up in advance to prevent intermediate potential input.

7.3.6 Level Interrupt

■ Level Interrupt (External interrupts 2, 3 and 4)

Level interrupt is the function which can select the input level High or input level Low for the signal input from the external interrupt input pin, then generate interrupts at the selected level. This interrupt can return the LSI from STANDBY mode.

■ Level Interrupt Example (External Interrupt 2, 3 and 4)

External interrupt 2 (IRQ2) is generated at the High-level of the input signal from pin P22.

The table below shows a setup example of IRQ2.

Setup Procedure	Description
(1) Set external interrupt IRQCNT (0x03FD0) bp2: P22EN = 1	(1) Set the P22EN flag of external interrupt pin set register (IRQCNT) to "1" to set P22 to an external interrupt.
(2) Specify the interrupt valid edge IRQ2ICR (0x03FE4) bp5: REDG2 = 1	(2) Set the REDG2 flag of external interrupt 0 control register (IRQ2ICR) to "0" and specify the rising edge as the valid edge.
(3) Specify the interrupt valid input LVLMD (0x03FD7) bp1: EXLVL2 = 1	(3) Set the EXLVL flag of external interrupt valid input switching control register (LVLMD) to "1" to specify the interrupt valid input level as the level interrupt (High-level).
(4) Enable the level interrupt LVLMD (0x03FD7) bp0: LEVEN2 = 1	(4) Set the LEVEN2 flag of external interrupt valid input switching control register (LVLMD) to "1" to specify the interrupt valid input level as the level interrupt (High-level).
(5) Set the interrupt level IRQ2ICR (0x03FE4) bp7 to 6: IRQ2LV1 to 0 = 10	(5) Set the interrupt priority level in the IRQ2LV1 to 0 flag of IRQ2ICR register. If the interrupt request flag (IRQ2IR) has already been set, make sure to clear it. [Chapter 7 7.1.4 Maskable Interrupt Control Register Setup]
(6) Enable the interrupt IRQ2ICR (0x03FE4) bp1: IRQ2IE = 1	(6) Set the IRQ2IE flag of IRQ2ICR register to "1" enable the interrupt.

External interrupt 2 is generated at the High-level of the input signal from P22.



Set the external interrupt valid input level equal to the polarity of the interrupt active edge. When the external interrupt valid input level is High-level, the interrupt active edge is rising edge. When the external interrupt valid input level is Low-level, the interrupt active edge is falling edge.



The interrupt request flag may have been set when the interrupt edge is switched. The interrupt active edge must be specified before the interrupts are enabled.



When using the level interrupt function, an interrupt may be generated again after completion of the interrupt process program.

When using the level interrupt function, disable the external interrupt enable flag in the interrupt process program.



"Edge interrupt selection" by the STB_EDG_IRQn flag of edge interrupt in STANDBY control register (STB_EDG) and "Level" settings by the EXLENn flag of external interrupt valid input control register (LVLMD) cannot be used at the same time. If using them together, an unintended interrupt may be generated.



"Masking operation" by the STB_MSK_IRQn flag of interrupt in STANDBY control register (STB_MSK) and "Level" settings by the EXLENn flag of external interrupt valid input control register (LVLMD) cannot be used at the same time. If using them together, an unintended interrupt may be generated.

7.3.7 Key Input Interrupt

■ Key Input Interrupt

This LSI can set the pins of Port 7 (P70 to P77) as the key input pins for each bit. Key input interrupt is the function which can generate interrupts at the falling edge if at least one key input pin is a "Low" level.

Also, when rising edge is selected by external interrupt 4 control register (REDG4 flag of IRQ4ICR register), the interrupts can be generated at the rising edge if at least one key input pin is a "High" level.



For external interrupt 0 to 4/key interrupt in STANDBY mode, the interrupt generation factor differs depending on the value of edge interrupt in STANDBY control register (STB_EDG) and interrupt in STANDBY control register (STB_MSK).
[Chapter 7 7.3.9 External Interrupt Operation in STANDBY Mode].



Key input pins need to be pulled up in advance to prevent intermediate potential input.

■ Key Input Interrupt Setup Example

After setting pins P70 to P70 of Port 7 as the key input, external interrupt 4 (IRQ4) is generated when key is input (Low-level). The following shows an example setup procedure.

Setup Procedure	Description
(1) Set the key input to input P7DIR (0x03E97) bp7 to 0: P5DIR7 to 0 = 00000000	(1) Set the P7DIR7 to 0 flags of Port 7 direction control register (P7DIR) to "00000000" and set pins P70 to P73 to the input pin.
(2) Set the pull-up resistor P7PLU (0x03EA7) bp7 to 0: P7PLU7 to 0 = 00001111	(2) Set the P7PLU7 to 0 flags of Port 7 pull-up resistor control register (P7PLU) to "00001111" to add pull-up resistors to pins P70 to P73.
(3) Select the key input interrupt KEYT3_1IMD (0x03F3E) bp7: KEYT3_1SEL = 1	(3) Set the KEYT3SEL flag of key interrupt register (KEYT3_1IMD) to "1" to select the key interrupt as the interrupt source.
(4) Select the key input pin KEYT3_1IMD (0x03F3E) bp3 to 0: KEYT3_1EN3 to 0 = 1111	(4) Set the KEYT3_1EN3 to 0 flags of key interrupt register (KEYT3_1IMD) to "1111" to set pins P70 to P73 as the key input pins.
(5) Specify the interrupt active edge EDGDT (0x03F1E) bp7: EDGSEL7 = 0 bp0: EDGSEL0 = 0 IRQ4ICR (0x03FE6) bp5: REDG0 = 0	(5) Set the EDGSEL7 and 0 flags of EDGDT register to "0" to set the interrupt edge operation to the interrupt active edge. Set the REDG0 flag of IRQ4ICR register to "0" to specify falling edge as the interrupt active edge.
(6) Set the interrupt level IRQ4ICR (0x03FE6) bp7 to 6: IRQ4LV1 to 0 = 10	(6) Set the interrupt level by setting the IRQ4LV1 to 0 flags of IRQ4ICR register. If the interrupt request flag (IRQ4IR) has been already set, make sure to clear it. [Chapter 7 7.1.4 Maskable Interrupt Control Register Setup]
(7) Enable interrupts IRQ4ICR (0x03FE6) bp1: IRQ4IE = 1	(7) Set the IRQ4IE flag of IRQ4ICR register to "1" to enable the interrupt.

*Above (3) and (4) can be set at the same time.

If at least one signal input from pins P70 to P73 is showing a "Low" level, external interrupt 4 is generated at the falling edge.



The key input must be set before interrupts are enabled.



When "key input edge detection circuit bit common" is selected by the EDGSEL0 flag, an interrupt request signal is generated by inputting OR of the match detection circuit output for the valid key input. During the match detection by a single valid key input, no interrupt request signal is generated even if a match is detected by another valid key input.



When "key input edge detection circuit bit independent" is selected by the EDGSEL0 flag, the edge determination circuit is used for each valid key input. As a result, an interrupt request signal is generated every time a valid key input match is detected.



If selecting the both edge interrupt by the EDGSEL7 flag of both edge interrupt control register (EDGDT), the following edge key interrupts are generated.

(When EDGSEL0 flag = 0: key input edge detection circuit bit common)

OR of the enable key input match detection circuit is determined with both edges and interrupt request signal is generated.

(When EDGSEL0 flag = 1: key input edge detection circuit bit independent)

The both edge determination is executed for the match detection circuit output every valid key input and interrupt request signal is generated.

* Match detection circuit outputs "High" by external interrupt 4 control register (REDG4 flag of IRQ4ICR register) under the following condition;

In falling edge setup: "Low" level match

In rising edge setup: "High" level match

7.3.8 AC Zero-Cross Detection

This LSI has AC zero-cross detection circuit. Pins P20/ACZ0 and P21/ACZ1 are the input pin of AC zero-cross detection circuit. AC zero-cross detection circuit outputs a "High" level when the input level is at the middle and outputs a "Low" level at other level.

■ AC Zero-Cross Detection (External Interrupt 0)

AC zero-cross detection is set by setting the P20IM or P21IM flag of AC zero-cross detection interrupt control register (ACZCTR) to "1". Also, if the AC zero-cross detection signal becomes "High" level, it is possible to return from STANDBY mode by generating the external interrupt.

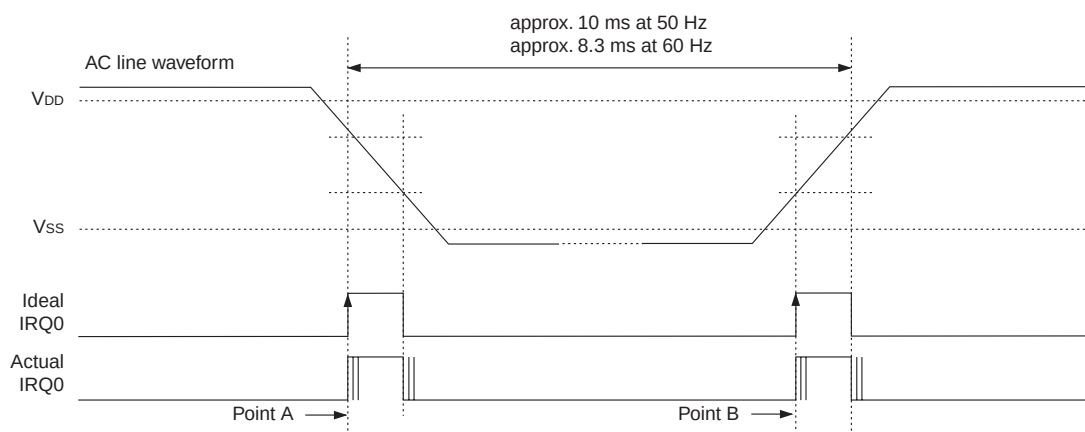


Figure:7.3.11 AC Line Waveform and IRQ Generation Timing

Actual IRQn interrupt request is generated several times at the crossing between the AC line waveform and the intermediate level (Point A and Point B). So, filtering operation by program is needed.



Interrupt requests are generated at the rising edge of AC zero-cross detection signal.



Interrupt requests are output at "High" level of AC zero-cross detection signal during STANDBY mode.

■ AC Zero-Cross Detection Setup Example (External Interrupt 1)

AC zero-cross detection generates IRQ1 interrupt request by using pin P21/ACZ1.

If the input level signal which is input from pin P21/ACZ1 crosses with the intermediate level, external interrupt 1 is generated.

The following is an example of setup procedures.

Setup Procedure	Description
(1) Set external interrupt IRQCNT (0x03FD0) bp1: P21EN = 1	(1) Set the P21EN flag of external interrupt pin setting register (IRQCNT) to "1" to set P21 as an external interrupt.
(2) Select the AC zero-cross detection signal ACZCTR (0x03F2E) bp7: P21IM = 1	(2) Set the P21IM flag of AC zero-cross detection control register (ACZCTR) to "1" to select the AC zero-cross detection signal as the external interrupt 1 generation factor.
(3) Set the interrupt level IRQ1ICR (0x03FE3) bp7 to 6: IRQ1LV1 to 0 = 10	(3) Set the interrupt level by setting the IRQ1LV1 to 0 flags of IRQ1ICR register. If the interrupt request flag (IRQ1IR) has already been set, make sure to clear it. [Chapter 7 7.1.4 Maskable Interrupt Control Register Setup]
(4) Enable the interrupt IRQ1ICR (0x03FE3) bp1: IRQ1IE = 1	(4) Set the IRQ1IE flag of IRQ1ICR register to "1" to enable the interrupt.

7.3.9 External Interrupt Operation in STANDBY Mode

This LSI can return from STANDBY mode (HALT0/HALT1/HALT2/STOP0/STOP1) with an interrupt from external interrupt input 0 to 4/key interrupt. The interrupt generation factor of external interrupt 0 to 4/key interrupt, differs between in CPU OPERATION mode and STANDBY mode.

For external interrupt 0 to 4/key interrupt in STANDBY mode, the interrupt generation can be selected by the STB_EDG_IRQn flag of edge interrupt in STANDBY control register (STB_EDG) and the STB_MSK_IRQn flag of interrupt in STANDBY control register (STB_MSK).



For CPU OPERATION mode and STANDBY mode, refer to [Chapter 3 Clock Control].

The external interrupt generation factors in CPU OPERATION mode/STANDBY mode depending on the combination with the STB_EDG_IRQn flag, the STB_MSK_IRQn flag and the EXLENn flag of external interrupt valid input control register (LVLMD).

Table:7.3.3 External Interrupt Generation Factors Before and After Transition to STANDBY Mode

External interrupt control flag			Interrupt generation factor	
STB_EDG_IRQn	STB_MSK_IRQn	EXLENn	CPU OPERATION mode	STANDBY mode
0	0	0	Edge interrupt	Level interrupt
0	0	1	Level interrupt	Level interrupt
1	0	0	Edge interrupt	Edge interrupt
0	1	0	Masking operation	Edge interrupt
1	1	0	Masking operation	Edge interrupt
Other combination			Setting prohibited	



"Edge interrupt selection" by the STB_EDG_IRQn flag of edge interrupt in STANDBY control register (STB_EDG) and "Level" settings by the EXLENn flag of external interrupt valid input control register (LVLMD) cannot be used at the same time. If using them together, an unintended interrupt may be generated.



"Masking operation" by the STB_MSK_IRQn flag of interrupt in STANDBY control register (STB_MSK) and "Level" settings by the EXLENn flag of external interrupt valid input control register (LVLMD) cannot be used at the same time. If using them together, an unintended interrupt may be generated.

■ Level Interrupt in STANDBY Mode

When "Level interrupt" is selected as the interrupt generation factor of external interrupt 0 to 4/key interrupt, an interrupt is generated if the external interrupt input reaches the level set by the REDGn flag of external interrupt n control register (IRQnICR). When [falling edge ("Low" level)] is selected by the REDGn flag, an interrupt is generated if the interrupt input reaches "Low" level in STANDBY mode. If the interrupt input has been at "Low" level before the transition to STANDBY mode, an interrupt is generated right after the transition and the LSI returns from STANDBY mode.

When [rising edge ("High" level)] is selected by the REDGn flag, an interrupt is generated if the interrupt input reaches "High" level in STANDBY mode. If the interrupt input has been at "High" level before the transition to STANDBY mode, an interrupt is generated right after the transition and the LSI returns from STANDBY mode.

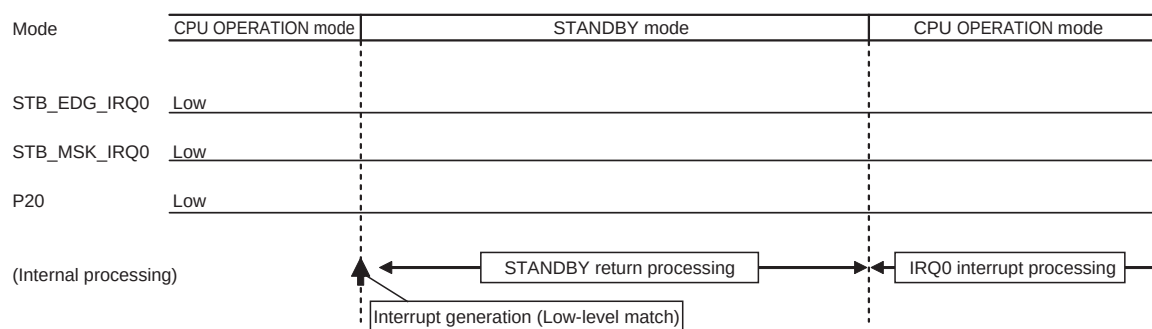


Figure:7.3.12 Timing 1 of P20 Interrupt Generation during STANDBY Mode
Level Interrupt (EXLEN0 = 0, REDG0 = 0)



For level interrupt, regardless of the value set to the EDGSELn flag of both edges interrupt control register (EDGDT), an interrupt is generated when the interrupt input reaches the level set by the REDGn flag of external interrupt n control register (IRQnICR).

■ Edge Interrupt in STANDBY Mode

When "Edge interrupt" is selected as the interrupt generation factor of external interrupt 0 to 4/key interrupt, an interrupt is generated if the external interrupt input reaches the edge set by the REDGn flag of external interrupt n control register (IRQnICR) and the EDGSELn flag of both edges interrupt control register (EDGDT).

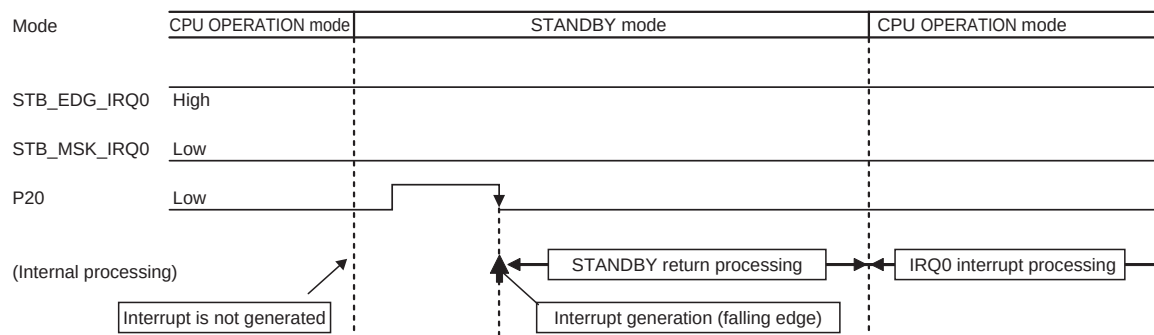


Figure:7.3.13 Timing 2 of P20 Interrupt Generation during STANDBY Mode
Edge Interrupt (EXLEN0 = 0, REDG0 = 0)

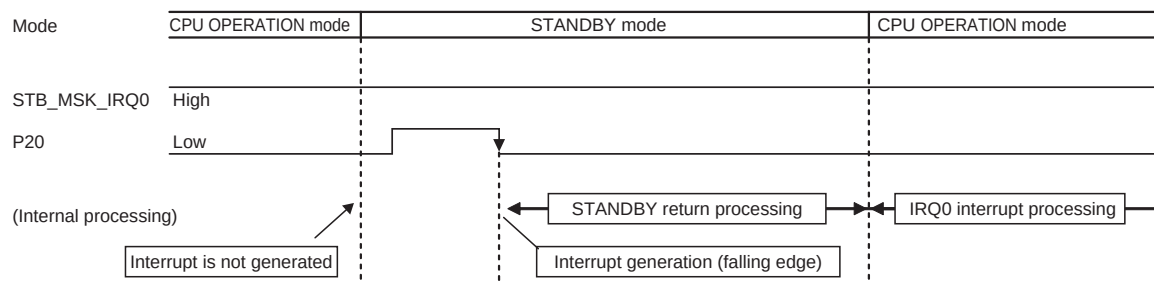


Figure:7.3.14 Timing 3 of P20 Interrupt Generation during STANDBY Mode
Edge Interrupt (EXLEN0 = 0, REDG0 = 0)



For edge interrupt, an interrupt is generated on the edge set by the REDGn flag of external interrupt n control register (IRQnICR) and the EDGESELn flag of both edge interrupt control register (EDGDT).



When "Mask operation" is selected by the STB_MSK_IRQn flag of interrupt in STANDBY control register (STB_MSK), regardless of the setting value of the STB_EDG_IRQn flag of edge interrupt in STANDBY control register (STB_EDG), the interrupt generation factor in STANDBY mode will be "Edge interrupt".



When the STB_MSK_IRQn flag of interrupt in STANDBY control register (STB_MSK) is set and the LSI returns from STANDBY mode by the external interrupt IRQn generation (including retention request), the STB_MSK_IRQn flag is automatically turned to "0".



When the STB_MSK_IRQn flag of interrupt in STANDBY control register (STB_MSK) is set and the LSI returns from STANDBY mode by the external interrupt IRQn generation (including external interrupts except IRQn), the STB_MSK_IRQn flag remains set but the mask operation is disabled. It is necessary to clear all the flags of STB_MSK register after return from STANDBY mode.

■ Masking Operation in CPU OPERATION Mode

When "Masking operation" is selected as the interrupt generation factor of the external interrupts 0 to 4/key interrupt, no interrupt is generated even if the external interrupt input reaches the edge set by the REDGn flag of external interrupt n control register (IRQnICR) and the EDGSELn flag of both edges interrupt control register (EDGDT). However, the interrupt request at that time is retained in the internal circuit (request retention). If a request retention is generated before the transition to STANDBY mode, an interrupt is generated by the interrupt request retained after the transition to STANDBY mode and the LSI returns from STANDBY mode.

This function can prevent STANDBY mode return factors generation before the transition to STANDBY mode when the external interrupts 0 to 4/key interrupt are used as return factors from STANDBY mode.

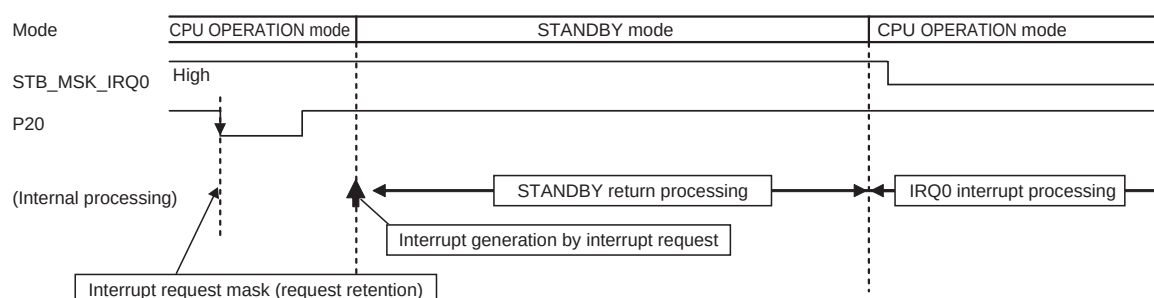


Figure:7.3.15 Timing 4 of P20 Interrupt Generation during STANDBY Mode
Masking Operation (EXLEN0 = 0, REDG0 = 0)



When the STB_MSK_IRQn flag of interrupt in STANDBY control register (STB_MSK) is set and the LSI returns from STANDBY mode by the external interrupt IRQn generation (including retention request), the STB_MSK_IRQn flag is automatically cleared to "Low".



When the STB_MSK_IRQn flag of interrupt in STANDBY control register (STB_MSK) is set and the LSI returns from STANDBY mode by the external interrupt IRQn generation (including external interrupts except IRQn), the STB_MSK_IRQn flag remains set but the mask operation is disabled. It is necessary to clear all the flags of STB_MSK register after return from STANDBY mode.

■ External Interrupt Operation in STANDBY Mode (External Interrupt 0 to 4/Key Interrupt)

The LSI can return from STANDBY mode by external interrupts.

The interrupt in STANDBY mode can be selected from either edge interrupt or level interrupt by edge interrupt in STANDBY control register (STB_EDG).

When the level interrupt is selected during STANDBY mode, an interrupt is generated when the level of the external interrupt pin reaches the value set in the external interrupt valid edge specification flag. Therefore, pay attention to the value set in the external interrupt valid edge specification flag and the level of the corresponding external interrupt pin during the transition to STANDBY mode. When the value of the external interrupt valid edge specification flag reaches the corresponding external interrupt, the LSI returns from STANDBY mode immediately.

■ Return by External Interrupt in STANDBY Mode Setup Example 1

The LSI returns from STOP mode by the external interrupt 0 (IRQ0) generated by the "Low" level signal input from the external interrupt 0 pin.

The following shows an example of setup procedure.

Setup Procedure	Description
(1) Specify interrupt valid edge IRQ0ICR (0x03FE2) bp5: REDG0 = 0	(1) Set the REDG0 of external interrupt 0 control register (IRQ0ICR) to "0" to specify falling edge as the interrupt active edge.
(2) Set the external interrupt pin Pull-up the external interrupt 0 pin in advance.	(2) The value of the REDG0 flag of IRQ0ICR register differs from the level of external interrupt 0 pin.
(3) Set the external interrupt IRQCNT (0x03FD0) bp0: P20EN = 1	(3) Set the P20EN flag of external interrupt pin setting register (IRQCNT) to "1" to set P20 as the external interrupt.
(4) Set the interrupt level IRQ0ICR (0x03FE2) bp7 to 6: IRQ0LV1 to 0 = 10	(4) Set the interrupt level by setting the IRQ0LV1 to 0 flags of IRQ0ICR register. If an interrupt request flag (IRQ0IR) has already been set, make sure to clear it.
(5) Enable interrupts IRQ0ICR (0x03FE2) bp1: IRQ0IE = 1	(5) Set the IRQ0IE flag of IRQ0ICR register to "1" to enable interrupts.
(6) Set STOP mode CPUM (0x03F00) bp3: STOP = 1	(6) Set the STOP flag of CPU mode control register (CPUM) to "1" to transit to STOP mode. [Chapter 3 3.3.3 STANDBY Mode]

If the "Low" level signal is input to the external interrupt 0 pin, then, the value of the external interrupt valid edge specification flag (REDG0) and the level of external interrupt 0 pin are matched, the external interrupt 0 is accepted and LSI returns from STOP mode.



The LSI returns from STOP mode after oscillation stabilization wait time set by oscillation stabilization wait control register (DLYCTR) has passed since the acceptance of an external interrupt. [Chapter 2 2.7.2 Oscillation Stabilization Wait time]

■ External Edge Interrupt Operation in STANDBY Mode (External Interrupts 0 to 4/Key Interrupt)

The LSI can return from STANDBY mode by external interrupts.

The interrupt in STANDBY can be selected from either edge interrupt or level interrupt by edge interrupt in STANDBY control register (STB_EDG).

When the edge interrupt is selected during STANDBY mode, an interrupt is generated at the edge set in the external interrupt valid edge specification flag. Therefore, even if the external interrupt reaches the value of the corresponding external interrupt valid edge specification flag, the LSI does not return from STANDBY mode and an interrupt is generated by the active edge input after that. When the both edges interrupt is selected by both edges interrupt control register (EDGDT), an interrupt is generated at the both edges of the external interrupt input signal.

■ Return by External Interrupt in STANDBY Mode Setup Example 2

The LSI returns from STOP mode by the external interrupt 0 (IRQ0) generated by the falling edge input from the external interrupt 0 pin.

The following shows an example of setup procedure.

Setup Procedure	Description
(1) Specify interrupt valid edge IRQ0ICR (0x03FE2) bp5: REDG0 = 0	(1) Set the REDG0 of external interrupt 0 control register (IRQ0ICR) to "0" to specify falling edge as the interrupt active edge.
(2) Set the external interrupt pin Pull-up the external interrupt 0 pin in advance.	(2) The value of the REDG0 flag of IRQ0ICR register (IRQCNT) differs from the level of the external interrupt 0 pin.
(3) Set the external interrupt IRQCNT(0x03FD0) bp0: P20EN = 1	(3) Set the P20EN flag of external interrupt pin setting register (IRQCNT) to "1" to set P20 as the external interrupt.
(4) Set the interrupt level IRQ0ICR (0x03FE2) bp7 to 6: IRQ0LV1 to 0 = 10	(4) Set the interrupt level by setting the IRQ0LV1 to 0 flags of IRQ0ICR register. If an interrupt request flag (IRQ0IR) has already been set, make sure to clear it.
(5) Set the interrupt operation in STANDBY mode STB_MSK (0x03F5F) bp0: STB_MSK_IRQ0 = 1	(5) Set the STB_MSK_IRQ0 flag of STB_MSK register so that an interrupt is generated after the transition to STANDBY mode.
(6) Enable interrupts IRQ0ICR (0x03FE2) bp1: IRQ0IE = 1	(6) Set the IRQ0IE flag of IRQ0ICR register to "1" to enable interrupts.
(7) Set STOP mode CPUM (0x03F00) bp3: STOP = 1	(7) Set the STOP flag of CPU mode control register (CPUM) to "1" to transit to STOP mode. [Chapter 3 3.3.3 STANDBY Mode]

If the "Low" level signal is input to the external interrupt 0 pin, then, the value of the external interrupt valid edge specification flag (REDG0) and the level of external interrupt 0 pin are matched, the external interrupt 0 is accepted and LSI returns from STOP mode.



The LSI returns from STOP mode after oscillation stabilization wait time set by oscillation stabilization wait control register (DLYCTR) has passed since the acceptance of an external interrupt. [Chapter 2 2.7.2 Oscillation Stabilization Wait time]



To change the value of external interrupt pin setting register (IRQCNT), noise filter n control register (NFnCTR) and external interrupt valid input control register (LVLMD), clear both edge interrupt in STANDBY control register (STB_EDG) and interrupt in STANDBY control register (STB_MSK) to "00".



"Edge interrupt selection" by the STB_EDG_IRQn flag of edge interrupt in STANDBY control register cannot be used with "Level" set by the EXLENn flag of external interrupt valid input control register (LVLMD). If using them at the same time, an unintended interrupt may be generated.

■ External Interrupt Operation in STANDBY Mode (External Interrupts 0 to 4/Key Interrupt)
(Control of interrupt which is return factor before the transition to STANDBY mode)

This setup example shows the operation to prevent that an interrupt to be the return factor from STANDBY mode is input after the transition to STANDBY mode, and the LSI cannot return from STANDBY mode.

The LSI can return from STANDBY mode by external interrupts. When the interrupt selection is set by interrupt in STANDBY control register (STB_MSK) after the transition to STANDBY mode, the external interrupt factor generated before the transition to STANDBY mode can be generated after the transition to STANDBY mode. In this case, an interrupt is accepted after the transition to STANDBY mode and the LSI returns from STANDBY mode immediately.

■ Return by External Interrupt in STANDBY Mode Setup Example 3

This setup example is used to prevent the following cases;

LSI cannot return from STANDBY mode even if an interrupt to be the return factor from STANDBY mode is input just before the transition to STANDBY mode.

Setup Procedure	Description
(1) Specify interrupt valid edge IRQ0ICR (0x03FE2) bp5: REDG0 = 0	(1) Set the REDG0 of external interrupt 0 control register (IRQ0ICR) to "0" to specify falling edge as the interrupt active edge.
(2) Set the external interrupt pin Pull-up the external interrupt 0 pin in advance.	(2) The value of the REDG0 flag of IRQ0ICR register (IRQCNT) differs from the level of the external interrupt 0 pin.
(3) Set the external interrupt IRQCNT(0x03FD0) bp0: P20EN = 1	(3) Set the P20EN flag of external interrupt pin setting register (IRQCNT) to "1" to set P20 as the external interrupt.
(4) Set the interrupt level IRQ0ICR (0x03FE2) bp7 to 6: IRQ0LV1 to 0 = 10	(4) Set the interrupt level by setting the IRQ0LV1 to 0 flags of IRQ0ICR register. If an interrupt request flag (IRQ0IR) has already been set, make sure to clear it.
(5) Set the interrupt operation in STANDBY mode STB_MSK (0x03F5F) bp0: STB_MSK_IRQ0 = 1	(5) Set the STB_MSK_IRQ0 flag of STB_MSK register so that an interrupt is generated after the transition to STANDBY mode.
(6) Enable interrupts IRQ0ICR (0x03FE2) bp1: IRQ0IE = 1	(6) Set the IRQ0IE flag of IRQ0ICR register to "1" to enable interrupts.
(7) Set STOP mode CPUM (0x03F00) bp3: STOP = 1	(7) Set the STOP flag of CPU mode control register (CPUM) to "1" to transit to STOP mode. [Chapter 3 3.3.3 STANDBY Mode]

If the "Low" level signal is input to the external interrupt 0 pin, then, the value of the external interrupt valid edge specification flag (REDG0) and the level of external interrupt 0 pin are matched, the external interrupt 0 is accepted and LSI returns from STOP mode.



The LSI returns from STOP mode after oscillation stabilization wait time set by oscillation stabilization wait control register (DLYCTR) has passed since the acceptance of an external interrupt. [Chapter 2 2.7.2 Oscillation Stabilization Wait time]



To use these settings indicated in the section, interrupts in STANDBY mode are edge interrupts regardless of edge interrupt in STANDBY control register (STB_EDG).



To change the value of external interrupt pin setting register (IRQCNT), noise filter n control register (NFnCTR) and external interrupt valid input control register (LVLMD), clear both edge interrupt in STANDBY control register (STB_EDG) and interrupt in STANDBY control register (STB_MSK) to "00".



"Mask operation" by the STB_MSK_IRQn flag of interrupt in STANDBY control register (STB_MSK) cannot be used with "Level" set by the EXLENn flag of external interrupt valid input control register (LVLMD). If using them at the same time, an unintended interrupt may be generated.



When the STB_MSK_IRQn flag of interrupt in STANDBY control register (STB_MSK) is set and the LSI returns from STANDBY mode by the external interrupt IRQn generation (including retention request) after the transition to STANDBY mode, the STB_MSK_IRQn flag is automatically cleared to "Low".



When the STB_MSK_IRQn flag of interrupt in STANDBY control register (STB_MSK) is set and the LSI returns from STANDBY mode by the external interrupt IRQn generation (including external interrupts except IRQn) after the transition to STANDBY mode, the STB_MSK_IRQn flag remains set but the mask operation is disabled. It is necessary to clear all the flags of STB_MSK register after return from STANDBY mode.

7.3.10 Noise Filter

■ Noise Filter (External Interrupts 0 to 4)

Noise filter reduces noise by sampling the input waveform from the external interrupt pins (IRQ0 to 4). There are 8 types of sampling cycle (fpll, fpll/2⁵, fpll/2⁶, fpll/2⁷, fpll/2⁸, fpll/2⁹, fpll/2¹⁰, fslow).

■ Noise Filter Selection (External Interrupts 0 to 4)

Noise filter can be selected by setting the NFnEN1 flag of noise filter control register (NFnCTR) to "1".

Table:7.3.4 Noise Removal Function

NFnEN1	IRQn input (P2n)
0	Noise filter OFF
1	Noise filter ON

■ Sampling Cycle Setup (External Interrupts 0 to 4)

The sampling cycle of noise removal function can be set by the NFnSCK2 to 0 flags of NFnCTR register.

Table:7.3.5 Sampling Cycle / Time of Noise Removal Function

NFnSCK2 to 0	Sampling cycle	fpll=10 MHz, fx=32 kHz	
000	fpll	10 MHz	100 ns
001	fpll/2 ⁵	312.5 kHz	3.2 μs
010	fpll/2 ⁶	156.25 kHz	6.4 μs
011	fpll/2 ⁷	78.12 kHz	12.8 μs
100	fpll/2 ⁸	39.06 kHz	25.6 μs
101	fpll/2 ⁹	19.53 kHz	51.20 μs
110	fpll/2 ¹⁰	9.76 kHz	102.40 μs
111	fslow	32 kHz	31.25 μs

■ Noise Removal Function Operation (External Interrupts 0 to 4)

After sampling the input signal to the external interrupt pins (IRQ0 to 4) with the set sampling time, if the same level is received for three times continuously, that level is sent to the LSI. If the same level is not received for three times continuously, the previous level is sent.

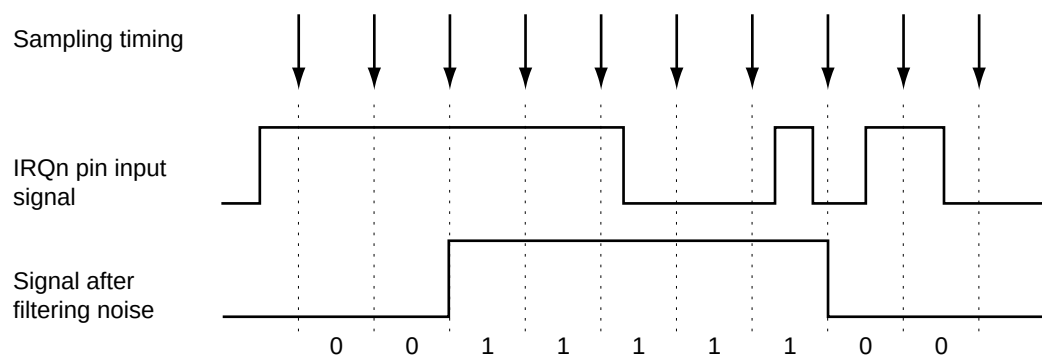


Figure:7.3.16 Operation of the Noise Removal Function



Noise filter cannot be used in STOP mode and HALT mode.
Set the NFnEN1 flag to "0" when using IRQ0 to 4 for returning from STANDBY mode.

■ Noise Filter Setup Example (External Interrupt 0)

Noise removal function is added to the input signal from P20 pin to generate the external interrupt 0 (IRQ0) at the rising edge. The sampling clock is set to fpll, and the operation state is fpll = 10 MHz. The following shows an example of setup procedures.

Setup Procedure	Description
(1) External interrupt setup IRQCNT (0x03F3D0) bp0: P20EN = 1	(1) Set the P20EN flag of external interrupt setup register (IRQCNT) to "1" to set P20 to the external interrupt.
(2) Specify the interrupt valid edge IRQ0ICR (0x03FE2) bp5: REDG0 = 1	(2) Set the REDG0 flag of external interrupt 0 control register (IRQ0ICR) to "1" to specify the interrupt valid edge to the rising edge.
(3) Select the sampling clock NF0CTR (0x03FD1) bp7 to 5: NF0SCK2 to 0 = 000	(3) Select the sampling clock to fpll by the NF0SCK2 to 0 flags of noise filter 0 control register (NF0CTR).
(4) Set the noise filter operation NF0CTR (0x03FD1) bp4: NF0EN1 = 1	(4) Set the NF0EN1 flag of NF0CTR register to "1" to add the noise filter operation.
(5) Set the interrupt level IRQ0ICR (0x03FE2) bp7 to 6: IRQ0LV1 to 0 = 10	(5) Set the interrupt level by setting the IRQ0LV1 to 0 flags of IRQ0ICR register. If the interrupt request flag (IRQ0IR) has already been set, make sure to clear it. [Chapter 7 7.1.4 Maskable Interrupt Control Register Setup]
(6) Enable the interrupt IRQ0ICR (0x03FE2) bp1: IRQ0IE = 1	(6) Set the IRQ0IE flag of IRQ0ICR register to "1" to enable the interrupt.

*Above (2) and (3) can be set at the same time.

The input signal from pin P20 outputs the interrupt factor at the edge based on the programmable active edge after passing through the noise filter.



The noise filter should be set before interrupts are enabled.



External interrupt pins need to be pulled up in advance to prevent intermediate potential input.

8.1 Overview

8.1.1 I/O Port

MN101EF76K has a total of 104 pins, including the shared special function pins which are allocated to the I/O ports (Port 0, Port 1, Port 2, Port 3, Port 4, Port 5, Port 6, Port 7, Port 8, Port 9, Port A, Port B, Port C and Port D).

MN101EF57G has a total of 70 pins, including the shared special function pins which are allocated to the I/O ports (Port 0, Port 2, Port 3, Port 4, Port 5, Port 6, Port 7, Port 8, Port 9, Port A and Port B).

MN101EF56K has a total of 90 pins, including the shared special function pins which are allocated to the I/O ports (Port 0, Port 1, Port 2, Port 3, Port 4, Port 5, Port 6, Port 7, Port 8, Port 9, Port A and Port B).

8.1.2 I/O Port State At Reset

Table:8.1.1 I/O Port State At Reset (Single Chip Mode)

Port	I/O mode	Pull-up/Pull-down resistor	I/O port/Special function
Port 0	Input mode	No pull-up/pull-down resistor	I/O port
Port 1	Input mode	No pull-up/pull-down resistor	I/O port
Port 2	Input mode	P27: Pull-up resistor Others: No pull-up resistor	I/O port
Port 3	Input mode	No pull-up resistor	I/O port
Port 4	Input mode	No pull-up/pull-down resistor	I/O port
Port 5	Input mode	No pull-up/pull-down resistor	I/O port
Port 6	Input mode	No pull-up resistor	I/O port
Port 7	Input mode	No pull-up/pull-down resistor	I/O port
Port 8	Input mode	No pull-up resistor	I/O port
Port 9	Input mode	No pull-up resistor	I/O port
Port A	Input mode	No pull-up resistor	I/O port
Port B	Input mode	No pull-up resistor	I/O port
Port C	Input mode	No pull-up resistor	I/O port
Port D	Input mode	No pull-up resistor	I/O port



The values of pull-up/pull-down resistors should be calculated in the following ways based on the electrical characteristics in LSI User's Manual of each model.

How to determine a pull-up resistor value:

ex) When pins maintain at the “Low” level value which guarantee the performance based on the electrical characteristics,

$V_{DD5} = 5\text{ V}$, $V_{IN} = V_{SS}$ and input current is Min = $-50\text{ }\mu\text{A}$, Typ = $-100\text{ }\mu\text{A}$, Max = $-500\text{ }\mu\text{A}$
(minus (-) refer to the current passing from microcontroller).

The resistor value of Typ = $50\text{ k}\Omega$ can converted based on the values above.

However, the value may vary significantly depending on the temperature.

At temperature variation from $-40\text{ }^{\circ}\text{C}$ to $85\text{ }^{\circ}\text{C}$,
the resistor value may vary from Min = $10\text{ k}\Omega$ to Max = $100\text{ k}\Omega$.

How to determine a pull-down resistor value:

ex) When pins maintain at the “High” level value which guarantee the performance based on the electrical characteristics,

$V_{DD5} = 5\text{ V}$, $V_{IN} = V_{DD5}$ and input current is Min = $50\text{ }\mu\text{A}$, Typ = $100\text{ }\mu\text{A}$, Max = $500\text{ }\mu\text{A}$

The resistor value of Typ = $50\text{ k}\Omega$ can converted based on the values above.

However, the value may vary significantly depending on the temperature.

At temperature variation from $-40\text{ }^{\circ}\text{C}$ to $85\text{ }^{\circ}\text{C}$,
the resistor value may vary from Min = $10\text{ k}\Omega$ to Max = $100\text{ k}\Omega$.

8.2 Registers

Each port is controlled by the data output register (PnOUT), the data input register (PnIN), the I/O direction control register (PnDIR), the pull-up resistor control register (PnPLU) or the pull-up/pull-down resistor control register (SELUD, PnPLUD) and registers that control special function pin (PnOMD, PnIMD, PnSYO, PnSEV, PnCNT, EXADV, PnODC). The following table shows the list of registers.

Table:8.2.1 Port Control Registers

Table remarks ✓: With function -: Without function

Register	Address	R/W	Function	Page	MN101 EF76K	MN101 EF57G	MN101 EF56K
P0OUT	0x03E70	R/W	Port 0 output register	VIII-8	✓	✓	✓
P0IN	0x03E80	R	Port 0 input register	VIII-9	✓	✓	✓
P0DIR	0x03E90	R/W	Port 0 direction control register	VIII-9	✓	✓	✓
P0PLUD	0x03EA0	R/W	Port 0 pull-up/pull-down resistor control register	VIII-9	✓	✓	✓
P0OMD1	0x03EB0	R/W	Port 0 output mode register 1	VIII-10	✓	✓	✓
P0OMD2	0x03EC0	R/W	Port 0 output mode register 2	VIII-10	✓	✓	✓
P0ODC	0x03EF0	R/W	Port 0 Nch open-drain control register	VIII-11	✓	✓	✓
P0LED	0x03EE0	R/W	Port 0 LED control register	VIII-12	✓	✓	✓
SELUD	0x03EAF	R/W	Pull-up/pull-down resistor selection register	VIII-13	✓	✓	✓
P1OUT	0x03E70	R/W	Port 1 output register	VIII-32	✓	-	✓
P1IN	0x03E81	R	Port 1 input register	VIII-33	✓	-	✓
P1DIR	0x03E91	R/W	Port 1 direction control register	VIII-33	✓	-	✓
P1PLUD	0x03EA1	R/W	Port 1 pull-up/pull-down resistor control register	VIII-33	✓	-	✓
P1OMD1	0x03EB1	R/W	Port 1 output mode register 1	VIII-34	✓	-	✓
P1OMD2	0x03EC1	R/W	Port 1 output mode register 2	VIII-35	✓	-	✓
P1IMD	0x03ED1	R/W	Port 1 input mode register	VIII-35	✓	-	✓
SELUD	0x03EAF	R/W	Pull-up/pull-down resistor selection register	VIII-13	✓	✓	✓
P2OUT	0x03E72	R/W	Port 2 output register	VIII-45	✓	✓	✓
P2IN	0x03E82	R	Port 2 input register	VIII-45	✓	✓	✓
P2DIR	0x03E92	R/W	Port 2 direction control register	VIII-46	✓	✓	✓
P2PLU	0x03EA2	R/W	Port 2 pull-up resistor control register	VIII-46	✓	✓	✓
P3OUT	0x03E73	R/W	Port 3 output register	VIII-55	✓	✓	✓
P3IN	0x03E83	R	Port 3 input register	VIII-56	✓	✓	✓
P3DIR	0x03E93	R/W	Port 3 direction control register	VIII-57	✓	✓	✓
P3PLU	0x03EA3	R/W	Port 3 pull-up resistor control register	VIII-58	✓	✓	✓
P3IMD	0x03EC3	R/W	Port 3 input mode register	VIII-58	-	✓	-
P3ODC	0x03EF3	R/W	Port 3 Nch open-drain control register	VIII-59	✓	✓	✓
P4OUT	0x03E74	R/W	Port 4 output register	VIII-72	✓	✓	✓
P4IN	0x03E84	R	Port 4 input register	VIII-73	✓	✓	✓
P4DIR	0x03E94	R/W	Port 4 direction control register	VIII-74	✓	✓	✓
P4PLUD	0x03EA4	R/W	Port 4 pull-up/pull-down resistor control register	VIII-75	✓	✓	✓
P4ODC	0x03EF4	R/W	Port 4 Nch open-drain control register	VIII-76	✓	✓	✓
SELUD	0x03EAF	R/W	Pull-up/pull-down resistor selection register	VIII-77	✓	✓	✓
P5OUT	0x03E75	R/W	Port 5 output register	VIII-93	✓	✓	✓
P5IN	0x03E85	R	Port 5 input register	VIII-93	✓	✓	✓
P5DIR	0x03E95	R/W	Port 5 direction control register	VIII-94	✓	✓	✓
P5PLUD	0x03EA5	R/W	Port 5 pull-up/pull-down resistor control register	VIII-94	✓	✓	✓
P5OMD	0x03EB5	R/W	Port 5 output mode register	VIII-95	✓	✓	✓
P5ODC	0x03EF5	R/W	Port 5 Nch open-drain control register	VIII-94	✓	✓	✓

Table remarks ✓: With function -: Without function

Register	Address	R/W	Function	Page	MN101 EF76K	MN101 EF57G	MN101 EF56K
SEUD	0x03EAF	R/W	Pull-up/pull-down resistor selection register	VIII-96	✓	✓	✓
P6OUT	0x03E76	R/W	Port 6 output register	VIII-111	✓	✓	✓
P6IN	0x03E86	R	Port 6 input register	VIII-112	✓	✓	✓
P6DIR	0x03E96	R/W	Port 6 direction control register	VIII-113	✓	✓	✓
P6PLU	0x03EA6	R/W	Port 6 pull-up resistor control register	VIII-114	✓	✓	✓
P6OMD	0x03EB6	R/W	Port 6 output mode register	VIII-115	✓	✓	✓
P6ODC	0x03EF6	R/W	Port 6 Nch open-drain control register	VIII-115	✓	✓	✓
P7OUT	0x03E77	R/W	Port 7 output register	VIII-131	✓	✓	✓
P7IN	0x03E87	R	Port 7 input register	VIII-131	✓	✓	✓
P7DIR	0x03E97	R/W	Port 7 direction control register	VIII-132	✓	✓	✓
P7PLU	0x03EA7	R/W	Port 7 pull-up/pull-down resistor control register	VIII-132	✓	✓	✓
P7ODC	0x03EF7	R/W	Port 7 Nch open-drain control register	VIII-133	✓	✓	✓
SEUD	0x03EAF	R/W	Pull-up/pull-down resistor selection register	VIII-134	✓	✓	✓
P8OUT	0x03E78	R/W	Port 8 output register	VIII-153	✓	✓	✓
P8IN	0x03E88	R	Port 8 input register	VIII-153	✓	✓	✓
P8DIR	0x03E98	R/W	Port 8 direction control register	VIII-154	✓	✓	✓
P8PLU	0x03EA8	R/W	Port 8 pull-up resistor control register	VIII-154	✓	✓	✓
P8OMD	0x03EB8	R/W	Port 8 output mode register	VIII-155	✓	✓	✓
P8SYO	0x03EF8	R/W	Port 8 synchronous output control register	VIII-156	✓	✓	✓
P8SEV	0x03EF9	R/W	Port 8 synchronous output event selection register	VIII-157	✓	✓	✓
P9OUT	0x03E79	R/W	Port 9 output register	VIII-175	✓	✓	✓
P9IN	0x03E89	R	Port 9 input register	VIII-176	✓	✓	✓
P9DIR	0x03E99	R/W	Port 9 direction control register	VIII-177	✓	✓	✓
P9PLU	0x03EA9	R/W	Port 9 pull-up resistor control register	VIII-178	✓	✓	✓
P9IMD	0x03EC9	R/W	Port 9 input mode register	VIII-179	✓	✓	✓
PAOUT	0x03E7A	R/W	Port A output register	VIII-191	✓	✓	✓
PAIN	0x03E8A	R	Port A input register	VIII-191	✓	✓	✓
PADIR	0x03E9A	R/W	Port A direction control register	VIII-192	✓	✓	✓
PAPLU	0x03EAA	R/W	Port A pull-up resistor control register	VIII-192	✓	✓	✓
PAOMD1	0x03EBA	R/W	Port A output mode register 1	VIII-193	✓	✓	✓
PAIMD	0x03ECA	R/W	Port A input mode register	VIII-194	✓	✓	✓
PACNT	0x03EDA	R/W	Port A output control register	VIII-195	✓	✓	✓
SWCNT	0x03E8F	R/W	0.54V _{DD5} input selection register 1	VIII-195	✓	✓	✓
PBOUT	0x03E7B	R/W	Port B output register	VIII-211	✓	✓	✓
PBIN	0x03E8B	R	Port B input register	VIII-212	✓	✓	✓
PBDIR	0x03E9B	R/W	Port B direction control register	VIII-213	✓	✓	✓
PBPLU	0x03EA8	R/W	Port B pull-up resistor control register	VIII-214	✓	✓	✓
PBIMD	0x03EC8	R/W	Port B input mode register	VIII-215	✓	✓	✓
PCOUT	0x03E7C	R/W	Port C output register	VIII-211	✓	-	-
PCIN	0x03E8C	R	Port C input register	VIII-212	✓	-	-
PCDIR	0x03E9C	R/W	Port C direction control register	VIII-213	✓	-	-
PCPLU	0x03EAC	R/W	Port C pull-up resistor control register	VIII-214	✓	-	-
PDOUT	0x03E7D	R/W	Port D output register	VIII-215	✓	-	-
PDIN	0x03E8D	R/W	Port D input register	VIII-211	✓	-	-
PDDIR	0x03E9D	R	Port D direction control register	VIII-212	✓	-	-
PDPLU	0x03EAD	R/W	Port D pull-up resistor control register	VIII-213	✓	-	-
PDOMD	0x03EBD	R/W	Port D output mode register	VIII-214	✓	-	-

R/W: Readable/Writable

R: Read only

8.3 Port 0

8.3.1 Description

■ General-purpose Port Setup

To output data to pins, set the control flag of Port 0 direction control register (P0DIR) to "1" and write the value of Port 0 output register (P0OUT).

To read input data of pins, set the control flag of Port 0 direction control register (P0DIR) to "0" and read the value of Port 0 input register (P0IN).

Each bit can be set individually as either an input or output by Port 0 direction control register (P0DIR). The control flag of P0DIR register is set to "1" for output mode, and "0" for input mode.

Each bit can be set individually to add a pull-up (or pull-down) resistor, by Port 0 pull-up/pull-down resistor control register (P0PLUD). Set the control flag of P0PLUD register to "1" to add a pull-up (or pull-down) resistor.

Port 0 can be selected to add a pull-up or pull-down resistor by bp0 of the pull-up/pull-down resistor selection register (SELUD).

For P03 and P04 each bit can be selected individually as Nch open-drain output by Port 0 Nch open-drain control register (P0ODC). The control flag of Port 0 Nch open-drain control register (P0ODC) is set to "1" for Nch open-drain output, and "0" for push-pull output.

Table:8.3.1 Port 0 Functions

Table remarks -: Without function

MN101EF76K/MN101EF56K				MN101EF57G			
Pin	Special function			Pin	Special function		
P00	LED0	OCD_DATA		P00	LED0	OCD_DATA	
P01	LED1	TM9IOB	OCD_CLK	P01	LED1	TM9IOB	OCD_CLK
P02	LED2	TM7IOB	SBI0A	P02	LED2	TM7IOB	SBI0A
P03	LED3	TM8IOB	SBO0A	P03	LED3	TM8IOB	SBO0A
P04	LED4	TM0IOB		P04	LED4	TM0IOB	
P05	LED5	SBO3A	TXD3A	P05	LED5	-	-
P06	LED6	SBI3A	RXD3A	P06	LED6	-	-
P07	LED7	DA_D	SBT3A	P07	LED7	DA_B	-

■ Special Function Pin Setup

P01 is also used as the I/O pin for Timer 9.

P02 is also used as the I/O pin for Timer 7.

P03 is also used as the I/O pin for Timer 8.

P04 is also used as the I/O pin for Timer 0 and Timer 2.

P02 is also used as the input pin for Serial Interface 0 reception data and UART0 reception data. When the SC0BIS flag of Serial Interface 0 mode register 0 (SC0MD0) is "1", P02 is the serial data input pin.

P03 is also used as the I/O pin for Serial Interface 0 transmission/reception data and the output pin of UART0 transmission data. When the SC0SBOS flag of Serial Interface 0 mode register 0 (SC0MD0) is "1", P03 is the serial data I/O pin. Push-pull output or Nch open-drain output can be selected by setting Port 0 Nch open-drain control register (P0ODC).

P04 is also used as the I/O pin for Serial Interface 0 clock. When the SC0SBTS flag of Serial Interface 0 mode register 0 (SC0MD0) is "1", P04 is the I/O pin of the serial clock. Push-pull output or Nch open-drain output can be selected by setting Port 0 Nch open-drain control register (P0ODC).

P05 is also used as the I/O pin for Serial Interface 3 transmission/reception data and the output pin of UART3 transmission data. When the SC3SBOS flag of Serial Interface 3 mode register 1 (SC3MD1) is "1", P05 is the serial data I/O pin. Push-pull output or Nch open-drain output can be selected by setting Port 0 Nch open-drain control register (P0ODC).

P06 is also used as the input pin for Serial Interface 3 reception data and UART3 reception data. When the SC3BIS flag of Serial Interface 3 mode register 1 (SC3MD1) is "1", P06 is the serial data input pin.

P07 is also used as the I/O pin for Serial Interface 3 clock. When the SC3SBTS flag of Serial Interface 3 mode register 1 (SC3MD1) is "1", P07 is the I/O pin of the serial clock. Push-pull output or Nch open-drain output can be selected by setting Port 0 Nch open-drain control register (P0ODC).

For P01 to P04 each bit can be set individually as either timer output or port output by Port 0 output mode register 1 (P0OMD1). The control flag of Port 0 output mode register 1 (P0OMD1) is set to "1" for special function output, and "0" for general-purpose port.

When P04 is used as timer output by P0OMD1 register, P04 can be set as either Timer 0 output or Timer 2 output by Port0 output mode register 2 (P0OMD2).

P00 to P07 are also used as LED0 to LED7 output pins.

Output mode of each bit can be set individually by Port 0 LED control register (P0LED). When Port 0 LED control register (P0LED) is set to "1", P00 to P07 are the large current output pins (Nch-Tr. side), and when set to "0", P00 to P05 are used as the normal current output pins.

The general-purpose port can output a large current in combination with the setting of Port 0 LED control register (P0LED).

P07 is also used as the analog output pin for D/A.

8.3.2 Registers

Table:8.3.2 shows registers that control Port 0.

Table:8.3.2 Port 0 Control Register

Table remarks √: With function -: Without function

Registers	Address	R/W	Function	Page	MN101 EF76K	MN101 EF57G	MN101 EF56K
P0OUT	0x03E70	R/W	Port 0 output register	VIII-8	√	√	√
P0IN	0x03E80	R	Port 0 input register	VIII-9	√	√	√
P0DIR	0x03E90	R/W	Port 0 direction control register	VIII-9	√	√	√
P0PLUD	0x03EA0	R/W	Port 0 pull-up/pull-down resistor control register	VIII-9	√	√	√
P0OMD1	0x03EB0	R/W	Port 0 output mode register 1	VIII-10	√	√	√
P0OMD2	0x03EC0	R/W	Port 0 output mode register 2	VIII-10	√	√	√
P0ODC	0x03EF0	R/W	Port 0 Nch open-drain control register	VIII-11	√	√	√
P0LED	0x03EE0	R/W	Port 0 LED control register	VIII-12	√	√	√
SELUD	0x03EAF	R/W	Pull-up/pull-down resistor selection register	VIII-13	√	√	√

R/W: Readable/Writable

R: Read only

■ Port 0 Output Register (P0OUT: 0x03E70)

bp	7	6	5	4	3	2	1	0
Flag	P0OUT7	P0OUT6	P0OUT5	P0OUT4	P0OUT3	P0OUT2	P0OUT1	P0OUT0
At reset	X	X	X	X	X	X	X	X
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description
7 to 0	P0OUT7 to 0	Output data 0: Output "Low" (V_{SS} -level) 1: Output "High" (V_{DD5} -level)

■ Port 0 Input Register (P0IN: 0x03E80)

bp	7	6	5	4	3	2	1	0
Flag	P0IN7	P0IN6	P0IN5	P0IN4	P0IN3	P0IN2	P0IN1	P0IN0
At reset	X	X	X	X	X	X	X	X
Access	R	R	R	R	R	R	R	R

bp	Flag	Description
7 to 0	P0IN7 to 0	Input data 0: Pin is "Low" (V_{SS} -level) 1: Pin is "High" (V_{DD5} -level)

■ Port 0 Direction Control Register (P0DIR: 0x03E90)

bp	7	6	5	4	3	2	1	0
Flag	P0DIR7	P0DIR6	P0DIR5	P0DIR4	P0DIR3	P0DIR2	P0DIR1	P0DIR0
At reset	0	0	0	0	0	0	0	0
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description
7 to 0	P0DIR5 to 0	I/O mode selection 0: Input mode 1: Output mode

■ Port 0 Pull-up/pull-down Resistor Control Register (P0PLUD: 0x03EA0)

bp	7	6	5	4	3	2	1	0
Flag	P0PLUD7	P0PLUD6	P0PLUD5	P0PLUD4	P0PLUD3	P0PLUD2	P0PLUD1	P0PLUD0
At reset	0	0	0	0	0	0	0	0
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description
7 to 0	P0PLUD7 to 0	Pull-up/pull-down resistor selection 0: Not added 1: Added

■ Port 0 Output Mode Register 1(P0OMD1: 0x03EB0)

bp	7	6	5	4	3	2	1	0
Flag	-	-	-	P0OMD14	P0OMD13	P0OMD12	P0OMD11	-
At reset	-	-	-	0	0	0	0	-
Access	-	-	-	R/W	R/W	R/W	R/W	-

bp	Flag	Description
7 to 5	-	-
4	P0OMD14	I/O port or (TM2IOB or TM0IOB) selection 0: P04 1: TM2IOB or TM0IOB
3	P0OMD13	I/O port or TM8IOB selection 0: P03 1: TM8IOB
2	P0OMD12	I/O port or TM7IOB selection 0: P02 1: TM7IOB
1	P0OMD11	I/O port or TM9IOB selection 0: P01 1: TM9IOB
0	-	-

■ Port 0 Output Mode Register 2 (P0OMD2: 0x03EC0)

bp	7	6	5	4	3	2	1	0
Flag	-	-	-	P0OMD24	-	-	-	-
At reset	-	-	-	0	-	-	-	-
Access	-	-	-	R/W	-	-	-	-

bp	Flag	Description
7 to 5	-	-
4	P0OMD24	P04 special function selection 0: TM0IOB 1: TM2IOB
3 to 0	-	-

■ Port 0 Nch Open-drain Control Register P0ODC: 0x03EF0)

<MN101EF76K/MN101EF56K>

bp	7	6	5	4	3	2	1	0
Flag	P0ODC7	-	P0ODC5	P0ODC4	P0ODC3	-	-	-
At reset	0	-	0	0	0	-	-	-
Access	R/W	-	R/W	R/W	R/W	-	-	-

bp	Flag	Description
7	P0ODC7	Nch open-drain output selection 0: Push-pull output 1: Nch open-drain output
6	-	-
5 to 3	P0ODC5 to 3	Nch open-drain output selection 0: Push-pull output 1: Nch open-drain output
2 to 0	-	-

■ Port 0 Nch Open-drain Control Register P0ODC: 0x03EF0)

<MN101EF57G>

bp	7	6	5	4	3	2	1	0
Flag	Reserved	-	Reserved	P0ODC4	P0ODC3	-	-	-
At reset	0	-	0	0	0	-	-	-
Access	R/W	-	R/W	R/W	R/W	-	-	-

bp	Flag	Description
7	Reserved	Always set to "0".
6	-	-
5	Reserved	Always set to "0".
4 to 3	P0ODC4 to 3	Nch open-drain output selection 0: Push-pull output 1: Nch open-drain output
2 to 0	-	-

■ Port 0 LED Control Register (POLED: 0x03EE0)

bp	7	6	5	4	3	2	1	0
Flag	POLED7	POLED6	POLED5	POLED4	POLED3	POLED2	POLED1	POLED0
At reset	0	0	0	0	0	0	0	0
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description
7	POLED7	LED7 (large current output) selection 0: Normal output 1: LED7 (large current output)
6	POLED6	LED6 (large current output) selection 0: Normal output 1: LED6 (large current output)
5	POLED5	LED5 (large current output) selection 0: Normal output 1: LED5 (large current output)
4	POLED4	LED4 (large current output) selection 0: Normal output 1: LED4 (large current output)
3	POLED3	LED3 (large current output) selection 0: Normal output 1: LED3 (large current output)
2	POLED2	LED2 (large current output) selection 0: Normal output 1: LED2 (large current output)
1	POLED1	LED1 (large current output) selection 0: Normal output 1: LED1 (large current output)
0	POLED0	LED0 (large current output) selection 0: Normal output 1: LED0 (large current output)

■ Pull-up/Pull-down Resistor Selection Register (SELUD: 0x03EAF)

<MN101EF76K/MN101EF56K>

bp	7	6	5	4	3	2	1	0
Flag	SELUD7	-	SELUD5	SELUD4	-	-	SELUD1	SELUD0
At reset	0	-	0	0	-	-	0	0
Access	R/W	-	R/W	R/W	-	-	R/W	R/W

bp	Flag	Description
7	SELUD7	Pull-up/pull-down selection (Port 7) 0: Pull-up 1: Pull-down
6	-	-
5	SELUD5	Pull-up/pull-down selection (Port 5) 0: Pull-up 1: Pull-down
4	SELUD4	Pull-up/pull-down selection (Port 4) 0: Pull-up 1: Pull-down
3 to 2	-	-
1	SELUD1	Pull-up/pull-down selection (Port 1) 0: Pull-up 1: Pull-down
0	SELUD0	Pull-up/pull-down selection (Port 0) 0: Pull-up 1: Pull-down

<MN101EF57G>

bp	7	6	5	4	3	2	1	0
Flag	SELUD7	-	SELUD5	SELUD4	-	-	-	SELUD0
At reset	0	-	0	0	-	-	-	0
Access	R/W	-	R/W	R/W	-	-	-	R/W

bp	Flag	Description
7	SELUD7	Pull-up/pull-down selection (Port 7) 0: Pull-up 1: Pull-down
6	-	-
5	SELUD5	Pull-up/pull-down selection (Port 5) 0: Pull-up 1: Pull-down
4	SELUD4	Pull-up/pull-down selection (Port 4) 0: Pull-up 1: Pull-down
3 to 1	-	-
0	SELUD0	Pull-up/pull-down selection (Port 0) 0: Pull-up 1: Pull-down

8.3.3 Block Diagram (MN101EF76K/MN101EF56K)

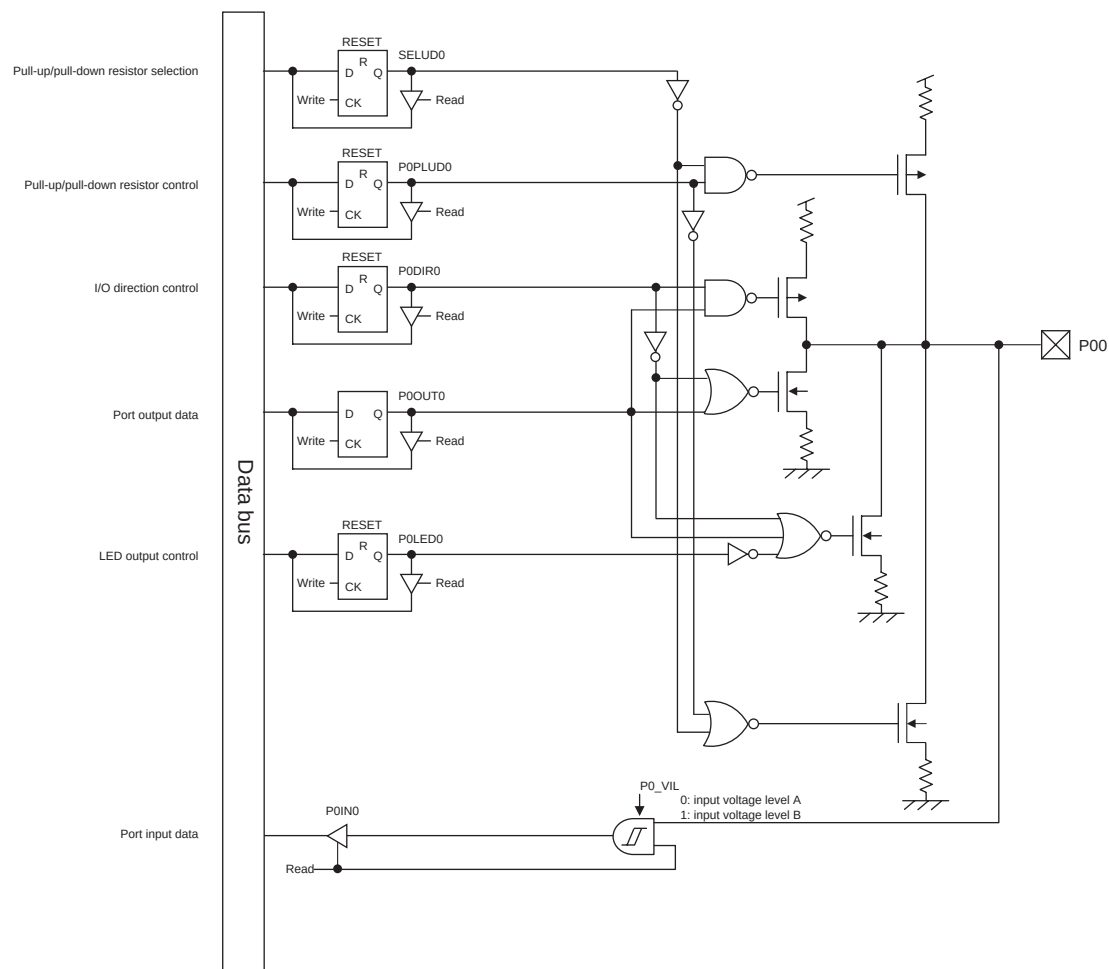


Figure:8.3.1 P00 Block Diagram (MN101EF76K/MN101EF56K)

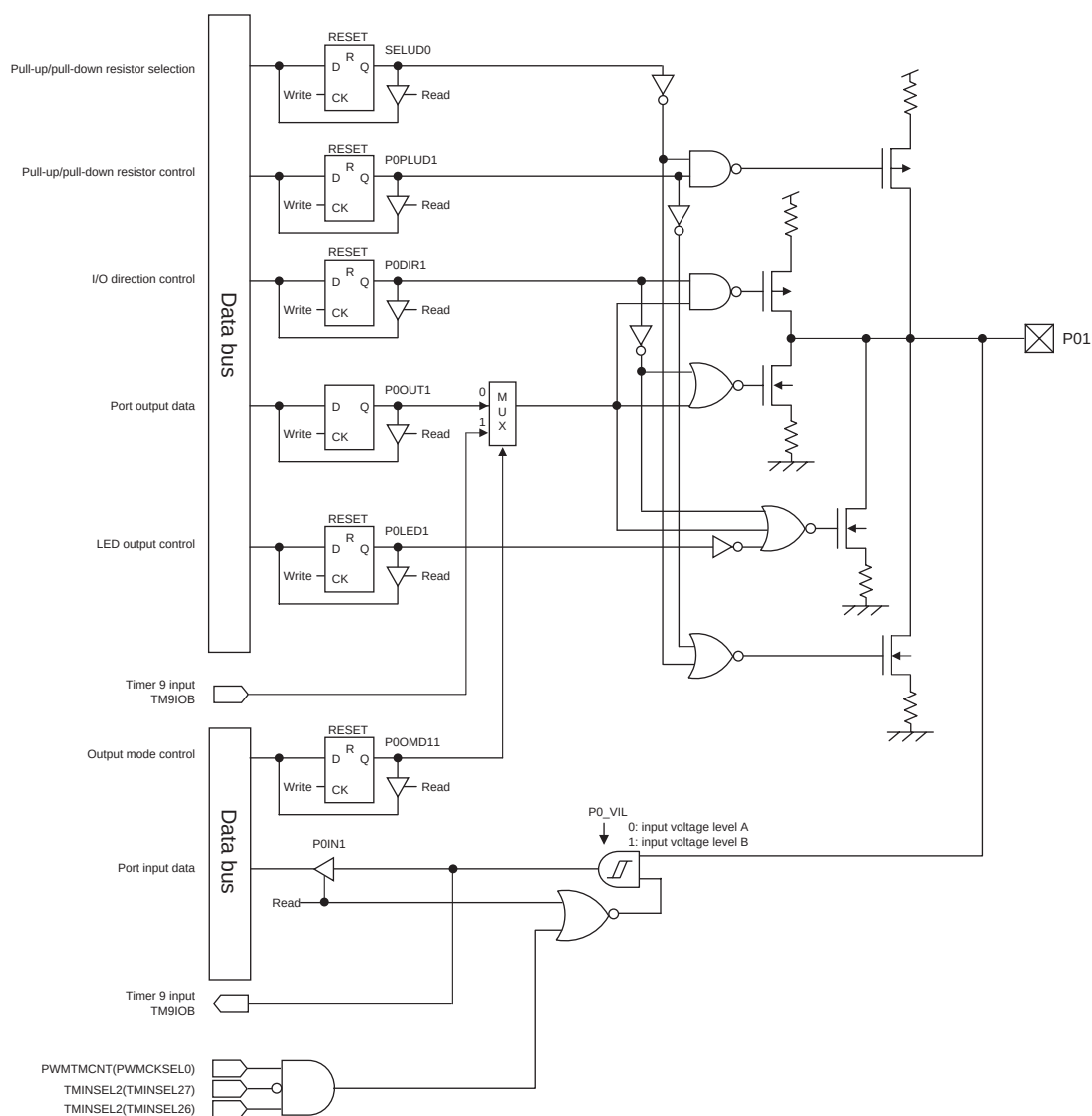


Figure:8.3.2 P01 Block Diagram (MN101EF76K/MN101EF56K)

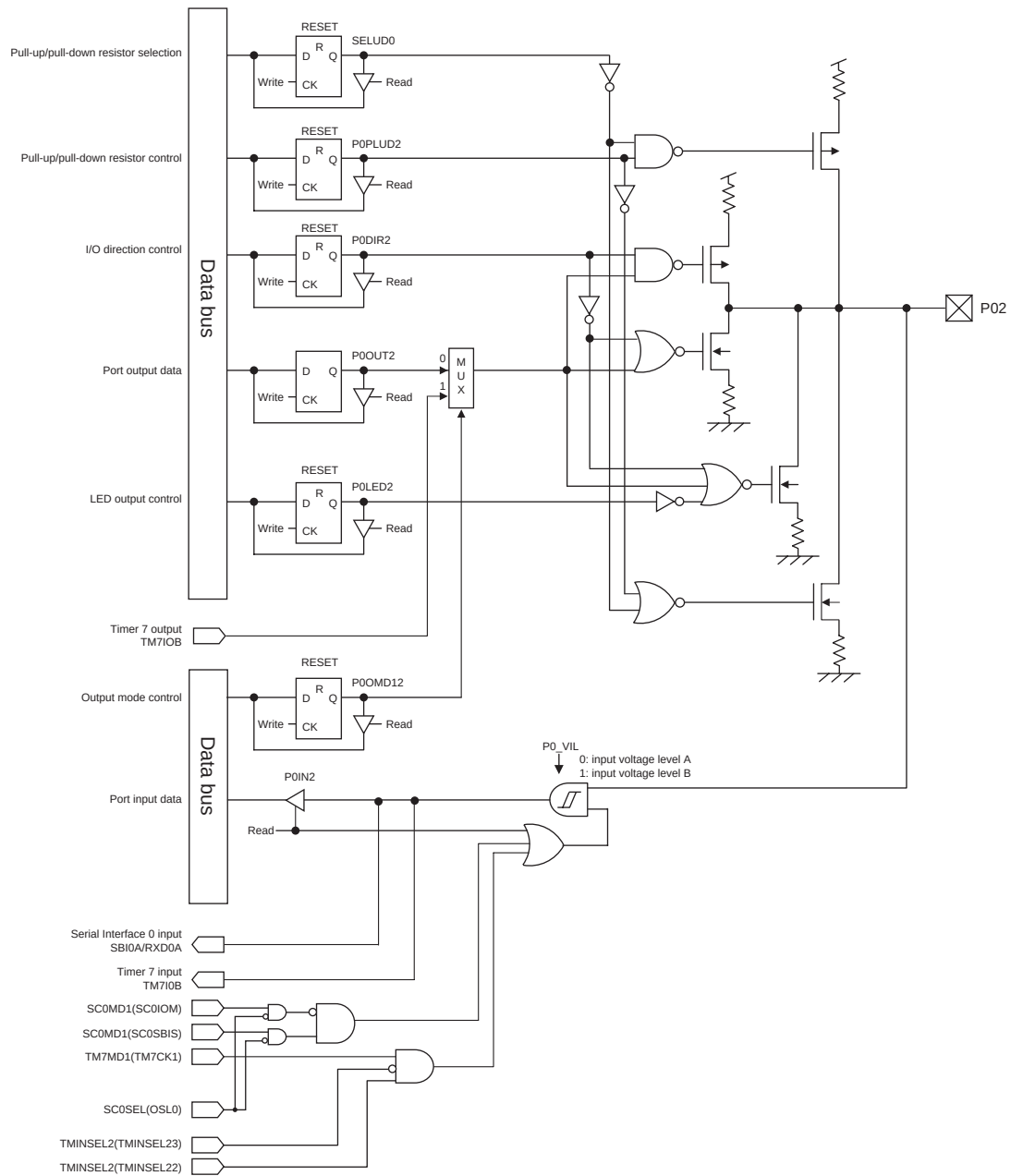


Figure:8.3.3 P02 Block Diagram (MN101EF76K/MN101EF56K)

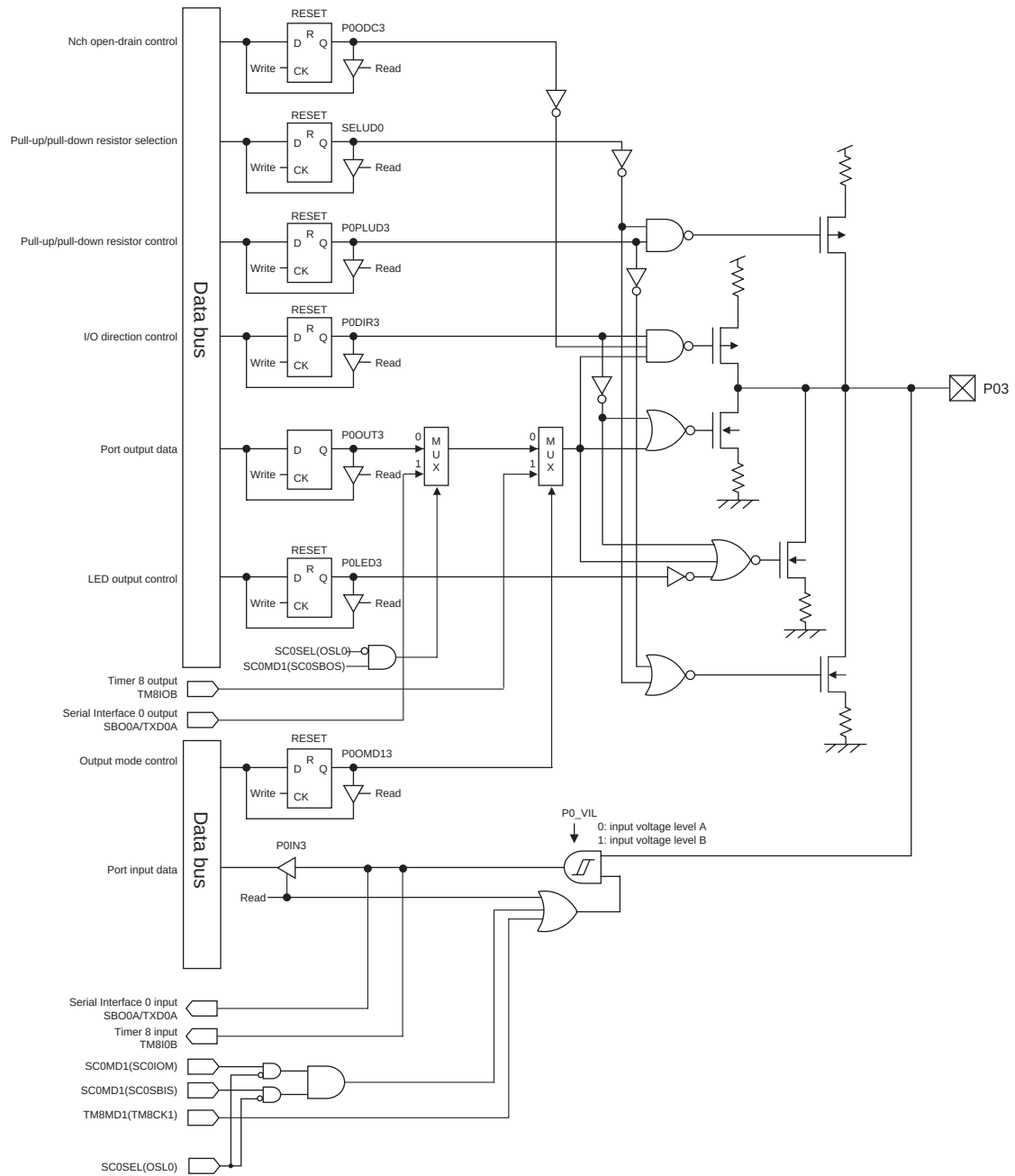


Figure:8.3.4 P03 Block Diagram (MN101EF76K/MN101EF56K)

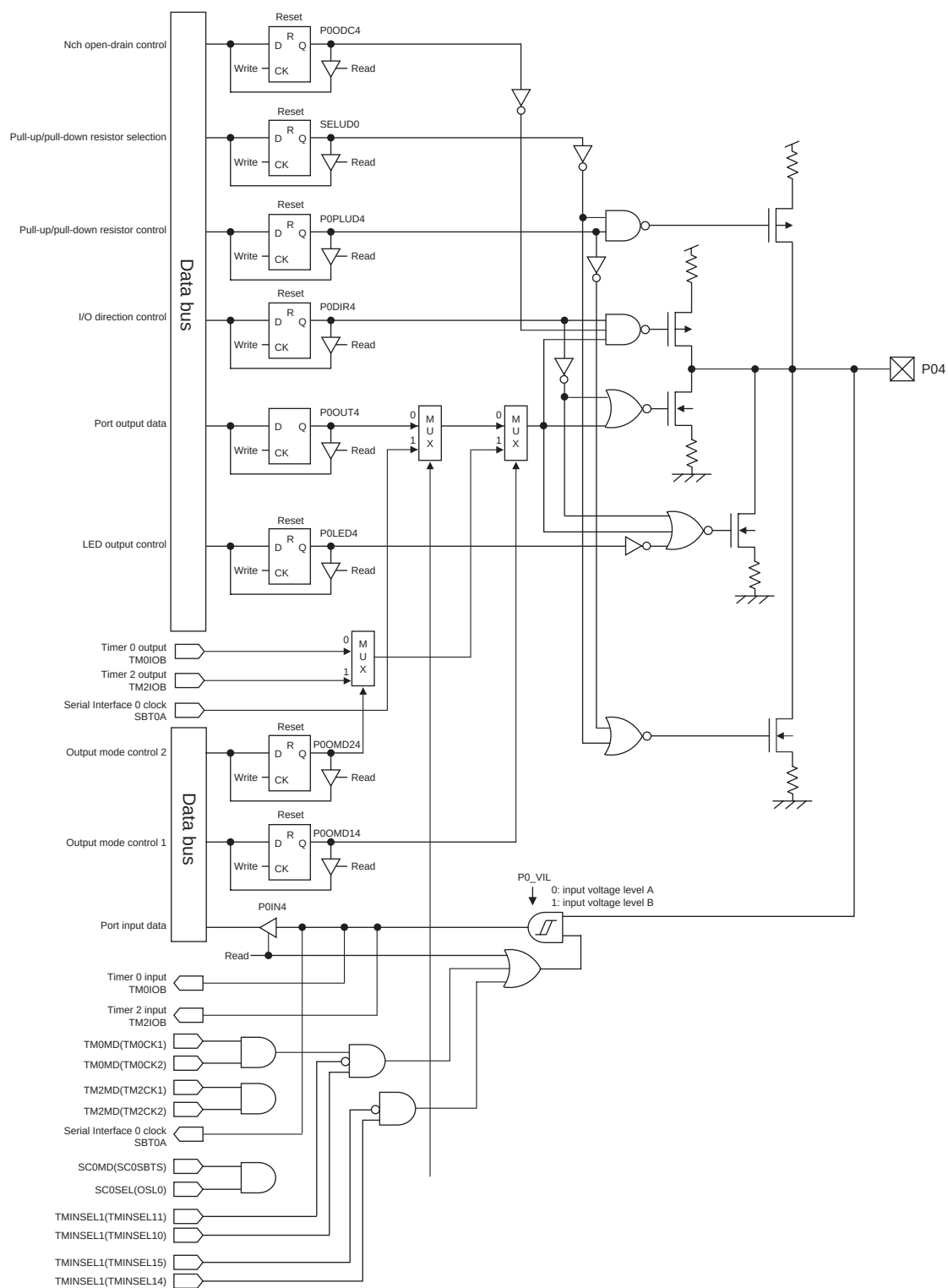
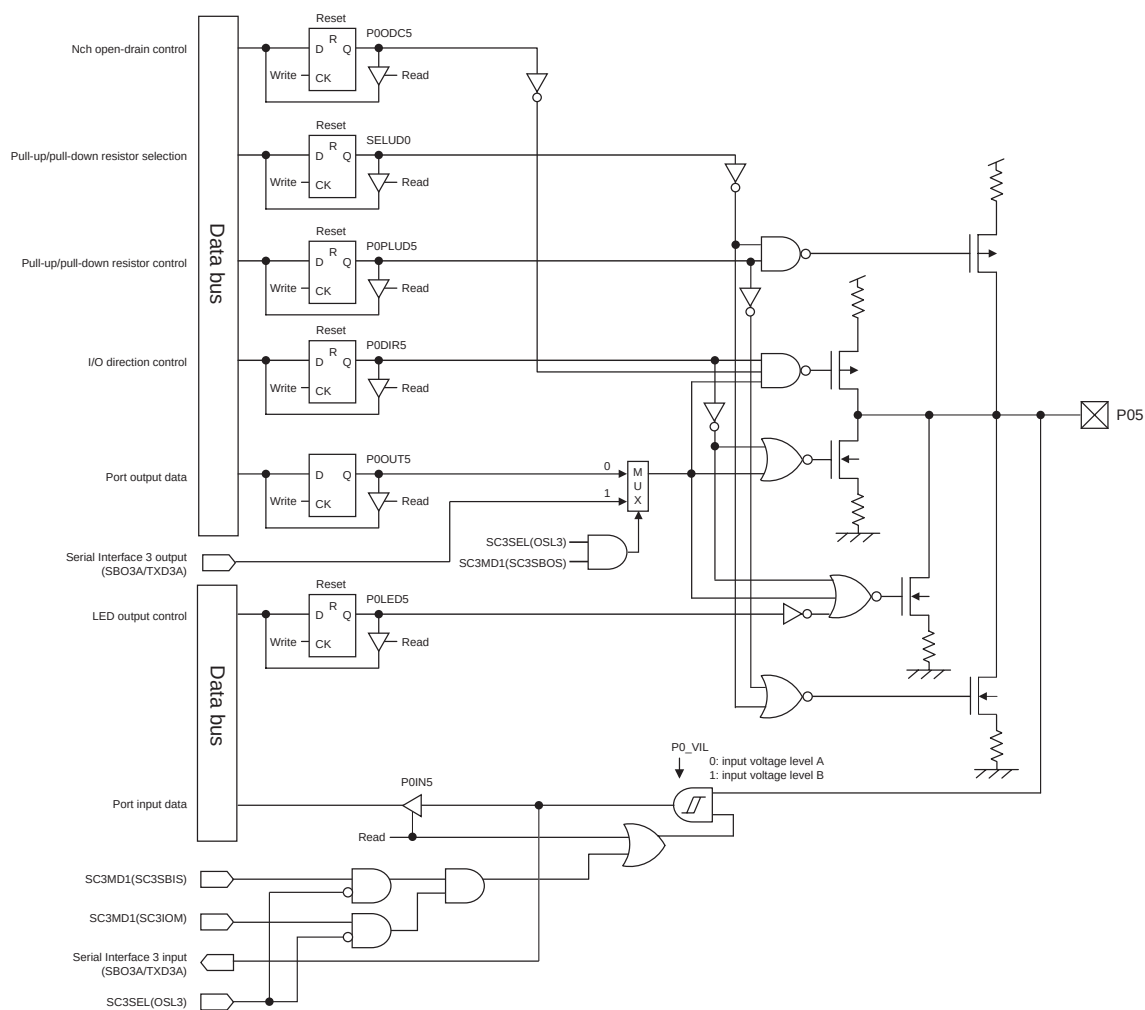


Figure:8.3.5 P04 Block Diagram (MN101EF76K/MN101EF56K)



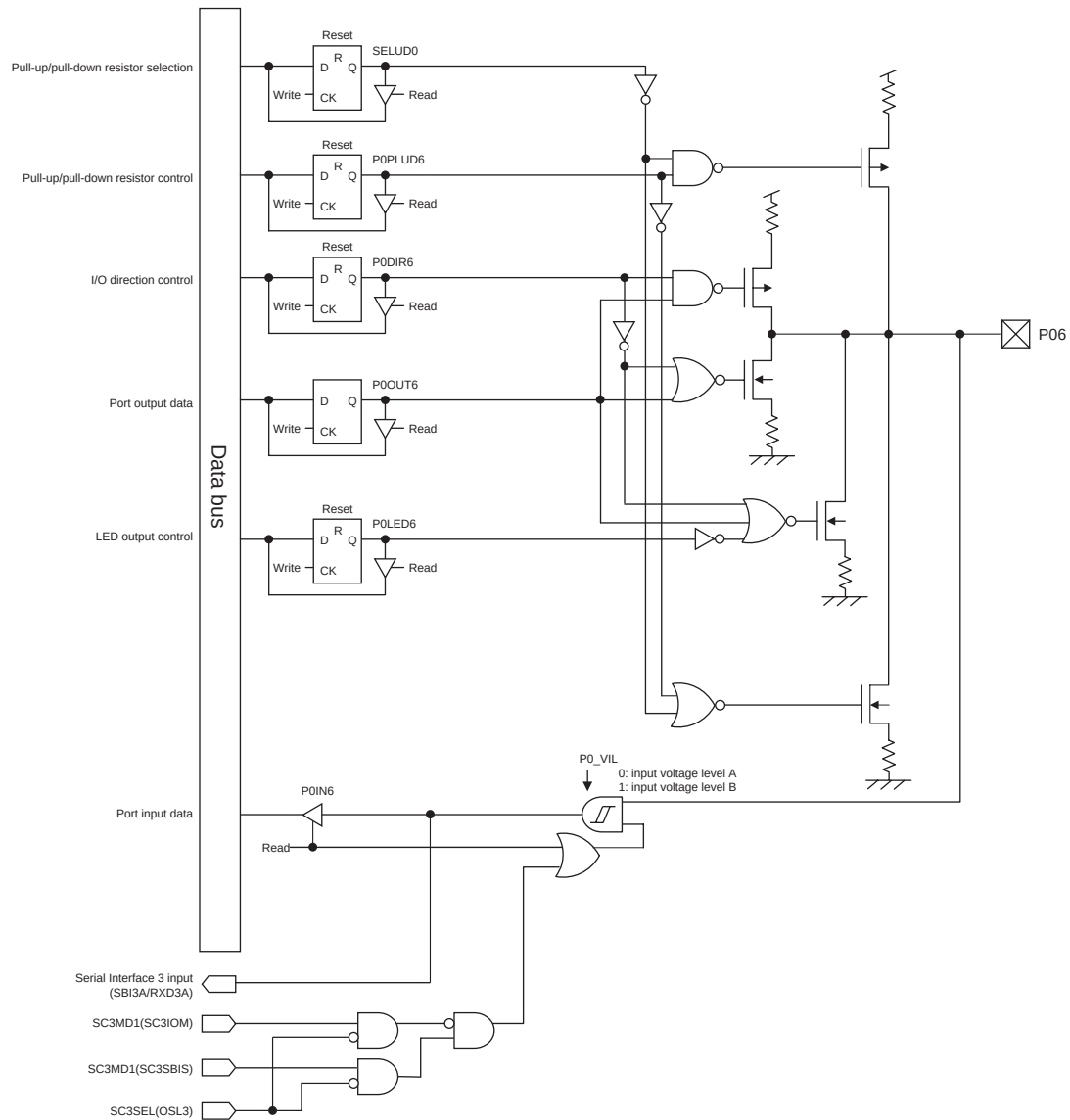


Figure:8.3.7 P06 Block Diagram (MN101EF76K/MN101EF56K)

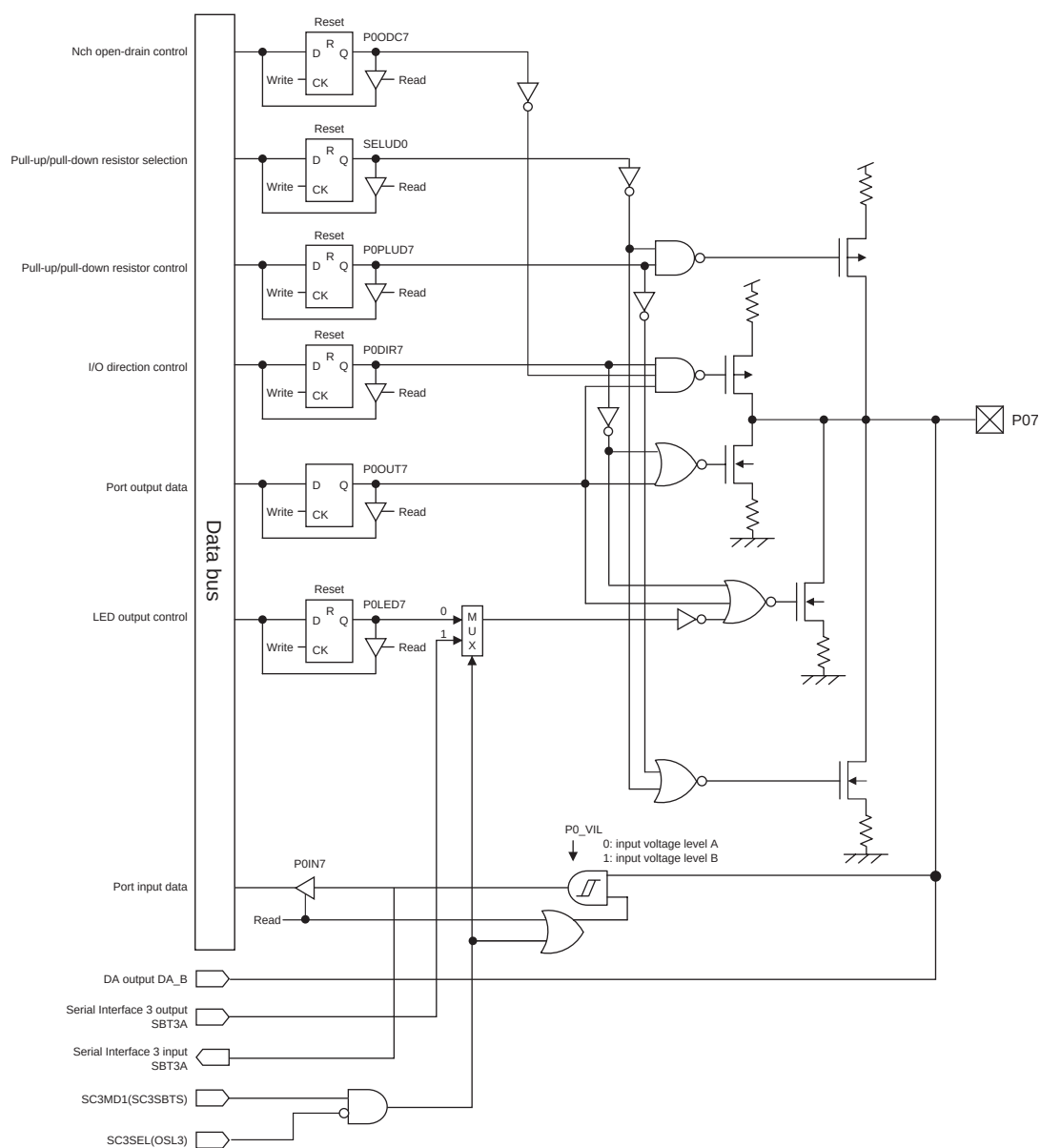


Figure:8.3.8 P07 Block Diagram (MN101EF76K/MN101EF56K)

8.3.4 Block Diagram (MN101EF57G)

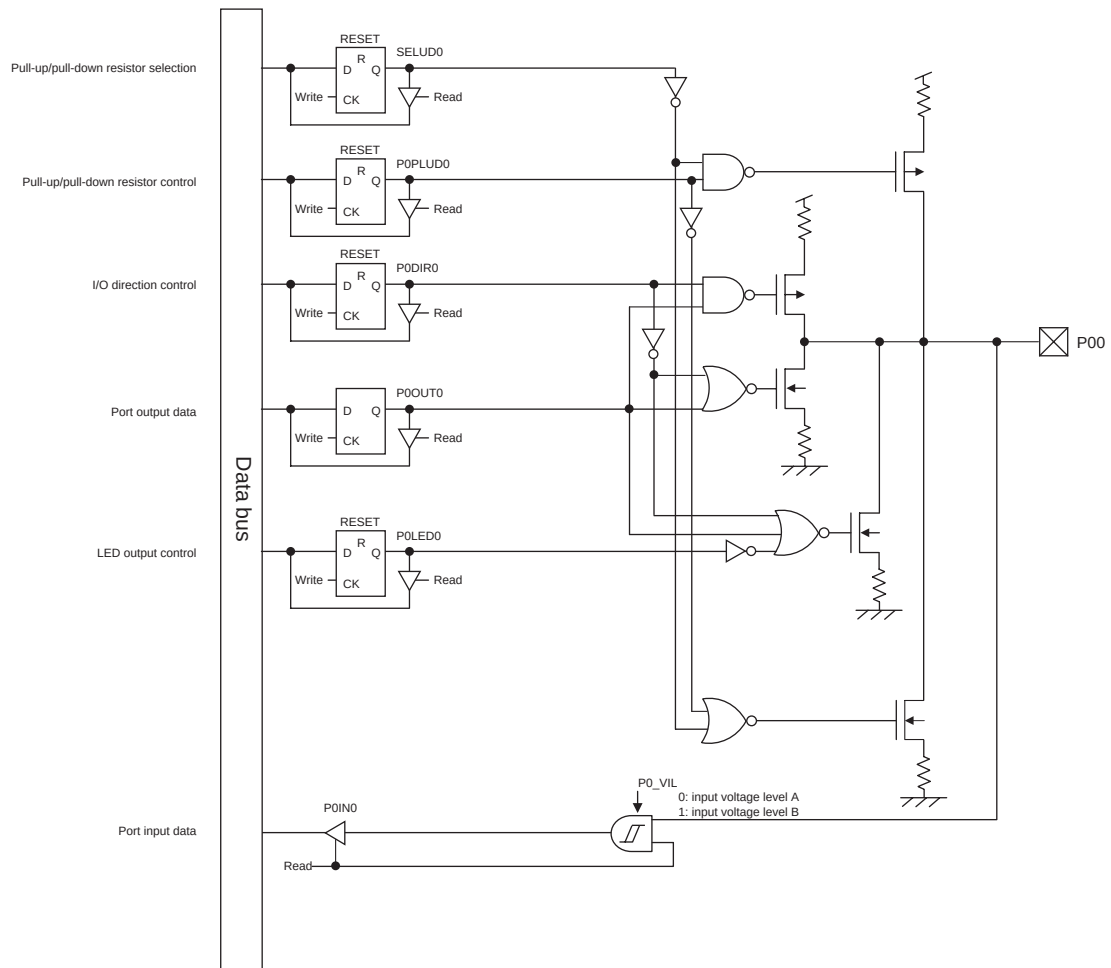


Figure:8.3.9 P00 Block Diagram (MN101EF57G)

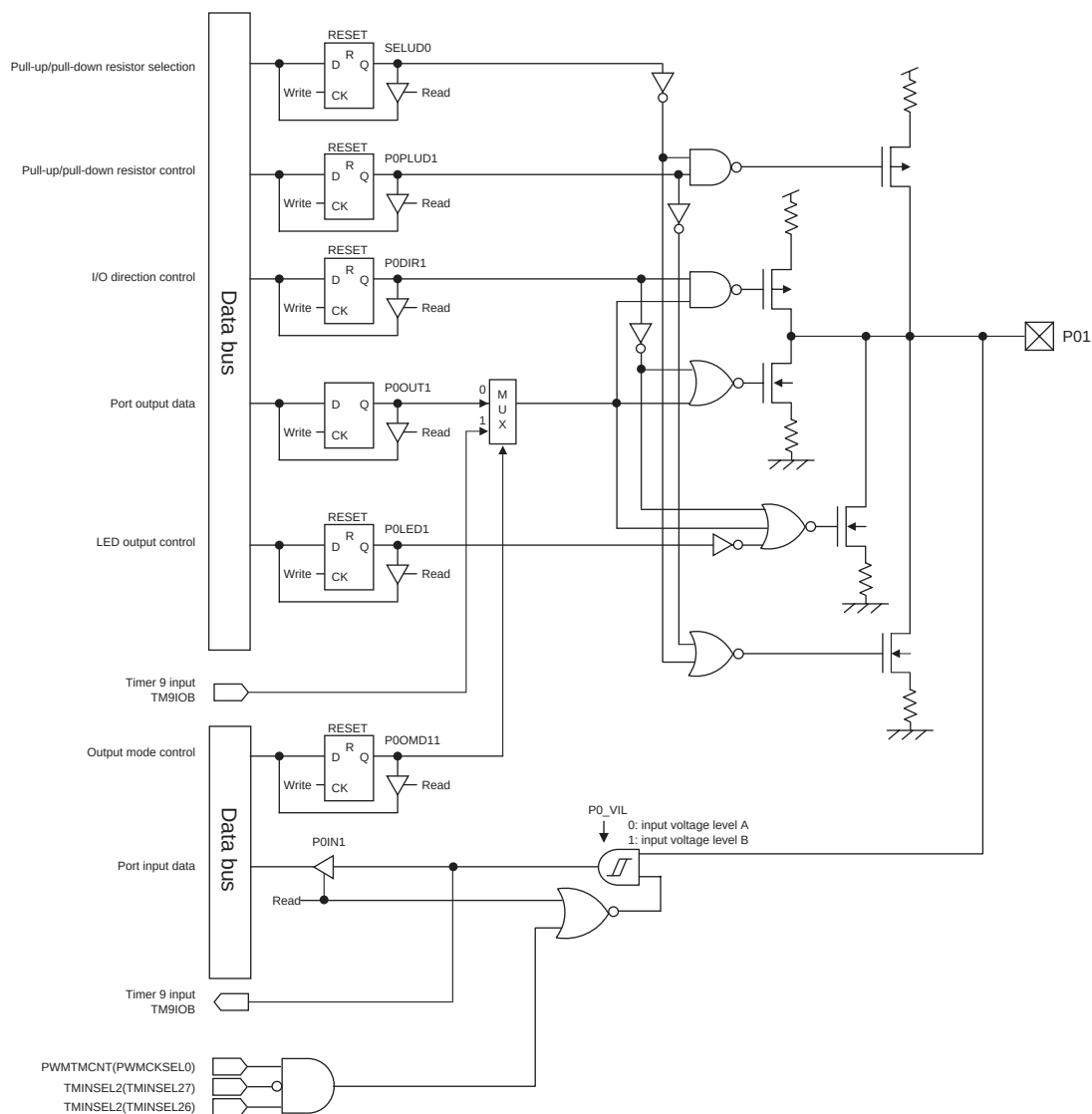


Figure:8.3.10 P01 Block Diagram (MN101EF57G)

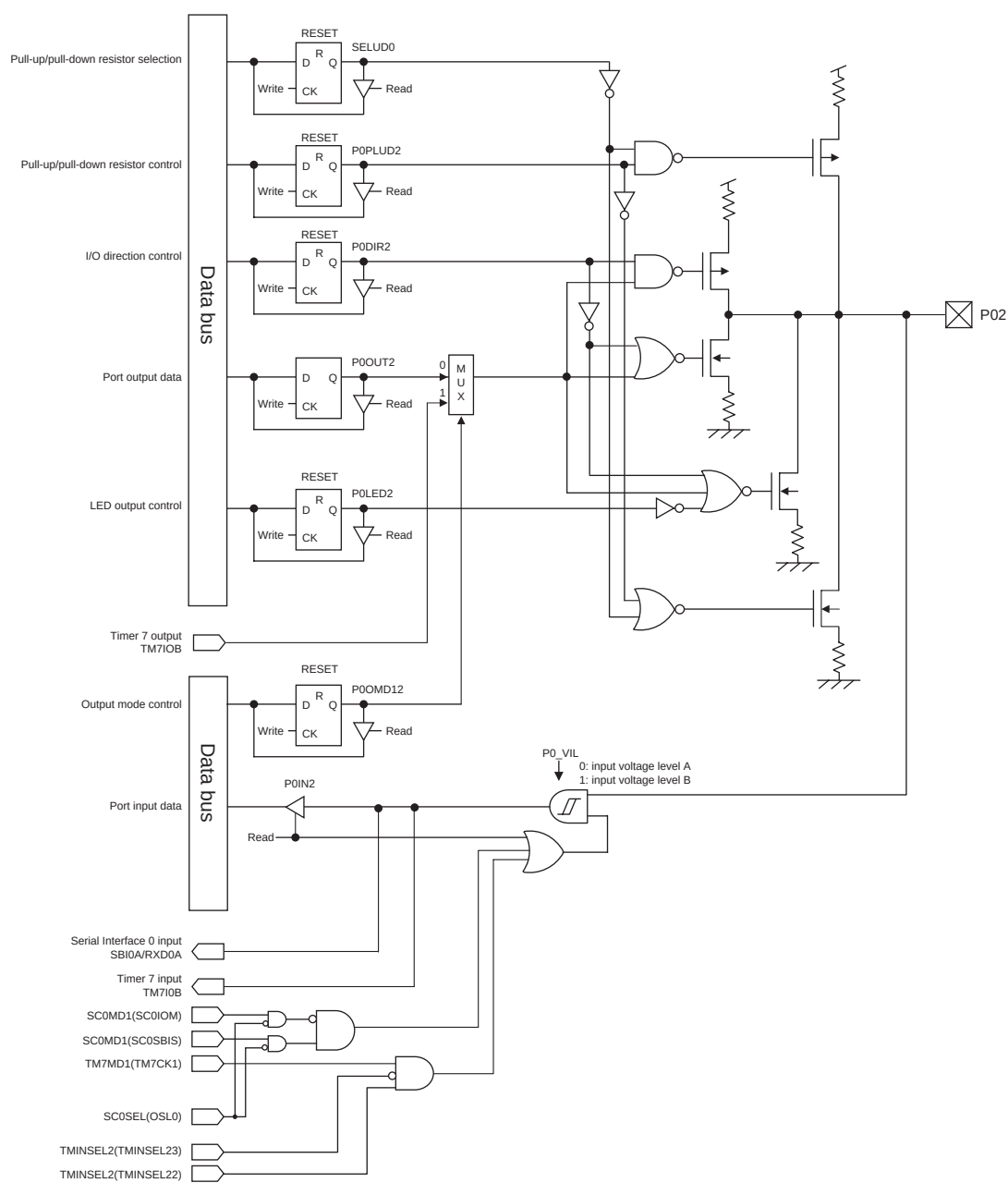


Figure:8.3.11 P02 Block Diagram (MN101EF57G)

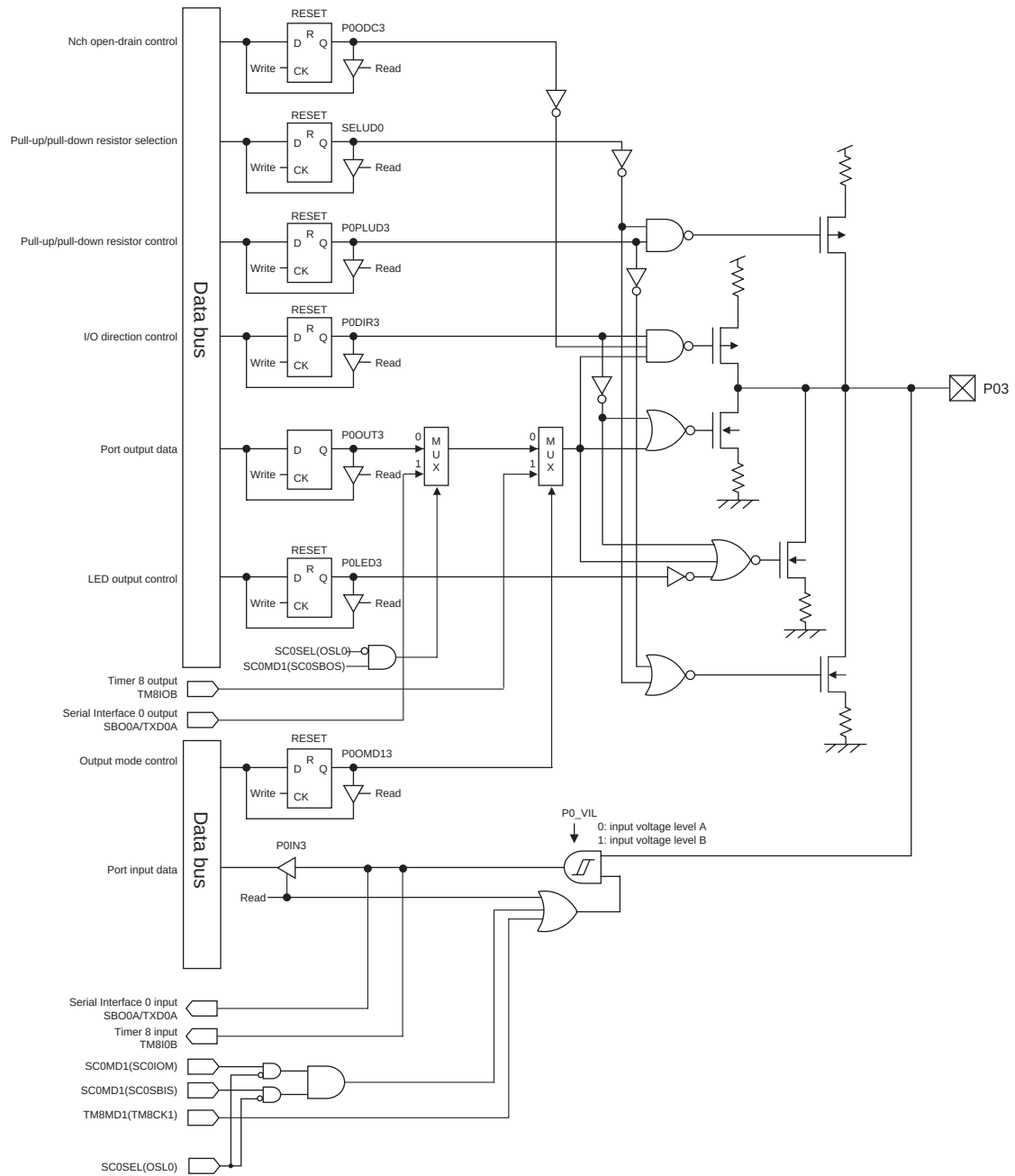


Figure:8.3.12 P03 Block Diagram (MN101EF57G)

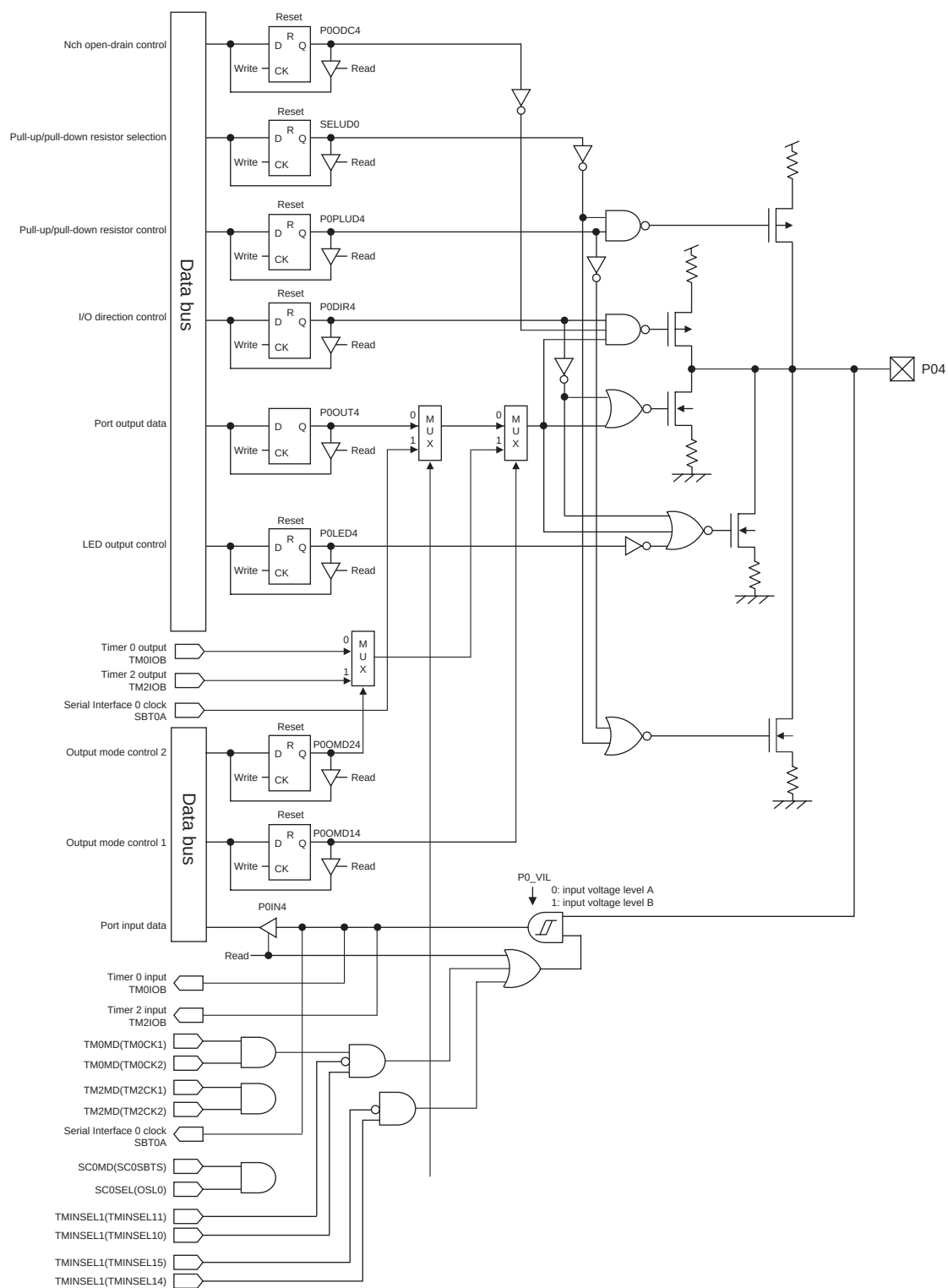


Figure:8.3.13 P04 Block Diagram (MN101EF57G)

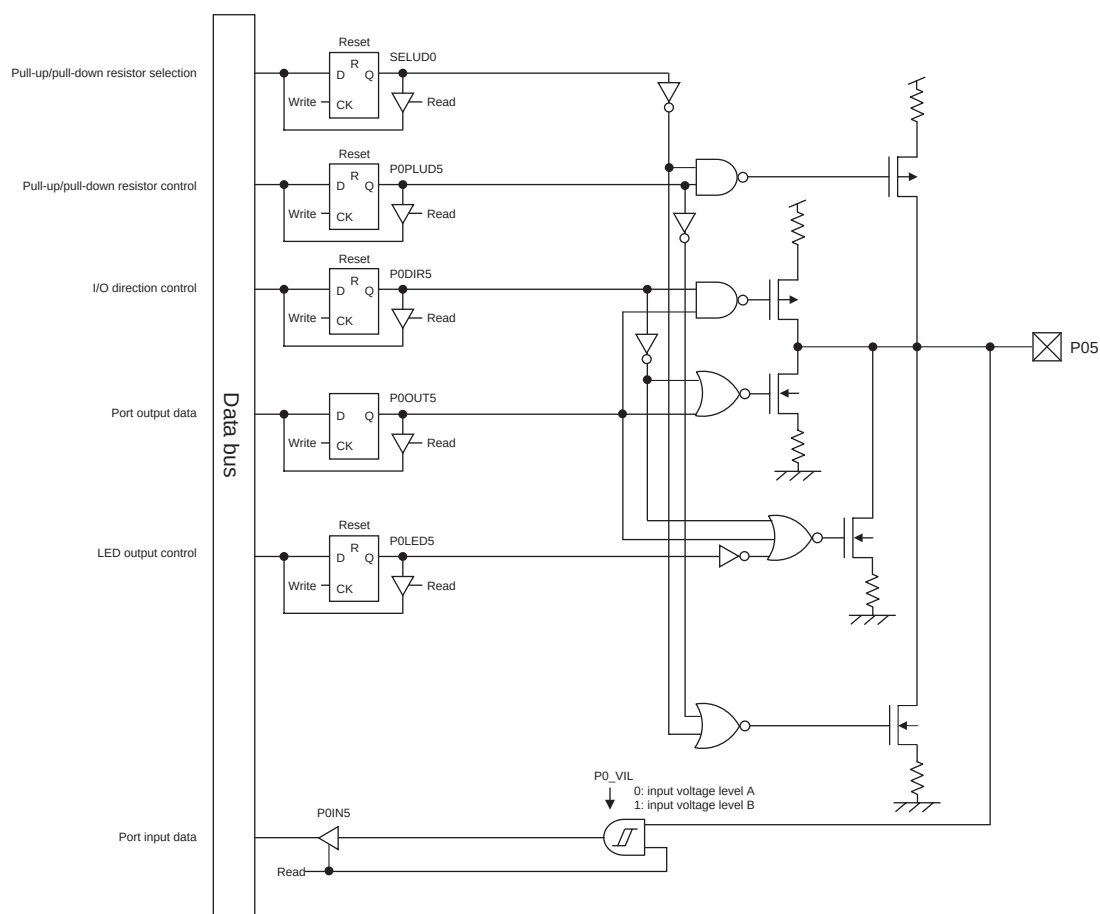


Figure:8.3.14 P05 Block Diagram (MN101EF57G)

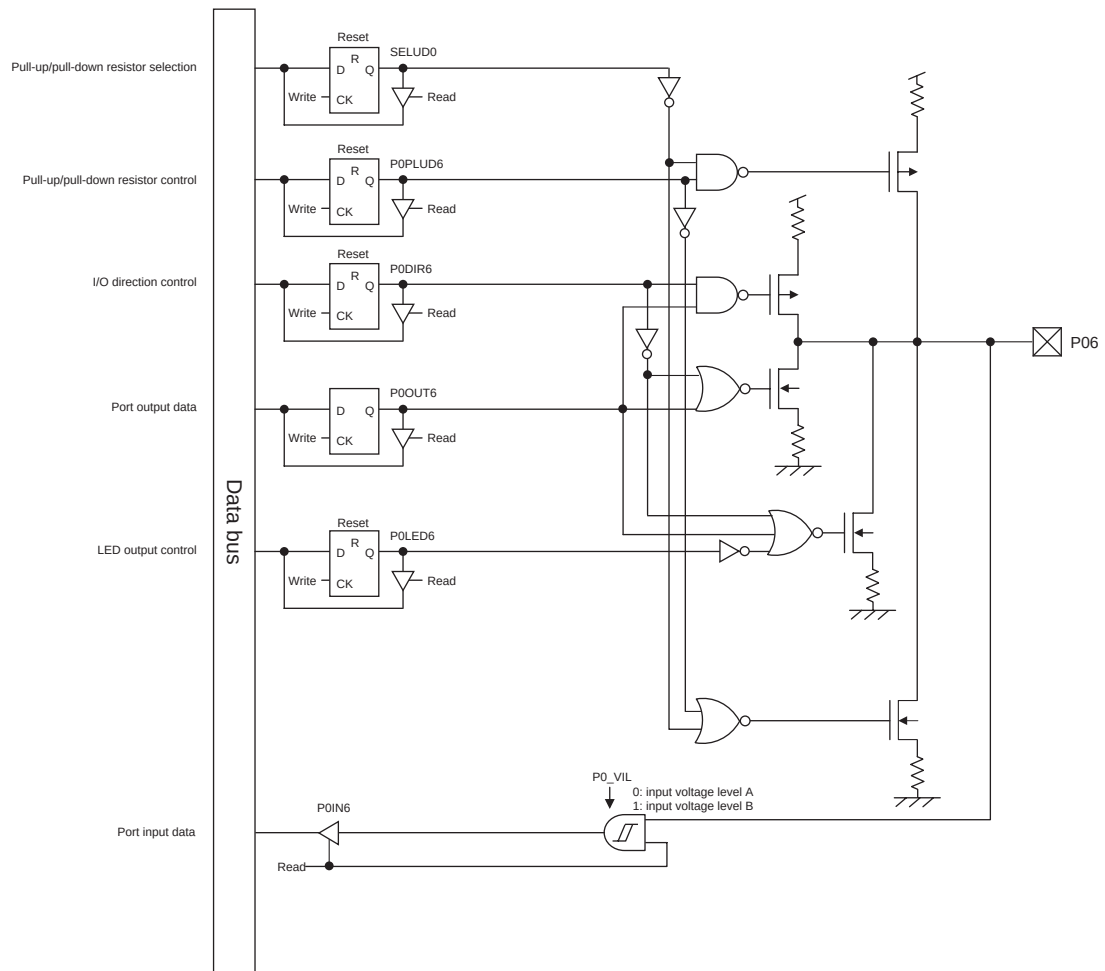


Figure:8.3.15 P06 Block Diagram (MN101EF57G)

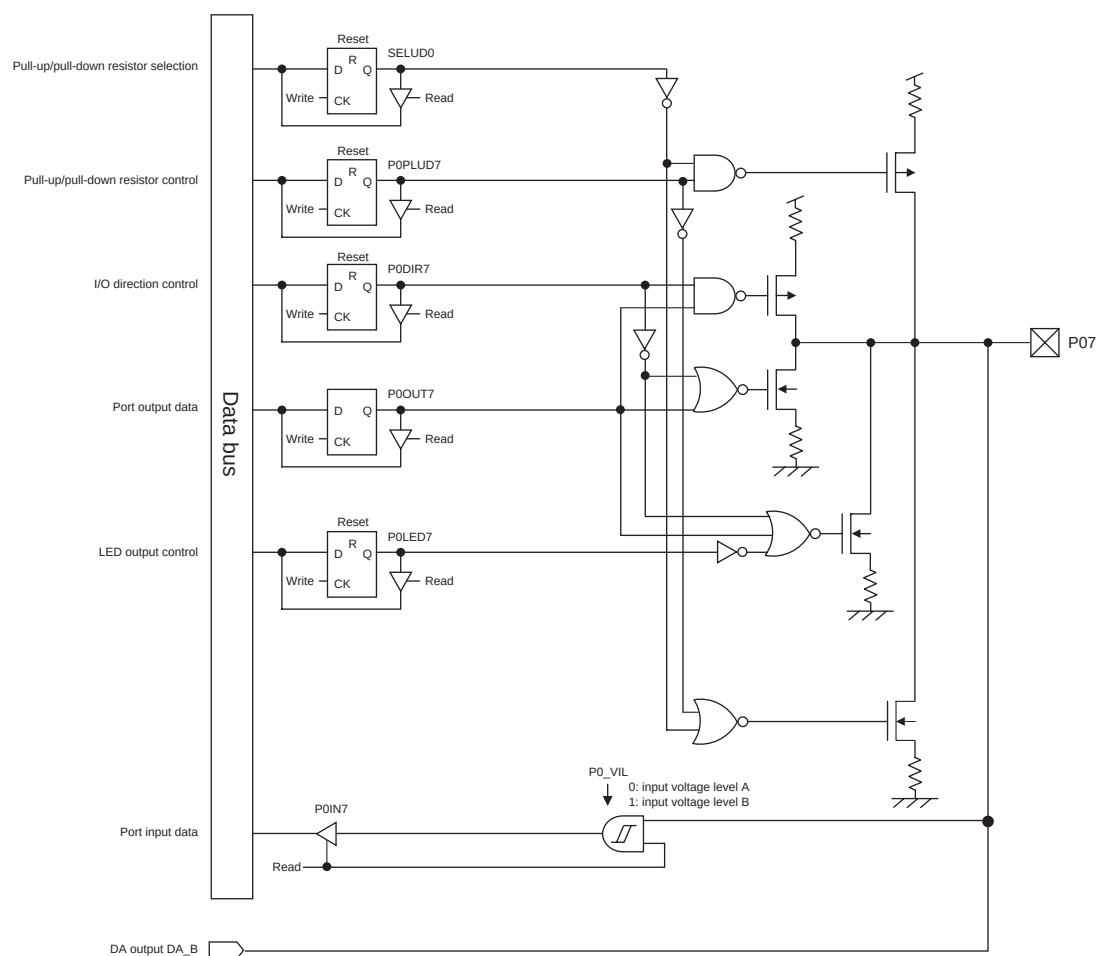


Figure:8.3.16 P07 Block Diagram (MN101EF57G)

8.4 Port 1

8.4.1 Description

■ General-purpose Port Setup

To output data to pins, set the control flag of Port 1 direction control register (P1DIR) to "1" and write the value of Port 1 output register (P1OUT).

To read input data of pins, set the control flag of Port 1 direction control register (P1DIR) to "0" and read the value of Port 1 input register (P1IN).

Each bit can be set individually as either an input or output by Port 1 direction control register (P1DIR). The control flag of P1DIR register is set to "1" for output mode, and "0" for input mode.

Each bit can be set individually to add a pull-up (or pull-down) resistor by Port 1 pull-up/pull-down resistor control register (P1PLUD). Set the control flag of P1PLUD register to "1" to add a pull-up (or pull-down) resistor.

Port 1 can be selected to add a pull-up or pull-down resistor by bp1 of the pull-up/pull-down resistor selection register (SELUD).

Each bit can be set individually to control input mode by Port 1 input mode control register (P1IMD).

The control flag of P1IMD register is set to "1" for using as the input port of special function data, and to "0" for using as general purpose port. When using as the special function input pin, undefined value is read from P1IN register.

Table:8.4.1 Port 1 Functions

Table remarks -: Without function

MN101EF76K/MN101EF56K				MN101EF57G			
Pin	Special function			Pin	Special function		
P10	SEG45	AN20	TM0IOC	-	-	-	-
P11	SEG44	AN21	TM2IOC	-	-	-	-
P12	SEG43	AN22	TM1IOC	-	-	-	-
P13	SEG42	AN23	TM3IOC	-	-	-	-
P14	SEG41	TM4IOC		-	-	-	
P15	SEG40	TM7IOC	BUZZERB	-	-	-	-
P16	SEG39	TM8IOC	NBUZZERB	-	-	-	-

■ Special Function Pin Setup

P10 to P16 are also used as LCD segment output pins.

LCD segment output pin is selected by setting the corresponding flag of LCD output control register n (LCCTRn) to "1".

Port and segment can be switched by one bit. When segment output pin is selected, this pin is forced to be output mode and pull-up resistor is not added.

P10 to P13 are also used as analog input pins.

Port 1 input mode register (P1IMD) can select the input mode by one bit. When using as the analog input pin, undefined value is read from P1IN register.

P10 is also used as the I/O pin for Timer 0.

P11 is also used as the I/O pin for Timer 2.

P12 is also used as the I/O pin for Timer 1.

P13 is also used as the I/O pin for Timer 3.

P14 is also used as the I/O pin for Timer 4.

P15 is also used as the I/O pin for Timer 7 or the output pin for Buzzer.

P16 is also used as the I/O pin for Timer 8 or the output pin for Reverse Buzzer.

Each bit can be set individually as either special function or general purpose port by Port 1 output mode register 1 (P1OMD1). The corresponding flag of P1OMD1 register is set to "1" for special function, and "0" for general-purpose port.

P15 to P16 are used as either Timer I/O or Buzzer output by Port 1 output mode register 2 (P1OMD2). The corresponding flag of P1OMD2 register is set to "0" for Timer I/O, and "1" for Buzzer output.

8.4.2 Registers

Table:8.4.2 shows registers that control Port 1.

Table:8.4.2 Port 1 Control Register

Table remarks √: With function -: Without function

Registers	Address	R/W	Function	Page	MN101 EF76K	MN101 EF57G	MN101 EF56K
P1OUT	0x03E71	R/W	Port 1 output register	VIII-32	√	-	√
P1IN	0x03E81	R	Port 1 input register	VIII-33	√	-	√
P1DIR	0x03E91	R/W	Port 1 direction control register	VIII-33	√	-	√
P1PLUD	0x03EA1	R/W	Port 1 pull-up/pull-down resistor control register	VIII-33	√	-	√
P1OMD1	0x03EB1	R/W	Port 1 output mode register 1	VIII-34	√	-	√
P1OMD2	0x03EC1	R/W	Port 1 output mode register 2	VIII-35	√	-	√
P1IMD	0x03ED1	R/W	Port 1 input mode register	VIII-35	√	-	√
SELUD	0x03EAF	R/W	Pull-up/pull-down resistor selection register	VIII-36	√	√	√

R/W: Readable/Writable

R: Read only

■ Port 1 Output Register (P1OUT: 0x03E71)

<MN101EF76K/MN101EF56K>

bp	7	6	5	4	3	2	1	0
Flag	-	P1OUT6	P1OUT5	P1OUT4	P1OUT3	P1OUT2	P1OUT1	P1OUT0
At reset	-	X	X	X	X	X	X	X
Access	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description
7	-	-
6 to 0	P1OUT6 to 0	Output data 0: Output "Low" (V_{SS} -level) 1: Output "High" (V_{DD5} -level)

■ Port 1 Input Register (P1IN: 0x03E81)

<MN101EF76K/MN101EF56K>

bp	7	6	5	4	3	2	1	0
Flag	-	P1IN6	P1IN5	P1IN4	P1IN3	P1IN2	P1IN1	P1IN0
At reset	-	X	X	X	X	X	X	X
Access	-	R	R	R	R	R	R	R

bp	Flag	Description
7	-	-
6 to 0	P1IN6 to 0	Input data 0: Pin is "Low" (V_{SS} -level) 1: Pin is "High" (V_{DD5} -level)

■ Port 1 Direction Control Register (P1DIR: 0x03E91)

<MN101EF76K/MN101EF56K>

bp	7	6	5	4	3	2	1	0
Flag	-	P1DIR6	P1DIR5	P1DIR4	P1DIR3	P1DIR2	P1DIR1	P1DIR0
At reset	-	0	0	0	0	0	0	0
Access	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description
7	-	-
6 to 0	P1DIR6 to 0	I/O mode selection 0: Input mode 1: Output mode

■ Port 1 Pull-up/Pull-down Resistor Control Register (P1PLUD: 0x03EA1)

<MN101EF76K/MN101EF56K>

bp	7	6	5	4	3	2	1	0
Flag	-	P1PLUD6	P1PLUD5	P1PLUD4	P1PLUD3	P1PLUD2	P1PLUD1	P1PLUD0
At reset	-	0	0	0	0	0	0	0
Access	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description
7	-	-
6 to 0	P1PLUD6 to 0	Pull-up/pull-down resistor selection 0: Not added 1: Added

■ Port 1 Output Mode Register 1 (P1OMD1: 0x03EB1)

<MN101EF76K/MN101EF56K>

bp	7	6	5	4	3	2	1	0
Flag	-	P1OMD16	P1OMD15	P1OMD14	P1OMD13	P1OMD12	P1OMD11	P1OMD10
At reset	-	0	0	0	0	0	0	0
Access	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description
7	-	-
6	P1OMD16	I/O port or special function selection 0: P16 1: TM8IOC or NBUZZERB
5	P1OMD15	I/O port or special function selection 0: P15 1: TM7IOC or BUZZERB
4	P1OMD14	I/O port or special function selection 0: P14 1: TM4IOC
3	P1OMD13	I/O port or special function selection 0: P13 1: TM3IOC
2	P1OMD12	I/O port or special function selection 0: P12 1: TM2IOC
1	P1OMD11	I/O port or special function selection 0: P11 1: TM1IOC
0	P1OMD10	I/O port or special function selection 0: P10 1: TM0IOC

■ Port 1 Output Mode Register 2 (P1OMD2: 0x03EC1)

<MN101EF76K/MN101EF56K>

bp	7	6	5	4	3	2	1	0
Flag	-	P1OMD26	P1OMD25	-	-	-	-	-
At reset	-	0	0	-	-	-	-	-
Access	-	R/W	R/W	-	-	-	-	-

bp	Flag	Description
7	-	-
6	P1OMD26	P16 special function selection (available at P1IMD16 = 1) 0: TM8IOC 1: NBUZZERB
5	P1OMD25	I/O port or special function selection (available at P1IMD15 = 1) 0: TM7IOC 1: BUZZERB
4 to 0	-	-

■ Port 1 Input Mode Register (P1IMD: 0x03ED1)

<MN101EF76K/MN101EF56K>

bp	7	6	5	4	3	2	1	0
Flag	-	-	-	-	P1IMD3	P1IMD2	P1IMD1	P1IMD0
At reset	-	-	-	-	0	0	0	0
Access	-	-	-	-	R/W	R/W	R/W	R/W

bp	Flag	Description
7 to 4	-	-
3	P1IMD3	I/O port or analog input selection 0: P13 1: AN23
2	P1IMD2	I/O port or analog input selection 0: P12 1: AN22
1	P1IMD1	I/O port or analog input selection 0: P11 1: AN21
0	P1IMD0	I/O port or analog input selection 0: P10 1: AN20

■ Pull-up/Pull-down Resistor Selection Register (SELUD: 0x03EAF)

<MN101EF76K/MN101EF56K>

bp	7	6	5	4	3	2	1	0
Flag	SELUD7	-	SELUD5	SELUD4	-	-	SELUD1	SELUD0
At reset	0	-	0	0	-	-	0	0
Access	R/W	-	R/W	R/W	-	-	R/W	R/W

bp	Flag	Description
7	SELUD7	Pull-up/pull-down selection (Port 7) 0: Pull-up 1: Pull-down
6	-	-
5	SELUD5	Pull-up/pull-down selection (Port 5) 0: Pull-up 1: Pull-down
4	SELUD4	Pull-up/pull-down selection (Port 4) 0: Pull-up 1: Pull-down
3 to 2	-	-
1	SELUD1	Pull-up/pull-down selection (Port 1) 0: Pull-up 1: Pull-down
0	SELUD0	Pull-up/pull-down selection (Port 0) 0: Pull-up 1: Pull-down

■ Pull-up/Pull-down Resistor Selection Register (SELUD: 0x03EAF)

<MN101EF57G>

bp	7	6	5	4	3	2	1	0
Flag	SELUD7	-	SELUD5	SELUD4	-	-	-	SELUD0
At reset	0	-	0	0	-	-	-	0
Access	R/W	-	R/W	R/W	-	-	-	R/W

bp	Flag	Description
7	SELUD7	Pull-up/pull-down selection (Port 7) 0: Pull-up 1: Pull-down
6	-	-
5	SELUD5	Pull-up/pull-down selection (Port 5) 0: Pull-up 1: Pull-down
4	SELUD4	Pull-up/pull-down selection (Port 4) 0: Pull-up 1: Pull-down
3 to 1	-	-
0	SELUD0	Pull-up/pull-down selection (Port 0) 0: Pull-up 1: Pull-down

8.4.3 Block Diagram (MN101EF76K/MN101EF56K)

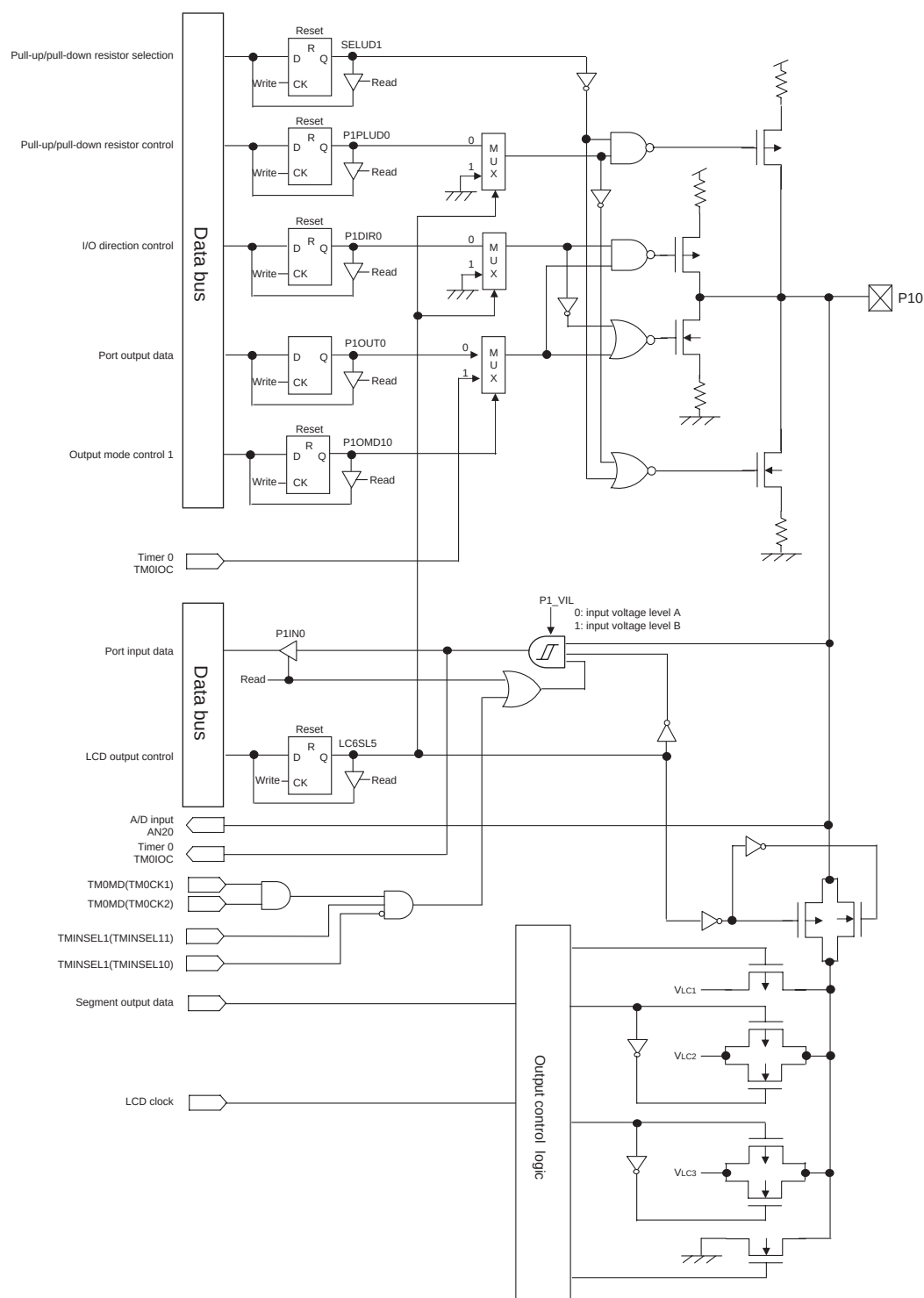


Figure:8.4.1 P10 Block Diagram (MN101EF76K/MN101EF56K)

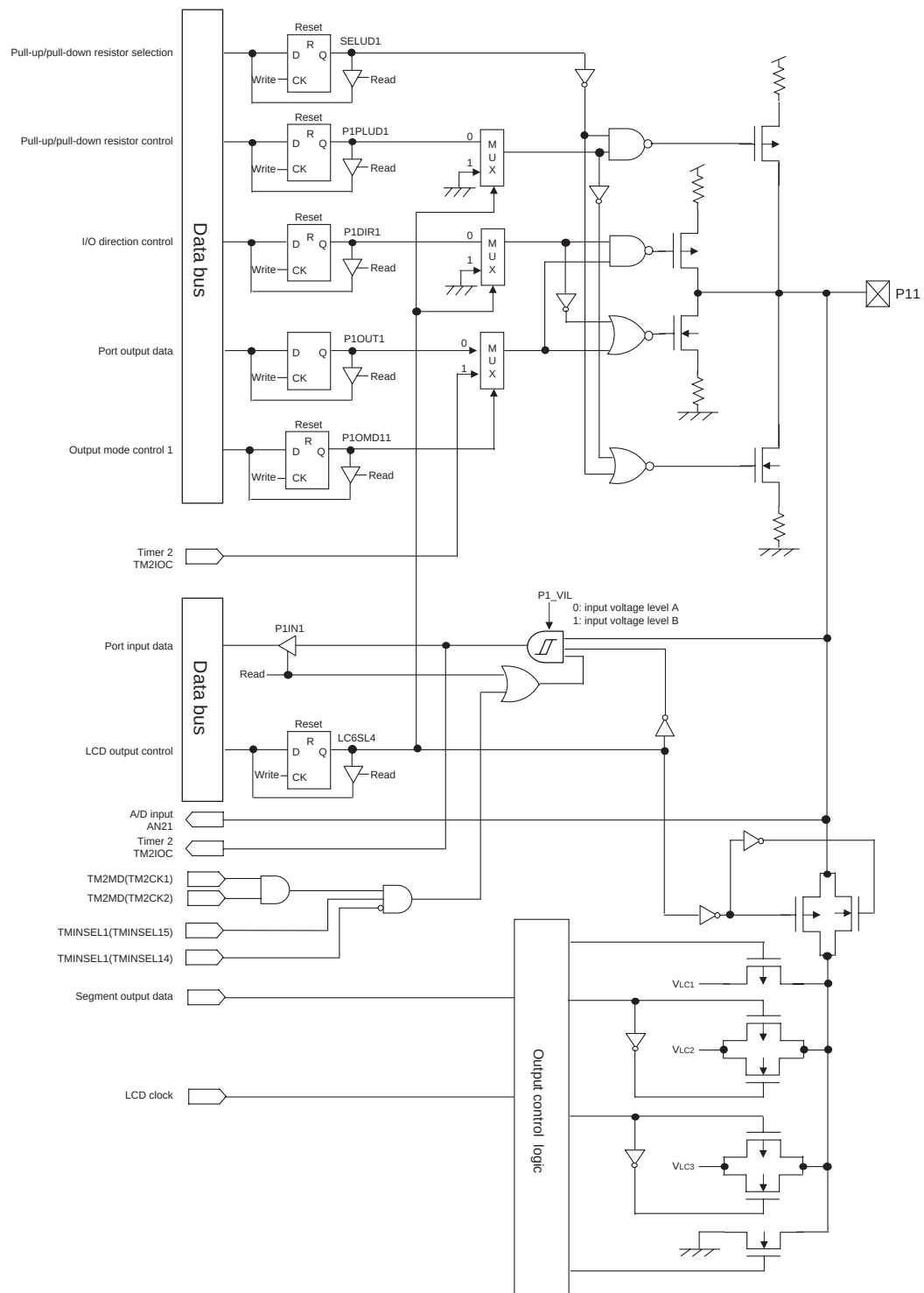


Figure:8.4.2 P11 Block Diagram (MN101EF76K/MN101EF56K)

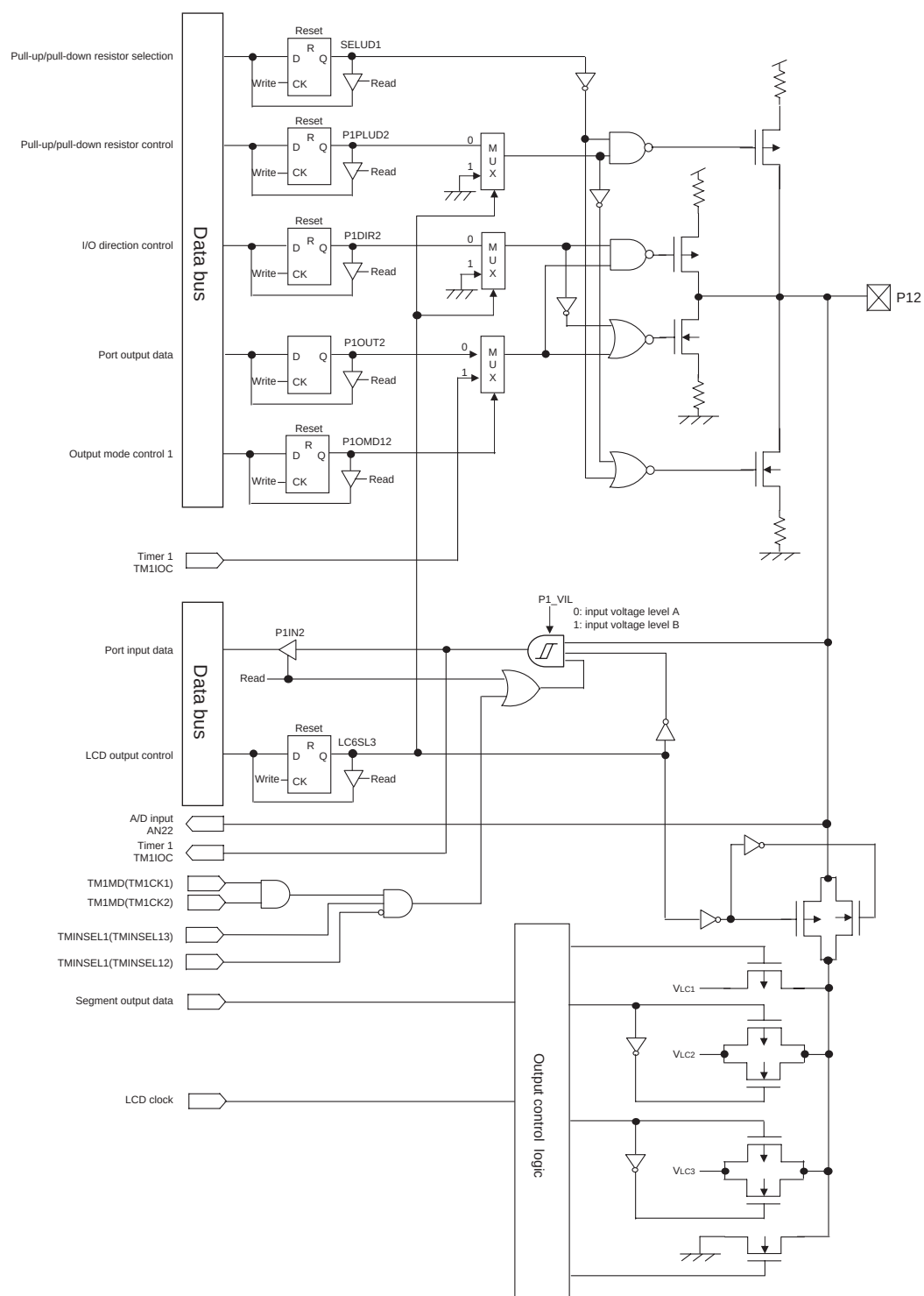


Figure:8.4.3 P12 Block Diagram (MN101EF76K/MN101EF56K)

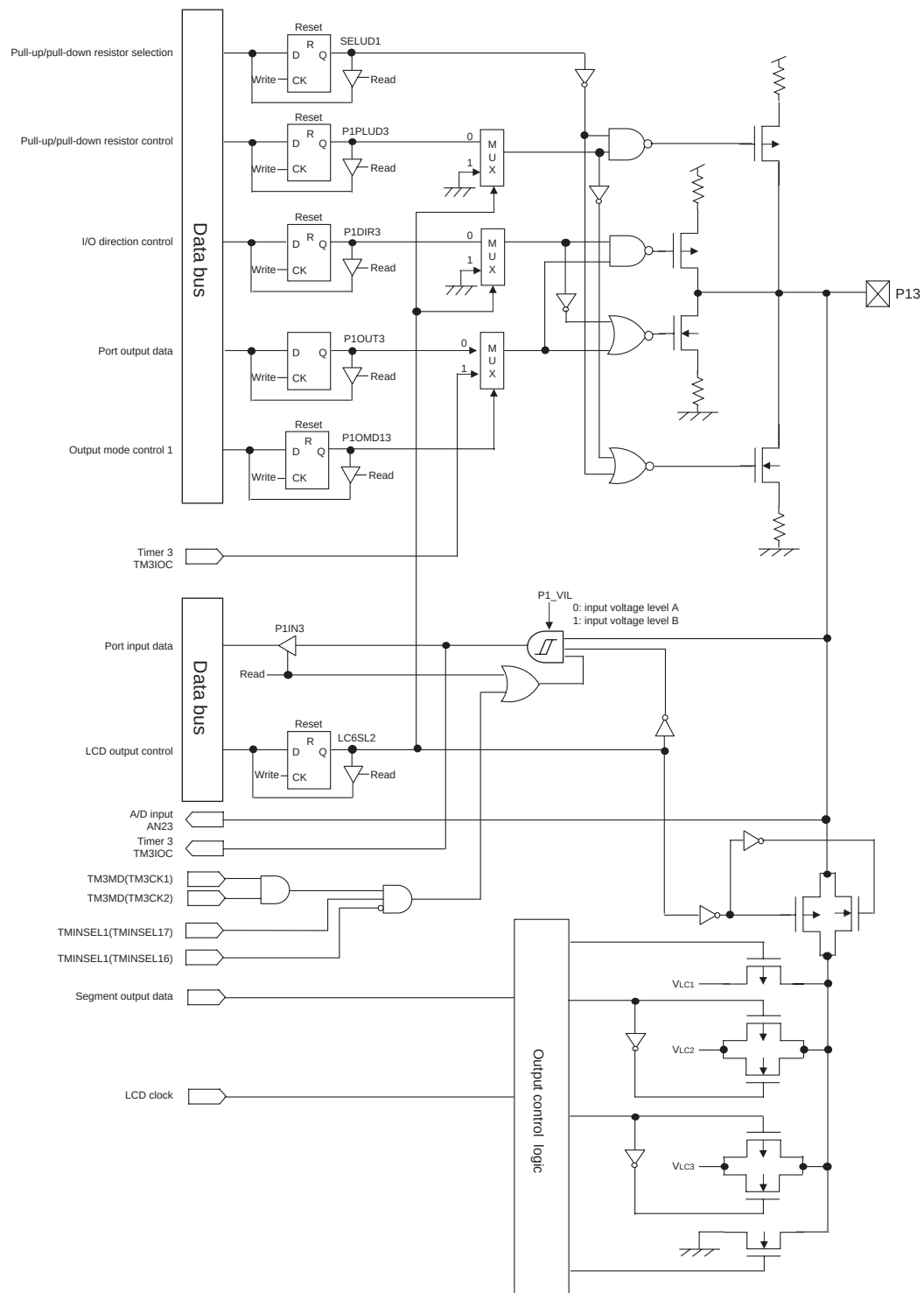


Figure:8.4.4 P13 Block Diagram (MN101EF76K/MN101EF56K)

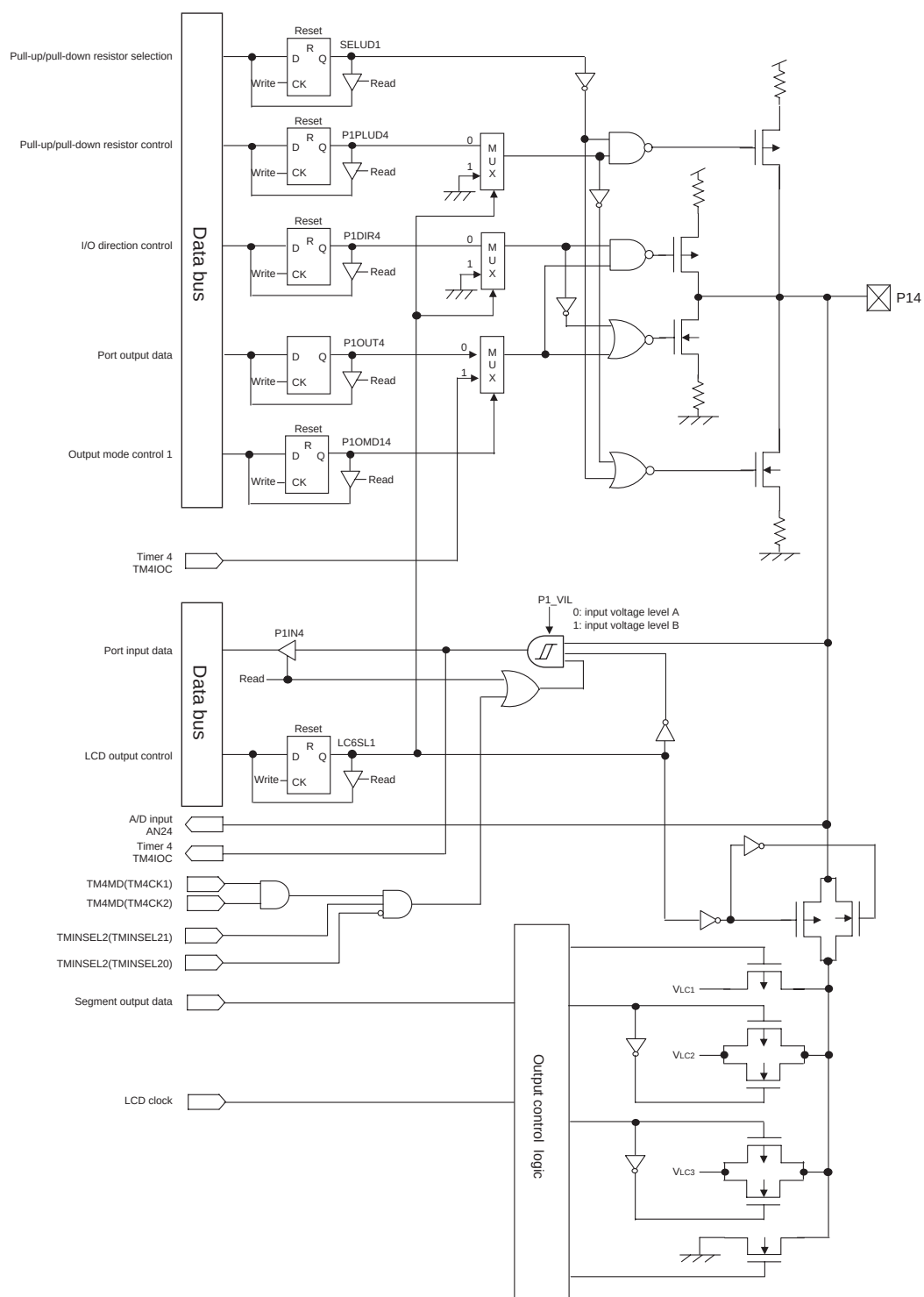


Figure:8.4.5 P14 Block Diagram (MN101EF76K/MN101EF56K)

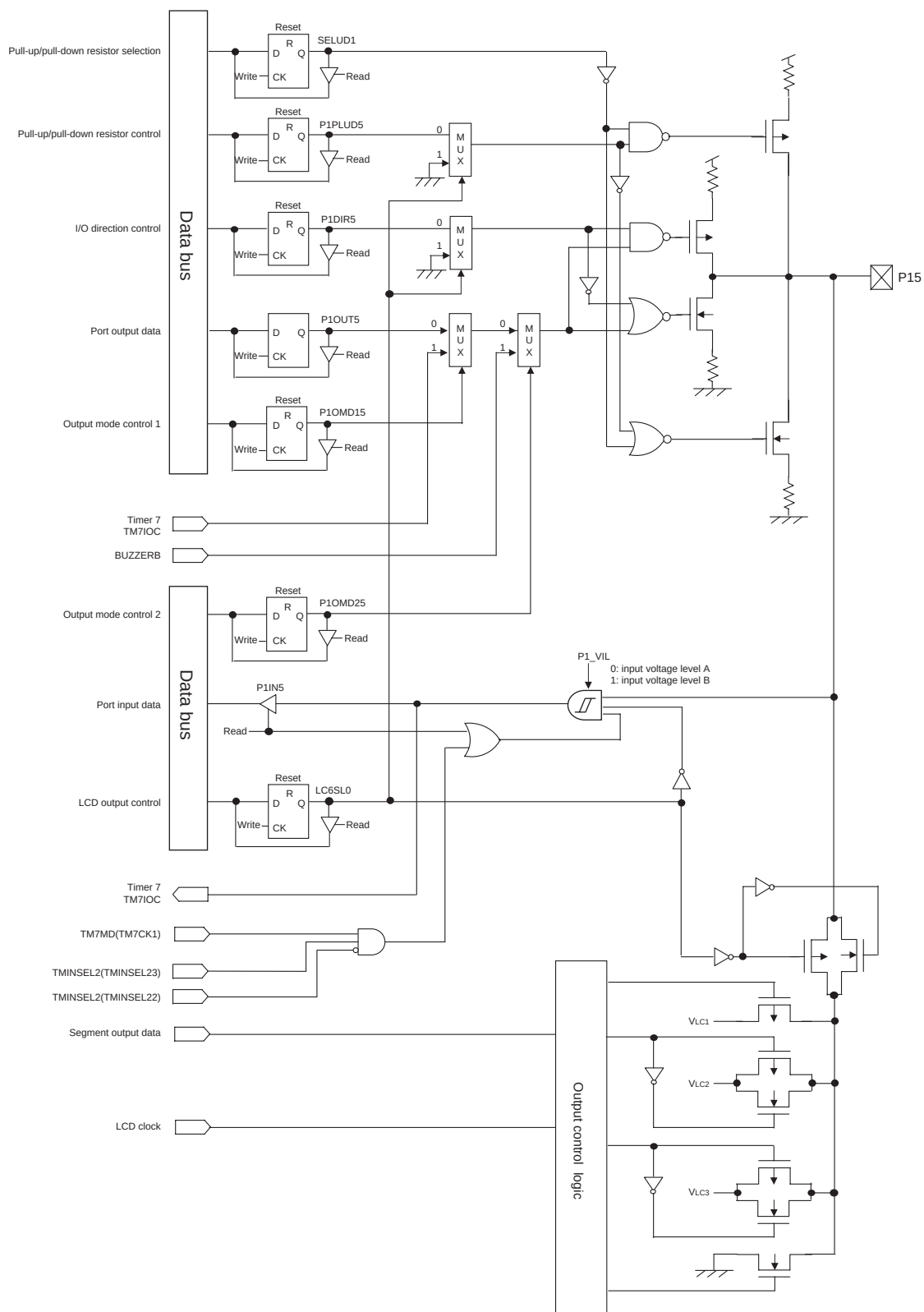


Figure:8.4.6 P15 Block Diagram (MN101EF76K/MN101EF56K)

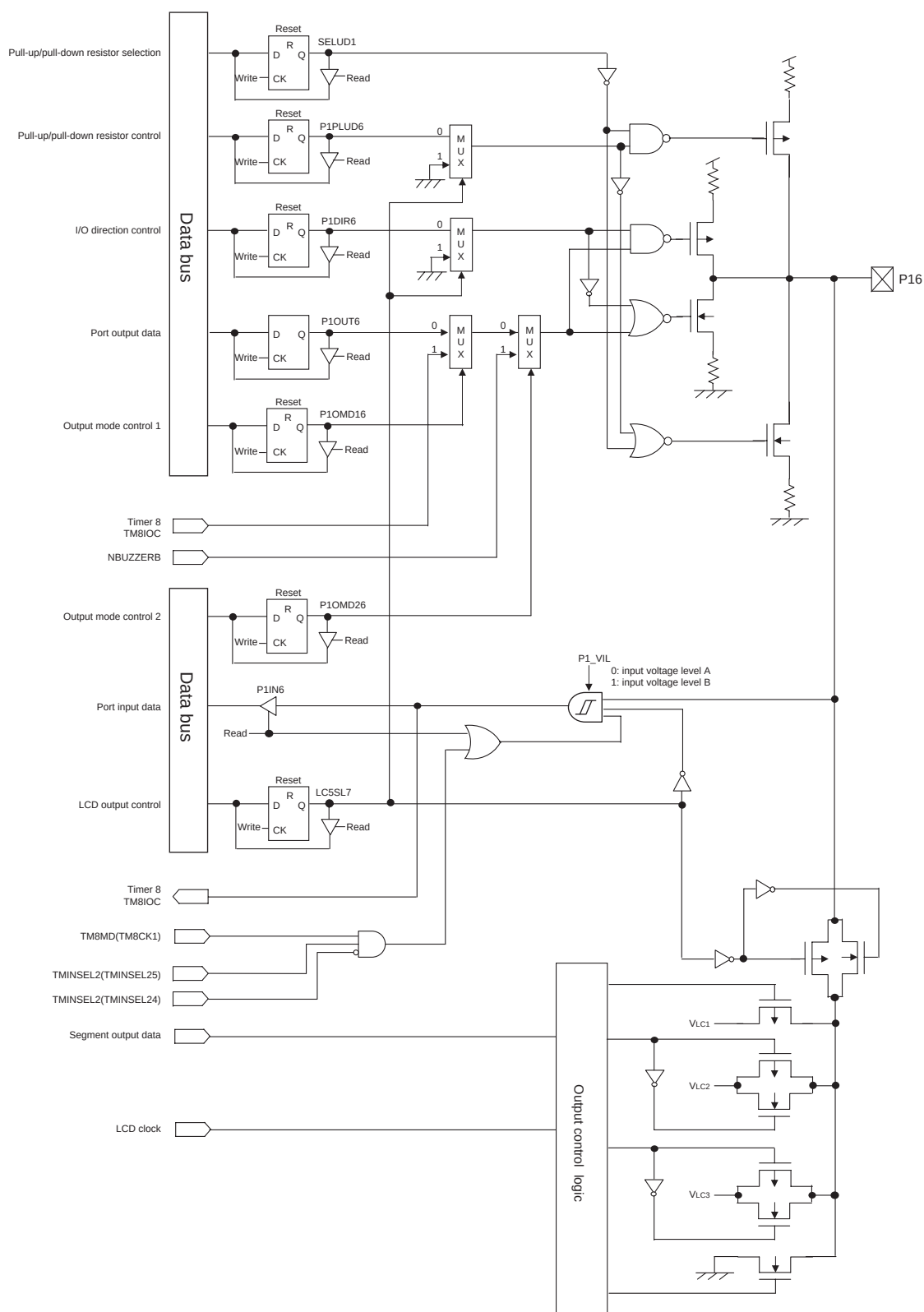


Figure:8.4.7 P16 Block Diagram (MN101EF76K/MN101EF56K)

8.5 Port 2

8.5.1 Description

■ General-purpose Port Setup

To output data to pins, set the control flag of Port 2 direction control register (P2DIR) to "1" and write the value of Port 2 output register (P2OUT).

To read input data of pins, set the control flag of Port 2 direction control register (P2DIR) to "0" and read the value of Port 2 input register (P2IN). Each bit can be set individually as either an input or output by Port 2 direction control register (P2DIR). The control flag of Port 2 direction control register (P2DIR) is set to "1" for output mode, and "0" for input mode.

Each bit can be set individually to add a pull-up resistor by Port 2 pull-up resistor control register (P2PLU). Set the control flag of P2PLU register to "1" to add a pull-up resistor.

P27 is the reset pin. To add soft reset, write "0" to the P2OUT7 flag of Port 2 output register (P2OUT). Pull-up resistor is always added to P27.

Table:8.5.1 Port 1 Functions

Table remarks -: Without function

MN101EF76K			MN101EF57G/MN101EF56K		
Pin	Special function		Pin	Special function	
P20	IRQ0A	ACZ0	P20	IRQ0	ACZ0
P21	IRQ1A	ACZ1	P21	IRQ1	ACZ1
P22	IRQ2A		P22	IRQ2	
P23	IRQ3A		P23	IRQ3	
P24	IRQ4A		P24	IRQ4	
P25	OSC1		P25	OSC1	
P26	OSC2		P26	OSC2	
P27	NRST		P27	NRST	

■ Special Function Pin Setup

P20 to P24 are also used as the external interrupt pin.

P20 and P21 are also used as the input pin of AC zero-cross. To use AC zero-cross detection, set both bp7 and bp3 of the AC zero-cross detector interrupt register (ACZCTR) to "1" and read the value of Port 2 input register (P2IN).

P25 is also used as the input pin of external high-speed oscillation.

P26 is also used as the output pin of external high-speed oscillation.

To use P25/P26 as external oscillation pin, set both bp0 and bp1 of the high-speed oscillation control register (OSCCNT) to "1" to select external high-speed oscillation as operating clock.

8.5.2 Registers

Port 2 shows registers that control Port

Table:8.5.2 Port 2 Control Register

Table remarks √: With function -: Without function

Registers	Address	R/W	Function	Page	MN101 EF76K	MN101 EF57G	MN101 EF56K
P2OUT	0x03E72	R/W	Port 2 output register	VIII-45	√	√	√
P2IN	0x03E82	R	Port 2 input register	VIII-45	√	√	√
P2DIR	0x03E92	R/W	Port 2 direction control register	VIII-46	√	√	√
P2PLU	0x03EA2	R/W	Port 2 pull-up resistor control register	VIII-46	√	√	√

R/W: Readable/Writable

R: Read only

■ Port 2 Output Register (P2OUT: 0x03E72)

bp	7	6	5	4	3	2	1	0
Flag	P2OUT7	P2OUT6	P2OUT5	P2OUT4	P2OUT3	P2OUT2	P2OUT1	P2OUT0
At reset	1	X	X	X	X	X	X	X
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description
7	P2OUT7	Output data (Reset output) 0: Output "Low" (V_{SS} -level) 1: Output "High" (V_{DD5} -level)
6 to 0	P2OUT6 to 0	Output data 0: Output "Low" (V_{SS} -level) 1: Output "High" (V_{DD5} -level)

■ Port 2 Input Register (P2IN: 0x03E82)

bp	7	6	5	4	3	2	1	0
Flag	P2IN7	P2IN6	P2IN5	P2IN4	P2IN3	P2IN2	P2IN1	P2IN0
At reset	X	X	X	X	X	X	X	X
Access	R	R	R	R	R	R	R	R

bp	Flag	Description
7 to 0	P2IN7 to 0	Input data 0: Pin is "Low" (V_{SS} -level) 1: Pin is "High" (V_{DD5} -level)

■ Port 2 Direction Control Register (P2DIR: 0x03E92)

bp	7	6	5	4	3	2	1	0
Flag	-	P2DIR6	P2DIR5	P2DIR4	P2DIR3	P2DIR2	P2DIR1	P2DIR0
At reset	-	0	0	0	0	0	0	0
Access	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description
7	-	-
6 to 0	P2DIR6 to 0	I/O mode selection 0: Input mode 1: Output mode

■ Port 2 Pull-up Resistor Control Register (P2PLU: 0x03EA2)

bp	7	6	5	4	3	2	1	0
Flag	-	P2PLU6	P2PLU5	P2PLU4	P2PLU3	P2PLU2	P2PLU1	P2PLU0
At reset	-	0	0	0	0	0	0	0
Access	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description
7	-	-
6 to 0	P2PLU6 to 0	Pull-up resistor selection 0: Not added 1: Added

8.5.3 Block Diagram (MN101EF76K)

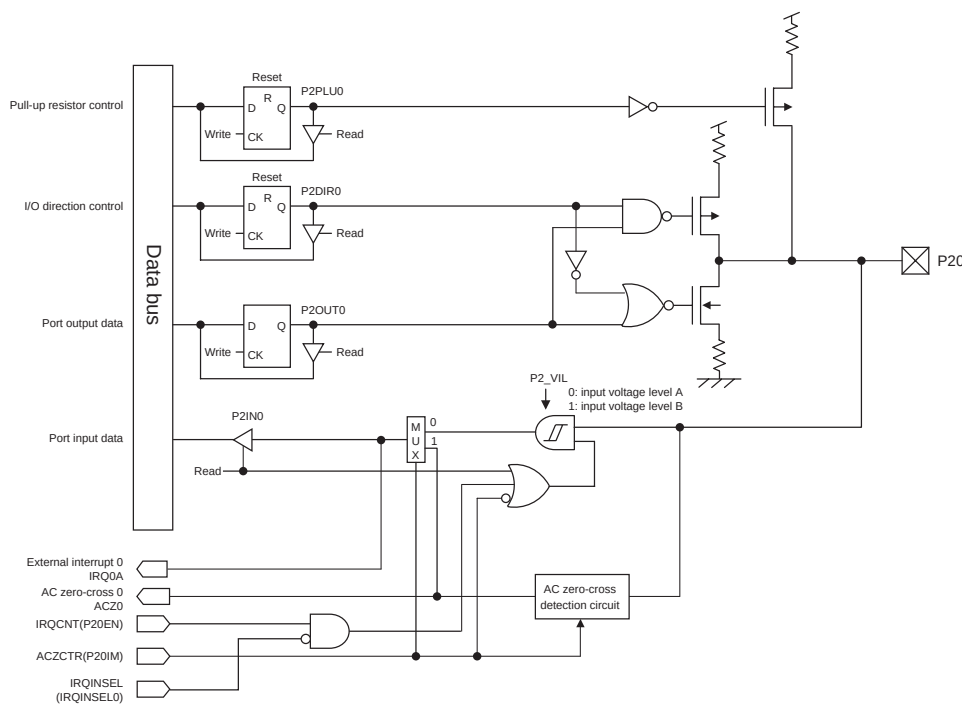


Figure:8.5.1 P20 Block Diagram (MN101EF76K)

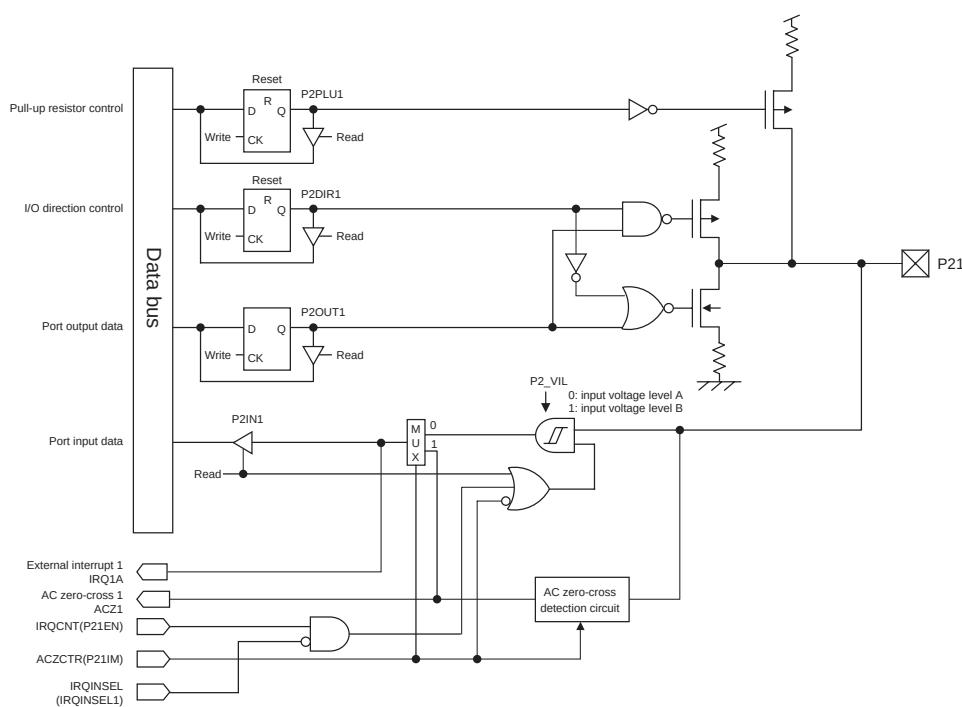


Figure:8.5.2 P21 Block Diagram (MN101EF76K)

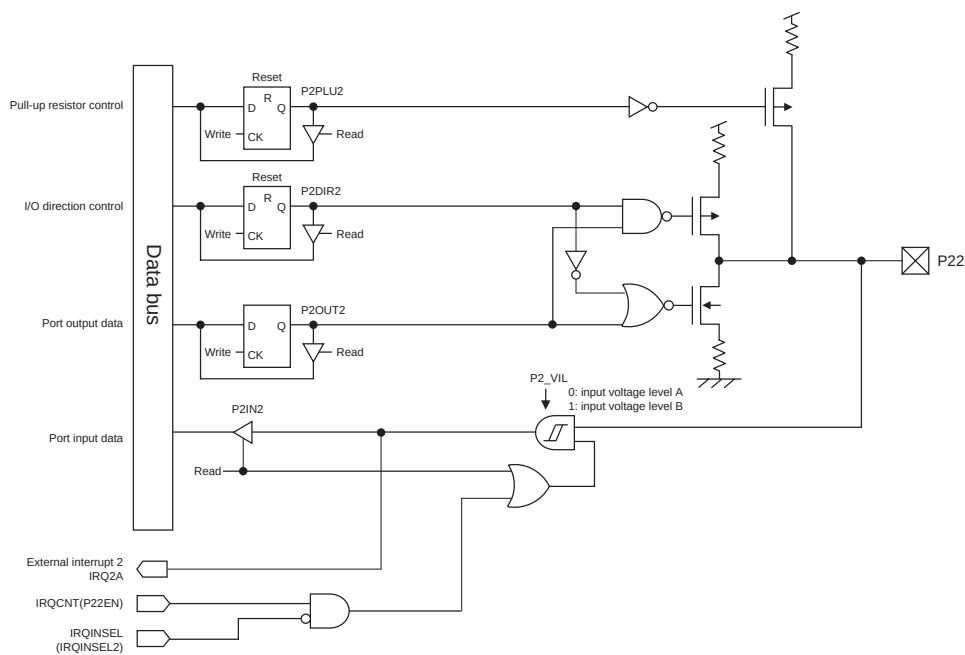


Figure:8.5.3 P22 Block Diagram (MN101EF76K)

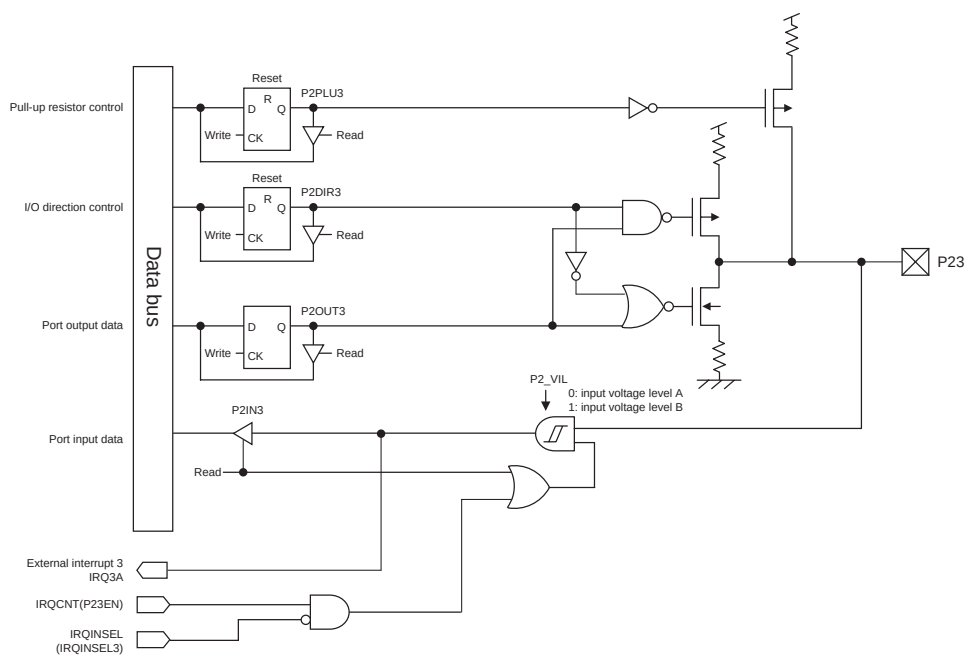


Figure:8.5.4 P23 Block Diagram (MN101EF76K)

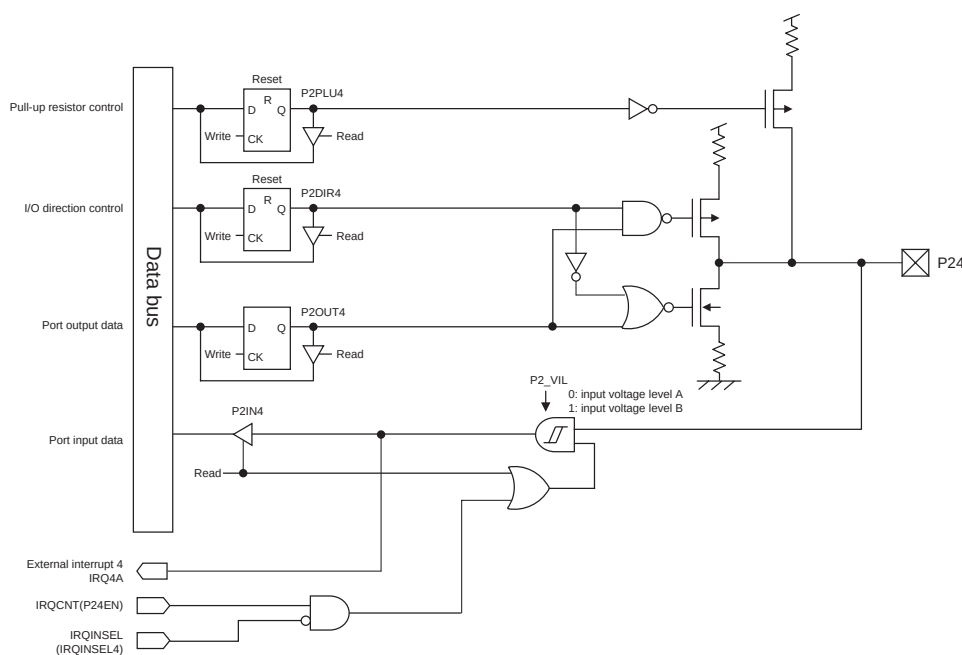


Figure:8.5.5 P24 Block Diagram (MN101EF76K)

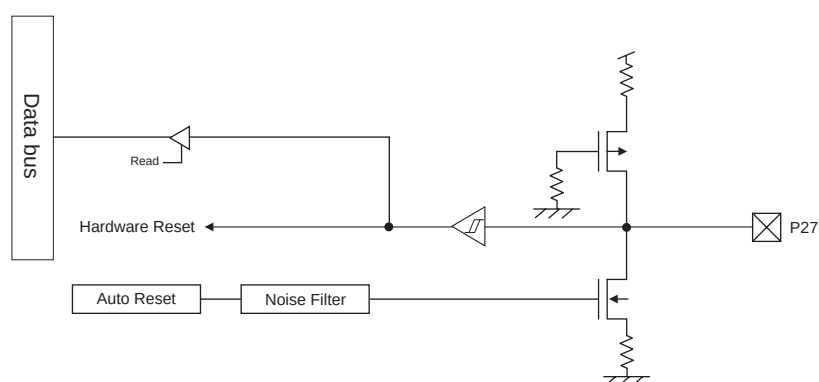


Figure:8.5.6 P27 Block Diagram (MN101EF76K)

8.5.4 Block Diagram (MN101EF57G/MN101EF56K)

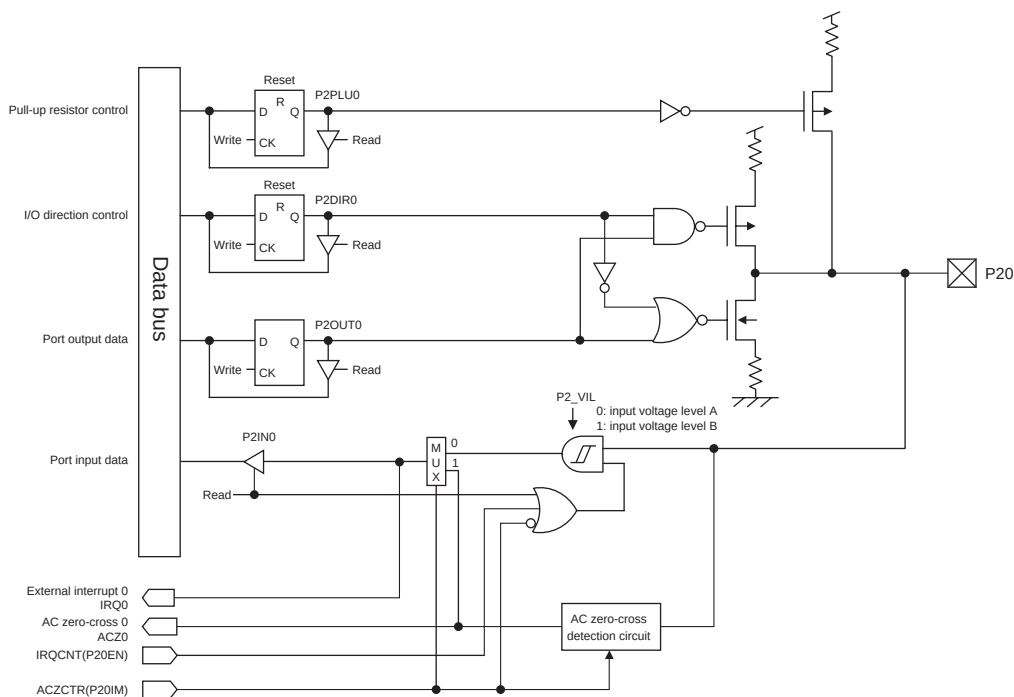


Figure:8.5.7 P20 Block Diagram (MN101EF57G/MN101EF56K)

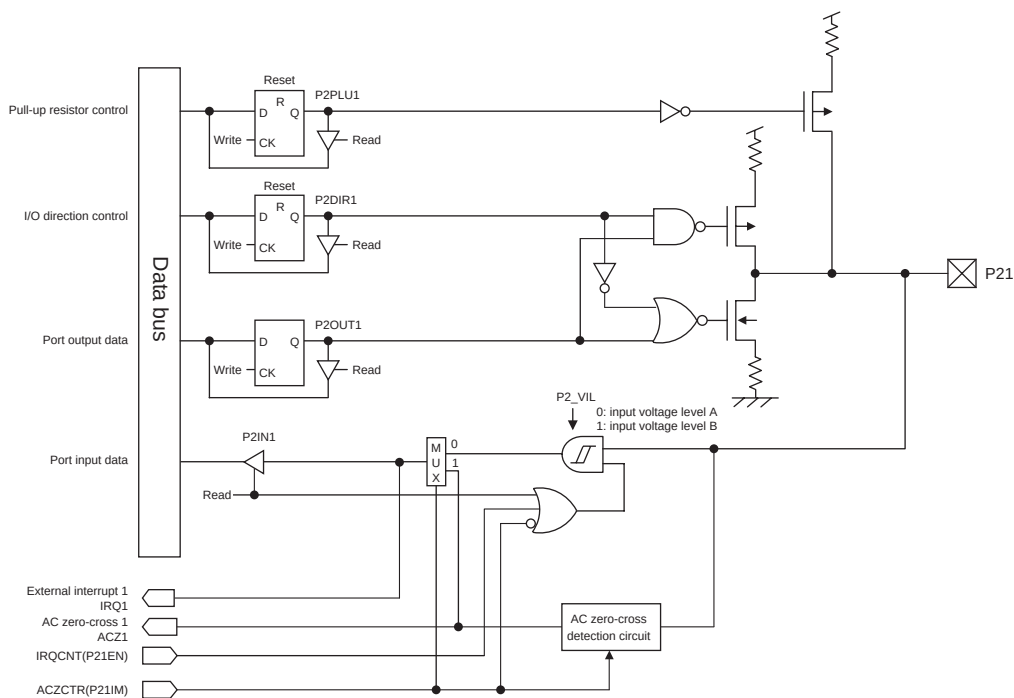


Figure:8.5.8 P21 Block Diagram (MN101EF57G/MN101EF56K)

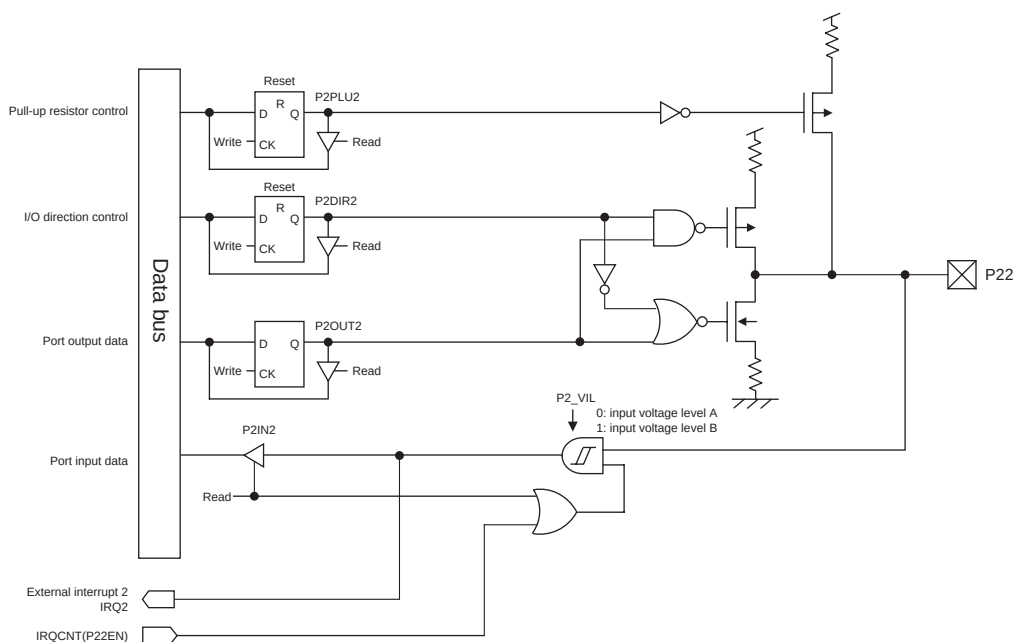


Figure:8.5.9 P22 Block Diagram (MN101EF57G/MN101EF56K)

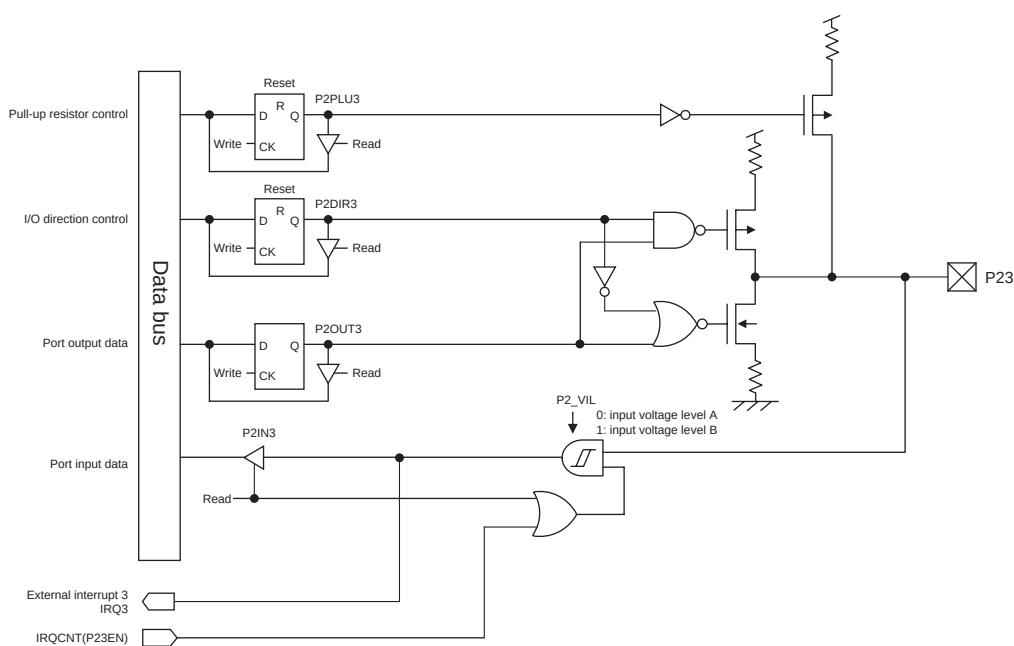


Figure:8.5.10 P23 Block Diagram (MN101EF57G/MN101EF56K)

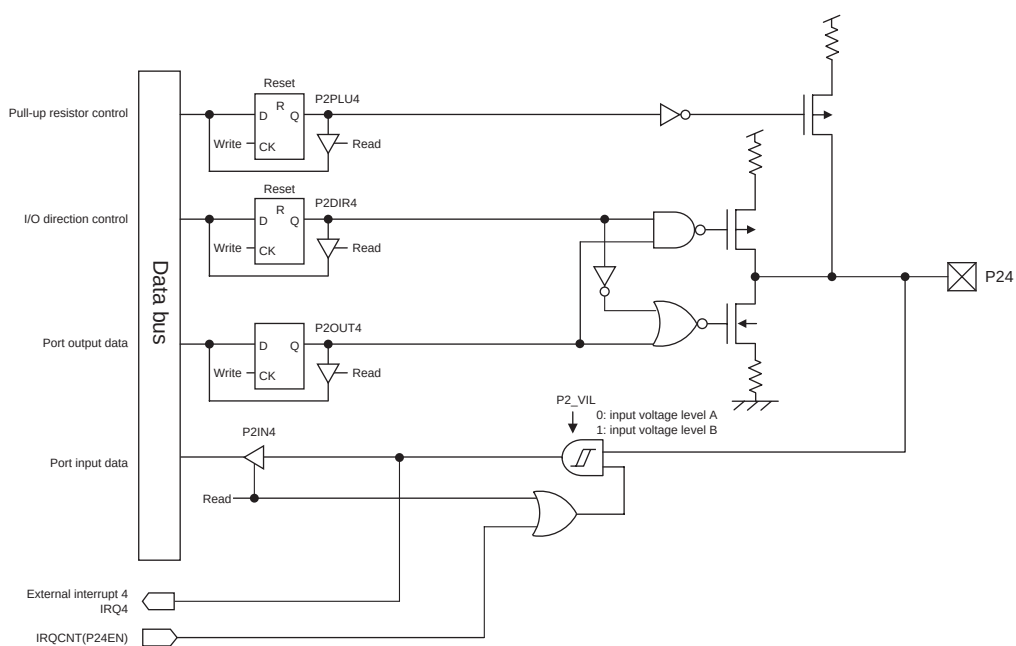


Figure:8.5.11 P24 Block Diagram (MN101EF57G/MN101EF56K)

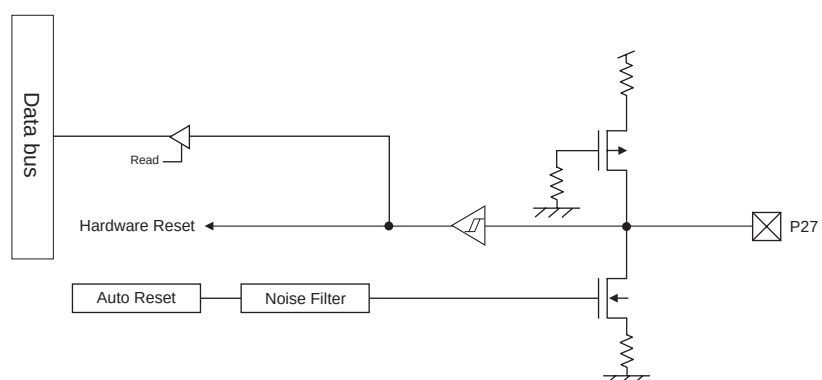


Figure:8.5.12 P27 Block Diagram (MN101EF57G/MN101EF56K)

8.6 Port 3

8.6.1 Description

■ General-purpose Port Setup

To output data to pins, set the control flag of Port 3 direction control register (P3DIR) to "1" and write the value of Port 3 output register (P3OUT).

To read input data of pins, set the control flag of Port 3 direction control register (P3DIR) to "0" and read the value of Port 3 input register (P3IN).

Each bit can be set individually as either an input or output by Port 3 direction control register (P3DIR). The control flag of P3DIR register is set to "1" for output mode, and "0" for input mode.

Each bit can be set individually to add a pull-up resistor by Port 3 pull-up resistor control register (P3PLU). Set the control flag of P3PLU register to "1" to add a pull-up resistor.

Each bit can be selected individually as Nch open-drain output by Port 3 Nch open-drain control register (P3ODC). The control flag of P3ODC register is set to "1" for Nch open-drain output, and "0" for push-pull output.

Each bit can be set individually to control input mode by Port 3 input mode control register (P3IMD). The control flag of P3IMD register is set to "1" for using as the input port of special function data, and to "0" for using as general purpose port. When using as the special function input pin, undefined value is read from P3IN register.

Table:8.6.1 Port 3 Functions

Table remarks -: Without function

MN101EF76K/MN101EF56K					MN101EF57G				
Pin	Special function				Pin	Special function			
P30	SEG38		SBO2B	TXD2B	-	-		-	-
P31	SEG37		SBI2B	RXD2B	-	-		-	-
P32	SEG36		SBT2B		-	-		-	
P33	SEG35	-	SBO4B	SDA4B	P33	SEG30	AN15	SBO4B	SDA4B
P34	SEG34		SBT4B	SCL4B	P34	SEG29		SBT4B	SCL4B
P35	SEG33		SBI4B		P35	SEG28		SBI4B	
P36	SEG32				-	-			

■ Special Function Pin Setup

P30 to P36 are also used as LCD segment output pins.

LCD segment output pin is selected by setting the corresponding flag of LCD output control register n (LCCTRn) to "1".

Port and segment can be switched by one bit. When segment output pin is selected, this pin is forced to be output mode and pull-up resistor is not added.

P33 is also used as analog input pin. Port 3 input mode register (P3IMD) can select the input mode by one bit. When using as the analog input pin, undefined value is read from P3IN register.

P30 is also used as the I/O pin for Serial Interface 2 transmission/reception data and the output pin of UART2 transmission data. When the SC2SBOS flag of Serial Interface 2 mode register 1 (SC2MD1) is "1", P30 is the serial data I/O pin. Push-pull output or Nch open-drain output can be selected by setting Port 3 Nch open-drain control register (P3ODC).

P31 is also used as the input pin for Serial Interface 2 reception data and UART2 reception data. When the SC2BIS flag of Serial Interface 2 mode register 1 (SC2MD1) is "1", P31 is the serial data input pin.

P32 is also used as the I/O pin for Serial Interface 2 clock. When the SC2SBTS flag of Serial Interface 2 mode register 1 (SC2MD1) is "1", P32 is the I/O pin of the serial clock. Push-pull output or Nch open-drain output can be selected by setting Port 3 Nch open-drain control register (P3ODC).

P33 is also used as the I/O pin for Serial Interface 4 transmission/reception data and the I/O pin of IIC4 transmission/reception data. When the SC4SBOS flag of Serial Interface 4 mode register 1 (SC4MD1) is "1", P33 is the serial data I/O pin. Push-pull output or Nch open-drain output can be selected by setting Port 3 Nch open-drain control register (P3ODC).

P34 is also used as the I/O pin for Serial Interface 4 clock and the output pin of IIC4 clock. When the SC4SBTS flag of Serial Interface 4 mode register 1 (SC4MD1) is "1", P34 is the I/O pin of the serial clock. Push-pull output or Nch open-drain output can be selected by setting Port 3 Nch open-drain control register (P3ODC).

P34 is also used as the I/O pin of serial 4 clock and the output pin of IIC4 clock. When the SC4SBTS flag of the serial interface 4 mode register 1 (SC4MD1) is "1", P34 is the I/O pin of the serial clock. Push-pull output or Nch open-drain output can be selected by setting the Port 3 Nch open-drain control register (P3ODC).

P35 is also used as the input pin of Serial Interface 4 reception data. When the SC4SBIS flag of the Serial Interface 4 mode register 1 (SC4MD1) is "1", P35 is the serial data input pin.

8.6.2 Registers

Table:8.3.2 shows registers that control Port 3.

Table:8.6.2 Port 3 Control Register

Table remarks √: With function -: Without function

Registers	Address	R/W	Function	Page	MN101 EF76K	MN101 EF57G	MN101 EF556K
P3OUT	0x03E73	R/W	Port 3 output register	VIII-55	√	√	√
P3IN	0x03E83	R	Port 3 input register	VIII-56	√	√	√
P3DIR	0x03E93	R/W	Port 3 direction control register	VIII-57	√	√	√
P3PLU	0x03EA3	R/W	Port 3 pull-up resistor control register	VIII-58	√	√	√
P3IMD	0x03EC3	R/W	Port 3 input mode register	VIII-58	-	√	-
P3ODC	0x03EF3	R/W	Port 3 Nch open-drain control register	VIII-59	√	√	√

R/W: Readable/Writable

R: Read only

■ Port 3 Output Register (P3OUT: 0x03E73)

<MN101EF76K/MN101EF56K>

bp	7	6	5	4	3	2	1	0
Flag	-	P3OUT6	P3OUT5	P3OUT4	P3OUT3	P3OUT2	P3OUT1	P3OUT0
At reset	-	X	X	X	X	X	X	X
Access	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description
7	-	-
6 to 0	P3OUT6 to 0	Output data 0: Output "Low" (V_{SS} -level) 1: Output "High" (V_{DD5} -level)

<MN101EF57G>

bp	7	6	5	4	3	2	1	0
Flag	-	-	P3OUT5	P3OUT4	P3OUT3	-	-	-
At reset	-	-	X	X	X	-	-	-
Access	-	-	R/W	R/W	R/W	-	-	-

bp	Flag	Description
7 to 6	-	-
5 to 3	P3OUT5 to 3	Output data 0: Output "Low" (V_{SS} -level) 1: Output "High" (V_{DD5} -level)
2 to 0	-	-

■ Port 3 Input Register (P3IN: 0x03E83)

<MN101EF76K/MN101EF56K>

bp	7	6	5	4	3	2	1	0
Flag	-	P3IN6	P3IN5	P3IN4	P3IN3	P3IN2	P3IN1	P3IN0
At reset	-	X	X	X	X	X	X	X
Access	-	R	R	R	R	R	R	R

bp	Flag	Description
7	-	-
6 to 0	P3IN6 to 0	Input data 0: Pin is "Low" (V_{SS} -level) 1: Pin is "High" (V_{DD5} -level)

<MN101EF57G>

bp	7	6	5	4	3	2	1	0
Flag	-	-	P3IN5	P3IN4	P3IN3	-	-	-
At reset	-	-	X	X	X	-	-	-
Access	-	-	R	R	R	-	-	-

bp	Flag	Description
7 to 6	-	-
5 to 3	P3IN5 to 3	Input data 0: Pin is "Low" (V_{SS} -level) 1: Pin is "High" (V_{DD5} -level)
2 to 0	-	-

■ Port 3 Direction Control Register (P3DIR: 0x03E93)

<MN101EF76K/MN101EF56K>

bp	7	6	5	4	3	2	1	0
Flag	-	P3DIR6	P3DIR5	P3DIR4	P3DIR3	P3DIR2	P3DIR1	P3DIR0
At reset	-	0	0	0	0	0	0	0
Access	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description
7	-	-
6 to 0	P3DIR6 to 0	I/O mode selection 0: Input mode 1: Output mode

<MN101EF57G>

bp	7	6	5	4	3	2	1	0
Flag	-	-	P3DIR5	P3DIR4	P3DIR3	-	-	-
At reset	-	-	0	0	0	-	-	-
Access	-	-	R/W	R/W	R/W	-	-	-

bp	Flag	Description
7 to 6	-	-
5 to 3	P3DIR5 to 3	I/O mode selection 0: Input mode 1: Output mode
2 to 0	-	-

■ Port 3 Pull-up Resistor Control Register (P3PLU: 0x03EA3)

<MN101EF76K/MN101EF56K>

bp	7	6	5	4	3	2	1	0
Flag	-	P3PLU6	P3PLU5	P3PLU4	P3PLU3	P3PLU2	P3PLU1	P3PLU0
At reset	-	0	0	0	0	0	0	0
Access	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description
7	-	-
6 to 0	P3PLU5 to 3	Pull-up resistor selection 0: Not added 1: Added

<MN101EF57G>

bp	7	6	5	4	3	2	1	0
Flag	-	-	P3PLU5	P3PLU4	P3PLU3	-	-	-
At reset	-	-	0	0	0	-	-	-
Access	-	-	R/W	R/W	R/W	-	-	-

bp	Flag	Description
7 to 6	-	-
5 to 3	P3PLU5 to 3	Pull-up resistor selection 0: Not added 1: Added
2 to 0	-	-

■ Port 3 Input Mode Register (P3IMD: 0x03EC3)

<MN101EF57G>

bp	7	6	5	4	3	2	1	0
Flag	-	-	-	-	P3IMD3	-	-	-
At reset	-	-	-	-	0	-	-	-
Access	-	-	-	-	R/W	-	-	-

bp	Flag	Description
7 to 4	-	-
3	P3IMD3	I/O port or analog input selection 0: P33 1: AN15
2 to 0	-	-

■ Port 3 Nch Open-drain Control Register P3ODC: 0x03EF3)

<MN101EF76K/MN101EF56K>

bp	7	6	5	4	3	2	1	0
Flag	-	-	-	P3ODC4	P3ODC3	P3ODC2	-	P3ODC0
At reset	-	-	-	0	0	0	-	0
Access	-	-	-	R/W	R/W	R/W	-	R/W

bp	Flag	Description
7 to 5	-	-
4 to 2	P3ODC4 to 2	Nch open-drain output selection 0: Push-pull output 1: Nch open-drain output
1	-	-
0	P3ODC0	Nch open-drain output selection 0: Push-pull output 1: Nch open-drain output

<MN101EF57G>

bp	7	6	5	4	3	2	1	0
Flag	-	-	-	P3ODC4	P3ODC3	-	-	-
At reset	-	-	-	0	0	-	-	-
Access	-	-	-	R/W	R/W	-	-	-

bp	Flag	Description
7 to 5	-	-
4 to 3	P3ODC4 to 3	Nch open-drain output selection 0: Push-pull output 1: Nch open-drain output
2 to 0	-	-

8.6.3 Block Diagram (MN101EF76K/MN101EF56K)

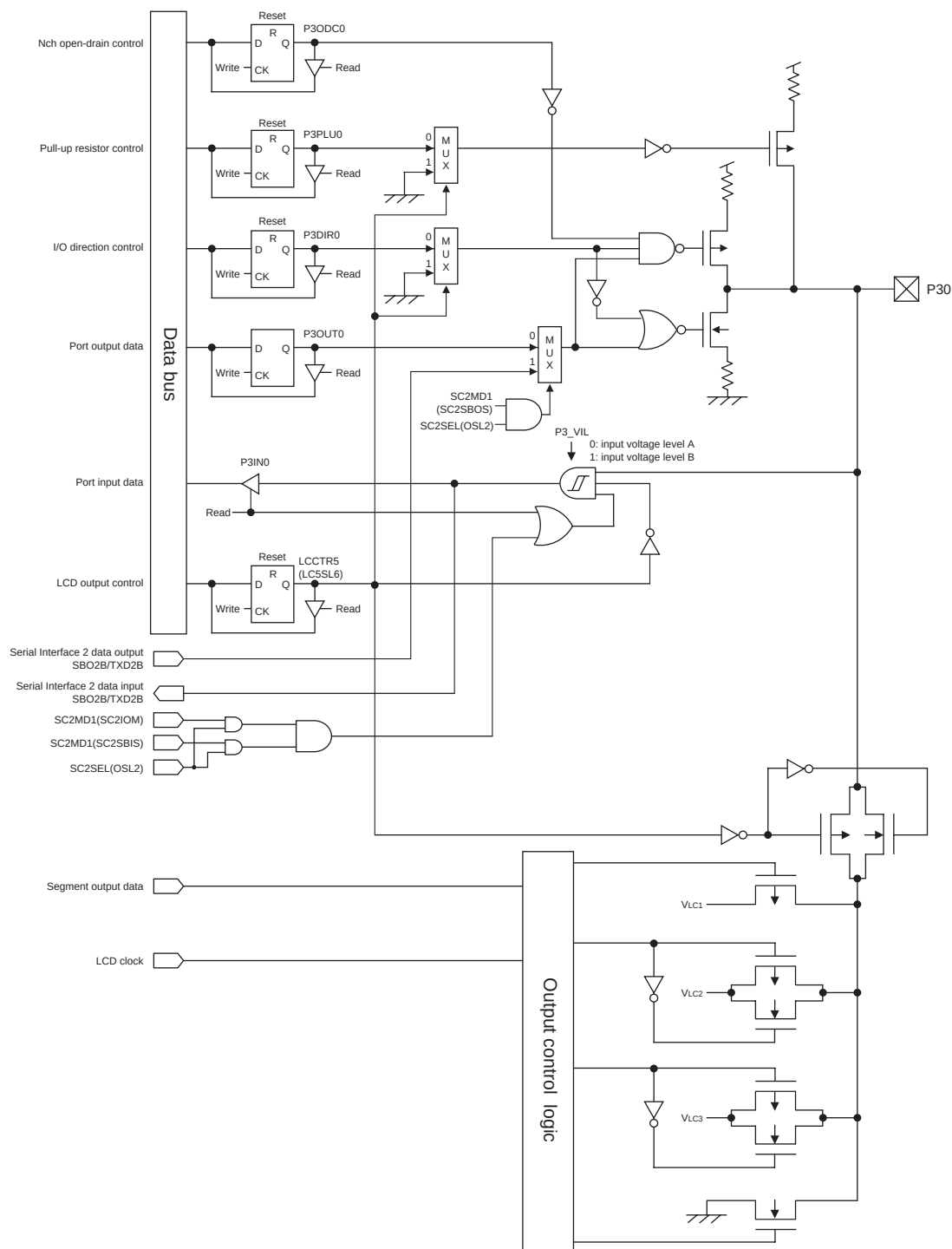


Figure:8.6.1 P30 Block Diagram (MN101EF76K/MN101EF56K)

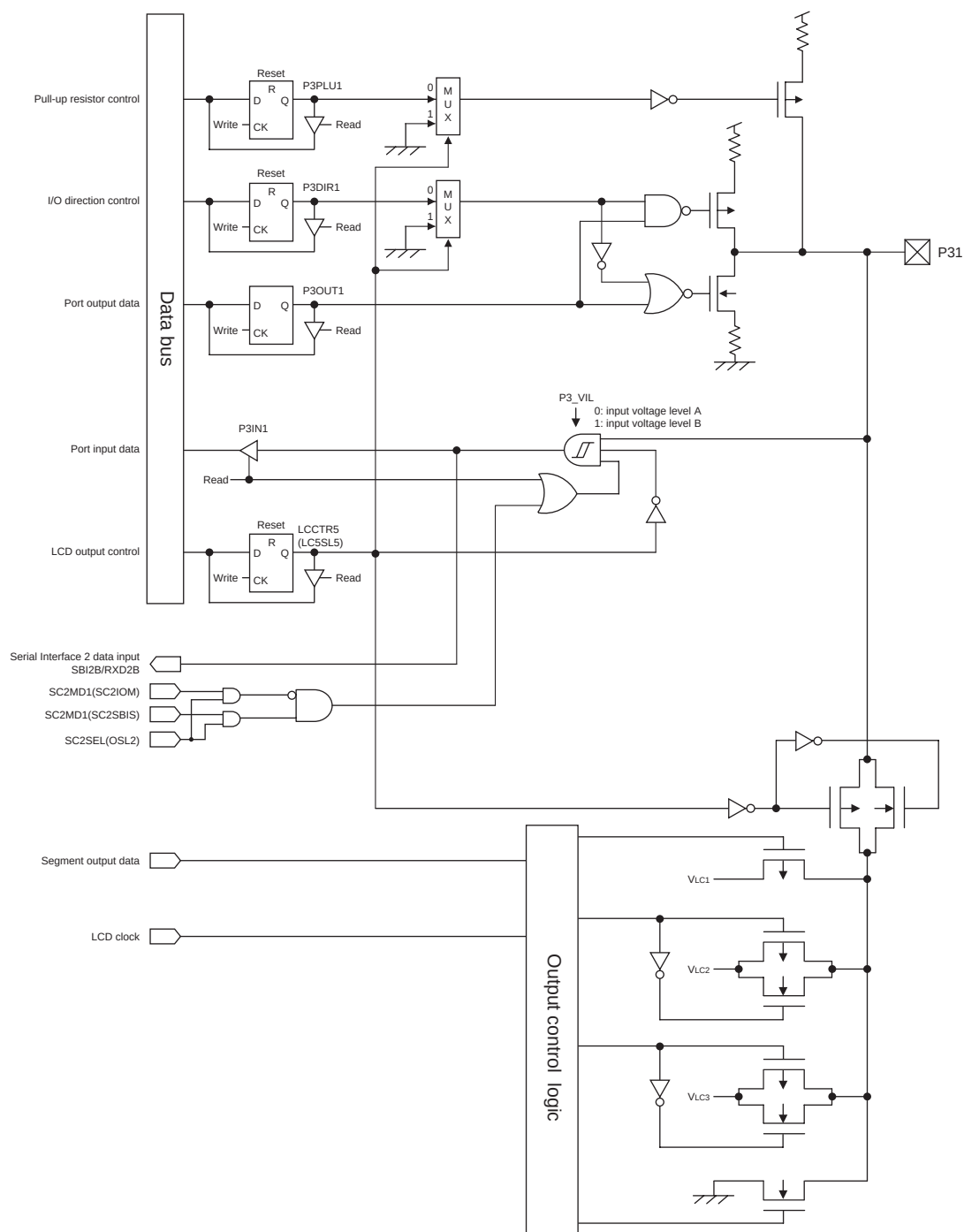


Figure:8.6.2 P31 Block Diagram (MN101EF76K/MN101EF56K)

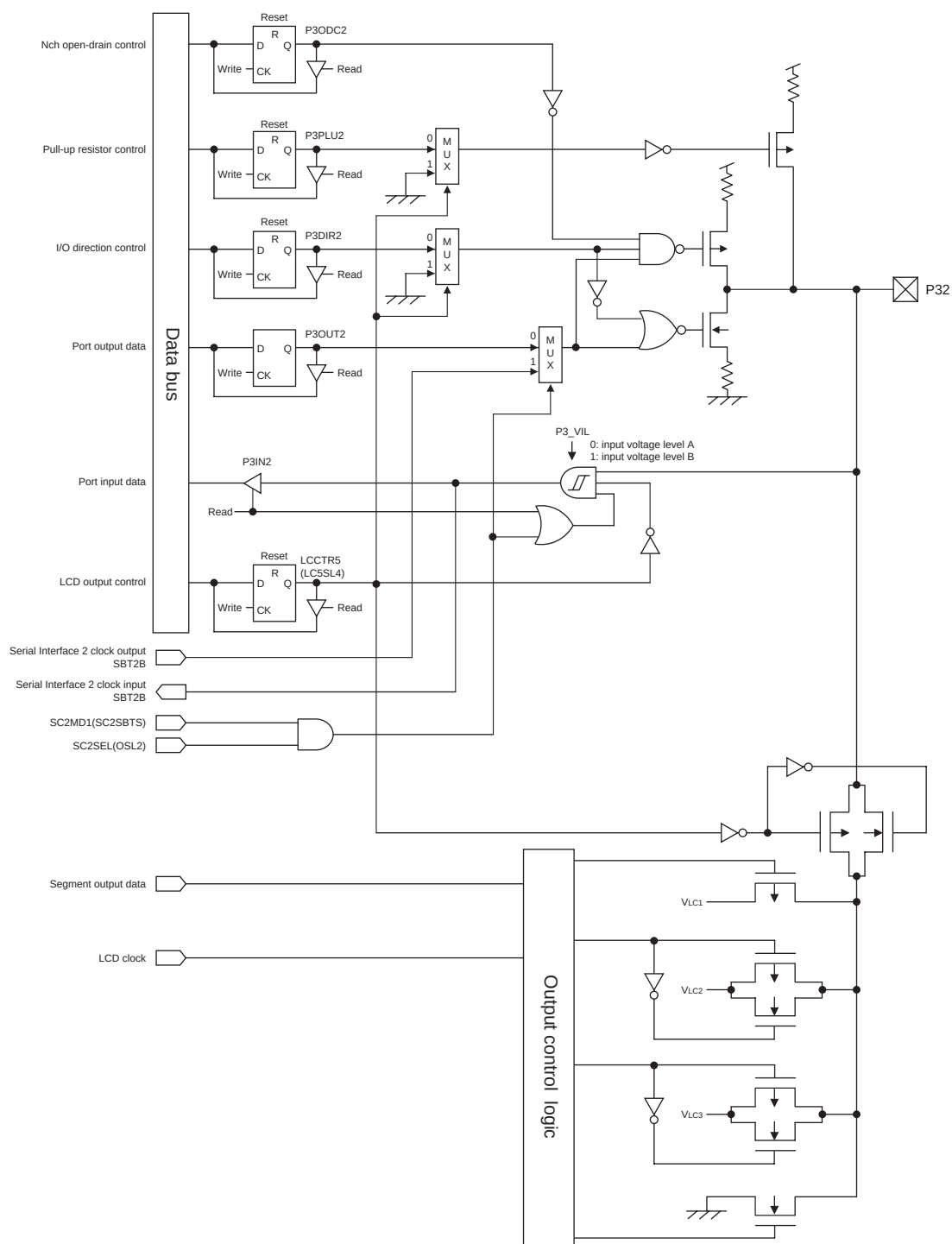


Figure:8.6.3 P32 Block Diagram (MN101EF76K/MN101EF56K)

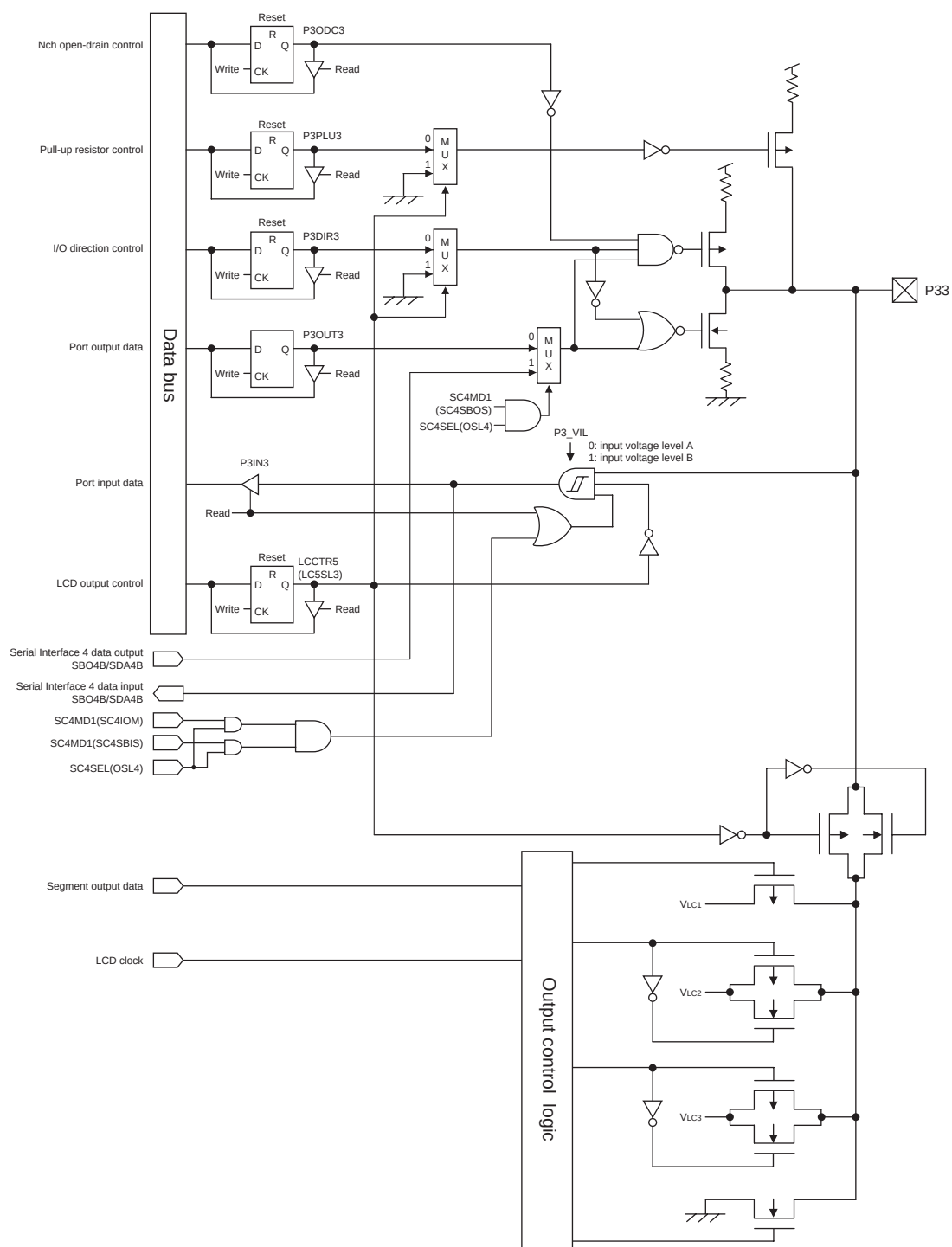


Figure:8.6.4 P33 Block Diagram (MN101EF76K/MN101EF56K)

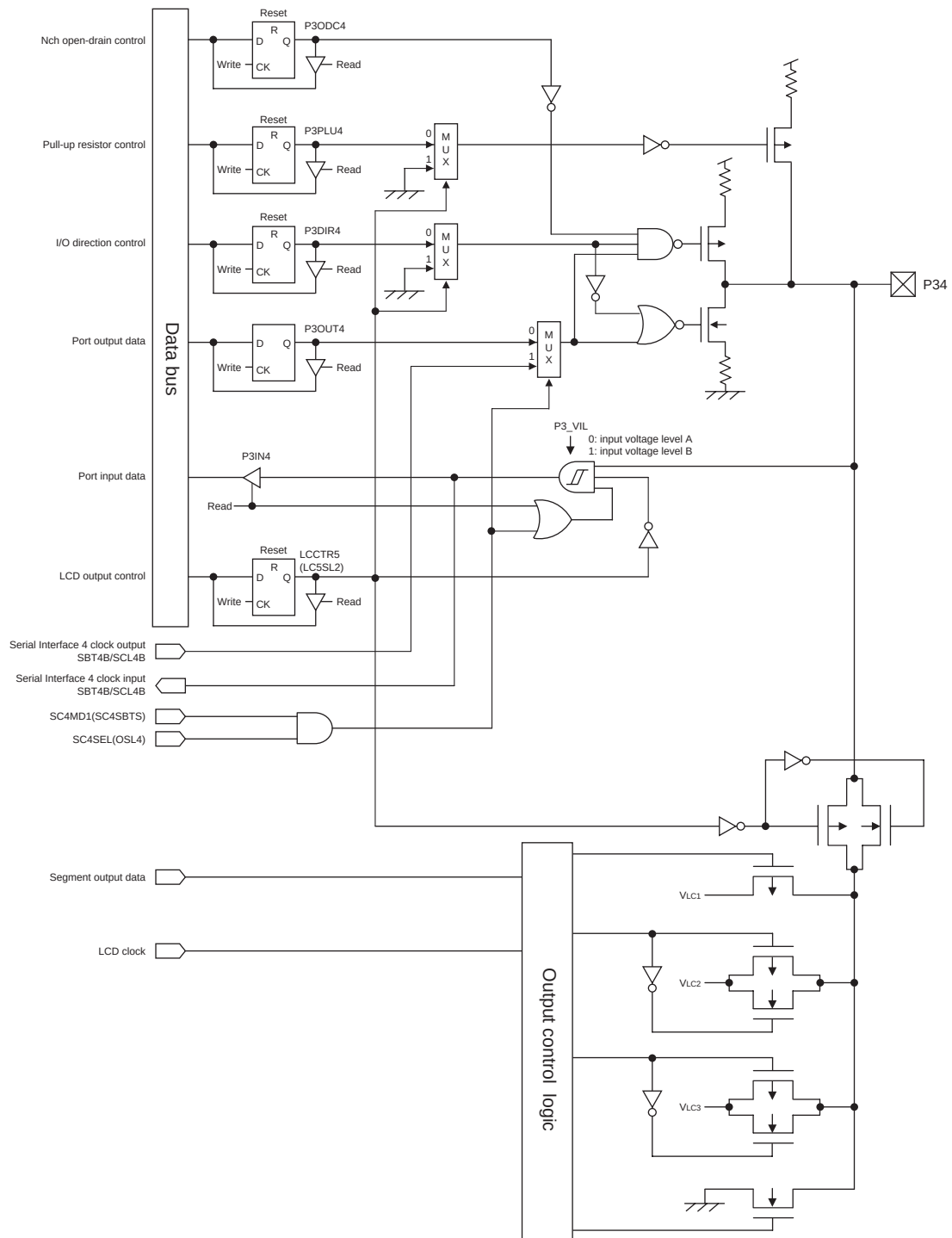


Figure:8.6.5 P34 Block Diagram (MN101EF76K/MN101EF56K)

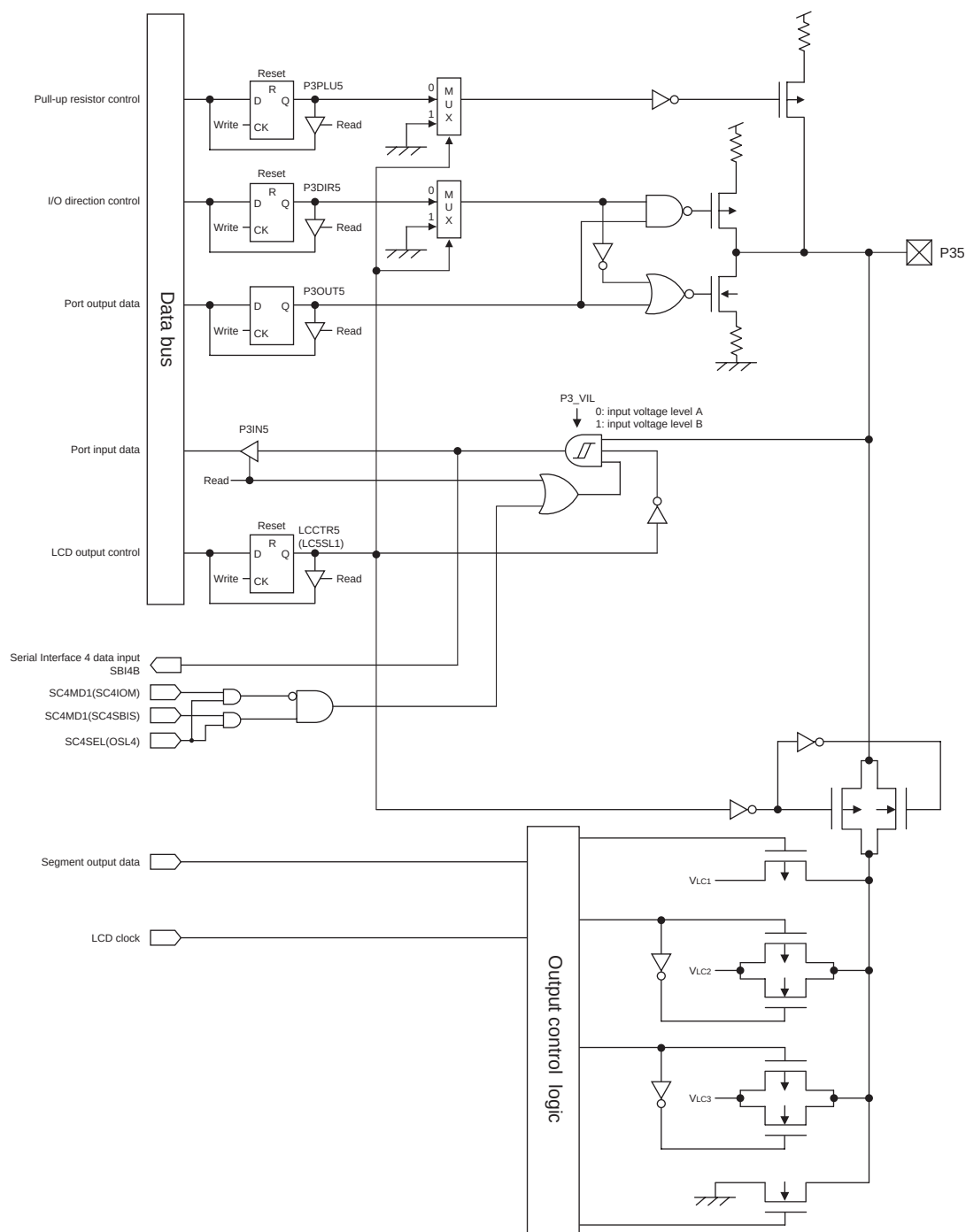


Figure:8.6.6 P35 Block Diagram (MN101EF76K/MN101EF56K)

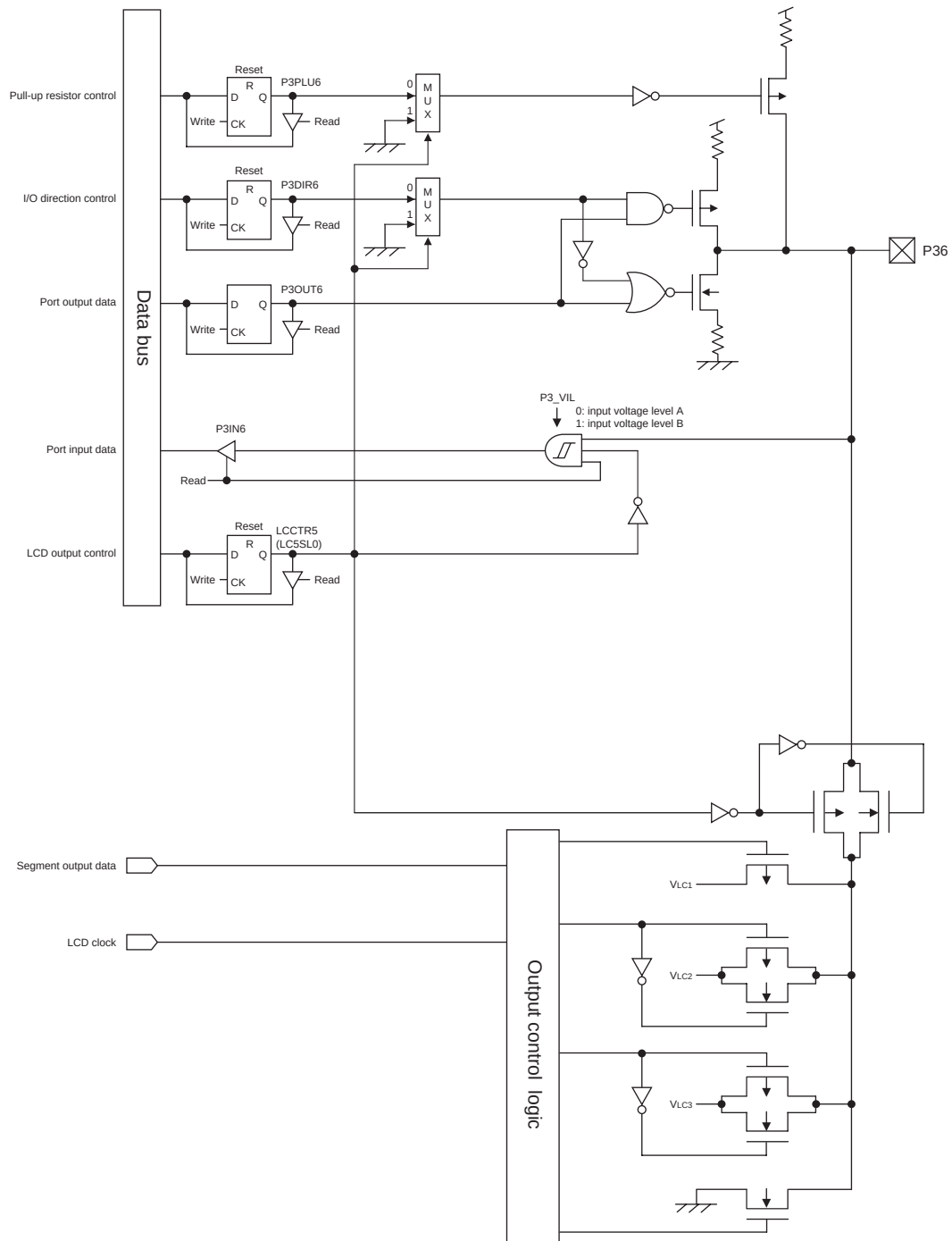


Figure:8.6.7 P36 Block Diagram (MN101EF76K/MN101EF56K)

8.6.4 Block Diagram (MN101EF57G)

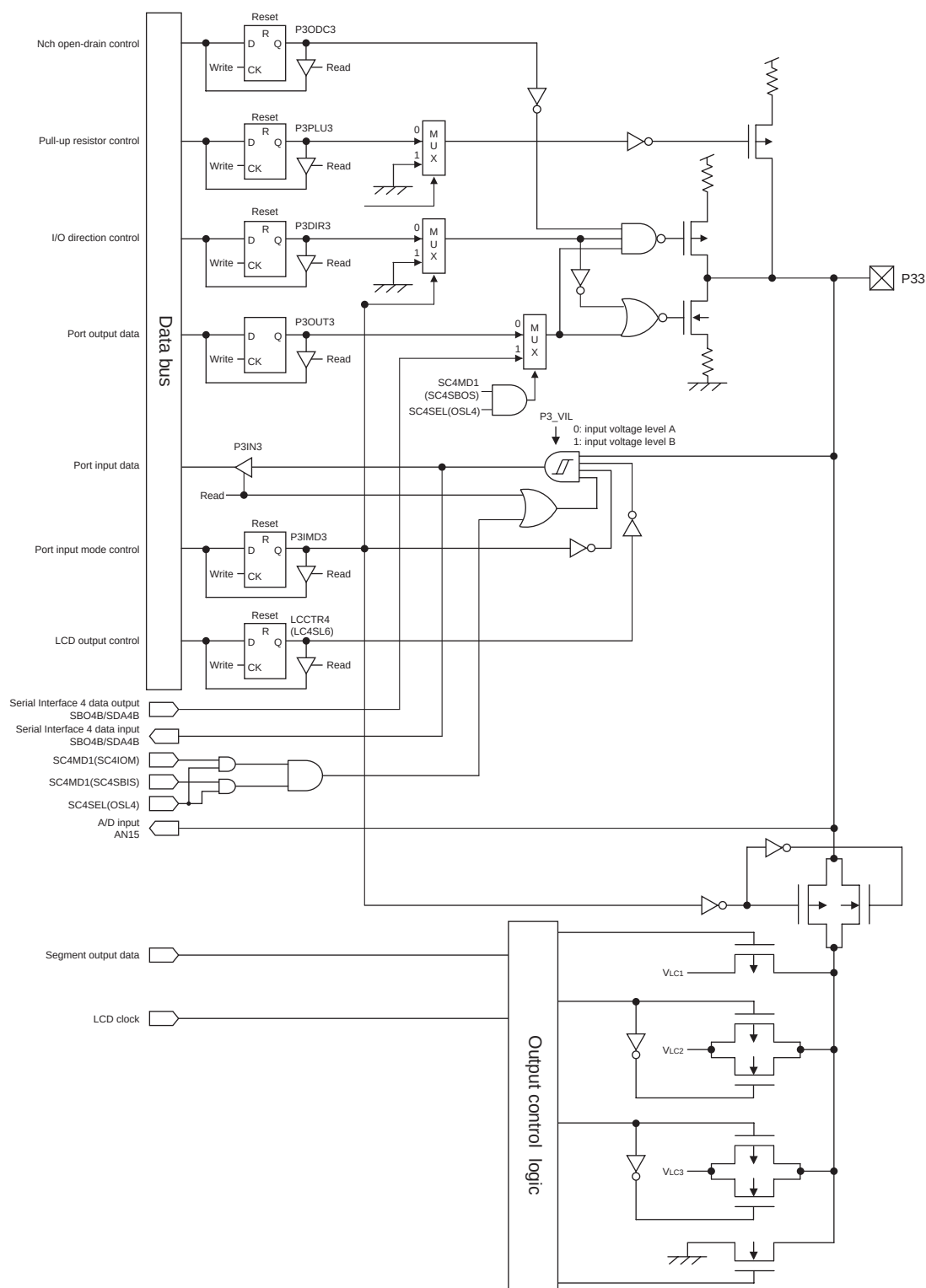


Figure:8.6.8 P33 Block Diagram (MN101EF57G)

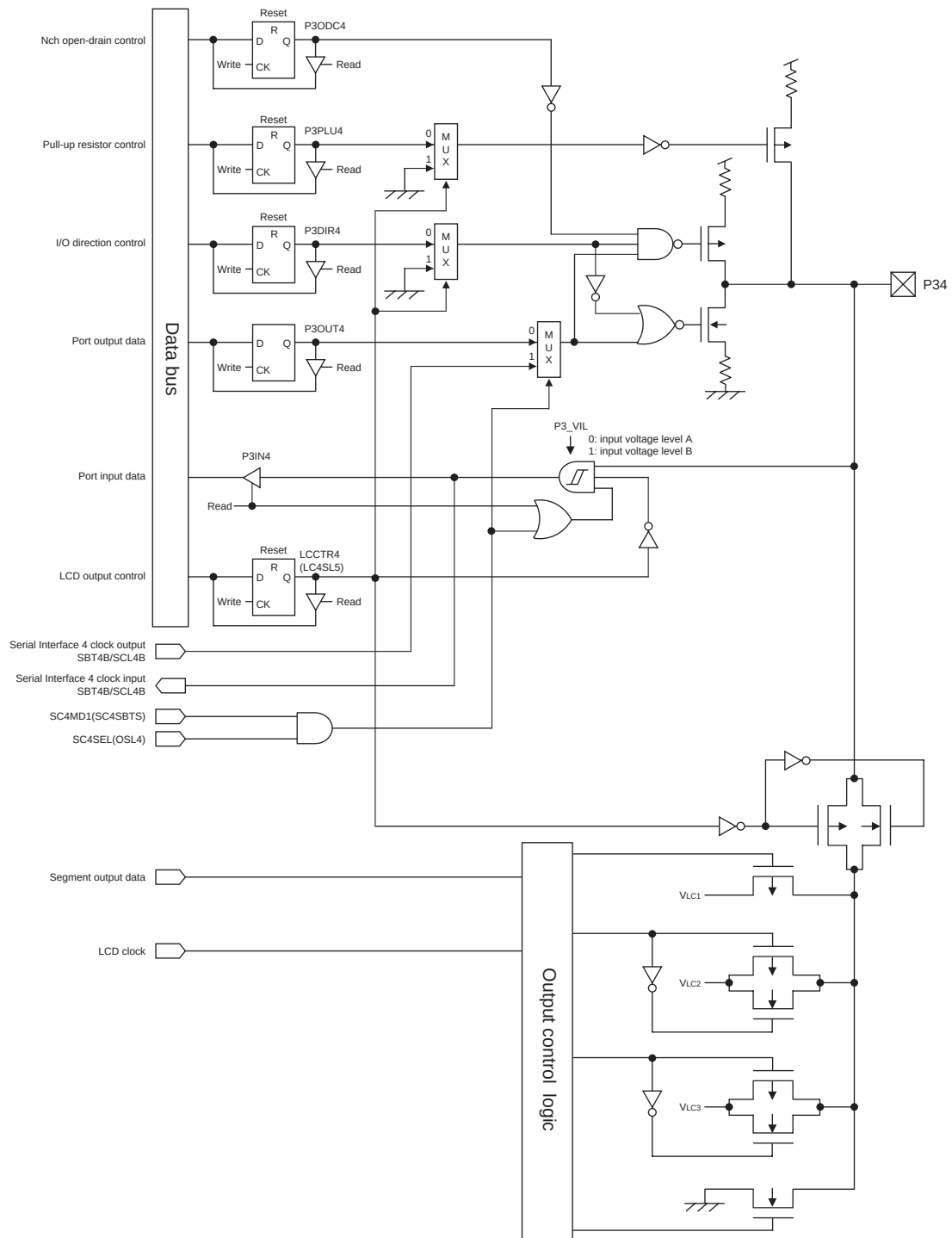


Figure:8.6.9 P34 Block Diagram (MN101EF57G)

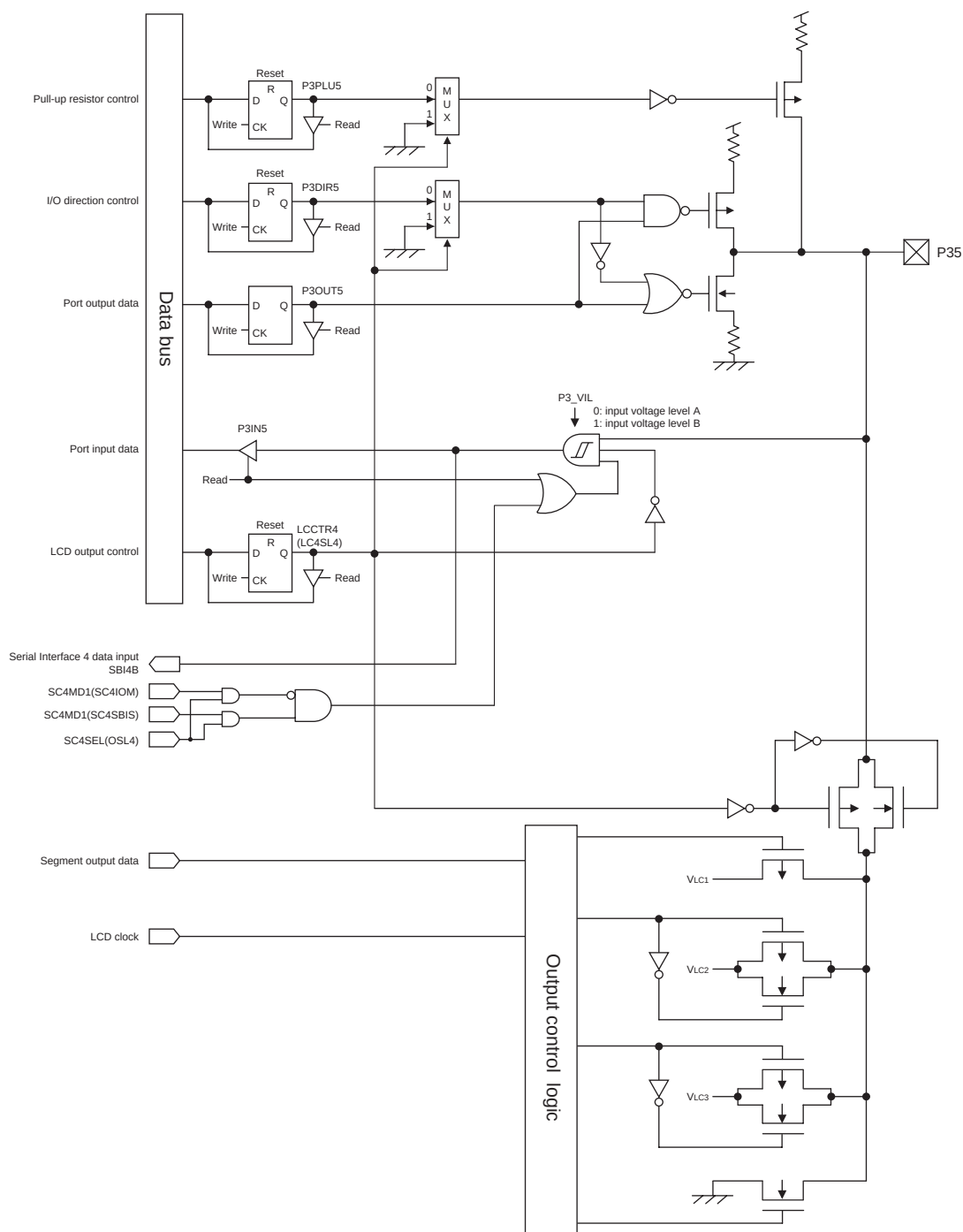


Figure:8.6.10 P35 Block Diagram (MN101EF57G)

8.7 Port 4

8.7.1 Description

■ General-purpose Port Setup

To output data to pin, set the control flag of Port 4 direction control register (P4DIR) to "1" and write the value of Port 4 output register (P4OUT).

To read input data of pins, set the control flag of Port 4 direction control register (P4DIR) to "0" and read the value of Port 4 input register (P4IN). Each bit can be set individually as either an input or output by Port 4 direction control register (P4DIR). The control flag of Port 4 direction control register (P4DIR) is set to "1" for output mode, and "0" for input mode.

Each bit can be set individually to add a pull-up (or pull-down) resistor by Port 4 pull-up/pull-down resistor control register (P4PLUD). Set the control flag of P4PLUD register to "1" to add pull-up (or pull-down) resistor.

Port 4 can be selected to add a pull-up or pull-down resistor by SELUD4 flag of the pull-up/pull-down resistor selection register (SELUD).

Each bit can be selected individually as Nch open-drain output by Port 4 Nch open-drain control register (P4ODC). The control flag of P4ODC register is set to "1" for Nch open-drain output, and "0" for push-pull output.

Table:8.7.1 Port 4 Functions

Table remarks -: Without function

MN101EF76K/MN101EF56K				MN101EF57G			
Pin	Special function			Pin	Special function		
P40	SEG31	SBO3B	TXD3B	-	-	-	-
P41	SEG30	SBI3B	RXD3B	-	-	-	-
P42	SEG29	SBT3B		-	-	-	
P43	SEG28	SBO0B	TXD0B	P43	SEG27	SBO0B	TXD0B
P44	SEG27	SBI0B	RXD0B	P44	SEG26	SBI0B	RXD0B
P45	SEG26	SBT0B		P45	SEG25	SBT0B	
P46	SEG25			P46	SEG24		
P47	SEG24			P47	SEG23		

■ Special Function Pin Setup

P40 to P47 are also used as LCD segment output pins.

LCD segment output pin is selected by setting the corresponding flag of LCD output control register n (LCCTRn) to "1".

Port and segment can be switched by one bit. When segment output pin is selected, this pin is forced to be output mode and pull-up resistor is not added.

P40 is also used as the I/O pin for Serial Interface 3 transmission/reception data and the output pin of UART3 transmission data. When the SC3SBOS flag of Serial Interface 3 mode register 1 (SC3MD1) is "1", P40 is the serial data I/O pin. Push-pull output or Nch open-drain output can be selected by setting Port 4 Nch open-drain control register (P4ODC).

P41 is also used as the input pin for Serial Interface 3 reception data and UART3 reception data. When the SC3BIS flag of Serial Interface 3 mode register 1 (SC3MD1) is "1", P41 is the serial data input pin.

P42 is also used as the I/O pin for Serial Interface 3 clock. When the SC3SBTS flag of Serial Interface 3 mode register 1 (SC3MD1) is "1", P42 is the I/O pin of the serial clock. Push-pull output or Nch open-drain output can be selected by setting Port 4 Nch open-drain control register (P4ODC).

P43 is also used as the I/O pin for Serial Interface 0 transmission/reception data and the output pin of UART0 transmission data. When the SC0SBOS flag of Serial Interface 0 mode register 1 (SC0MD1) is "1", P43 is the serial data I/O pin. Push-pull output or Nch open-drain output can be selected by setting Port 4 Nch open-drain control register (P4ODC).

P44 is also used as the input pin for Serial Interface 0 reception data and UART0 reception data. When the SC0BIS flag of Serial Interface 0 mode register 1 (SC0MD1) is "1", P44 is the serial data input pin.

P45 is also used as the I/O pin for Serial Interface 0 clock. When the SC0STBS flag of Serial Interface 0 mode register 1 (SC0MD1) is "1", P45 is the I/O pin of the serial clock. Push-pull output or Nch open-drain output can be selected by setting Port 4 Nch open-drain control register (P4ODC).

8.7.2 Registers

Table:8.3.2 shows registers that control Port 4.

Table:8.7.2 Port 4 Control Register

Table remarks √: With function -: Without function

Registers	Address	R/W	Function	Page	MN101 EF76K	MN101 EF57G	MN101 EF56K
P4OUT	0x03E74	R/W	Port 4 output register	VIII-72	√	√	√
P4IN	0x03E84	R	Port 4 input register	VIII-73	√	√	√
P4DIR	0x03E94	R/W	Port 4 direction control register	VIII-74	√	√	√
P4PLUD	0x03EA4	R/W	Port 4 pull-up/pull-down resistor control register	VIII-75	√	√	√
P4ODC	0x03EF4	R/W	Port 4 Nch open-drain control register	VIII-76	√	√	√
SELUD	0x03EAF	R/W	Pull-up/pull-down resistor selection register	VIII-77	√	√	√

R/W: Readable/Writable

R: Read only

■ Port 4 Output Register (P4OUT: 0x03E74)

<MN101EF76K/MN101EF56K>

bp	7	6	5	4	3	2	1	0
Flag	P4OUT7	P4OUT6	P4OUT5	P4OUT4	P4OUT3	P4OUT2	P4OUT1	P4OUT0
At reset	X	X	X	X	X	X	X	X
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description
7 to 0	P4OUT7 to 0	Output data 0: Output "Low" (V_{SS} -level) 1: Output "High" (V_{DD5} -level)

<MN101EF57G>

bp	7	6	5	4	3	2	1	0
Flag	P4OUT7	P4OUT6	P4OUT5	P4OUT4	P4OUT3	-	-	-
At reset	X	X	X	X	X	-	-	-
Access	R/W	R/W	R/W	R/W	R/W	-	-	-

bp	Flag	Description
7 to 3	P4OUT7 to 3	Output data 0: Output "Low" (V_{SS} -level) 1: Output "High" (V_{DD5} -level)
2 to 0	-	-

■ Port 4 Input Register (P4IN: 0x03E84)

<MN101EF76K/MN101EF56K>

bp	7	6	5	4	3	2	1	0
Flag	P4IN7	P4IN6	P4IN5	P4IN4	P4IN3	P4IN2	P4IN1	P4IN0
At reset	X	X	X	X	X	X	X	X
Access	R	R	R	R	R	R	R	R

bp	Flag	Description
7 to 0	P4IN7 to 0	Input data 0: Pin is "Low" (V_{SS} -level) 1: Pin is "High" (V_{DD5} -level)

<MN101EF57G>

bp	7	6	5	4	3	2	1	0
Flag	P4IN7	P4IN6	P4IN5	P4IN4	P4IN3	-	-	-
At reset	X	X	X	X	X	-	-	-
Access	R	R	R	R	R	-	-	-

bp	Flag	Description
7 to 3	P4IN7 to 3	Input data 0: Pin is "Low" (V_{SS} -level) 1: Pin is "High" (V_{DD5} -level)
2 to 0	-	-

■ Port 4 Direction Control Register (P4DIR: 0x03E94)

<MN101EF76K/MN101EF56K>

bp	7	6	5	4	3	2	1	0
Flag	P4DIR7	P4DIR6	P4DIR5	P4DIR4	P4DIR3	P4DIR2	P4DIR1	P4DIR0
At reset	0	0	0	0	0	0	0	0
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description
7 to 0	P4DIR7 to 0	I/O mode selection 0: Input mode 1: Output mode

<MN101EF57G>

bp	7	6	5	4	3	2	1	0
Flag	P4DIR7	P4DIR6	P4DIR5	P4DIR4	P4DIR3	-	-	-
At reset	0	0	0	0	0	-	-	-
Access	R/W	R/W	R/W	R/W	R/W	-	-	-

bp	Flag	Description
7 to 3	P4DIR7 to 3	I/O mode selection 0: Input mode 1: Output mode
2 to 0	-	-

■ Port 4 Pull-up/Pull-down Resistor Control Register (P4PLUD: 0x03EA4)

<MN101EF76K/MN101EF56K>

bp	7	6	5	4	3	2	1	0
Flag	P4PLUD7	P4PLUD6	P4PLUD5	P4PLUD4	P4PLUD3	P4PLUD2	P4PLUD1	P4PLUD0
At reset	0	0	0	0	0	0	0	0
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description
7 to 0	P4PLUD7 to 0	Pull-up/Pull-down resistor selection 0: Not added 1: Added

<MN101EF57G>

bp	7	6	5	4	3	2	1	0
Flag	P4PLUD7	P4PLUD6	P4PLUD5	P4PLUD4	P4PLUD3	-	-	-
At reset	0	0	0	0	0	-	-	-
Access	R/W	R/W	R/W	R/W	R/W	-	-	-

bp	Flag	Description
7 to 3	P4PLUD7 to 3	Pull-up/Pull-down resistor selection 0: Not added 1: Added
2 to 0	-	-

■ Port 4 Nch Open-drain Control Register (P4ODC: 0x03EF4)

<MN101EF76K/MN101EF56K>

bp	7	6	5	4	3	2	1	0
Flag	-	-	P4ODC5	-	P4ODC3	P4ODC2	-	P4ODC0
At reset	-	-	0	-	0	0	-	0
Access	-	-	R/W	-	R/W	R/W	-	R/W

bp	Flag	Description
7 to 6	-	-
5	P4ODC5	Nch open-drain output selection 0: Push-pull output 1: Nch open-drain output
4	-	-
3 to 2	P4ODC3	Nch open-drain output selection 0: Push-pull output 1: Nch open-drain output
1	-	-
0	P4ODC0	Nch open-drain output selection 0: Push-pull output 1: Nch open-drain output

<MN101EF57G>

bp	7	6	5	4	3	2	1	0
Flag	-	-	P4ODC5	-	P4ODC3	-	-	-
At reset	-	-	0	-	0	-	-	-
Access	-	-	R/W	-	R/W	-	-	-

bp	Flag	Description
7 to 6	-	-
5	P4ODC5	Nch open-drain output selection 0: Push-pull output 1: Nch open-drain output
4	-	-
3	P4ODC3	Nch open-drain output selection 0: Push-pull output 1: Nch open-drain output
2 to 0	-	-

■ Pull-up/pull-down Resistor Selection Register (SELUD: 0x03EAF)

<MN101EF76K/MN101EF56K>

bp	7	6	5	4	3	2	1	0
Flag	SELUD7	-	SELUD5	SELUD4	-	-	SELUD1	SELUD0
At reset	0	-	0	0	-	-	0	0
Access	R/W	-	R/W	R/W	-	-	R/W	R/W

bp	Flag	Description
7	SELUD7	Pull-up/pull-down selection (Port 7) 0: Pull-up 1: Pull-down
6	-	-
5	SELUD5	Pull-up/pull-down selection (Port 5) 0: Pull-up 1: Pull-down
4	SELUD4	Pull-up/pull-down selection (Port 4) 0: Pull-up 1: Pull-down
3 to 2	-	-
1	SELUD0	Pull-up/pull-down selection (Port 1) 0: Pull-up 1: Pull-down
0	SELUD0	Pull-up/pull-down selection (Port 0) 0: Pull-up 1: Pull-down

<MN101EF57G>

bp	7	6	5	4	3	2	1	0
Flag	SELUD7	-	SELUD5	SELUD4	-	-	-	SELUD0
At reset	0	-	0	0	-	-	-	0
Access	R/W	-	R/W	R/W	-	-	-	R/W

bp	Flag	Description
7	SELUD7	Pull-up/pull-down selection (Port 7) 0: Pull-up 1: Pull-down
6	-	-
5	SELUD5	Pull-up/pull-down selection (Port 5) 0: Pull-up 1: Pull-down
4	SELUD4	Pull-up/pull-down selection (Port 4) 0: Pull-up 1: Pull-down
3 to 1	-	-
0	SELUD0	Pull-up/pull-down selection (Port 0) 0: Pull-up 1: Pull-down

8.7.3 Block Diagram (MN101EF76K/MN101EF56K)

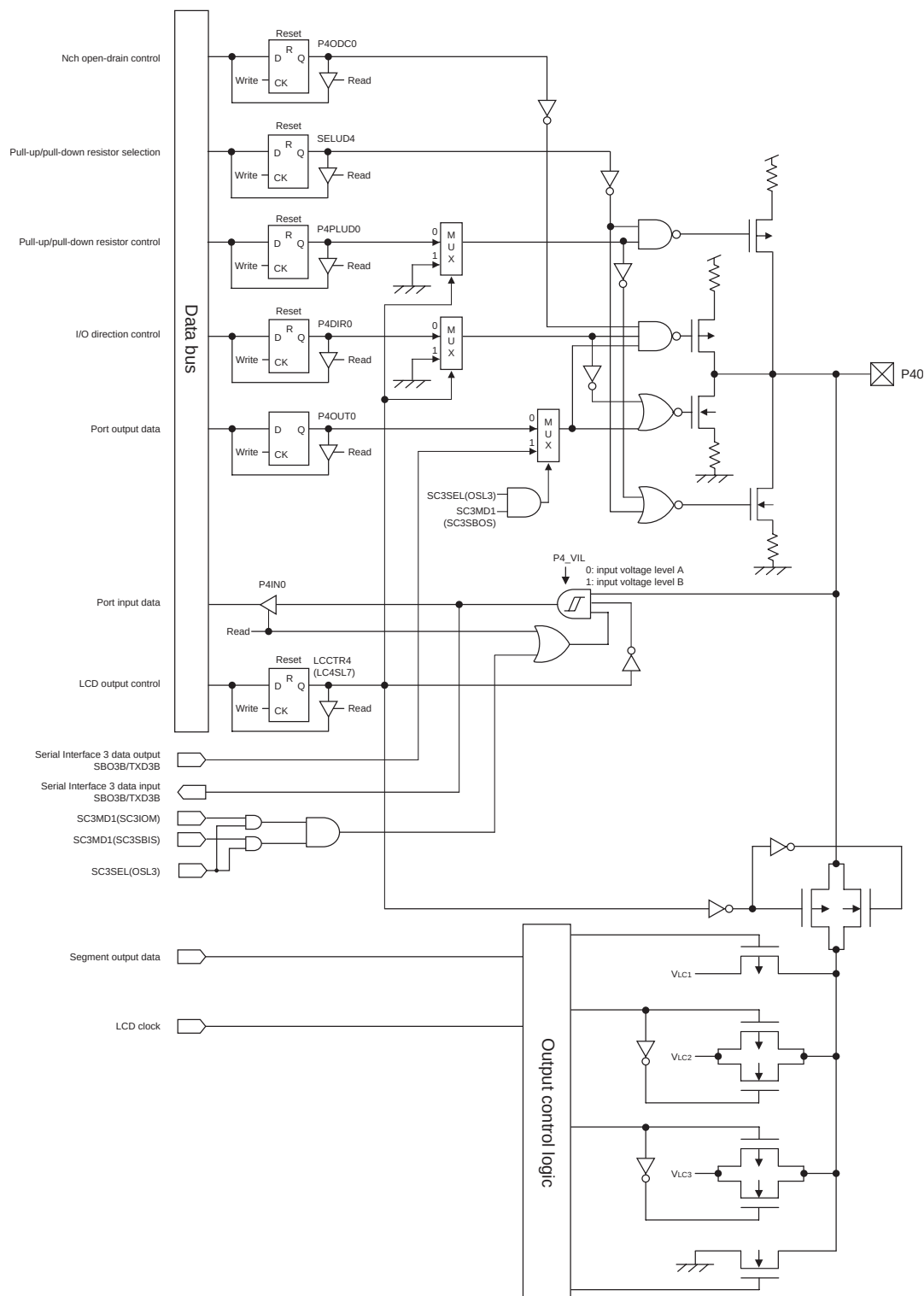


Figure:8.7.1 P40 Block Diagram (MN101EF76K/MN101EF56K)

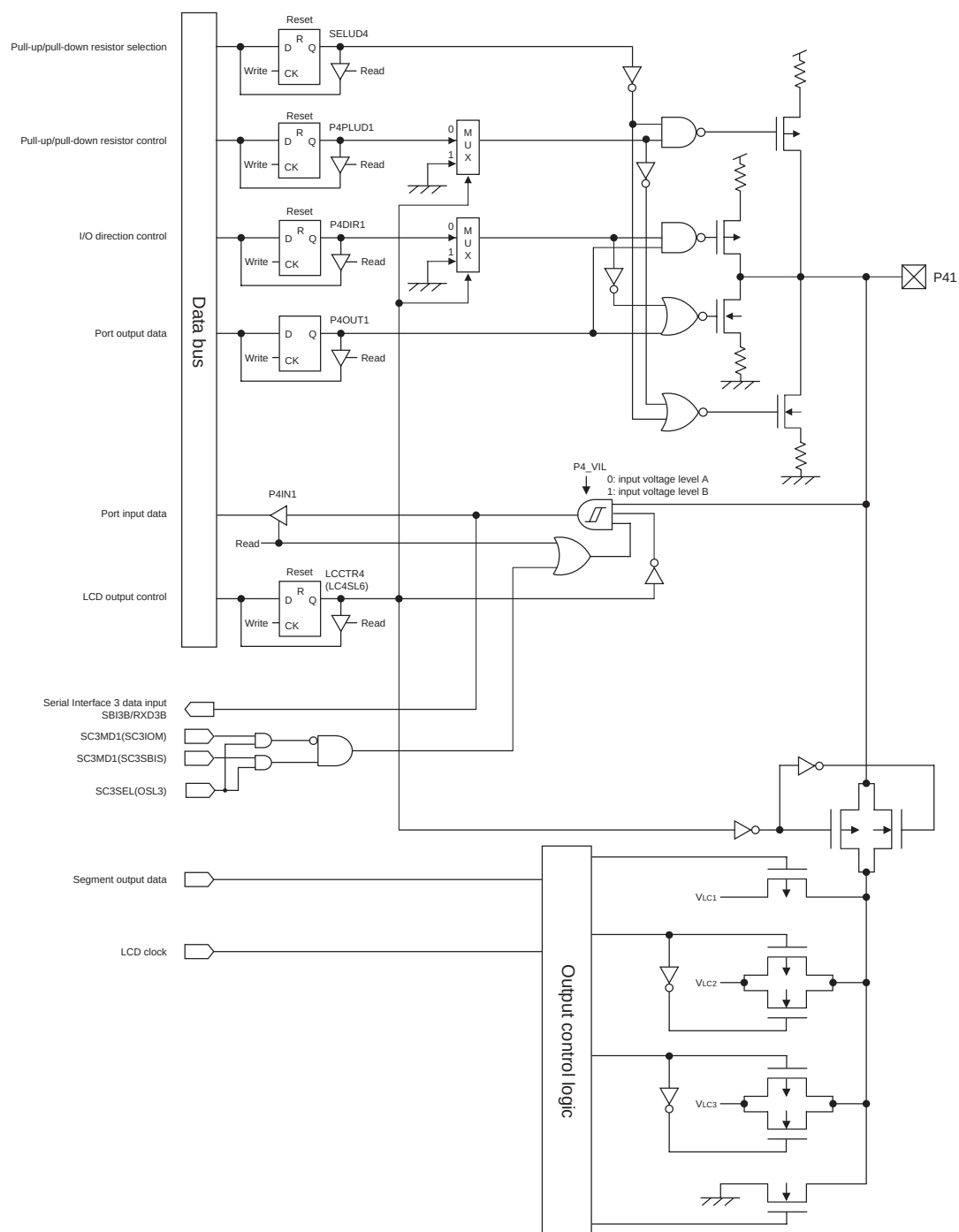


Figure:8.7.2 P41 Block Diagram (MN101EF76K/MN101EF56K)

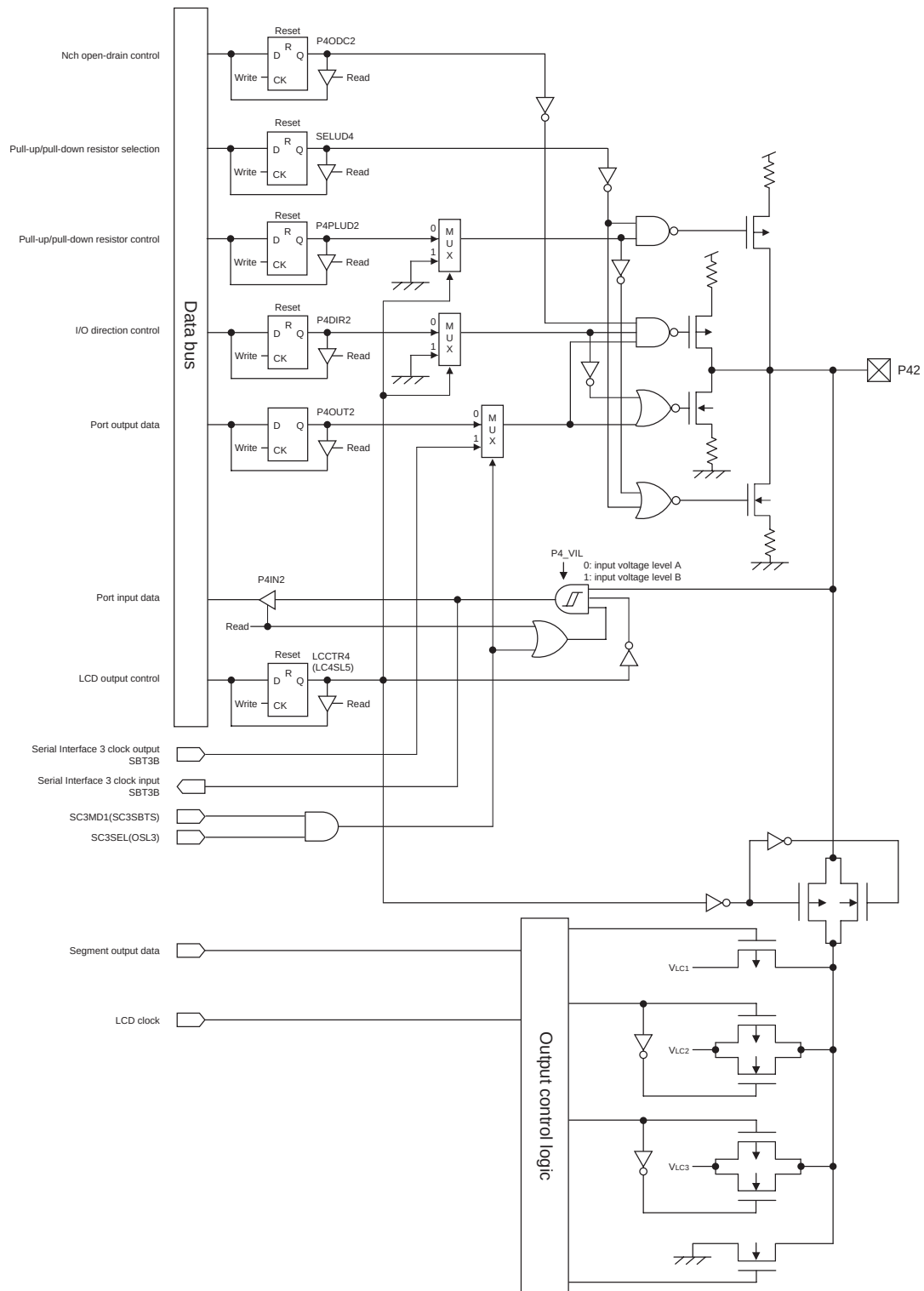


Figure:8.7.3 P42 Block Diagram (MN101EF76K/MN101EF56K)

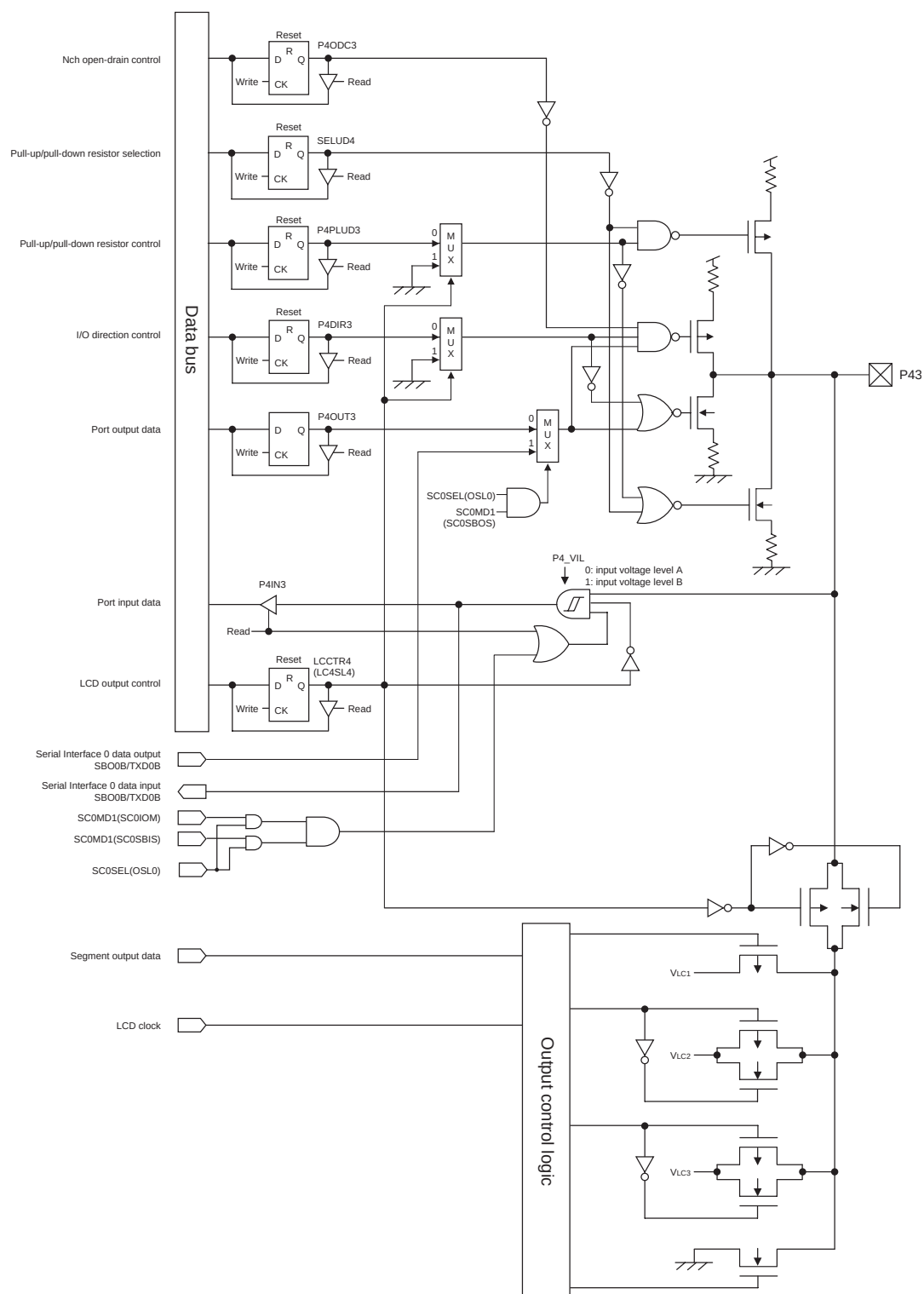


Figure:8.7.4 P43 Block Diagram (MN101EF76K/MN101EF56K)

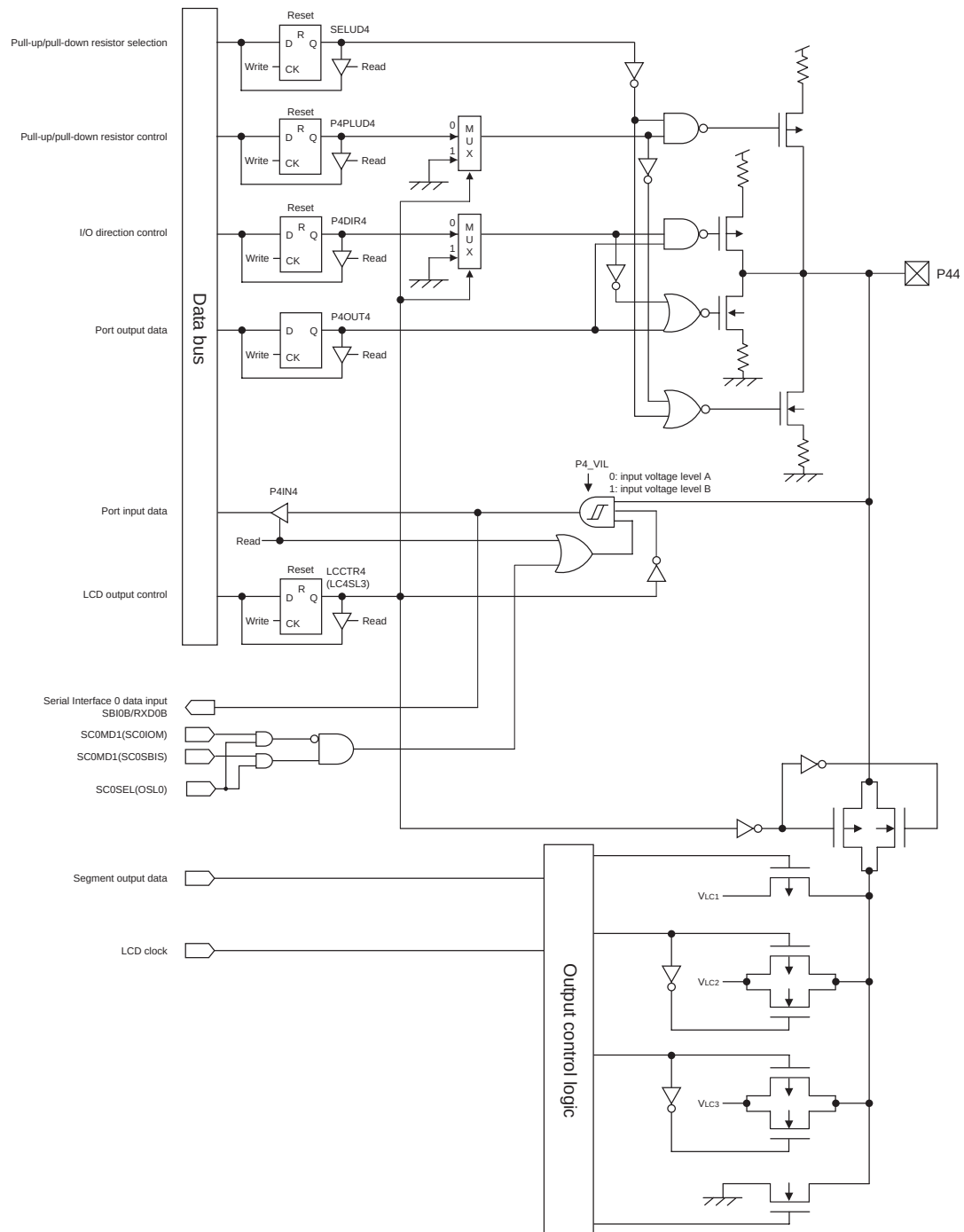


Figure:8.7.5 P44 Block Diagram (MN101EF76K/MN101EF56K)

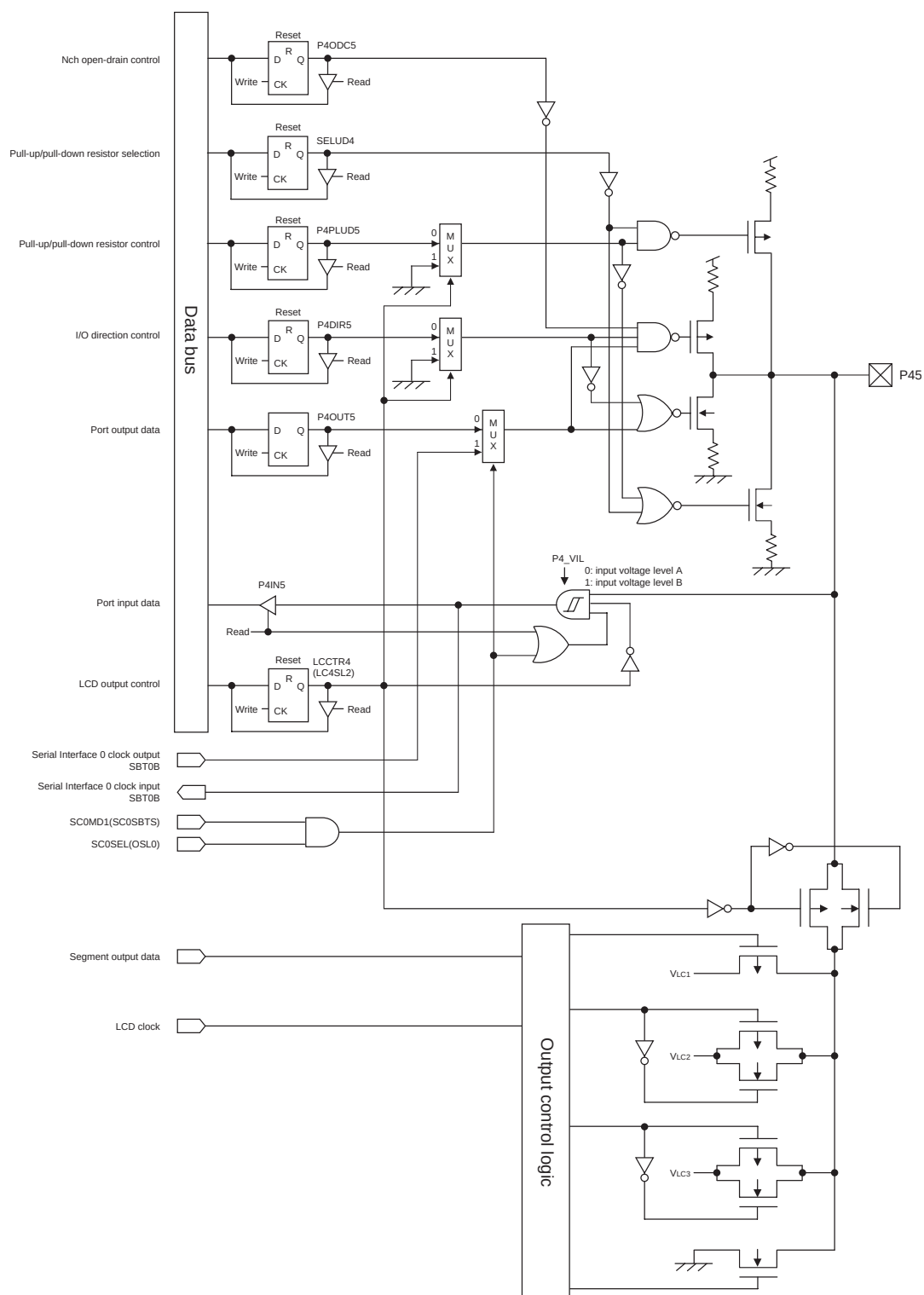


Figure:8.7.6 P45 Block Diagram (MN101EF76K/MN101EF56K)

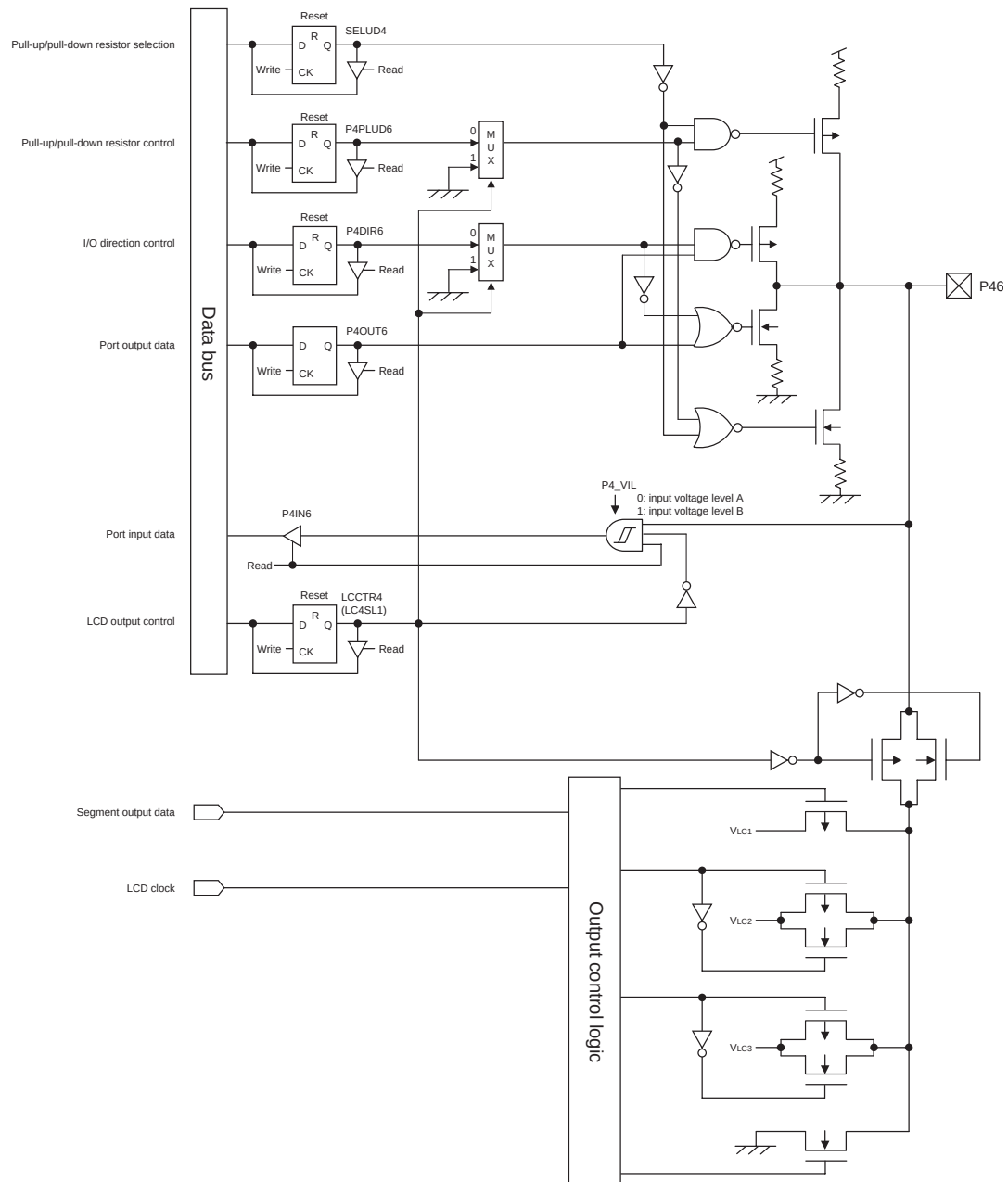


Figure:8.7.7 P46 Block Diagram (MN101EF76K/MN101EF56K)

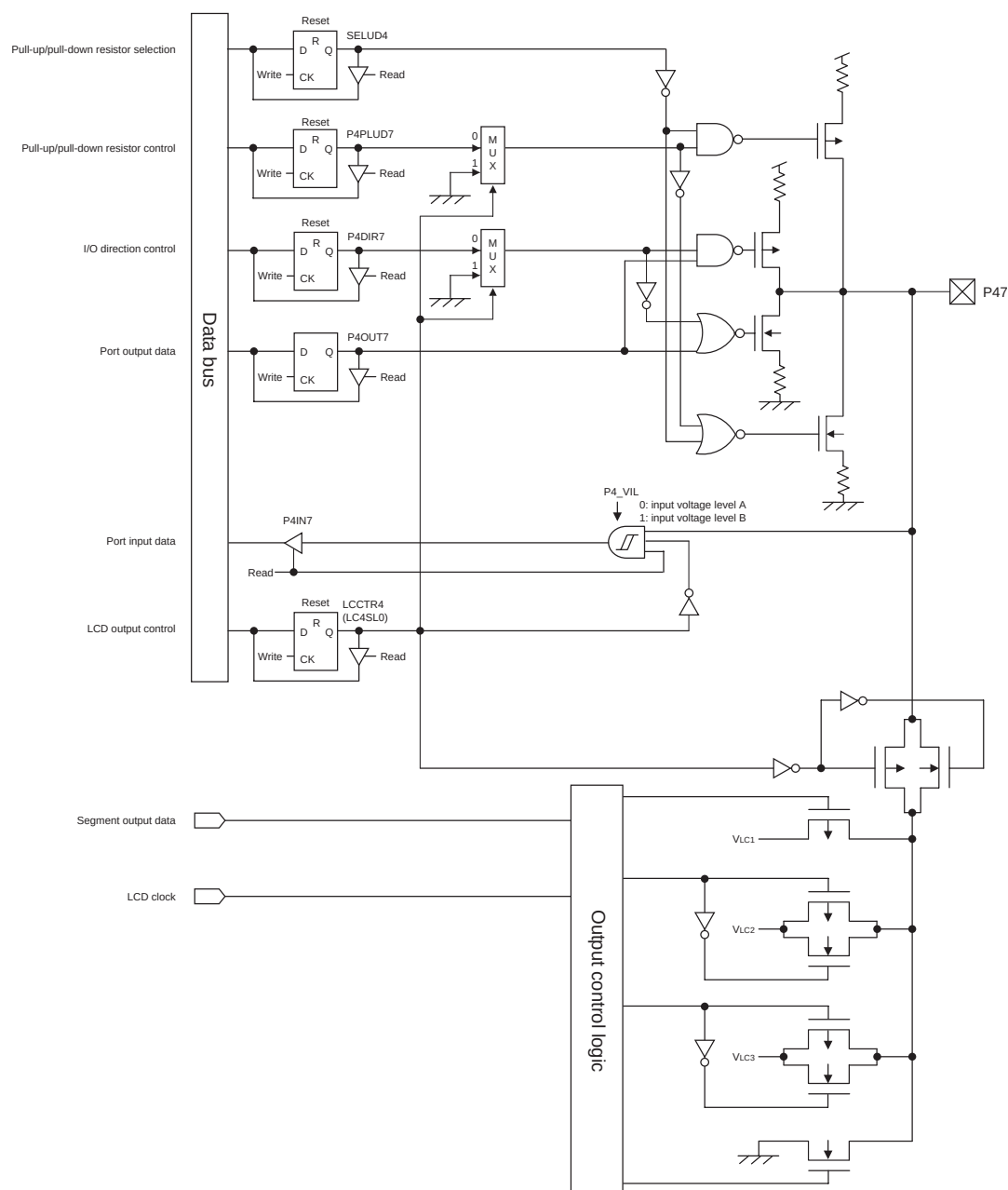


Figure:8.7.8 P47 Block Diagram (MN101EF76K/MN101EF56K)

8.7.4 Block Diagram (MN101EF57G)

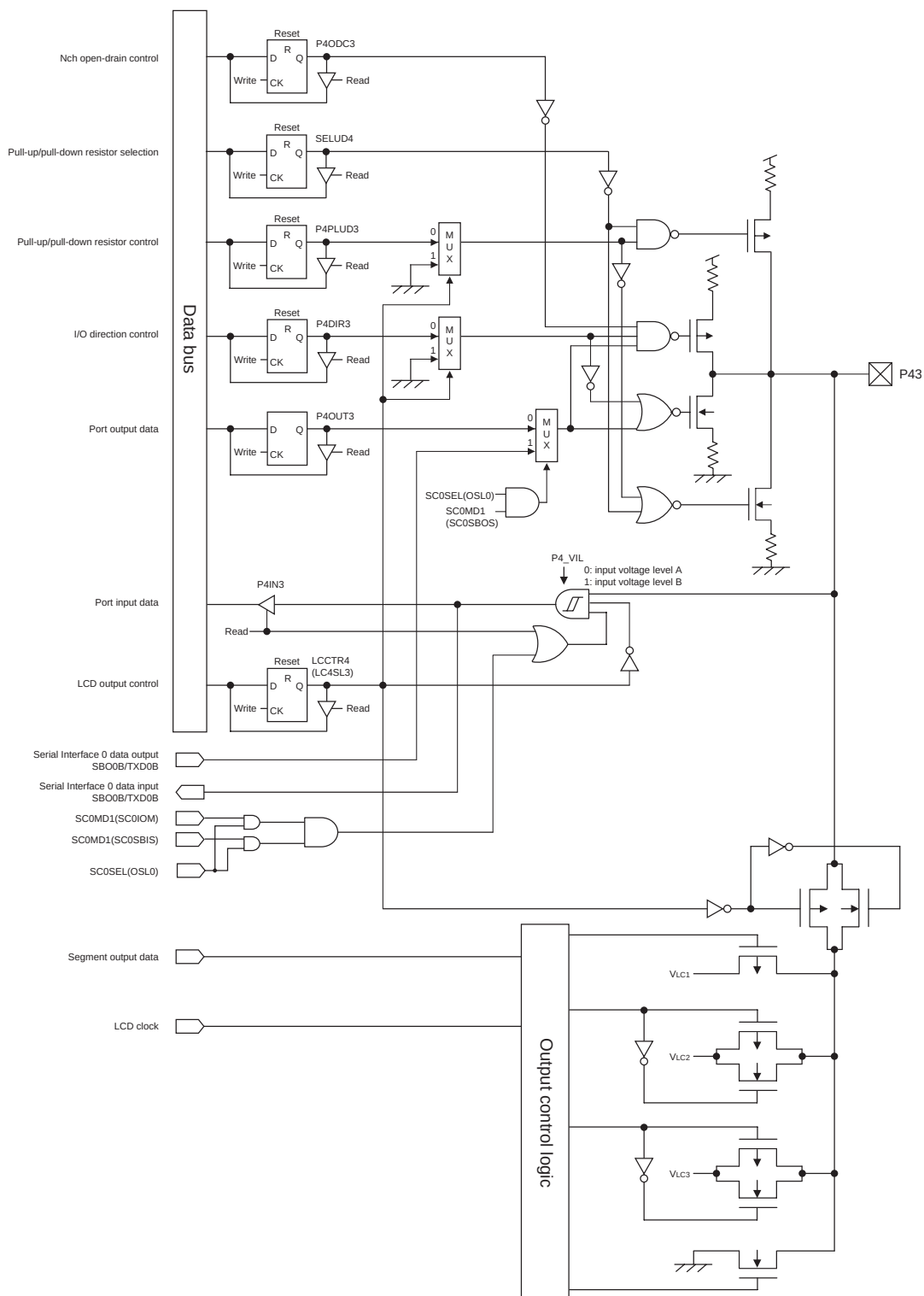


Figure:8.7.9 P43 Block Diagram (MN101EF57G)

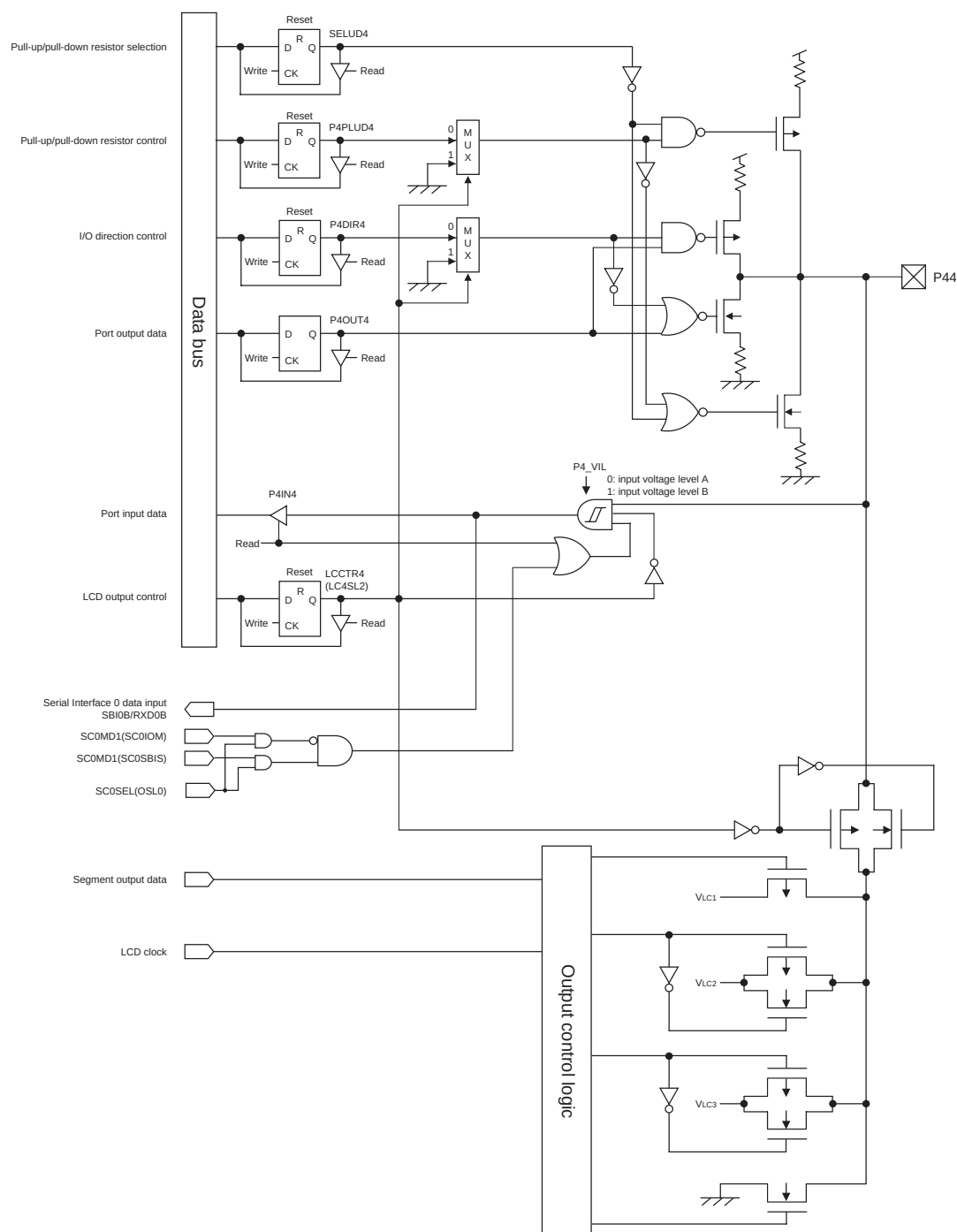


Figure:8.7.10 P44 Block Diagram (MN101EF57G)

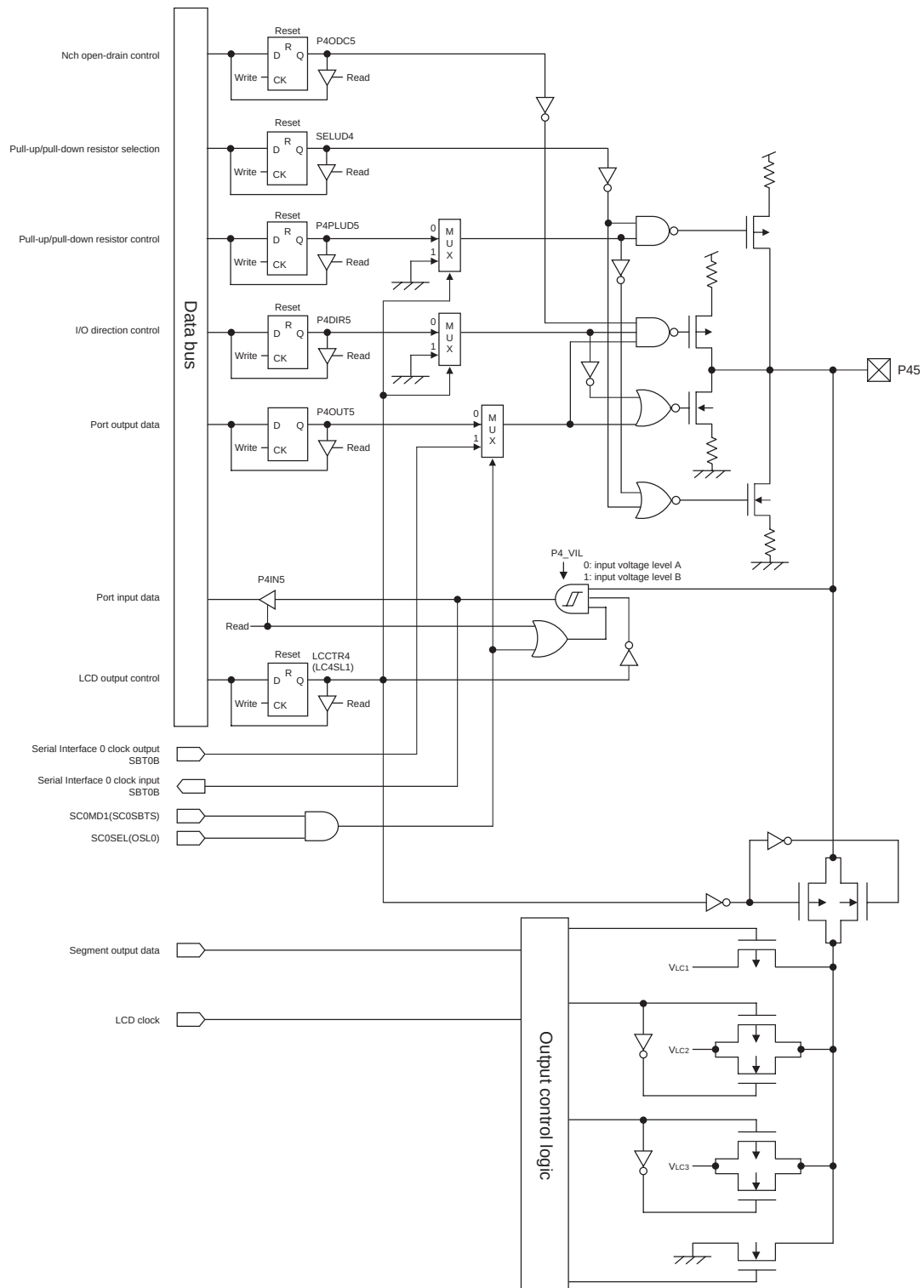


Figure:8.7.11 P45 Block Diagram (MN101EF57G)

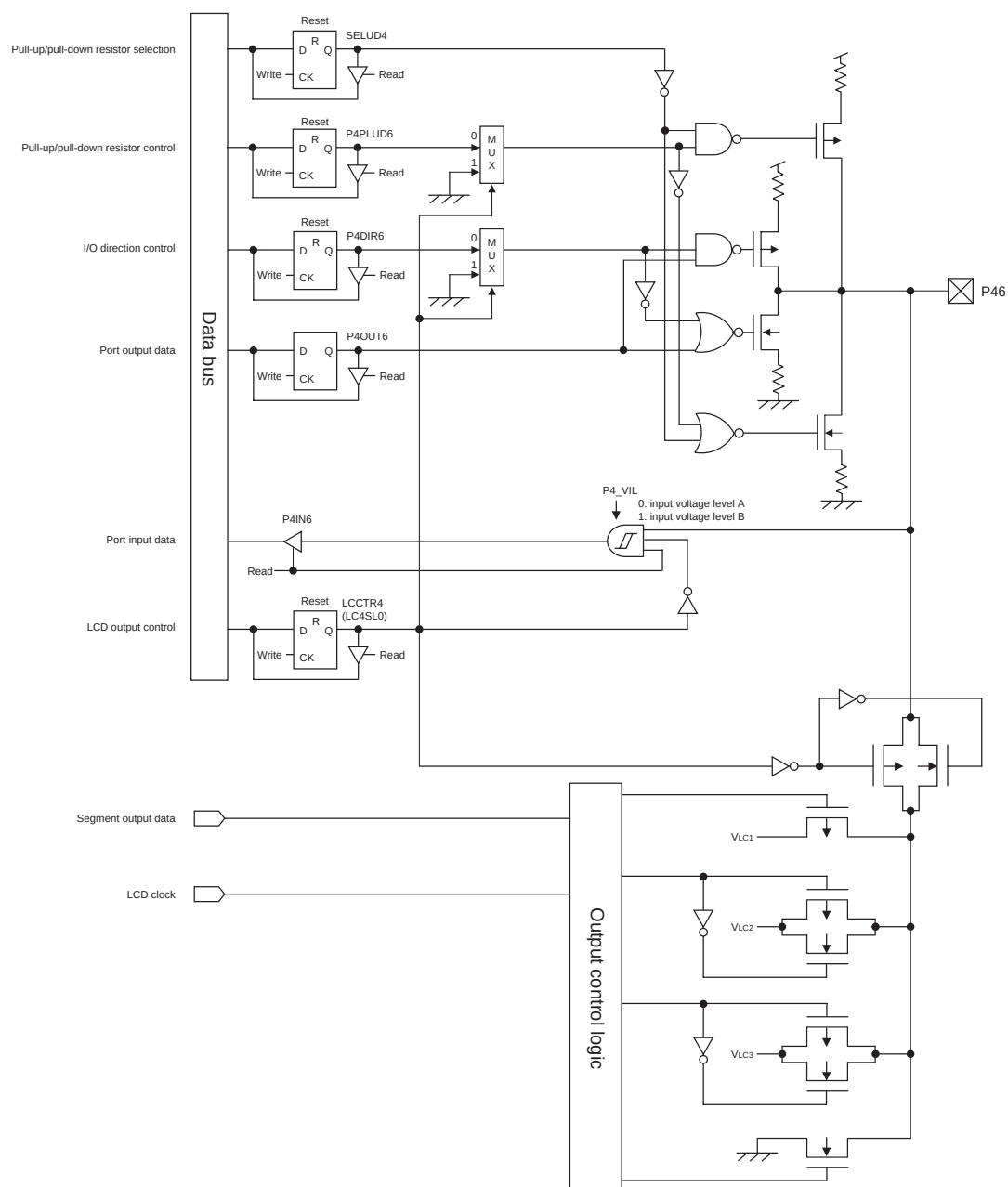


Figure:8.7.12 P46 Block Diagram (MN101EF57G)

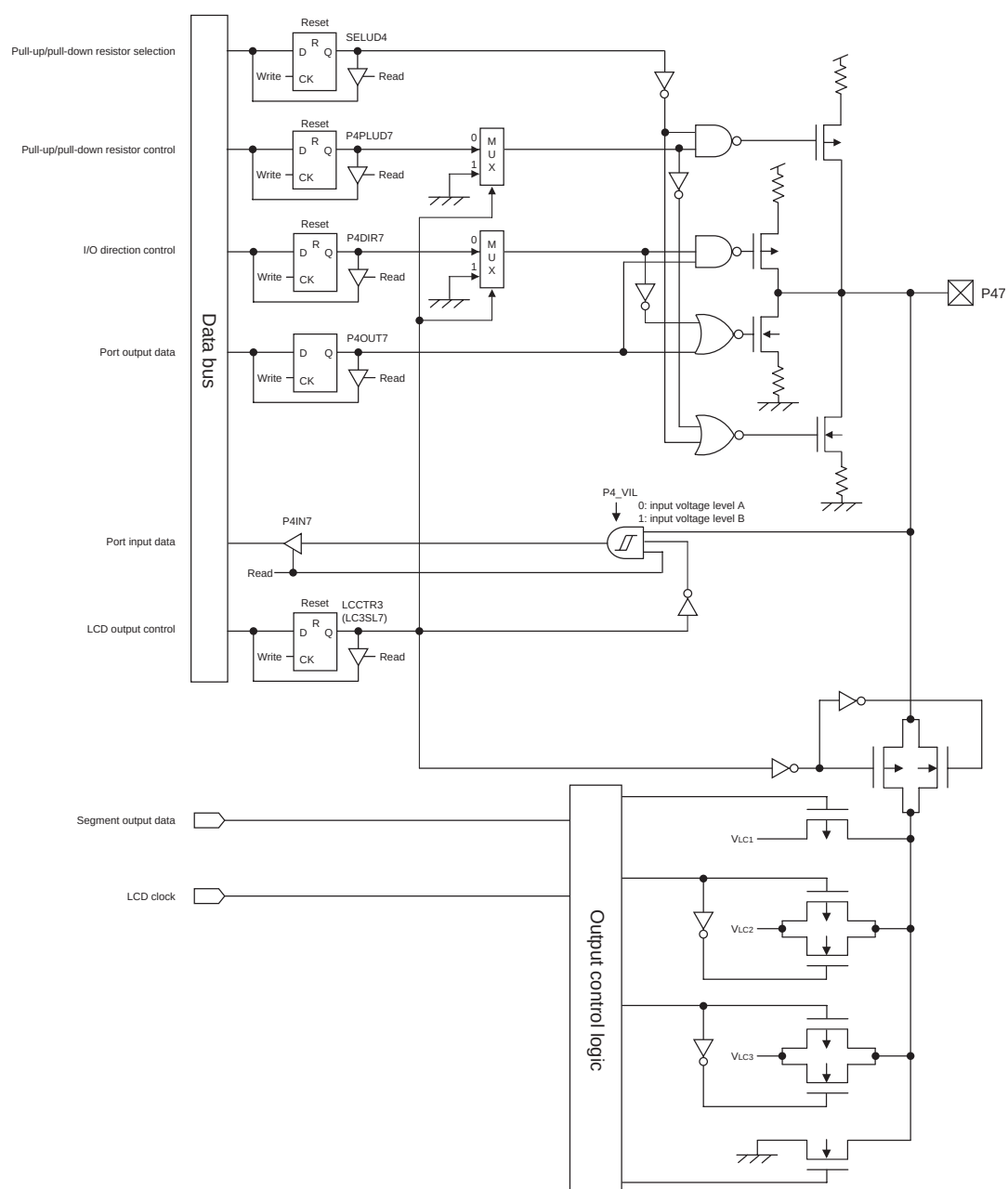


Figure:8.7.13 P47 Block Diagram (MN101EF57G)

8.8 Port 5

8.8.1 Description

■ General-purpose Port Setup

To output data to pin, set the control flag of Port 5 direction control register (P5DIR) to "1" and write the value of Port 5 output register (P5OUT).

To read input data of pin, set the control flag of Port 5 direction control register (P5DIR) to "0" and read the value of Port 5 input register (P5IN).

Each bit can be set individually as either an input or output by Port 5 direction control register (P5DIR). The control flag of P5DIR register is set to "1" for output mode, and "0" for input mode.

Each bit can be set individually to add a pull-up (or pull-down) resistor by Port 5 pull-up/pull-down resistor control register (P5PLUD). Set the control flag of P5PLUD register to "1" to add pull-up (or pull-down) resistor.

Port 5 can be selected to add a pull-up or pull-down resistor by SELUD5 flag of the pull-up/pull-down resistor selection register (SELUD).

Each bit can be selected individually as Nch open-drain output by Port 5 Nch open-drain control register (P5ODC). The control flag of P5ODC register is set to "1" for Nch open-drain output, and "0" for push-pull output.

Table:8.8.1 Port 5 Functions

Table remarks -: Without function

MN101EF76K/MN101EF56K				MN101EF57G			
Pin	Special function			Pin	Special function		
P50	SEG16	SBO1A	TXD1A	P50	SEG15	SBO1A	TXD1A
P51	SEG17	SBI1A	RXD1A	P51	SEG16	SBI1A	RXD1A
P52	SEG18	SBT1A		P52	SEG17	SBT1A	
P53	SEG19			P53	SEG18		
P54	SEG20			P54	SEG19		
P55	SEG21			P55	SEG20		
P56	SEG22	NBUZZERA		P56	SEG21	NBUZZERA	
P57	SEG23	BUZZERA		P57	SEG22	BUZZERA	

■ Special Function Pin Setup

P57 is also used as Buzzer A output pin.

P56 is also used as Reverse Buzzer A output pin.

Each bit can be set individually as either buzzer output or port output by Port 5 output mode register (P5OMD). When the corresponding flag of P5OMD register is set to "1", P57 and P56 output Buzzer/Reverse Buzzer, and when set to "0", P57 and P56 are used as the general purpose port.

P50 to P57 are also used as LCD segment output pins.

LCD segment output pin is selected by setting the corresponding flag of LCD output control register n (LCCTRn) to "1".

Port and segment can be switched by one bit. When segment output pin is selected, this pin is forced to be output mode and pull-up resistor is not added.

P50 is also used as the I/O pin for Serial Interface 1 transmission/reception data and the output pin of UART1 transmission data. When the SC1SBOS flag of Serial Interface 1 mode register 1 (SC1MD1) is "1", P50 is the serial data I/O pin. Push-pull output or Nch open-drain output can be selected by setting Port 5 Nch open-drain control register (P5ODC).

P51 is also used as the input pin for Serial Interface 1 reception data and UART1 reception data. When the SC1BIS flag of Serial Interface 1 mode register 1 (SC1MD1) is "1", P51 is the serial data input pin.

P52 is also used as the I/O pin for Serial Interface 1 clock. When the SC1SBTS flag of Serial Interface 1 mode register 1 (SC1MD1) is "1", P52 is the I/O pin of the serial clock. Push-pull output or Nch open-drain output can be selected by setting Port 5 Nch open-drain control register (P5ODC).

8.8.2 Registers

Table:8.8.2 shows registers that control Port 5.

Table:8.8.2 Port 5 Control Register

Table remarks √: With function -: Without function

Registers	Address	R/W	Function	Page	MN101 EF76K	MN101 EF57G	MN101 EF56K
P5OUT	0x03E75	R/W	Port 5 output register	VIII-93	√	√	√
P5IN	0x03E85	R	Port 5 input register	VIII-93	√	√	√
P5DIR	0x03E95	R/W	Port 5 direction control register	VIII-94	√	√	√
P5PLUD	0x03EA5	R/W	Port 5 pull-up/pull-down resistor control register	VIII-94	√	√	√
P5OMD	0x03EB5	R/W	Port 5 output mode register	VIII-95	√	√	√
P5ODC	0x03EF5	R/W	Port 5 Nch open-drain control register	VIII-94	√	√	√
SELUD	0x03EAF	R/W	Pull-up/pull-down resistor selection register	VIII-96	√	√	√

R/W: Readable/Writable

R: Read only

■ Port 5 Output Register (P5OUT: 0x03E75)

bp	7	6	5	4	3	2	1	0
Flag	P5OUT7	P5OUT6	P5OUT5	P5OUT4	P5OUT3	P5OUT2	P5OUT1	P5OUT0
At reset	X	X	X	X	X	X	X	X
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description
7 to 0	P5OUT7 to 0	Output data 0: Output "Low" (V_{SS} -level) 1: Output "High" (V_{DD5} -level)

■ Port 5 Input Register (P5IN: 0x03E85)

bp	7	6	5	4	3	2	1	0
Flag	P5IN7	P5IN6	P5IN5	P5IN4	P5IN3	P5IN2	P5IN1	P5IN0
At reset	X	X	X	X	X	X	X	X
Access	R	R	R	R	R	R	R	R

bp	Flag	Description
7 to 0	P5IN7 to 0	Input data 0: Pin is "Low" (V_{SS} -level) 1: Pin is "High" (V_{DD5} -level)

■ Port 5 Direction Control Register (P5DIR: 0x03E95)

bp	7	6	5	4	3	2	1	0
Flag	P5DIR7	P5DIR6	P5DIR5	P5DIR4	P5DIR3	P5DIR2	P5DIR1	P5DIR0
At reset	0	0	0	0	0	0	0	0
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description
7 to 0	P5DIR7 to 0	I/O mode selection 0: Input mode 1: Output mode

■ Port 5 Pull-up/pull-down Resistor Control Register (P5PLUD: 0x03EA5)

bp	7	6	5	4	3	2	1	0
Flag	P5PLUD7	P5PLUD6	P5PLUD5	P5PLUD4	P5PLUD3	P5PLUD2	P5PLUD1	P5PLUD0
At reset	0	0	0	0	0	0	0	0
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description
7 to 0	P5PLUD7 to 0	Pull-up/pull-down resistor selection 0: Not added 1: Added

■ Port 5 Nch Open-drain Control Register P5ODC: 0x03EF5)

bp	7	6	5	4	3	2	1	0
Flag	-	-	-	-	-	P5ODC2	-	P5ODC0
At reset	-	-	-	-	-	0	-	0
Access	-	-	-	-	-	R/W	-	R/W

bp	Flag	Description
7 to 3	-	-
2	P5ODC2	Nch open-drain output selection 0: Push-pull output 1: Nch open-drain output
1	-	-
0	P5ODC0	Nch open-drain output selection 0: Push-pull output 1: Nch open-drain output

■ Port 5 Output Mode Register (P5OMD: 0x3ECB5)

bp	7	6	5	4	3	2	1	0
Flag	P5OMD7	P5OMD6	-	-	-	-	-	-
At reset	0	0	-	-	-	-	-	-
Access	R/W	R/W	-	-	-	-	-	-

bp	Flag	Description
7	P5IMD7	I/O port or BUZZER selection 0: P57 1: BUZZERA
6	P5IMD6	I/O port or NBUZZER selection 0: P56 1: NBUZZERA
5 to 0	-	-

■ Pull-up/pull-down Resistor Selection Register (SELUD: 0x03EAF)

<MN101EF76K/MN101EF56K>

bp	7	6	5	4	3	2	1	0
Flag	SELUD7	-	SELUD5	SELUD4	-	-	SELUD1	SELUD0
At reset	0	-	0	0	-	-	0	0
Access	R/W	-	R/W	R/W	-	-	R/W	R/W

bp	Flag	Description
7	SELUD7	Pull-up/pull-down selection (Port 7) 0: Pull-up 1: Pull-down
6	-	-
5	SELUD5	Pull-up/pull-down selection (Port 5) 0: Pull-up 1: Pull-down
4	SELUD4	Pull-up/pull-down selection (Port 4) 0: Pull-up 1: Pull-down
3 to 2	-	-
1	SELUD0	Pull-up/pull-down selection (Port 1) 0: Pull-up 1: Pull-down
0	SELUD0	Pull-up/pull-down selection (Port 0) 0: Pull-up 1: Pull-down

<MN101EF57G>

bp	7	6	5	4	3	2	1	0
Flag	SELUD7	-	SELUD5	SELUD4	-	-	-	SELUD0
At reset	0	-	0	0	-	-	-	0
Access	R/W	-	R/W	R/W	-	-	-	R/W

bp	Flag	Description
7	SELUD7	Pull-up/pull-down selection (Port 7) 0: Pull-up 1: Pull-down
6	-	-
5	SELUD5	Pull-up/pull-down selection (Port 5) 0: Pull-up 1: Pull-down
4	SELUD4	Pull-up/pull-down selection (Port 4) 0: Pull-up 1: Pull-down
3 to 1	-	-
0	SELUD0	Pull-up/pull-down selection (Port 0) 0: Pull-up 1: Pull-down

8.8.3 Block Diagram (MN101EF76K/MN101EF56K)

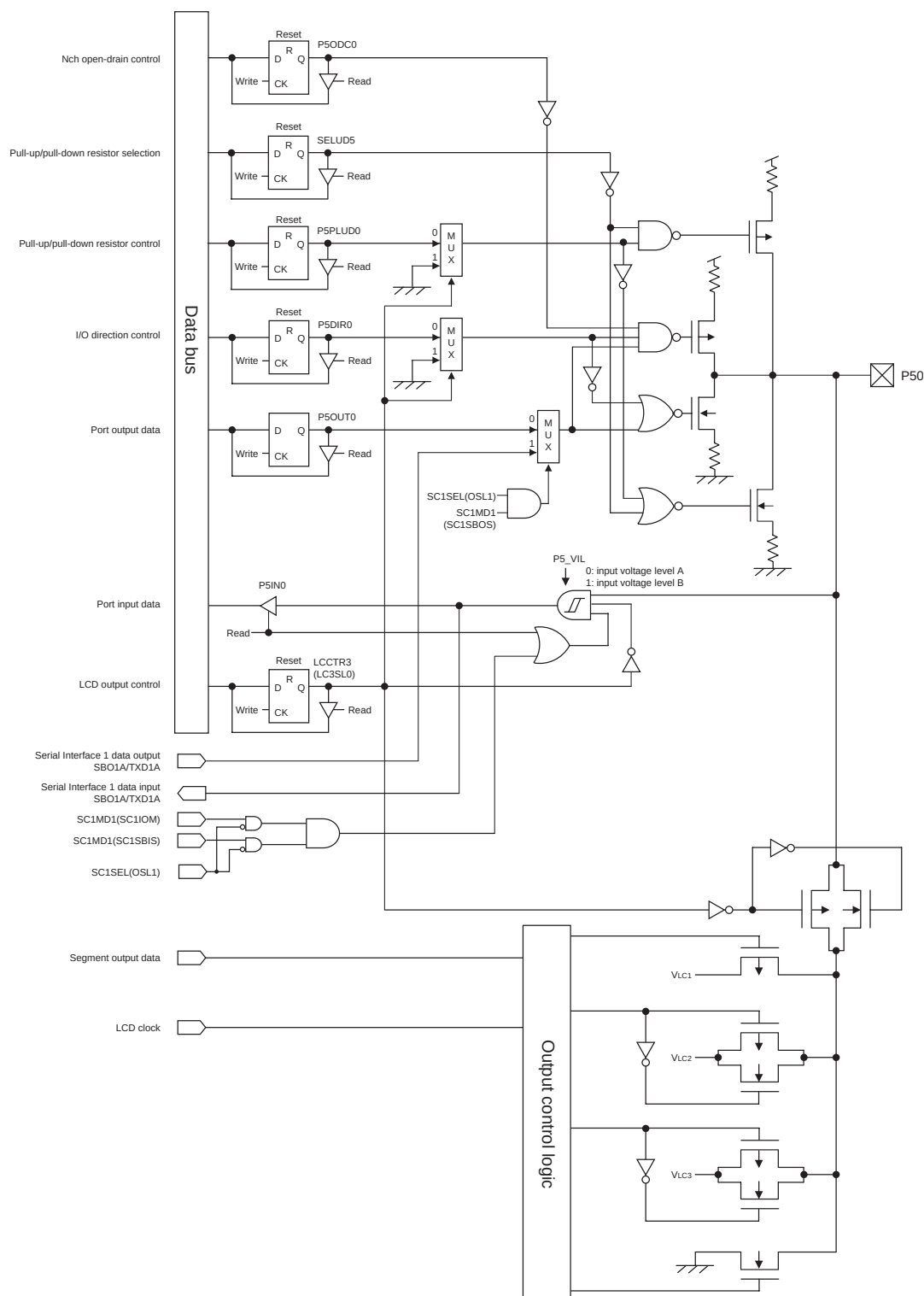


Figure:8.8.1 P50 Block Diagram (MN101EF76K/MN101EF56K)

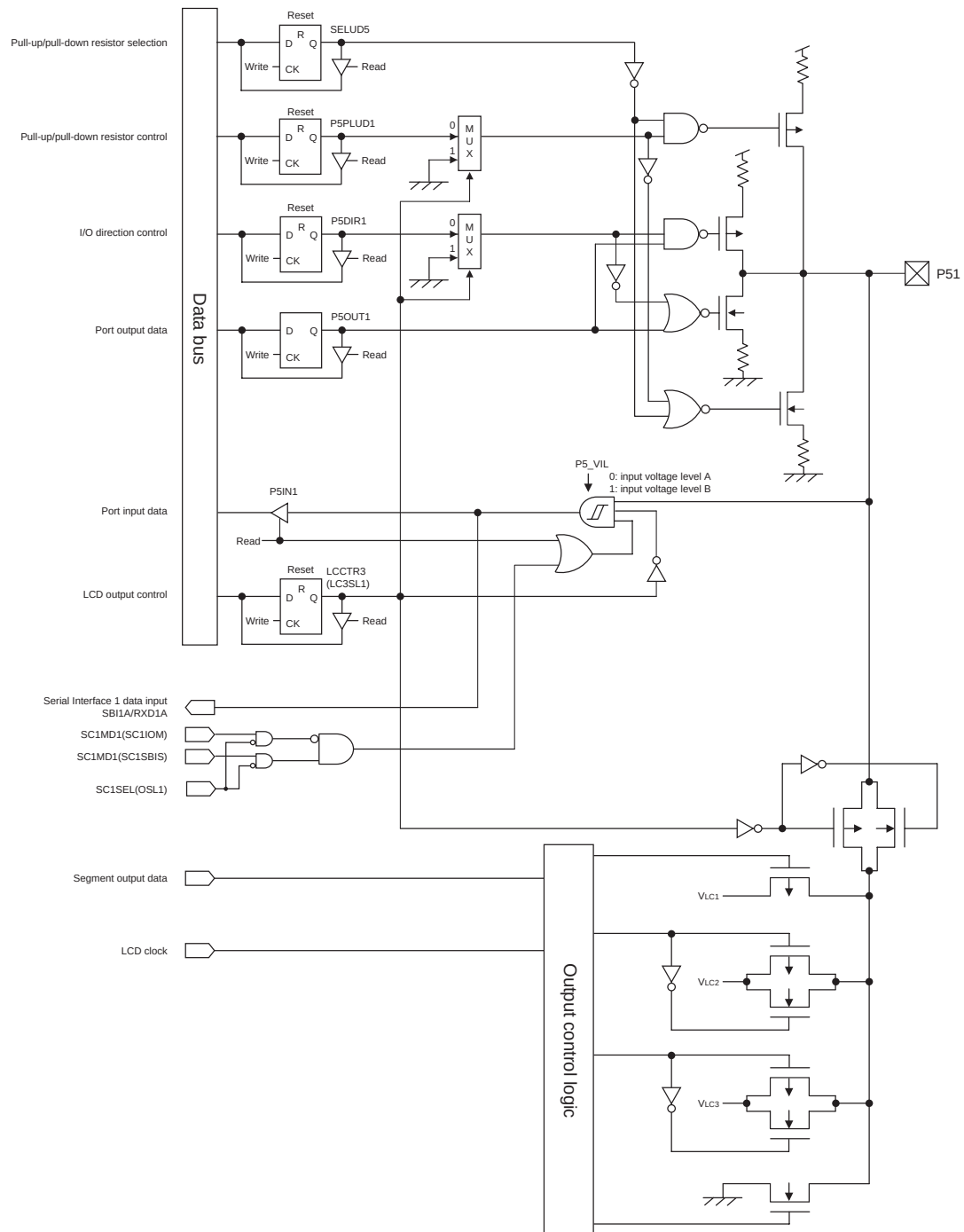


Figure:8.8.2 P51 Block Diagram (MN101EF76K/MN101EF56K)

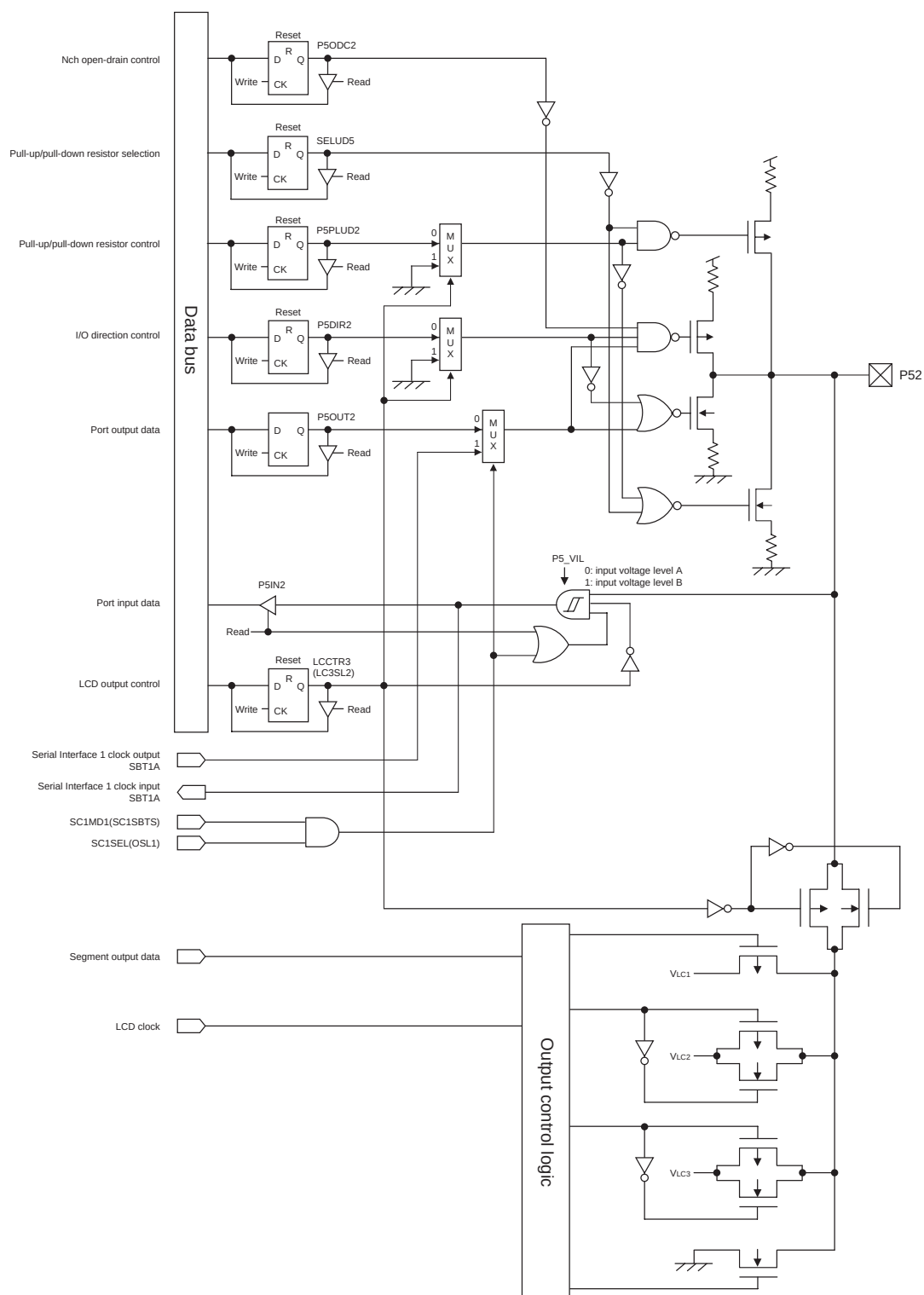


Figure:8.8.3 P52 Block Diagram (MN101EF76K/MN101EF56K)

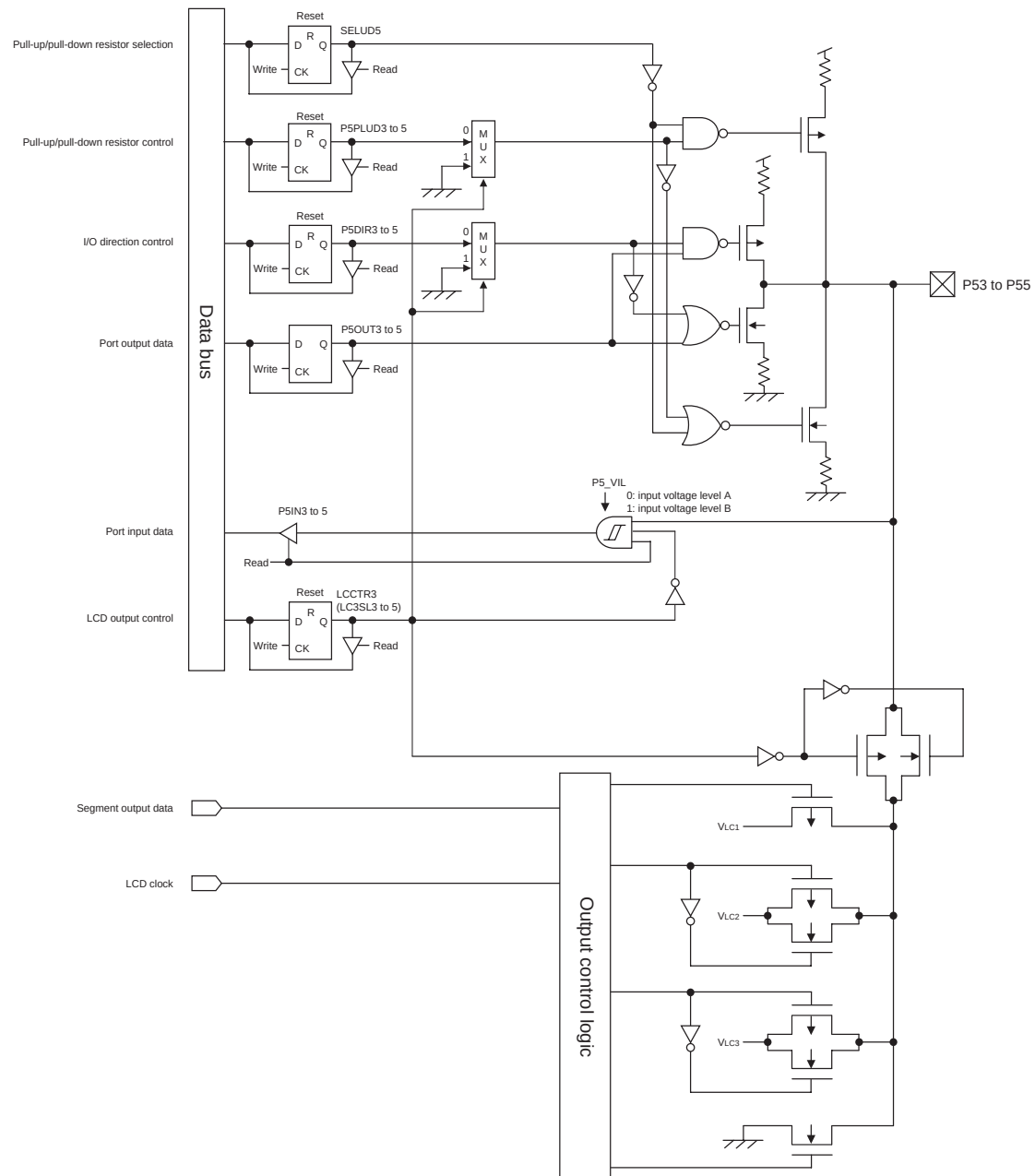


Figure:8.8.4 P53 to P55 Block Diagram (MN101EF76K/MN101EF56K)

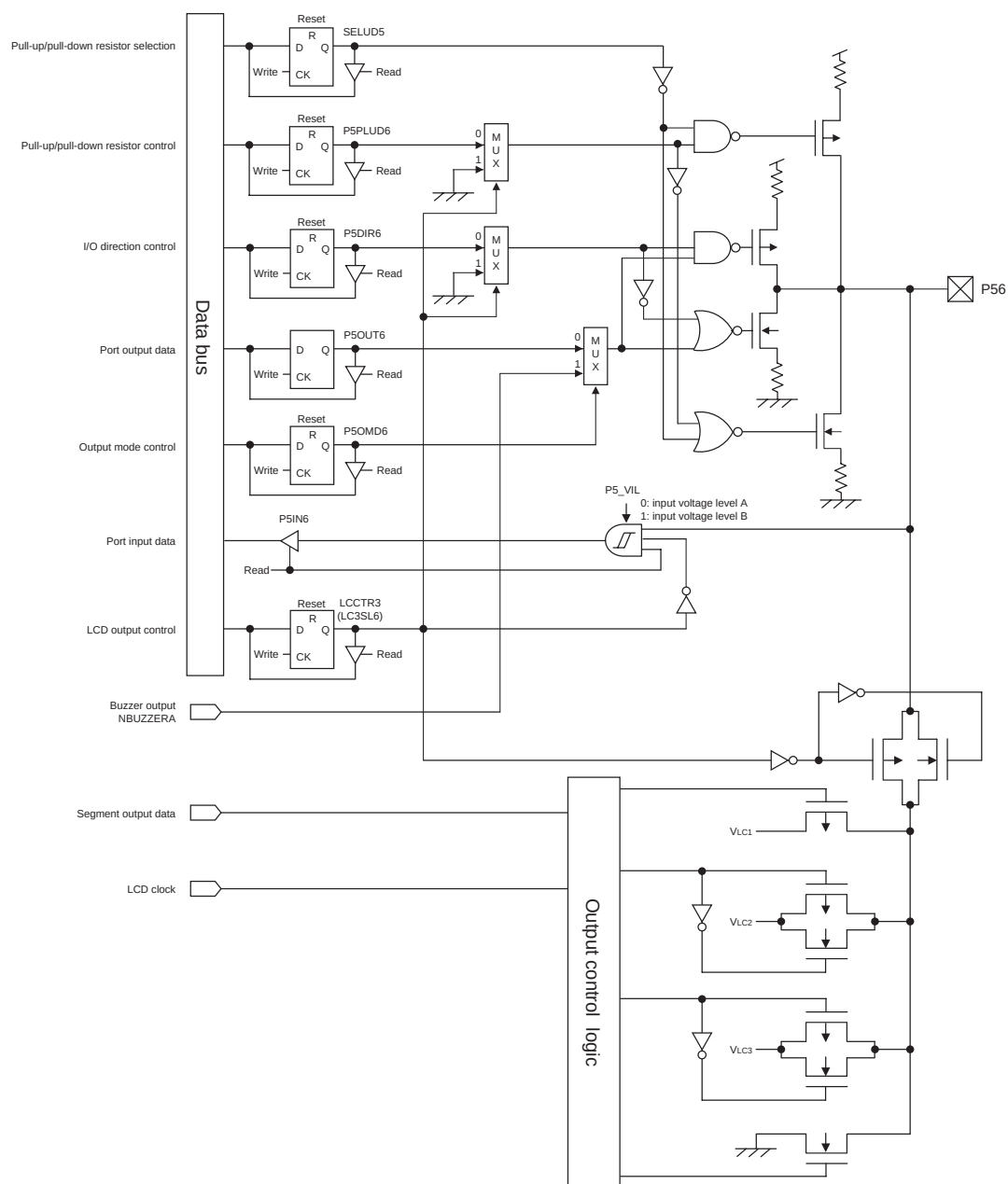


Figure:8.8.5 P56 Block Diagram (MN101EF76K/MN101EF56K)

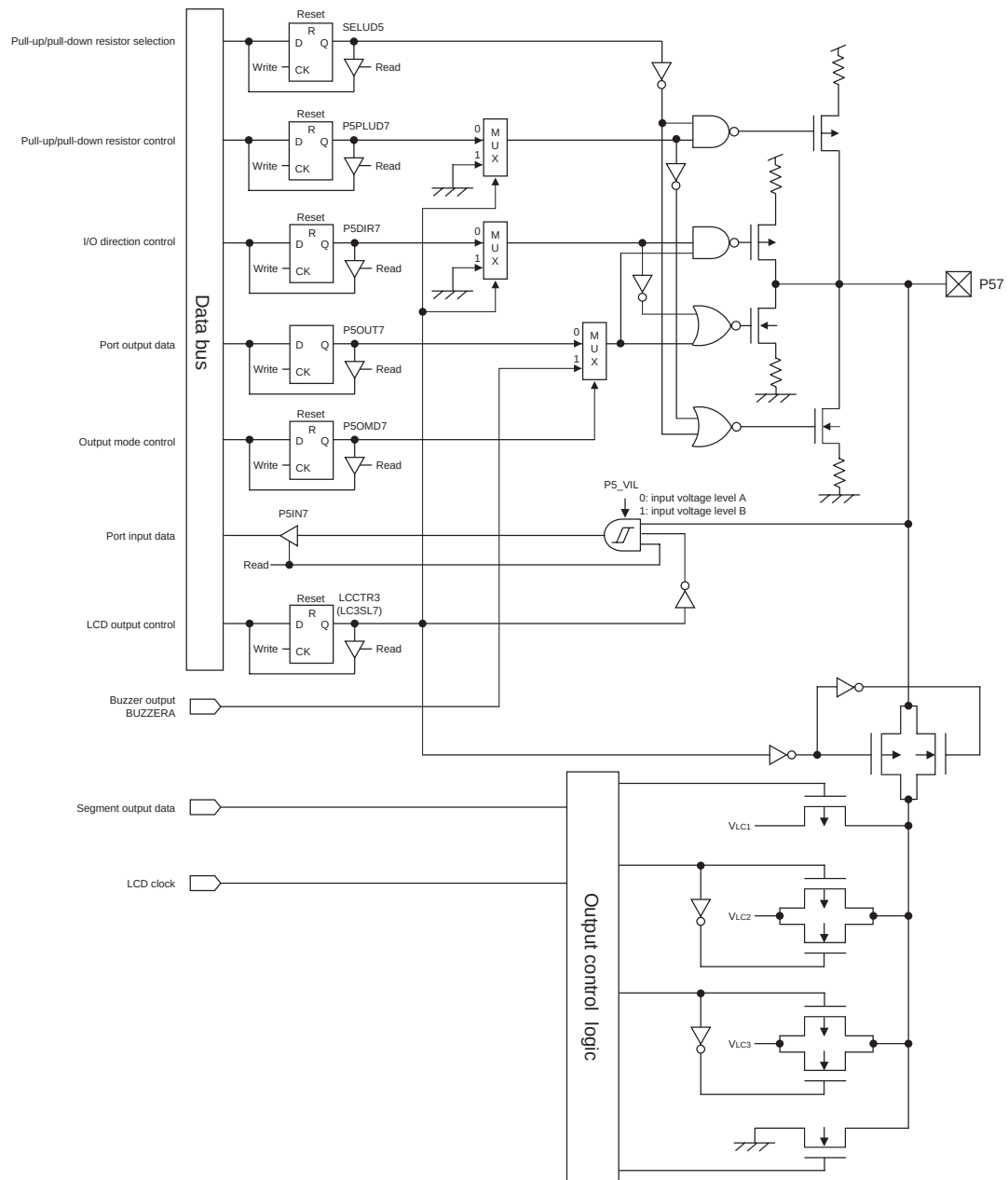


Figure:8.8.6 P57 Block Diagram (MN101EF76K/MN101EF56K)

8.8.4 Block Diagram (MN101EF57G)

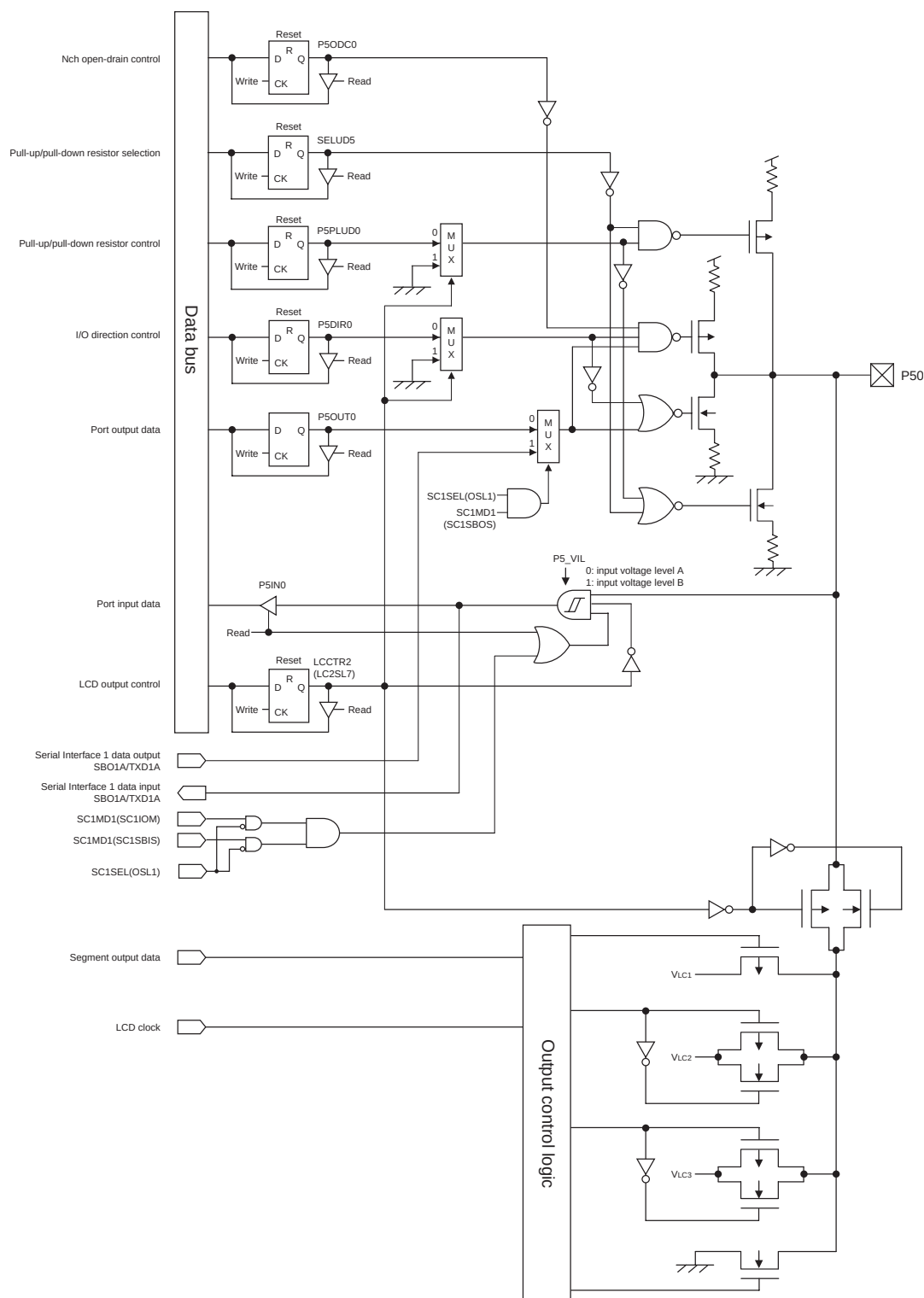


Figure:8.8.7 P50 Block Diagram (MN101EF57G)

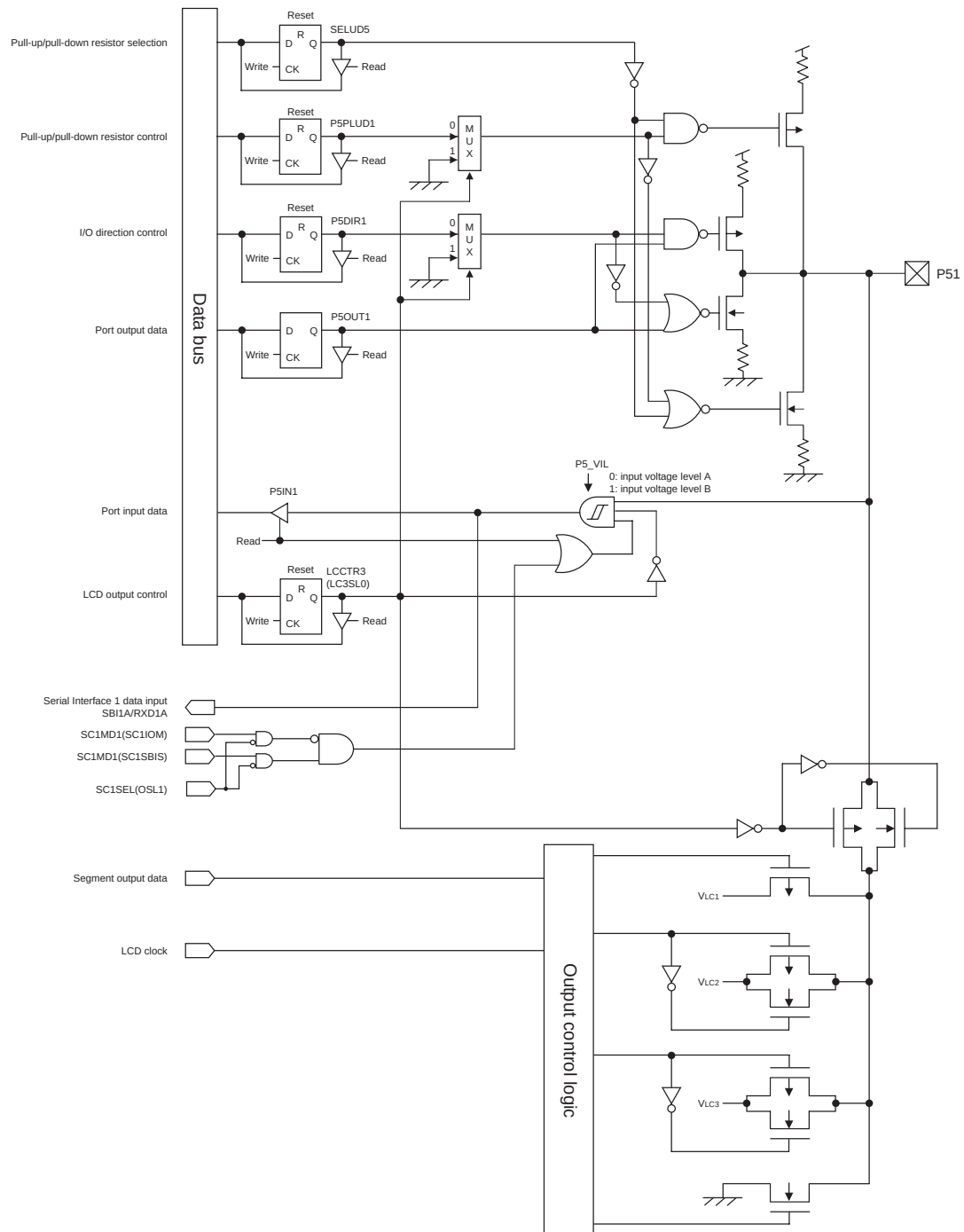


Figure:8.8.8 P51 Block Diagram (MN101EF57G)

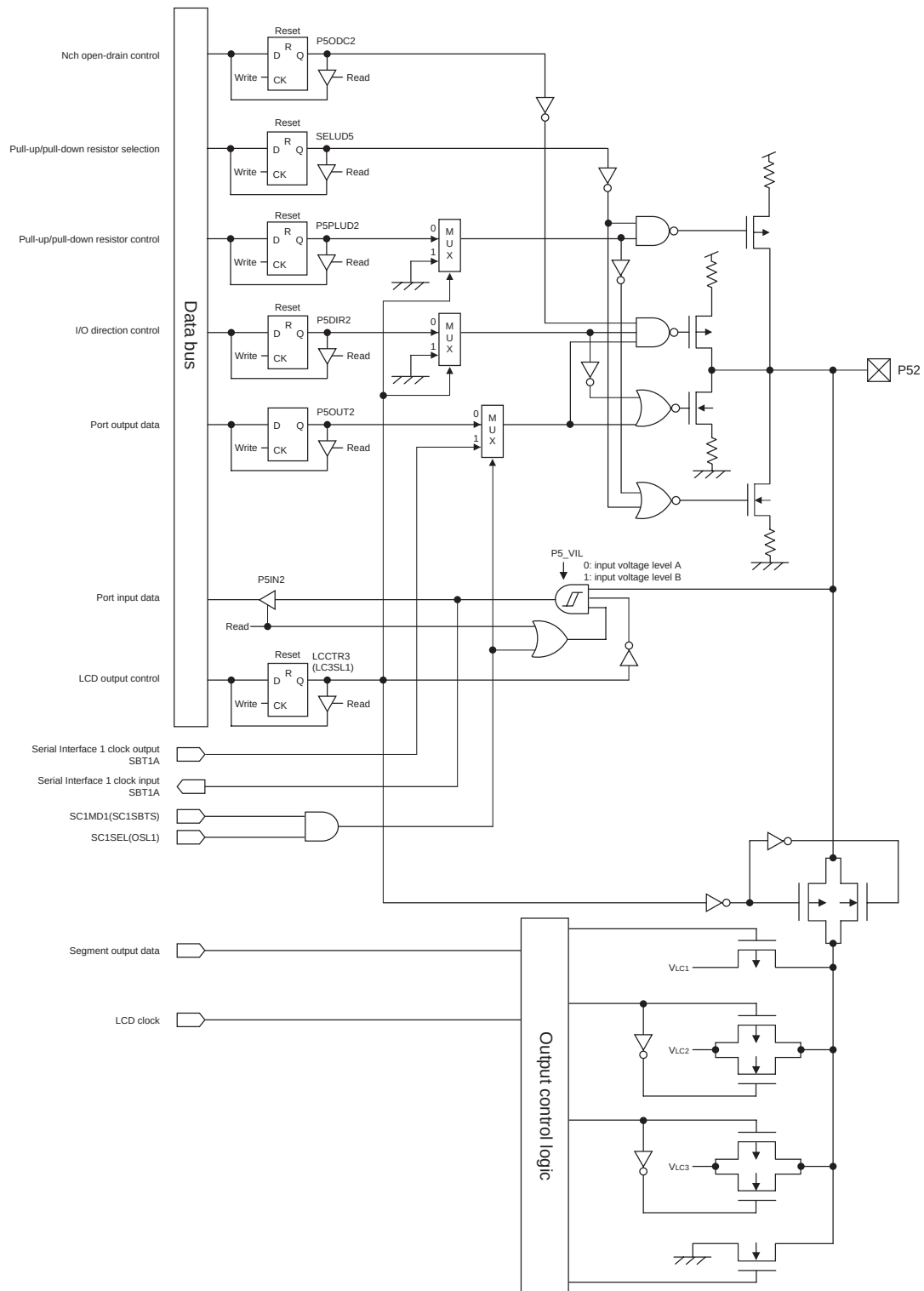


Figure:8.8.9 P52 Block Diagram (MN101EF57G)

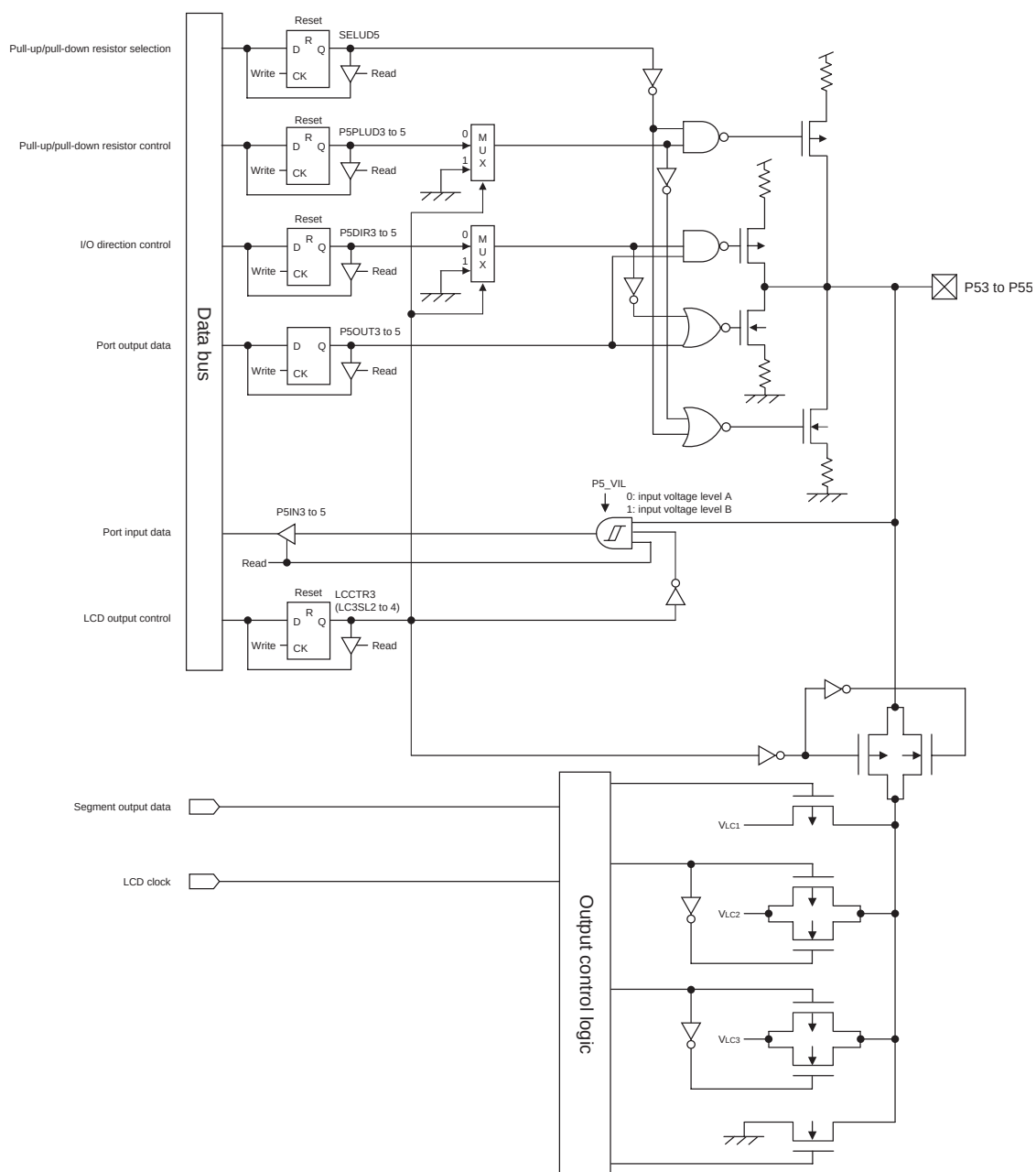


Figure:8.8.10 P53 to P55 Block Diagram (MN101EF57G)

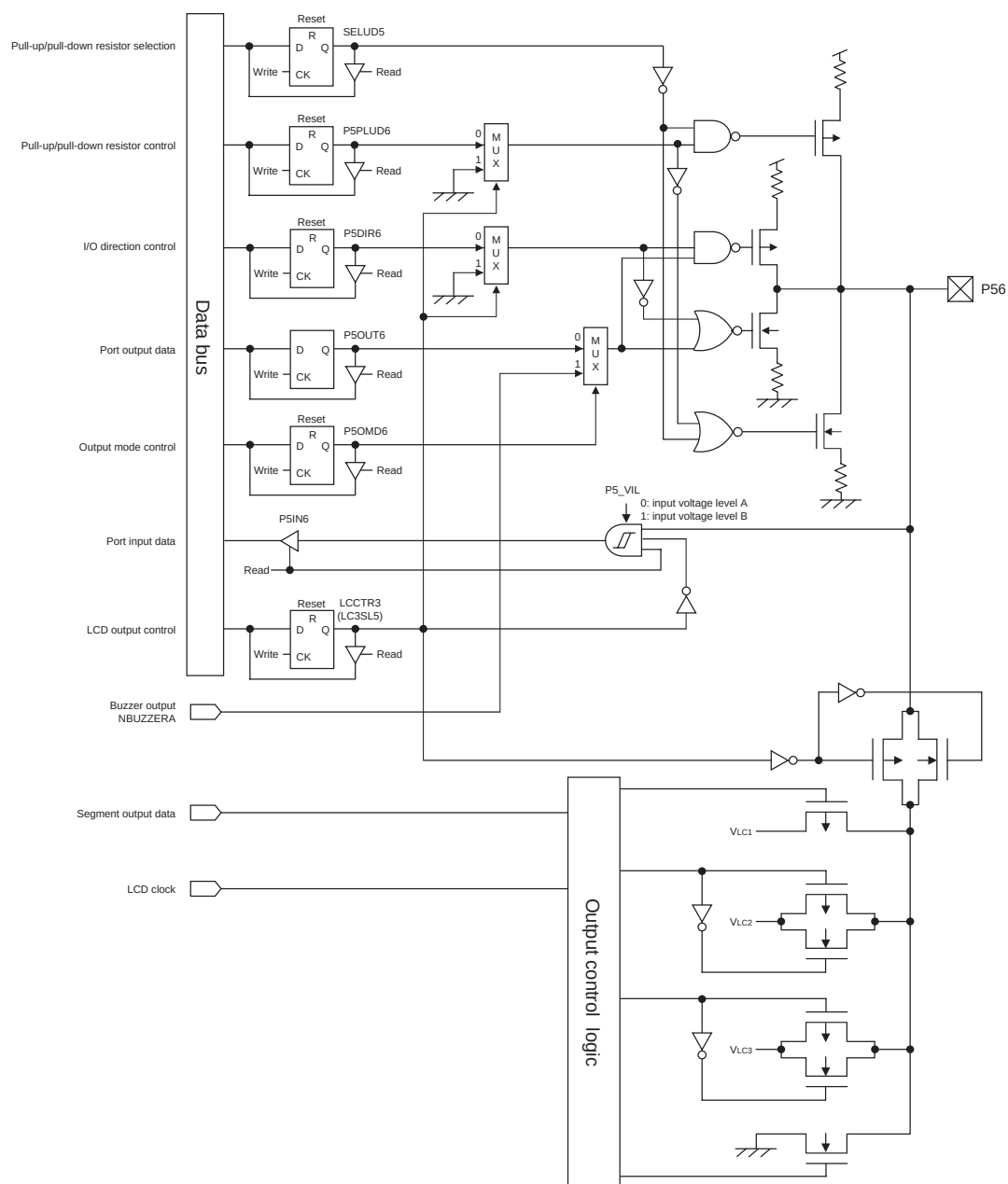


Figure:8.8.11 P56 Block Diagram (MN101EF57G)

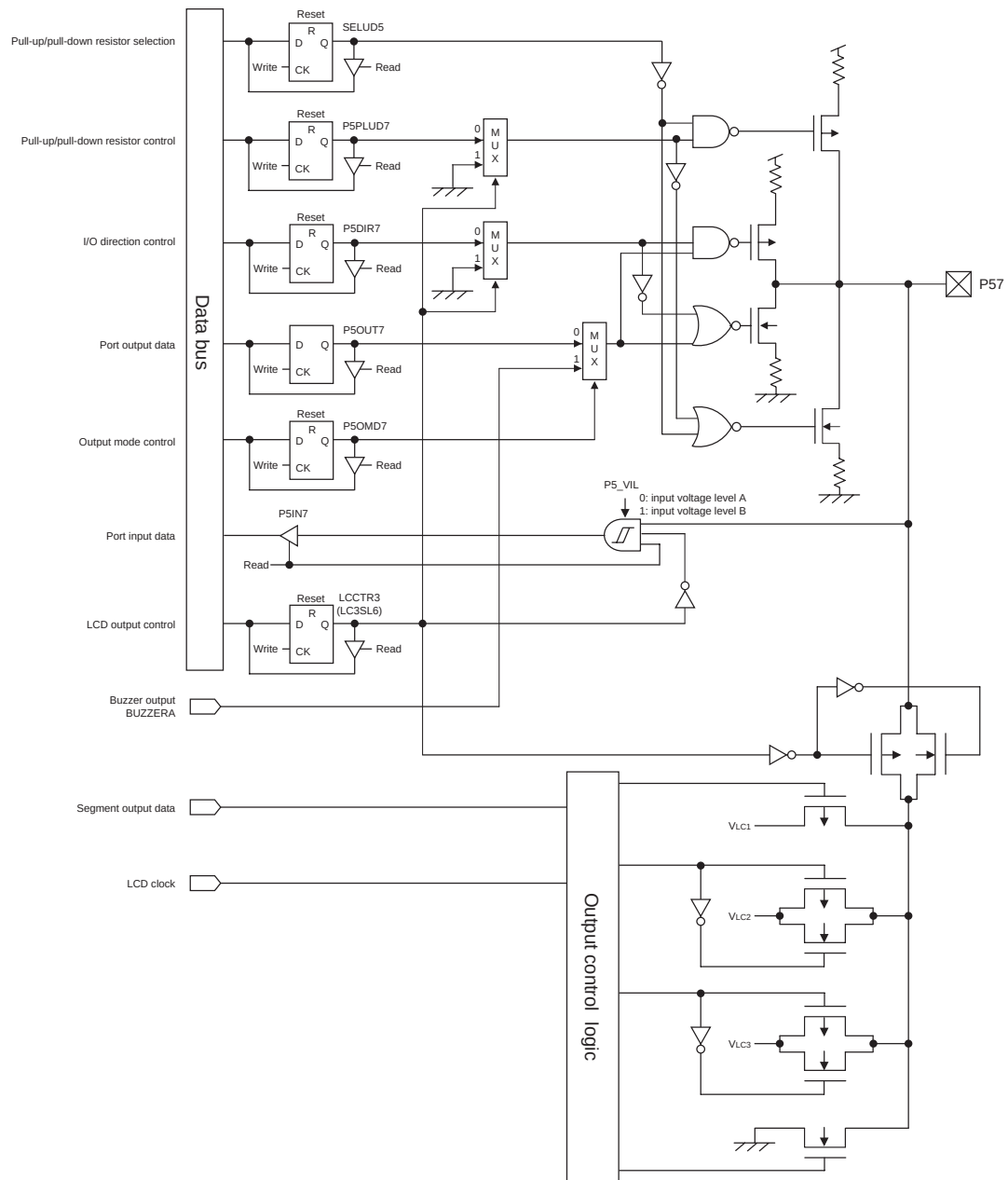


Figure:8.8.12 P57 Block Diagram (MN101EF57G)

8.9 Port 6

8.9.1 Description

■ General-purpose Port Setup

To output data to pin, set the control flag of Port 6 direction control register (P6DIR) to "1" and write the value of Port 6 output register (P6OUT).

To read input data of pin, set the control flag of Port 6 direction control register (P6DIR) to "0" and read the value of Port 6 input register (P6IN).

Each bit can be set individually as either an input or output by Port 6 direction control register (P6DIR). The control flag of Port 6 direction control register (P6DIR) is set to "1" for output mode, and "0" for input mode.

Each bit can be set individually to add a pull-up resistor by Port 6 pull-up resistor control register (P6PLU). Set the control flag of P6PLU register to "1" to add pull-up resistor.

Each bit can be selected individually as Nch open-drain output by Port 6 Nch open-drain control register (P6ODC). The control flag of P6ODC register is set to "1" for Nch open-drain output, and "0" for push-pull output.

Table:8.9.1 Port 6 Functions

Table remarks -: Without function

MN101EF76K/MN101EF56K				MN101EF57G			
Pin	Special function			Pin	Special function		
P61	SEG15	DA_C		P61	-	-	
P62	SEG14	TM1IOB		P62	SEG14	TM1IOB	
P63	SEG13	TM3IOB		P63	SEG13	TM3IOB	
P64	SEG12	TM4IOB		P64	SEG12	TM4IOB	
P65	SEG11	SB02A	TXD2A	P65	SEG11	SB02A	TXD2A
P66	SEG10	SBI2A	RXD2A	P66	SEG10	SBI2A	RXD2A
P67	SEG9	SBT2A		P67	SEG9	SBT2A	

■ Special Function Pin Setup

P61 is also used as the analog output pin for D/A.

P61 to P67 are also used as LCD segment output pins.

LCD segment output pin is selected by setting the corresponding flag of LCD output control register n (LCCTRn) to "1".

Port and segment can be switched by one bit. When segment output pin is selected, this pin is forced to be output mode and pull-up resistor is not added.

P62 is also used as the I/O pin for Timer 1.

P63 is also used as the I/O pin for Timer 3.

P64 is also used as the I/O pin for Timer 4.

P65 is also used as the I/O pin for Serial Interface 2 transmission/reception data and the output pin of UART2 transmission data. When the SC2SBOS flag of Serial Interface 2 mode register 1 (SC2MD1) is "1", P65 is the serial data I/O pin. Push-pull output or Nch open-drain output can be selected by setting Port 6 Nch open-drain control register (P6ODC).

P66 is also used as the input pin for Serial Interface 2 reception data and UART2 reception data. When the SC2BIS flag of Serial Interface 2 mode register 1 (SC2MD1) is "1", P66 is the serial data input pin.

P67 is also used as the I/O pin for Serial Interface 2 clock. When the SC2SBTS flag of Serial Interface 2 mode register 1 (SC2MD1) is "1", P67 is the I/O pin of the serial clock. Push-pull output or Nch open-drain output can be selected by setting Port 6 Nch open-drain control register (P6ODC).

For P62 to P63 each bit can be set individually as either timer output or port output by Port 6 output mode register (P6OMD). The corresponding flag of P6OMD register is set to "1" for special function output, and "0" for general-purpose port.

8.9.2 Registers

Table:8.9.2 shows registers that control Port 6.

Table:8.9.2 Port 6 Control Register

Table remarks √: With function -: Without function

Registers	Address	R/W	Function	Page	MN101 EF76K	MN101 EF57G	MN101 EF56K
P6OUT	0x03E76	R/W	Port 6 output register	VIII-111	√	√	√
P6IN	0x03E86	R	Port 6 input register	VIII-112	√	√	√
P6DIR	0x03E96	R/W	Port 6 direction control register	VIII-113	√	√	√
P6PLU	0x03EA6	R/W	Port 6 pull-up resistor control register	VIII-114	√	√	√
P6OMD	0x03EB6	R/W	Port 6 input mode register	VIII-115	√	√	√
P6ODC	0x03EF6	R/W	Port 6 Nch open-drain control register	VIII-115	√	√	√

R/W: Readable/Writable

R: Read only

■ Port 6 Output Register (P6OUT: 0x03E76)

<MN101EF76K/MN101EF56K>

bp	7	6	5	4	3	2	1	0
Flag	P6OUT7	P6OUT6	P6OUT5	P6OUT4	P6OUT3	P6OUT2	P6OUT1	-
At reset	X	X	X	X	X	X	X	-
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	-

bp	Flag	Description
7 to 1	P6OUT7 to 1	Output data 0: Output "Low" (V_{SS} -level) 1: Output "High" (V_{DD5} -level)
0	-	-

<MN101EF56G>

bp	7	6	5	4	3	2	1	0
Flag	P6OUT7	P6OUT6	P6OUT5	P6OUT4	P6OUT3	P6OUT2	-	-
At reset	X	X	X	X	X	X	-	-
Access	R/W	R/W	R/W	R/W	R/W	R/W	-	-

bp	Flag	Description
7 to 2	P6OUT7 to 2	Output data 0: Output "Low" (V_{SS} -level) 1: Output "High" (V_{DD5} -level)
1 to 0	-	-

■ Port 6 Input Register (P6IN: 0x03E86)

<MN101EF76K/MN101EF56K>

bp	7	6	5	4	3	2	1	0
Flag	P6IN7	P6IN6	P6IN5	P6IN4	P6IN3	P6IN2	P6IN1	-
At reset	X	X	X	X	X	X	X	-
Access	R	R	R	R	R	R	R	-

bp	Flag	Description
7 to 1	P6IN7 to 1	Input data 0: Pin is "Low" (V_{SS} -level) 1: Pin is "High" (V_{DD5} -level)
0	-	-

<MN101EF57G>

bp	7	6	5	4	3	2	1	0
Flag	P6IN7	P6IN6	P6IN5	P6IN4	P6IN3	P6IN2	-	-
At reset	X	X	X	X	X	X	-	-
Access	R	R	R	R	R	R	-	-

bp	Flag	Description
7 to 2	P6IN7 to 2	Input data 0: Pin is "Low" (V_{SS} -level) 1: Pin is "High" (V_{DD5} -level)
1 to 0	-	-

■ Port 6 Direction Control Register (P6DIR: 0x03E96)

<MN101EF76K/MN101EF56K>

bp	7	6	5	4	3	2	1	0
Flag	P6DIR7	P6DIR6	P6DIR5	P6DIR4	P6DIR3	P6DIR2	P6DIR1	-
At reset	0	0	0	0	0	0	0	-
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	-

bp	Flag	Description
7 to 1	P6DIR7 to 1	I/O mode selection 0: Input mode 1: Output mode
0	-	-

<MN101EF57G>

bp	7	6	5	4	3	2	1	0
Flag	P6DIR7	P6DIR6	P6DIR5	P6DIR4	P6DIR3	P6DIR2	-	-
At reset	0	0	0	0	0	0	-	-
Access	R/W	R/W	R/W	R/W	R/W	R/W	-	-

bp	Flag	Description
7 to 2	P6DIR7 to 2	I/O mode selection 0: Input mode 1: Output mode
1 to 0	-	-

■ Port 6 Pull-up Resistor Control Register (P6PLU: 0x03EA6)

<MN101EF76K/MN101EF56K>

bp	7	6	5	4	3	2	1	0
Flag	P6PLU7	P6PLU6	P6PLU5	P6PLU4	P6PLU3	P6PLU2	P6PLU1	-
At reset	0	0	0	0	0	0	0	-
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	-

bp	Flag	Description
7 to 1	P6PLU7 to 1	Pull-up resistor selection 0: Not added 1: Added
0	-	-

<MN101EF57G>

bp	7	6	5	4	3	2	1	0
Flag	P6PLU7	P6PLU6	P6PLU5	P6PLU4	P6PLU3	P6PLU2	-	-
At reset	0	0	0	0	0	0	-	-
Access	R/W	R/W	R/W	R/W	R/W	R/W	-	-

bp	Flag	Description
7 to 2	P6PLU7 to 2	Pull-up resistor selection 0: Not added 1: Added
1 to 0	-	-

■ Port 6 Output Mode Register (P6OMD: 0x03EB6)

bp	7	6	5	4	3	2	1	0
Flag	-	-	-	P6OMD4	P6OMD3	P6OMD2	-	-
At reset	-	-	-	0	0	0	-	-
Access	-	-	-	R/W	R/W	R/W	-	-

bp	Flag	Description
7 to 5	-	-
4	P6OMD4	I/O port or TM4IOB selection 0: P64 1: TM4IOB
3	P6OMD3	I/O port or TM3IOB selection 0: P63 1: TM3IOB
2	P6OMD2	I/O port or TM1IOB selection 0: P62 1: TM1IOB
1 to 0	-	-

■ Port 6 Nch Open-drain Control Register P6ODC: 0x03EF6)

bp	7	6	5	4	3	2	1	0
Flag	P6ODC7	-	P6ODC5	-	-	-	-	-
At reset	0	-	0	-	-	-	-	-
Access	R/W	-	R/W	-	-	-	-	-

bp	Flag	Description
7	P6ODC7	Nch open-drain output selection 0: Push-pull output 1: Nch open-drain output
6	-	-
5	P6ODC5	Nch open-drain output selection 0: Push-pull output 1: Nch open-drain output
4 to 0	-	-

8.9.3 Block Diagram (MN101EF76K/MN101EF56K)

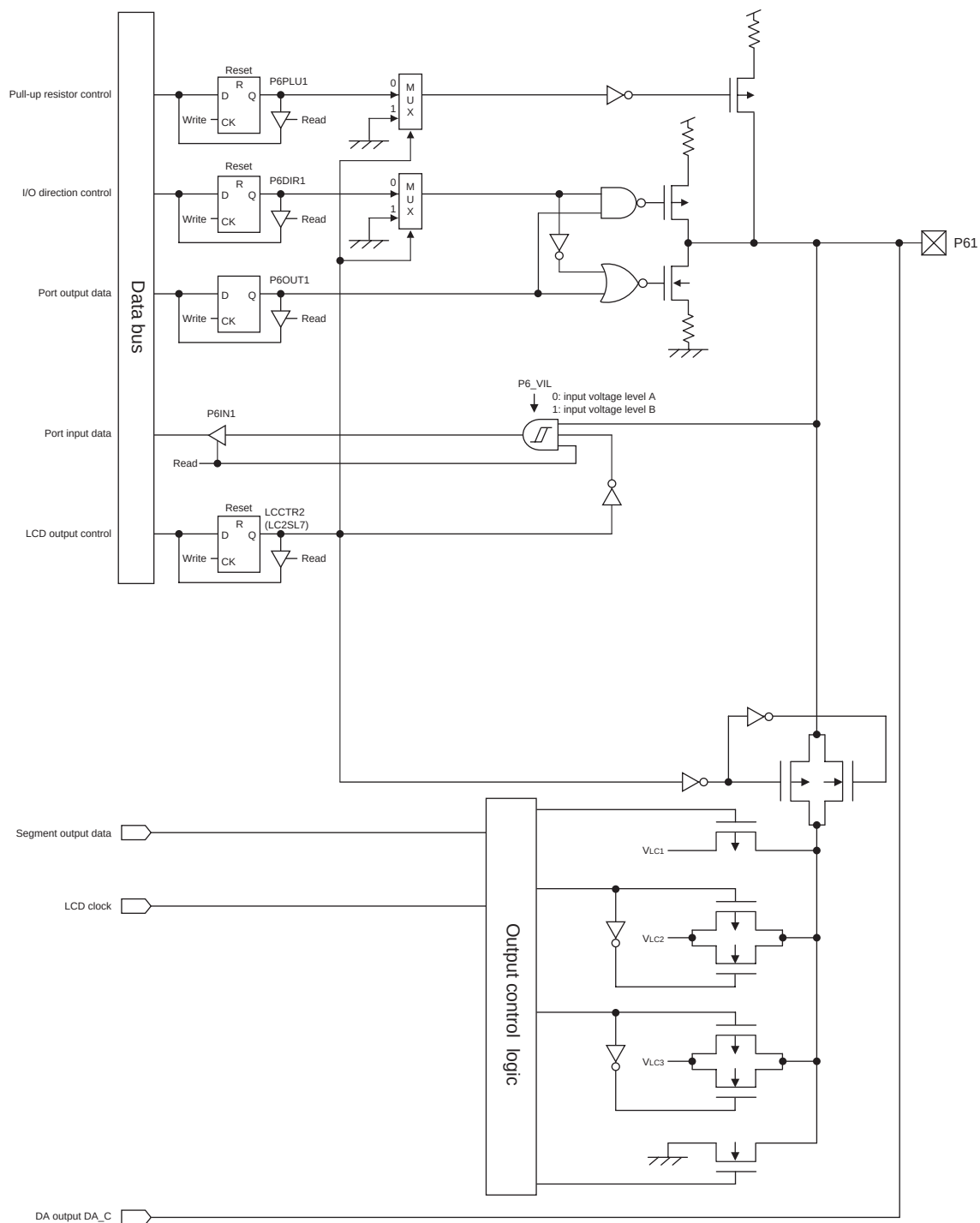


Figure:8.9.1 P61 Block Diagram (MN101EF76K/MN101EF56K)

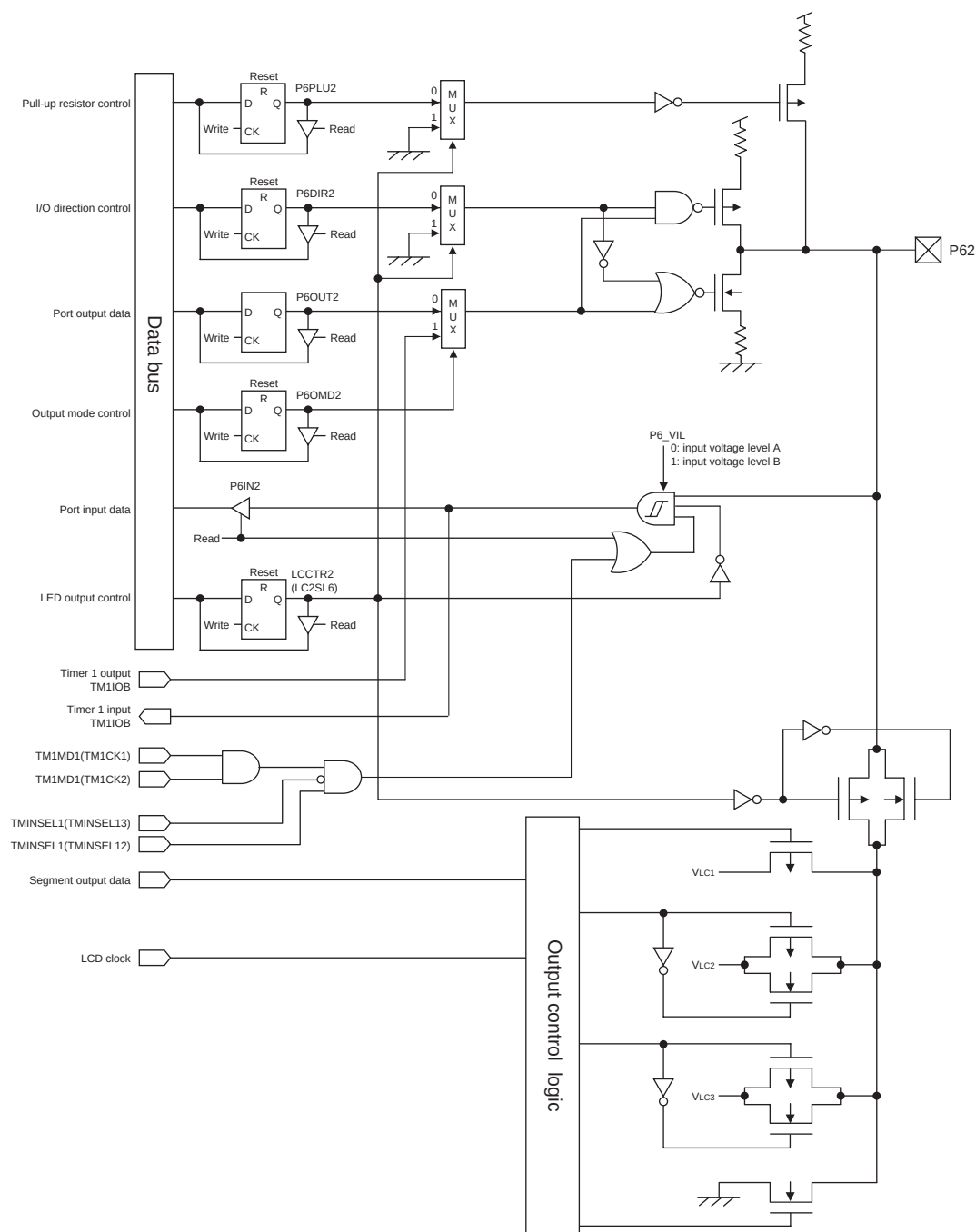


Figure:8.9.2 P62 Block Diagram (MN101EF76K/MN101EF56K)

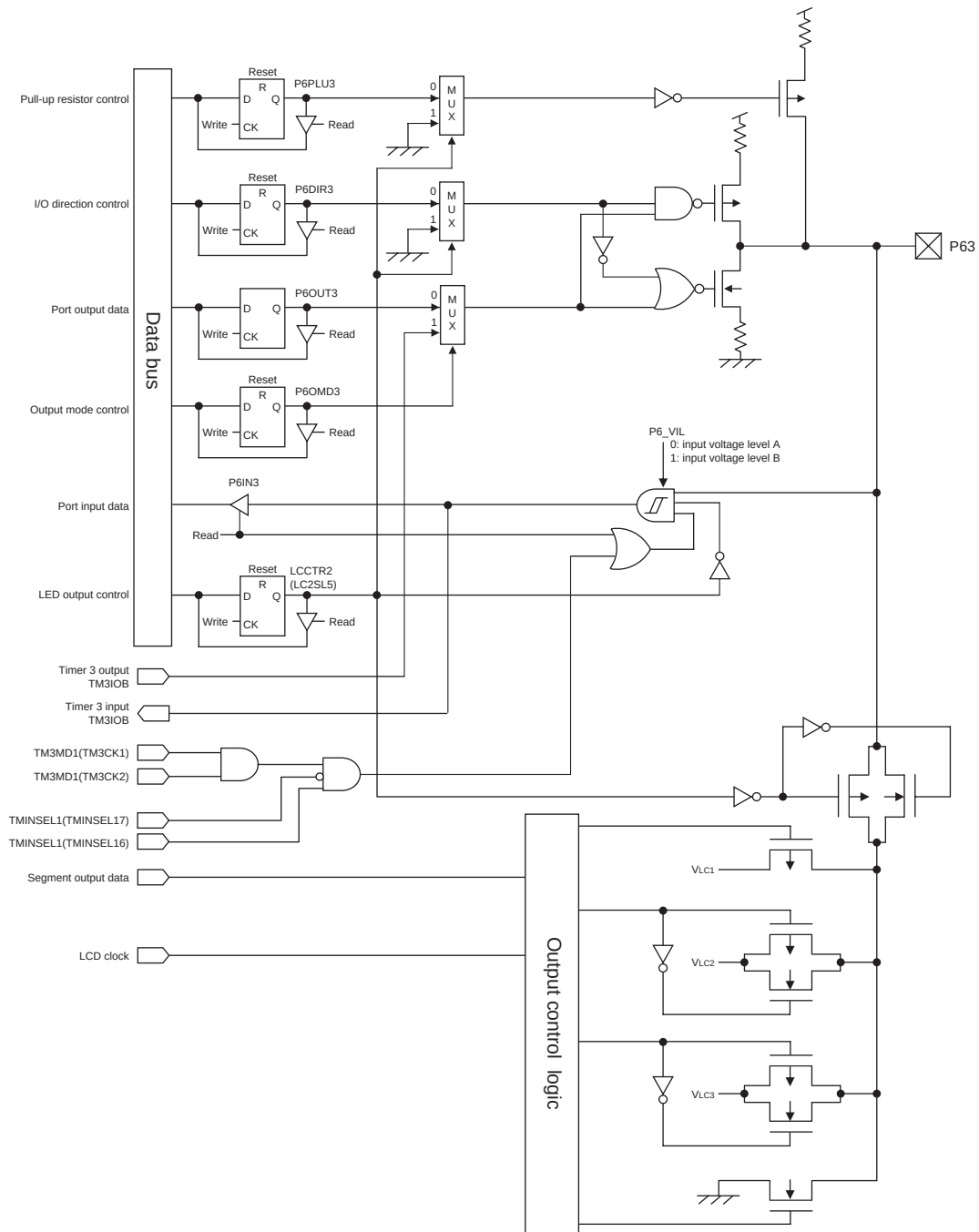


Figure:8.9.3 P63 Block Diagram (MN101EF76K/MN101EF56K)

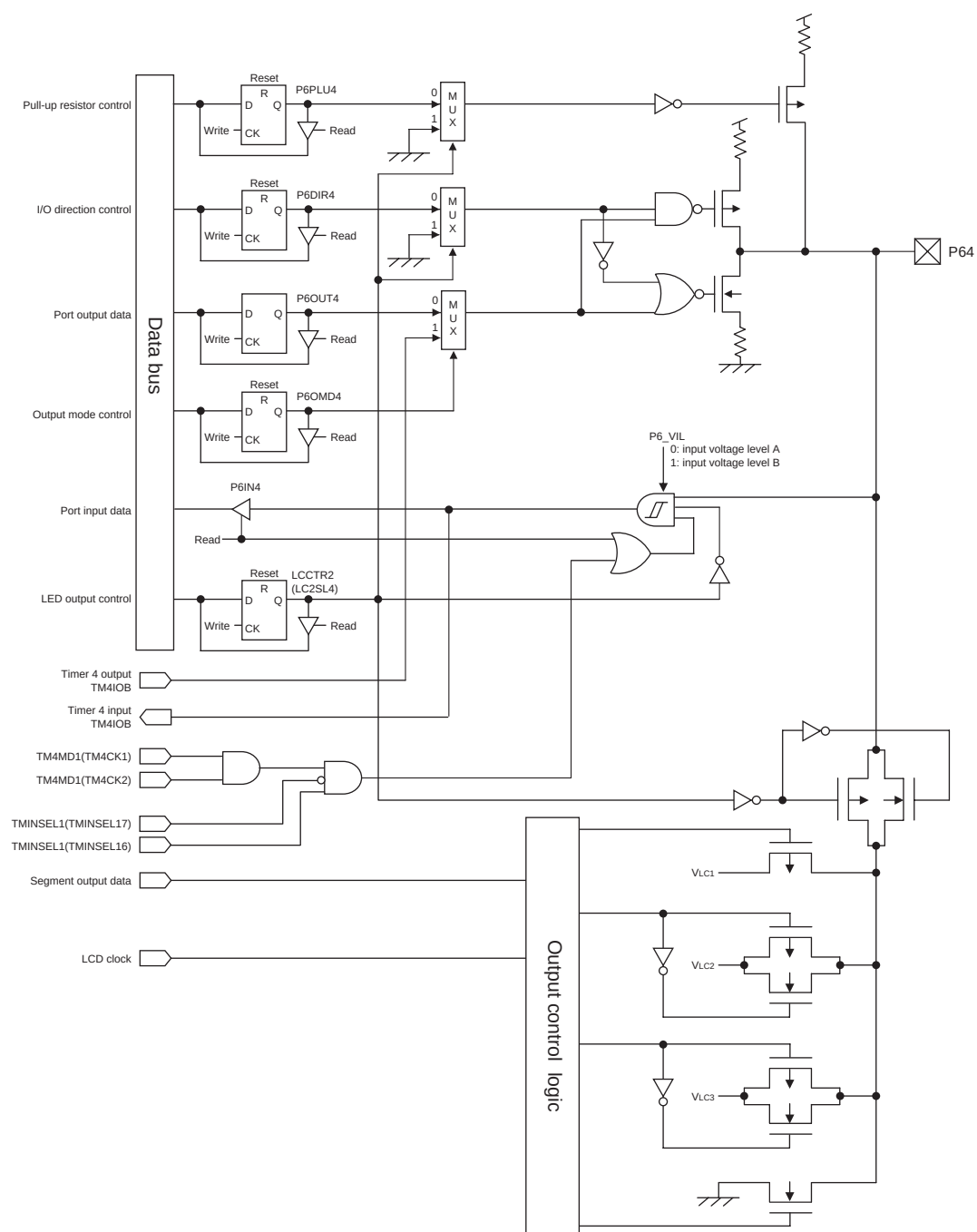


Figure:8.9.4 P64 Block Diagram (MN101EF76K/MN101EF56K)

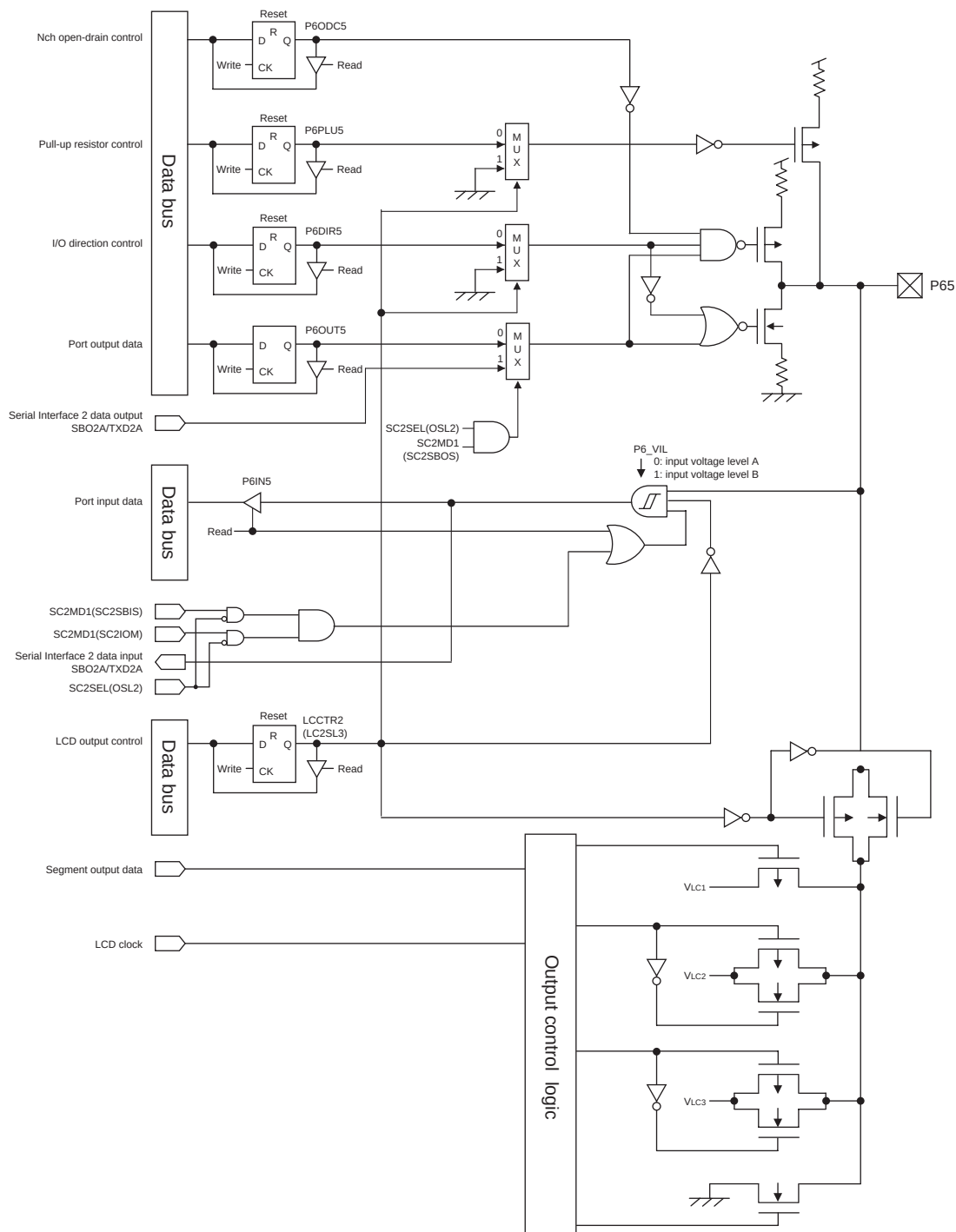


Figure:8.9.5 P65 Block Diagram (MN101EF76K/MN101EF56K)

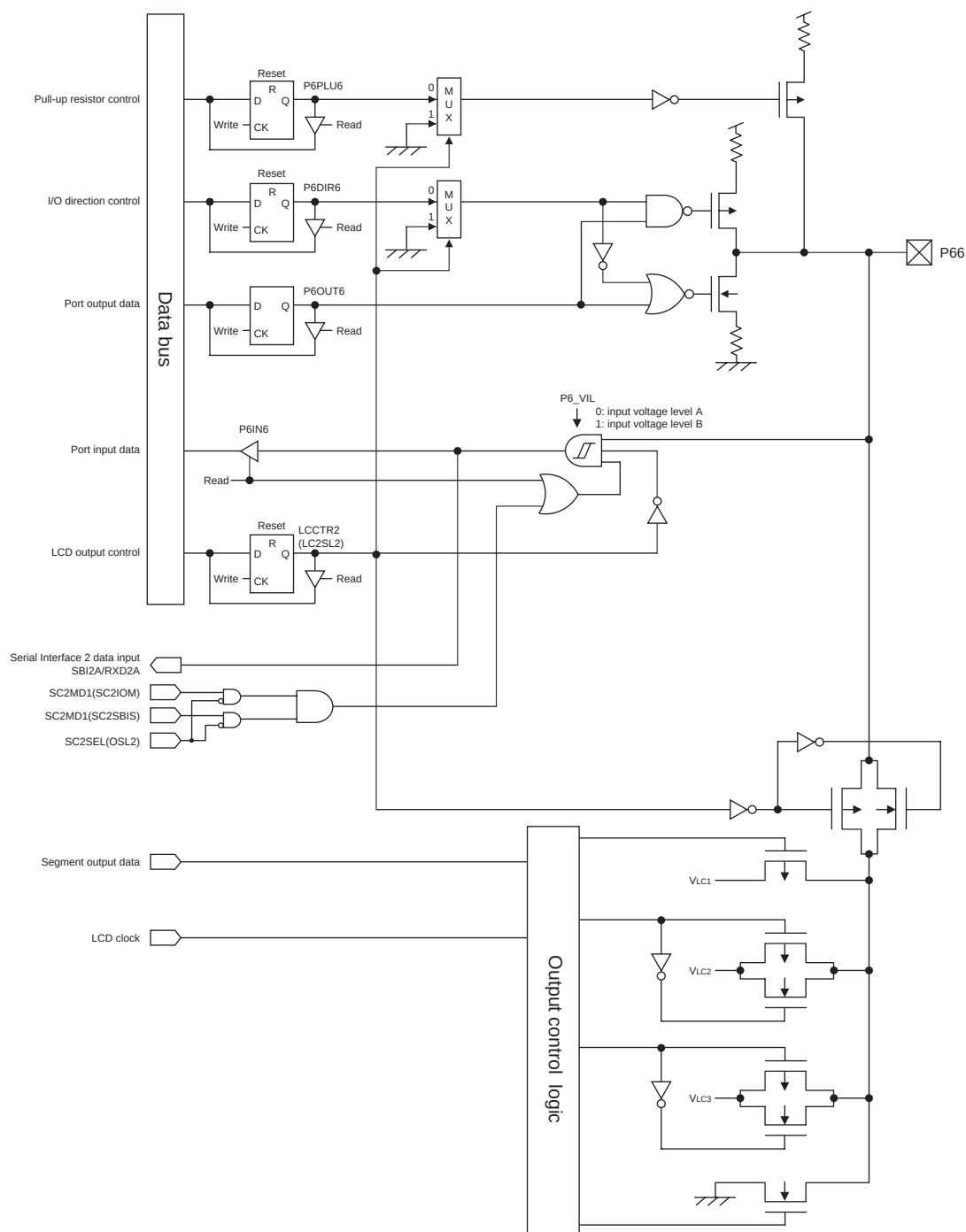


Figure:8.9.6 P66 Block Diagram (MN101EF76K/MN101EF56K)

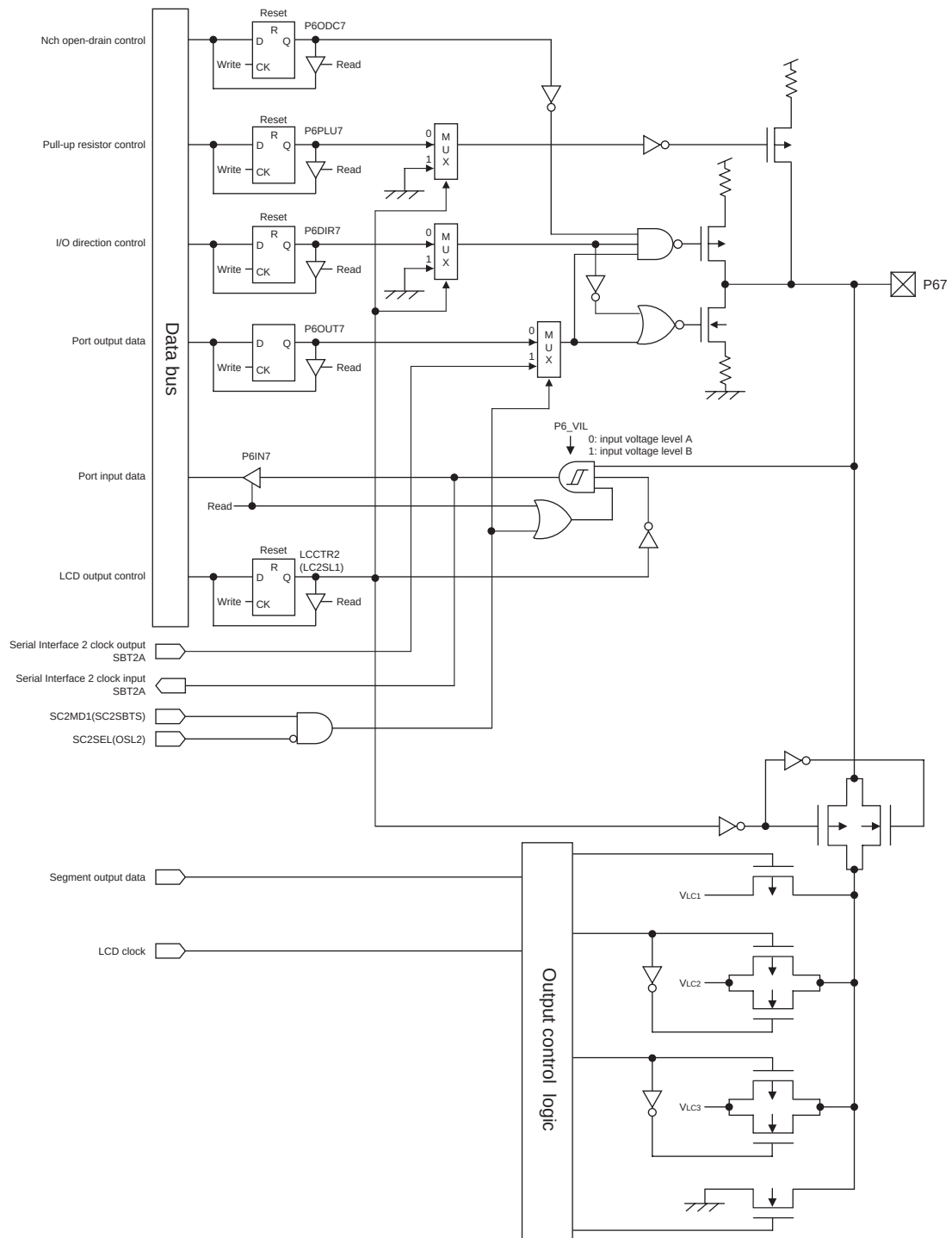


Figure:8.9.7 P67 Block Diagram (MN101EF76K/MN101EF56K)

8.9.4 Block Diagram (MN101EF57G)

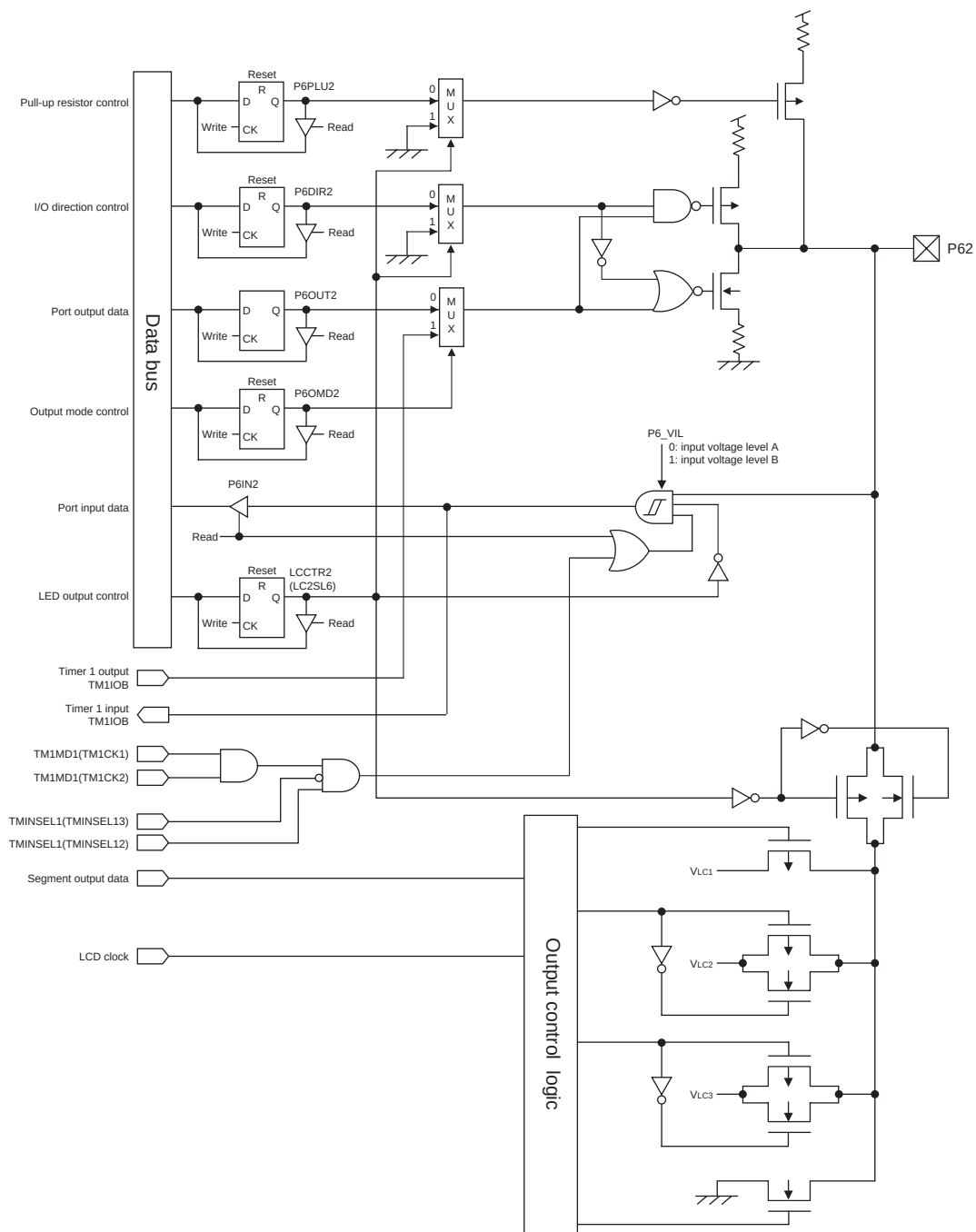


Figure:8.9.8 P62 Block Diagram (MN101EF57G)

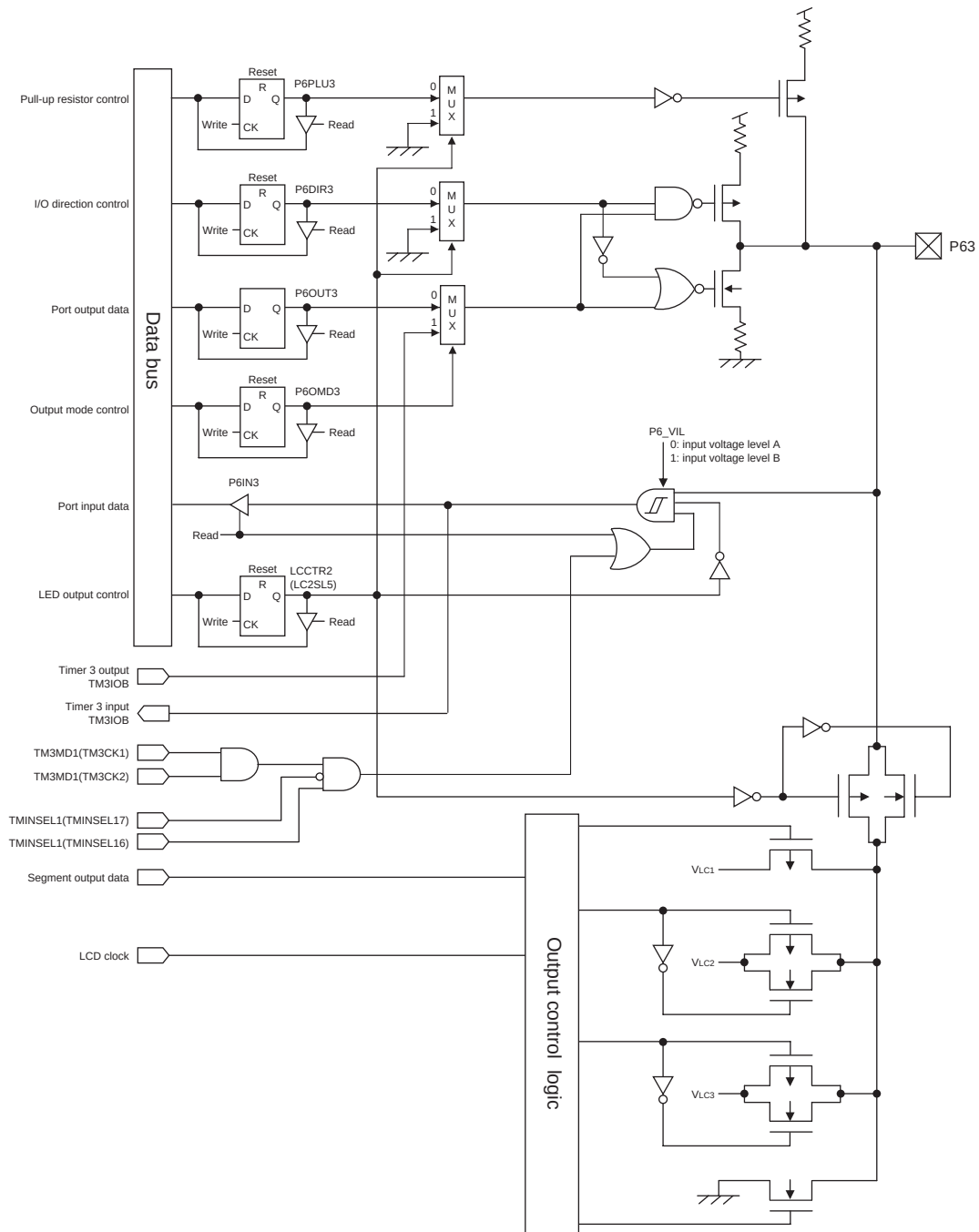


Figure:8.9.9 P63 Block Diagram (MN101EF57G)

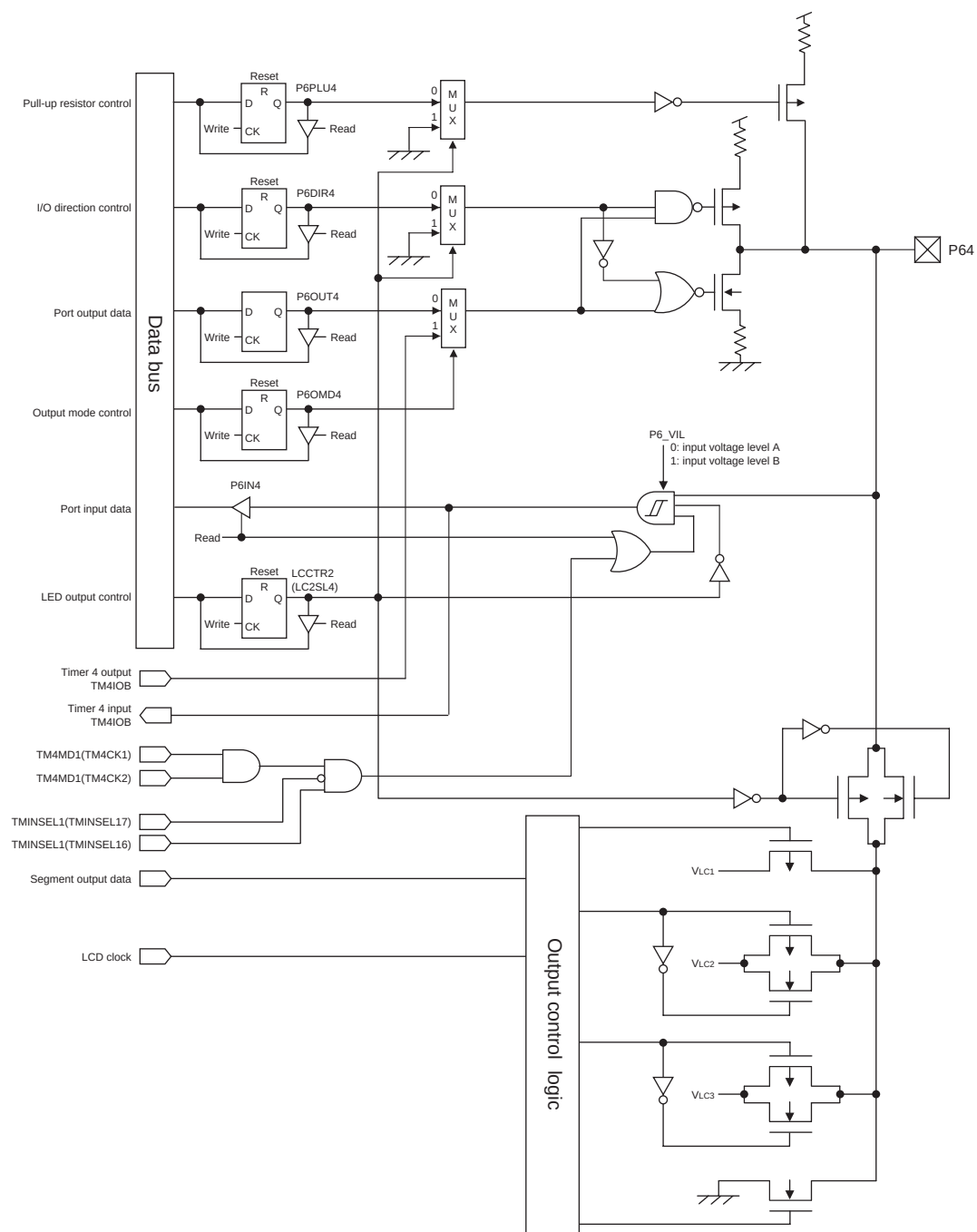


Figure:8.9.10 P64 Block Diagram (MN101EF57G)

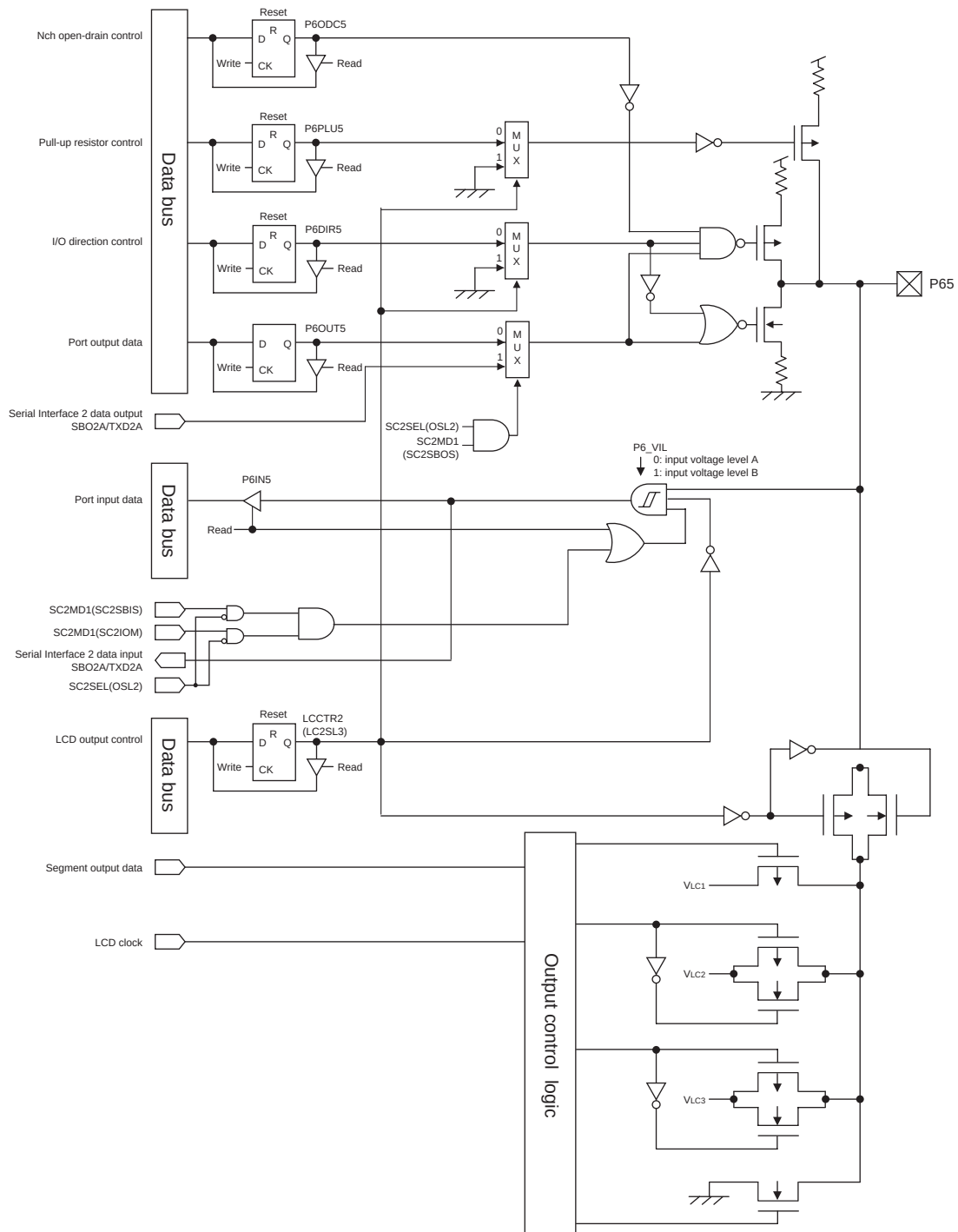
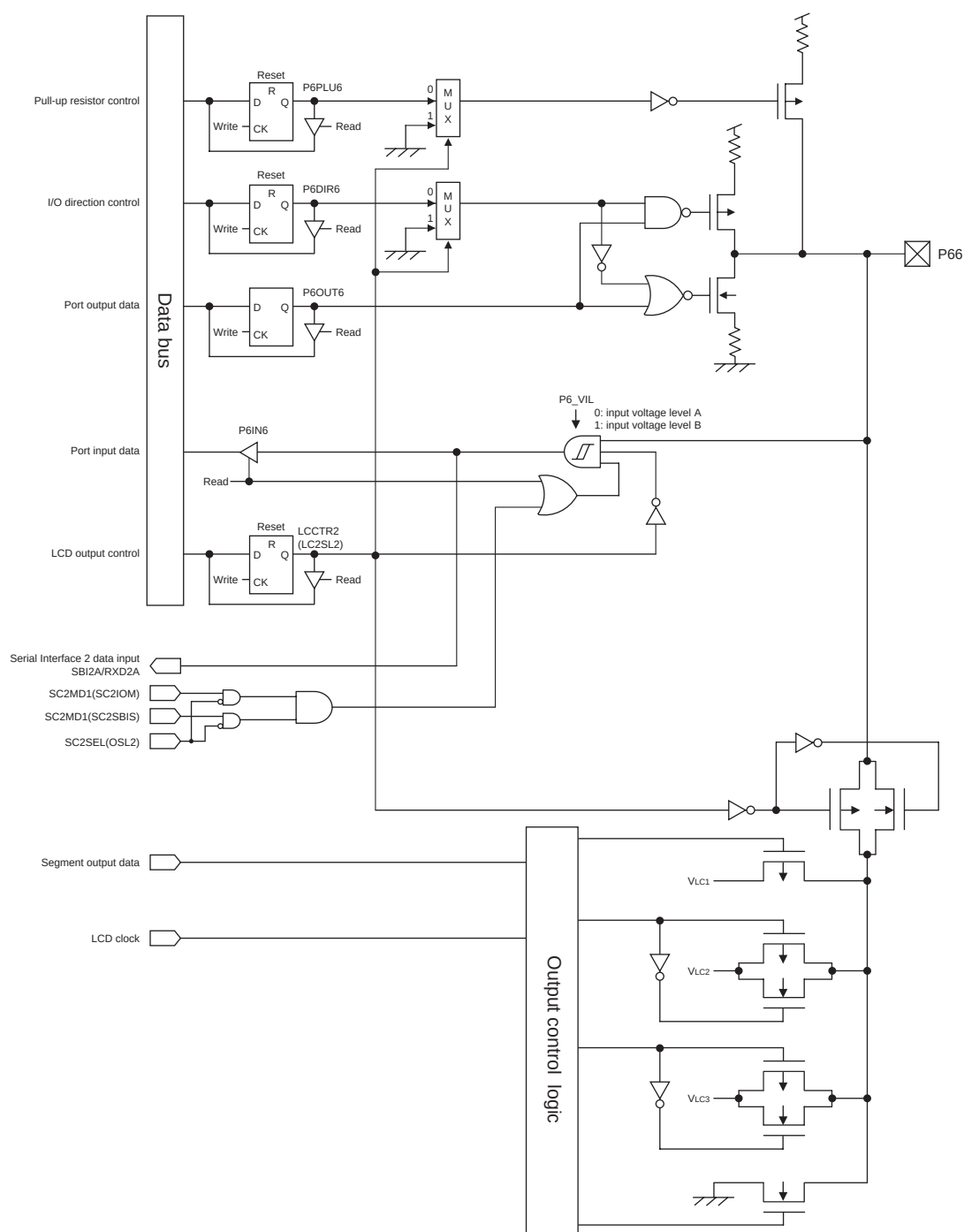


Figure:8.9.11 P65 Block Diagram (MN101EF57G)



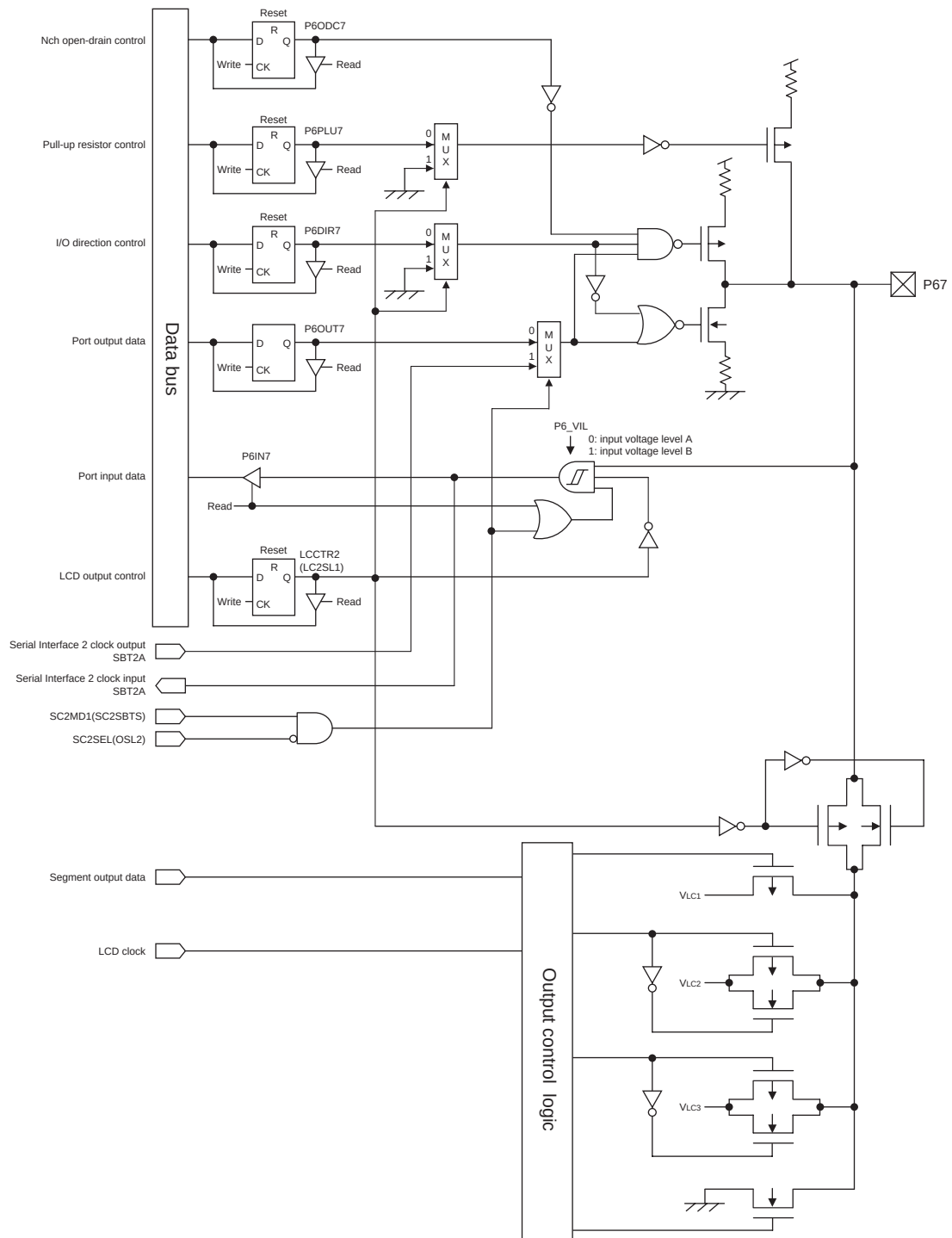


Figure:8.9.13 P67 Block Diagram (MN101EF57G)

8.10 Port 7

8.10.1 Description

■ General-purpose Port Setup

To output data to pin, set the control flag of Port 7 direction control register (P7DIR) to "1" and write the value of Port 7 output register (P7OUT).

To read input data of pin, set the control flag of Port 7 direction control register (P7DIR) to "0" and read the value of Port 7 input register (P7IN).

Each bit can be set individually as either an input or output by Port 7 direction control register (P7DIR). The control flag of P7DIR register is set to "1" for output mode, and "0" for input mode.

Each bit can be set individually to add a pull-up (or pull-down) resistor by Port 7 pull-up/pull-down resistor control register (P7PLUD). Set the control flag of P7PLUD register to "1" to add pull-up (or pull-down) resistor.

Port 7 can be selected to add a pull-up or pull-down resistor by SELUD7 flag of the pull-up/pull-down resistor selection register (SELUD).

Each bit can be selected individually as Nch open-drain output by Port 7 Nch open-drain control register (P7ODC). The control flag of P7ODC register is set to "1" for Nch open-drain output, and "0" for push-pull output.

Table:8.10.1 Port 7 Functions

Table remarks -: Without function

MN101EF76K/MN101EF56K					MN101EF57G				
Pin	Special function				Pin	Special function			
P70	SEG8	KEY0	SBI4A		P70	SEG8	KEY0	SBI4A	
P71	SEG7	KEY1	SBO4A	SDA4A	P71	SEG7	KEY1	SBO4A	SDA4A
P72	SEG6	KEY2	SBT4A	SCL4A	P72	SEG6	KEY2	SBT4A	SCL4A
P73	SEG5	KEY3			P73	SEG5	KEY3		
P74	SEG4	KEY4			P74	SEG4	KEY4		
P75	SEG3	KEY5	SBO1B	TXD1B	P75	SEG3	KEY5	SBO1B	TXD1B
P76	SEG2	KEY6	SBI1B	RXD1B	P76	SEG2	KEY6	SBI1B	RXD1B
P77	SEG1	KEY7	SBT1B		P77	SEG1	KEY7	SBT1B	

■ Special Function Pin Setup

P70 to P77 are also used as LCD segment output pins.

LCD segment output pin is selected by setting the corresponding flag of LCD output control register n (LCCTRn) to "1".

Port and segment can be switched by one bit. When segment output pin is selected, this pin is forced to be output mode and pull-up/pull-down resistor is not added.

P70 to 77 are also used as the input pin for KEY interrupt.

P70 is also used as the input pin for Serial Interface 4 reception data. When the SC4BIS flag of Serial Interface 4 mode register 1 (SC4MD1) is "1", P70 is the serial data input pin.

P71 is also used as the I/O pin for Serial Interface 4 transmission/reception data and the I/O pin for IIC4 transmission/reception data. When the SC4SBOS flag of Serial Interface 4 mode register 1 (SC4MD1) is "1", P71 is the serial data I/O pin. Push-pull output or Nch open-drain output can be selected by setting Port 7 Nch open-drain control register (P7ODC).

P72 is also used as the I/O pin for Serial Interface 4 clock and the output pin for IIC4 clock. When the SC4STBS flag of the Serial Interface 4 mode register 1 (SC4MD1) is "1", P72 is the the I/O pin of the serial clock. Push-pull output or Nch open-drain output can be selected by setting the Port 7 Nch open-drain control register (P7ODC).

P75 is also used as the I/O pin for Serial Interface 1 transmission/reception data and the output pin for UART1 transmission data. When the SC1SBOS flag of Serial Interface 1 mode register 1 (SC1MD1) is "1", P75 is the serial data I/O pin. Push-pull output or Nch open-drain output can be selected by setting Port 7 Nch open-drain control register (P7ODC).

P76 is also used as the input pin for Serial Interface 1 reception data and UART1 reception data. When the SC1BIS flag of Serial Interface 1 mode register 1 (SC1MD1) is "1", P76 is the serial data input pin.

P77 is also used as the I/O pin for Serial Interface 1 clock. When the SC1SBTS flag of Serial Interface 1 mode register 1 (SC1MD1) is "1", P77 is the I/O pin of the serial clock. Push-pull output or Nch open-drain output can be selected by setting Port 7 Nch open-drain control register (P7ODC).

8.10.2 Registers

Table:8.10.2 shows registers that control Port 7.

Table:8.10.2 Port 7 Control Register

Table remarks √: With function -: Without function

Registers	Address	R/W	Function	Page	MN101 EF76K	MN101 EF57G	MN101 EF56K
P7OUT	0x03E77	R/W	Port 7 output register	VIII-131	√	√	√
P7IN	0x03E87	R	Port 7 input register	VIII-131	√	√	√
P7DIR	0x03E97	R/W	Port 7 direction control register	VIII-132	√	√	√
P7PLUD	0x03EA7	R/W	Port 7 pull-up/pull-down resistor control register	VIII-132	√	√	√
P7ODC	0x03EF7	R/W	Port 7 Nch open-drain control register	VIII-133	√	√	√
SELUD	0x03EAF	R/W	Pull-up/pull-down resistor selection register	VIII-134	√	√	√

R/W: Readable/Writable

R: Read only

■ Port 7 Output Register (P7OUT: 0x03E77)

bp	7	6	5	4	3	2	1	0
Flag	P7OUT7	P7OUT6	P7OUT5	P7OUT4	P7OUT3	P7OUT2	P7OUT1	P7OUT0
At reset	X	X	X	X	X	X	X	X
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description
7 to 0	P7OUT7 to 0	Output data 0: Output "Low" (V_{SS} -level) 1: Output "High" (V_{DD5} -level)

■ Port 7 Input Register (P7IN: 0x03E87)

bp	7	6	5	4	3	2	1	0
Flag	P7IN7	P7IN6	P7IN5	P7IN4	P7IN3	P7IN2	P7IN1	P7IN0
At reset	X	X	X	X	X	X	X	X
Access	R	R	R	R	R	R	R	R

bp	Flag	Description
7 to 0	P7IN7 to 0	Input data 0: Pin is "Low" (V_{SS} -level) 1: Pin is "High" (V_{DD5} -level)

■ Port 7 Direction Control Register (P7DIR: 0x03E97)

bp	7	6	5	4	3	2	1	0
Flag	P7DIR7	P7DIR6	P7DIR5	P7DIR4	P7DIR3	P7DIR2	P7DIR1	P7DIR0
At reset	0	0	0	0	0	0	0	0
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description
7 to 0	P7DIR7 to 0	I/O mode selection 0: Input mode 1: Output mode

■ Port 7 Pull-up/pull-down Resistor Control Register (P7PLUD: 0x03EA7)

bp	7	6	5	4	3	2	1	0
Flag	P7PLUD7	P7PLUD6	P7PLUD5	P7PLUD4	P7PLUD3	P7PLUD2	P7PLUD1	P7PLUD0
At reset	0	0	0	0	0	0	0	0
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description
7 to 0	P7PLUD7 to 0	Pull-up/pull-down resistor selection 0: Not added 1: Added

■ Port 7 Nch Open-drain Control Register (P7ODC: 0x03EF7)

bp	7	6	5	4	3	2	1	0
Flag	P7ODC7	-	P7ODC5	-	-	P7ODC2	P7ODC1	-
At reset	0	-	0	-	-	0	0	-
Access	R/W	-	R/W	-	-	R/W	R/W	-

bp	Flag	Description
7	P7ODC7	Nch open-drain output selection 0: Push-pull output 1: Nch open-drain output
6	-	-
5	P7ODC5	Nch open-drain output selection 0: Push-pull output 1: Nch open-drain output
4 to 3	-	-
2 to 1	P7ODC2 to 1	Nch open-drain output selection 0: Push-pull output 1: Nch open-drain output
0	-	-

■ Pull-up/pull-down Resistor Selection Register (SELUD: 0x03EAF)

<MN101EF76K/MN101EF56K>

bp	7	6	5	4	3	2	1	0
Flag	SELUD7	-	SELUD5	SELUD4	-	-	SELUD1	SELUD0
At reset	0	-	0	0	-	-	0	0
Access	R/W	-	R/W	R/W	-	-	R/W	R/W

bp	Flag	Description
7	SELUD7	Pull-up/pull-down selection (Port 7) 0: Pull-up 1: Pull-down
6	-	-
5	SELUD5	Pull-up/pull-down selection (Port 5) 0: Pull-up 1: Pull-down
4	SELUD4	Pull-up/pull-down selection (Port 4) 0: Pull-up 1: Pull-down
3 to 2	-	-
1	SELUD0	Pull-up/pull-down selection (Port 1) 0: Pull-up 1: Pull-down
0	SELUD0	Pull-up/pull-down selection (Port 0) 0: Pull-up 1: Pull-down

<MN101EF57G>

bp	7	6	5	4	3	2	1	0
Flag	SELUD7	-	SELUD5	SELUD4	-	-	-	SELUD0
At reset	0	-	0	0	-	-	-	0
Access	R/W	-	R/W	R/W	-	-	-	R/W

bp	Flag	Description
7	SELUD7	Pull-up/pull-down selection (Port 7) 0: Pull-up 1: Pull-down
6	-	-
5	SELUD5	Pull-up/pull-down selection (Port 5) 0: Pull-up 1: Pull-down
4	SELUD4	Pull-up/pull-down selection (Port 4) 0: Pull-up 1: Pull-down
3 to 1	-	-
0	SELUD0	Pull-up/pull-down selection (Port 0) 0: Pull-up 1: Pull-down

8.10.3 Block Diagram (MN101EF76K/MN101EF56K)

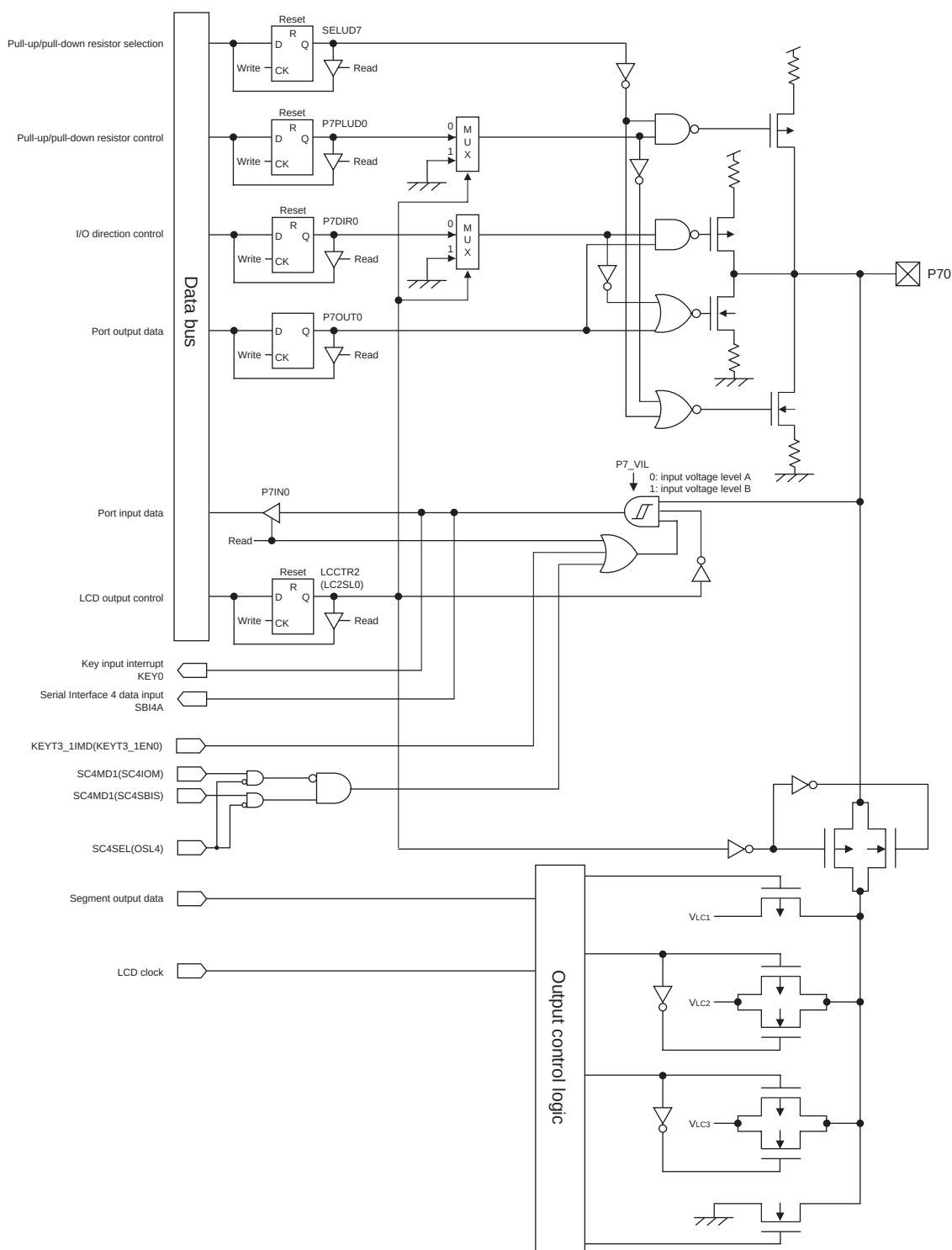


Figure:8.10.1 P70 Block Diagram (MN101EF76K/MN101EF56K)

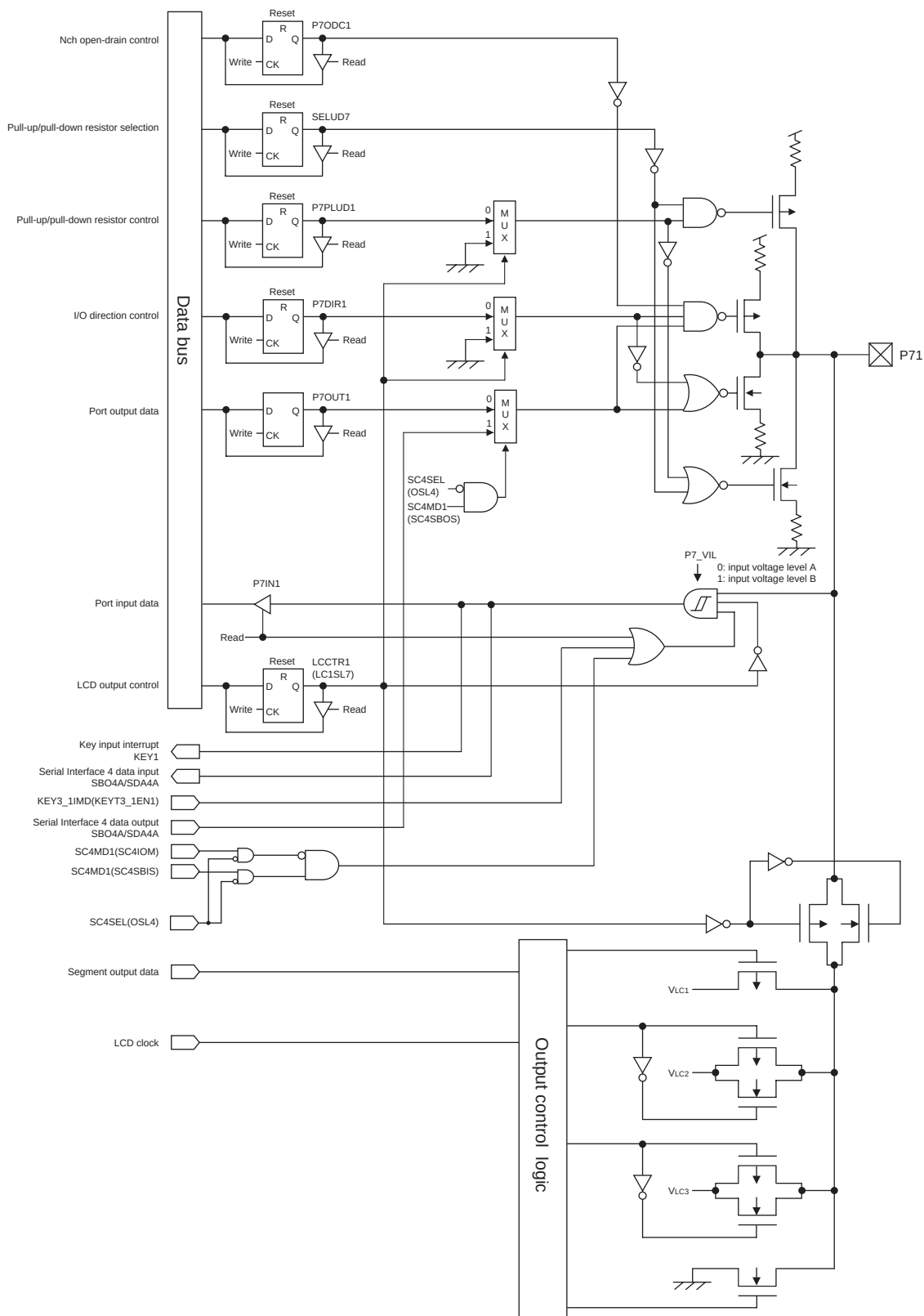


Figure:8.10.2 P71 Block Diagram (MN101EF76K/MN101EF56K)

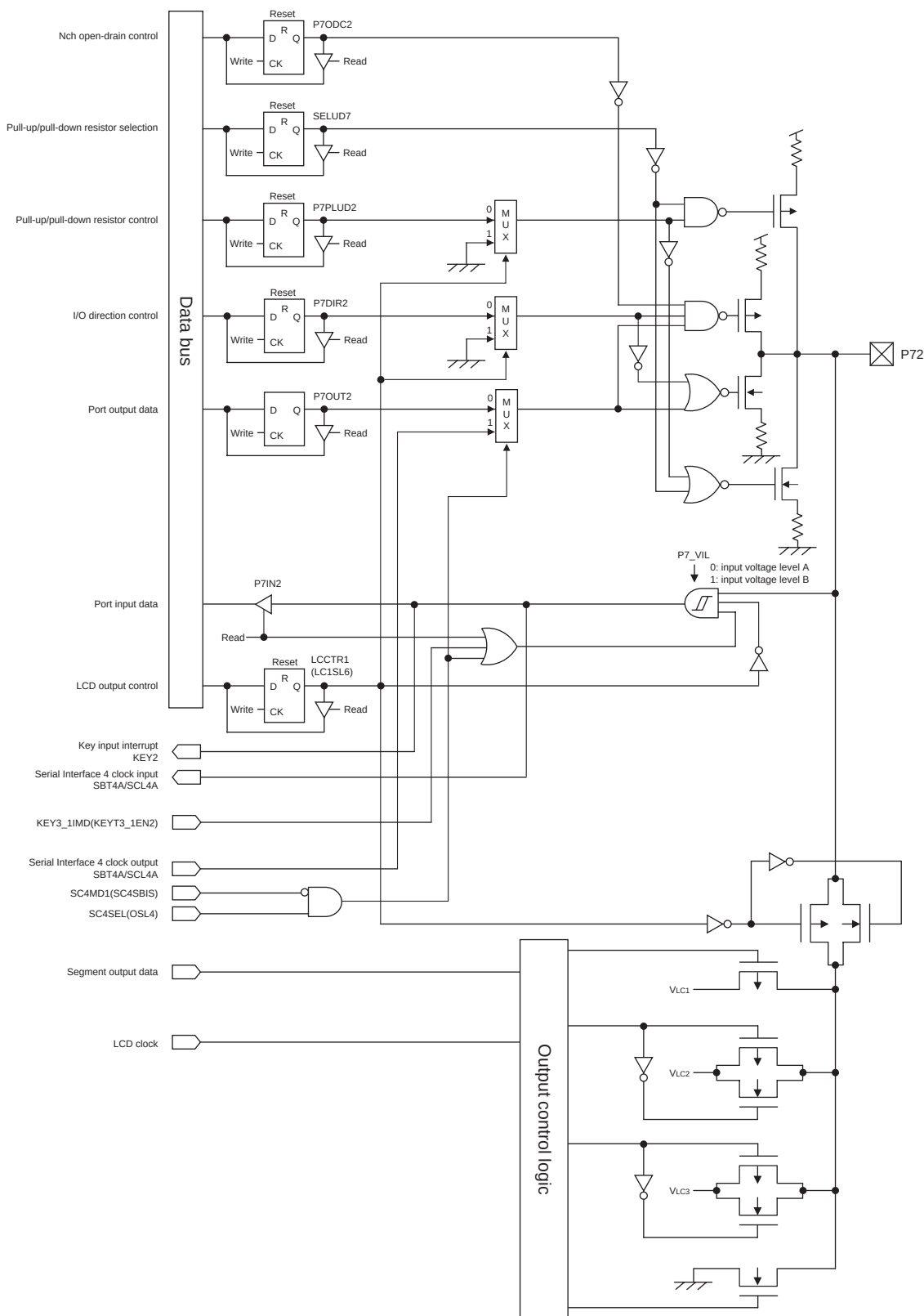


Figure:8.10.3 P72 Block Diagram (MN101EF76K/MN101EF56K)

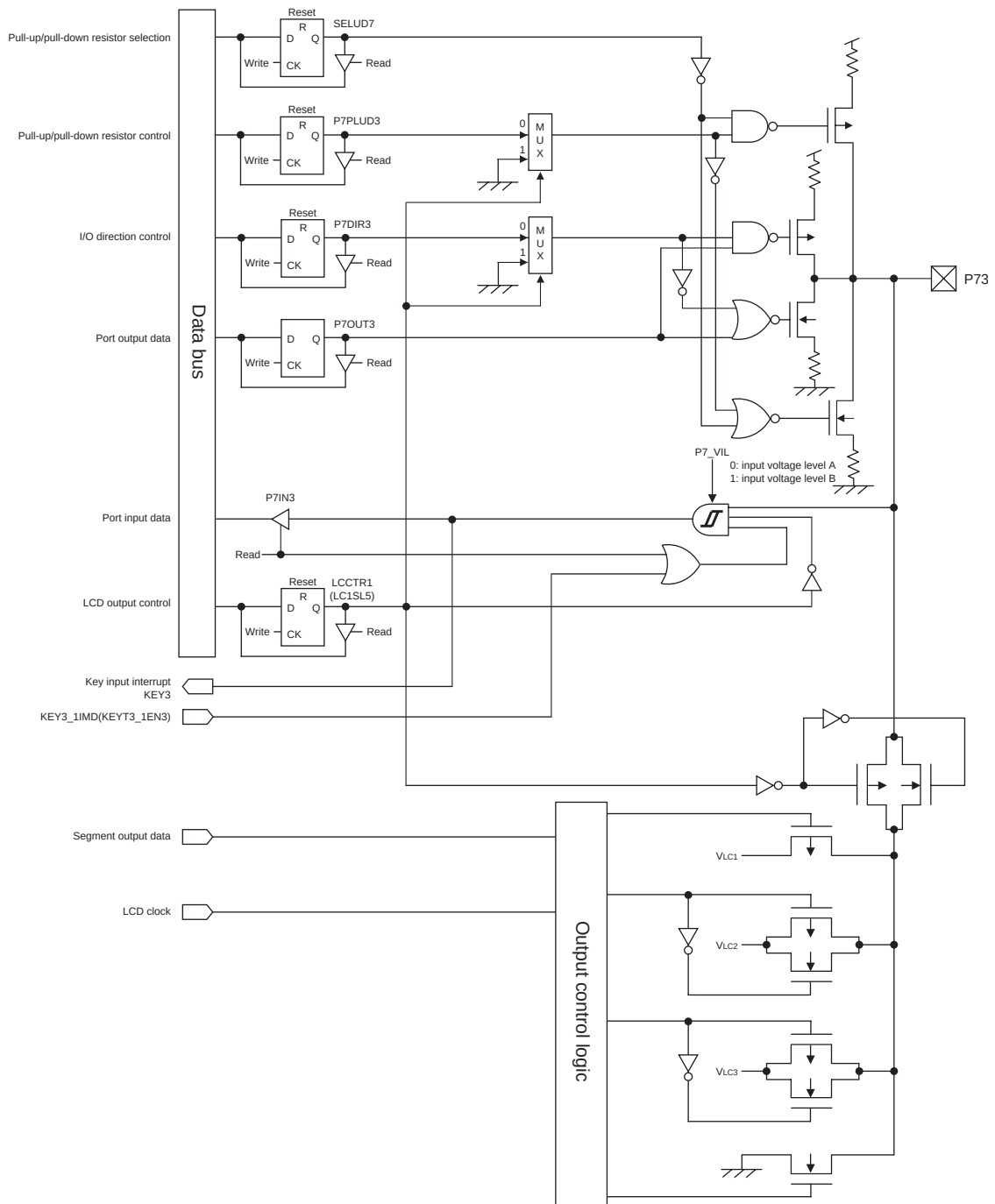


Figure:8.10.4 P73 Block Diagram (MN101EF76K/MN101EF56K)

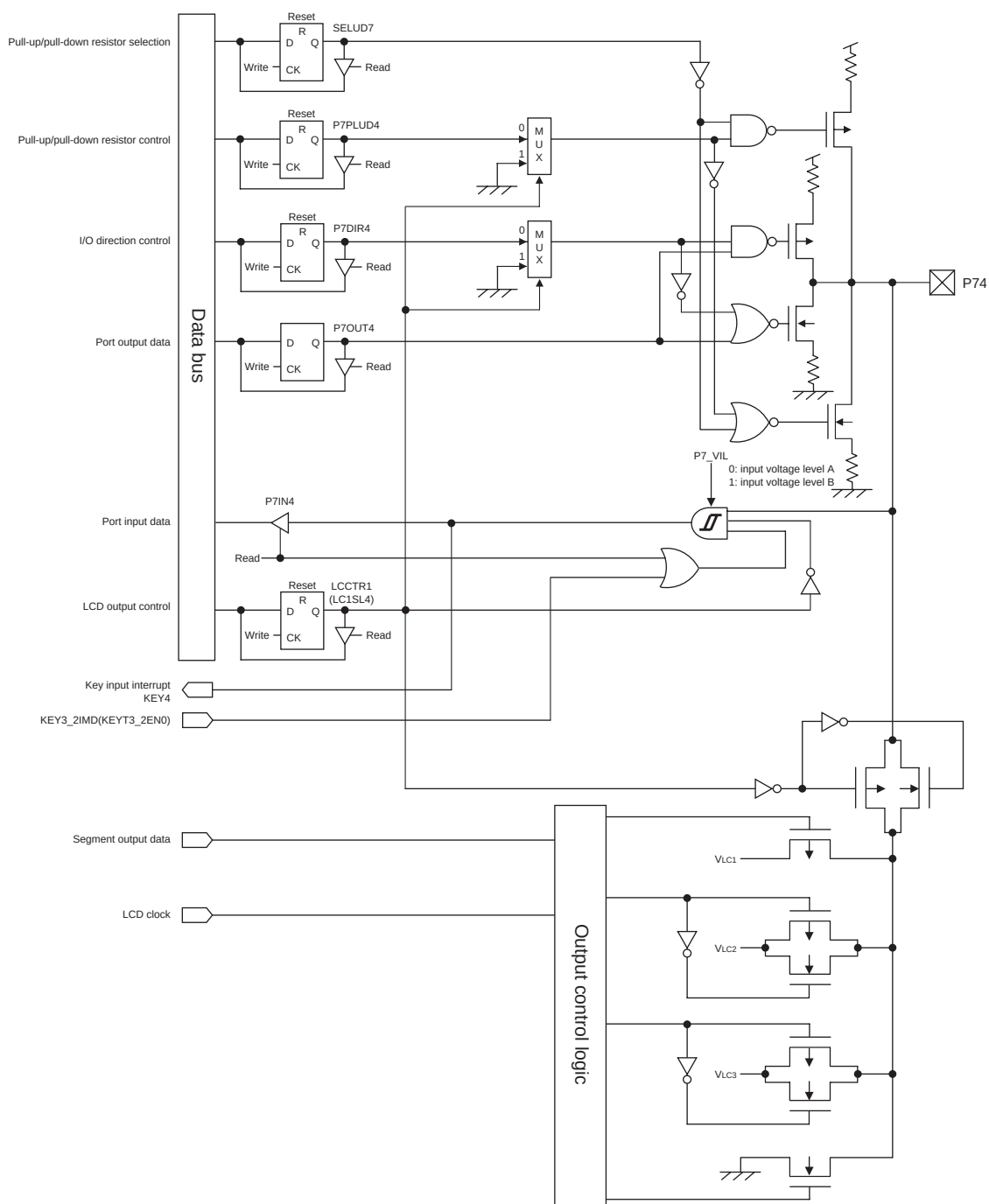


Figure:8.10.5 P74 Block Diagram (MN101EF76K/MN101EF56K)

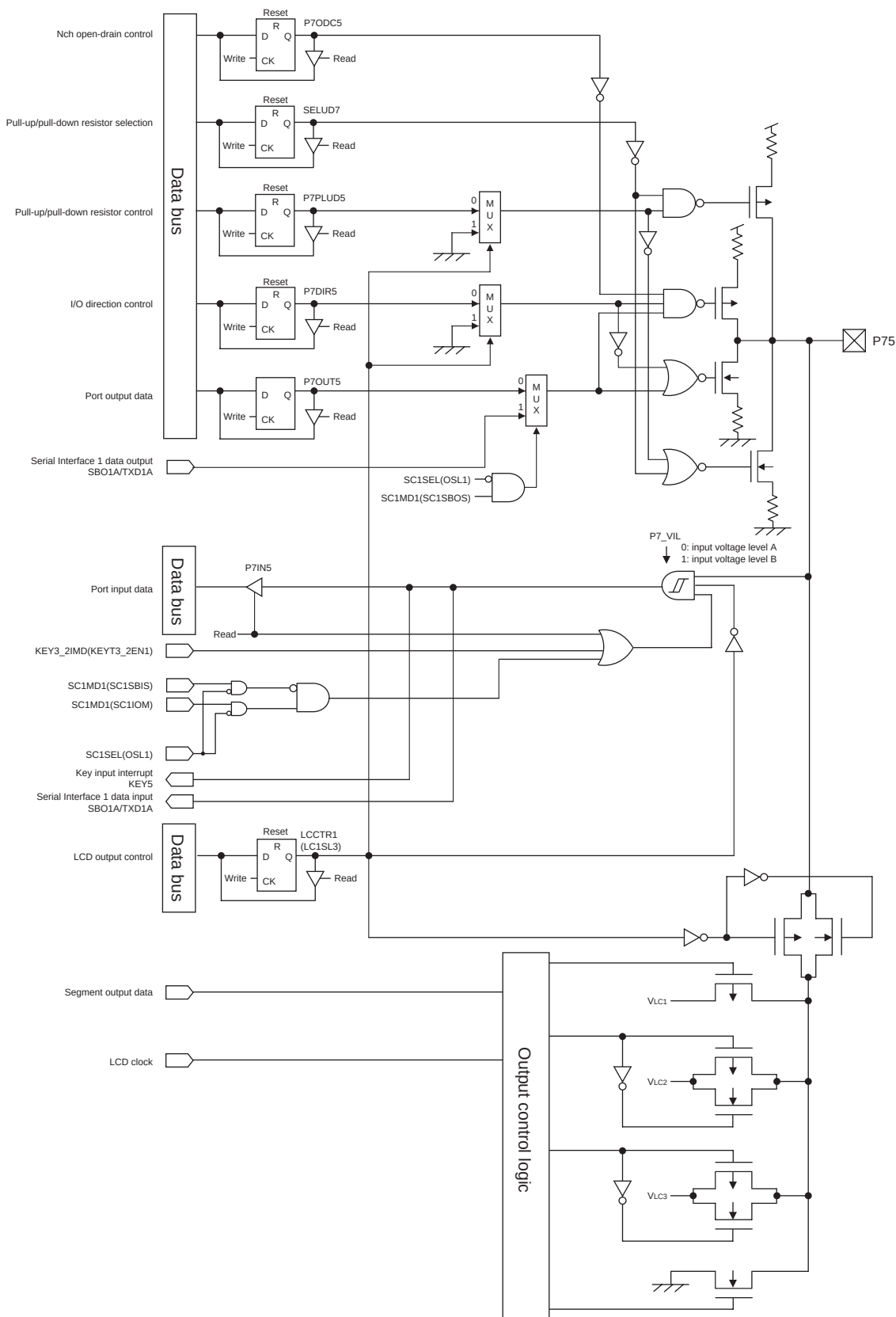


Figure:8.10.6 P75 Block Diagram (MN101EF76K/MN101EF56K)

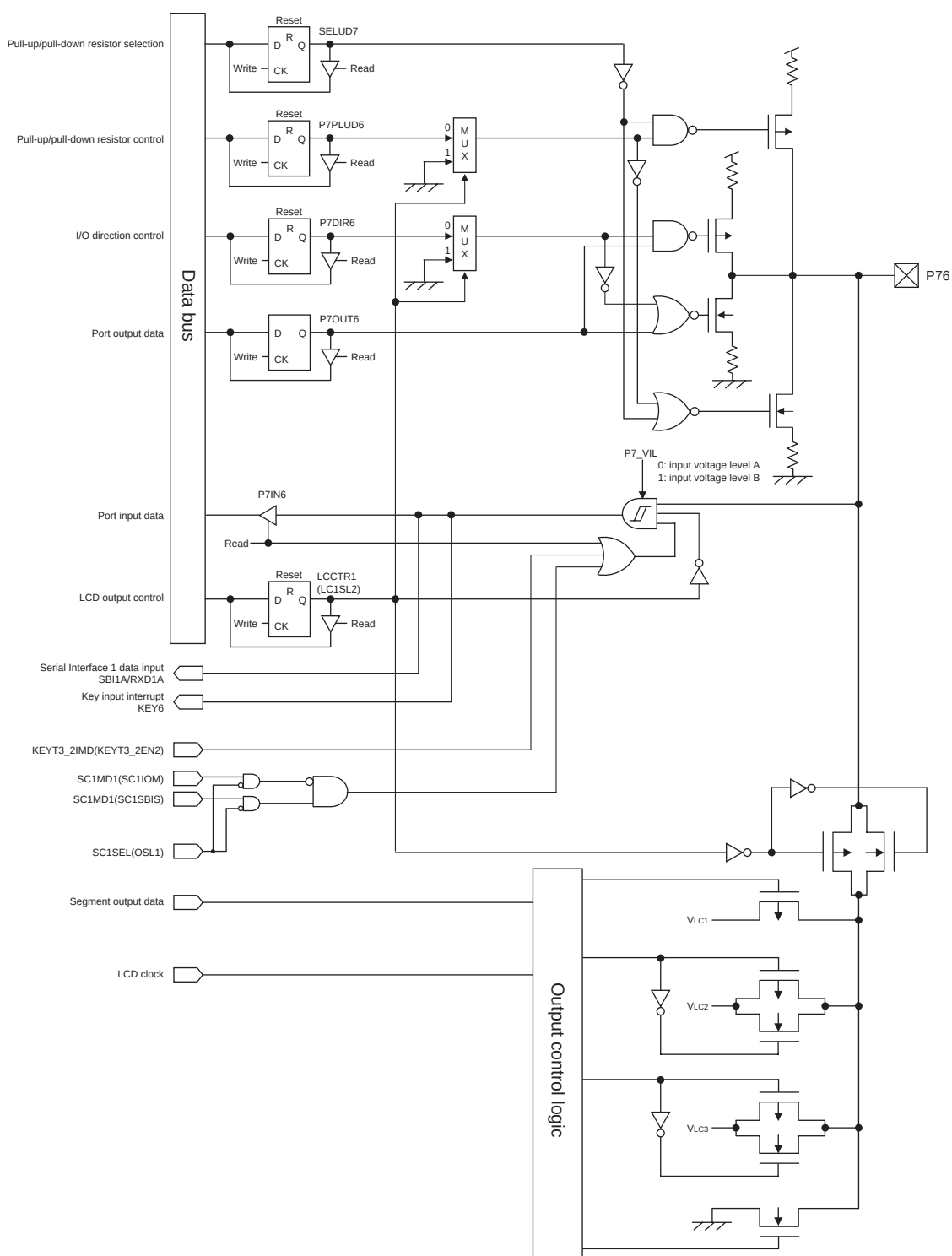


Figure:8.10.7 P76 Block Diagram (MN101EF76K/MN101EF56K)

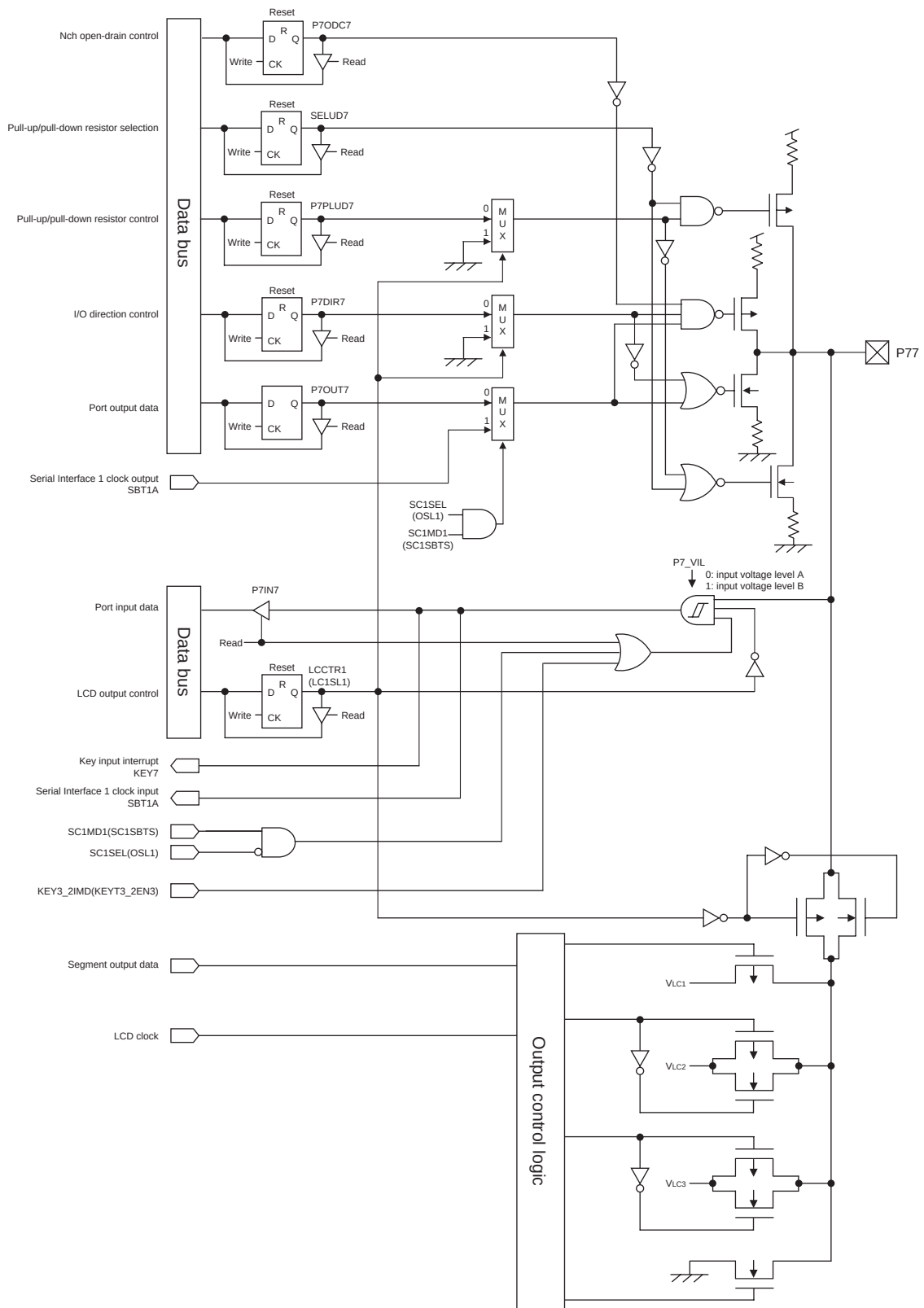


Figure:8.10.8 P77 Block Diagram (MN101EF76K/MN101EF56K)

8.10.4 Block Diagram (MN101EF57G)

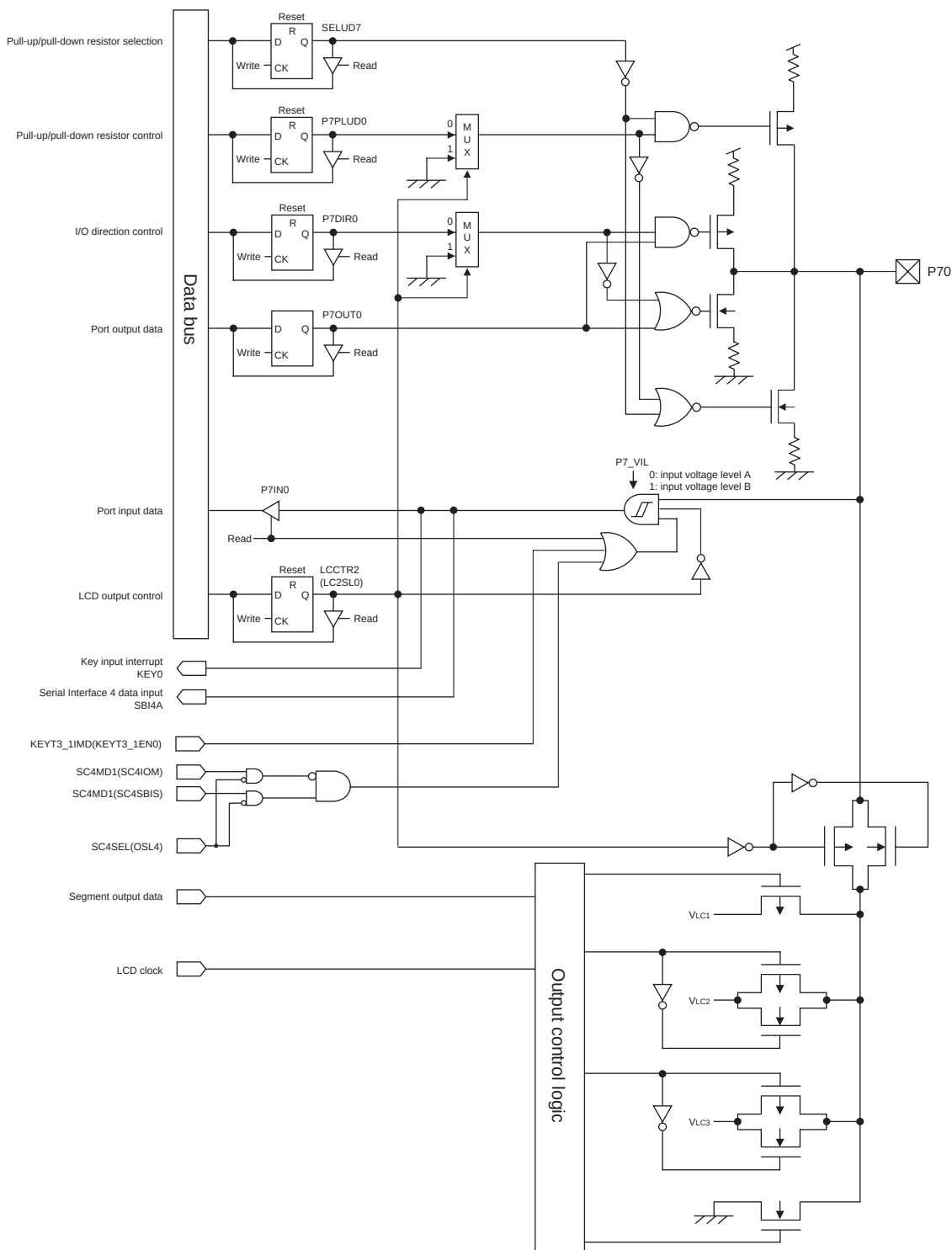


Figure:8.10.9 P70 Block Diagram (MN101EF57G)

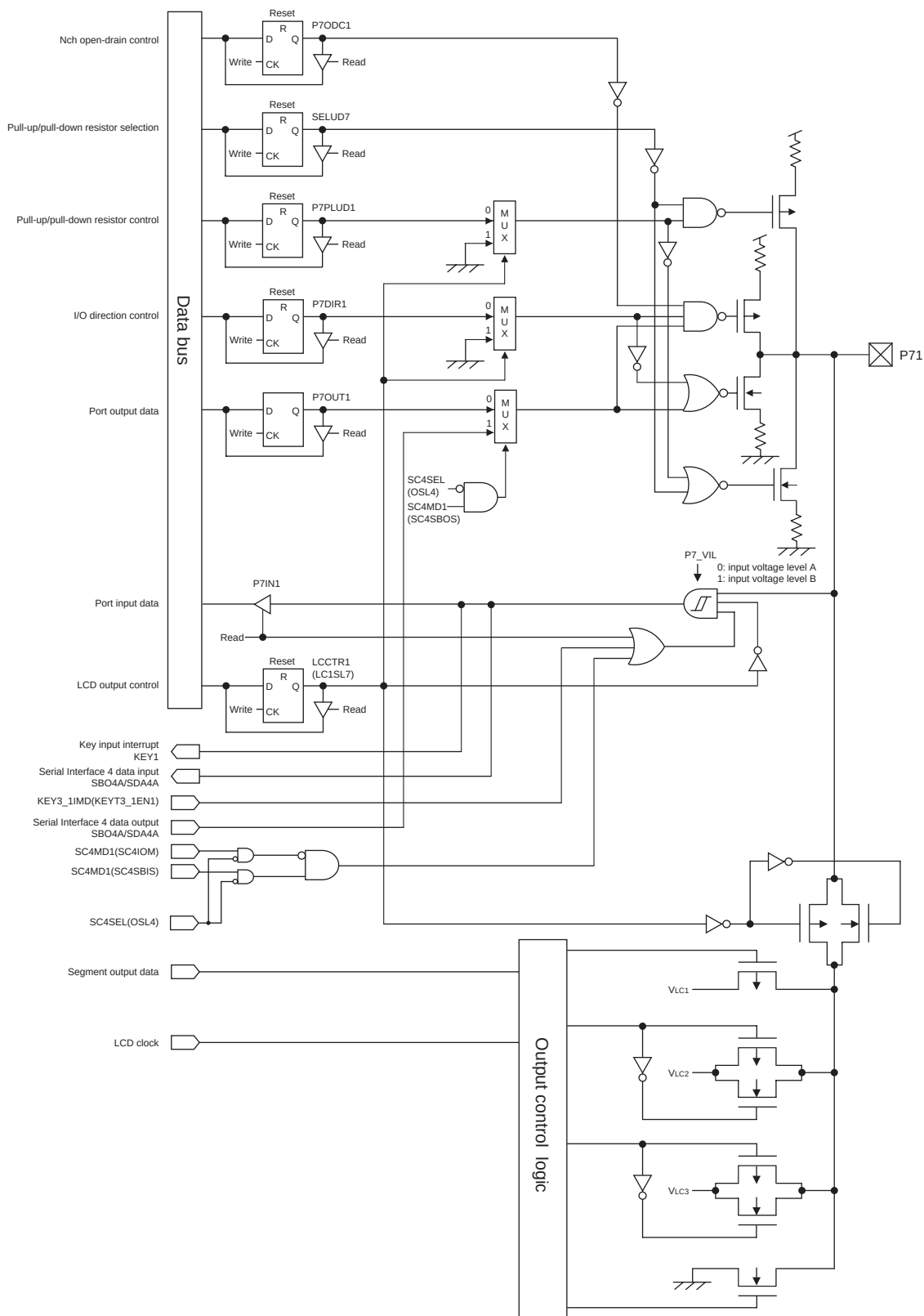


Figure:8.10.10 P71 Block Diagram (MN101EF57G)

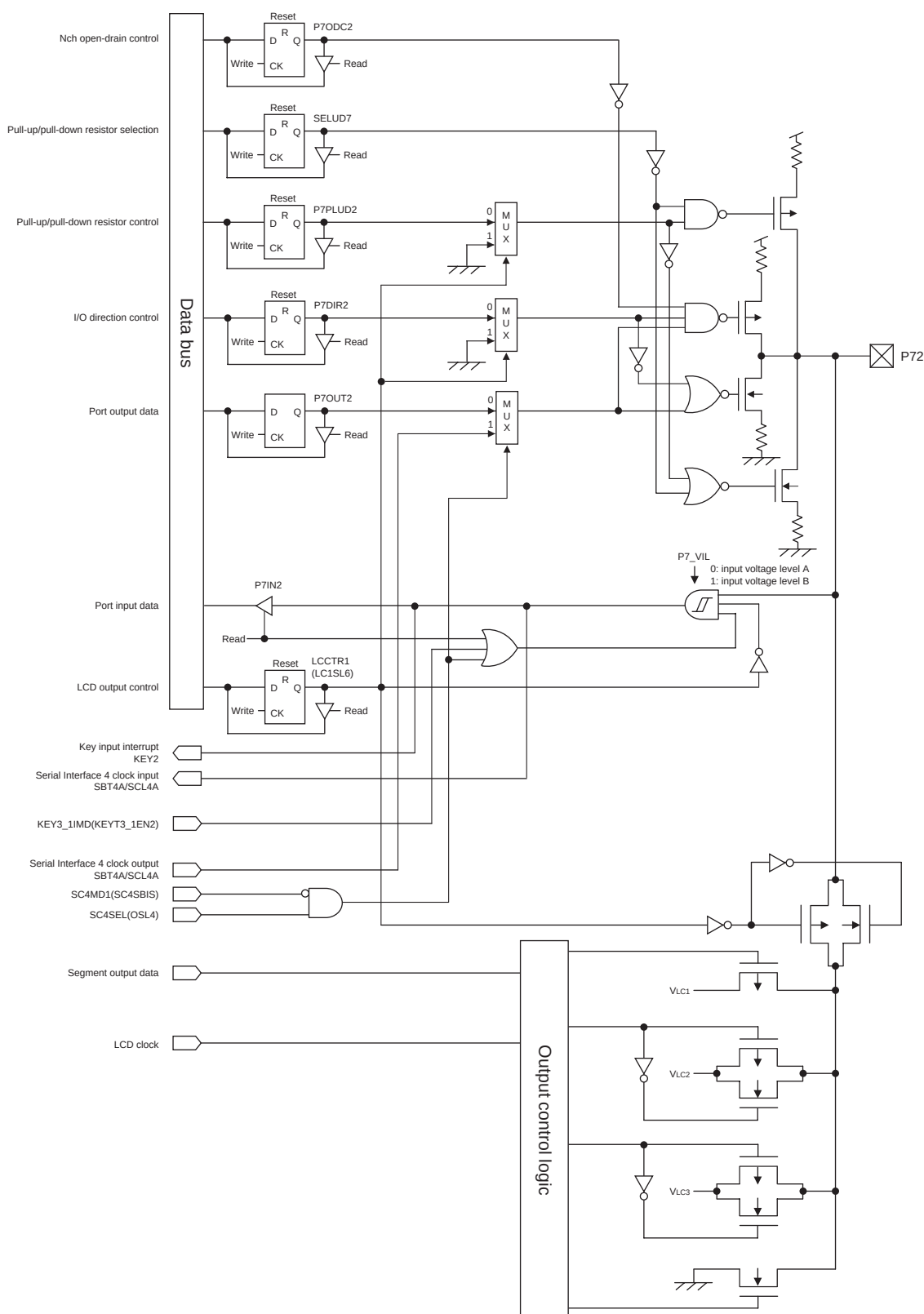


Figure:8.10.11 P72 Block Diagram (MN101EF57G)

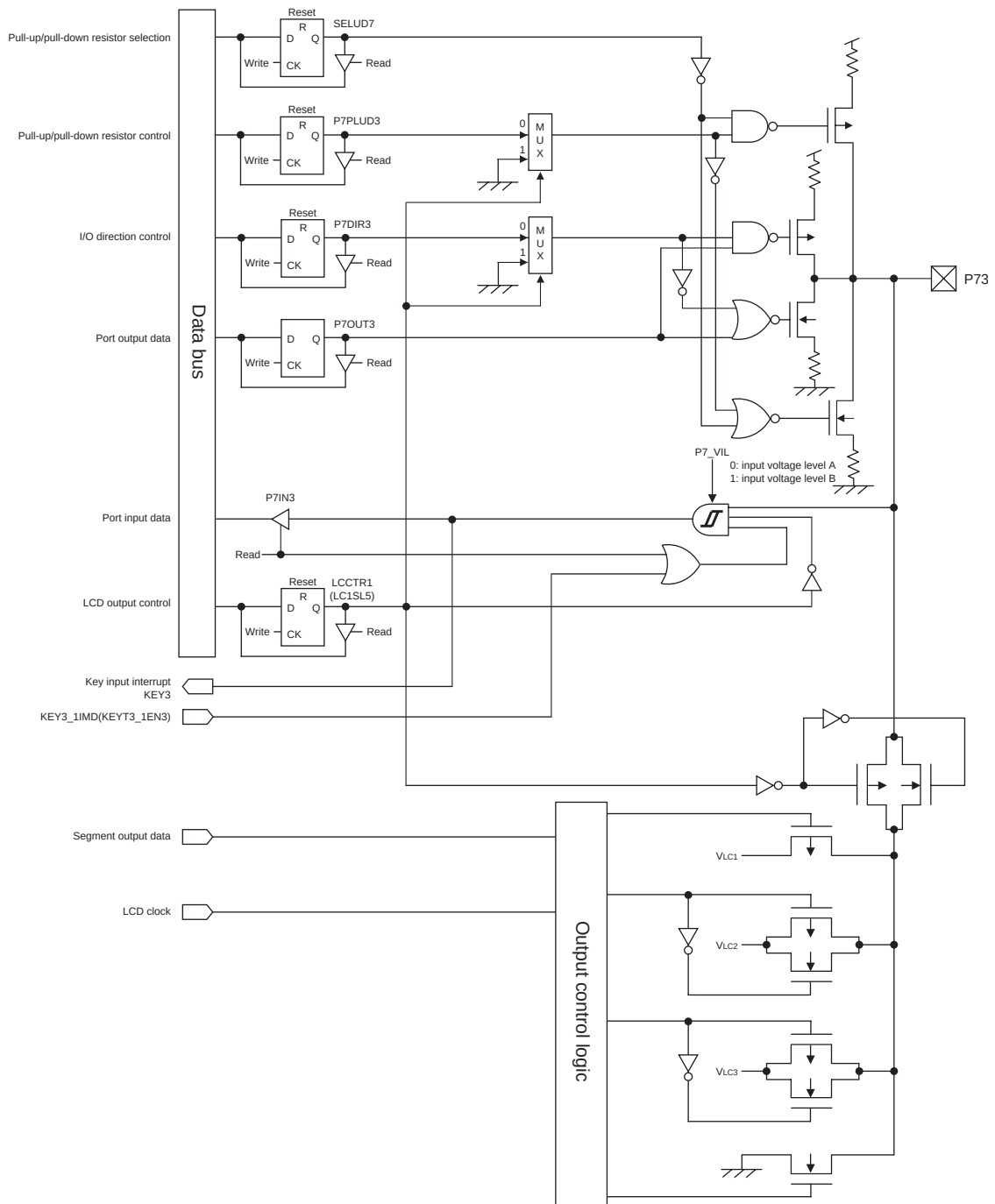


Figure:8.10.12 P73 Block Diagram (MN101EF57G)

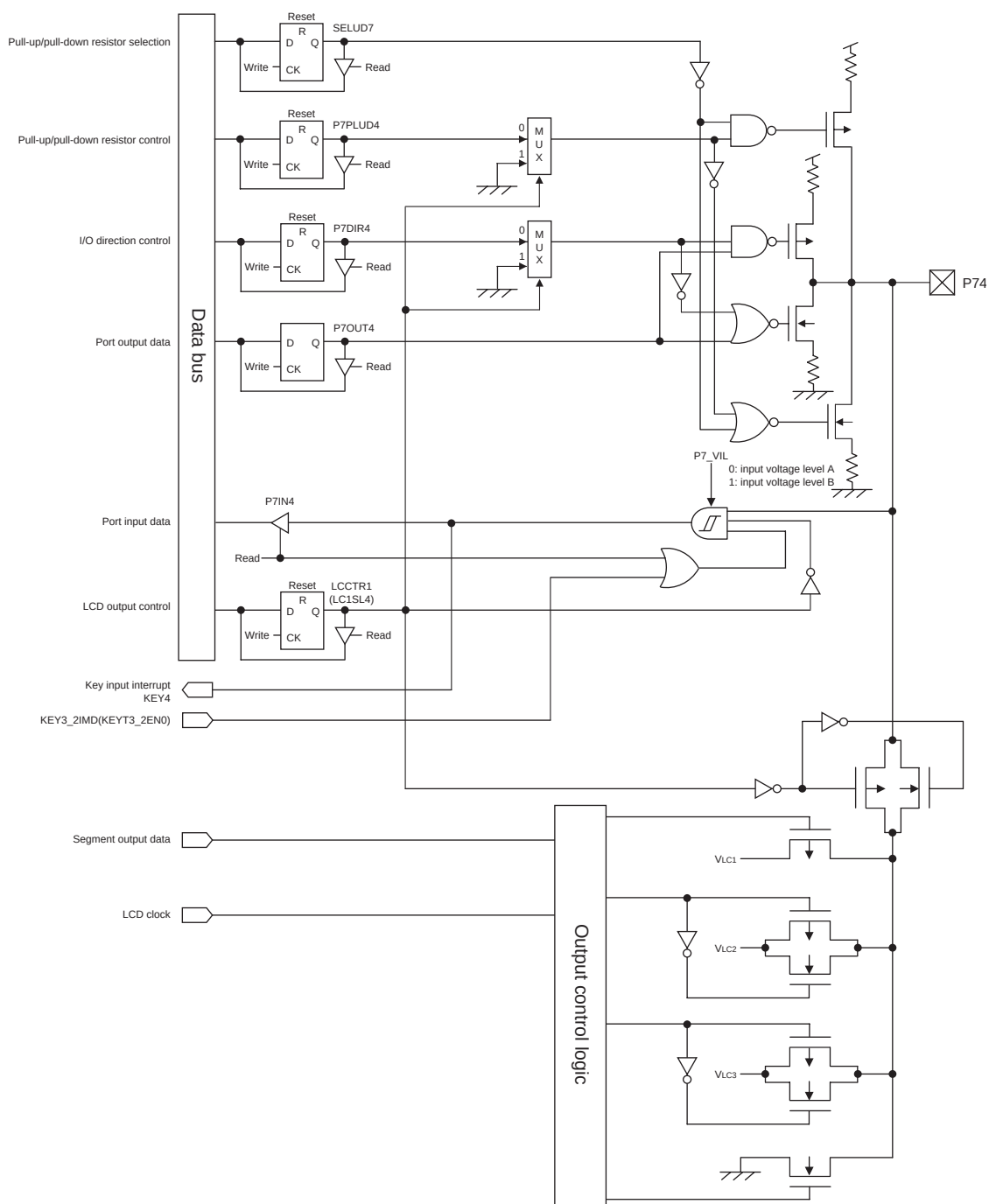


Figure:8.10.13 P74 Block Diagram (MN101EF57G)

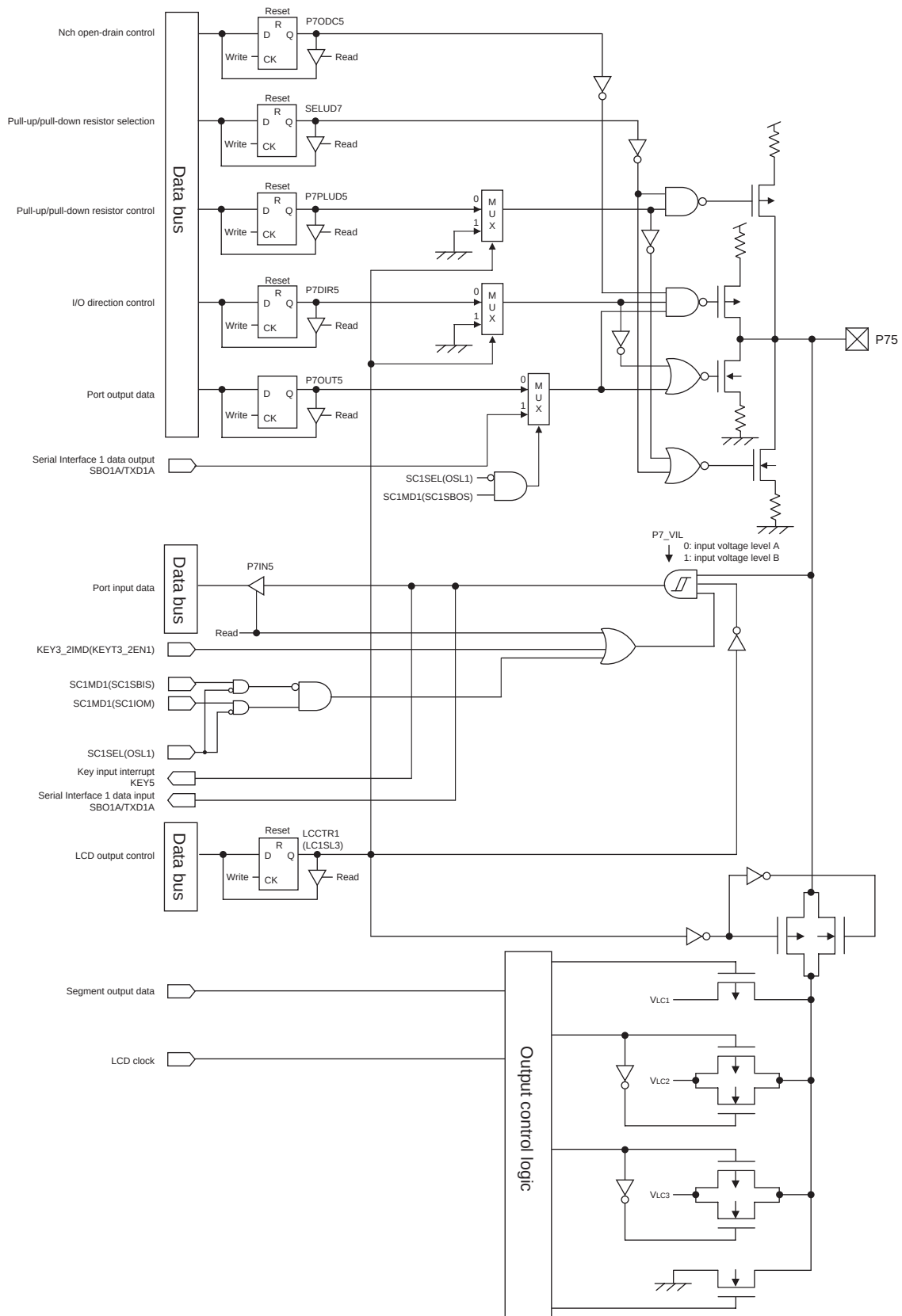


Figure:8.10.14 P75 Block Diagram (MN101EF57G)

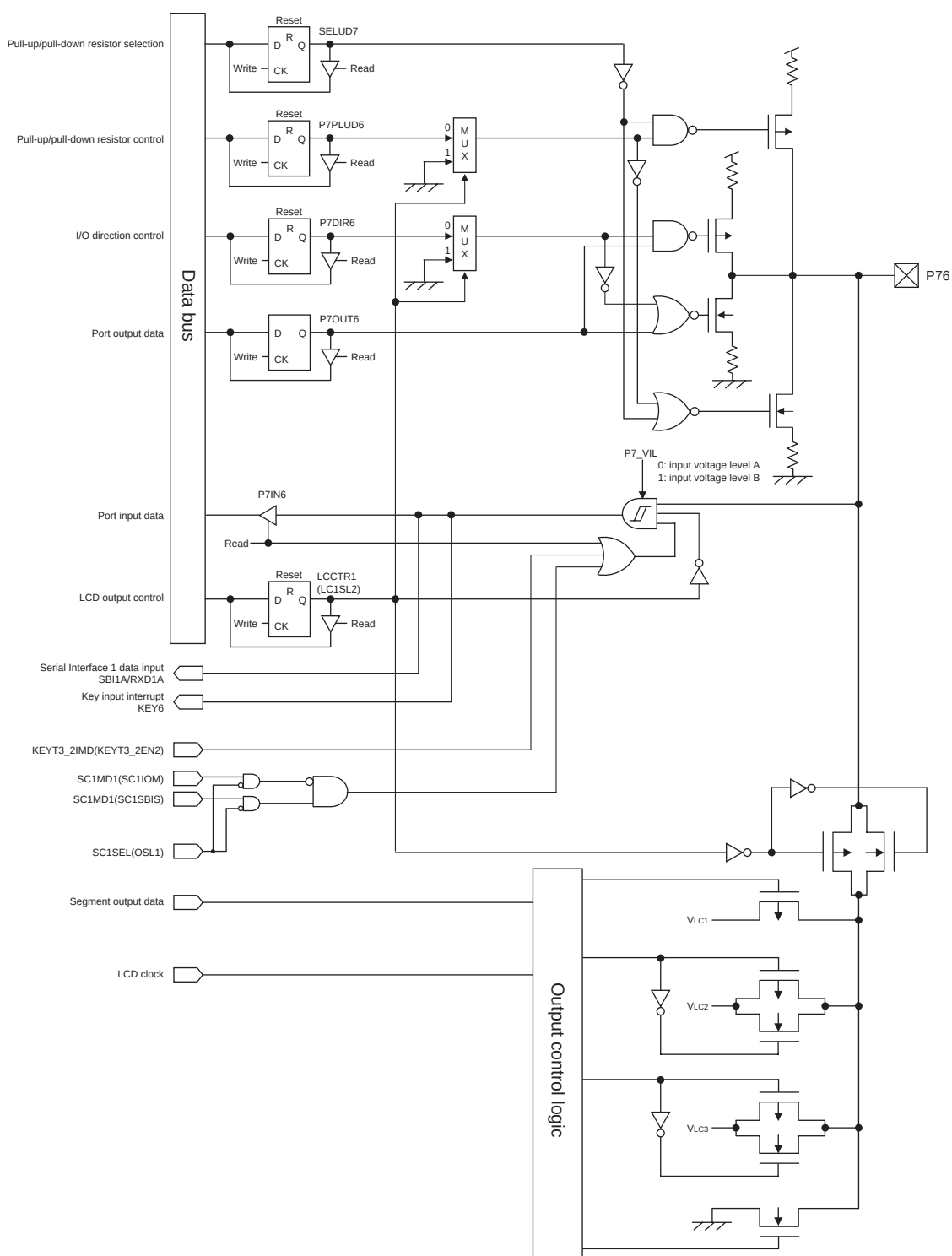


Figure:8.10.15 P76 Block Diagram (MN101EF57G)

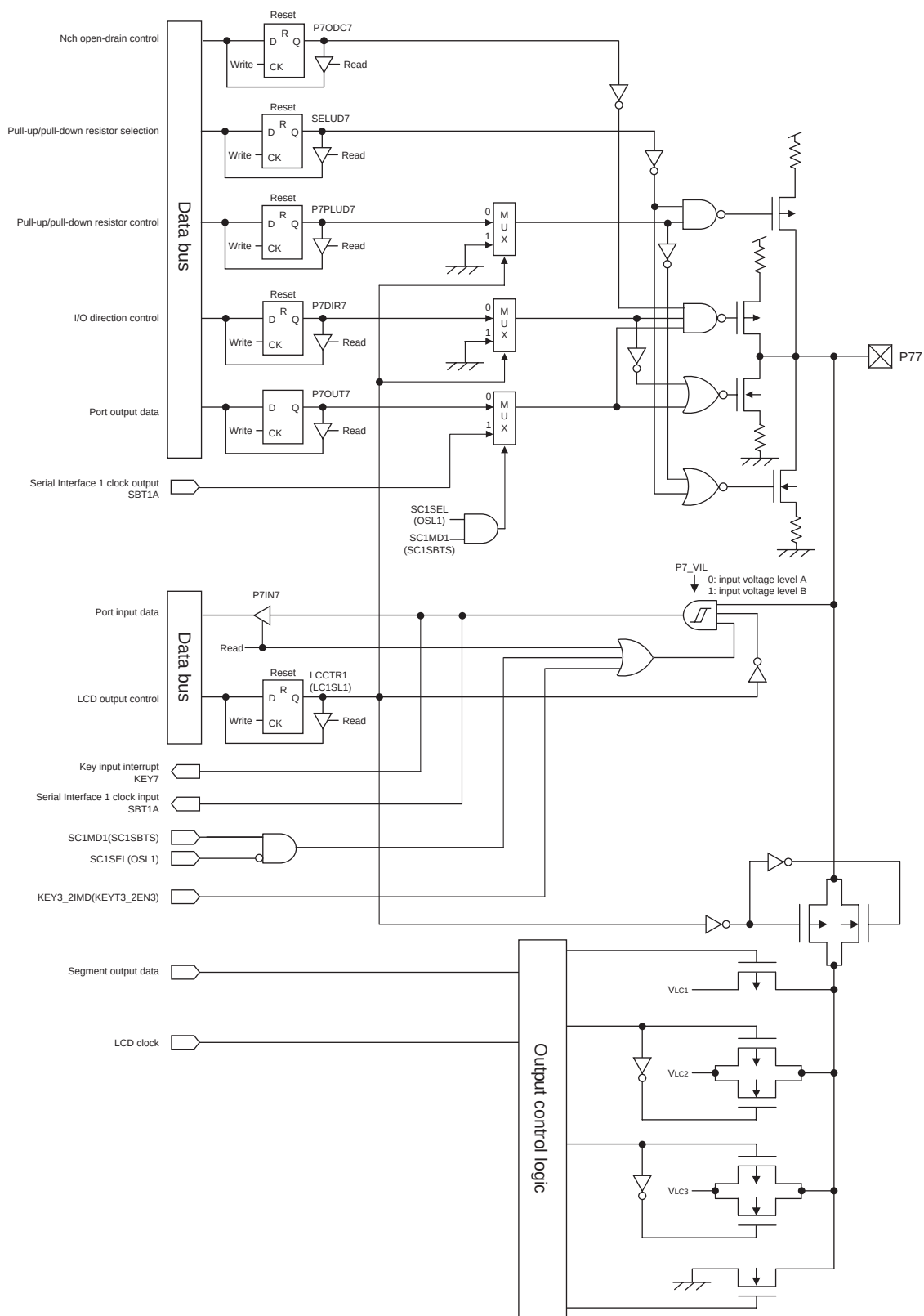


Figure:8.10.16 P77 Block Diagram (MN101EF57G)

8.11 Port 8

8.11.1 Description

■ General-purpose Port Setup

To output data to pin, set the control flag of Port 8 direction control register (P8DIR) to "1" and write the value of Port 8 output register (P8OUT).

To read input data of pin, set the control flag of Port 8 direction control register (P8DIR) to "0" and read the value of Port 8 input register (P8IN).

Each bit can be set individually as either an input or output by Port 8 direction control register (P8DIR). The control flag of P8DIR register is set to "1" for output mode, and "0" for input mode.

Each bit can be set individually to add a pull-up resistor by Port 8 pull-up resistor control register (P8PLU). Set the control flag of P8PLU register to "1" to add pull-up resistor.

Each bit can be set individually as synchronous output by Port 8 synchronous output control register (P8SYO). The control flag of P8SYO register is set to "1" for synchronous output, and "0" for general-purpose port.

Table:8.11.1 Port 8 Functions

Table remarks -: Without function

MN101EF76K				MN101EF57G/MN101EF56K			
Pin	Special function			Pin	Special function		
P80	SEG0	SDO0	TM9OD0A	P80	SEG0	SDO0	TM9OD0
P81	COM0	SDO1	TM9OD1A	P81	COM0	SDO1	TM9OD1
P82	COM1	SDO2	TM9OD2A	P82	COM1	SDO2	TM9OD2
P83	COM2	SDO3	TM9OD3A	P83	COM2	SDO3	TM9OD3
P84	COM3	SDO4	TM9OD4A	P84	COM3	SDO4	TM9OD4
P85	VLCD3	SDO5	TM9OD5A	P85	VLCD3	SDO5	TM9OD5
P86	VLCD2	SDO6		P86	VLCD2	SDO6	
P87	VLCD1	SDO7		P87	VLCD1	SDO7	

■ Special Function Pin Setup

For P80 to P85, each bit can be set individually as either timer 9 output or port output by Port 8 output mode register (P8OMD). The control flag of P8OMD register is set to "1" for special function data output, and "0" for general-purpose port.

P80 is also used as LCD segment output pin.

LCD segment output pin is selected by setting the corresponding flag of LCD output control register 1 (LCCTR1) to "1".

Port and segment can be switched by one bit. When segment output pin is selected, this pin is forced to be output mode and pull-up resistor is not added.

P81 to 84 are also used as the LCD common output pins.

COM0 to COM3 pins are selected by setting the corresponding flags of LCD output control register 0 (LCCTR0). When common output pins are selected, these pins are forced to be output mode and pull-up resistors are not added.

P85 to 87 are also used as the power supply pins of LCD driver circuit VLC3 to VLC1.

VLC3 to VLC1 pins are selected by setting the corresponding flags of LCD output control register 0 (LCCTR0). When VLC3 to VLC1 pins are selected, pull-up resistors are not added.

For P80 to 87, each bit can be set to select synchronous output by Port 8 synchronous output control register (P8SYO). The control flag of P8SYO register is set to "1" for synchronous output, "0" for general-purpose port. The event to generate synchronous outputs is selected by Port 8 synchronous output event selection register (P8SEV). The corresponding flags of P8SEV register are set to "00" for IRQ, "01" for TM7IRQ, "10" for TM2IRQ, and "11" for TM1IRQ.

8.11.2 Registers

Table:8.11.2 shows registers that control Port 8.

Table:8.11.2 Port 8 Control Register

Table remarks √: With function -: Without function

Registers	Address	R/W	Function	Page	MN101 EF76K	MN101 EF57G	MN101 EF56K
P8OUT	0x03E78	R/W	Port 8 output register	VIII-153	√	√	√
P8IN	0x03E88	R	Port 8 input register	VIII-153	√	√	√
P8DIR	0x03E98	R/W	Port 8 direction control register	VIII-154	√	√	√
P8PLU	0x03EA8	R/W	Port 8 pull-up resistor control register	VIII-154	√	√	√
P8OMD	0x03EB8	R/W	Port 8 output mode register	VIII-155	√	√	√
P8SYO	0x03EF8	R/W	Port 8 synchronous output control register	VIII-156	√	√	√
P8SEV	0x03EF9	R/W	Port 8 synchronous output event selection register	VIII-157	√	√	√

R/W: Readable/Writable

R: Read only

■ Port 8 Output Register (P8OUT: 0x03E78)

bp	7	6	5	4	3	2	1	0
Flag	P8OUT7	P8OUT6	P8OUT5	P8OUT4	P8OUT3	P8OUT2	P8OUT1	P8OUT0
At reset	X	X	X	X	X	X	X	X
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description
7 to 0	P8OUT7 to 0	Output data 0: Output "Low" (V_{SS} -level) 1: Output "High" (V_{DD5} -level)

■ Port 8 Input Register (P8IN: 0x03E88)

bp	7	6	5	4	3	2	1	0
Flag	P8IN7	P8IN6	P8IN5	P8IN4	P8IN3	P8IN2	P8IN1	P8IN0
At reset	X	X	X	X	X	X	X	X
Access	R	R	R	R	R	R	R	R

bp	Flag	Description
7 to 0	P8IN7 to 0	Input data 0: Pin is "Low" (V_{SS} -level) 1: Pin is "High" (V_{DD5} -level)

■ Port 8 Direction Control Register (P8DIR: 0x03E98)

bp	7	6	5	4	3	2	1	0
Flag	P8DIR7	P8DIR6	P8DIR5	P8DIR4	P8DIR3	P8DIR2	P8DIR1	P8DIR0
At reset	0	0	0	0	0	0	0	0
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description
7 to 0	P8DIR7 to 0	I/O mode selection 0: Input mode 1: Output mode

■ Port 8 Pull-up Resistor Control Register (P8PLU: 0x03EA8)

bp	7	6	5	4	3	2	1	0
Flag	P8PLU7	P8PLU6	P8PLU5	P8PLU4	P8PLU3	P8PLU2	P8PLU1	P8PLU0
At reset	0	0	0	0	0	0	0	0
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description
7 to 0	P8PLU7 to 0	Pull-up resistor selection 0: Not added 1: Added

■ Port 8 Output Mode Register (P8OMD: 0x03EB8)

<MN101EF76K>

bp	7	6	5	4	3	2	1	0
Flag	-	-	P8OMD5	P8OMD4	P8OMD3	P8OMD2	P8OMD1	P8OMD0
At reset	-	-	0	0	0	0	0	0
Access	-	-	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description
7 to 6	-	-
5	P8OMD5	I/O port or timer output selection 0: P85 1: TM9OD5A
4	P8OMD4	I/O port or timer output selection 0: P84 1: TM9OD4A
3	P8OMD3	I/O port or timer output selection 0: P83 1: TM9OD3A
2	P8OMD2	I/O port or timer output selection 0: P82 1: TM9OD2A
1	P8OMD1	I/O port or timer output selection 0: P81 1: TM9OD1A
0	P8OMD0	I/O port or timer output selection 0: P80 1: TM9OD0A

■ Port 8 Output Mode Register (P8OMD: 0x03EB8)

<MN101EF57G/MN101EF56K>

bp	7	6	5	4	3	2	1	0
Flag	-	-	P8OMD5	P8OMD4	P8OMD3	P8OMD2	P8OMD1	P8OMD0
At reset	-	-	0	0	0	0	0	0
Access	-	-	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description
7 to 6	-	-
5	P8OMD5	I/O port or timer output selection 0: P85 1: TM9OD5
4	P8OMD4	I/O port or timer output selection 0: P84 1: TM9OD4
3	P8OMD3	I/O port or timer output selection 0: P83 1: TM9OD3
2	P8OMD2	I/O port or timer output selection 0: P82 1: TM9OD2
1	P8OMD1	I/O port or timer output selection 0: P81 1: TM9OD1
0	P8OMD0	I/O port or timer output selection 0: P80 1: TM9OD0

■ Port 8 Synchronous Output Control Register (P8SYO: 0x03EF8)

bp	7	6	5	4	3	2	1	0
Flag	P8SYO7	P8SYO6	P8SYO5	P8SYO4	P8SYO3	P8SYO2	P8SYO1	P8SYO0
At reset	0	0	0	0	0	0	0	0
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description
7 to 0	P8SYO7 to 0	Synchronous output selection 0: I/O port 1: Synchronous output

■ Port 8 Synchronous Output Event Selection Register (P8SEV: 0x03EF9)

bp	7	6	5	4	3	2	1	0
Flag	-	-	-	-	-	-	P8SEV1	P8SEV0
At reset	-	-	-	-	-	-	0	0
Access	-	-	-	-	-	-	R/W	R/W

bp	Flag	Description
7 to 2	-	-
1 to 0	P8SEV1 to 0	Synchronous output event selection 00: IRQ2 01: TM7IRQ 10: TM2IRQ 11: TM1IRQ

8.11.3 Block Diagram (MN101EF76K)

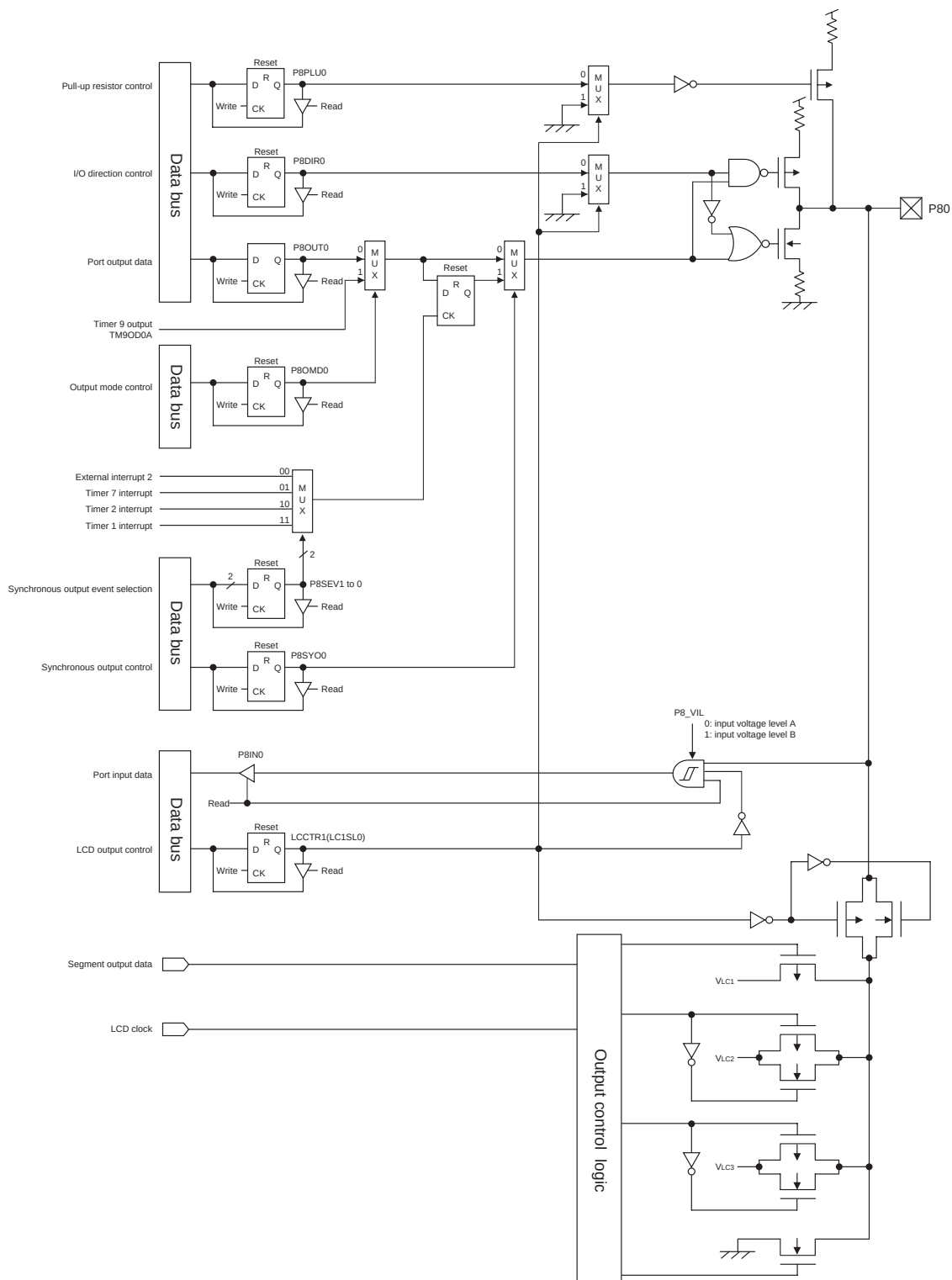


Figure:8.11.1 P80 Block Diagram (MN101EF76K)

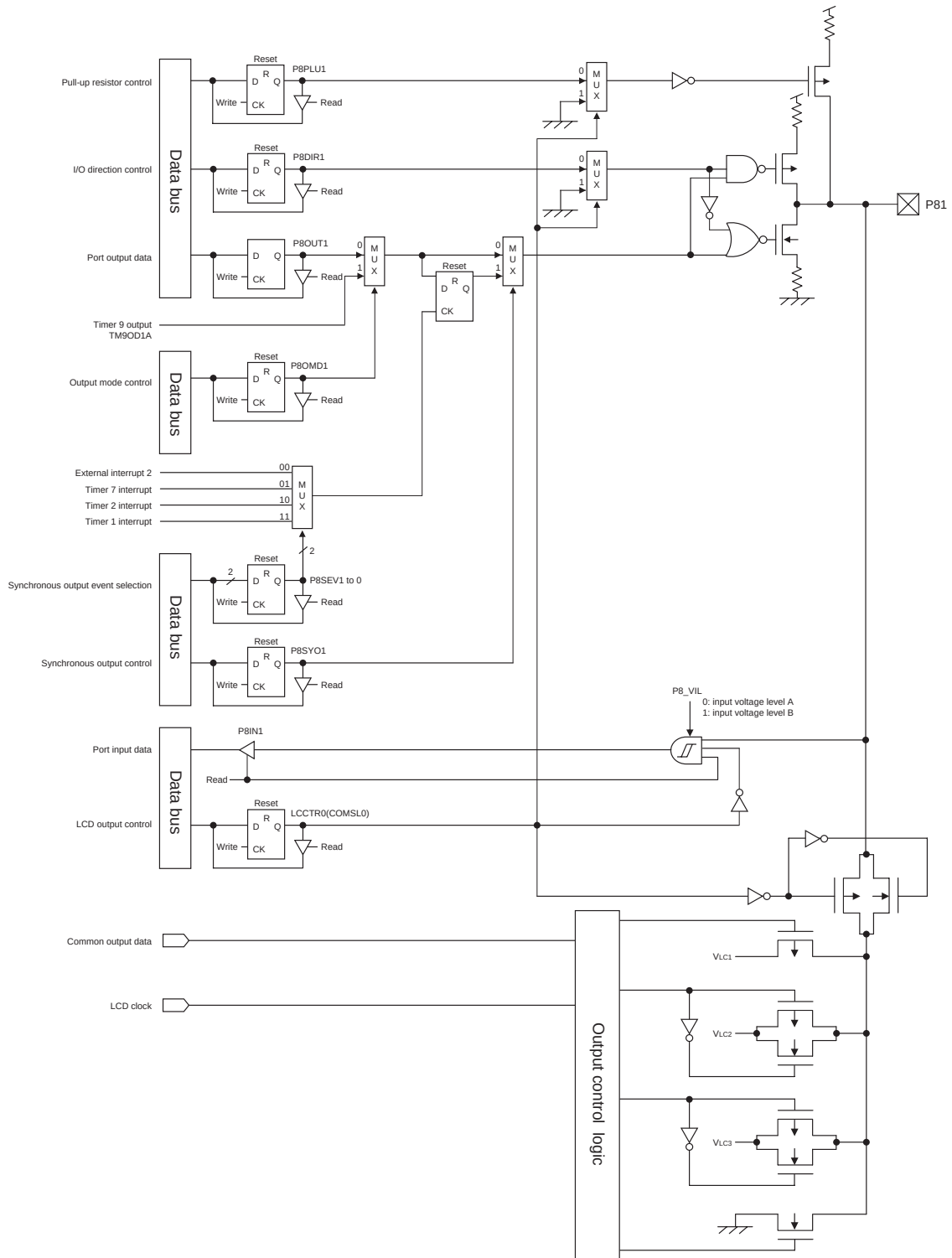


Figure:8.11.2 P81 Block Diagram (MN101EF76K)

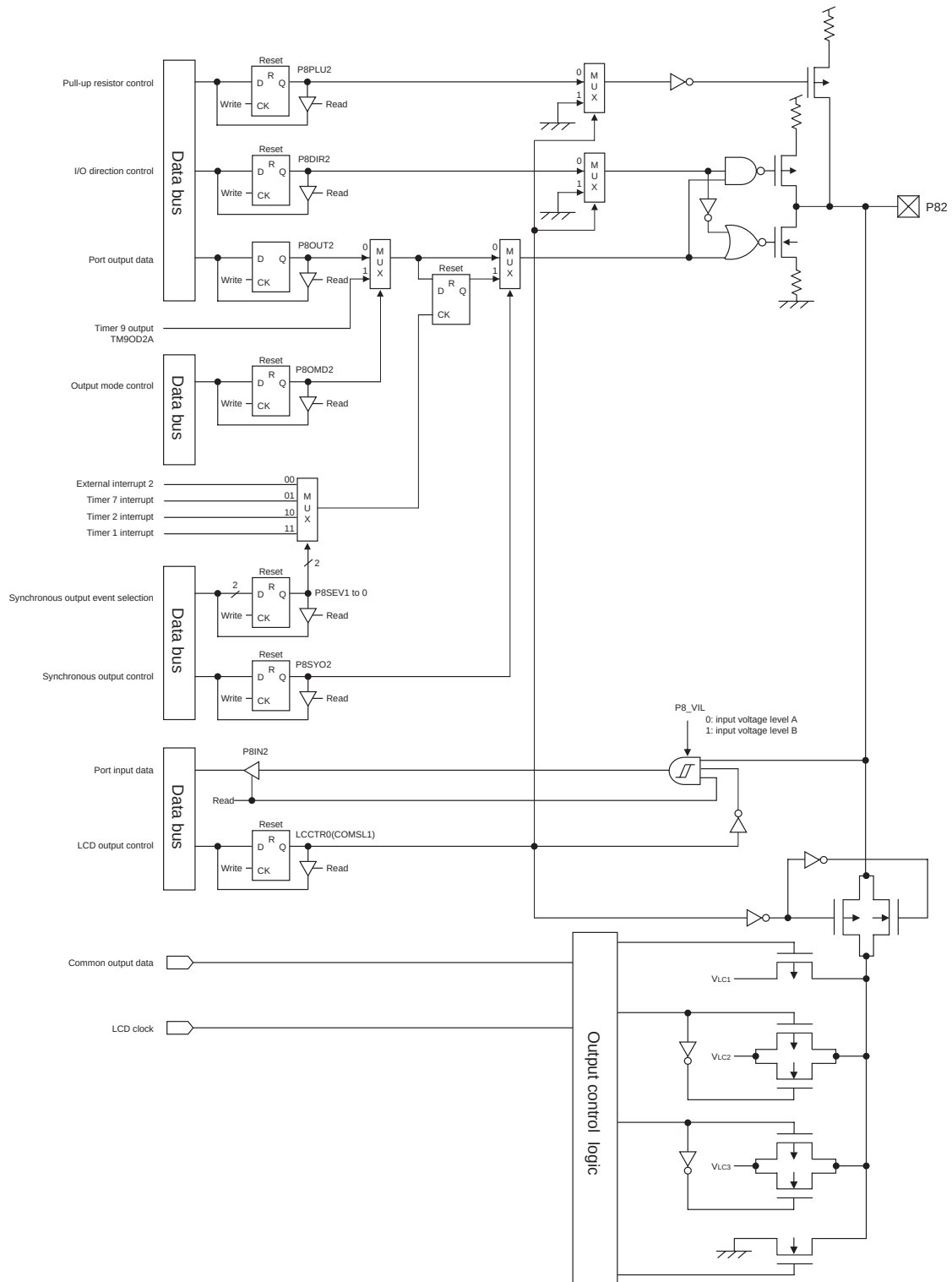


Figure:8.11.3 P82 Block Diagram (MN101EF76K)

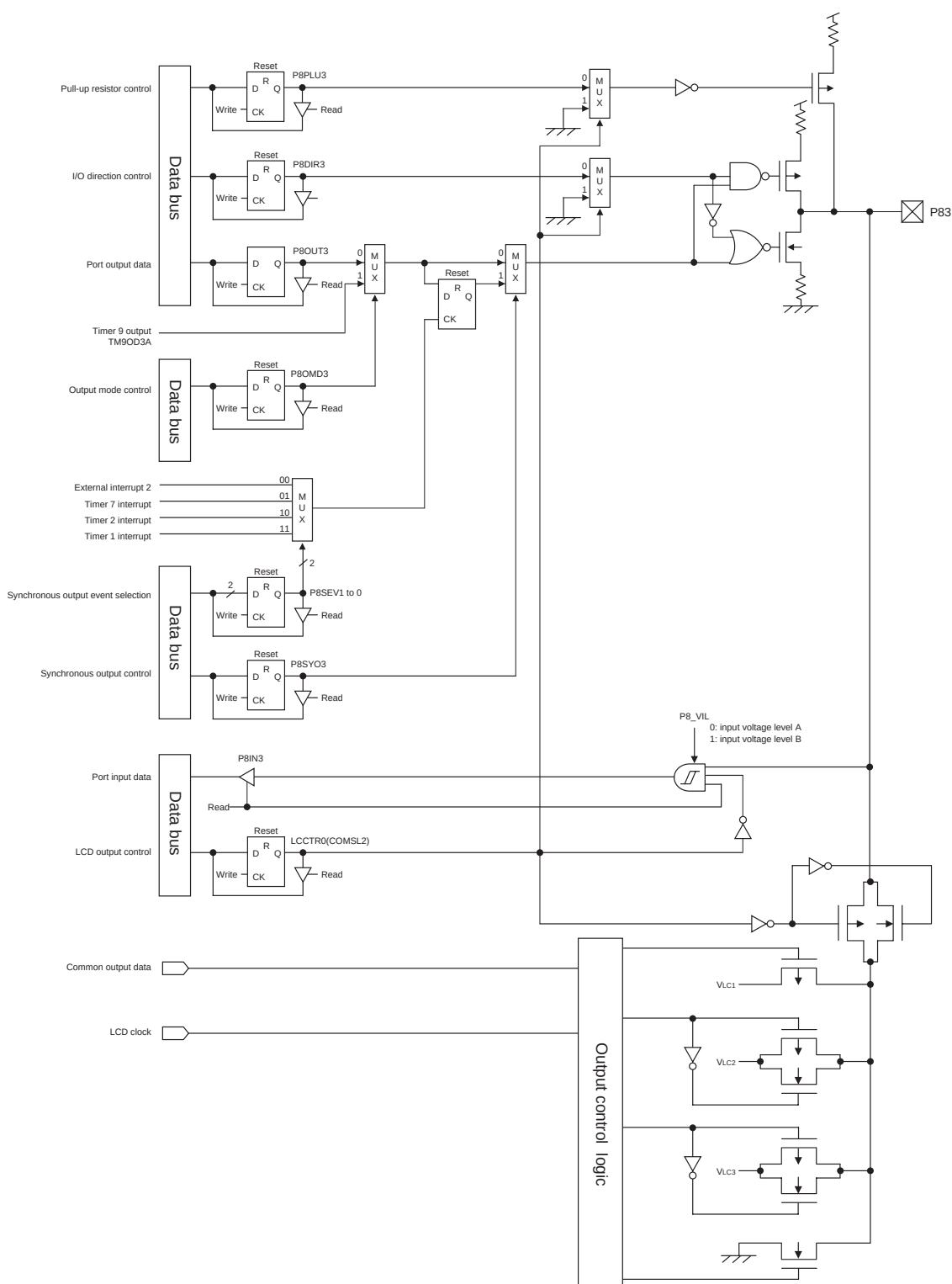


Figure:8.11.4 P83 Block Diagram (MN101EF76K)

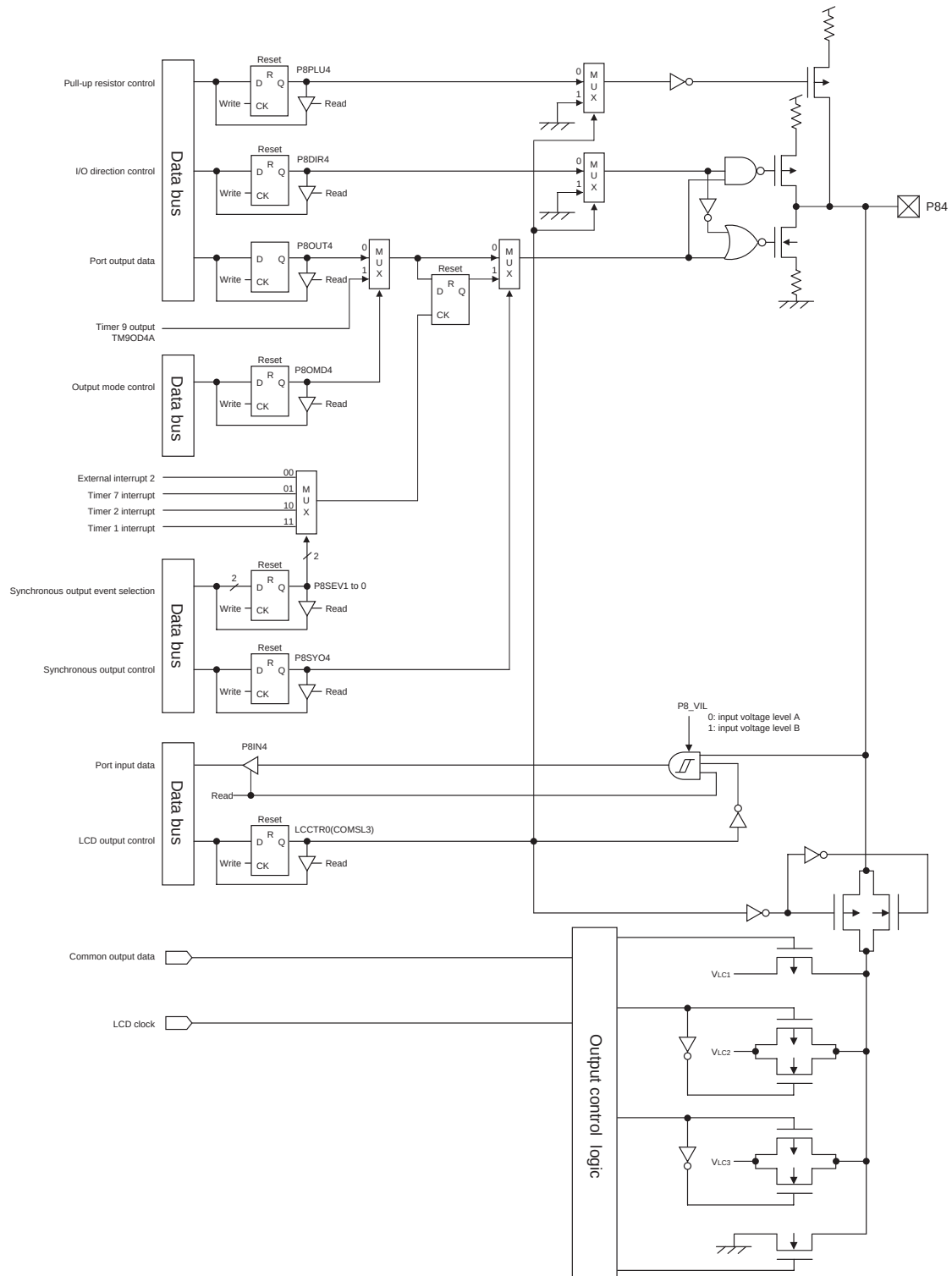


Figure:8.11.5 P84 Block Diagram (MN101EF76K)

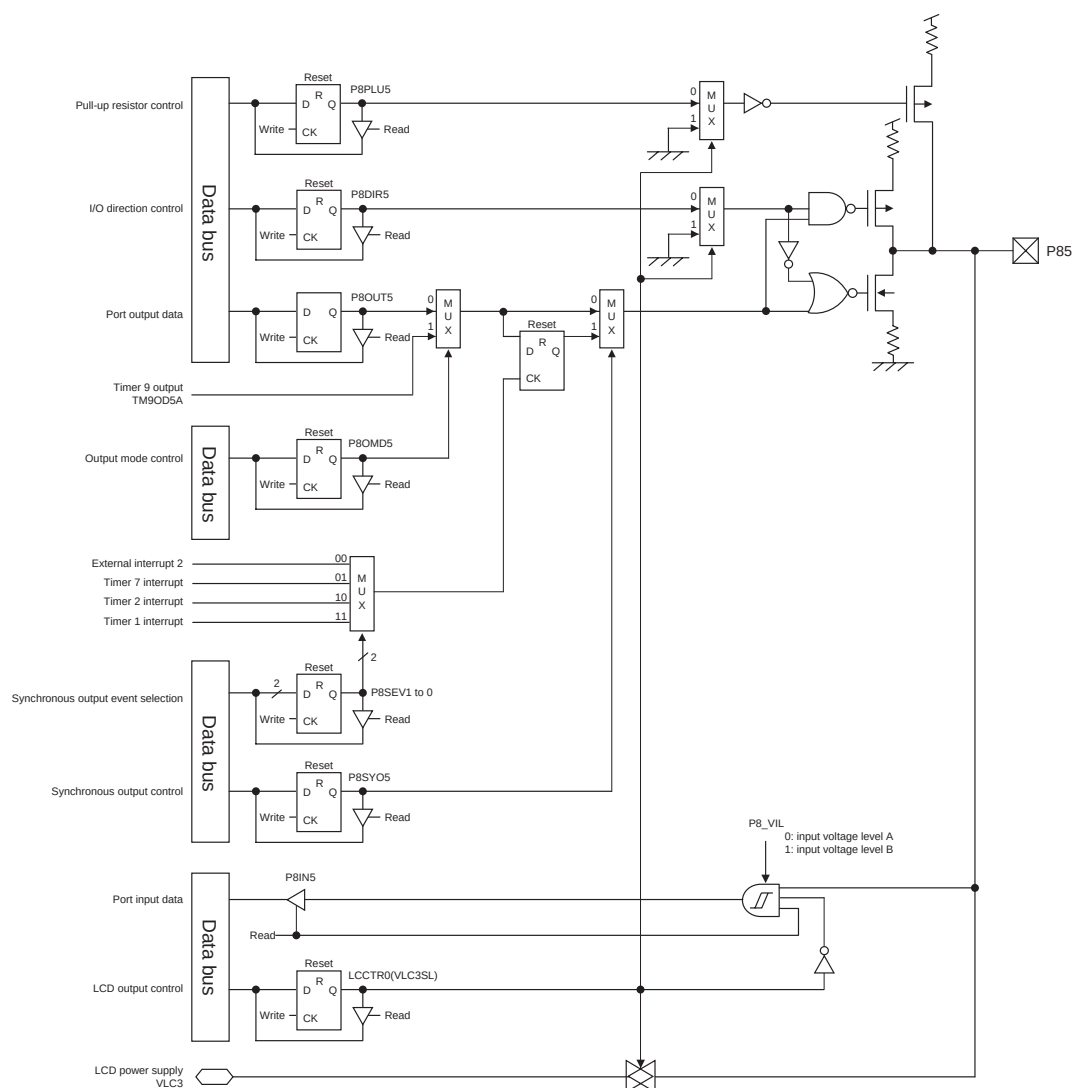


Figure:8.11.6 P85 Block Diagram (MN101EF76K)

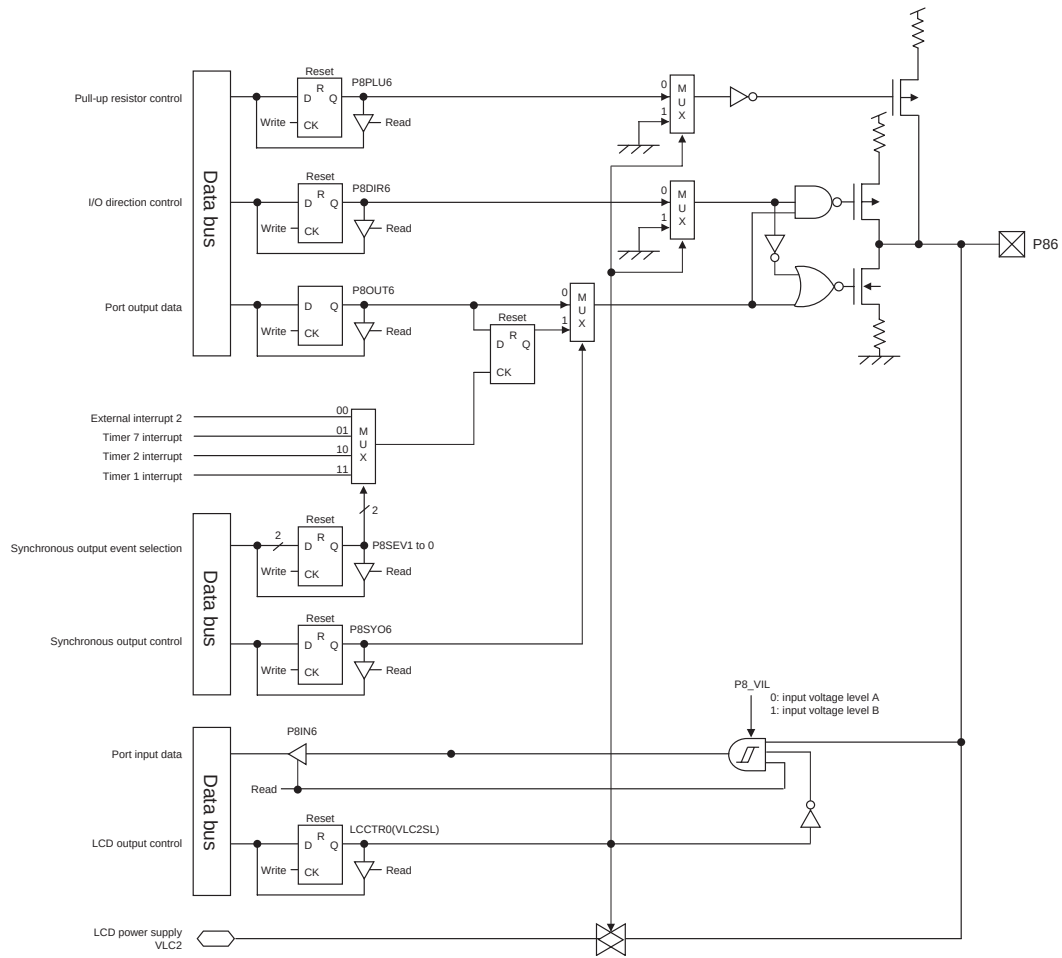


Figure:8.11.7 P86 Block Diagram (MN101EF76K)

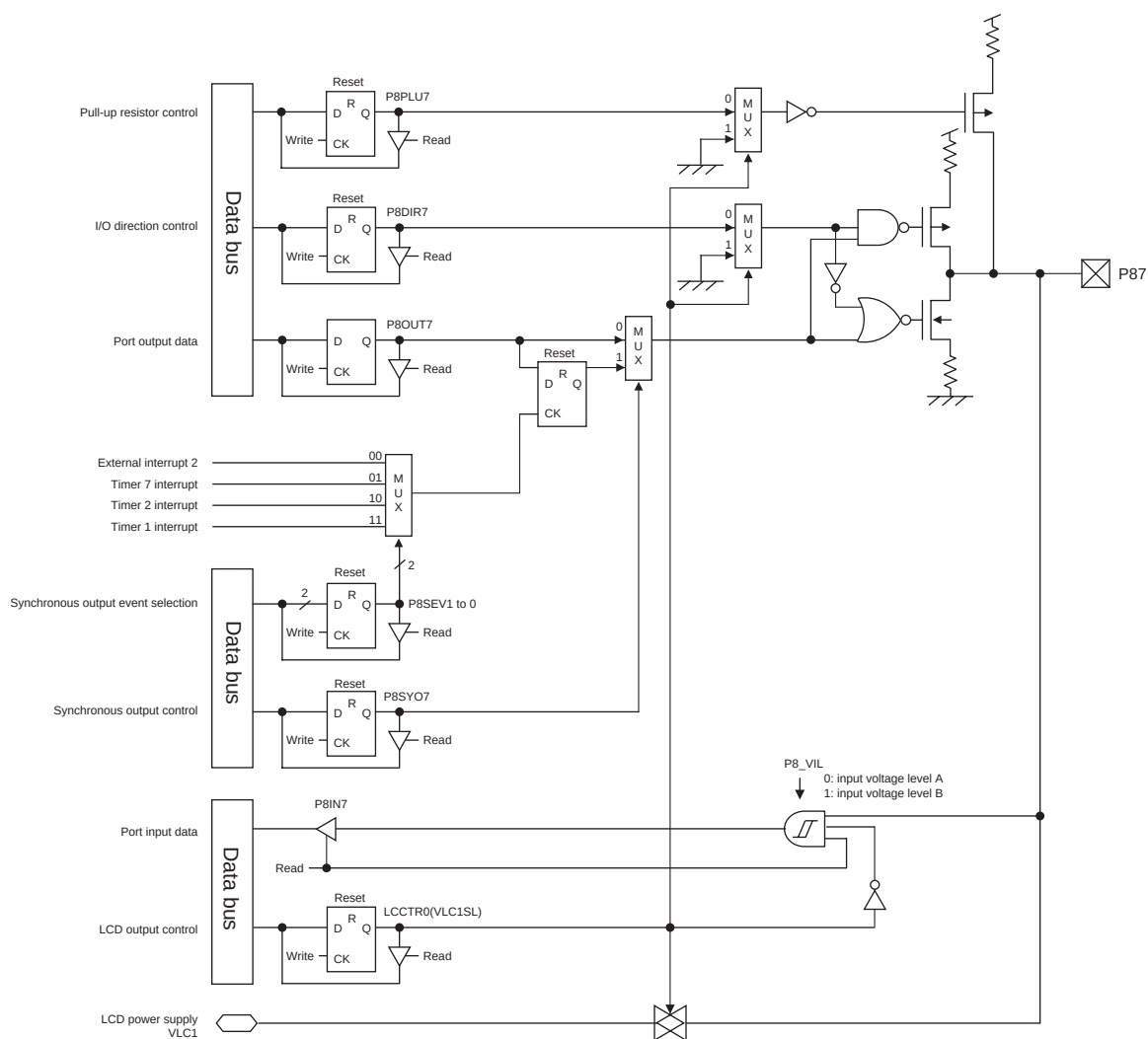


Figure:8.11.8 P87 Block Diagram (MN101EF76K)

8.11.4 Block Diagram (MN101EF57G/MN101EF56K)

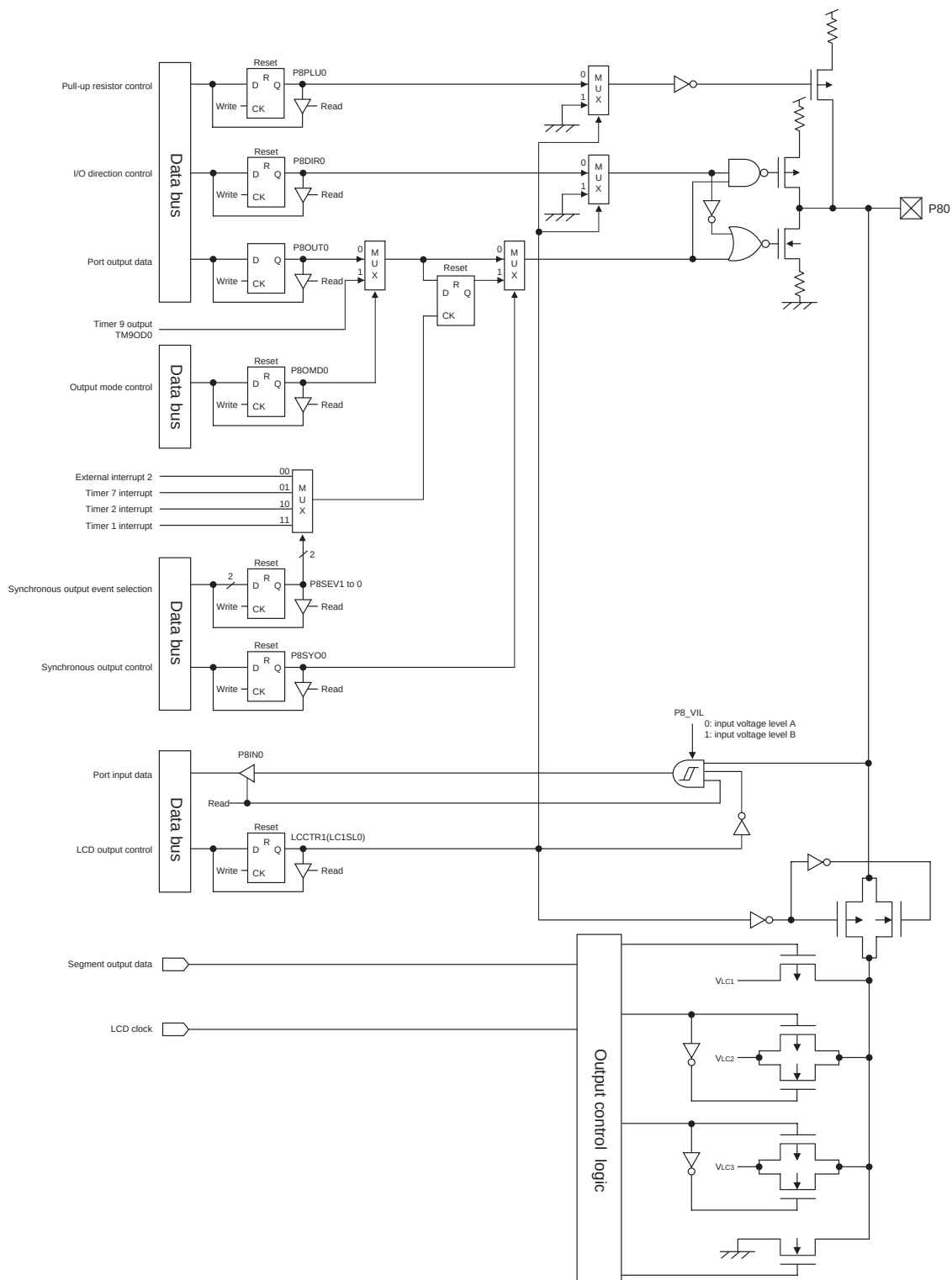


Figure:8.11.9 P80 Block Diagram (MN101EF57G/MN101EF56K)

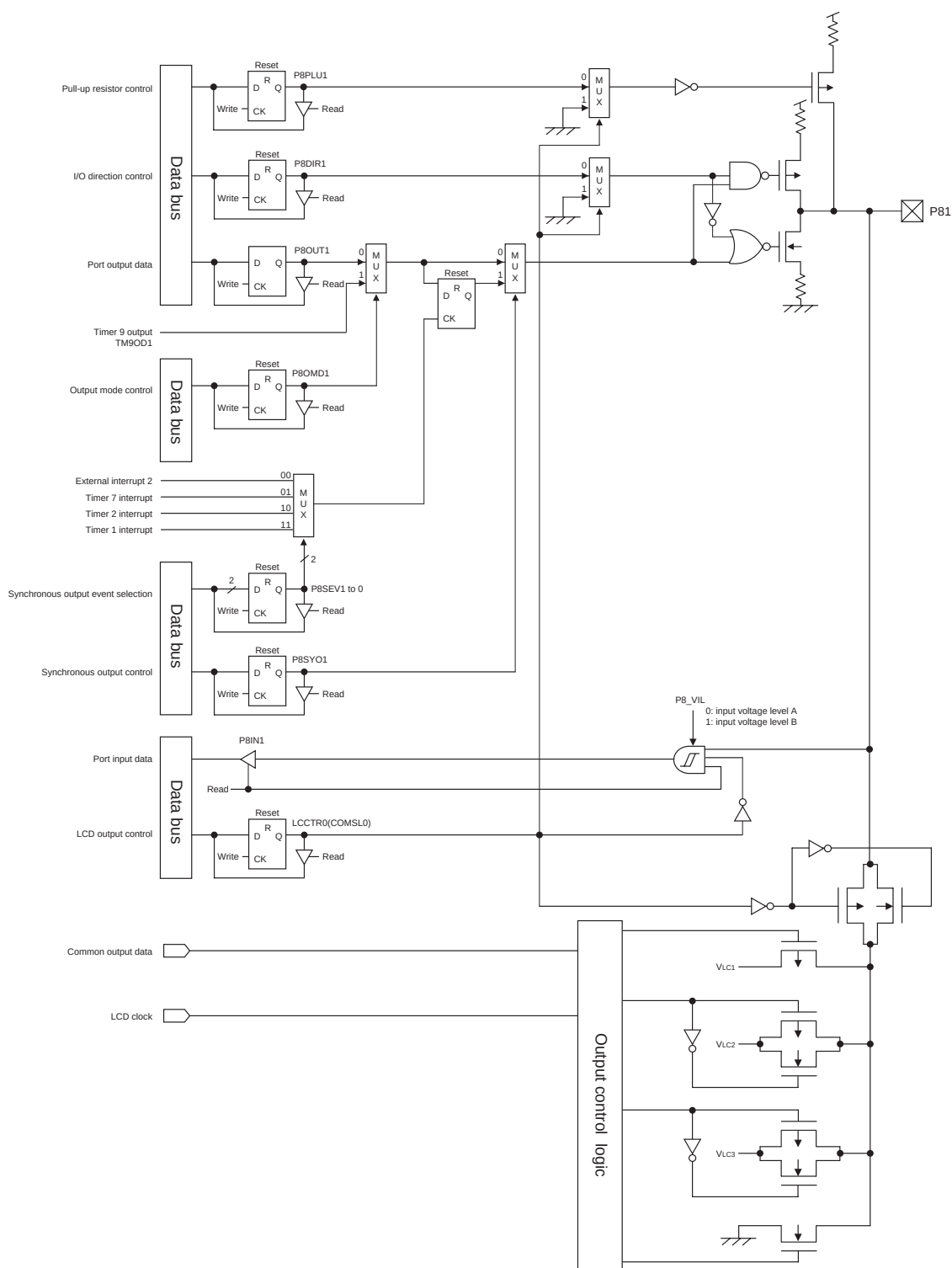


Figure:8.11.10 P81 Block Diagram (MN101EF57G/MN101EF56K)

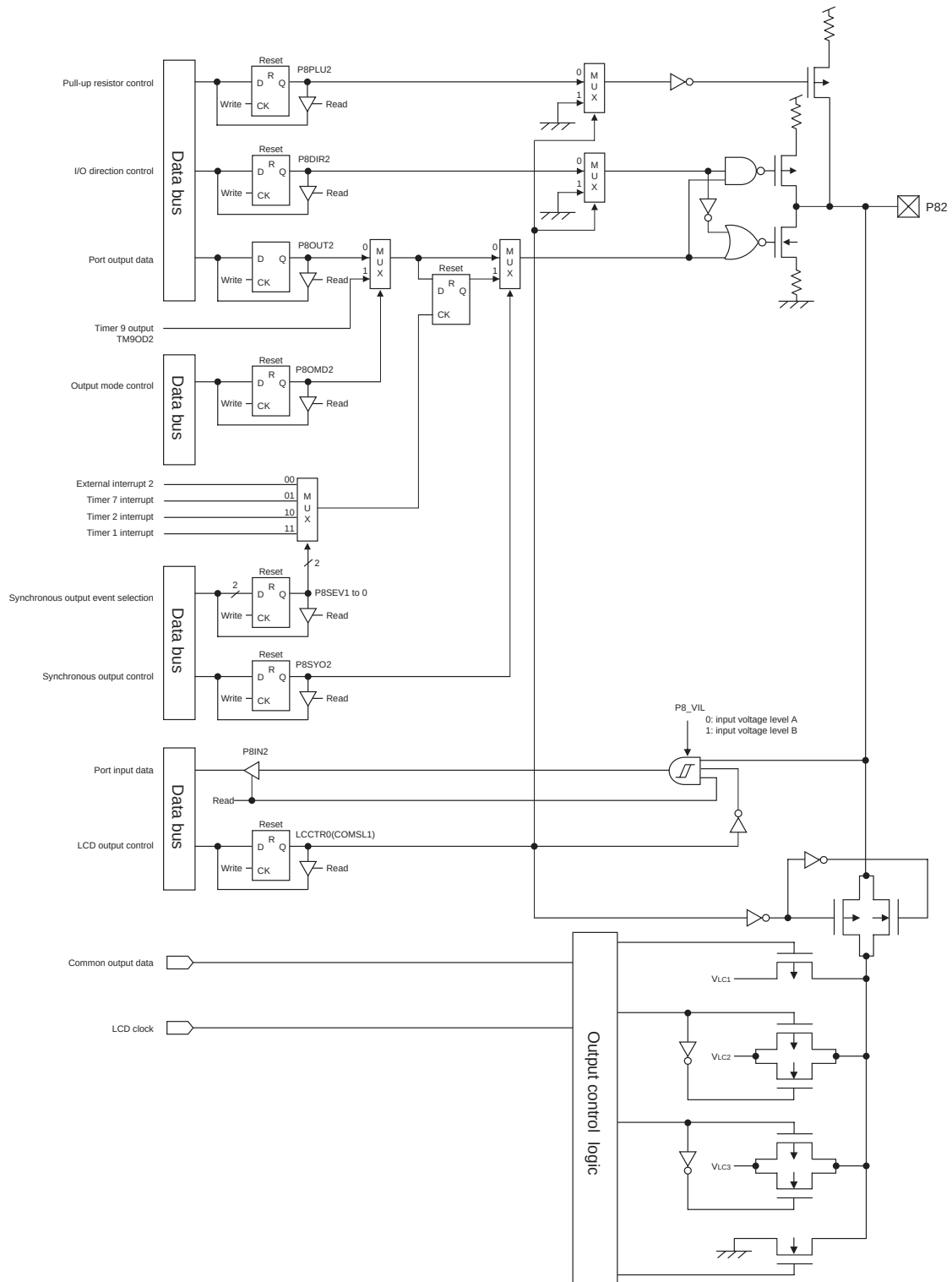


Figure:8.11.11 P82 Block Diagram (MN101EF57G/MN101EF56K)

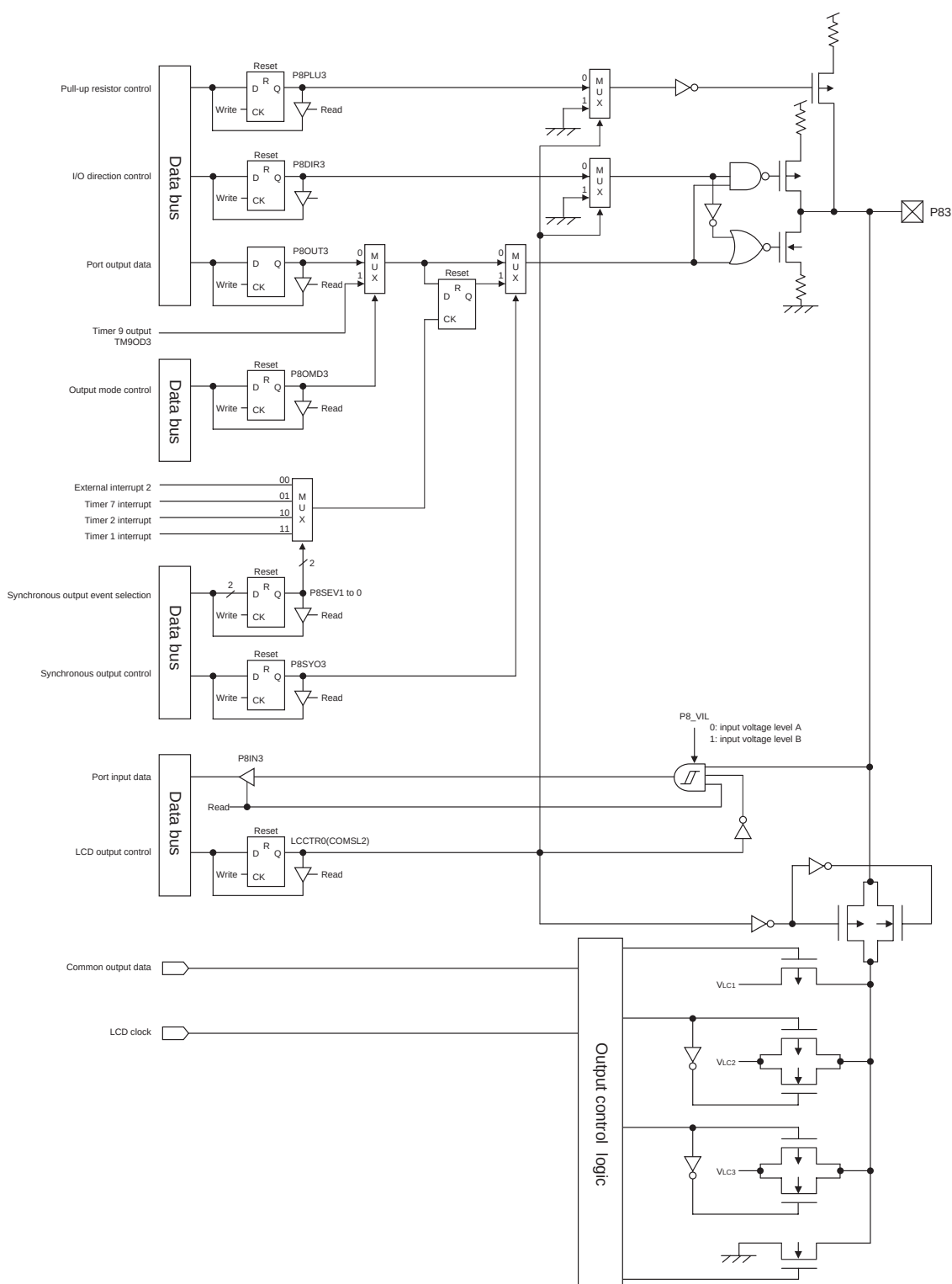


Figure:8.11.12 P83 Block Diagram (MN101EF57G/MN101EF56K)

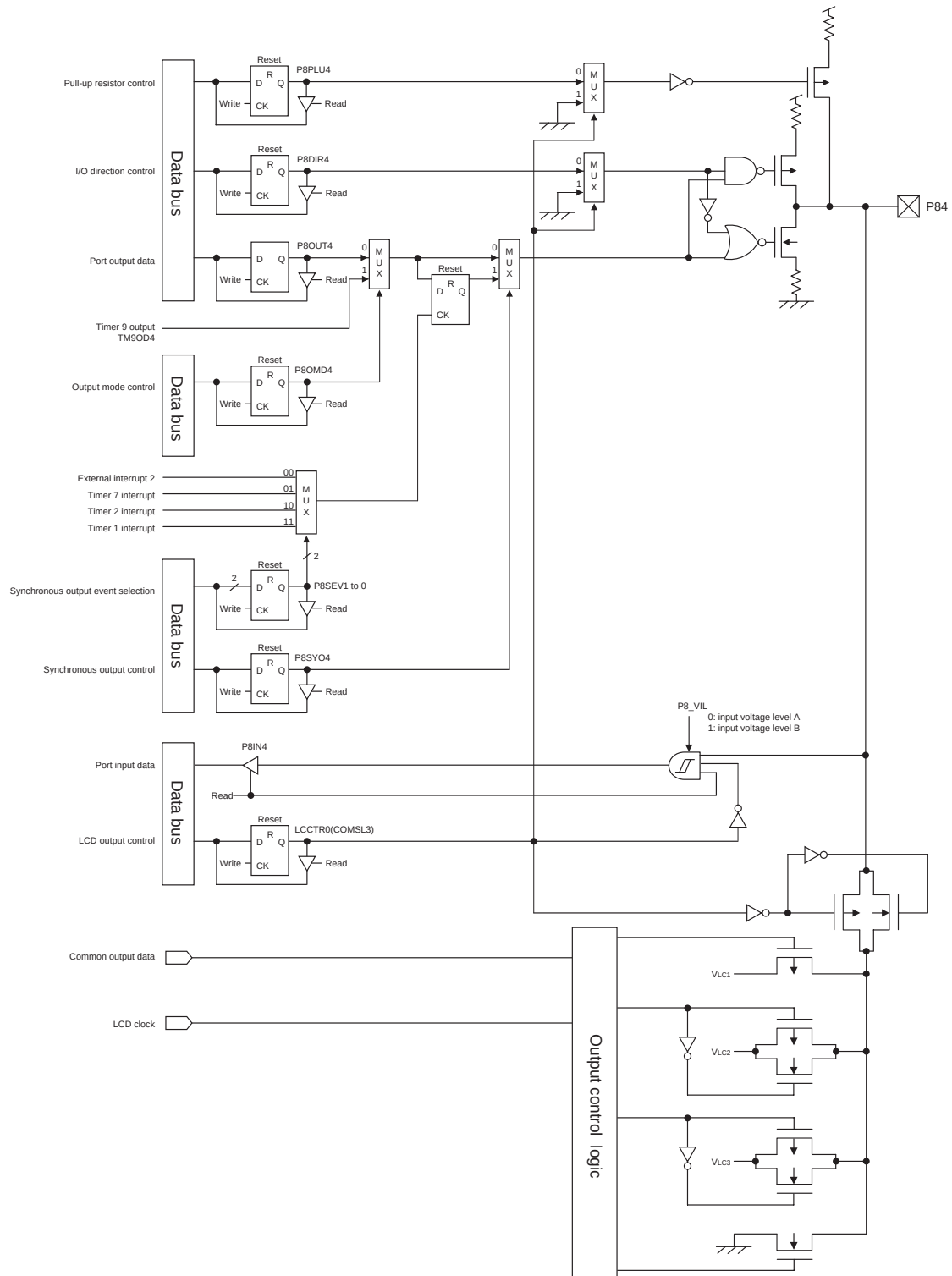


Figure:8.11.13 P84 Block Diagram (MN101EF57G/MN101EF56K)

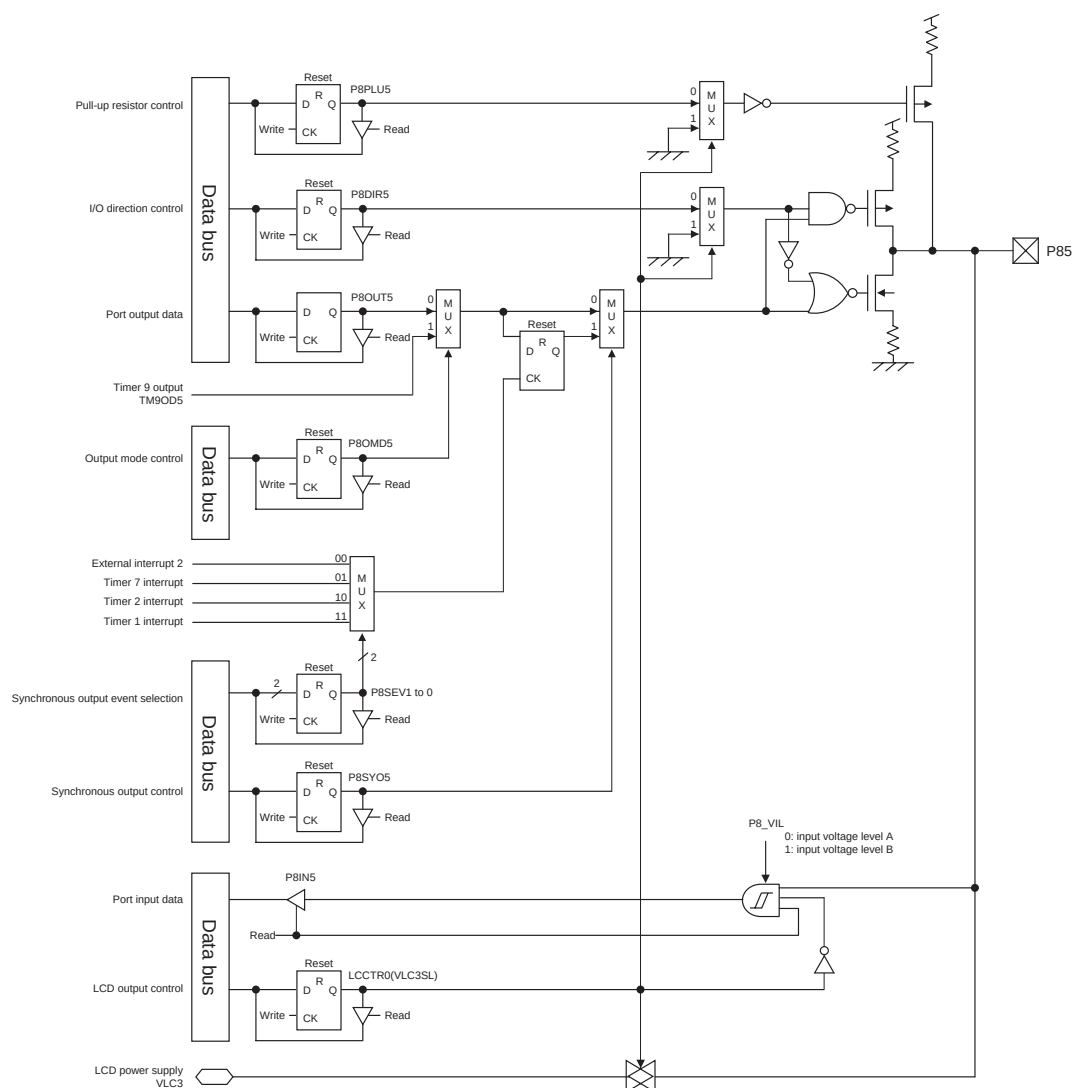


Figure:8.11.14 P85 Block Diagram (MN101EF57G/MN101EF56K)

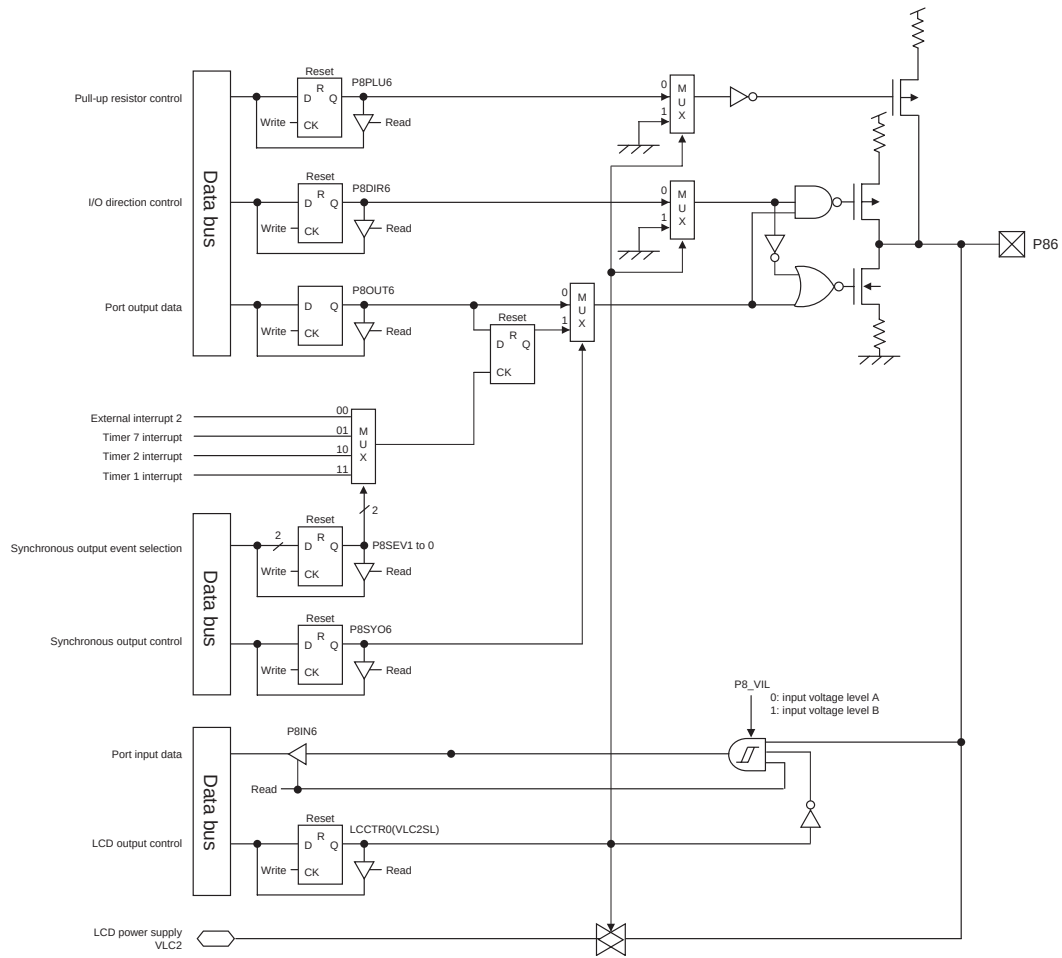


Figure:8.11.15 P86 Block Diagram (MN101EF57G/MN101EF56K)

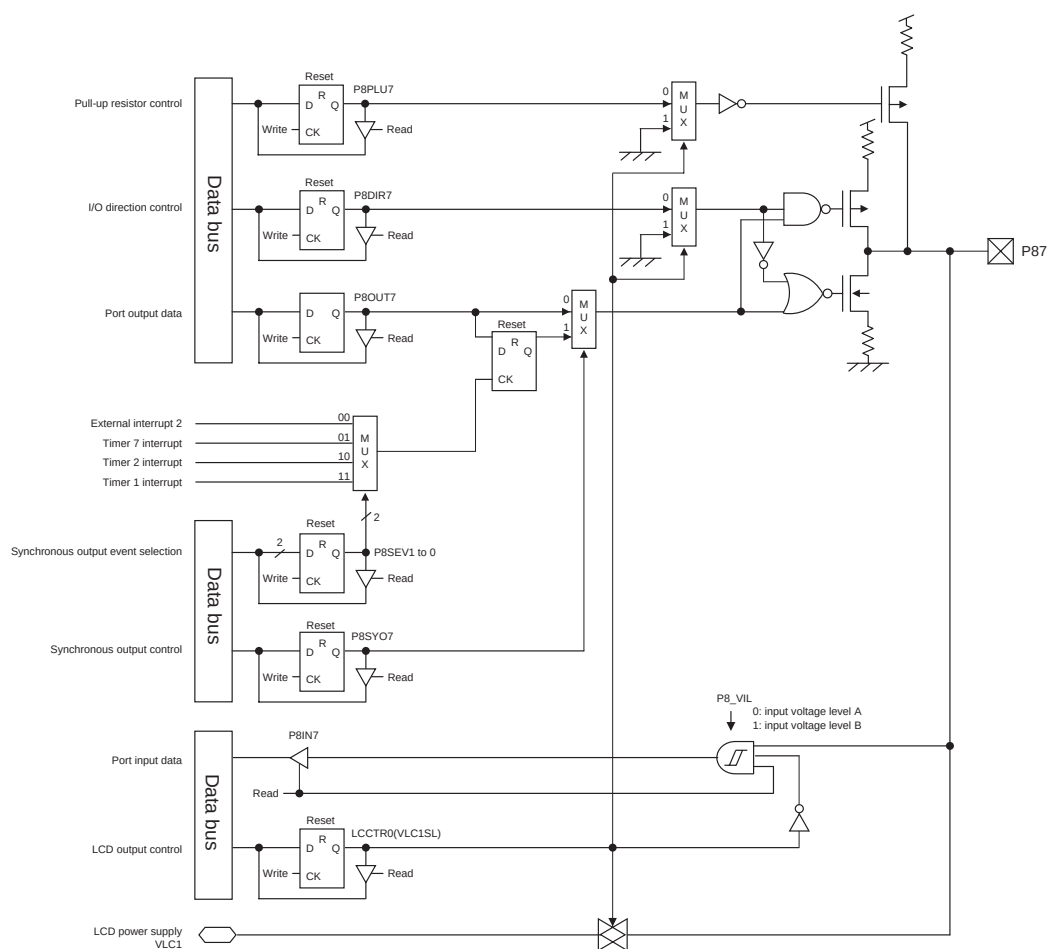


Figure:8.11.16 P87 Block Diagram (MN101EF57G/MN101EF56K)

8.12 Port 9

8.12.1 Description

■ General-purpose Port Setup

To output data to pin, set the control flag of Port 9 direction control register (P9DIR) to "1" and write the value of Port 9 output register (P9OUT).

To read input data of pin, set the control flag of Port 9 direction control register (P9DIR) to "0" and read the value of Port 9 input register (P9IN).

Each bit can be set individually as either an input or output by Port 9 direction control register (P9DIR). The control flag of P9DIR register is set to "1" for output mode, and "0" for input mode.

Each bit can be set individually to add a pull-up resistor by Port 9 pull-up resistor control register (P9PLU). Set the control flag of P9PLU register to "1" to add pull-up resistor.

Table:8.12.1 Port 9 Functions

Table remarks -: Without function

MN101EF76K/MN101EF56K				MN101EF57G			
Pin	Special function			Pin	Special function		
P90	XI			P90	XI		
P91	XO			P91	XO		
P92	SEG46	AN19		P92	SEG31	AN14	
P93	SEG47	AN18		P93	SEG32	AN13	
P94	SEG48	AN17	DA_B	P94	SEG33	AN12	DA_A
P95	SEG49	AN16	DA_A	P95	-	-	-

■ Special Function Pin Setup

P92 to P95 are also used as LCD segment output pins.

LCD segment output pin is selected by setting the corresponding flag of LCD output control register n (LCCTRn) to "1".

Port and segment can be switched by one bit. When segment output pin is selected, this pin is forced to be output mode and pull-up resistor is not added.

P92 to P95 are also used as analog input pins.

Port 9 input mode register (P9IMD) can select the input mode by one bit. When using as the analog input pin, undefined value is read from Port 9 input register (P9IN).

P90 and P91 are also used as low-speed oscillator.

When the SOSCNT flag of the external low-speed oscillation control register (OSCSCNT) is "1", P90 and P91 can be used as low-speed oscillator.

P94 and P95 are also used as the analog output pins for D/A.

8.12.2 Registers

Table:8.12.2 shows registers that control Port 9.

Table:8.12.2 Port 9 Control Register

Table remarks √: With function -: Without function

Registers	Address	R/W	Function	Page	MN101 EF76K	MN101 EF57G	MN101 EF56K
P9OUT	0x03E79	R/W	Port 9 output register	VIII-175	√	√	√
P9IN	0x03E89	R	Port 9 input register	VIII-176	√	√	√
P9DIR	0x03E99	R/W	Port 9 direction control register	VIII-177	√	√	√
P9PLU	0x03EA9	R/W	Port 9 pull-up resistor control register	VIII-178	√	√	√
P9IMD	0x03EC9	R/W	Port 9 input mode register	VIII-179	√	√	√

R/W: Readable/Writable

R: Read only

■ Port 9 Output Register (P9OUT: 0x03E79)

<MN101EF76K/MN101EF56K>

bp	7	6	5	4	3	2	1	0
Flag	-	-	P9OUT5	P9OUT4	P9OUT3	P9OUT2	P9OUT1	P9OUT0
At reset	-	-	X	X	X	X	X	X
Access	-	-	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description
7 to 6	-	-
5 to 0	P9OUT5 to 0	Output data 0: Output "Low" (V_{SS} -level) 1: Output "High" (V_{DD5} -level)

<MN101EF57G>

bp	7	6	5	4	3	2	1	0
Flag	-	-	-	P9OUT4	P9OUT3	P9OUT2	P9OUT1	P9OUT0
At reset	-	-	-	X	X	X	X	X
Access	-	-	-	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description
7 to 5	-	-
4 to 0	P9OUT4 to 0	Output data 0: Output "Low" (V_{SS} -level) 1: Output "High" (V_{DD5} -level)

■ Port 9 Input Register (P9IN: 0x03E89)

<MN101EF76K/MN101EF56K>

bp	7	6	5	4	3	2	1	0
Flag	-	-	P9IN5	P9IN4	P9IN3	P9IN2	P9IN1	P9IN0
At reset	-	-	X	X	X	X	X	X
Access	-	-	R	R	R	R	R	R

bp	Flag	Description
7 to 6	-	-
5 to 0	P9IN5 to 0	Input data 0: Pin is "Low" (V_{SS} -level) 1: Pin is "High" (V_{DD5} -level)

<MN101EF57G>

bp	7	6	5	4	3	2	1	0
Flag	-	-	-	P9IN4	P9IN3	P9IN2	P9IN1	P9IN0
At reset	-	-	-	X	X	X	X	X
Access	-	-	-	R	R	R	R	R

bp	Flag	Description
7 to 5	-	-
4 to 0	P9IN4 to 0	Input data 0: Pin is "Low" (V_{SS} -level) 1: Pin is "High" (V_{DD5} -level)

■ Port 9 Direction Control Register (P9DIR: 0x03E99)

<MN101EF76K/MN101EF56K>

bp	7	6	5	4	3	2	1	0
Flag	-	-	P9DIR5	P9DIR4	P9DIR3	P9DIR2	P9DIR1	P9DIR0
At reset	-	-	0	0	0	0	0	0
Access	-	-	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description
7 to 6	-	-
5 to 0	P9DIR5 to 0	I/O mode selection 0: Input mode 1: Output mode

<MN101EF57G>

bp	7	6	5	4	3	2	1	0
Flag	-	-	-	P9DIR4	P9DIR3	P9DIR2	P9DIR1	P9DIR0
At reset	-	-	-	0	0	0	0	0
Access	-	-	-	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description
7 to 5	-	-
4 to 0	P9DIR4 to 0	I/O mode selection 0: Input mode 1: Output mode

■ Port 9 Pull-up Resistor Control Register (P9PLU: 0x03EA9)

<MN101EF76K/MN101EF56K>

bp	7	6	5	4	3	2	1	0
Flag	-	-	P9PLU5	P9PLU4	P9PLU3	P9PLU2	P9PLU1	P9PLU0
At reset	-	-	0	0	0	0	0	0
Access	-	-	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description
7 to 6	-	-
5 to 0	P9PLU5 to 0	Pull-up resistor selection 0: Not added 1: Added

<MN101EF57G>

bp	7	6	5	4	3	2	1	0
Flag	-	-	-	P9PLU4	P9PLU3	P9PLU2	P9PLU1	P9PLU0
At reset	-	-	-	0	0	0	0	0
Access	-	-	-	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description
7 to 5	-	-
4 to 0	P9PLU4 to 0	Pull-up resistor selection 0: Not added 1: Added

■ Port 9 Input Mode Register (P9IMD: 0x03EC9)

<MN101EF76K/MN101EF56K>

bp	7	6	5	4	3	2	1	0
Flag	-	-	P9IMD5	P9IMD4	P9IMD3	P9IMD2	-	-
At reset	-	-	0	0	0	0	-	-
Access	-	-	R/W	R/W	R/W	R/W	-	-

bp	Flag	Description
7 to 6	-	-
5	P9IMD5	I/O port or analog input selection 0: P95 1: AN16
4	P9IMD4	I/O port or analog input selection 0: P94 1: AN17
3	P9IMD3	I/O port or analog input selection 0: P93 1: AN18
2	P9IMD2	I/O port or analog input selection 0: P92 1: AN19
1 to 0	-	-

<MN101EF57G>

bp	7	6	5	4	3	2	1	0
Flag	-	-	-	P9IMD4	P9IMD3	P9IMD2	-	-
At reset	-	-	-	0	0	0	-	-
Access	-	-	-	R/W	R/W	R/W	-	-

bp	Flag	Description
7 to 5	-	-
4	P9IMD4	I/O port or analog input selection 0: P94 1: AN12
3	P9IMD3	I/O port or analog input selection 0: P93 1: AN13
2	P9IMD2	I/O port or analog input selection 0: P92 1: AN14
1 to 0	-	-

8.12.3 Block Diagram (MN101EF76K/MN101EF56K)

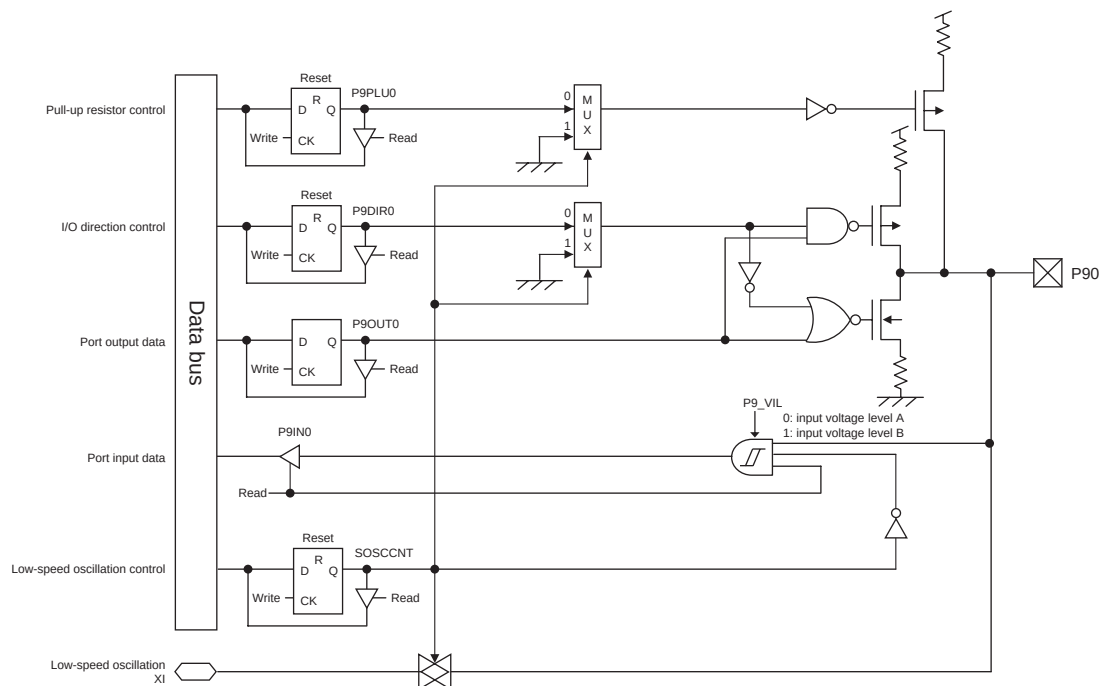


Figure:8.12.1 P90 Block Diagram (MN101EF76K/MN101EF56K)

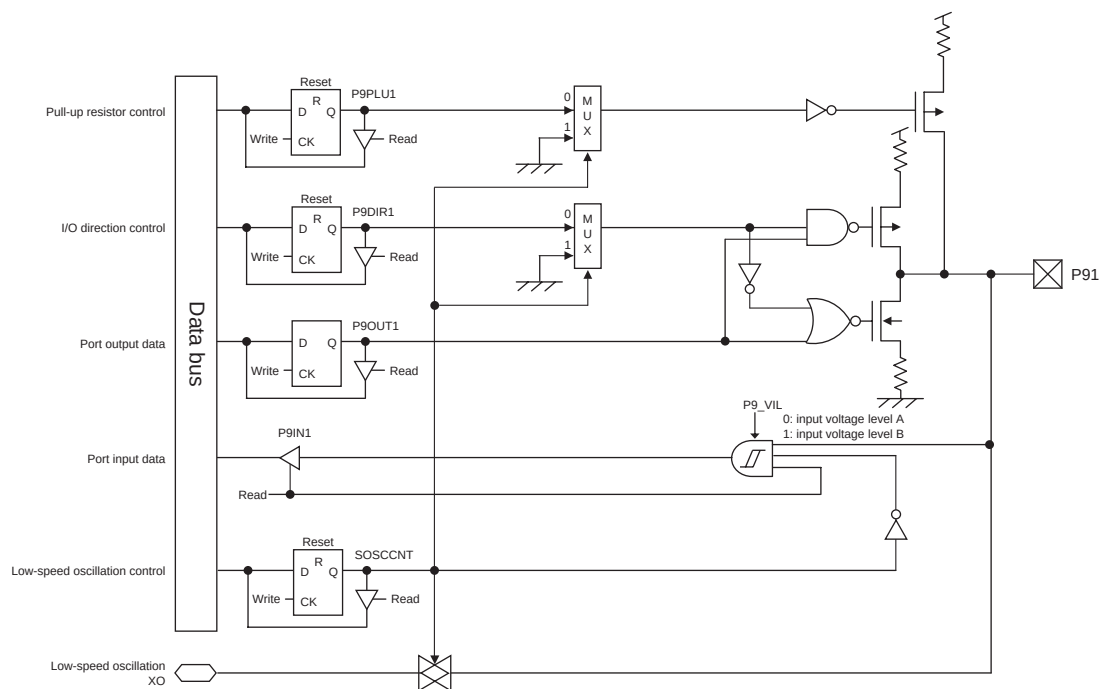


Figure:8.12.2 P91 Block Diagram (MN101EF76K/MN101EF56K)

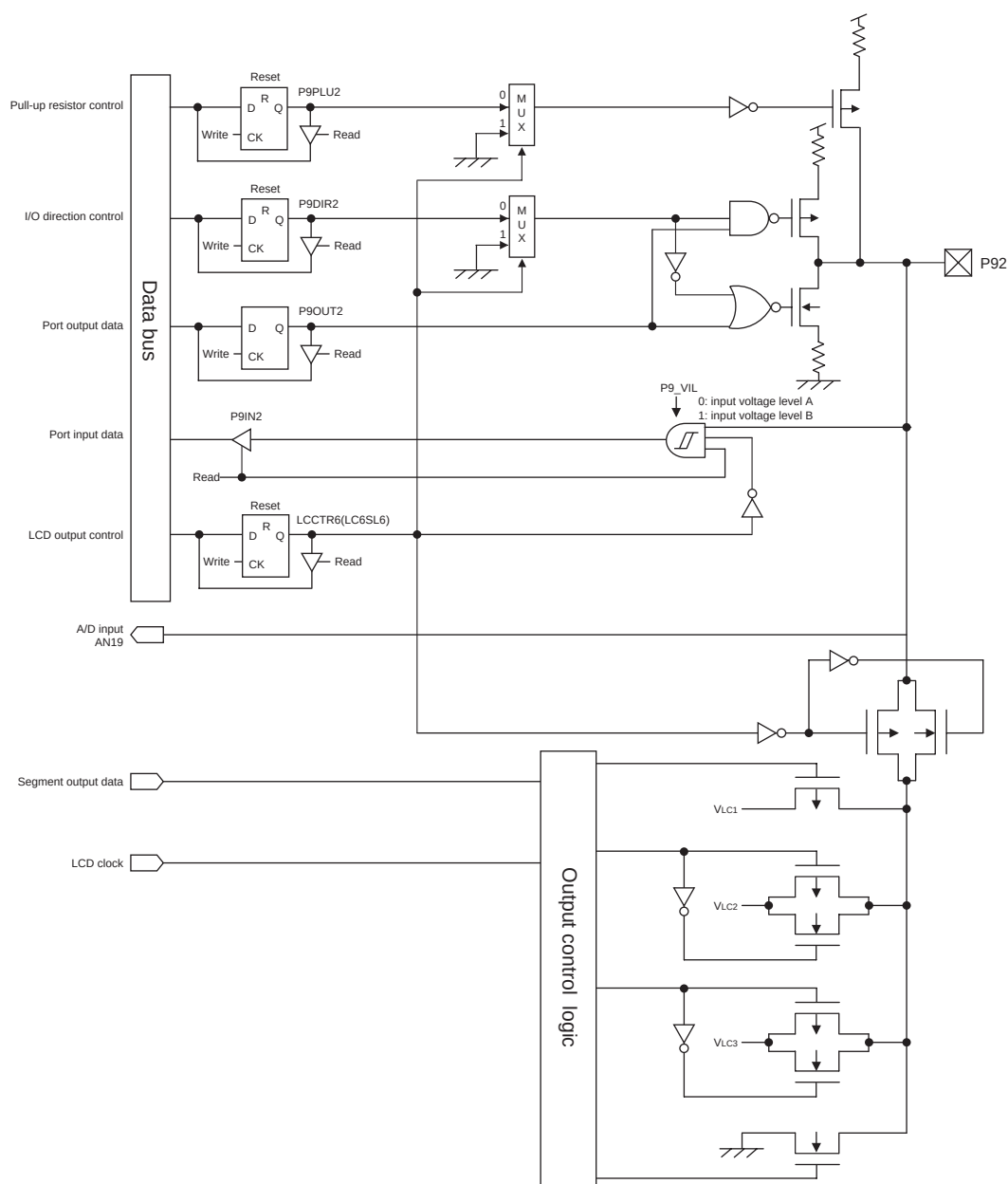


Figure:8.12.3 P92 Block Diagram (MN101EF76K/MN101EF56K)

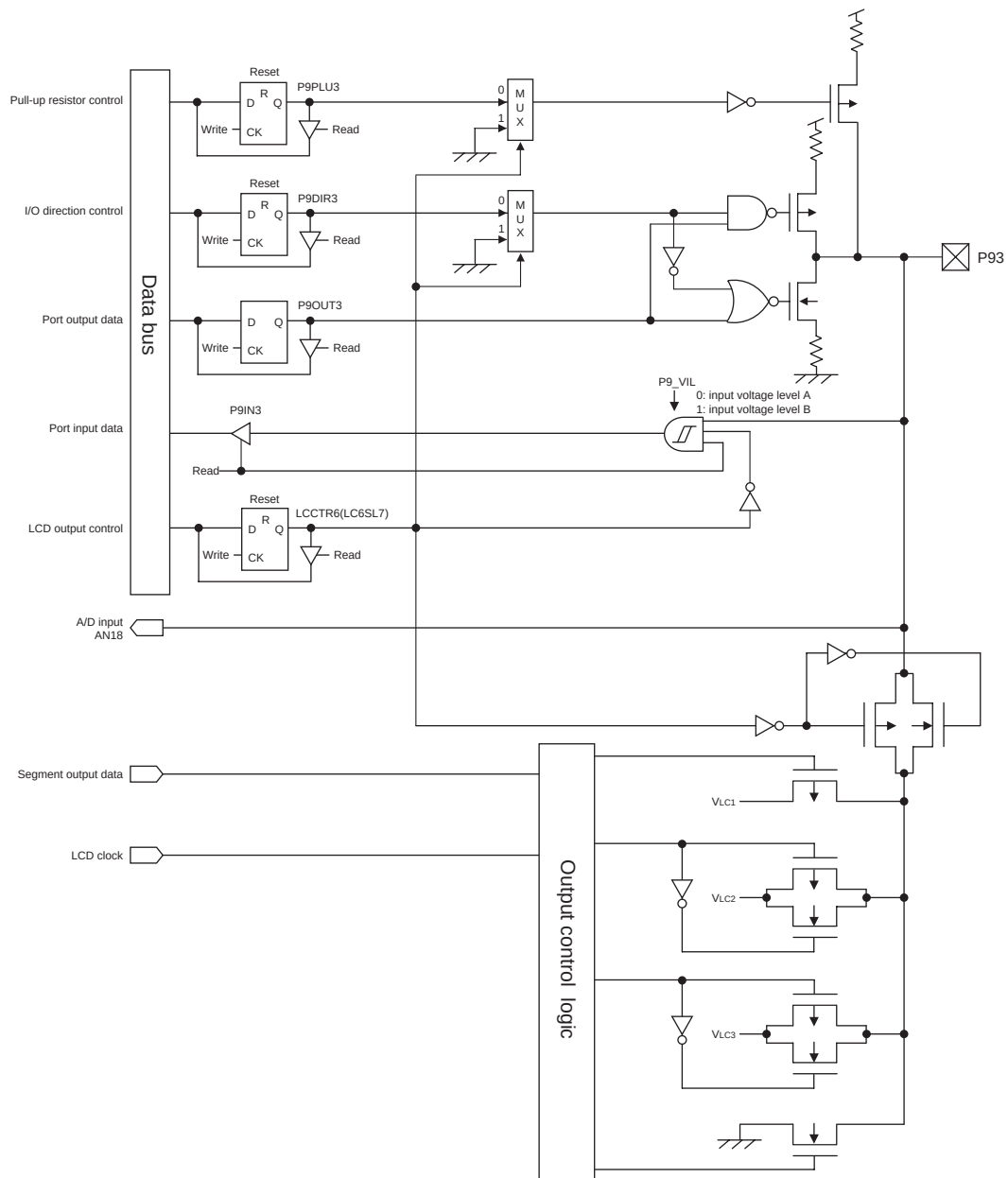


Figure:8.12.4 P93 Block Diagram (MN101EF76K/MN101EF56K)

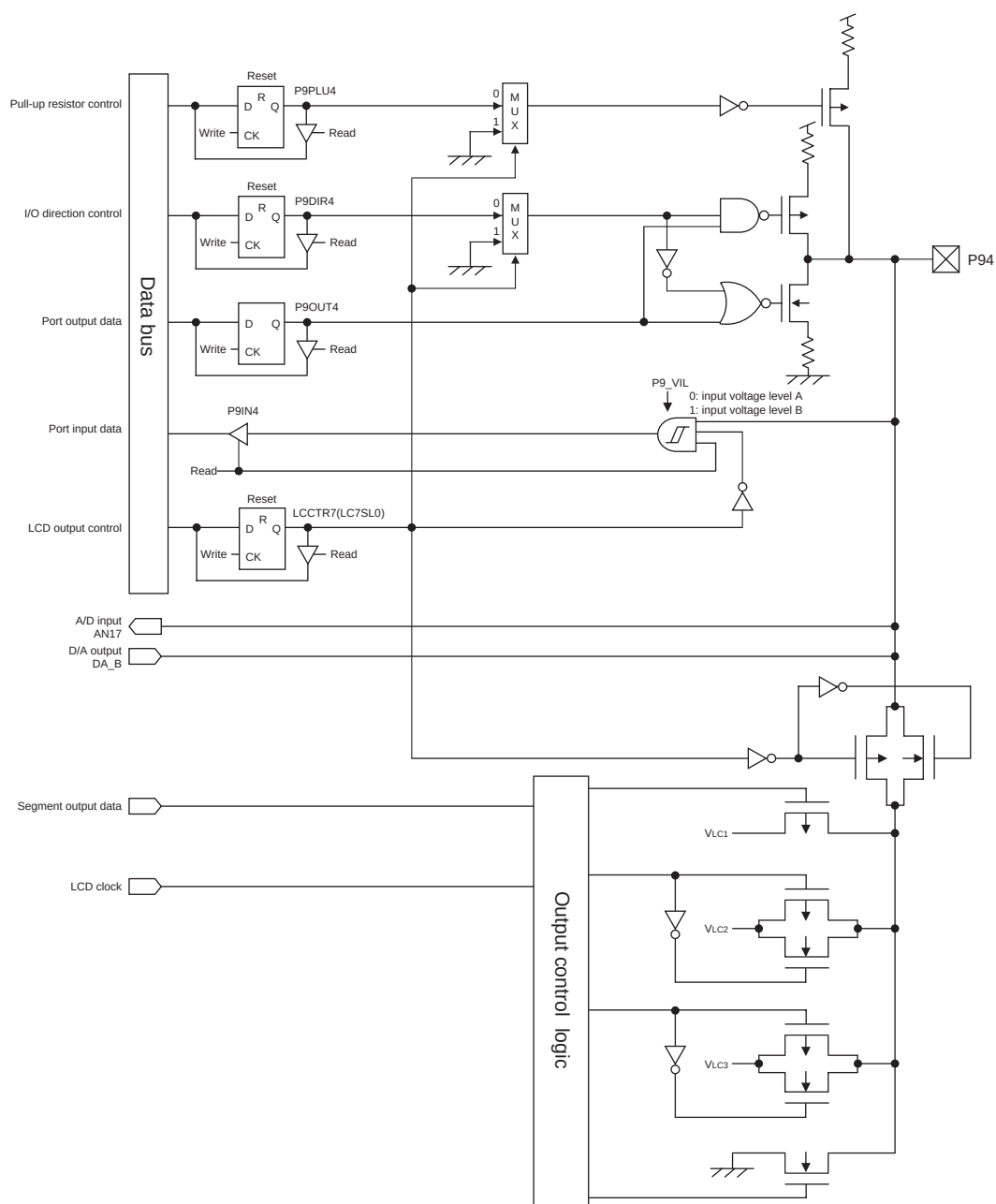


Figure:8.12.5 P94 Block Diagram (MN101EF76K/MN101EF56K)

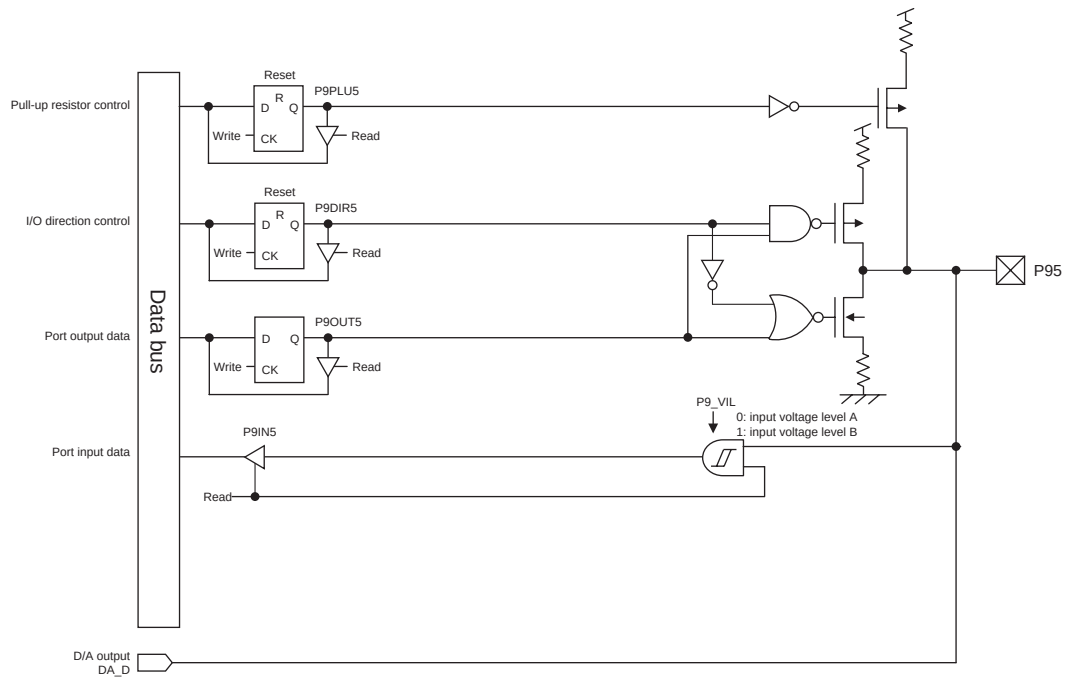


Figure:8.12.6 P95 Block Diagram (MN101EF76K/MN101EF56K)

8.12.4 Block Diagram (MN101EF57G)

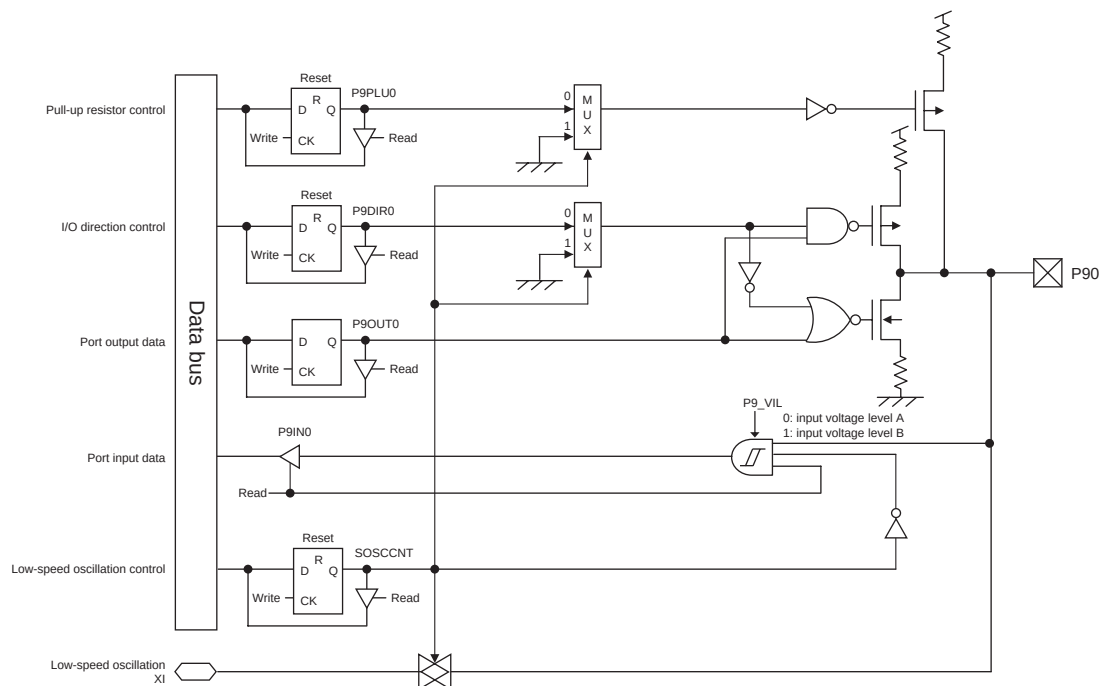


Figure:8.12.7 P90 Block Diagram (MN101EF57G)

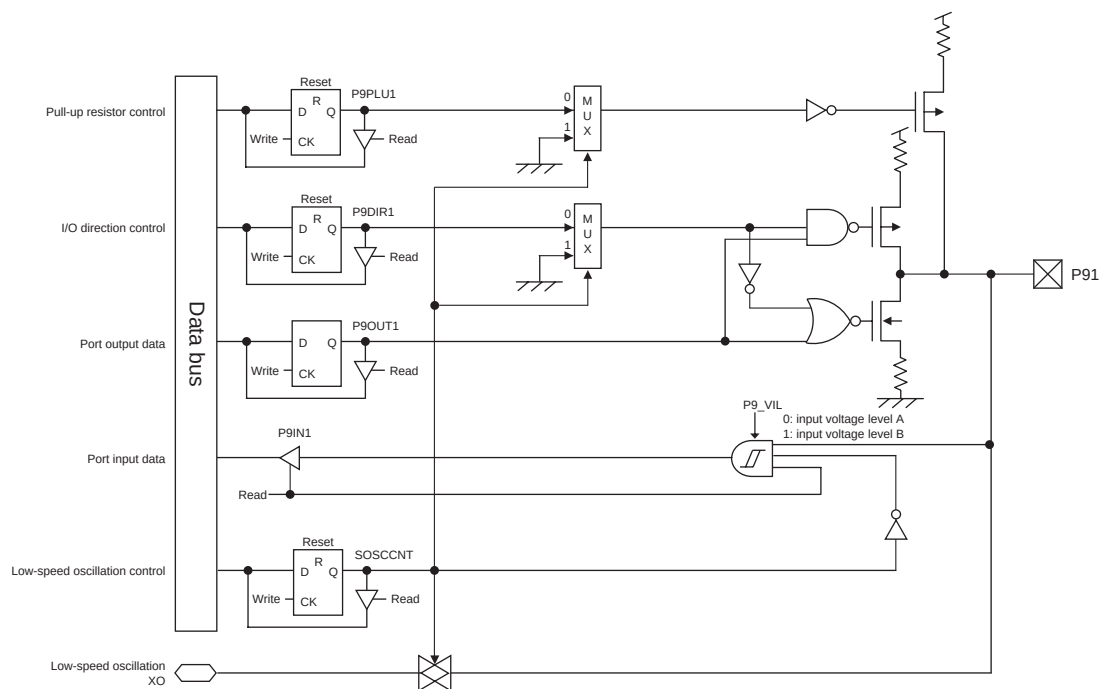


Figure:8.12.8 P91 Block Diagram (MN101EF57G)

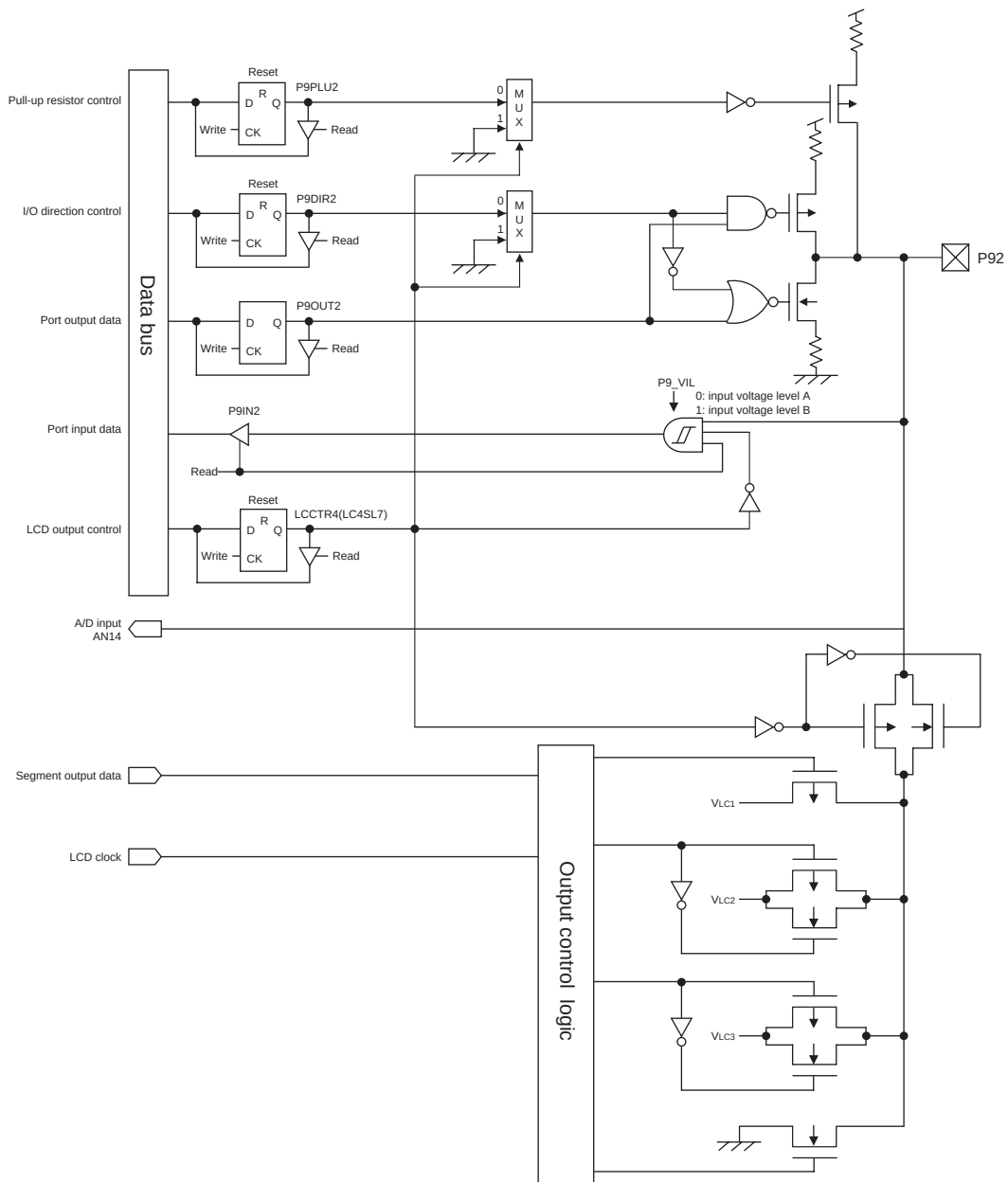


Figure:8.12.9 P92 Block Diagram (MN101EF57G)

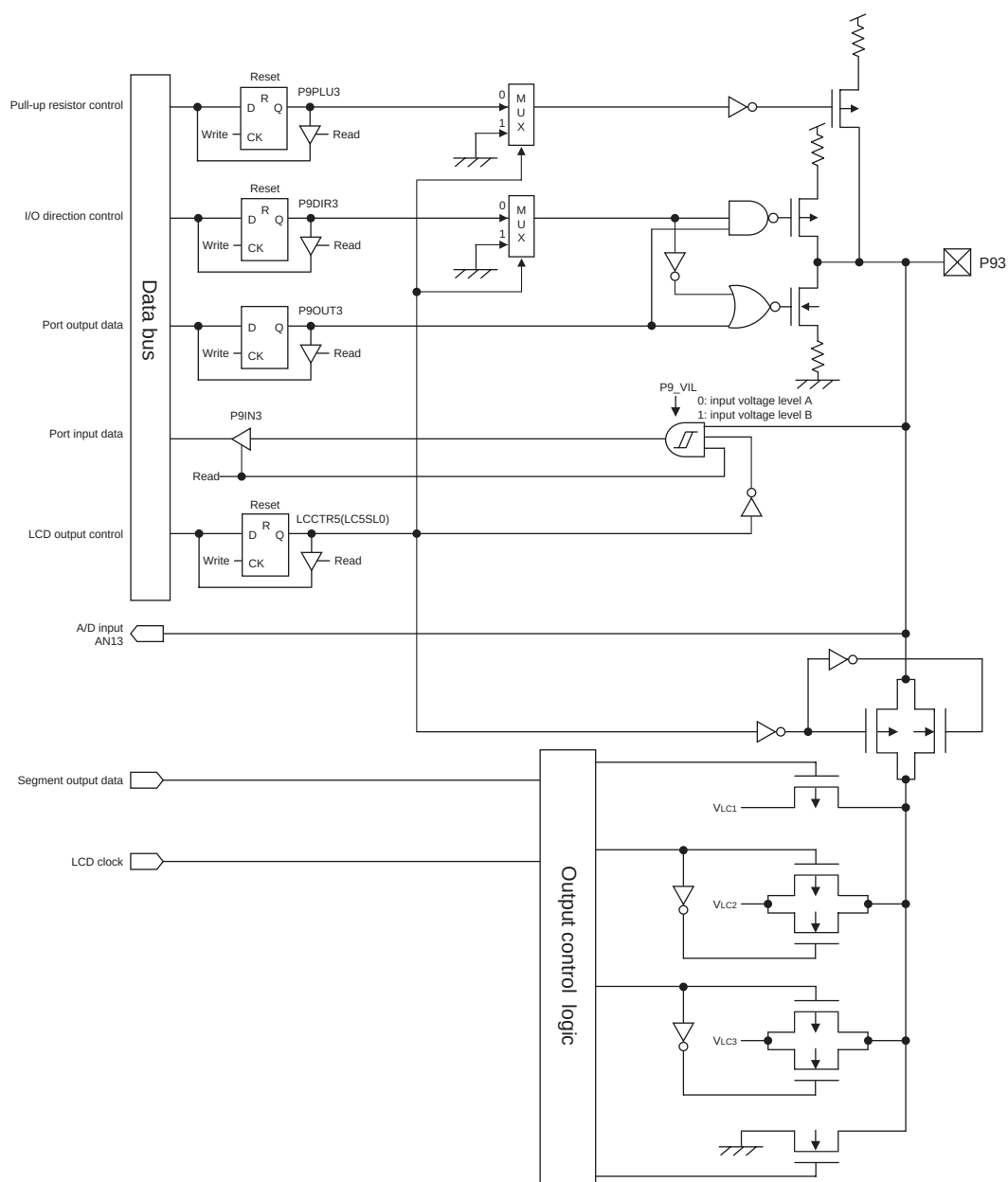


Figure:8.12.10 P93 Block Diagram (MN101EF57G)

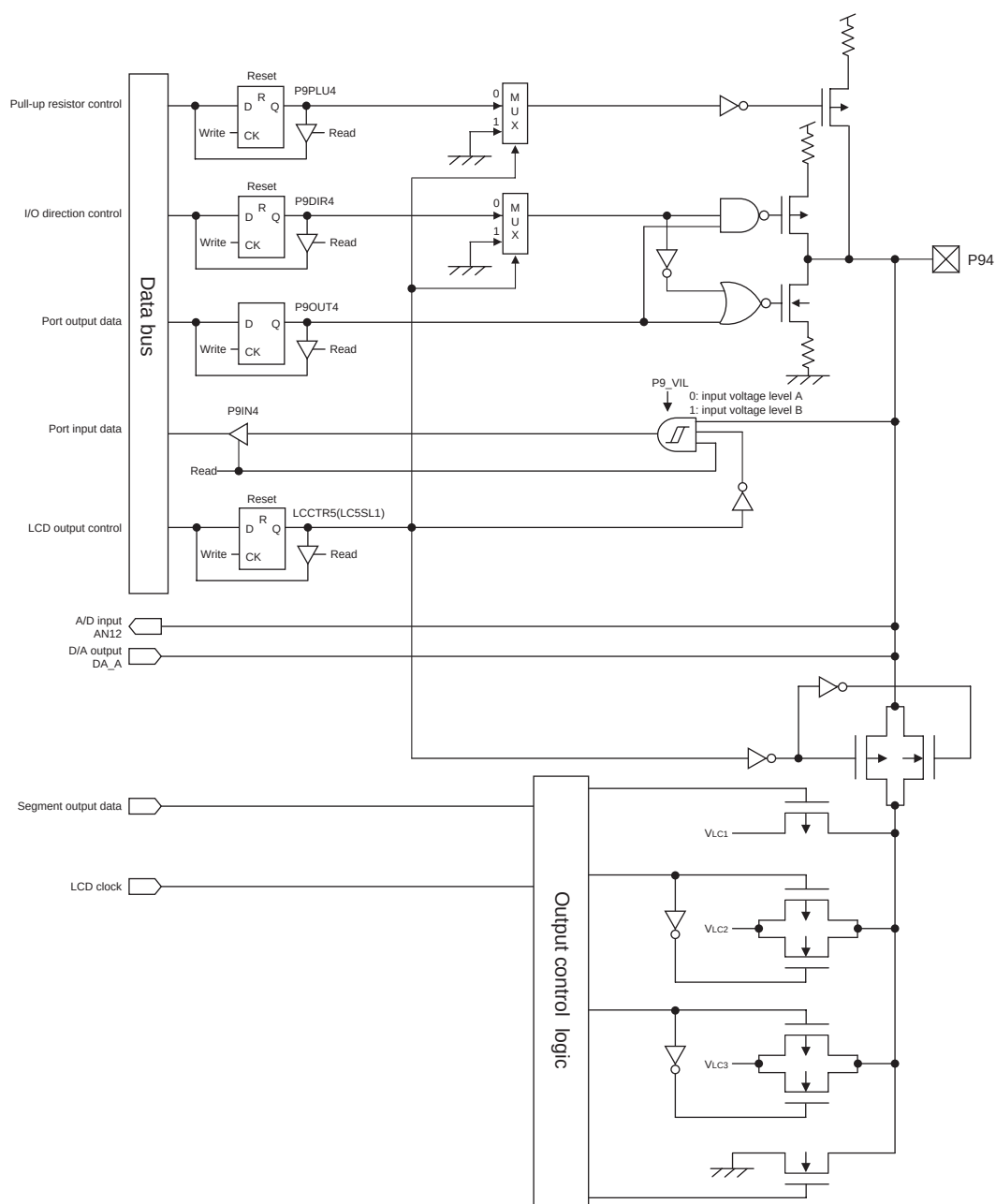


Figure:8.12.11 P94 Block Diagram (MN101EF57G)

8.13 Port A

8.13.1 Description

■ General-purpose Port Setup

To output data to pin, set the control flag of Port A direction control register (PADIR) to "1" and write the value of Port A output register (PAOUT).

To read input data of pin, set the control flag of Port A direction control register (PADIR) to "0" and read the value of Port A input register (PAIN).

Each bit can be set individually as either an input or output by Port A direction control register (PADIR). The control flag of PADIR register is set to "1" for output mode, and "0" for input mode.

Each bit can be set individually to add a pull-up resistor by Port A pull-up resistor control register (PAPLU). Set the control flag of PAPLU register to "1" to add pull-up resistor.

Each bit can be set individually to control input mode by Port A input mode control register (PAIMD). The control flag of PAIMD register is set to "1" for using as the input port of special function data, and to "0" for using as general purpose port. When using as the special function input pin, undefined value is read from PAIN register.

Table:8.13.1 Port A Functions

Table remarks -: Without function

MN101EF76K/MN101EF56K				MN101EF57G			
Pin	Special function			Pin	Special function		
PA0	-	AN0	TM0IOA	PA0	SEG38	AN0	TM0IOA
PA1	-	AN1	TM1IOA	PA1	SEG39	AN1	TM1IOA
PA2	-	AN2	TM2IOA	PA2	SEG40	AN2	TM2IOA
PA3		AN3	TM3IOA	PA3		AN3	TM3IOA
PA4		AN4	TM4IOA	PA4		AN4	TM4IOA
PA5		AN5	TM5IOA	PA5		AN5	TM5IOA
PA6		AN6	TM6IOA	PA6		AN6	TM6IOA
PA7		AN7	TM7IOA	PA7		AN7	TM7IOA

■ Special Function Pin Setup

PA0 to PA2 are also used as LCD segment output pins.

LCD segment output pin is selected by setting the corresponding flag of LCD output control register n (LCCTRn) to "1".

Port and segment can be switched by one bit. When segment output pin is selected, this pin is forced to be output mode and pull-up resistor is not added.

PA0, PA2 and PA5 have real-time output function of controlling to switch the output of pins. When a falling edge event of External Interrupt 0 occurs, the output is switched to three status, "0", "1" and "high impedance (Hi-Z)". The real-time output control changes the output status by synchronizing interrupt events without software intervention.

PA0 to PA7 are also used as analog input pins.

Port A input mode register (PAIMD) can select the input mode by one bit. When using as the analog input pin, undefined value is read from Port A input register (PAIN).

PA0 to PA7 are also used as the I/O pins for Timer 0 to Timer 6 and Timer 9.

Each bit can be set individually as either timer output or general purpose port by Port A output mode register 1 (PAOMD1). The corresponding flag of PAOMD1 register is set to "1" for timer output, and "0" for general-purpose port.

8.13.2 Registers

Table:8.13.2 shows registers that control Port A.

Table:8.13.2 Port A Control Register

Table remarks √: With function -: Without function

Registers	Address	R/W	Function	Page	MN101 EF76K	MN101 EF57G	MN101 EF56K
PAOUT	0x03E7A	R/W	Port A output register	VIII-191	√	√	√
PAIN	0x03E8A	R	Port A input register	VIII-191	√	√	√
PADIR	0x03E9A	R/W	Port A direction control register	VIII-192	√	√	√
PAPLU	0x03EAA	R/W	Port A pull-up resistor control register	VIII-192	√	√	√
PAOMD1	0x03EBA	R/W	Port A output mode register 1	VIII-193	√	√	√
PAIMD	0x03ECA	R/W	Port A input mode register	VIII-194	√	√	√
PACNT	0x03EDA	R/W	Port A output control register	VIII-195	√	√	√
SWCNT	0x03E8F	R/W	0.54 V _{DD5} input selection register 1	VIII-195	√	√	√

R/W: Readable/Writable

R: Read only

■ Port A Output Register (PAOUT: 0x03E7A)

bp	7	6	5	4	3	2	1	0
Flag	PAOUT7	PAOUT6	PAOUT5	PAOUT4	PAOUT3	PAOUT2	PAOUT1	PAOUT0
At reset	X	X	X	X	X	X	X	X
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description
7 to 0	PAOUT7 to 0	Output data 0: Output "Low" (V _{SS} -level) 1: Output "High" (V _{DD5} -level)

■ Port A Input Register (PAIN: 0x03E8A)

bp	7	6	5	4	3	2	1	0
Flag	PAIN7	PAIN6	PAIN5	PAIN4	PAIN3	PAIN2	PAIN1	PAIN0
At reset	X	X	X	X	X	X	X	X
Access	R	R	R	R	R	R	R	R

bp	Flag	Description
7 to 0	PAIN7 to 0	Input data 0: Pin is "Low" (V _{SS} -level) 1: Pin is "High" (V _{DD5} -level)

■ Port A Direction Control Register (PADIR: 0x03E9A)

bp	7	6	5	4	3	2	1	0
Flag	PADIR7	PADIR6	PADIR5	PADIR4	PADIR3	PADIR2	PADIR1	PADIR0
At reset	0	0	0	0	0	0	0	0
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description
7 to 0	PADIR7 to 0	I/O mode selection 0: Input mode 1: Output mode

■ Port A Pull-up Resistor Control Register (PAPLU: 0x03EAA)

bp	7	6	5	4	3	2	1	0
Flag	PAPLU7	PAPLU6	PAPLU5	PAPLU4	PAPLU3	PAPLU2	PAPLU1	PAPLU0
At reset	0	0	0	0	0	0	0	0
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description
7 to 0	PAPLU7 to 0	Pull-up resistor selection 0: Not added 1: Added

■ Port A Output Mode Register 1 (PAOMD1: 0x03EBA)

bp	7	6	5	4	3	2	1	0
Flag	PAOMD7	PAOMD6	PAOMD5	PAOMD4	PAOMD3	PAOMD2	PAOMD1	PAOMD0
At reset	0	0	0	0	0	0	0	0
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description
7	PAOMD7	I/O port or timer output selection 0: PA7 1: TM9IOA
6	PAOMD6	I/O port or timer output selection 0: PA6 1: TM8IOA
5	PAOMD5	I/O port or timer output selection 0: PA5 1: TM7IOA
4	PAOMD4	I/O port or timer output selection 0: PA4 1: TM4IOA
3	PAOMD3	I/O port or timer output selection 0: PA3 1: TM3IOA
2	PAOMD2	I/O port or timer output selection 0: PA2 1: TM2IOA
1	PAOMD1	I/O port or timer output selection 0: PA1 1: TM1IOA
0	PAOMD0	I/O port or timer output selection 0: PA0 1: TM0IOA

■ Port A Input Mode Register (PAIMD: 0x03ECA)

bp	7	6	5	4	3	2	1	0
Flag	PAIMD7	PAIMD6	PAIMD5	PAIMD4	PAIMD3	PAIMD2	PAIMD1	PAIMD0
At reset	0	0	0	0	0	0	0	0
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description
7	PAIMD7	I/O port or analog input selection 0: PA7 1: AN7
6	PAIMD6	I/O port or analog input selection 0: PA6 1: AN6
5	PAIMD5	I/O port or analog input selection 0: PA5 1: AN5
4	PAIMD4	I/O port or analog input selection 0: PA4 1: AN4
3	PAIMD3	I/O port or analog input selection 0: PA3 1: AN3
2	PAIMD2	I/O port or analog input selection 0: PA2 1: AN2
1	PAIMD1	I/O port or analog input selection 0: PA1 1: AN1
0	PAIMD0	I/O port or analog input selection 0: PA0 1: AN0

■ Port A Output Control Register (PACNT: 0x03EDA)

bp	7	6	5	4	3	2	1	0
Flag	-	-	PACNT5	PACNT4	PACNT3	PACNT2	PACNT1	PACNT0
At reset	-	-	0	0	0	0	0	0
Access	-	-	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description
7 to 6	-	-
5 to 4	PACNT5 to 4	PA5 real-time control (IRQ0 event) 00: I/O port (real-time control disabled) 01: "High" output 10: "Low" output 11: "Hi-Z" output
3 to 2	PACNT3 to 2	PA2 real-time control (IRQ0 event) 00: I/O port (real-time control disabled) 01: "High" output 10: "Low" output 11: "Hi-Z" output
1 to 0	PACNT1 to 0	PA0 real-time control (IRQ0 event) 00: I/O port (real-time control disabled) 01: "High" output 10: "Low" output 11: "Hi-Z" output

■ 0.54 V_{DD5} Input Selection Register 1 (SWCNT: 0x3E8F)

bp	7	6	5	4	3	2	1	0
Flag	-	-	-	-	-	SWCNT_A	-	-
At reset	-	-	-	-	-	0	-	-
Access	-	-	-	-	-	R/W	-	-

bp	Flag	Description
7 to 3	-	-
2	SWCNT_A	Port A input voltage level selection 0: Input voltage level A/B 1: Input voltage level C
1 to 0	-	-



Refer to [Chapter 8 8.19 Input Voltage Level Switching Function] for input voltage level A/B.

8.13.3 Block Diagram (MN101EF76K/MN101EF56K)

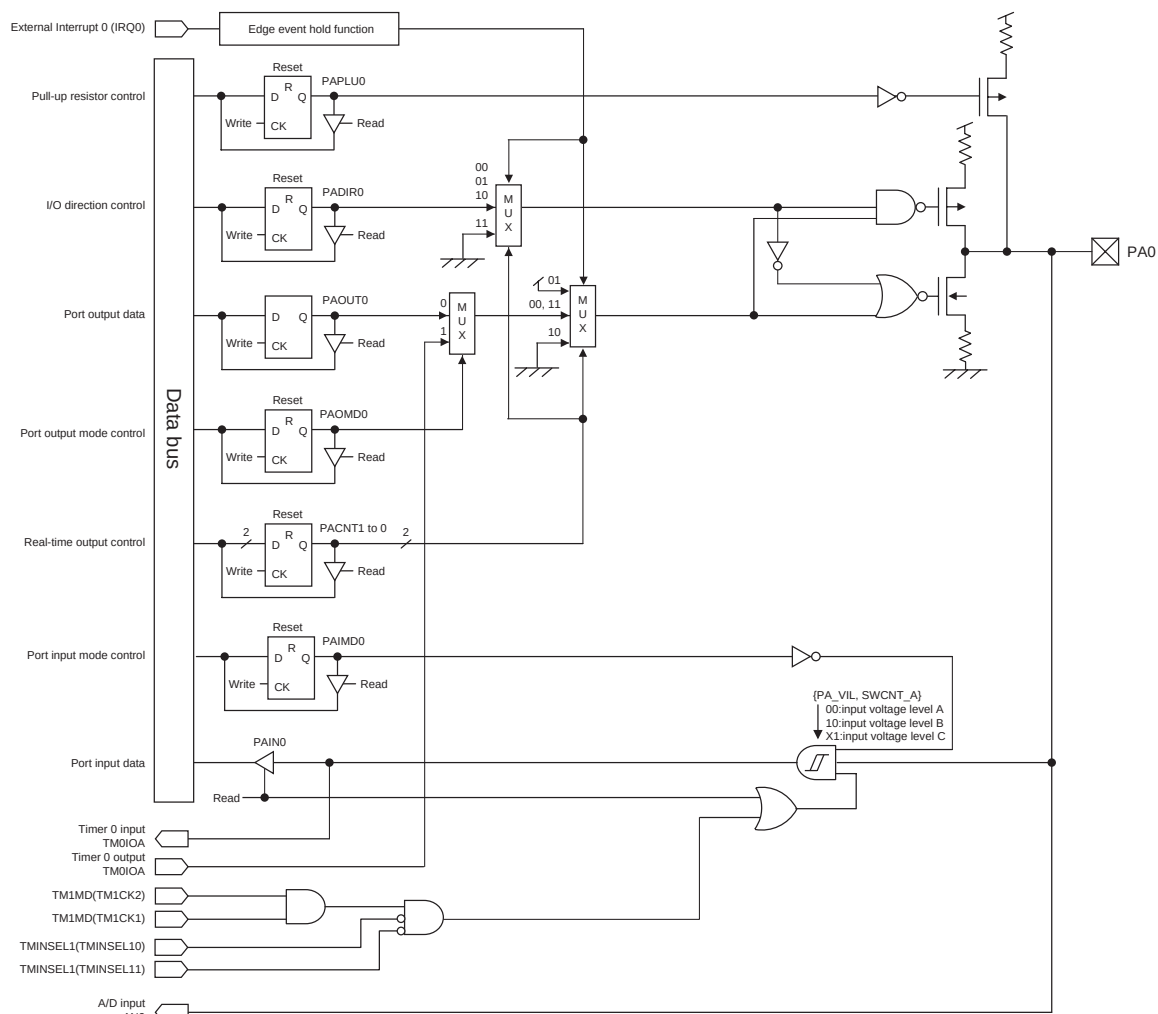


Figure:8.13.1 PA0 Block Diagram (MN101EF76K/MN101EF56K)

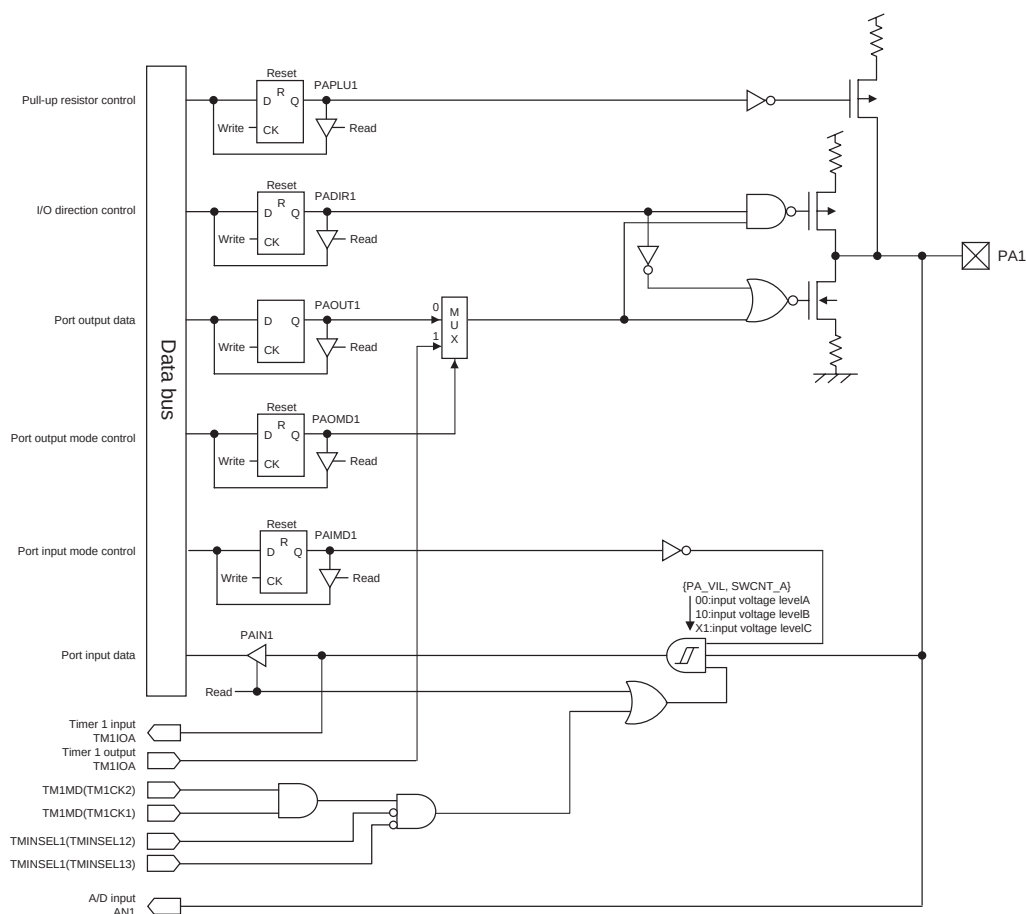


Figure:8.13.2 PA1 Block Diagram (MN101EF76K/MN101EF56K)

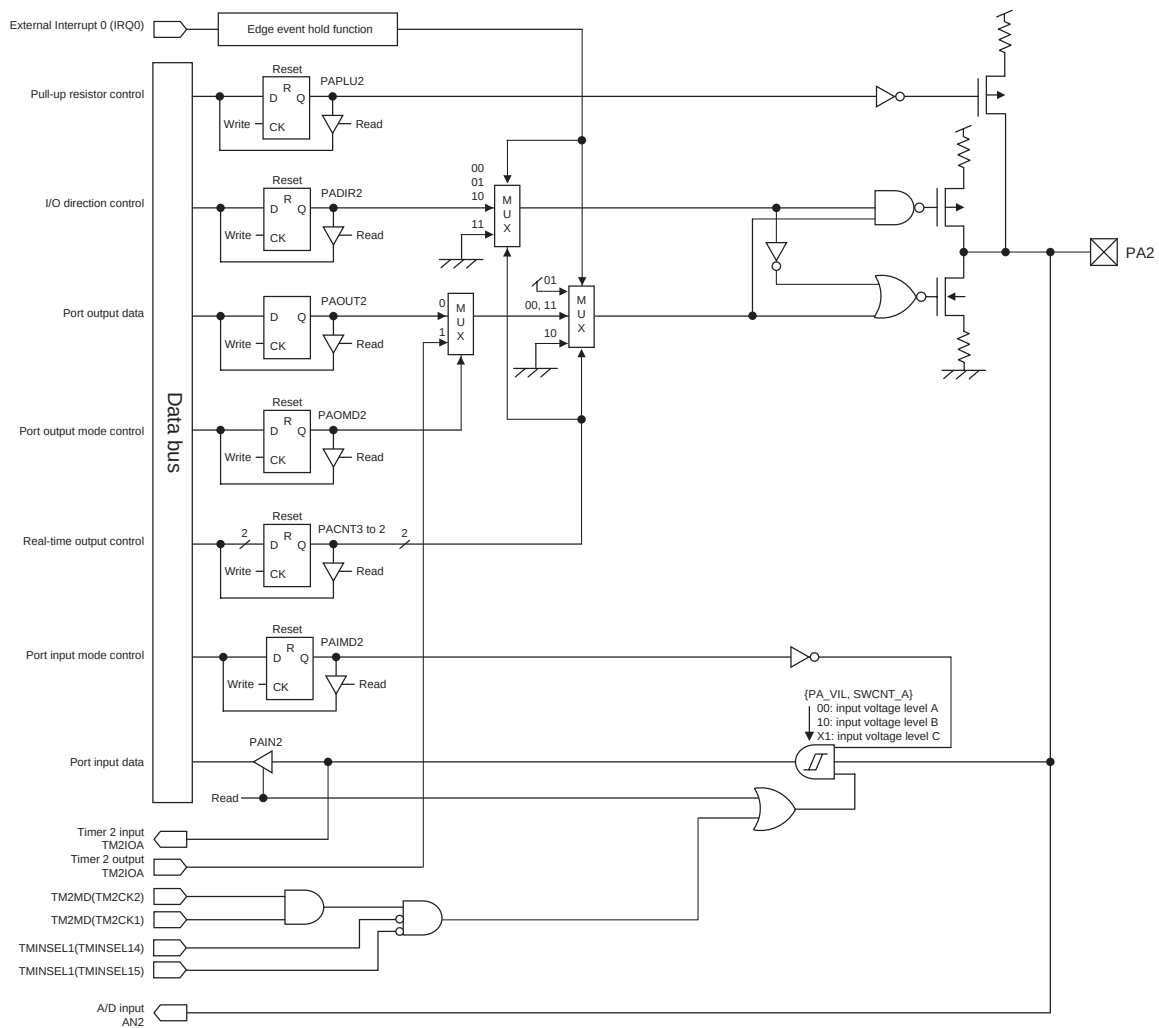


Figure:8.13.3 PA2 Block Diagram (MN101EF76K/MN101EF56K)

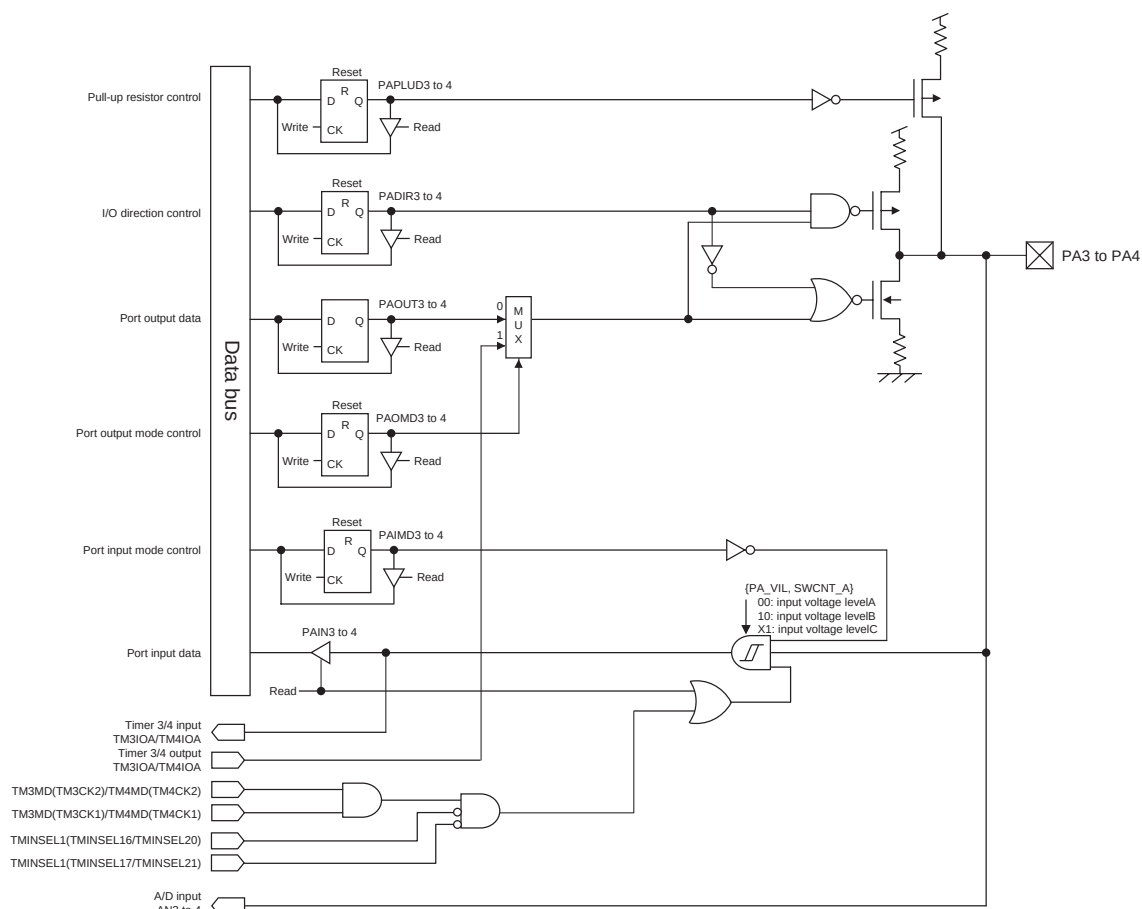


Figure:8.13.4 PA3 to PA4 Block Diagram (MN101EF76K/MN101EF56K)

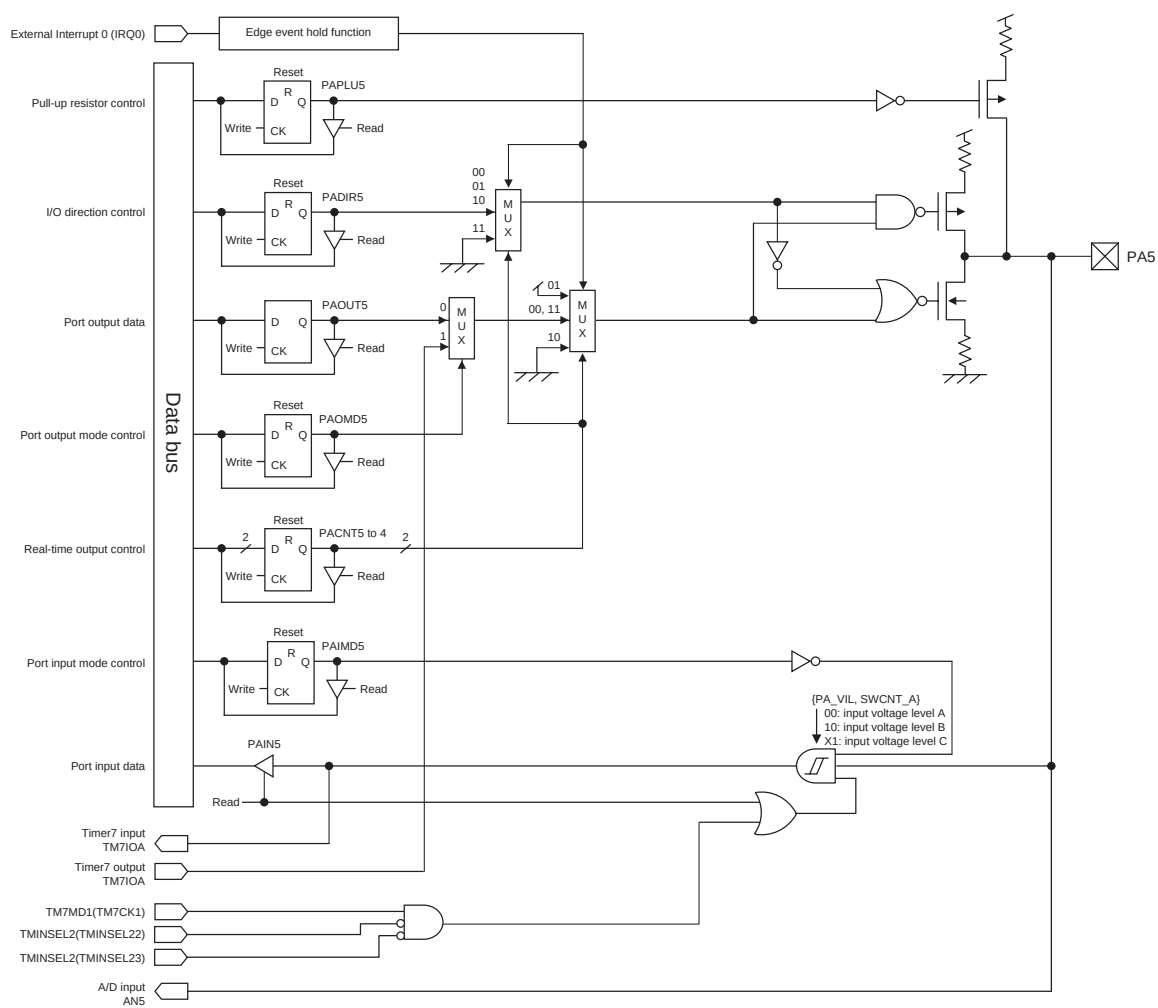


Figure:8.13.5 PA5 Block Diagram (MN101EF76K/MN101EF56K)

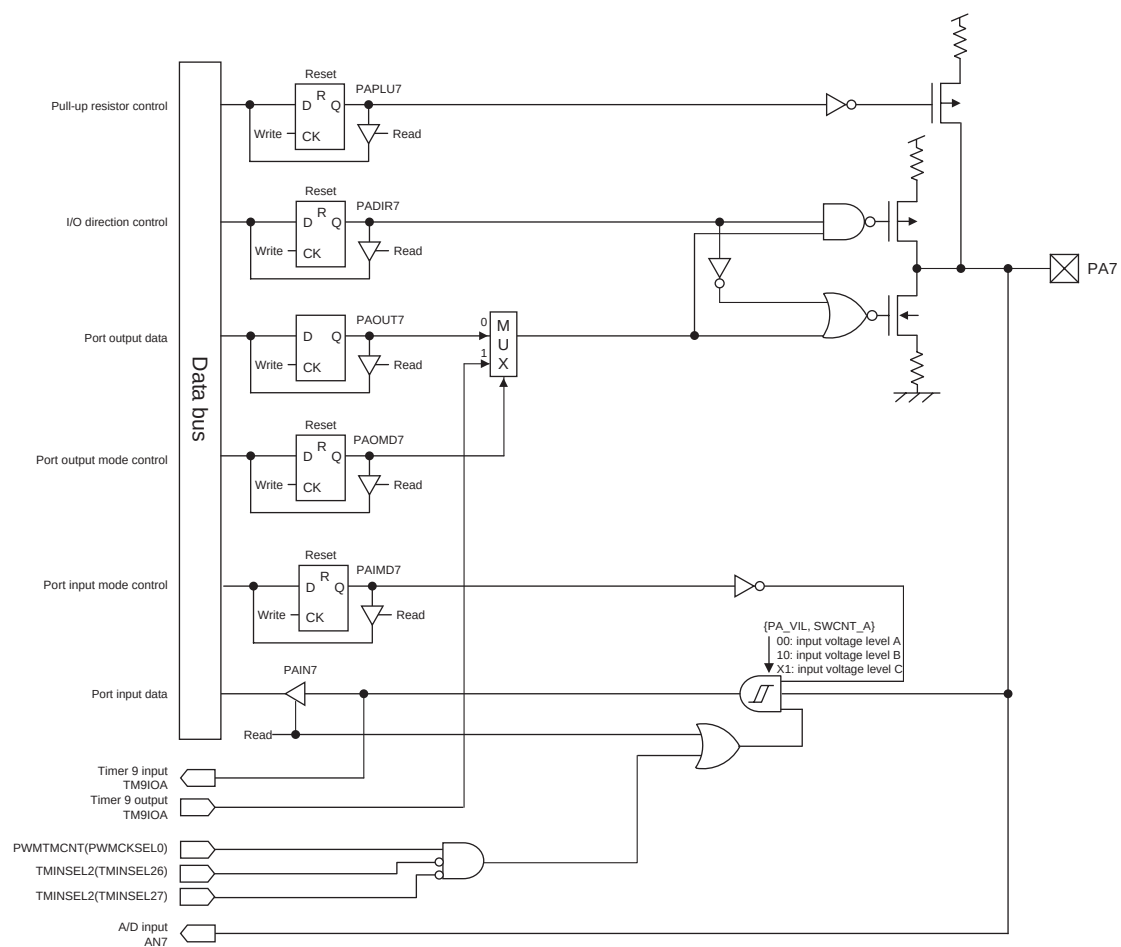


Figure:8.13.7 PA7 Block Diagram (MN101EF76K/MN101EF56K)

8.13.4 Block Diagram (MN101EF57G)

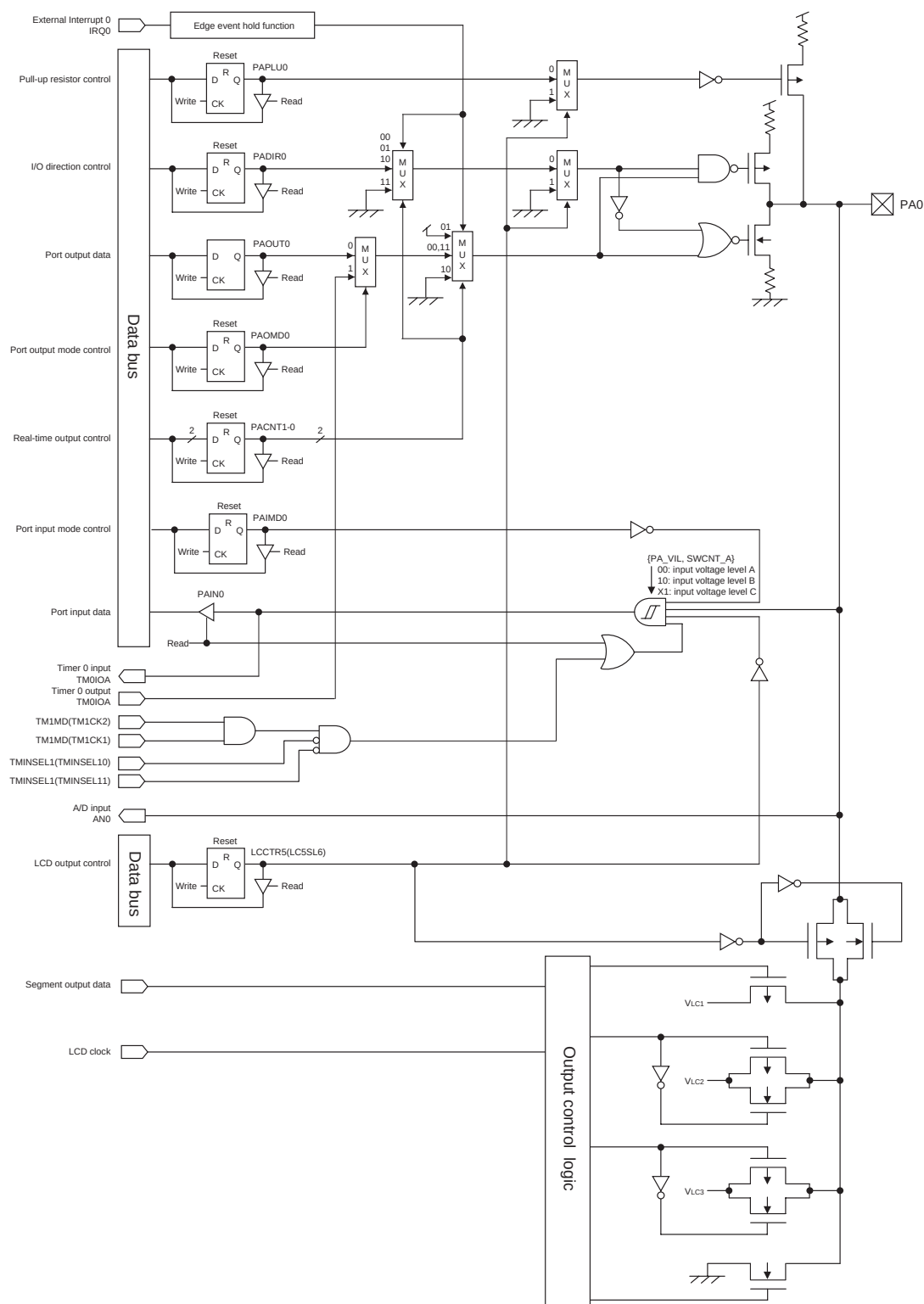


Figure:8.13.8 PA0 Block Diagram (MN101EF57G)

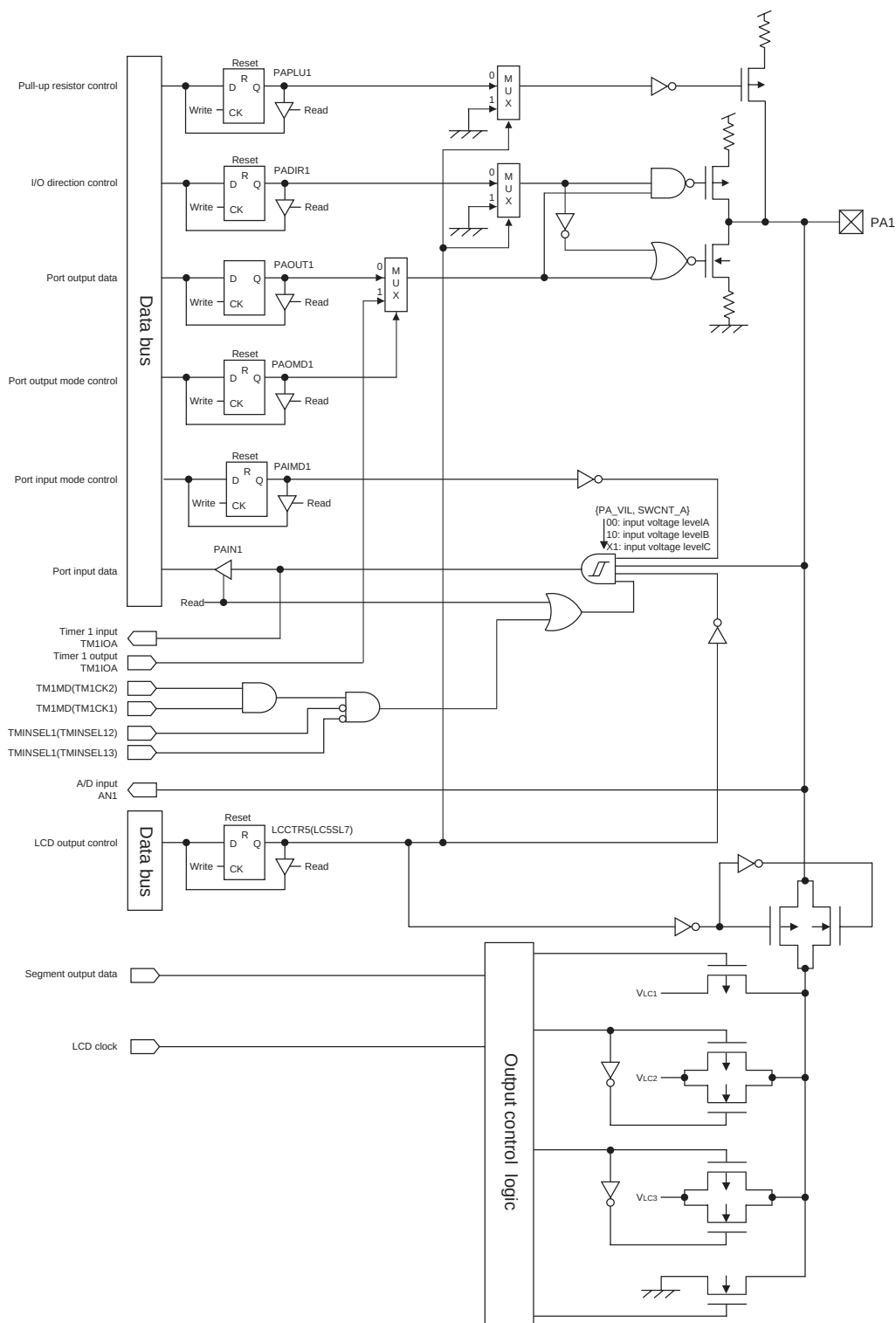


Figure:8.13.9 PA1 Block Diagram (MN101EF57G)

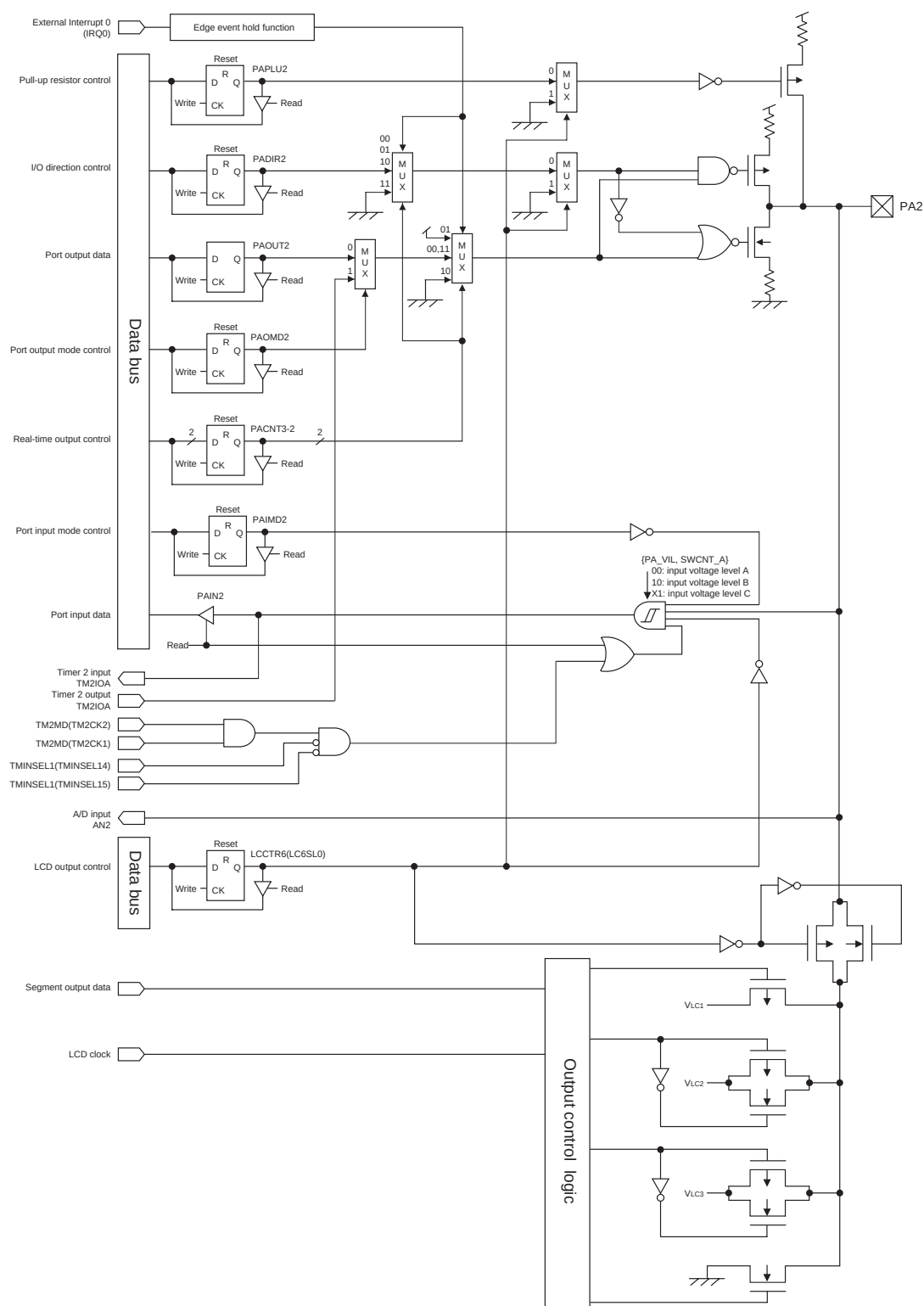


Figure:8.13.10 PA2 Block Diagram (MN101EF57G)

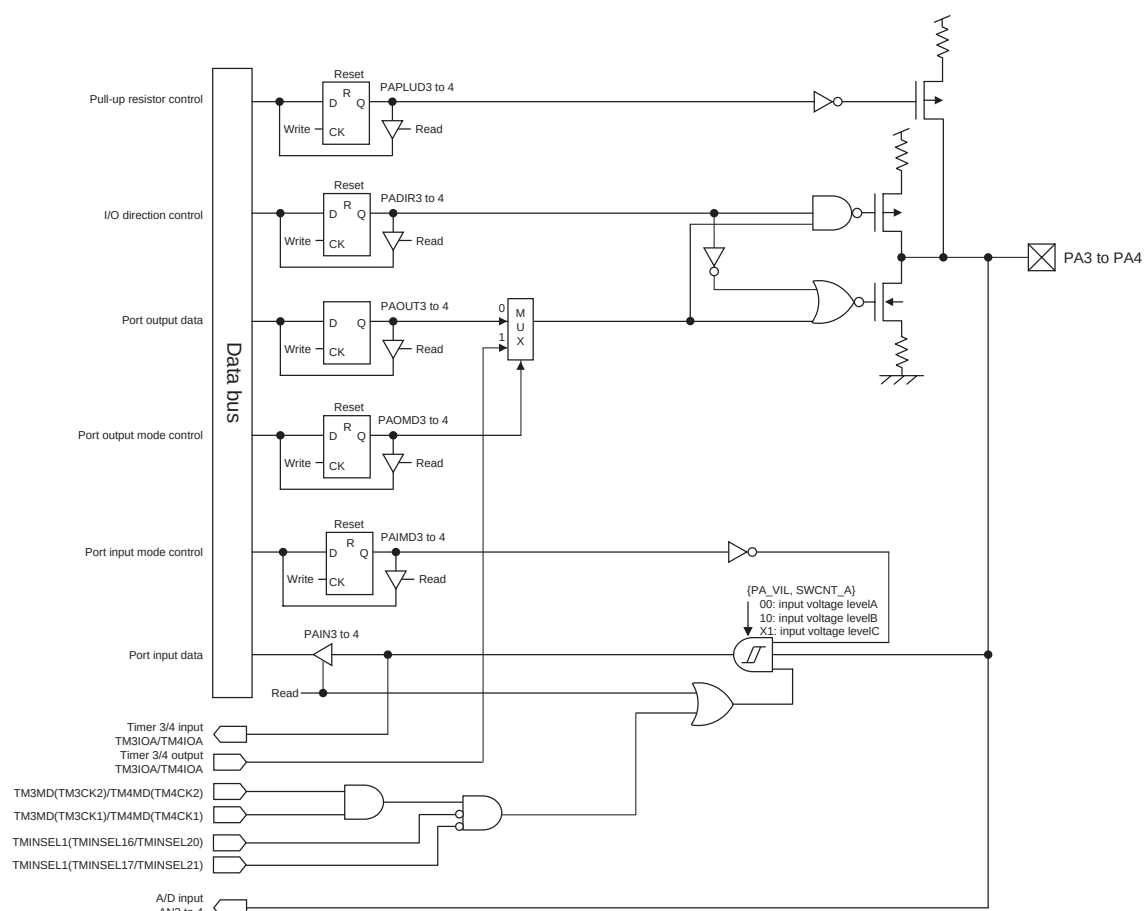


Figure:8.13.11 PA3 to PA4 Block Diagram (MN101EF57G)

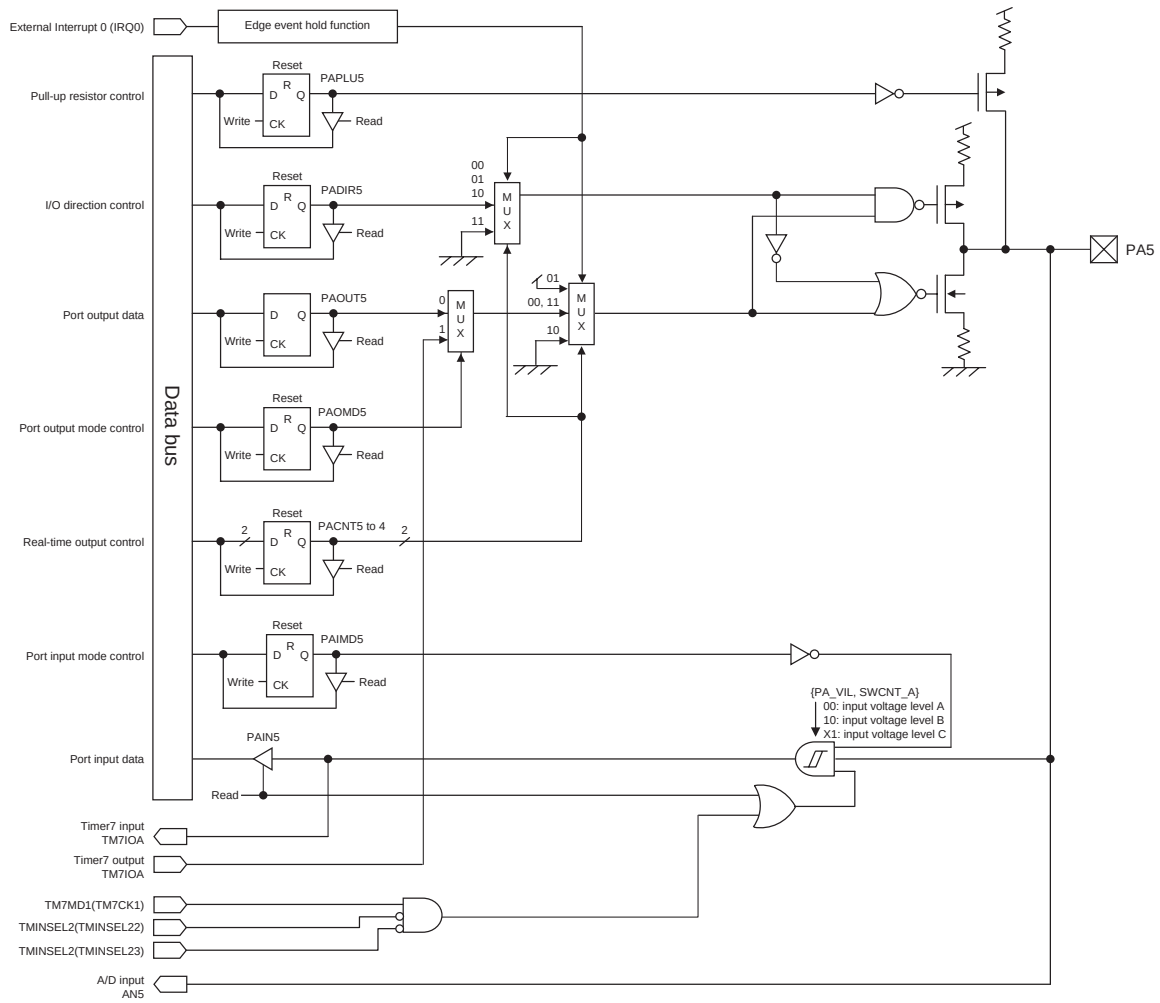


Figure:8.13.12 PA5 Block Diagram (MN101EF57G)

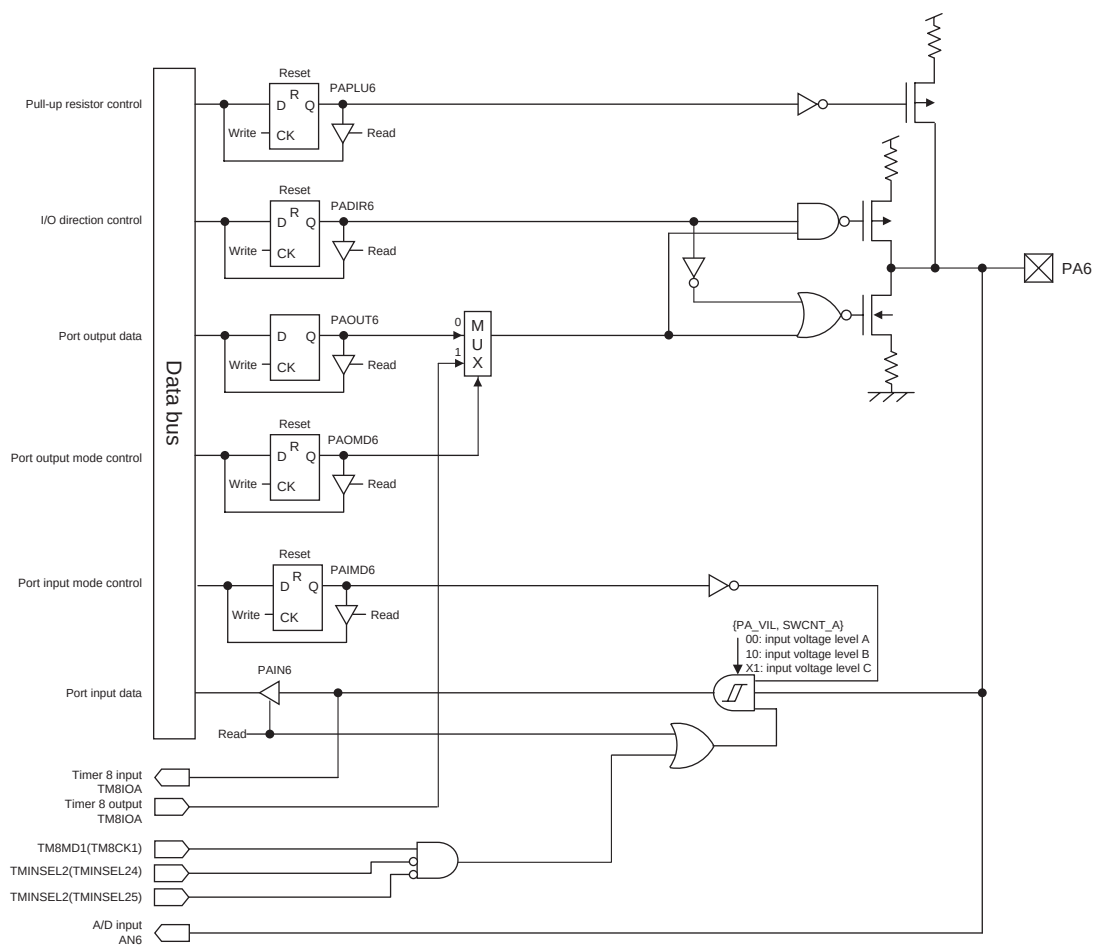


Figure:8.13.13 PA6 Block Diagram (MN101EF57G)

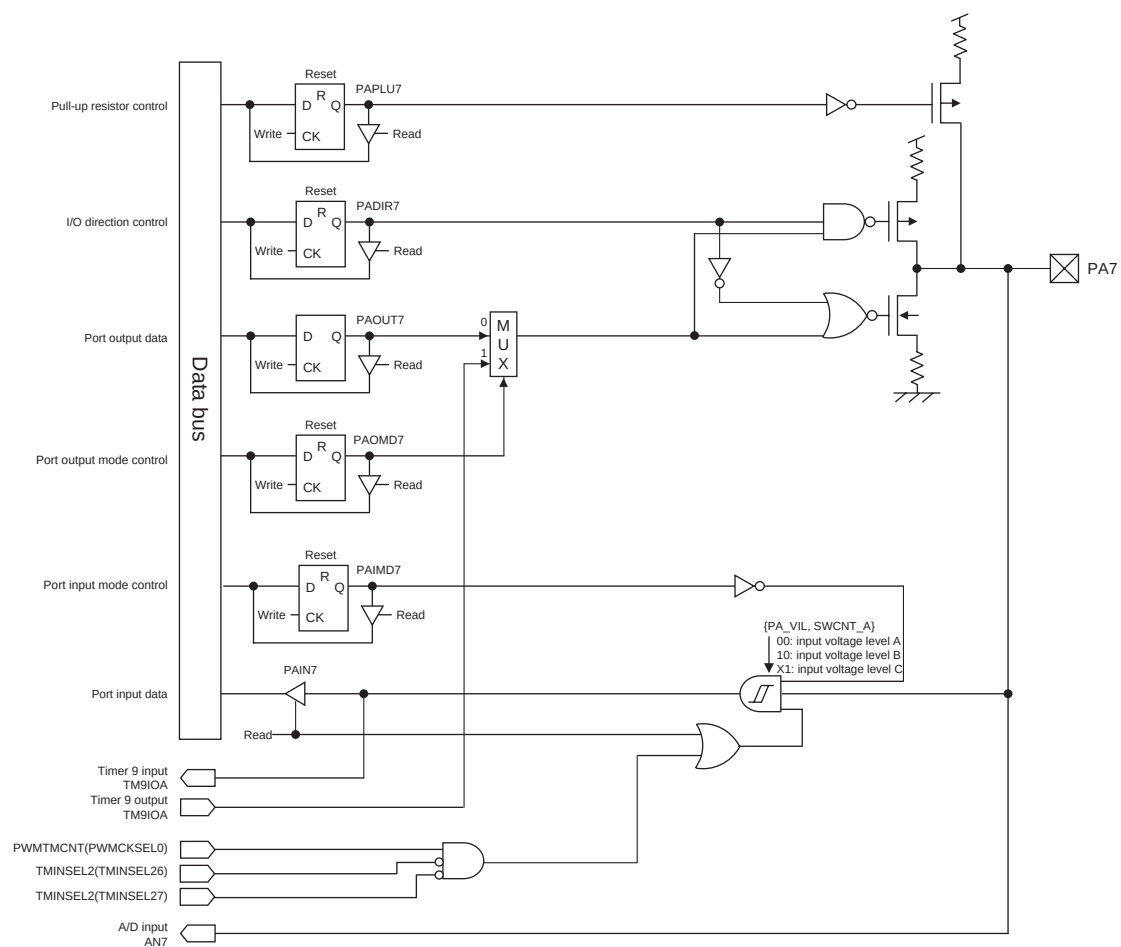


Figure:8.13.14 PA7 Block Diagram (MN101EF57G)

8.14 Port B

8.14.1 Description

■ General-purpose Port Setup

To output data to pin, set the control flag of Port B direction control register (PBDIR) to "1" and write the value of Port B output register (PBOUR).

To read input data of pin, set the control flag of Port B direction control register (PBDIR) to "0" and read the value of Port B input register (PBIN).

Each bit can be set individually as either an input or output by Port B direction control register (PBDIR). The control flag of PBDIR register is set to "1" for output mode, and "0" for input mode.

Each bit can be set individually to add a pull-up resistor by Port B pull-up resistor control register (PBPLU). Set the control flag of PBPLU register to "1" to add pull-up resistor.

Each bit can be set individually to control input mode by Port B input mode control register (PBIMD). The control flag of PBIMD register is set to "1" for using as the input port of special function data, and to "0" for using as general purpose port. When using as the special function input pin, undefined value is read from PBIN register.

Table:8.14.1 Port B Functions

Table remarks -: Without function

MN101EF76K/MN101EF56K			MN101EF57G		
Pin	Special function		Pin	Special function	
PB0	-	AN8	PB0	SEG37	AN8
PB1	-	AN9	PB1	SEG36	AN9
PB2	-	AN10	PB2	SEG35	AN10
PB3	SEG54	AN11	PB3	SEG34	AN11
PB4	SEG53	AN12	-	-	-
PB5	SEG52	AN13	-	-	-
PB6	SEG51	AN14	-	-	-
PB7	SEG50	AN15	-	-	-

■ Special Function Pin Setup

PB0 to PB7 are also used as LCD segment output pins.

LCD segment output pin is selected by setting the corresponding flag of LCD output control register n (LCCTRn) to "1".

Port and segment can be switched by one bit. When segment output pin is selected, this pin is forced to be output mode and pull-up resistor is not added.

PB0 to PB7 are also used as analog input pins.

Port B input mode register (PBIMD) can select the input mode by one bit. When using as the analog input pin, undefined value is read from Port B input register (PBIN).

8.14.2 Registers

Table:8.14.2 shows registers that control Port B.

Table:8.14.2 Port B Control Register

Table remarks √: With function -: Without function

Registers	Address	R/W	Function	Page	MN101 EF76K	MN101 EF57G	MN101 EF56K
PBOUT	0x03E7B	R/W	Port B output register	VIII-211	√	√	√
PBIN	0x03E8B	R	Port B input register	VIII-212	√	√	√
PBDIR	0x03E9B	R/W	Port B direction control register	VIII-213	√	√	√
PBPLU	0x03EAB	R/W	Port B pull-up resistor control register	VIII-214	√	√	√
PBIMD	0x03ECB	R/W	Port B input mode register	VIII-215	√	√	√

R/W: Readable/Writable

R: Read only

■ Port B Output Register (PBOUT: 0x03E7B)

<MN101EF76K/MN101EF56K>

bp	7	6	5	4	3	2	1	0
Flag	PBOUT7	PBOUT6	PBOUT5	PBOUT4	PBOUT3	PBOUT2	PBOUT1	PBOUT0
At reset	X	X	X	X	X	X	X	X
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description
7 to 0	PBOUT7 to 0	Output data 0: Output "Low" (V_{SS} -level) 1: Output "High" (V_{DD5} -level)

<MN101EF57G>

bp	7	6	5	4	3	2	1	0
Flag	-	-	-	-	PBOUT3	PBOUT2	PBOUT1	PBOUT0
At reset	-	-	-	-	X	X	X	X
Access	-	-	-	-	R/W	R/W	R/W	R/W

bp	Flag	Description
7 to 4	-	-
3 to 0	PBOUT3 to 0	Output data 0: Output "Low" (V_{SS} -level) 1: Output "High" (V_{DD5} -level)

■ Port B Input Register (PBIN: 0x03E8B)

<MN101EF76K/MN101EF56K>

bp	7	6	5	4	3	2	1	0
Flag	PBIN7	PBIN6	PBIN5	PBIN4	PBIN3	PBIN2	PBIN1	PBIN0
At reset	X	X	X	X	X	X	X	X
Access	R	R	R	R	R	R	R	R

bp	Flag	Description
7 to 0	PBIN7 to 0	Input data 0: Pin is "Low" (V_{SS} -level) 1: Pin is "High" (V_{DD5} -level)

<MN101EF57G>

bp	7	6	5	4	3	2	1	0
Flag	-	-	-	-	PBIN3	PBIN2	PBIN1	PBIN0
At reset	-	-	-	-	X	X	X	X
Access	-	-	-	-	R	R	R	R

bp	Flag	Description
7 to 4	-	-
3 to 0	PBIN3 to 0	Input data 0: Pin is "Low" (V_{SS} -level) 1: Pin is "High" (V_{DD5} -level)

■ Port B Direction Control Register (PBDIR: 0x03E9B)

<MN101EF76K/MN101EF56K>

bp	7	6	5	4	3	2	1	0
Flag	PBDIR7	PBDIR6	PBDIR5	PBDIR4	PBDIR3	PBDIR2	PBDIR1	PBDIR0
At reset	0	0	0	0	0	0	0	0
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description
7 to 0	PBDIR7 to 0	I/O mode selection 0: Input mode 1: Output mode

<MN101EF57G>

bp	7	6	5	4	3	2	1	0
Flag	-	-	-	-	PBDIR3	PBDIR2	PBDIR1	PBDIR0
At reset	-	-	-	-	0	0	0	0
Access	-	-	-	-	R/W	R/W	R/W	R/W

bp	Flag	Description
7 to 4	-	-
3 to 0	PBDIR3 to 0	I/O mode selection 0: Input mode 1: Output mode

■ Port B Pull-up Resistor Control Register (PBPLU: 0x03EAB)

<MN101EF76K/MN101EF56K>

bp	7	6	5	4	3	2	1	0
Flag	PBPLU7	PBPLU6	PBPLU5	PBPLU4	PBPLU3	PBPLU2	PBPLU1	PBPLU0
At reset	0	0	0	0	0	0	0	0
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description
7 to 0	PBPLU7 to 0	Pull-up resistor selection 0: Not added 1: Added

<MN101EF57G>

bp	7	6	5	4	3	2	1	0
Flag	-	-	-	-	PBPLU3	PBPLU2	PBPLU1	PBPLU0
At reset	-	-	-	-	0	0	0	0
Access	-	-	-	-	R/W	R/W	R/W	R/W

bp	Flag	Description
7 to 4	-	-
3 to 0	PBPLU3 to 0	Pull-up resistor selection 0: Not added 1: Added

■ Port B Input Mode Register (PBIMD: 0x03ECB)

<MN101EF76K/MN101EF56K>

bp	7	6	5	4	3	2	1	0
Flag	PBIMD7	PBIMD6	PBIMD5	PBIMD4	PBIMD3	PBIMD2	PBIMD1	PBIMD0
At reset	0	0	0	0	0	0	0	0
Access	R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description
7	PBIMD7	I/O port or analog input selection 0: PB7 1: AN15
6	PBIMD6	I/O port or analog input selection 0: PB6 1: AN14
5	PBIMD5	I/O port or analog input selection 0: PB5 1: AN13
4	PBIMD4	I/O port or analog input selection 0: PB4 1: AN12
3	PBIMD3	I/O port or analog input selection 0: PB3 1: AN11
2	PBIMD2	I/O port or analog input selection 0: PB2 1: AN10
1	PBIMD1	I/O port or analog input selection 0: PB1 1: AN9
0	PBIMD0	I/O port or analog input selection 0: PB0 1: AN8

■ Port B Input Mode Register (PBIMD: 0x03ECB)

<MN101EF57G>

bp	7	6	5	4	3	2	1	0
Flag	-	-	-	-	PBIMD3	PBIMD2	PBIMD1	PBIMD0
At reset	-	-	-	-	0	0	0	0
Access	-	-	-	-	R/W	R/W	R/W	R/W

bp	Flag	Description
7 to 4	-	-
3	PBIMD3	I/O port or analog input selection 0: PB3 1: AN11
2	PBIMD2	I/O port or analog input selection 0: PB2 1: AN10
1	PBIMD1	I/O port or analog input selection 0: PB1 1: AN9
0	PBIMD0	I/O port or analog input selection 0: PB0 1: AN8

8.14.3 Block Diagram (MN101EF76K/MN101EF56K)

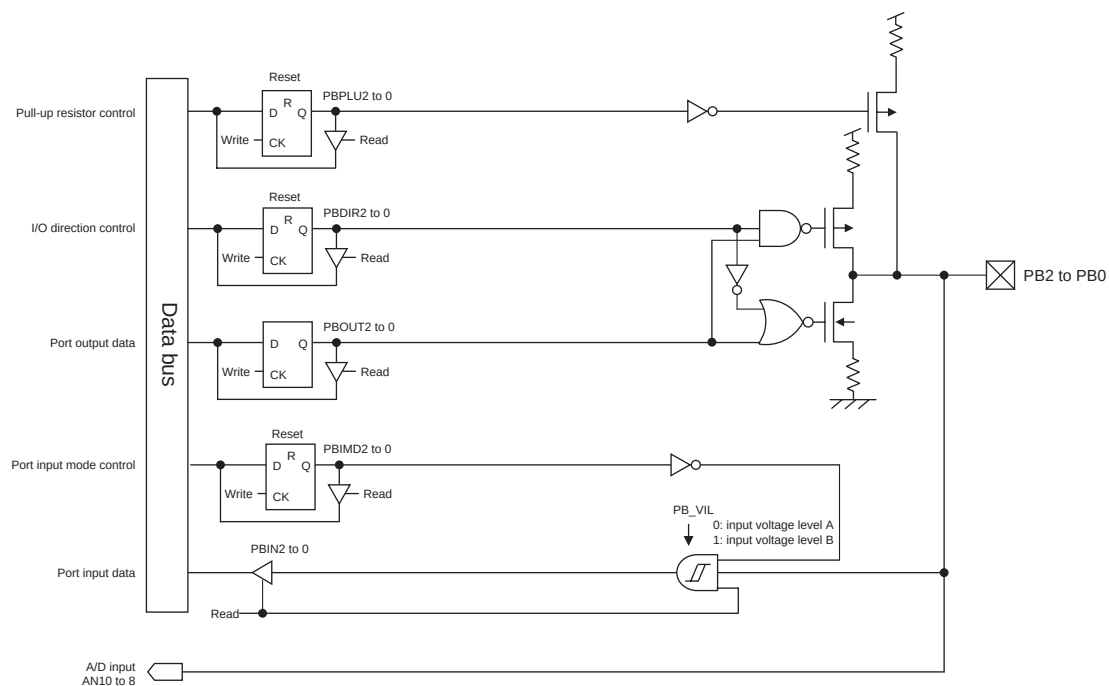


Figure:8.14.1 PB2 to PB0 Block Diagram (MN101EF76K/MN101EF56K)

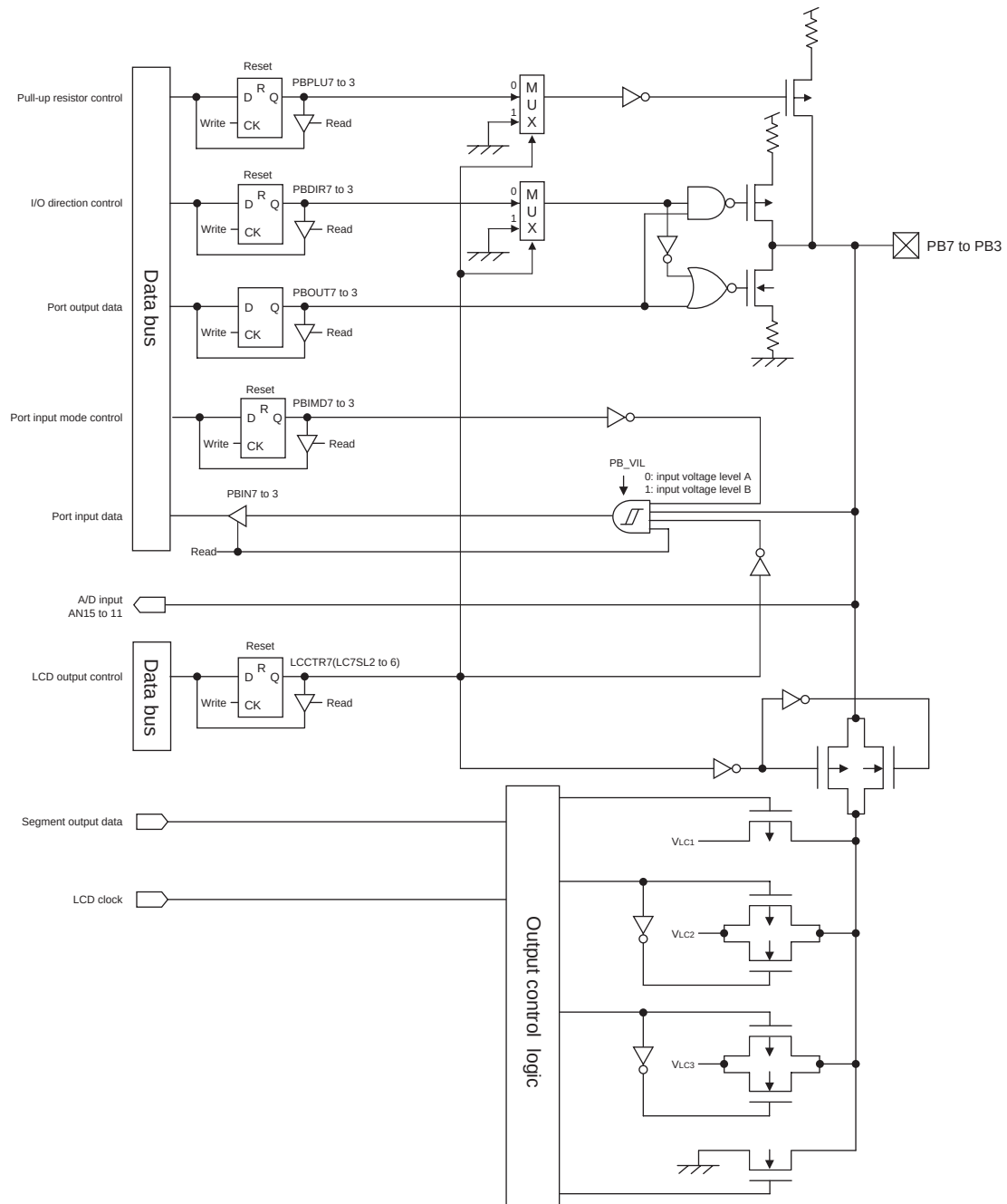


Figure:8.14.2 PB7 to PB3 Block Diagram (MN101EF76K/MN101EF56K)

8.14.4 Block Diagram (MN101EF57G)

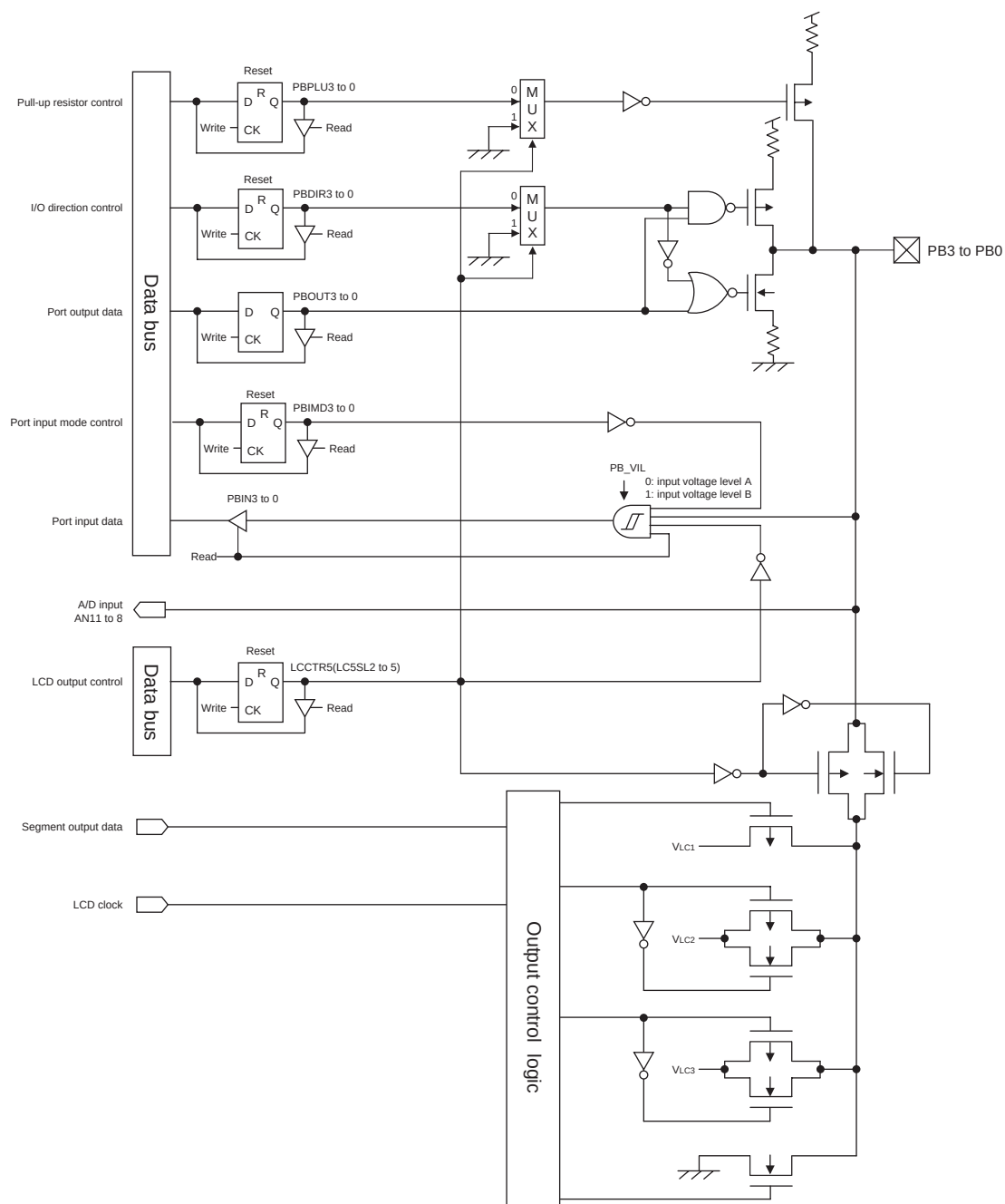


Figure:8.14.3 PB3 to PB0 Block Diagram (MN101EF57G)

8.15 Port C

8.15.1 Description

■ General-purpose Port Setup

To output data to pins, set the control flag of Port C direction control register (PCDIR) to "1" and write the value of Port C output register (PCOUT).

To read input data of pins, set the control flag of Port C direction control register (PCDIR) to "0" and read the value of Port C input register (PCIN).

Each bit can be set individually as either an input or output by Port C direction control register (PCDIR). The control flag of Port C direction control register (PCDIR) is set to "1" for output mode, and "0" for input mode.

Each bit can be set individually to add a pull-up resistor by Port C pull-up resistor control register (PCPLU). Set the corresponding flag of PCPLU register to "1" to add a pull-up resistor.

Table:8.15.1 Port C Functions

Table remarks -: Without function

MN101EF76K		MN101EF57G/MN101EF56K	
Pin	Special function	Pin	Special function
PC0	IRQ0B	-	-
PC1	IRQ1B	-	-
PC2	IRQ2B	-	-
PC3	IRQ3B	-	-
PC4	IRQ4B	-	-
PC5		-	
PC6		-	

■ Special Function Pin Setup

PC0 to PC4 are also used as the external interrupt pin.

8.15.2 Registers

Table:8.15.2 shows registers that control Port C.

Table:8.15.2 Port C Control Register

Table remarks √: With function -: Without function

Registers	Address	R/W	Function	Page	MN101 EF76K	MN101 EF57G	MN101 EF56K
PCOUT	0x03E7C	R/W	Port C output register	VIII-221	√	-	-
PCIN	0x03E8C	R	Port C input register	VIII-221	√	-	-
PCDIR	0x03E9C	R/W	Port C direction control register	VIII-222	√	-	-
PCPLU	0x03EAC	R/W	Port C pull-up resistor control register	VIII-222	√	-	-

R/W: Readable/Writable

R: Read only

■ Port C Output Register (PCOUT: 0x03E7C)

<MN101EF76K>

bp	7	6	5	4	3	2	1	0
Flag	-	PCOUT6	PCOUT5	PCOUT4	PCOUT3	PCOUT2	PCOUT1	PCOUT0
At reset	-	X	X	X	X	X	X	X
Access	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description
7	-	-
6 to 0	PCOUT6 to 0	Output data 0: Output "Low" (V_{SS} -level) 1: Output "High" (V_{DD5} -level)

■ Port C Input Register (PCIN: 0x03E8C)

<MN101EF76K>

bp	7	6	5	4	3	2	1	0
Flag	-	PCIN6	PCIN5	PCIN4	PCIN3	PCIN2	PCIN1	PCIN0
At reset	-	X	X	X	X	X	X	X
Access	-	R	R	R	R	R	R	R

bp	Flag	Description
7	-	-
6 to 0	PCIN6 to 0	Input data 0: Pin is "Low" (V_{SS} -level) 1: Pin is "High" (V_{DD5} -level)

■ Port C Direction Control Register (PCDIR: 0x03E9C)

<MN101EF76K>

bp	7	6	5	4	3	2	1	0
Flag	-	PCDIR6	PCDIR5	PCDIR4	PCDIR3	PCDIR2	PCDIR1	PCDIR0
At reset	-	0	0	0	0	0	0	0
Access	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description
7	-	-
6 to 0	PCDIR6 to 0	I/O mode selection 0: Input mode 1: Output mode

■ Port C Pull-up Resistor Control Register (PCPLU: 0x03EAC)

<MN101EF76K>

bp	7	6	5	4	3	2	1	0
Flag	-	PCPLU6	PCPLU5	PCPLU4	PCPLU3	PCPLU2	PCPLU1	PCPLU0
At reset	-	0	0	0	0	0	0	0
Access	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description
7	-	-
6 to 0	PCPLU6 to 0	Pull-up resistor selection 0: Not added 1: Added

8.15.3 Block Diagram (MN101EF76K)

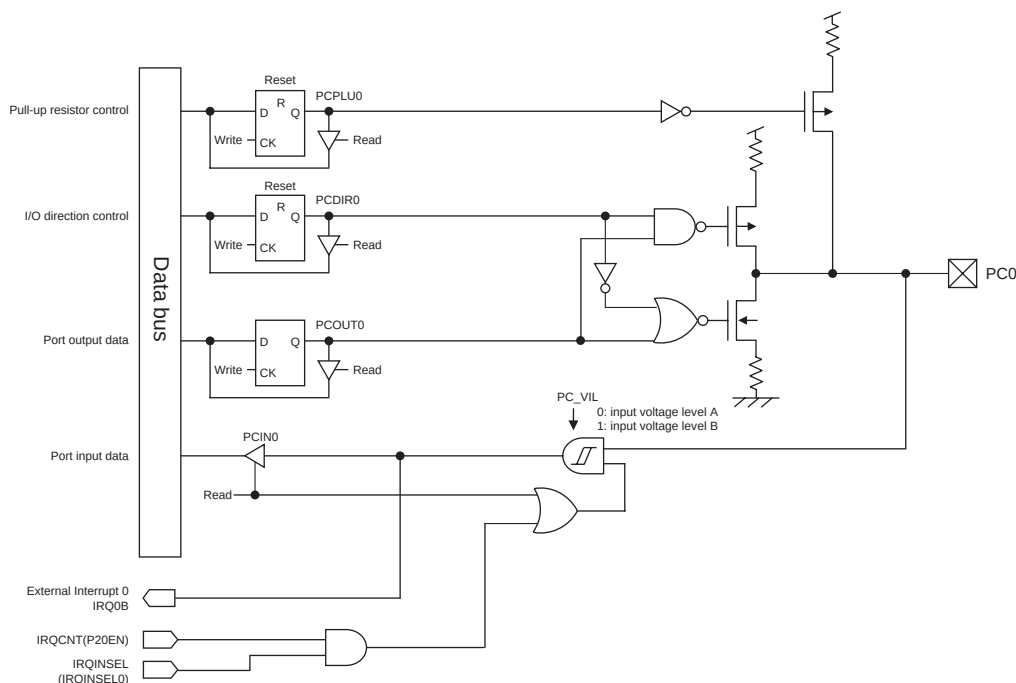


Figure:8.15.1 PC0 Block Diagram (MN101EF76K)

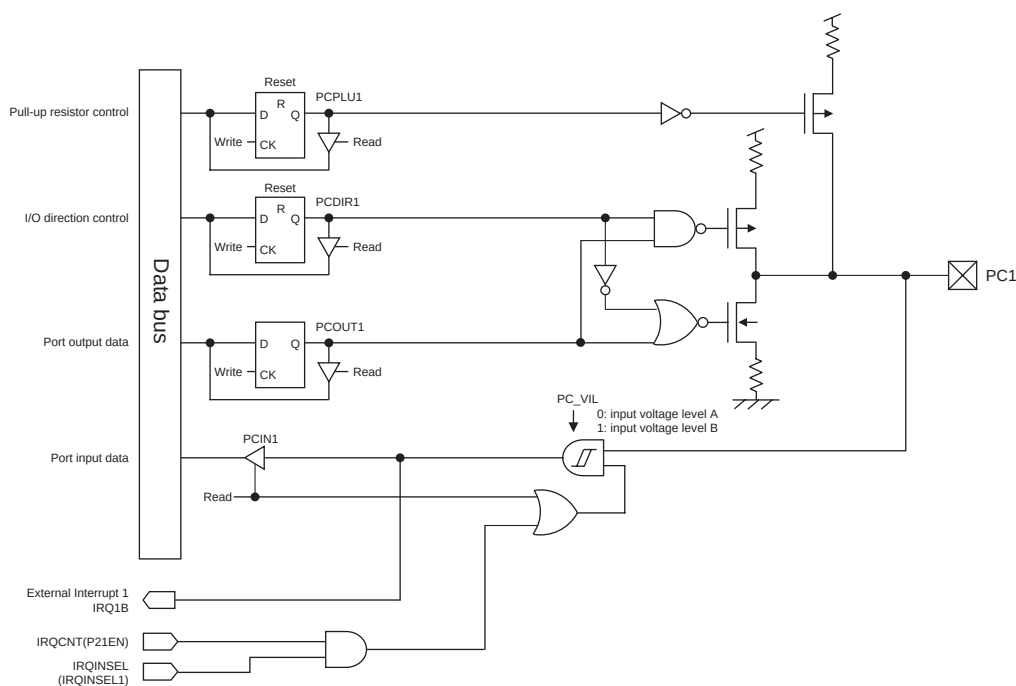


Figure:8.15.2 PC1 Block Diagram (MN101EF76K)

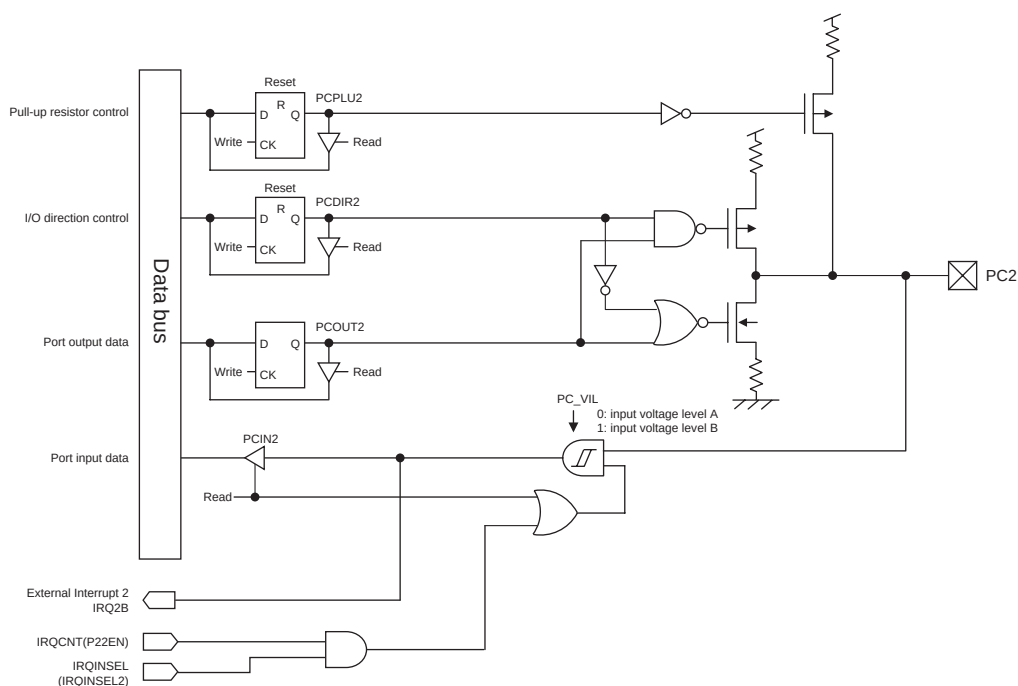


Figure:8.15.3 PC2 Block Diagram (MN101EF76K)

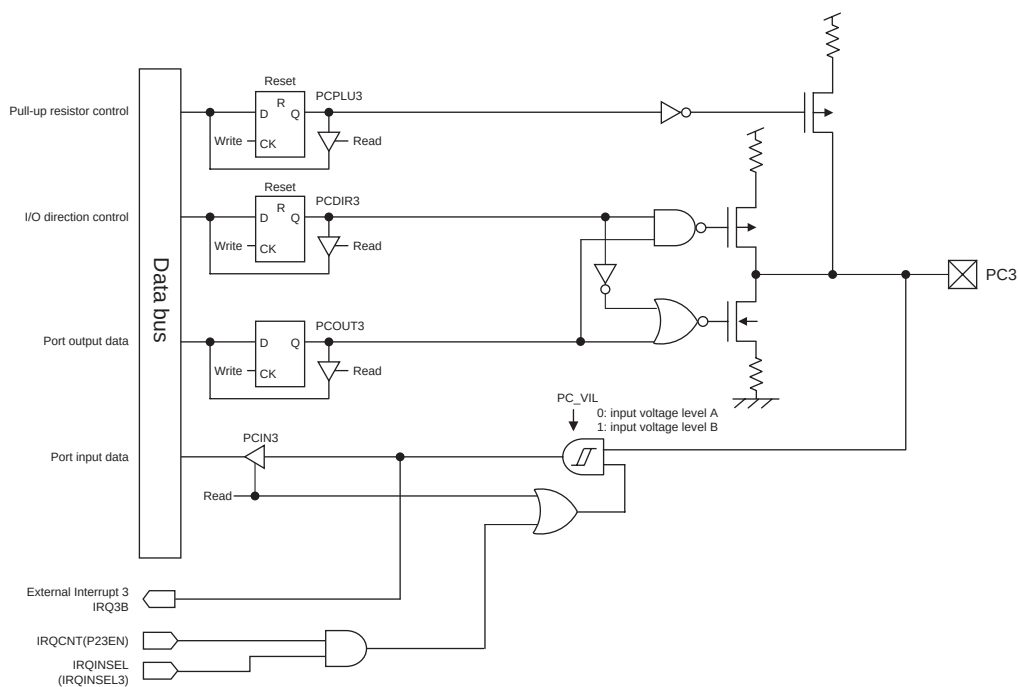


Figure:8.15.4 PC3 Block Diagram (MN101EF76K)

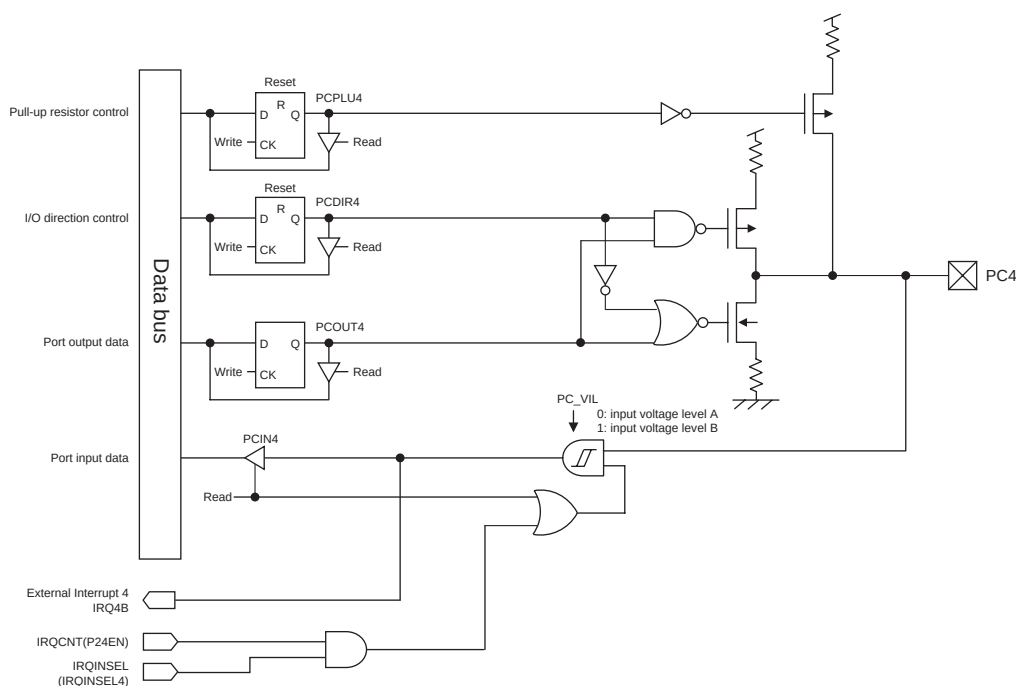


Figure:8.15.5 PC4 Block Diagram (MN101EF76K)

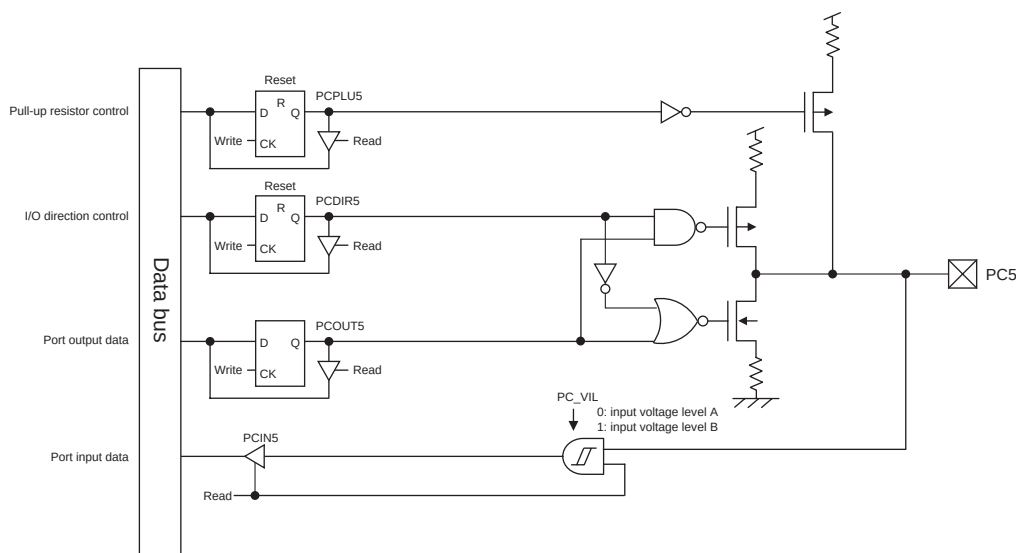


Figure:8.15.6 PC5 Block Diagram (MN101EF76K)

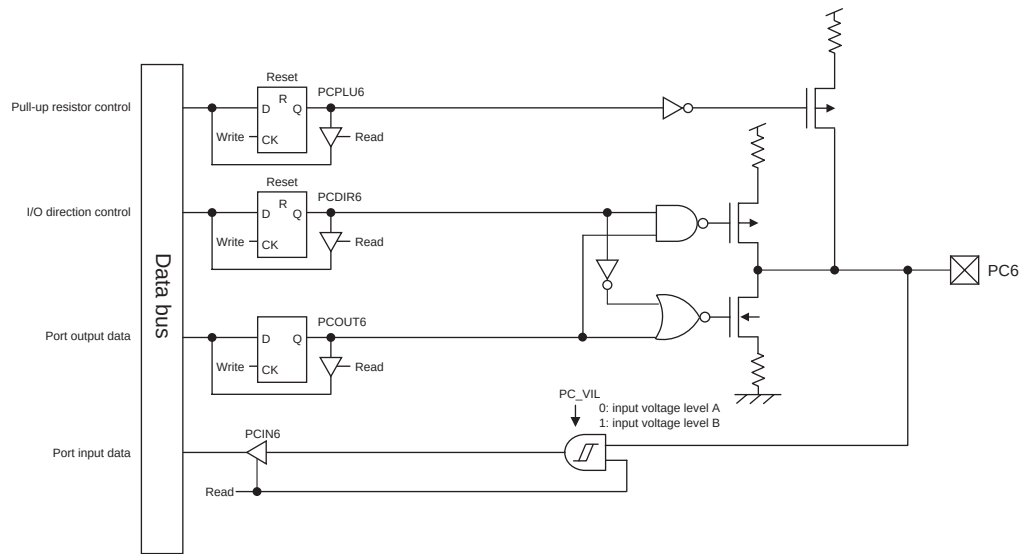


Figure:8.15.7 PC6 Block Diagram (MN101EF76K)

8.16 Port D

8.16.1 Description

■ General-purpose Port Setup

To output data to pin, set the control flag of Port D direction control register (PDDIR) to "1" and write the value of Port D output register (PDOUT).

To read input data of pin, set the control flag of Port D direction control register (PDDIR) to "0" and read the value of Port D input register (PDIN).

Each bit can be set individually as either an input or output by Port D direction control register (PDDIR). The control flag of PDDIR register is set to "1" for output mode, and "0" for input mode.

Each bit can be set individually to add a pull-up resistor by Port D pull-up resistor control register (PDPLU). Set the control flag of PDPLU register to "1" to add pull-up resistor.

Table:8.16.1 Port D Functions

Table remarks -: Without function

MN101EF76K		MN101EF57G/MN101EF56K	
Pin	Special function	Pin	Special function
PD0	TM9OD0B	-	-
PD1	TM9OD1B	-	-
PD2	TM9OD2B	-	-
PD3	TM9OD3B	-	-
PD4	TM9OD4B	-	-
PD5	TM9OD5B	-	-
PD6		-	

■ Special Function Pin Setup

PD0 to PD5 are also used as the output pins for Timer 9.

Each bit can be set individually as either timer output or general purpose port by Port D output mode register (PDOMD). The corresponding flag of PDOMD register is set to "1" for timer output, and "0" for general-purpose port.

8.16.2 Registers

Table:8.16.2 shows registers that control Port D.

Table:8.16.2 Port D Control Register

Table remarks √: With function -: Without function

Registers	Address	R/W	Function	Page	MN101 EF76K	MN101 EF57G	MN101 EF56K
PDOUT	0x03E7D	R/W	Port D output register	VIII-228	√	-	-
PDIN	0x03E8D	R	Port D input register	VIII-228	√	-	-
PDDIR	0x03E9D	R/W	Port D direction control register	VIII-229	√	-	-
PDPLU	0x03EAD	R/W	Port D pull-up resistor control register	VIII-229	√	-	-
PDOMD	0x03EBD	R/W	Port D output mode register	VIII-230	√	-	-

R/W: Readable/Writable

R: Read only

■ Port D Output Register (PDOUT: 0x03E7D)

<MN101EF76K>

bp	7	6	5	4	3	2	1	0
Flag	-	PDOUT6	PDOUT5	PDOUT4	PDOUT3	PDOUT2	PDOUT1	PDOUT0
At reset	-	X	X	X	X	X	X	X
Access	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description
7	-	-
6 to 0	PDOUT6 to 0	Output data 0: Output "Low" (V_{SS} -level) 1: Output "High" (V_{DD5} -level)

■ Port D Input Register (PDIN: 0x03E8D)

<MN101EF76K>

bp	7	6	5	4	3	2	1	0
Flag	-	PDIN6	PDIN5	PDIN4	PDIN3	PDIN2	PDIN1	PDIN0
At reset	-	X	X	X	X	X	X	X
Access	-	R	R	R	R	R	R	R

bp	Flag	Description
7	-	-
6 to 0	PDIN6 to 0	Input data 0: Pin is "Low" (V_{SS} -level) 1: Pin is "High" (V_{DD5} -level)

■ Port D Direction Control Register (PDDIR: 0x03E9D)

<MN101EF76K>

bp	7	6	5	4	3	2	1	0
Flag	-	PDDIR6	PDDIR5	PDDIR4	PDDIR3	PDDIR2	PDDIR1	PDDIR0
At reset	-	0	0	0	0	0	0	0
Access	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description
7	-	-
6 to 0	PDDIR6 to 0	I/O mode selection 0: Input mode 1: Output mode

■ Port D Pull-up Resistor Control Register (PDPLU: 0x03EAD)

<MN101EF76K>

bp	7	6	5	4	3	2	1	0
Flag	-	PDPLU6	PDPLU5	PDPLU4	PDPLU3	PDPLU2	PDPLU1	PDPLU0
At reset	-	0	0	0	0	0	0	0
Access	-	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description
7	-	-
6 to 0	PDPLU6 to 0	Pull-up resistor selection 0: Not added 1: Added

■ Port D Output Mode Register (PDOMD: 0x03EBD)

<MN101EF76K>

bp	7	6	5	4	3	2	1	0
Flag	-	-	PDOMD5	PDOMD4	PDOMD3	PDOMD2	PDOMD1	PDOMD0
At reset	-	-	0	0	0	0	0	0
Access	-	-	R/W	R/W	R/W	R/W	R/W	R/W

bp	Flag	Description
7 to 6	-	-
5	PDOMD5	I/O port or timer output selection 0: PD5 1: TM9OD5B
4	PDOMD4	I/O port or timer output selection 0: PD4 1: TM9OD4B
3	PDOMD3	I/O port or timer output selection 0: PD3 1: TM9OD3B
2	PDOMD2	I/O port or timer output selection 0: PD2 1: TM9OD2B
1	PDOMD1	I/O port or timer output selection 0: PD1 1: TM9OD1B
0	PDOMD0	I/O port or timer output selection 0: PD0 1: TM9OD0B

8.16.3 Block Diagram (MN101EF76K)

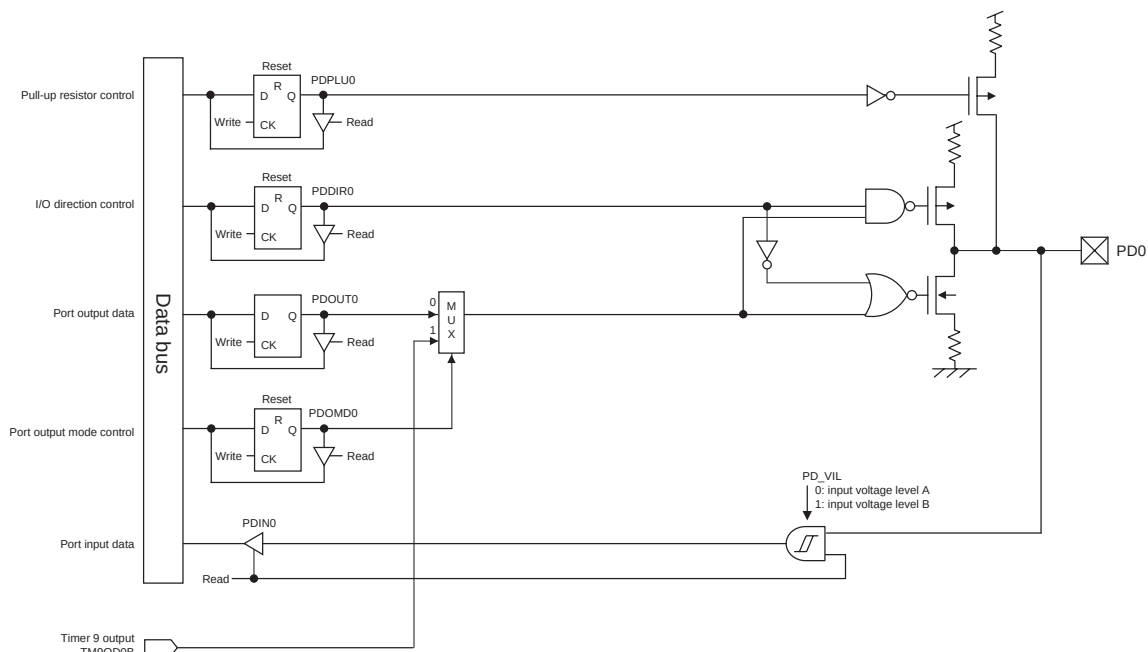


Figure:8.16.1 PD0 Block Diagram (MN101EF76K)

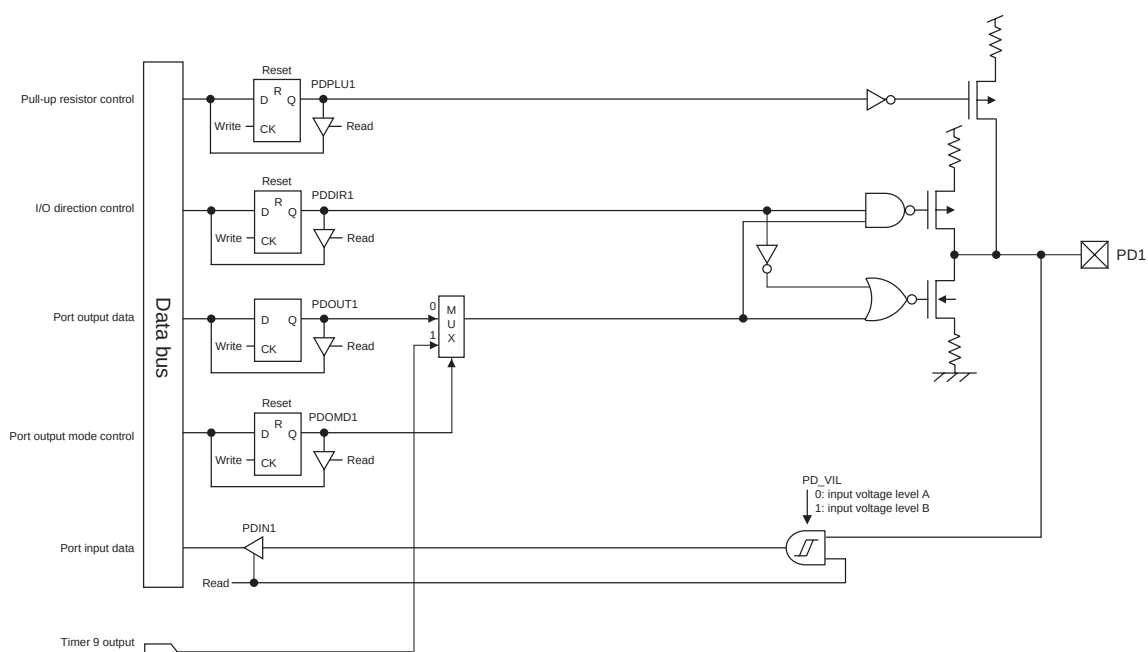


Figure:8.16.2 PD1 Block Diagram (MN101EF76K)

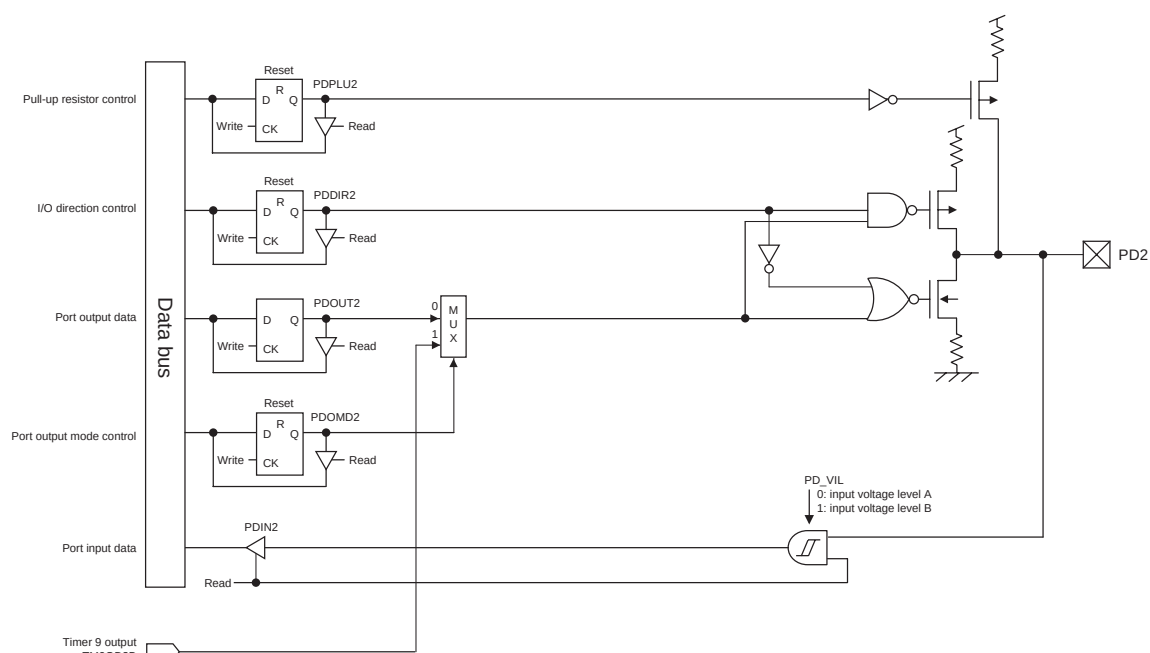


Figure:8.16.3 PD2 Block Diagram (MN101EF76K)

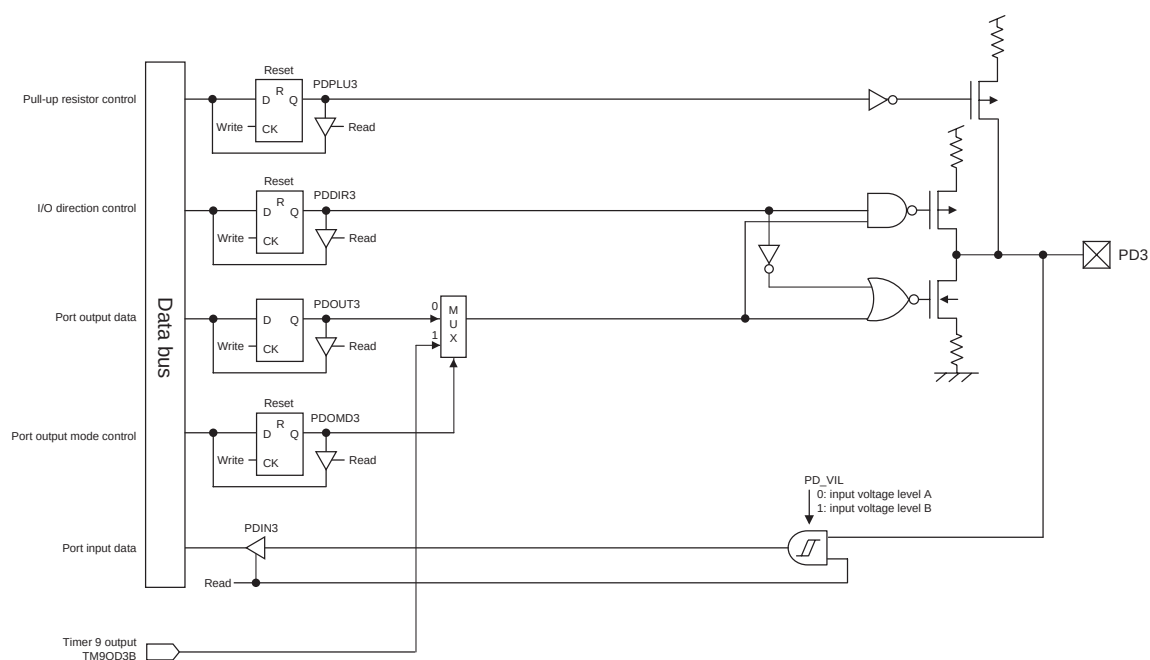


Figure:8.16.4 PD3 Block Diagram (MN101EF76K)

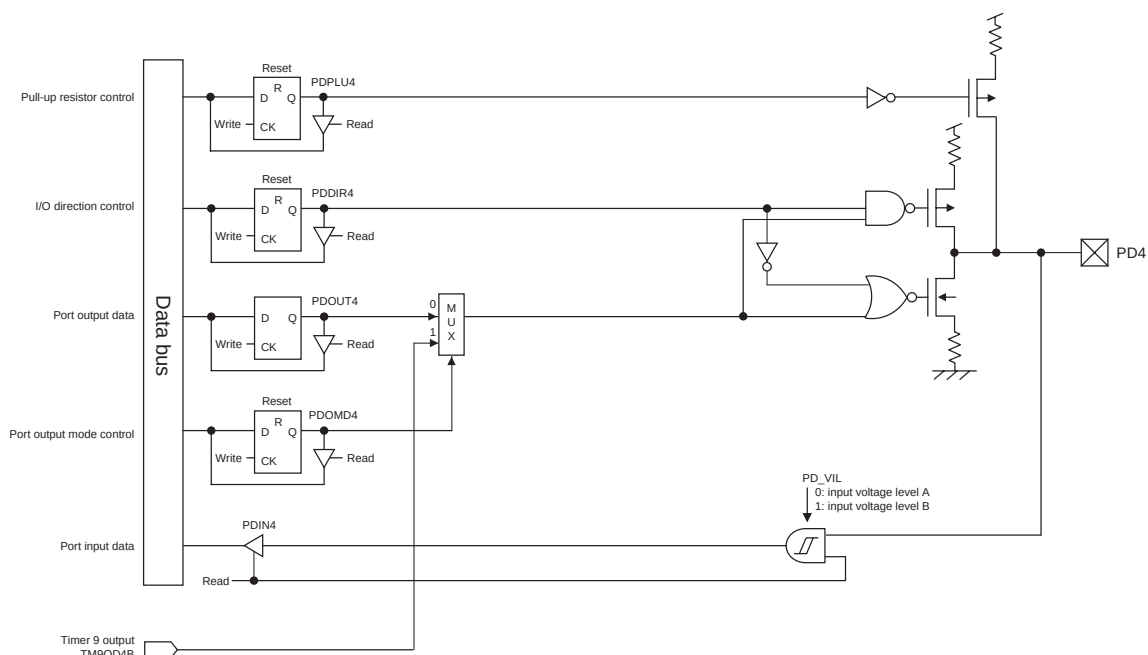


Figure:8.16.5 PD4 Block Diagram (MN101EF76K)

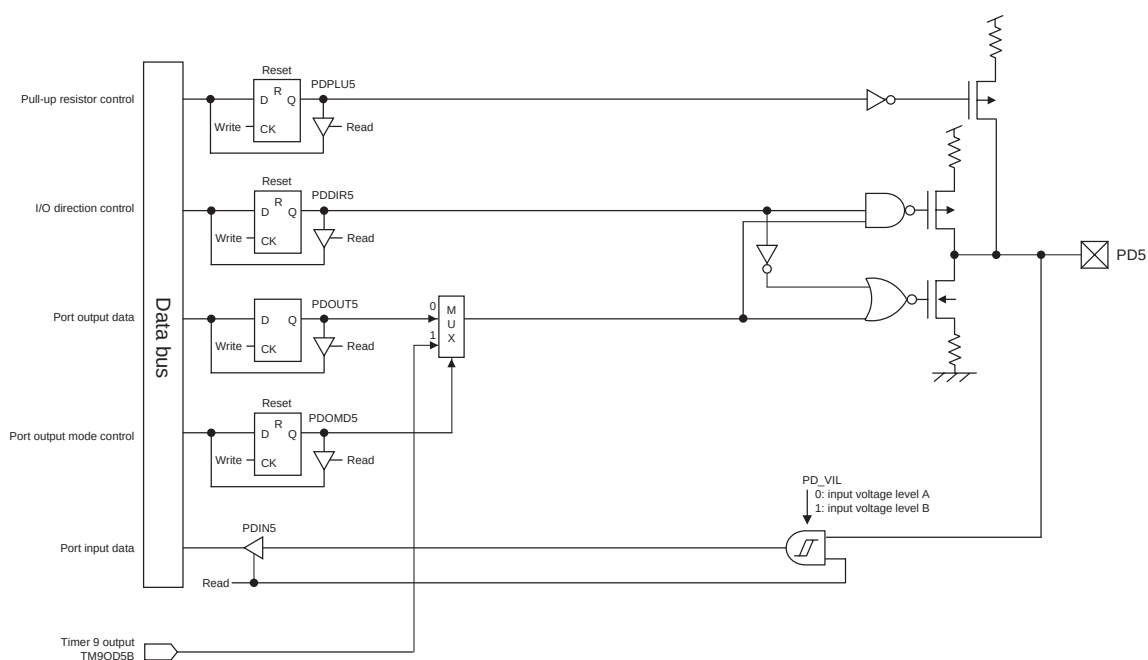


Figure:8.16.6 PD5 Block Diagram (MN101EF76K)

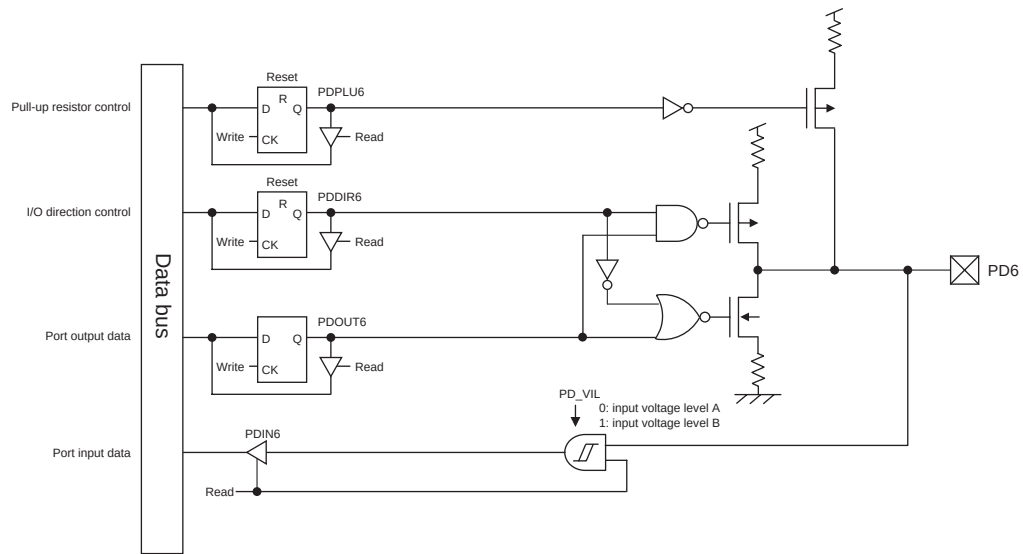


Figure:8.16.7 PD6 Block Diagram (MN101EF76K)

8.17 Real Time Output Control

PA0, PA2 and PA5 include real-time output control function which switches the output of pins during the falling edge event of External Interrupt 0 pin (P20/IRQ0).

The real time control functions change the output signals (PWM output and timer pulse output) synchronous with the external event without software intervention. It switches between three output values, "0", "1" and "high-impedance (Hi-Z)", when an event occurs.

8.17.1 Registers

Table:8.17.1 shows the real-time output control registers of Port A.

Table:8.17.1 Real Time Output Control Registers

Register	Address	R/W	Function	Page
PAOUT	0x03E7A	R/W	Port A output register	VIII-191
PADIR	0x03E9A	R/W	Port A direction control register	VIII-192
PAPLU	0x03EAA	R/W	Port A pull-up resistor control register	VIII-192
PAOMD1	0x03EB6	R/W	Port A output mode register 1	VIII-193
PACNT	0x03EF6	R/W	Port A output control register	VIII-195

8.17.2 Operation

■ Real Time Output Pin Setup

The real time output pin is configured in Port A output control register (PACNT). It selects between PA0, PA2 and PA5 and each bit can be specified individually. Select the output mode in Port A direction control register (PADIR).

At the falling edge event of External Interrupt 0 pin (P20/IRQ0), the pin output switches between the three values "0", "1" and "high-impedance (Hi-Z)". When high-impedance is output, the port is in input state.

The real-time control functions change the timer output signal (PWM output and timer pulse output) synchronous with the external event. It also functions as the normal port output.

When I/O port (real-time control disabled) is selected in PACNT register, the value will not be changed even if a switching event occurs. To use these pins as general-purpose ports, this mode should be set.

■ Real-Time Output Control Operation

After setting PACNT register, the functions selected in Port A output mode register (PAOMD) are output to the pin until a falling edge is generated at External Interrupt 0 pin (P20/IRQ0).

When a falling edge of External Interrupt 0 is generated, the pin output switches to the set value. The falling edge event is held by the edge event hold function shown in Figure:8.13.1, Figure:8.13.3, Figure:8.13.5 and the setup value of PACNT register will be held until the information is cleared.

■ Real-Time Output Release (Clearance of Edge Event Hold Function)

After event occurrence, when data is written in PAOUT register, the information of the edge event hold function is cleared and all the outputs of pins are recovered to the output data before the event occurs. When an event occurs again, the data switches to the setup value of PACNT register. To cancel the real-time control, set PACNT register to I/O port (real-time control disabled).



Regardless of the settings in External Interrupt 0 control register (IRQ0ICR), only falling edge is valid for IRQ0.



When the real time output control function is used, write in Port A output control register (PACNT) to clear the information of the edge event hold function in advance.

■ Timing of Real Time Output Control (Port A)

PACNT set value: "0" (Low) output

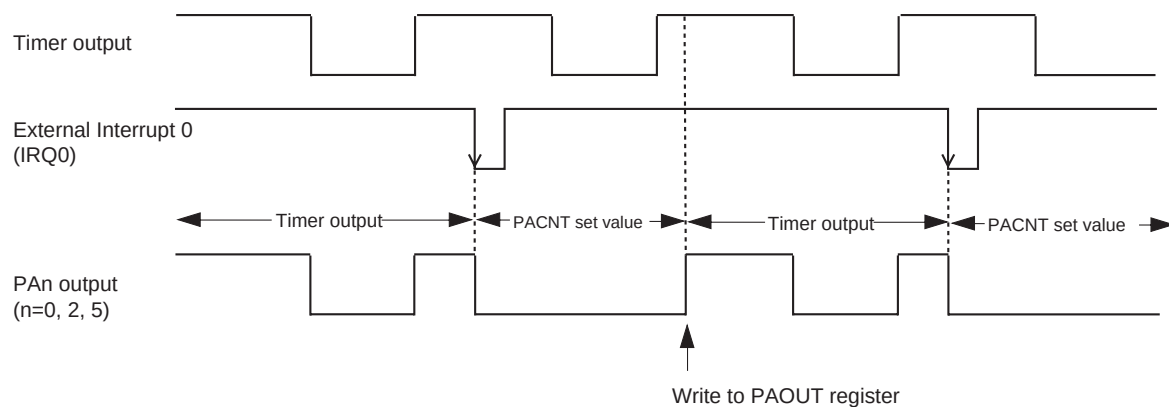


Figure:8.17.1 Timing of Real Time Output Control

8.18 Synchronous Output

Port 8 has the synchronous output function that outputs appropriate data to pins without software intervention during the occurrence of a specified event. Synchronous event can be selected from External Interrupt 2 (P22/IRQ2), Timer 1 interrupt, Timer 2 interrupt and Timer 7 interrupt signals.

8.18.1 Registers

Table:8.18.1 shows Port 8 synchronous output control registers.

Table:8.18.1 Synchronous Output Control Register

Register	Address	R/W	Function	Page
P8OUT	0x03E78	R/W	Port 8 output register	VIII-153
P8DIR	0x03F38	R/W	Port 8 direction control register	VIII-154
P8PLU	0x03F48	R/W	Port 8 pull-up resistor control register	VIII-154
P8SYO	0x03EF7	R/W	Port 8 synchronous output control register	VIII-156
P8SEV	0x03EF8	R/W	Port 8 synchronous output event selection register	VIII-157

8.18.2 Operation

■ Synchronous Output Setup

Port 8 synchronous output pins for each bit are set by Port 8 synchronous output control register (P8SYO). The synchronous output event is selected by Port 8 synchronous output event selection register (P8SEV).

When External Interrupt 2 (IRQ2) is selected, this pin synchronizes with falling edges regardless of the IRQ2ICR edge specification.

■ Synchronous Output Operation

When P8SYO register is set to disable the synchronous output (I/O port), Port 8 functions as a general purpose port. When synchronous output is disabled in Port 8, the value same as Port 8 output register (P8OUT) is always loaded to the synchronous output value storage register.

After the output mode is selected in Port 8 direction control register (P8DIR), the value of the synchronous output value storage register will be output when the synchronous output is enabled by P8SYO register. Unless the synchronous output event set in P8SEV register occurs, the synchronous output value storage register holds its value after the synchronous output is enabled.

P8OUT register stores the value which should be output from pins after the synchronous output event is occurred. When the synchronous output event set in P8SEV register occurs, the data of the synchronous output storage register is switched to the data of P8OUT register, and the pin output changes.



Set the initial value of the synchronous output in Port 8 output register (P8OUT) before enabled by the synchronous output control register (P8SY0).

■ Port 8 Synchronous Output (External Interrupt 2 (IRQ2))

Figure:8.18.1 shows the synchronous output timing when the synchronous output event is set to External Interrupt 2. The value of P8OUT register is output synchronous with IRQ2 falling edge.

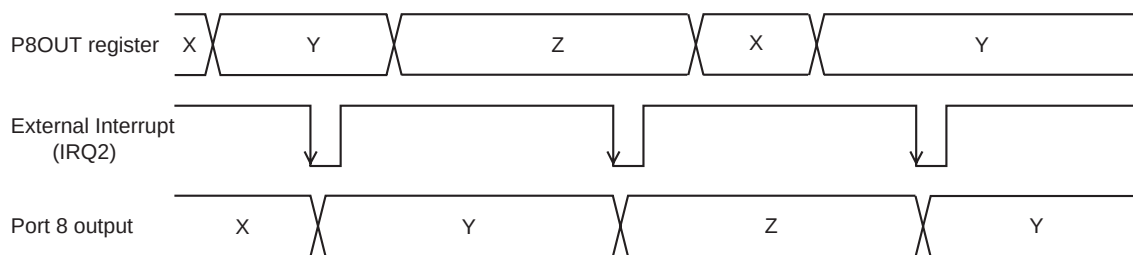


Figure:8.18.1 Synchronous Output Timing by Event Generation (IRQ2)

■ Port 8 Synchronous Output (Timer 1, Timer 2 and Timer 7)

The timer interrupt flag TMnIRQ is generated when the setting value of the binary counter reaches the compare register. Port 8 output latched data is output from Port 8 when TMnIRQ flag is rising.

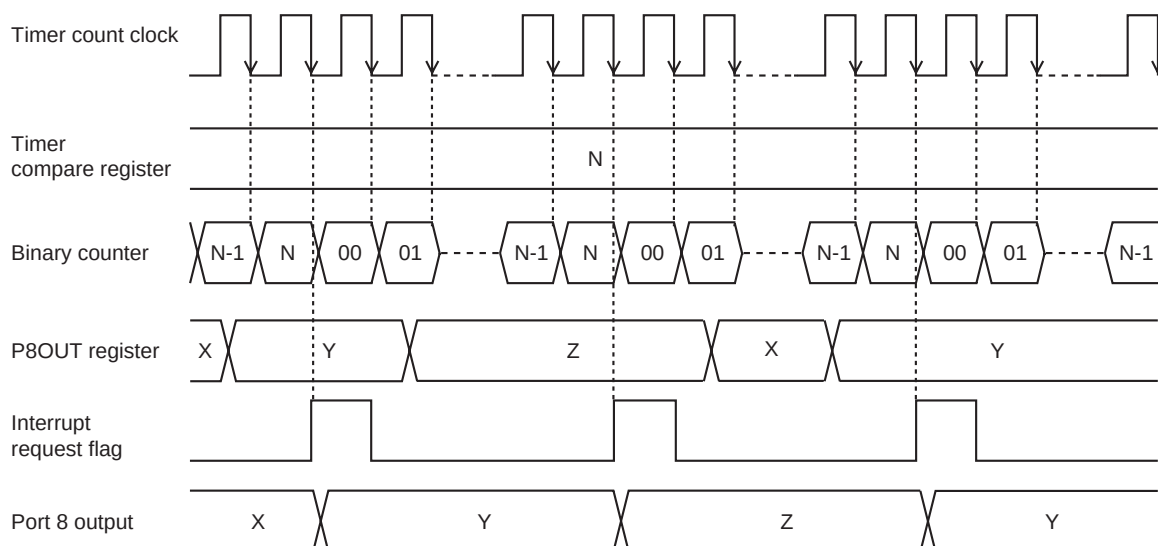


Figure:8.18.2 Synchronous Output Timing by Event Generation (Timer 1, Timer 2 and Timer 7)

8.19 Input Voltage Level Switching Function

8.19.1 Overview

This LSI can switch the levels to detect the input voltage with the pin input voltage switching flash option (FLOP1 to 0) and the pin input voltage level switching register (SWCNT). For detailed setting of this function, refer to [Chapter 8 8.19.2 Input Voltage Level Switching Flash Option] and [Chapter 8 8.19.3 Input Voltage Level Switching Register].

Table:8.19.1 Input Voltage Level Switching Flash Option

Register	Address	R/W	Function	Page
FLOP0	0x040C0	R/W	Flash option 0	VIII-241
FLOP1	0x040C1	R/W	Flash option 1	VIII-242

Table:8.19.2 Input Voltage Level Switching Register

Register	Address	R/W	Function	Page
SWCNT	0x03E8F	R/W	0.54 V _{DD5} input control register	VIII-243

8.19.2 Input Voltage Level Switching Flash Option

This LSI allocates 0x040C0 and 0x040C1 of memory area as flash option area.
When power-on or internal reset is released after oscillation stabilization wait time, the flash option area is referred and the setting value of this area is automatically read out by hardware to set the functions.
When using this LSI, please always configure the flash option settings.

Flash option in this LSI is allocated as indicated below.
Refer to [Chapter 8 8.19.4 Input Voltage Level] for the input voltage level A/B.

■ Flash Option 0 (FLOP0: 0x040C0)

bp	7	6	5	4	3	2	1	0
Flag	P4_VIL	P3_VIL	P2_VIL	-	P0_VIL	WD2OP2	WD2OP1	WD2OP0

bp	Flag	Description
7	P4_VIL	Select Port 4 input voltage level 1: Input voltage level A 0: Input voltage level B
6	P3_VIL	Select Port 3 input voltage level 1: Input voltage level A 0: Input voltage level B
5	P2_VIL	Select Port 2 input voltage level 1: Input voltage level A 0: Input voltage level B
4	-	-
3	P0_VIL	Select Port 0 input voltage level 1: Input voltage level A 0: Input voltage level B
2 to 0	WD2OP2 to 0	Setting control of WDT2 operation mode setting register (WD2MDSEL) 101 : Setting enabled for WD2MDSEL register Other than 101 : Setting disabled for other than WD2MDSEL register (Watchdog timer 2 is always operating.)

■ Flash Option 1 (FLOP1: 0x040C1)

bp	7	6	5	4	3	2	1	0
Flag	-	PB_VIL	PA_VIL	P9_VIL	P8_VIL	P7_VIL	P6_VIL	P5_VIL

bp	Flag	Description
7	-	-
6	PB_VIL	Select Port B input voltage level 1: Input voltage level A 0: Input voltage level B
5	PA_VIL	Select Port A input voltage level 1: Input voltage level A 0: Input voltage level B
4	P9_VIL	Select Port 9 input voltage level 1: Input voltage level A 0: Input voltage level B
3	P8_VIL	Select Port 8 input voltage level 1: Input voltage level A 0: Input voltage level B
2	P7_VIL	Select Port 7 input voltage level 1: Input voltage level A 0: Input voltage level B
1	P6_VIL	Select Port 6 input voltage level 1: Input voltage level A 0: Input voltage level B
0	P5_VIL	Select Port 5 input voltage level 1: Input voltage level A 0: Input voltage level B

8.19.3 Input Voltage Level Switching Register

Port A can be switched the input voltage level by the 0.54 V_{DD5} input control register1.
Refer to [Chapter 8 8.19.4 Input Voltage Level] for the input voltage level A, B and C.

- 0.54 V_{DD5} input control register (SWCNT: 0x03E8F)

bp	7	6	5	4	3	2	1	0
Flag	-	-	-	-	-	SWCNT_A	-	-
At reset	-	-	-	-	-	0	-	-
Access	-	-	-	-	-	R/W	-	-

bp	Flag	Description
7 to 3	-	-
2	SWCNT_A	Port A input voltage level selection 0: Input voltage level A/B 1: Input voltage level C
1 to 0	-	-



Refer to [Chapter 8 8.19.2 Input Voltage Level Switching Flash Option] for switching of the input voltage level A/B.

8.19.4 Input Voltage Level

This LSI can switch the levels to detect input voltage with the pin input voltage switching flash option (FLOP1 to 0) and the pin input voltage level switching register (SWCNT).

For detailed setting of this function, refer to [Chapter 8 8.19.2 Input Voltage Level Switching Flash Option] and [Chapter 8 8.19.3 Input Voltage Level Switching Register].

■ Input Voltage Level

Ta = -40 °C to +85 °C

	Parameter	Conditions	Rating			Unit
			MIN	TYP	MAX	

P0 to PB

Input voltage level A	Input high voltage	$V_{DD5} = 1.8 \text{ to } 5.5 \text{ V}$	$0.8 V_{DD5}$		V_{DD5}	V
	Input low voltage		0		$0.2 V_{DD5}$	
Input voltage level B	Input high voltage	$V_{DD5} = 4.5 \text{ to } 5.5 \text{ V}$	$0.8 V_{DD5}$		V_{DD5}	V
	Input low voltage		0		$0.45 V_{DD5}$	

PA

Input voltage level C	Input high voltage	$V_{DD5} = 1.8 \text{ to } 5.5 \text{ V}$	$0.54 V_{DD5}$		V_{DD5}	V
	Input low voltage		0		$0.2 V_{DD5}$	

Record of Changes

The following shows the changes in the publication of "MN101EF56K/57G/76K LSI User's Manual" (From 3rd Edition dated in January, 2012 to the 3rd Edition 1st Printing dated in February, 2014)

Page	Line	Definition	Former Edition (3.0)	New Edition (3.1)
I-3	Package	Specification addition	-	Package(TQFP080-P-1212F) is added in MN101EF57G.
I-13	Package	Specification addition	80LQFP Top View	80LQFP/ <u>TQFP</u> Top View
I-73	Package	Specification addition	-	Package code: TQFP080-P-1212F (MN101EF57G) is added.

Remark: Definition in the above table is classified according to the content of changes as follows.

Error correction, Description change, Description addition, Description deletion: For the description of LSI manual.

Specification change, Specification addition, Specification deletion: For microcontroller's specification.

The following shows the changes in the publication of "MN101EF56K/57G/76K LSI User's Manual" (From 2nd Edition dated in September, 2011 to the 3rd Edition dated in January, 2012.)

Page	Line	Definition	Former Edition (2.0)	New Edition (3.0)
-	All Chapters	Description addition	-	Added the specification for MN101EF56K.
I-2 I-47	Note	Description change	...When using <u>ICE version</u> , it is...	...When using <u>In-circuit Emulator</u> , it is...
I-10	- Ports	Error correction	MN101EF76K Timer I/O <u>28</u> External interrupt <u>5</u>	MN101EF76K Timer I/O <u>34</u> External interrupt <u>10</u>
I-26	Table:1.3.4	Description addition	-	Added OCD_DATA and OSC_CLK in MN101EF76K.
I-35	Table:1.3.5	Description addition	-	Added OCD_DATA and OSC_CLK in MN101EF57G.
I-59	*17	Description change	...When using <u>ICE version</u> , it is...	...When using <u>In-circuit Emulator</u> , it is...
VII-3	Table:7.1.1	Error correction	Table address IVBM=0 0x04008 to <u>0x04074</u> IVBM=1 0x00108 to <u>0x00174</u>	Table address IVBM=0 0x04008 to <u>0x04078</u> IVBM=1 0x00108 to <u>0x00178</u>
VII-52	Note 2	Description addition	-	<u>When the external interrupt is not used, set IROnIE flag to "0"...</u>
	Note 3	Description addition	-	<u>The interrupt request flag may have been set when the interrupt edge is switched...</u>
VIII-2	8.1.1	Description addition	MN101EF76K has...which are allocated to the I/O ports (Port 0, Port 2...	MN101EF76K has...which are allocated to the I/O ports (Port 0, <u>Port 1</u> , Port 2...
	Table:8.1.1	Description addition	-	<u>Port 1 Input mod...</u>
VIII-4	Table:8.2.1	Description addition	-	MN101EF76K: <u>P1OUT, P1IN, P1DIR...</u>
		Description deletion	MN101EF76K: <u>P3IMD...</u>	MN101EF76K: <u>-</u>
IX-3	Note 1	Description addition	...Pin <u>A and B</u> , " <u>A</u> " and " <u>B</u> " of the pin names are omitted.	...Pin <u>A, B and C</u> , " <u>A</u> ", " <u>B</u> ", and " <u>C</u> " of the pin names are omitted.
IX-4	Table:9.1.2	Description addition	-	Event count <u>TMnIOC input</u> Timer pulse output <u>TMnIOC output</u> PWM output <u>TMnIOC output</u>
IX-21	Note 2	Description addition	-	<u>Under the following conditions, be sure to select a system clock (fs) as a clock source of timers to set compare register of Timer n...</u>
XI-14	Note	Description addition	-	<u>Under the following conditions, be sure to select a system clock (fs) as a clock source of timer to set preset register 1 and 2 of Timer 7 and Timer 8....</u>
XI-39	Table:11.4.1	Description addition	-	Timer 7 Event input <u>TM7IOC input</u> <u>TM8IOC input</u>
XI-46	Table:11.5.1	Description addition	-	Timer 7 Pulse output pin <u>TM7IOC output</u> <u>TM8IOC output</u>
XII-18	Note	Description addition	-	<u>Be sure to set the IROSEL2 to IROSEL0 flags before setting the other flags in PWMOFFL register.</u>
XII-26 XII-28	Note 3	Description addition	-	<u>Under the following conditions, be sure to set PWMCKSL1 to 0 flags of PWM timer operation control register to "00".</u>
XVIII-3	Table:18.1.3	Description deletion	Serial Interface 2 <u>SBO2B(P30), SBI2B(P31), SBT2B(P32), TXD2B(P30), RXD2B(P31)</u>	-

Page	Line	Definition	Former Edition (2.0)	New Edition (3.0)
XVIII-4	Table:18.1.4	Description addition	Serial Interface 0, 1, 2 and 3 Selection of transfer clock division ratio Divide by 8 Divide by16	Serial Interface 0, 1, 2 and 3 Selection of transfer clock division ratio <u>No dividing</u> Divide by 8 Divide by16
XVIII-12	Figure:18.1.6	Description deletion	<u>SBO2B/TXD2B/P30, SBI2B/RXD2B/P31, SBT2B/P32</u>	-
XVIII-35	SC4MD0	Description change	bp: 5 Flag: SC4DEM Description: Operation selection after detecting <u>instability of IIC slave communication data</u> 0: <u>Disable communication</u> 1: <u>Enable communciation</u>	bp: 5 Flag: SC4DEM Description: Operation selection after detecting <u>communication data instability in IIC slave communication mode</u> 0: <u>Start condition is not detected</u> 1: <u>Start condition is detected</u>
XVIII-38	SC4MD3	Description deletion	bp: 5 Flag: SC4STPC Description: Stop condition generation request flag in IIC communication <u>*1</u>	bp: 5 Flag: SC4STPC Description: Stop condition generation request flag in IIC communication
		Description change	bp: 0 Flag: SC4ACKO Description: At transmission mode (<u>SC4REX=0</u>) <u>ACK</u> bit detection flag At reception mode (<u>SC4REX=1</u>) <u>ACK</u> bit level specification <u>*2</u>	bp: 0 Flag: SC4ACKO Description: At transmission <u>ACK/NACK</u> bit detection flag At reception <u>ACK/NACK</u> bit transmission flag
	Note 1 (Note 1 and 2: Former Edition)	Description change	Note 1: <u>*1: Only "1" can be written. In addition, write is disabled when a forced reset...</u> Note 2: <u>Set data to SC4MD3 register with a MOV instruction according to each bit setting...</u>	Note 1: <u>SC4STPC and SC4ACKO flags are updated by hardware. Thus, when values of these flags are read out, the values might be...</u>
XVIII-40	Note	Description addition	-	Regarding SC4ORE flag: when "1" is written, the value of this flag... when "0" is written, the value of this flag...
XVIII-41	SC4STR1	Description deletion	bp: 6 Flag: SC4ABT_LST Description: Arbitration lost detection flag <u>*1</u> bp: 0 Flag: SC4DATA_ERR Description: Communication error detection flag <u>*1</u>	bp: 6 Flag: SC4ABT_LST Description: Arbitration lost detection flag bp: 0 Flag: SC4DATA_ERR Description: Communication error detection flag
			<u>*1 Only "0" can be written.</u>	<u>Regarding SC4ABT_LST flag and SC4DATA_ERR flag:</u> <u>when "1" is written, the value of this flag...</u> <u>when "0" is written, the value of this flag...</u>
XVIII-48	■ Overrun Error and...	Description addition	■ Overrun Error and Error Monitor Flag (<u>Serial Interface 0, 1 and 2</u>)	■ Overrun Error and Error Monitor Flag (<u>Serial Interface 0, 1, 2 and 3</u>)
XVIII-73	Table:18.4.5	Description addition	<u>Serial Interface 0, 1 and 2</u> Setup: <u>Serial Interface 0, 1 and 2</u> clock source... Register: SC0MD3, SC1MD3	<u>Serial Interface 0, 1, 2 and 3</u> Setup: <u>Serial Interface 0, 1, 2 and 3</u> clock source... Register: SC0MD3, SC1MD3, <u>SC2MD3, SC3MD3</u>
			-	When the SCnCMD flag of SCnMD1 register is set to "1"...
XVIII-79	Figure:18.4.4 Figure:18.4.5	Description deletion	-	The timing from a data setup to a startbit output is changed.
XVIII-87	Hint 3	Description addition	-	When the SCnCMD flag of SCnMD1 register is set to "1"...

Page	Line	Definition	Former Edition (2.0)		New Edition (3.0)	
XVIII-93	■ Communication Data Instability...	Description change	<u>When it detects a change in the data bus (SDA) while...the subsequent start condition and communication data instability cannot be detected</u>		<u>When a serial circuit detects a changes in the data bus (SDA4 pin) while...the circuit can detect start conditions and starts communication even if the SC4DATA_ERR flag is "1"</u>	
XXI-2	Table:21.1.2	Description deletion	<u>Input Range</u>		-	
XXII-4	22.1.4	Error correction	LCD output control registers <u>1 to 6 (LCCTR1 to LCCTR6)</u>		LCD output control register <u>1 to 7 (LCCTR1 to LCCTR7)</u>	
XXIV-30	Table:24.7.3	Error correction	Command library Program_1ms_Lib Erase_1ms_Lib	Address <u>0x6F889</u> <u>0x6F893</u>	Command library Program_1ms_Lib Erase_1ms_Lib	Address <u>0x6F899</u> <u>0x6F8A3</u>

Remark: Definition in the above table is classified according to the content of changes as follows.

Error correction, Description change, Description addition, Description deletion: For the description of LSI manual.

Specification change, Specification addition, Specification deletion: For microcontroller's specification.

The following shows the changes in the publication of "MN101EF57G/76K LSI User's Manual" (From 1st Edition dated in April, 2009 to the 2nd Edition dated in September, 2011.)

Page	Line	Definition	Former Edition (1.0)	New Edition (2.0)
-	All Chapters	Description addition	-	Added the specification for MN101EF76K.
-	All Chapters	Error correction	PAPLUD Port A pull-up/ <u>pull-down</u> resistor control register	PAPLU Port A pull-up resistor control register
-	All Chapters	Description deletion	-	Deleted SELUD2 (pull-up/pull-down resistor selection register 2).
-	-	Description deletion	-	Deleted reset pin of PnOUT registers.
I-2	22	Description deletion	The package is 80-pin, LQFP.	-
I-5	22	Description deletion	16-bit cascade connection (connected with timer 0), <u>timer synchronous output</u>	16-bit cascade connection (connected with timer 0)
I-28	Table: 1.3.4 <NRST>	Error correction	...it is recommended that a discharge diode be placed between NRST and <u>VDD</u> .	...it is recommended that a discharge diode be placed between NRST and <u>VDD5</u> .
I-30	Table: 1.3.4 <PA0 to PA7>	Description deletion	...A pull-up/ <u>pull-down</u> resistor for each bit can be selected individually by the PAPLUD register. <u>A pull-up/down resistor connection for each port can be selected individually by the SELUD2 register.(A pull-up/pull-down can not be mixed.)</u> At reset,...	...A pull-up resistor for each bit can be selected individually by the PAPLU register. At reset,...
I-32	Table: 1.3.4 <VREF+>	Error correction	...This pin is generally used as VREF+ = <u>VDD</u> .	...This pin is generally used as VREF+ = <u>VDD5</u> .
I-34	Table: 1.3.4 <MMOD>	Description change	Fix this pin at VSS level.	Set to VDD5-level or VSS-level.
I-35	Note	Error correction	...it is necessary to connect the pull-up resistor on the <u>substrate</u> .	...it is necessary to connect the pull-up resistor on the <u>circuit board</u> .
I-38	1, 2	Description addition	-	This LSI manual describes standard specifications. When....
I-41	B16, B18 Figure: 1.5.2, 1.5.3	Description change	<u>f_{hosc-1}</u> , <u>f_{sosc-1}</u>	<u>f_{hosc1}</u> , <u>f_{sosc1}</u>
I-42	B20, B25	Description change	<u>f_{hosc-2}</u> , <u>f_{sosc-2}</u>	<u>f_{hosc2}</u> , <u>f_{sosc2}</u>
I-44	C9, C10	Error correction	frc=20 MHz [Double-speed mode: fs= <u>fosc</u>]	frc=20 MHz [Double-speed mode: fs= <u>frc</u>]
I-46	*17	Error correction	...,it is necessary to connect the pull-up resistor on the <u>substrate</u> .	...,it is necessary to connect the pull-up resistor on the <u>circuit board</u> .
I-49	C68 to C72	Description change	Condition: <u>VDD5 = 5.0 V (Figure: 1.5.6)</u>	Condition: <u>Figure: 1.5.6</u>
I-49	C74	Error correction	Display output pin 2 SEG0 to <u>SEG31</u>	Display output pin 2 MN101EF57G: SEG0 to <u>SEG40</u>
I-49	*20	Error correction	<u>However</u> , COM0 to COM3 are also used as <u>P84 to P87</u> .	COM0 to COM3 are also used as <u>P81 to P84</u> .
I-49	*22	Description change	<u>Summation of 3 resistors among</u> V _{LC1} and...V _{SS} .	<u>Total of 3 resistor values between</u> V _{LC1} and...V _{SS} , <u>respectively</u> .
I-51	1.5.5	Description change	E13 Reference voltage pin input leakage current Condition: <u>Vref+ = OFF</u>	E13 Reference voltage pin input leakage current Condition: <u>Ladder resistance</u> OFF
I-51	1.5.5	Error correction	E14 Ladder resistance MIN:15 TYP: <u>35</u> MAX: 80	E14 Ladder resistance MIN:15 TYP: <u>40</u> MAX: 80

Page	Line	Definition	Former Edition (1.0)	New Edition (2.0)
I-54	1.5.8	Error correction	H21 Minimum pulse width Unit: <u>ms</u>	H21 Minimum pulse width Unit: <u>μs</u>
			H23 Consumption current in power supply voltage detection circuit Conditions: $V_{DD} = 5.0\text{ V}$	H23 Consumption current in power supply voltage detection circuit Conditions: $V_{DD5} = 5.0\text{ V}$
I-55	1.5.9	Description change	I. Internal High-speed Oscillation Circuit <u>I1, I2</u>	I. Internal High-speed Oscillation Circuit <u>I1 to I4</u> <u>*24 The specification values described in I3 and I4 are....</u>
I-56	1.5.10	Description addition	K. Flash EEPROM Program Conditions -	K. Flash EEPROM Program Conditions <u>*25 The specification value described in K1 is....</u>
I-58	Figure: 1.6.2	Description change	-	Replaced the figure for 80-pin LQFP package dimension.
I-60	Table: 1.7.1	Description change	-	Replaced the table for recommended oscillators.
I-63	2 to 5	Description change	Unused I/O pins should be set according to <u>pins'</u> condition at reset. If the output....	Unused I/O pins should be set according to <u>port</u> direction control. In the input mode....
	figure: 1.7.5	Description deletion	Figure: 1.7.5 <u>Unused I/O Pins (high impedance output at reset)</u>	Figure: 1.7.5 <u>Unused I/O Pins</u> <Deleted the figure for setting the unused Nch open-drain pins.>
	Table: 1.7.2	Description addition	-	Added the table: 1.7.2
I-65 I-66 I-67	Note	Description addition	-	<u>The circuit connected with the reset pin should be evaluated....</u>
I-65	5	Error correction	[The time of V_{DD18} is over 1.65 V (t_{VDD18})] <u>≤</u> [The time of Reset pin voltage should be low (t_{NRSTL})]	[The time of V_{DD18} is over 1.65 V (t_{VDD18})] <u>× 2 ≤</u> [The time of Reset pin voltage should be low (t_{NRSTL})]
I-66	Figure: 1.7.9	Description change	-	Changed the term of tATRSTH.
II-15	Note 2	Description deletion	<u>The MMOD pin should be fixed at "L" level.</u> Do not change the settings of the MMOD pin even after reset release.	Do not change the settings of MMOD pin even after reset is released.
	Hint	Description addition	-	For MMOD setup to rewrite the flash memory, refer to ...
II-23	3	Description addition	-	Address 0x03C00 to 0x03DEF are reserved.
II-24	Table: 2.2.5	Error correction	03EBF: <u>SELUD2</u>	03EBF: <u>Reserved</u>
II-25	Note 1	Description addition	-	<u>For details of the P4_VIL to P0_VIL flags....</u>
	Note 2	Description change	-	<Transferred from page II-26> <u>For details of the WD2OP2 to WD2OP0 flags....</u>
	Note 3	Description change	-	<Transferred from page II-26> <u>The flash option function is determined before internal reset release....</u>
II-26	Note	Description addition	-	<u>For details of the PB_VIL to P5_VIL flags....</u>
II-27 to II-35	Chapter 2 2.3	Description addition	-	Added Chapter 2 2.3 ROM Correction.
II-36	Figure: 2.4.1	Description deletion	-	Deleted ATC block.

Page	Line	Definition	Former Edition (1.0)	New Edition (2.0)
II-47	■Initiating a Reset	Error correction	<u>100</u> μs	<u>1</u> μs
	Note 2	Description deletion	<u>When NRST pin connected to low power voltage detection circuit that gives plus for...</u>	-
	Note 3	Description addition	-	<u>Refer to [Chapter 1 1.7.4 Power Supply] for reset control at power supply on.</u>
II-39 (Former Edition)	■Reset Sequence at Power-On	Description deletion	<Deleted the page II-39 in former edition> ■ <u>Reset Sequence at Power-On</u> <u>This LSI is equipped....</u> <u>Note....</u> <u>Figure: 2.6.2</u>	-
II-50	Note 1	Error correction	...which is set by <u>DLY1-0</u> flags...	...which is set by <u>DLYS1 and DLYS0</u> flags...
III-2	4 to 19	Description addition	-	1) ... <u>At reset releasing, oscillation circuit is operated at 20 MHz.</u> 2) ... <u>At reset releasing, oscillation circuit is operated.</u> 3) ... <u>At reset releasing, oscillation circuit is stopped.</u> 4) ... <u>At reset releasing, oscillation circuit is stopped.</u> 5) ... <u>At reset releasing, PLL circuit is stopped.</u>
III-7	Figure: 3.1.2	Error correction	RCCNT bp2: <u>RCFSEL</u> OSCCNT bp4 to bp 7: <u>HOSCIV...</u>	RCCNT bp2: - OSCCNT bp4 to bp 7: <u>HOSCIV...</u>
III-10	OSCCNT	Error correction	bp 7 to 4 3 2 1 to 0 At reset - <u>1</u> <u>0</u> - Access - <u>R/W</u> <u>R/W</u> -	bp 7 to 4 3 2 1 to 0 At reset <u>0</u> - - <u>0</u> Access <u>R/W</u> - - <u>R/W</u>
III-11	Note 1	Error correction	..., wait <u>more than 30 μs</u> after the setting the RCON flag of the RCCNT register. Then <u>clear</u> the HOSCSEL flag <u>and change the operation clock.</u>	..., wait for <u>30 μs or longer</u> after setting the RCON flag of the RCCNT register <u>to "0"</u> . Then <u>set</u> the HOSCSEL flag <u>of the OSCCNT register to "0"</u> .
III-12	Note 4	Error correction	..., <u>set the RCON flag and wait more than 30 μs.</u> Then <u>clear</u> the HOSCSEL flag of the OSCCNT register.	..., <u>wait for 30 μs or longer after setting the RCON flag of RCCNT register to "0"</u> . Then <u>set</u> the HOSCSEL flag of OSCCNT register <u>to "0"</u> .
III-14	Note 4	Error correction	...,wait more than 200 μs after <u>the setting</u> the RCSON flag of the RCSCNT register. Then <u>clear</u> the SOSCSEL flag <u>and change the operation clock.</u>	...,wait more than 200 μs after <u>setting</u> the RCSON flag of RCSCNT register <u>to "0"</u> . Then <u>set</u> the SOSCSEL flag <u>of OSCSCNT register to "0"</u> .
III-16	RCSCNT	Description change	bp 7 Flag <u>Reserved</u> At reset <u>1</u> Access <u>R/W</u> bp Flag Description 7 <u>Reserved</u> <u>Always set to "1"</u> .	bp 7 Flag - At reset - Access - bp Flag Description 7 - -
III-16	*1	Description change	Rewriting is <u>enable</u> when <u>SOSCSEL flag of the OSCCNT register is "1"</u> and <u>WD2MD7-0 flags of the WD2MDSEL register are "00"</u> .	Rewriting is <u>enabled</u> when <u>WD2MD7 to WD2MD0 flags of WD2MDESEL register are "0x00"</u> .
III-16	Note 5	Description change	...,wait more than 200 μs after <u>clearing</u> the SOSCSEL flag of the OSCSCNT register <u>and switch the operation clock.</u>	...,wait more than 200 μs after <u>setting the RCSON flag of RCSCNT register to "0"</u> . Then <u>set</u> the SOSCSEL flag of OSCSCNT register <u>to "0"</u> .

Page	Line	Definition	Former Edition (1.0)	New Edition (2.0)
III-20	PLL CNT	Description change	PLLCK3-0 4'h0: 2 <u>multiplication</u> ... 4'h1: 2 <u>multiplication</u> ... 4'h2: 3 <u>multiplication</u> ... 4'h3: 4 <u>multiplication</u> ... 4'h4: 4 <u>multiplication</u> ... 4'h5: 5 <u>multiplication</u> ... 4'h6: 5 <u>multiplication</u> ... 4'h7: 6 <u>multiplication</u> ... 4'h8: 8 <u>multiplication</u> ... 4'h9: 10 <u>multiplication</u> ...	PLLCK3, PLLCK2, PLLCK1, PLLCK0 0000: Multiply by 2.... 0001: Multiply by 2.... 0010: Multiply by 3.... 0011: Multiply by 4.... 0100: Multiply by 4.... 0101: Multiply by 5.... 0110: Multiply by 5.... 0111: Multiply by 6.... 1000: Multiply by 8.... 1001: Multiply by 10.... 1010: <u>Setting prohibited</u> 1011: <u>Setting prohibited</u> 1100: <u>Setting prohibited</u> 1101: <u>Setting prohibited</u> 1110: <u>Setting prohibited</u> 1111: <u>Setting prohibited</u>
III-21	Note 3	Error correction	changing the <u>OSCSEL</u> flag of the OSCCNT register....	changing the <u>HOSCSEL</u> flag of OSCCNT register.
III-22	CPUM	Description addition	 1011: STOP1	 1011: STOP1 <u>Others: Setting prohibited</u>
III-25	Hint 3	Description addition	-	<u>On the condition of $f_s \leq 4$ MHz ($1.8\text{ V} \leq V_{DD5} < 2.7\text{ V}$), $f_s \leq 10$ MHz...</u>
III-27 III-29 to III-31	Hint	Description addition	Refer to [Chapter 6 Power Supply Voltage Detection Function] for...	<u>The frequency used as a system clock varies depending on...</u> Refer to [Chapter 6 Power Supply Voltage Detection] for...
III-29	■Transition from OSC mode to RC mode	Error correction	Transition from <u>RC</u> mode to <u>OSC</u> mode requires...	Transition from <u>OSC</u> mode to <u>RC</u> mode requires...
	Note 1	Description change	..., <u>set the RCON flag and wait 30 μs and over</u> . Then <u>clear</u> the HOSCSEL flag of the OSCCNT register.	..., <u>wait more than 30 μs after setting the RCON flag of RCCNT register to "0"</u> . Then <u>set</u> the HOSCSEL flag of OSCCNT register <u>to "0"</u> .
III-30	Program 3	Description change	 BSET (RCCNT)1 Set the internal.... <u>MOV X'8C', D0</u> <u>LOOP ADD -1, D0</u> <u>Loop to wait....</u> <u>BNE LOOP</u> BCLR (CPUM)0 Set NORMAL....	 BSET (RCCNT)1 Set the internal.... <u>NOP</u> BCLR (CPUM)0 Set NORMAL....
III-33	Note 1	Description change	After setting the RCSON flag of the RCSCNT register to set the internal low-speed oscillation <u>circuit</u> , transition to SLOW mode <u>is</u> executed after waiting more than 200 μ s.	After setting the RCSON flag of RCSCNT register <u>to "0"</u> to set the internal low-speed oscillation <u>operation</u> , transition to SLOW mode <u>should be</u> executed after waiting more than 200 μ s.
	Note 2	Error correction	After the reset release, do not transit <u>300 ms</u> to SLOW, HALT1 and STOP modes.	After the reset is released, wait for <u>30 ms or longer</u> to change the operation mode to SLOW, HALT1 or STOP.
III-34	Hint	Description deletion	When setting the SOSSEL flag of the OSCSCNT register to change....	-
	Note 2	Description addition	-	After the reset is released, wait for 30 ms or longer to change the operation mode to SLOW, HALT1 or STOP.
III-35	Program 8	Description deletion	 BSET (OSCSCNT)1 Set the low-speed... <u>BSET (OSCSCNT)1</u> <u>Set XI mode</u> MOV X'88', (PRTKEY) Release....	 BSET (OSCSCNT)1 Set the low-speed... MOV X'88', (PRTKEY) Release....
	Hint	Description deletion	When setting the SOSSEL flag of the OSCSCNT register to change....	-

Page	Line	Definition	Former Edition (1.0)	New Edition (2.0)
III-36	Note 1	Description change	..., <u>set the RCSON flag of the RCSCNT register and</u> wait more than 200 μ s. Then <u>clear</u> the SOSSEL flag.	..., wait more than 200 μ s <u>after setting the RCSON flag of RCSCNT register to "0"</u> . Then <u>set</u> the SOSSEL flag of OSCSCNT register to "0".
	Note 2	Error correction	When the SOSCCNT flag is cleared to set <u>P90/P91</u> to...	When the SOSCCNT flag is cleared to set <u>P25/P26</u> to...
III-39	Note 2	Error correction	After the reset releasing, do not transfer <u>300</u> ms to SLOW,....	After the reset is released, wait for <u>30 ms or longer</u> to change...
III-41	■Transition to STOP mode	Error correction	Program <u>Z</u>	Program <u>14</u>
IV-15	■Setup Example: Clock Transition Mode is Not Used	Error correction	Upper limit frequency: 27 <u>MHz</u> Lower limit frequency: <u>15 MHz</u> Setup Procedure (6).... bp7-0: MNLSET[7:0] = <u>0x0f</u> Description (5)....the upper limit frequency to 27 <u>MHz</u> . (6)....the lower limit frequency to <u>15 MHz</u> .	Upper limit frequency: 27 <u>kHz</u> Lower limit frequency: <u>1 kHz</u> Setup Procedure (6).... bp7 to 0: MNLSET[7:0] = <u>0x01</u> Description (5)....the upper limit frequency to 27 <u>kHz</u> . (6)....the lower limit frequency to <u>1 kHz</u> .
IV-19, IV-20	■Setup Example: Clock Transition Mode is Used	Error correction	Upper limit frequency: 27 <u>MHz</u> Lower limit frequency: 15 <u>MHz</u> Description (5)....the upper limit frequency to 27 <u>MHz</u> . (6)....the lower limit frequency to 15 <u>MHz</u> .	Upper limit frequency: 27 <u>kHz</u> Lower limit frequency: 15 <u>kHz</u> Description (5)....the upper limit frequency to 27 <u>kHz</u> . (6)....the lower limit frequency to 15 <u>kHz</u> .
V-2	1	Description change	This LSI has a register protect function to prevent writing errors <u>of</u> the control registers <u>of</u> fail-safe function....	This LSI has a register protect function to prevent writing errors <u>to</u> the control registers <u>as</u> fail-safe functions....
V-5	Figure: 5.3.1	Error correction	<u>NM</u> interrupt	<u>NMI</u> interrupt
VI-3	LVIMD	Error correction	bp Description 7 to 4 : 1 1: $V_{DD5} \geq$ detection voltage	bp Description 7 to 4 : <u>1000: 3.0 V</u> 1 1: $V_{DD5} \geq$ detection voltage
VI-4	Hint	Error correction	The detection voltage which can be monitored by the <u>LVIMON</u> flag is the voltage detected....	The detection voltage which can be monitored by the <u>LVION</u> flag is the voltage detected....
	Note 2	Error correction	The <u>LVIMON</u> flag and the <u>LVICR0</u> to 2 flags must be set when the LVION flag is "0".	The <u>LVIMOD</u> flag and the <u>LVICTR0</u> to <u>LVICTR2</u> flags must be set when the LVION flag is "0".
VI-5, 6	6.3.1	Error correction	<u>LVI1ICR</u>	<u>LVIICR</u>
VII-4	Figure: 7.1.1	Description change	<u>Vector 29</u>	<u>Vector 30</u>
VII-7	Figure: 7.1.3	Description change	<u>Vector 8</u>	<u>Vector 7</u>
VII-10	Figure: 7.1.3	Description change	PC7 to <u>1</u>	PC7 to <u>0</u>
VII-11	Note 2	Description change	when program counter (PC [<u>bit20 to bit17, bit0</u>]) are saved to the stack....	when program counter (PC [<u>bit19 to bit16, bitH</u>]) are saved to the stack....
VII-32	Note	Description deletion	Do not word access to the register denoted by *.	-
VII-43 to VII-52	Figure: 7.3.1 to Figure: 7.3.10	Description change	-	Deleted IRQCNT register. Added AC zero-cross function (IRQ0, 1). Corrected EDGDT register (IRQ2, 3, 4). bp2: EDGSEL2 bp3: EDGSEL3 bp5: EDGSEL5
VII-51	3	Description addition	-	The IRQCNT register should be operated when...
	Hint	Description addition	-	Set P2nEN flag to "0" to fix the input of external interrupts...
	Note	Description addition	-	To enable maskable interrupts after setting P2nEN flag...

Page	Line	Definition	Former Edition (1.0)	New Edition (2.0)
VII-53	5	Description addition	-	The NFnEN1 flags should be operated when...
	Note	Description addition	-	To enable maskable interrupts after setting NFnEN1 flag...
VII-54	3	Description addition	-	The ACZCTR register should be operated when...
	Note	Description addition	-	To enable maskable interrupts after setting P2nIM flag...
VII-57	5	Description addition	-	The KEYT3_1ENn flags should be operated when...
	Note 2	Description addition	-	To enable maskable interrupts after setting KEYT3_1ENn flag...
VII-58	4	Description addition	-	The KEYT3_2IMD register should be operated when...
	Note	Description addition	-	To enable maskable interrupts after setting KEYT3_2ENn flag...
VIII-67, 68	7.3.8	Description addition	-	Added [Chapter 7 7.3.8 AC Zero-Cross Detection]
VII-80	4	Description deletion	...the previous level is sent. <u>It means that only...</u>	...the previous level is sent.
VIII-2	Table:8.1.1	Error correction	Port A: No pull-up/ <u>pull-down</u> resistor	Port A: No pull-up resistor
VIII-5	Table:8.2.1	Error correction	<u>PAPLUD</u> : Port A pull-up/ <u>pull-down</u> resistor control register	<u>PAPLU</u> : Port A pull-up resistor control register
		Description deletion	SEUD2 register	-
VIII-6	Table:8.3.1	Description addition	-	Added Table:8.3.1
VIII-7	15 to 28	Description addition	-	P05 is also used as the I/O pin for Serial Interface 3... For P01 to P04 each bit...by Port0 output mode register 2 (P0OMD2).
VIII-51	Figure:8.5.7 Figure:8.5.8	Description addition	-	Added AC-zero-cross detection circuit and selector.
VIII-68	Figure:8.6.8	Description addition	-	Added Port input mode control
VIII-34	Figure:8.6.9	Description addition	"SC4BTS"	"SC4 <u>S</u> BTS"
VIII-111	20 to 22	Description addition	-	For P62 to P63 each bit can be set individually...
VIII-134	P7ODC	Error correction	P7ODC: <u>0x03EF5</u>	P7ODC: <u>0x03EF7</u>
VIII-172 to VIII-174	Figure:8.11.14 to Figure:8.11.16	Description addition	-	Added switch for VLC3 to VLC1
VIII-175	■Special Function Pin Setup	Error correction	P92 to <u>P94</u> are also used... P92 to <u>P94</u> are also used... <u>the value of the port 9 input register is read to be "0"</u> .	P92 to <u>P95</u> are also used... P92 to <u>P95</u> are also used... <u>undefined value is read from Port 9 input register (P9IN)</u> .
		Description addition	-	P94 and P95 are also used as the analog output pins for D/A.
VIII-185	Figure:8.12.8	Description addition	-	Added [Figure:8.12.8 P91 Block Diagram (MN101EF57G)]
VIII-189	10	Description deletion	... <u>port A can be selected to add pull-up or pull-down resistor by bp2 of the pull-up/pull-down resistor selection register 2(SELUD2).</u> <u>For port A</u> , each bit can...	... Each bit can...

Page	Line	Definition	Former Edition (1.0)	New Edition (2.0)
VIII-190	■Special Function Pin Setup	Error correction	PA0 to PA7 are also used... <u>the value of the port A input register is read to be "0"</u> .	PA0 to PA7 are also used... <u>undefined value is read from Port A input register (PAIN)</u> .
		Description change	PA0 is also used as the I/O pin of timer 0. ... PA7 is also used as the I/O pin of timer 9....	PA0 to PA7 are also used as the I/O pins for Timer 0 to Timer 6 and Timer 9...
VIII-191	Table:8.13.2	Description deletion	SELUD2 register	-
VIII-103 (Former Edition)	-	Error correction	■ Pull-UP/pull-down Resistor Selection Register 2 (SELUD2: 0x03EBF)	Deleted the page
VIII-203 to VIII-209	Figure:8.13.8 to Figure:8.13.14	Description deletion	-	Deleted Pull-up/pull down resistor selection
VIII-210	■Special Function Pin Setup	Error correction	PB0 to <u>PB3</u> are also used... PB0 to <u>PB3</u> are also used... <u>the value of the port B input register is read to be "0"</u> .	PB0 to <u>PB7</u> are also used... PB0 to <u>PB7</u> are also used... <u>undefined value is read from Port B input register (PBIN)</u> .
VIII-244	8.19.4 Input Voltage Level	Description change	■ Input Voltage Level P0 to <u>P9</u> ,PB	■ Input Voltage Level P0 to PB
		Error correction	Parameter MAX Input voltage level B: Input low voltage <u>0.5</u> V _{DD5}	Parameter MAX Input voltage level B: Input low voltage <u>0.45</u> V _{DD5}
IX-5	Table:9.1.3	Error correction	Timer 0 to Timer 4 Clock source <u>Simple timer A input</u>	Timer 0 to Timer 4 Clock source <u>Timer A output</u>
			Timer 1 Synchronous <u>TM1IO input</u> Timer 3 Synchronous <u>TM3IO input</u>	Timer 1 Synchronous <u>TM1IO input</u> Timer 3 Synchronous <u>TM3IO input</u>
IX-9 IX-12 IX-14	Figure:9.1.3 Figure:9.1.6 Figure:9.1.8	Error correction	CKnMD (<u>bp5-3</u>)	CKnMD (<u>bp6 to 4</u>)
IX-21	Note	Description addition	-	In double buffer configuration, when the clear timing of binary counter occurs...
IX-25 IX-27	Note	Description addition	-	To load data from a compare register to an internal register...
IX-42	Note 1	Description addition	-	MN101EF57G does not have double-buffer function...
IX-47	Note 3	Description change	In double buffer configuration, <u>the event input (TM2IO input, TM3IO input) is selected as...</u>	In double-buffer configuration, <u>the clock other than fs is selected as...</u>
IX-50	Note	Description addition	-	When the synchronous TMnIO input is selected as the count clock source...
IX-57	Figure:9.7.5	Error correction	CKnMD <u>bit 4, 5</u>	CKnMD <u>bit 6, 5</u>
IX-60	Table:9.9.1	Description change	· Timer 0: Serial interface 0, 2, 4 · Timer 1: Serial interface 0, 2, 4 · Timer 2: Serial interface 0, 2, 4	Replaced to Table:9.9.1
IX-64 IX-68 IX-70	Table:9.11.1 Table:9.11.2 Table:9.11.3	Description addition	-	Added "Timer pulse output"
IX-71	Note 2	Error correction	At 32-bit cascade connection, always set TM3CAS flag of TM3MD register to " <u>0</u> ".	At 32-bit cascade connection, always set TM3CAS flag of TM3MD register to " <u>1</u> ".
IX-72	Note	Description addition	-	In cascade connection (16-bit, 24-bit, 32-bit) with double-buffered compare register...
	Figure:9.11.1	Description addition	-	Added Figure:9.11.1

Page	Line	Definition	Former Edition (1.0)	New Edition (2.0)																		
XI-8	Figure:11.1.5	Error correction																				
XI-22	Note	Description change	When T7ICT1 to 0 flags <u>are set to timer interrupt</u> , the <u>timer</u> set by T7ICT2 flag of the TM7MD4 register is <u>set</u> as a capture trigger.	When T7ICT1 to 0 flags <u>select "timer interrupt"</u> , the <u>timer interrupt</u> set by T7ICT2 flag of TM7MD4 register is <u>used</u> as a capture trigger.																		
XI-27	Note	Description change	When T8ICT1 to 0 flags <u>are set to timer interrupt</u> , the <u>timer</u> set by T8ICT2 flag of the TM8MD4 register is <u>set</u> as a capture trigger.	When T8ICT1 to 0 flags <u>select "timer interrupt"</u> , the <u>timer interrupt</u> set by T8ICT2 flag of TM8MD4 register is <u>used</u> as a capture trigger.																		
XI-29	TM8MD4	Error correction	<table><tr><th>bp</th><th>Flag</th><th>Description</th></tr><tr><td>3</td><td><u>TM8SEL_B</u></td><td><u>B port selection</u></td></tr><tr><td>2</td><td><u>TM8SEL_A</u></td><td><u>A port selection</u></td></tr></table>	bp	Flag	Description	3	<u>TM8SEL_B</u>	<u>B port selection</u>	2	<u>TM8SEL_A</u>	<u>A port selection</u>	<table><tr><th>bp</th><th>Flag</th><th>Description</th></tr><tr><td>3</td><td><u>Reserved</u></td><td><u>Always set to "0"</u></td></tr><tr><td>2</td><td><u>TM8SEL_A</u></td><td><u>A port selection</u></td></tr></table>	bp	Flag	Description	3	<u>Reserved</u>	<u>Always set to "0"</u>	2	<u>TM8SEL_A</u>	<u>A port selection</u>
bp	Flag	Description																				
3	<u>TM8SEL_B</u>	<u>B port selection</u>																				
2	<u>TM8SEL_A</u>	<u>A port selection</u>																				
bp	Flag	Description																				
3	<u>Reserved</u>	<u>Always set to "0"</u>																				
2	<u>TM8SEL_A</u>	<u>A port selection</u>																				
XI-37	Note 3	Description change	When the binary counter is read during the timer operation, <u>it is treated as the data by every 8 bits...</u>	When binary counter is read during the timer operation, <u>the data is processed as 8-bit unit data...</u>																		
XI-40	Hint	Description change	<u>If the binary counter is read out during operation, incorrect data at counting up may be read...</u>	<u>When event input (TMnIO input) is selected as a count clock source, do not read out values...</u>																		
	Note 4 (Former Edition)	Description deletion	<u>If the TM7IO input is selected as the count clock source, procedure such as pull-up is...</u>	-																		
XI-57	Figure:11.7.4	Error correction	Figure:11.7.4 <u>Pulse Width Measurement of External Interrupt 0</u>	Figure:11.7.4 <u>Output Waveform of TM7IO Output Pin</u>																		
XI-66	3	Description change	...the capture trigger is synchronized with the <u>capture sampling clock</u> which is set by...	...the capture trigger synchronized with the <u>clock</u> which is set by...																		
XI-67	Note	Error correction	...When setting the <u>capture clock as the count clock</u> to execute the event count operation...	...When setting the <u>TMnCKSMP flag to "Count clock"</u> to execute the event count operation...																		
XI-96	Table:11.13.1	Description addition	-	Added "Capture function" and "Pulse width measurement"																		
XII-2	Table:12.1.1	Description addition	-	Added Table:12.1.1 Timer 9 Pin Functions																		
XII-9	PWMMDH	Error correction	<table><tr><th>bp</th><th>Flag</th><th>Access</th></tr><tr><td>2</td><td>DTSEL</td><td><u>R/W</u></td></tr></table>	bp	Flag	Access	2	DTSEL	<u>R/W</u>	<table><tr><th>bp</th><th>Flag</th><th>Access</th></tr><tr><td>2</td><td>DTSEL</td><td><u>W</u></td></tr></table>	bp	Flag	Access	2	DTSEL	<u>W</u>						
bp	Flag	Access																				
2	DTSEL	<u>R/W</u>																				
bp	Flag	Access																				
2	DTSEL	<u>W</u>																				
XII-9	Note 2	Description addition	-	DTSEL flag can only write it...																		
XII-25	12.3	Description change	<u>12.3 Operation</u> <u>12.3.1 16-bit Timer Operation</u> <u>12.3.1.1 16-bit Timer Operation</u>	12.3 16-bit Timer Operation 12.3.1 Operation -																		
XII-29	12.3.2	Description change	<u>12.3.1.2 Setup Example</u>	12.3.2 Setup Example																		
XII-30	12.4	Description change	<u>12.3.2 16-bit Event Count</u> <u>12.3.2.1 Operation</u>	12.4 16-bit Event Count 12.4.1 Operation																		
XII-32	12.4.2	Description change	<u>12.3.2.2 Setup Example</u>	12.4.2 Setup Example																		
XII-32	■ Event Count Setup Example	Error correction	Setup Procedure (3) Set the special function pin to input PADIR(0x03E9A) bp7: <u>PADIR0</u> = 0	Setup Procedure (3) Set the special function pin to input PADIR(0x03E9A) bp7: <u>PADIR7</u> = 0																		
XII-33	12.5	Description change	<u>12.3.3 16-bit Timer Pulse Output</u> <u>12.3.3.1 Operation</u>	12.5 16-bit Timer Pulse Output 12.5.1 Operation																		
XII-34	12.5.2	Description change	<u>12.3.3.2 Setup Example</u>	12.5.2 Setup Example																		

Page	Line	Definition	Former Edition (1.0)		New Edition (2.0)	
XVIII-28	SCnMD1 (SCnCKM)	Description addition	Description Transfer clock division selection 0: Not divided 1: Divided		Description <u>Clock synchronous (SCnCMD=0)</u> Transfer clock division selection 0: Not divided 1: Divided <u>UART (SCnCMD=1)</u> <u>Transfer clock is divided regardless of the setting value.</u> <u>The data read out is the setting value.</u>	
	Note 2	Description addition	-		Writing to watchdog timer 2 operation mode register is possible only once...	
XVIII-34	18.2.4	Error correction	This serial interface consists of 2 buffers and <u>7</u> registers.		This serial interface consists of 2 buffers and <u>8</u> registers.	
XVIII-39	SC4AD0 SC4AD1	Error correction	<u>SC4AD9</u> to <u>SC4AD0</u>		<u>SC4ADR9</u> to <u>SC4ADR0</u>	
XVIII-57	Note 3	Description change	<u>When the SCnIGC flag of clock synchronous serial reception interrupt control register (SCINTSEL) is...</u>		<u>SCnIGC flag of clock synchronous serial reception interrupt control register (SCINTSEL) is set to...</u>	
XVIII-63	Table:18.3.8	Error correction	Setup item Set to Serial data input selection <u>SBI1A</u>		Setup item Set to Serial data input selection <u>SBI1</u>	
XVIII-63 XVIII-66	Setup Procedure	Description change	Description (1) Set the SC1PSCE flag of the SC1MD3 register to "1" to select <u>"prescaler operation"</u> .		Description (1) Set the SC1PSCE flag of SC1MD3 register to "1" to select <u>"Enable count" for prescaler count control.</u>	
XVIII-86	Setup Procedure	Description addition	Setup Procedure (6)... Select the clock dividing SC1MD1 (0x03F22) bp3: SC1CKM =1 ...	Description (6)... Set the SC1CKM flag... The SC1MST flag...	Setup Procedure (6)... Select the clock dividing SC1MD1 (0x03F22) bp3: SC1CKM =1 <u>bp1: SC1DIV =0</u> ...	Description (6)... Set the SC1CKM flag... <u>Set the SC1DIV flag...</u> <u>clock division value...</u> The SC1MST flag...
	Setup Procedure	Description change	Setup Procedure (<u>1</u>) Set the baud rate timer	Description (<u>1</u>) Set the baud rate by the TMAMD register and...	Setup Procedure (<u>7</u>) Set the baud rate timer	Description (<u>7</u>) Set the baud rate by TMAMD register and...
XVIII-91	2nd line under Table:18.5.4	Error correction	<u>SC4ADm</u> flag of SC4ADn register		<u>SC4ADRm</u> flag of SC4ADn register	
XVIII-106	18.5.2	Error correction	Pin setup (flag setup) SC4MD1 register ... <u>SBT2</u> pin function selection	... <u>SBO2 (RXD2)</u> pin function selection	Pin setup (flag setup) SC4MD1 register ... <u>SBT4</u> pin function selection	... <u>SBO4 (RXD4)</u> pin function selection
XVIII-107 XVIII-110	Table:18.5.6 Table:18.5.7	Error correction	Setup item Set to Serial data input selection <u>SDA4A</u>		Setup item Set to Serial data input selection <u>SDA4</u>	
XVIII-107 XVIII-110	Setup Procedure	Description change	Description (1) Set the SC4PSCE flag of the SC4MD2 register to "1" to select <u>prescaler operation.</u>		Description (1) Set the SC4PSCE flag of SC4MD2 register to "1" to select <u>"Enable count" for prescaler count control.</u>	
XVIII-112		Error correction	Description (13) Communication completion interrupt (<u>SC4TIRO</u>) is generated...		Description (1) Serial 4 communication completion interrupt (<u>SC4IRQ</u>) is generated...	
XX-2	Table:20.1.1	Description addition	-		Added Table:20.1.1 A/D Converter Pin Functions	
XX-7	ANCTR0 bp2, 5 to 4	Error correction	Flag ANCK <u>5</u> ANCK <u>4</u> ANCK <u>2</u>	Description -	Flag ANCK <u>2</u> ANCK <u>1</u> ANCK <u>0</u>	Description <u>111: Setting prohibited</u>
XX-13	Note 4	Description change	<u>...Change the flags when A/D conversion is not executed or after forced stop.</u>		<u>...Set the ANLADE flag of A/D converter control register 0 (ANCTR0) to "0" to disable the operation before changing the flags.</u>	

Page	Line	Definition	Former Edition (1.0)	New Edition (2.0)
XX-14	Table:20.3.1	Error correction	A/D conversion cycle (T_{AD}) at low speed oscillation fslow=32.768 <u>MHz</u> 244.14 μ s 488.28 μ s 976.56 μ s 15.26 μ s 1953.12 μ s	A/D conversion cycle (T_{AD}) Low-speed operation fslow=32.768 <u>kHz</u> 244.14 μ s (<u>setting prohibited</u>) 488.28 μ s (<u>setting prohibited</u>) 976.56 μ s (<u>setting prohibited</u>) 15.26 μ s 1953.12 μ s (<u>setting prohibited</u>)
XX-21	Hint	Description addition	-	This LSI contains a sample hold capacitor (C) (approximately. 10 pF)...
	Table:20.3.4	Description addition	-	Added Table:20.3.4 Impedance of analog signal output circuit
	Figure:20.3.5	Description addition	-	Added Figure:20.3.5 Circuit Example
XXI-2	Table:21.1.1	Description addition	-	Added Table:21.1.1 D/A Converter Pin Functions
XXI-2	Table:21.1.2	Description addition	-	Input Range: V_{SS} to V_{REF+}
XXII-2	Table:22.1.1	Description deletion	Clock Source ... (LCDCLK) <u>Timer 0 to 4 or simple timer A selection</u>	Clock Source ... (LCDCLK) <u>Timer 0 to Timer 4</u>
XXII-5	Figure:22.1.1	Description deletion	<u>Timer A</u>	$\pm$ (Deleted Timer A input)
XXII-9	Note	Description addition	-	To use Timer n (n= 0 to 4) as the LCD clock source, set the LCDEN flag...
XXIV-2	Figure:24.1.1	Description addition	-	Added Figure:24.1.1 Sector Structure
XXIV-4	■ Parallel... ■ Serial... ■ User...	Description deletion	Sector 0 to 6 can be rewritten.	-
	■ BOOT...		Sector 0 to 5 can be rewritten.	-
XXIV-4	Table:24.1.2	Description change	Rewritable area: <u>Sector 0 to 6</u> Features: <u>-Activating and rewriting areas...</u>	Rewritable area: <u>All sectors</u> Features: <u>-The last sector is used as activating...</u>
XXIV-5	Table:24.1.4	Description change	Sector ... SA6 (also used as BOOT area) BOOT area is used MAIN area ... (<u>Rewriting Prohibited</u>)	Sector ... SA6 (also used as BOOT area) When BOOT area is in used MAIN area ... <u>0x23000 to 0x23FFF</u>
	1st line under Table:24.1.4	Description deletion	*1 Unwritable. Reading or instruction execution...	-
XXIV-6	24.1.3	Description addition	...MAIN area is to store user's program, and the area is rewritable. Reserved area is to store...	...MAIN area stores user program, and it is rewritable. <u>BOOT area activates BOOT mode microcontroller rewriting method, and it is rewritable.</u> Reserved area stores...

Page	Line	Definition	Former Edition (1.0)	New Edition (2.0)
XXIV-7	24.2	Description deletion	Parallel programmer rewriting is used to rewrite the...<u>For parallel programmer writing, sector 0 to 6 can be rewritten. The following rewriting methods are provided depending on the usage pattern of users.</u> ■ <u>General Usage</u> <u>(1) Entire area rewrite</u> ... ■ <u>When Using BOOT Mode Microcontroller...</u> <u>For BOOT mode microcontroller rewriting...</u> <u>(1) BOOT area rewriting in BOOT mode...</u> ... <u>(2) MAIN area rewrite in BOOT mode...</u> ...	Parallel programmer rewriting is used to rewrite the ...the usage pattern of users.
XXIV-7	Hint	Description change	If using entire area rewrite, batch rewriting...<u>However, when using entire area rewrite in...</u> The following description shows how to create a data file using debugger or <u>programmer</u> for the batch rewriting of MAIN area and BOOT area. (a) Using PanaXSeries... ... (b) Using <u>programmer</u> 1. Prepare data... 2. Write the data file for BOOT area <u>using BOOT area rewriting in BOOT mode microcontroller rewriting method.</u> 3. Write the data file for MAIN area <u>using MAIN area rewriting in BOOT mode microcontroller rewriting method.</u> 4. Read the data of the entire area <u>using entire area rewrite to write into file.</u>	To rewrite entire area, batch rewriting... The following description shows how to create a data file using debugger or <u>PX-FW2 programmer</u> for the batch rewriting of MAIN area and BOOT area. <u>For using the other programmer, it is...manufacturer.</u> (a) Using PanaXSeries... ... (b) Using <u>PX-FW2 programmer</u> 1. Prepare data... 2. Write the data file for BOOT area <u>in BOOT area.</u> 3. Write the data file for MAIN area <u>in BOOT area.</u> 4. Read the data of the entire area, <u>and write it into the file.</u>
XXIV-8	Figure:24.2.1	Description change	-	Replaced Figure:24.2.1 Parallel Programmer Rewriting Memory Map
XXIV-9	Table:24.2.1	Description change	Product ... Model MN101EF57G ... <u>AF8723</u>	Product ... Model MN101EF57G ... <u>AF9723</u>
XXIV-10	24.3	Description deletion	Serial programmer rewriting is used to rewrite...in the flash memory in advance. <u>For serial programmer writing, sector 0 to 6 can be rewritten. The following rewriting methods are provided depending on the usage pattern of users.</u> ■ <u>General Usage</u> <u>(1) Entire area rewrite</u> ... ■ <u>When Using BOOT Mode Microcontroller...</u> <u>For BOOT mode microcontroller rewriting...</u> <u>(1) BOOT area rewriting in BOOT mode...</u> ... <u>(2) MAIN area rewrite in BOOT mode...</u> ...	Serial programmer rewriting is used to rewrite...in the flash memory in advance.

Page	Line	Definition	Former Edition (1.0)	New Edition (2.0)
XXIV-10	Hint	Description change	If using entire area rewrite, batch rewriting... <u>However, when using entire area rewrite in...</u> The following description shows how to... (a) Using DebugFactory Builder ... (b) Using <u>programmer</u> 1. Prepare data... 2. Write the data file for BOOT area <u>using BOOT area rewriting in BOOT mode microcontroller rewriting method.</u> 3. Write the data file for MAIN area <u>using MAIN area rewriting in BOOT mode microcontroller rewriting method.</u> 4. Read the data of the entire area <u>using entire area rewrite to write into file.</u>	To rewrite entire area, batch rewriting... The following description shows how to... (a) Using DebugFactory Builder ... (b) Using <u>PX-FW2 programmer</u> 1. Prepare data... 2. Write the data file for BOOT area <u>in BOOT area.</u> 3. Write the data file for MAIN area <u>in MAIN area.</u> 4. Read the data of the entire area, <u>and write it into the file.</u>
XXIV-11	Figure:24.3.1	Description change	-	Replaced Figure:24.3.1 Serial Programmer Rewriting Memory Map
XXIV-14	24.4.2	Description change	<u>Writing to MN101EF57G, it uses the oscillator...</u> $Crst = \frac{\text{The oscillator stabilization time}}{(3 * \text{Pull-up resistor } R1)} \dots \text{Equation (1)}$	<u>After applying a reset to the target microcomputer...</u> $Crst = (250 * 10^{-3} - T_{wait}) / R1 \dots \text{Formula (1)}$
XXIV-15	24.4.3	Description change	The maximum output current from...<u>R_{upMin}</u> can be determined <u>from equation (2)</u> $R_{upMin} = \text{Operating supply voltage (V}_{DD5}) / 12 \text{ mA} \dots \text{Equation (2)}$	The maximum output current from...<u>R_{upRst}</u> can be determined <u>according to Formula (2)</u> $R_{upRst} = \text{Operating supply voltage (V}_{DD5}) / 12 \text{ mA} \dots \text{Formula (2)}$
XXIV-15	24.4.4	Description change	<u>If you want to insert a resistor with a large value...</u>	<u>If a resistor with a large value is inserted in series...</u>
XXIV-15	24.4.5	Description change	Find the maximum output current, <u>I_{OH}</u>, for the pins... $R_{upMin} = \text{Operating supply voltage (V}_{DD5}) / \text{Pin maximum output current (I}_{OH}) \dots \text{Equation (4)}$	Find the maximum output current, <u>I_{OL}</u>, for the pins... $R_{upMin} = \text{Operating supply voltage (V}_{DD5}) / \text{Pin maximum output current (I}_{OL}) \dots \text{Formula (4)}$
XXIV-18 XXIV-23	Figure:24.5.2 Figure:24.6.2	Error correction	[Internal RAM] Address: 0x00000 to <u>0x07FFE</u>: 6 Kbyte Address: 0xF0000 to <u>0xF7FFE</u>: 6 Kbyte(Mirror RAM)	[Internal RAM] Address: 0x00000 to <u>0x017FE</u>: 6 Kbyte Address: 0xF0000 to <u>0xF17FE</u>: 6 Kbyte(Mirror RAM)
XXIV-22	24.6 5 to 6	Description addition	...the user's program is destroyed <u>by mistake</u>. The LSI activates by releasing reset...	...the user program is destroyed <u>accidentally</u>. <u>In this rewriting method, the entire area can be...this method cannot be performed.</u> The LSI activates by releasing reset...
XXIV-23	5 to 6	Description change	...the command library becomes available to enable the rewriting of MAIN area (sector 0 to 5). ...The setting routine of the rewriting enable register <u>and the rewriting program</u> must be allocated in the internal RAM. <u>For BOOT area (sector 6) after changing...</u> After rewriting, when the activation mode...	...the command library becomes available to enable the rewriting of MAIN area (Sector 0 to 12[Sector 0 to 5]) <u>and BOOT area (Sector 13[Sector 6]).</u> ...The setting routine of the rewriting enable register must be allocated in the internal RAM. After rewriting, when the activation mode...
XXIV-24	Figure:24.6.3	Description addition	-	"BOOT area" is added in <Transfer the writing data> and <Execute the rewriting>.
	*1	Description deletion	...The setting routine of rewriting enable register <u>and the rewriting program</u> must be allocated in the internal RAM...	...The setting routine of rewriting enable register must be allocated in the internal RAM...
XXIV-24	24.6.2	Description deletion	<u>BOOT area (sector 6) is mapped in MAIN area and can be rewritten in user mode microcontroller rewriting.</u> If the sector 6 is rewritten <u>by mistake</u> , rewriting with...	If <u>BOOT area (Sector 13[Sector 6])</u> is rewritten <u>accidentally</u> , rewriting with...
XXIV-25	Table:24.7.2	Description addition	Activation Area Rewriting Area BOOT area <u>MAIN area</u>	Activation Area Rewriting Area BOOT area <u>MAIN area + BOOT area</u>
XXIV-28	24.7.4	Error correction	- The command library needs up to <u>613</u> byte as...	- The command library needs up to <u>623</u> byte as...

Page	Line	Definition	Former Edition (1.0)		New Edition (2.0)	
XXIV-28	24.7.4	Description addition	-		- To execute rewriting flash memory using... - When calling command library, it is required...	
XXIV-31	24.8	Description addition	-		Added 24.8 ROM Programming Service	
XXV-33	0x03F05 WD2CTR	Error correction	bit7	bit6	bit7	bit6
			-	-	<u>Reserved</u>	<u>Reserved</u>
			-	-	0	0
			-	-	<u>Always set to "0"</u>	<u>Always set to "0"</u>
XXV-39	0x03F7F TM8MD4	Error correction	bit3	bit2	bit3	bit2
			<u>TM8SEL_B</u>	<u>TM8SEL_A</u>	<u>Reserved</u>	<u>TM8SEL</u>
			0	0	0	0
			<u>B port timer8/ IGBT selection</u>	<u>A port timer8/ IGBT selection</u>	<u>Always set to "0"</u>	<u>Timer8/IGBT selection</u>

Remark: Definition in the above table is classified according to the content of changes as follows.

Error correction, Description change, Description addition, Description deletion: For the description of LSI manual.

Specification change, Specification addition, Specification deletion: For microcontroller's specification.

Inquiries

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